John B. Guerard Jr.
Anureet Saxena
Mustafa Gultekin

Quantitative Corporate Finance

Second Edition

Springer

Quantitative Corporate Finance

John B. Guerard Jr. • Anureet Saxena
Mustafa Gultekin

Quantitative Corporate Finance

Second Edition

Springer

John B. Guerard Jr.
McKinley Capital Management, LLC
Anchorage, AK, USA

Anureet Saxena
McKinley Capital Management, LLC
Stamford, CT, USA

Mustafa Gultekin
Kenan-Flagler Business School
University of North Carolina Chapel Hill
Chapel Hill, NC, USA

ISBN 978-3-030-43546-2 ISBN 978-3-030-43547-9 (eBook)
https://doi.org/10.1007/978-3-030-43547-9

© Springer Nature Switzerland AG 2007, 2021
This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors, and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Why a Second Edition of *Quantitative Corporate Finance*?

Eli Schwartz and I were very pleased with our *Quantitative Corporate Finance* text, published by Kluwer Academic Publishers in 2007. Eli was an economist and elder statesman, a sage, of 86. Eli had written a great book, *Corporate Finance*, published by St. Martin's in 1962. We combined my *Handbook of Financial Modeling* (coauthored with H.T. Vaught, for Probus, in 1989) and updated all materials for the Kluwer text. Eli was fond of telling me that when undergraduate economics students at MIT asked Robert Solow, the famed Nobel Prize Laureate, about what they needed to know to get a job on Wall Street, Professor Solow handed them a copy of Eli's book and told them to read and know the text. Eli earned his Ph.D. from Brown University and taught at Lehigh for 38 years. Eli was a visiting professor at the University of Pennsylvania, New York University, the London School of Economics, and Tel Aviv University. Eli published over 50 articles, including papers in the *American Economic Review*, *the Journal of Finance* (five), *Journal of Business, and the National Tax Journal.*

In 2010, Lehigh honored Eli with a festschrift for his 85th birthday. The festschrift included works by two Nobel laureates, Harry Markowitz and Robert Solow; Murray Weidenbaum, President Reagan's Chairman of the Council of Economic Advisors; John Hilley; several Lehigh colleagues; and one practitioner, me.

https://www2.lehigh.edu/news/schwartz-honored-with-festschrift

Nick Philipson, our publisher, attended the Lehigh celebration and an academic version of the festschrift was published by Springer, who had purchased Kluwer, in 2010. Nick made a great call to publish the book.

Eli Schwartz, 89, of Allentown, passed away in his home on August 31, 2010. Born in New York City, he was the husband of Renee (Kartiganer) Schwartz and they celebrated 62 years of marriage on August 29.

Eli was brilliant, a great conversationalist, and a very wise man. He is greatly missed.

There are other reasons for a second edition. First, the papers of the Schwartz festschrift are integrated into the revised edition. Second, our book was 99% complete when I became Director of Quantitative Research at McKinley Capital

Management (MCM) in Anchorage, Alaska. The final ten pages of chapter "Multifactor Risk Models and Portfolio Construction and Management" took me 18 months to complete. We successfully used the multifactor APT risk model featured in chapter "Multifactor Risk Models and Portfolio Construction and Management" for portfolio construction at McKinley for over 10 years (validated in the forthcoming *Wilmott Magazine* piece). Moreover, I taught in-house classes at McKinley Capital; 2-day, 15-hour seminars at Rutgers University, where I prepared the first edition of this book for my MBA classes; and the Georgia Institute of Technology Quantitative and Computational Finance (QCF) programs, where I recruited my MCM Quants. The Tech QCF Program is a top-ten Financial Engineering program. I grew up near Atlanta, earned my MSIM (Master's in Industrial Management) degree from Tech, and I am completely biased in hiring exceptionally bright and hard-working Tech QCF Quants. You will see references to Sundaram Chettiappan, Eli Krauklis, Manish Kumar, and Elaine Wang, all Tech QCF graduates with whom I published MCM Horse Race studies, to validate APT and Axioma risk models. These references are in chapter "Multifactor Risk Models and Portfolio Construction and Management." The extensions to chapters "Risk and Return of Equity and the Capital Asset Pricing Model" and "Multifactor Risk Models and Portfolio Construction and Management" reflect multifactor risk modeling at McKinley Capital, 2006–2019.

Third, Harry Markowitz and I are presenting an extension to our financial anomalies, stock selection, and equity modeling work of 1993 referenced in chapters "Risk and Return of Equity and the Capital Asset Pricing Model" and "Multifactor Risk Models and Portfolio Construction and Management" in October 2019, at the Q-Group meeting. The models we showcased in the first edition continue to be statistically significant. I will bet that there is a 65–75% chance that these stock selection models will continue to be statistically significant, passing the Markowitz-Xu (1994) Data Mining Corrections test, during the 2020–2029 time period, and will be reported in the third edition of the text. Fourth, a new edition allows me to use a more current text in my Tech and Rutgers 1 and 2-day classes. We have had 56,140 PDF downloads of our text, as reported by the Springer website, putting our book in the top 20%. My Springer book to honor Harry Markowitz, with 94,086 downloads, is my personal best. Finally, and most importantly, several major texts and associated papers have been published that compel our second edition. These texts are:

Castle, J., & Shepard, N. (2009). *The methodology and practice of econometrics.* Oxford, UK: Oxford University Press.

Castle, J., Doornik, J. A., & Hendry, D. F. (2013). Model selection in equations with many 'small' effects. *Oxford Bulletin of Economics and Statistics*, *75*, 6–22.

Castle, J., Clements, M. P., & Hendry, D.F. (2015). 'Robust approaches to forecasting', *International Journal of Forecasting*, *31*, 99–112.

Clements, M. P., & Hendry, D. F. (1998). *Forecasting economic time series*. Cambridge, UK: Cambridge University Press.

Connor, G., Goldberg, L., & Korajczyk, R. A. (2010). *Portfolio risk analysis*. Princeton: Princeton University Press.

Elton, E. J., Gruber, M. J. Brown, S. J., & Goetzman, W. N. (2007). *Modern portfolio theory and investment analysis* (7th ed.). Hoboken, NJ: John Wiley & Sons, Inc..

Dhrymes, P. J. (2017). *Introductory econometrics* (Revised ed.). New York: Springer.

Hendry, D. F., & Doornik, J. A. (2014). *Empirical model discovery and theory evaluation.* Cambridge, MA: MIT Press.

Levy, H. (2012). *The capital asset pricing model in the 21st century*. Cambridge: Cambridge University Press.

Lo, A. W., Mamaysky, H., & Wang, J. (2000). Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. *Journal of Finance*, *60*, 1705–1764.

Lo, A. (2017). *Adaptive markets*. Princeton: Princeton University Press.

Markowitz, H. M. (2013). *Risk-return analysis*. New York: McGraw-Hill. The first volume of a four-volume set on the risk-return trade-off.

Liu, L. M. (2006). *Time series forecasting and time series analysis.* Chicago: Scientific Computing Associates Corp.

Maronna, R. A., Martin, R. D., Yohai, V. J., & Salibian-Barrera, M. (2019). *Robust statistics: Theory and methods (with R)*. New York: Wiley.

Nikolopoulos, K. I., & Thomakos, D. D. (2019). Forecasting with the theta method. New York: Wiley.

Tsay, R. S. (2010). *Analysis of financial time series* (3rd ed.). New York: Wiley.

Tsay, R. S., & Chen, R. (2019). *Nonlinear time series analysis*. New York: Wiley.

In the years post our first edition, several good friends and coauthors passed. Eli Schwartz and Phoebus Dhrymes were over 80. Phoebus Dhrymes was a great Springer author of four major econometric texts. Eli and Phoebus were magnificent, highly independent thinkers, who left very high standards for us to try to reach. Bob Gillam, the founder of McKinley Capital, passed away last year at far too young an age, 72, and is missed. Bob taught me and others one of the greatest lessons an individual can teach young Americans: what it means to be an entrepreneur and keep the lights on in industry! Rob Gillam, our CEO/CIO, has big shoes to fill and big innovative ideas for McKinley.

As with the first edition, this text has been written, 95% of my own time, on Saturday and Sunday afternoons. To Julie, my wife of 38 years, I say thank you. Richard, Katherine, and Stephanie, my children in the first edition, are now real-world working millennials who give me great pride and hope for the future. To my coauthors of papers used in the second edition, Dimitrios Thomakos, Harry Markowitz, Ganlin Xu, Anureet Saxena, Shijie Deng, Andrew Mark, Rob Gillam, Sundaram Chettiappan, Eli Krauklis, Manish Kumar, and Elaine Wang, I say thank you. To my professors of Finance and Economics, Jim Vander Weide, Steve Maier, Bernell Stone, Henry Latane (deceased), and Jan Mossin (deceased) in Finance, and Ed Tower, Larry Moss, Rick Ashley, and Thomas Havrilesky (deceased) in Economics, I say thank you. Any errors remaining in this text are not their fault!

We gratefully acknowledge the typing and editorial assistance of Allison Capps.

Please enjoy the second edition. Eli is not with us, but I bet he would be very pleased!

As I send this book to press in January 2020, I am very pleased to acknowledge my two coauthors who joined me on the revised text, Anureet Saxena and Mustafa Gultekin. Anureet is brilliant, a Carnegie Mellon Ph.D. in Operations Research, with publications in *Mathematical Programming*, the *Journal of Portfolio Management*, *Journal of Investing*, and *Frontiers in Applied Mathematical and Statistics*. Mustafa, a Ph.D. in Finance, New York University, has published in the *Journal of Finance*, *Journal of Financial Economics*, the *Journal of Financial and Quantitative Analysis*, *Management Science*, and *Research in Finance*. Allison Capps helped us with editorial assistance and manuscript preparation. Thank you very much.

The goal of this book is the same goal as the first edition. We seek to introduce the reader to the world of Harry Markowitz, Bill Sharpe, Marty Gruber, and Jack Treynor. Quantitative financial economics should be fun, and properly educated (young) bright Quants can make a very good living. My Tech QCF graduates, 3 years post-graduation, should be earning in the top 10% of the income distributions before they are 30; in the top 5% within 5–10 years. Have you any Grey Poupon? But of course. My goal is that my Quants will be the victors of income-inequalities. As they say downhome, "to the victors go the spoils." Why eat yellow mustard on a hot dog when you can have Grey Poupon on a steak sandwich or Reuben? The fact that the models of chapters 8, 13, 14, 15, 21, and 22 of the first edition have been statistically significant, post-publication, was expected but nevertheless celebrated. I have similar expectations for these chapters for 2020–2029. Factor and model statistical significance and ensuring proper portfolio implementation are the only meaningful professional life measures for Quants.

As we are making revisions in our text, March 28, 2020, the business and political worlds are at odds on corporate stock buybacks, which we discuss in chapters "The Equity of the Corporation: Common and Preferred Stock" and "Multifactor Risk Models and Portfolio Construction and Management." Buybacks, a component, of corporate exports enhanced domestic and global portfolios during the 1997–2015 time periods. Whether buybacks have retarded corporate investment is a great question, one that requires a far greater effort and longer time period of analysis that afforded by our revised edition contract. Perhaps we will answer this question in a third edition.

Anchorage, AK, USA John Guerard
Labor Day, 9/2/2019

Contents

Introduction: Capital Formation, Risk, and the Corporation 1
Financial Mathematics and Theory 2
Growth and Survival of the Firm 2
Risk and Uncertainty Inherent in Finance 3
Types of Business Risk 3
Financial Risk 5
Division of Risk, Income, and Control 5
Profitability and Risk 8
Areas Covered in This Book 9
References 10

The Corporation and Other Forms of Business Organization 11
The Sole or Single Proprietorship 13
The Partnership 14
The Limited Partnership 16
The Corporation, Its Basic Characteristics 17
Chartering the Corporation 19
Administrative Organization 22
Major Rights of the Shareholders 24
The Advantages of the Corporate Form 26
References 27

The Corporation Balance Sheet 29
The Balance Sheet 29
Assets 30
Liabilities and Stockholder Equity 35
Book Value of Common Stock 40
Summary and Conclusions 47
References 52

The Annual Operating Statements: The Income Statement and Cash Flow Statement 53
Form and Content of the Income Statement 53
Retained Earnings vs. Dividends 59
Income Statement in the World of Business: IBM, Boeing, and Dominion Energy 60
Annual Cash Flow Statement 67
US Firm Cash Flow Analysis, 1971–2018 71
Summary 76
References 77

Financing Current Operations and Efficiency Ratio Analysis 79
Working Capital Concepts 79
Quantitative Working Capital Models: Cash Management 80
Sources of Net Working Capital 83
Working Capital Management and Efficiency Ratio Analysis 83
Financial Ratios and the Perceived Financial Health of Firms 87
The Time Series of Ratios in the United States, 1962–2017 89
Profitability Ratios in Industry 92
Limitations of Ratio Analysis 92
Working Capital Analysis and Granting Credit 93
A Summary of Ratio Analysis 95
Appendix 96
References 98

Financing Current Operations and the Cash Budget 99
Sources of Short-Term Financing 101
Trade Credit 102
Bank Credit 103
Other Forms of Short-Term Financing 107
The Cash Budget 110

Capital and New Issue Markets 115
The Secondary Markets 115
The Primary Market 118
The Originating House 119
The Underwriting Group 120
The Selling Group 120
Other Aspects of Investment Banking 120
Best Effort vs. Firm Commitment or Underwriting Basis 120
Initial Public Offerings (IPOs) 121
Expansion of a Privately Held Firm into a Public Corporation 122
The Problem of Control 123
Promotion of a Subsidiary by Parent Corporations 124
Formation of a Joint Subsidiary by Two or More Parent Companies 125
The SEC and the Flotation of New Issues 125

Secondary Flotations . . . 127
Issuing Securities Through Rights . . . 127
Stock Tenders . . . 129
Costs of Floating an Issue . . . 130
Regulation of the Capital Markets . . . 132
Securities Act of 1933 . . . 132
The Securities Exchange Act of 1934 . . . 133
Banking Act (Glass-Steagall Act) of 1933 . . . 134
Glass-Steagall Act Amended . . . 135
Retail Brokerage Houses . . . 135
Public Utility Act of 1935 . . . 135
The Maloney Amendment, 1938 . . . 136
The Investment Company Act of 1940 . . . 137
Sarbanes-Oxley Act of 2002 . . . 137
The Capital Market as a Source of Funds . . . 138
The Debate on the Optimal Organization of the Capital Market . . . 141
Capital Markets and Long-Term Economic Growth . . . 141
References . . . 142

The Equity of the Corporation: Common and Preferred Stock . . . 145
Common Stock . . . 145
Rewards to Common Shareholders . . . 147
The Corporate Sector: A Net Exporter of Funds . . . 149
Corporate Exports . . . 151
Definitions of the Value of Common Shares . . . 155
Stock Prices and Dividends: An Example . . . 159
Noncash-Paying Growth Shares . . . 162
Valuing a Dividend-Paying Growth Stock . . . 163
Illustration of the Valuation of a Dividend-Paying: Growth Stock—Shoreham Inc. . . . 163
Super Growth Cannot Be Infinite . . . 165
The Paradox of the Low Current Return on Growth Options . . . 166
Risk and Returns to Growth Investments . . . 166
The Cost of Capital to a Growth Firm . . . 167
The Cost of Common Stock Financing: The Norm . . . 167
Preferred Stock . . . 168
Features of Preferred Stock . . . 168
Rational for Preferred Stock Financing . . . 170
Convertible Preferred . . . 170
Protective Features on Preferred Shares . . . 172
Floating New Common Equity Issues . . . 173
Advantage of New Share Financing . . . 173
References . . . 174

Long-Term Debt 177
Bonds 177
Other Types of Long-Term Debt 181
Long-Term Lease 182
Lease Accounting: Recent Changes 185
The Cost of Debt Capital 185
Level and Structure of the Interest Rates 188
The Call Feature on Bonds 193
Convertible Bonds and Bonds with Warrants Attached 195
The Advantages and Disadvantages of Long-Term Debt 198
Malkiel's Bond Theorems 198
Retirement of Debt 200
Reference 208

Debt, Equity, the Optimal Financial Structure, and the Cost of Funds 209
A Most Misleading Relationship 210
Definition of Leverage: Profits and Financial Risk 210
Illustrations of Leverage: Return and Risk 211
Surrogate Evidence on the Development of "Optimum" Financial Structure 214
The Pure Theory of the Optimal Financial Structure 215
Modigliani and Miller: Constant Capital Costs 218
The Optimal Capital Structure and the M&M Hypothesis 222
Empirical Factors Influencing Financial Structures 223
Measures for Approximating Financial Risk 224
Outside Financing Capacity 227
References 231

Investing in Assets: Theory of Investment Decision-Making 233
Net Present Value and the Internal Rate of Return 234
Mutually Exclusive Projects 237
Difference in Project Size 239
Differing Duration of the Inflows 240
Lowest Annualized Total Costs 242
The Irrational Fixed Capital Budget 242
Operating Practice and the Internal Rate of Return 243
Real Investments and the Cost of Funds 243
Applying Investment Theory 243
CFO Practice 246
Current Costs of the "Optimum" Financial Mix 247
Adjusting the Capital Mix and Costs for Individual Project 247
Closing or Continuing Operations 248
Stability of Forecasts: Risk of the Investment 249
The Theory of Dealing with Risk 249

Risk-Adjusted Discount Rate . . . 250
Risk-Return Distribution . . . 250
Certainty Equivalence . . . 251
Maximum Loss and Reversibility . . . 252
Gross Uncertainty . . . 252
Market Risk . . . 252
The Effect of Taxes on the Financial Structure . . . 253
Costing the Components of the Financial Mix . . . 253
Investments Under Negative Interest Rates and Hyperinflation . . . 257
Summary . . . 259
Problems . . . 260
References . . . 260

Regression Analysis and Estimating Regression Models . . . 263
Estimating an Ordinary Least Square Regression Line . . . 264
Autocorrelation . . . 268
Estimating Multiple Regression Lines . . . 271
Influential Observations and Possible Outliers and the Application of Robust Regression . . . 273
The Conference Board Composite Index of Leading Economic Indicators and Real US GDP Growth: A Regression Example . . . 275
Summary and Conclusions . . . 282
Appendix 1: Least Angle Regression . . . 285
Appendix 2: The US Leading Economic Indicators . . . 287
BCI-01 Average Weekly Hours, Manufacturing . . . 287
BCI-05 Average Weekly Initial Claims for Unemployment Insurance . . . 287
BCI-08 Manufacturers' New Orders, Consumer Goods and Materials (in 1982 $) . . . 287
BCI-130 ISM New Order Index . . . 288
BCI-33 Manufacturers' New Orders, Non-defense Capital Goods Excl. Aircraft (in 1982 $) . . . 288
BCI-29 Building Permits, New Private Housing Units . . . 288
BCI-19 Stock Prices, 500 Common Stocks . . . 288
BCI-107 Leading Credit Index™ . . . 289
BCI-129 Interest Rate Spread, 10-Year Treasury Bonds Less Federal Funds . . . 289
BCI-125 Avg. Consumer Expectations for Business and Economic Conditions . . . 289
Regression Appendix 3 . . . 289
References . . . 300

Time Series Modeling and the Forecasting Effectiveness of the US Leading Economic Indicators . . . 301
Basic Statistical Properties of Economic Series . . . 302

The Autoregressive and Moving Average Processes 305
ARMA Model Identification in Practice 311
Leading Economic Indicators (LEI) and Real GDP Analysis:
The Statistical Evidence of the Initial Period, 1970–2002 315
Updating the Forecasting Effectiveness of Time Series Modeling:
Studies of the US Real GDP, 1959Q2–2018Q3 318
Forecasting Effectiveness of Time Series Modeling Using Autometrics
and SCA Model the US Unemployment Rate, 1959Q2–2018Q3 334
Causality Testing of the US Unemployment, 1959Q2–2018Q2 350
Summary and Conclusions 373
Time Series Analysis Appendix 1: Table 19 (Continued) with Identified
Outliers 374
Time Series Analysis Appendix 2: Table 22 (Continued) 376
References 379

Risk and Return of Equity and the Capital Asset Pricing Model 383
Calculating Holding Period Returns 384
Minimizing Risk 387
An Introduction to Modern Portfolio Theory 389
Expected Returns vs. Historic Mean Returns 403
Fundamental Analysis and Stock Selection 404
Modern Portfolio Theory and GPRD: An Example of Markowitz
Analysis 406
Further Estimations of a Composite Equity Valuation Model:
The Roles of Analysts Forecasts and Momentum in Stock
Selection 406
REG8 Model 414
REG9 Model 414
REG10 Model 415
Summary and Conclusions 420
Appendix: The Three-Asset Case 420
References 423

Multifactor Risk Models and Portfolio Construction
and Management 427
The Barra System 428
Barra Model Mathematics 429
Risk Prediction with MFMs 430
The Barra Multifactor Model and Analysts' Forecasts, Revisions,
and Breadth 433
Early Alternative Multi-Beta Risk Models 437
APT Approach 439
Applying the Blin and Blender APT Model 442
Applying the Blin and Blender APT Model, Barra, and Axioma:
The McKinley Capital Management (MCM) Horse Race Tests 444

Why Use the Axioma Statistical Model? . . . 445
Alpha Alignment Factor . . . 447
Axioma AAF Applications . . . 448
Summary and Conclusions . . . 449
Appendix 1: US-E3 Descriptor Definitions . . . 450
Volatility . . . 451
Momentum . . . 452
Size . . . 453
Size Nonlinearity . . . 453
Trading Activity . . . 453
Growth . . . 454
Earnings Yield . . . 455
Value . . . 456
Earnings Variability . . . 456
Leverage . . . 457
Currency Sensitivity . . . 458
Dividend Yield . . . 458
Non-estimation Universe Indicator . . . 458
Appendix 2: Factor Alignment Problems and Quantitative Portfolio Management . . . 459
References . . . 462

Options . . . 465
The Malkiel-Quandt Notation . . . 470
The Binominal Option Pricing Model . . . 472
The More Traditional Black and Scholes Option Pricing Model Derivation . . . 475
Black and Scholes Model Calculation . . . 478
The OPM and Corporate Liabilities . . . 480
References . . . 484

Real Options . . . 485
The Option to Delay a Project . . . 486
Implications of Viewing the Right to Delay a Project as an Option . . . 487
Abandonment Value . . . 487
Options in Investment Analysis/Capital Budgeting . . . 494
References . . . 494

Mergers and Acquisitions . . . 495
Noneconomic Motives for Combinations . . . 496
Holding Companies . . . 497
A Merger History of the United States . . . 498
Using an Accounting Basis . . . 501
The Economic Basis for Acquisitions . . . 502
Theories of Conglomerate Mergers . . . 502
Combinations Correcting Economic or Financial Imbalances . . . 507

Combinations Increasing Market Dominance 508
Combinations for Tax Advantages 509
The Larson-Gonedes Exchange Ratio Model 511
Valuation of a Merger Candidate 514
Testing for Synergism 517
Do Mergers Enhance Shareholder Wealth? 519
Divestment and Spinoff 521
Summary and Conclusions 522
References 522

Corporate Restructing: Leveraged Buyouts, Liquidations, Failures, and Bankruptcies 525
Voluntary Liquidation 529
A Liquidation Example 530
Remaining in Business 531
Failure 532
Informal Remedies 533
Bankruptcy 534
Bankruptcy Procedures 536
Priorities in Liquidation 536
Reorganization 538
Analysis of the Plan 541
Analysis of the Reorganization 542
Summary 543
References 543

Corporation Growth and Economic Growth and Stability 545
Factors in Economic Growth 545
Savings and Real Investment 547
Corporation Investment Spending and Economic Stability 548
Monetary Policy, the Cost of Capital, and the Firm Investment Process 550
Economic Growth and Firm Growth 551
Firm Growth and Economic Growth 551
References 553

International Business Finance 555
Currency Exchange Rates 555
International Diversification 560
International Stock Selection and Portfolio Construction and Optimization 563
International Corporate Finance Decisions 566
References 570

Management-Stockholder Relations: Is Optimal Behavior All That Is Necessary? 571
General Agreement and Potential Conflicts in Management and Control 573
Areas of Potential Conflict 573
Executive Compensation 574
Board of Directors 576
Stock Options 577
Bonuses 578
Risky Acquisitions 581
Turning Agents into Owners 582
The Diseconomies of Financial Scams 583
Insider Trading 583
Stockholder Remedies 585
To Whom Is Management Responsible? 586
Summary and Conclusions 587
Appendix: Socially Responsible Investing 587
From Raw to Normalized Score Definitions 591
Fama- and French-Weighted Portfolios 593
A Return to Optimized Portfolio Construction and Management 594
CTEF MVTaR with Normalized KLD Criteria 598
ESG Ratings' Definitions 600
References 602

Index 603

About the Authors

John B. Guerard Jr., PhD was the Director of Quantitative Research at McKinley Capital Management, in Anchorage, Alaska, while this book was penned. He has transitioned the McKinley Capital Management Scientific Advisory Board with Harry Markowitz, Ganlin Xu, Jose Menchero, and Rochester Cahan. He earned his AB in Economics from Duke University, MA in Economics from the University of Virginia, MSIM from the Georgia Institute of Technology, and PhD in Finance from the University of Texas, Austin. John taught at the McIntire School of Commerce, the University of Virginia, Lehigh University, and Rutgers University. He taught as an adjunct faculty member at the International University of Monaco, Rutgers University, and the University of Pennsylvania. He worked with the DAIS Group at Drexel, Burnham, Lambert, Daiwa Securities Trust Company, and Vantage Global Advisors and served on the Virtual Research team at GlobeFlex Capital. John co-managed a Japanese equity portfolio with Harry Markowitz at Daiwa Securities Trust Company. While serving as Director of Quantitative Research at Vantage Global Advisors (formerly MPT Associates), Mr. Guerard was awarded the first Moskowitz Prize for research in socially responsible investing. He has published several monographs, including the *Handbook of Financial Decision Making* (Probus Publishing Company, 1989, with H. T. Vaught), *Corporate Financial Policy and R&D Management* (Wiley, 2006, second edition), *Quantitative Corporate Finance* (Kluwer Academic Publishing, 2007, with Eli Schwartz), *Introduction to Financial Forecasting in Investment Analysis* (Springer, 2013), and *Portfolio and Investment Analysis using SAS* (SAS Press, 2019). John has edited *Advances in Mathematical Programming and Financial Planning* (JAI Press, 1987, with Kenneth Lawrence and Gary Reeves), *The Handbook of Portfolio Construction: Contemporary Applications of Markowitz Techniques* (Springer, 2010), *Portfolio Construction, Measurement, and Efficiency: Essays in Honor of Jack Treynor* (Springer, 2017), and *Handbook of Applied Investment Research* (Singapore: World Scientific Publishing Company, 2020, with W.T. Ziemba). He serves as an associate editor of the *Journal*

of Investing and *The International Journal of Forecasting*. Mr. Guerard has published research papers in *The International Journal of Forecasting, Management Science, IBM Journal of Research and Development, Research in Finance, Financial Analysts Journal, Journal of Investing, Journal of Forecasting, Research Policy*, and *Journal of the Operational Research Society*.

Anureet Saxena, PhD, CFA is the Director of Quantitative Research at McKinley Capital Management, in Anchorage, Alaska, and has been the member of the McKinley Capital Management Scientific Advisory Board since 2016. Prior to joining McKinley Capital Management, Dr. Saxena was a Portfolio Manager at Lazard Asset Management and held various positions at Allianz Global Investors, Qontigo (formerly Axioma Inc), and Assiduous Investment. Dr. Saxena has taught in the Quantitative and Computational Finance (QCF) program at Georgia Institute of Technology and Quantitative Methods program at Krannert School of Management and has published several papers in peer-reviewed journals such as *Mathematical Programming*, *Journal of Portfolio Management*, *Journal of Investment Management*, *Journal of Investing*, and *Journal of Risk*. Dr. Saxena is the winner of 2004 Egon Balas award for best paper in Algorithms, Combinatorics and Optimization, 2008 Gerald L. Thompson award for best dissertation in Management Science, and 2014 excellence in Economics scholarship by Krannert School of Management. He earned his B.Tech in Computer Science and Engineering from IIT Bombay, M.S and Ph.D in Management Science from Tepper School of Business (Carnegie Mellon University), M.S in Economics from Krannert School of Management (Purdue University), and a certificate in Accounting from UC Berkeley (extension program). Anureet is a CFA, CIPM, and CQF charter holder.

Mustafa N. Gültekin Investments, portfolio theory, asset pricing models, financial modeling, valuation, and risk management are the focus of Prof. Mustafa N. Gültekin's work. He has published in *Management Science, Journal of Financial Economics, Journal of Banking and Finance, Journal of Finance, The Review of Financial Studies, Journal of Business and Economic Statistics, Research in Finance, Journal of Investing*, and *Frontiers in Applied Mathematics and Statistics.* Other areas of expertise include international finance, mortgage-backed securities, and asset-liability management. He joined Kenan-Flagler Business School at the University of North Carolina at Chapel Hill as an assistant professor in 1985 and has served as Finance area Chair at the School.

Dr. Gültekin has taught PhD courses in Financial Theory, Portfolio Theory and Investments, Asset Pricing Models, and Quantitative Methods and Empirical Finance at UNC and other universities. He teaches Applied Investment Management (an active fund management course with $2M AUM), Financial Modeling, Valuation and Corporate Restructuring in the MBA, BSBA, and Master of Finance programs at UNC and other institutions.

Dr. Gültekin is a limited partner and senior advisor to Morning Meeting Inc., a financial modeling and consulting group, and associate at Yilmaz Advisory Group, a consulting group specializing in M&A, takeover, and valuation disputes. He has

served as a consultant to major corporations in the United States and abroad. He served as a consultant to the Blackethouse Group LLC and as consultant and director of research at the Community First Investment Risk Evaluation team of Community First Financial Group. He served on the board of Belltower Advisors LLC, a hedge fund; Clockworks Therapeutics LLC, a biotech company; and Ardic Tech, Inc., an ICT services and outsourcing company. He currently serves as a consultant to Jultz Co. LLC, a biotech investment search fund, and AI Analytics, an AI and machine learning consulting firm that specializes in quantitative modeling in investments.

Dr. Gültekin is the former President of the European Financial Management Association and the former Dean of the College of Administrative Sciences and Economics at Koç University in Istanbul. He also served as associate director of the Management Decision Laboratory at New York University and as a research scientist at Boğazici University in Turkey. He received his PhD in finance from New York University, his MA in operations management from Boğazici University, and his BS in physics from Middle East Technical University. He lives in Chapel Hill NC with his family.

Introduction: Capital Formation, Risk, and the Corporation



The corporation is the major institution for private capital formation in our economy. The corporate firm acquires funds from many different sources to purchase or hire economic resources, which are then used to produce marketable goods and services. Investors in the corporation expect to be rewarded for the use of their funds; they also take losses if the investment does not succeed. The study of corporation finance deals with the legal arrangement of the corporation (i.e., its structure as an economic institution), the instruments and institutions through which capital can be raised, the management of the flow of funds through the individual firm, and the methods of dividing the risks and returns among the various contributors of funds. Corporate finance represents the apogee of capitalism. Individuals own the means of production and are free to choose their activities, where they work, and what they buy and sell, and their product prices (Meltzer, 2012). Capitalism relies upon the rule of law and the protection of property rights. Capitalism embraces competition and rewards individuals that build value. There are successes and failures with capitalism. Meltzer (2012) reminds us that only capitalism increases growth and freedom. Wright (1951) discussed the development of capitalism in the growth of rice forms of Georgia and South Carolina in the 1790s, where farmers generated both tremendous profits and staggering losses. Wright offered that high net average rates of profits were necessary for growth. There is a risk-return tradeoff. We embrace the risk-return tradeoff for corporations and investors. The goal of corporate management is to maximize stockholder wealth. A major societal function of the firm is to accumulate capital, provide productive employment, and distribute wealth. The firm distributes wealth by compensating labor, paying interest on loans, and purchasing goods and services and accumulates capital by making investments in real productive facilities. The goal of corporate finance is to maximize the firm's stock price. We will discuss management-stockholder relations. We will discuss, in considerable detail, socially responsible investing and the firm. Companies can "do good while doing good." However, we firmly believe that management creates its corporate investment, dividend, research and development, acquisitions, and debt policies to maximize the stock price, which maximizes the market value of the firm.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_1

Financial Mathematics and Theory

The financial market is basically free of the frictions of imperfect competition. Securities of a given class and grade are largely homogenous, and the traders and investors do not have strong label or brand preferences. Because the markets are large and have a long history, there is a large mass of data that can be evaluated. This means that financial theory and derived mathematical models may be better evaluated, more appropriate, and better applied than elsewhere in the continuum of business and economic studies. We make extensive use of data in this text, using the data found on MSN (Money), FactSet (for Institutional Managers), Wharton Research Data Services (WRDS) Compustat, Global Compustat, IBES, and Center for Research in Security Prices (CRSP) databases. These databases are well known and respected sources of financial data. The investor or financial researcher can find 4 years on MSN, 10 or 20 years on FactSet, and 20 years of income statement and balance sheet data on an industrial Compustat database, and data for the 1950–2018 period with the WRDS Compustat data facilitates. We specifically selected Boeing, Dominion Energy, and IBM as firms to study with respect to their respective financial statements, ratios, valuation, and the cost of capital. Guerard and Vaught (1989) highlighted IBM and Dominion Resources (now Dominion Energy) and Guerard and Schuartz (2007) featured IBM, DuPont, and Dominion Resources in the first edition. These three firms are large, respected firms in their industries and are familiar to many readers of this text. The reader is introduced to regression and time series analysis to facilitate quantitative analysis, such as estimating betas, or measures of market risk, forecasting earnings per share, predicting stock rankings, and analyzing the predictive power of the leading economic indicators. Researchers have used financial data for well over 75 years to evaluate financial policies, such as merger (see Dewing, 1953), and to build financial planning models. Mao (1969), Carleton and McInnes (1982), Vander Weide and Maier (1985), and Pogue and Bussard (1972) are excellent examples of financial planning models.

Growth and Survival of the Firm

Finance focuses on the flow of values through the firm. Corporate finance is concerned with how the firm produces goods and services, generates cash flow, and generates returns for its investors. It explores the effects that different levels of the flow of values over time will have upon the complex legal and accounting entities making up the firm. It is interested in the relations between the legal owners and the various classes of creditors, and it explores the circumstances under which the claims of the original owners can grow and survive and be augmented, or, in contrast, those circumstances under which the legal claims of the owners must be forfeited to the claims of the creditors.

Risk and Uncertainty Inherent in Finance

The financial management has the tasks of minimizing the total cost of financial funds to the firm, providing adequate resources for expansion at a cost low enough to make it profitable at a risk low enough to maximize the firm's chance of survival and its stock price. Sometimes the problem of providing adequate funds at minimum financial costs can be separated from the problem of survival or risk and be dealt with simply as a problem in costing or economics. Where this is possible, however, it is usually because some earlier decision set the bounds of the problem. An earlier decision assumed the risks, and historically assumed risks can no more be discarded than historically assumed sunk costs. Whatever the initial approach, the heart of financial theory is the problem of risk and risk-bearing.[1] The flow of values generated by an initial commitment of productive resources cannot be exactly predicted. Risk is always two-sided. Asset prices may be better or lower than the original value of the assets invested. If the long-run flow of values is larger than estimated, the value of the owners' original investment is augmented; if, on the other hand, the flow of values falls below the original estimate, the owners will have to accept a lower valuation or a forfeiture of their claims. Should the stream of income fall low enough, even the creditors, depending on their legal position and status as risk-bearers, might have to accept some of the losses. One of the goals of this text is to acquaint the reader with the tools necessary to quantify risk and return.

The financial student must always grapple with the problem of uncertainty. The financial results of any single productive attempt are never sure. There is risk not only when a new enterprise is launched, but also for established firms as long as uncertainty exists continually and concurrently with the productive cycle.

Types of Business Risk

All other things equal, the element of risk intensifies the longer the production cycle. Risk varies directly with the length of time required from the first application of resources to the final appearance of the product. One type of time risk is of natural or physical origin. It includes fire, theft, flood, drought, and machinery breakdown. These risks can sometimes be insured against, if the probabilities of their recurrence are known, if they are independent events, and if society is sufficiently well organized to have set up the insurance mechanism. But the full losses entailed by these events are often uninsurable, for though the value of the physical assets may be recovered, lost time in production or sales may not be. Furthermore, a company may suffer a loss through disasters to other firms it relies on as suppliers or customers.

[1] Risk is used here in its broader sense to include the problems of uncertainty and not in its more precise definition, where it includes only those phenomena whose probabilities are reasonably known and which therefore can be insured against.

Insurance companies may not write regular policies against certain hazards (such as flood) or the insurance costs may be prohibitive.

Another type of time risk involves unexpected political or economic changes. A recession may occur, government policies may change, a rival product appears, or tastes may change before the process of recovering the investment is completed. This type of risk is double-edged; there is always the possibility that conditions may change in a favorable direction and lead to a greater return than anticipated.

There is a greater chance of the economy veering over a longer time span. Any productive organization utilizes many factors to which its commitment varies in length of time, and the risks to the firm depend partly upon the composition of these productive resources. Some resources, such as labor, may perhaps be withdrawn quickly; fixed plant or machinery, however, may have to be used for a long period before their value can be recovered. The complete production cycle for different resources (assets) varies, and for the firm, the composition of the resources it uses entails different degrees of risk.

Another risk is that the original estimate of demand or costs may be wrong. The possibility of error in either the forecast or in the original estimate of the economy as it currently exists can be minimized by competent research before production is launched. But error can only be minimized, never eliminated. Moreover, research is not costless; the outlay for research and its value in the reduction of hazards must be weighed against all the other possible outlays and their probable returns. The operational model involves comparing the marginal cost of additional research to the marginal value of the additional risk allayed. The optimum point is reached when the marginal cost of additional research equals the marginal value of the additional risk averted.

The costs of investigation are the reason many beginning small businesses make perfunctory estimates or forecasts. The small businessman cannot spend let us say, $20,000 to investigate the potential of an enterprise for which he has only $20,000 to invest. The initial estimate on returns and costs behind the launching of many small enterprises is likely to exist largely in the head of its founder.[2]

Another class of risks is called exploratory or technical risks. The exploration for mineral deposits is a risky enterprise; one can never be certain that the mineral will be found—or found in amounts sufficient to justify commercial exploitation. Similarly, a new product or a new method of production always involves uncertainty, for no one can be sure exactly how the innovation will behave when put into commercial production. No matter what precautions are taken, the characteristics of a product in the test tube or pilot plant often change unaccountably when the item is produced

[2]Even where resources may be more abundant the costs of testing or obtaining statistics should be weighed against the degree of certainty likely to be obtained. This concept as first elaborated by Abraham Wald, e.g., "Basic Ideas of a General Theory of Statistical Decision Rules," Proceedings of the International Congress of Mathematicians, Vol. I, 1950, has led to the development of sequential analysis and sampling techniques.

commercially. Neither the mine operators nor the innovators can be sure of the economic feasibility of their activity until production is launched.[3]

Financial Risk

We pay particular attention to financial risk in this text. Stockholder returns have been subject to larger variability than bond returns during the 1926–2018 period. One must discern between systematic risk, the risk of the market that cannot be diversified away, and total risk, as measured by the variance of stockholder returns. We estimate the stock beta, or measure of systematic risk, and illustrate how securities can be combined into portfolios to minimize portfolio risk to stockholders. Management calculates the firm's cost of issuing equity and its cost of capital. The cost of capital is the minimum acceptable rate of return for projects to be acceptable to enhance stockholder wealth. Management must decide whether the expansion of the firm is financed by using reinvesting corporate profits or by issuing debt or new shares of stock. The debt or equity decision should be made such that the firm's earnings per share (eps) are maximized.

Division of Risk, Income, and Control

Although risks and uncertainties are inherent in every business enterprise, different classes of investors in the firm do not shoulder the same degree of risk. The investor earns returns from bonds and stocks and incurs risk in the form of uncertainty of those returns. An investor in bonds earns returns from interest paid on the debt and possible bond price appreciation. The stock investor earns a return dependent upon the dividends paid on the shares of equity and the stock price appreciation. The interest paid on bonds is determined by the coupon rate of the bond, whereas dividends are paid from the earnings of the corporation. Firms may experience losses for several years and not pay dividends. Moreover, the firm's stock price may decline, as we saw from 2000 to 2003, such that total returns may be negative. Traditionally, the total return on stocks over time has exceeded the return on bonds, but with much greater variability of returns.

Modern finance has created different formal legal (or accounting) relationships that distinguish different classes of investors contributing to the funds of the business organization; the intensity of risk which each investor class bears varies considerably. The legal classifications on the liability and capital side of the balance sheet represent a division of risk. For example, the element of risk is less for current

[3]This differs in degree from the error or risk in forecasting demand. In this case, it is the characteristics of the production function that are uncertain.

creditors. They advance funds for only a few weeks or months. The possibility of deterioration in the credit position of the borrower is limited because of the short-term maturity of the loan. A trade creditor or banker making a short-term advance of credit funds usually looks to the flow of cash from current operations or from the liquidation of receivables or inventory to provide the repayment to the firm. Because of his minimal risk position, the short-term creditor generally offers funds at a low cost.

Moreover, because the risks are relatively short in duration, the supplier of current funds generally does not seek a voice in shaping company policies. Since the advance is temporary, the short-term creditor can do little, in any case, to enhance the probabilities of repayment at maturity. (Occasionally, however, if a firm is slow in meeting its current obligations, the trade creditors may combine in an informal committee to set terms of extended payment and to establish an operating framework for the firm until it is once more current.)

Usually, a distinction through superiority of claim and restrictive covenants is made between the long-term debt (bonds, funded debt, etc.) and the current liabilities. These distinctions are not inherent in the original legal-accounting position of the long-term creditor. At the start, a long-term creditor would have no more control over the policies of the firm than the current creditors, and in case of actual insolvency, the longer-term creditors do not initially receive a position superior to short-term creditors. In a failure, all general creditors share the option to collect from the general assets equally. Therefore, since uncertainty usually increases as the time period increases, the long-term creditors, committed to the enterprise for a considerable period, would be at an essential disadvantage. In the case of default, long-term creditors would have only an equal claim shared with the current creditors to the assets. The time risk could be so great that a mere priority of claim to the general assets against the owners, but not against the short-term creditors, would not be sufficient to induce creditors to lend at long-term without a wide differential in interest charges. If the long-term creditors were to invest funds on a basis of equality with the short-term lender, they might require so high a rate of interest that the earnings left to the owners would not be sufficient to induce them to take the remaining risk. This dilemma is solved by granting the long-term lender certain distinct advantages. In the original debt contract, the long-term creditors may be promised that certain financial policies will be followed; in this sense, the long-term creditors are given a limited amount of control over the financial policy of the firm. The long-term creditors may be granted a prior claim over other creditors to specific assets plus an additional undifferentiated claim against the general assets if the value of the pledged asset should prove inadequate. These concessions, coupled with a prior but limited claim to earnings, lower the risk to the long-term lender to the point where he is willing to lend at a usable price. The bondholders have a prior claim on assets relative to stockholders in failure.

The contractual right to a prior claim on earnings is the most important part of the guarantee to the long-term creditors. They are to be paid the contractual interest even though the total flow of earnings does not justify the original cost of the total assets (productive factors) dedicated to the enterprise. The lien against assets is a secondary

guarantee or option which comes into play only if the cash flow is insufficient to cover the contracted interest rate and the repayment of principal. The option to acquire the assets is nowhere a full guarantee against loss. The assets of a falling firm would bring full value only if they could be transferred to another enterprise or use where their expected earnings will be higher than that in the present firm. This is not a likely occurrence; nevertheless, the right to seize essential assets gives the long-term creditors a strong bargaining position in the event of a reorganization of the existing failed firm. In most instances, a lien or option against the assets provides a reinforcement of the claim against earnings (or cash flow) and not a guarantee against, only some hope of a limitation to losses.

Since no change in the formal accounting relationships can change the totality of risk or uncertainty, the privileged position of some financial contributors, creditors or preferred shareholders, can only heighten the risk of the other investors or shareholders. In financial practice, the most unsheltered position belongs to the residual owners (the shareholders), for they contract to use outside funds and they negotiate the terms under which the risks of the various creditors or preferred shareholders are reduced.[4] The owners' investment serves as a buffer to protect the contributions of the various creditors. There is no pro rata sharing of losses; if the business experiences an operational loss, the owners bear it. Should losses keep recurring, however, and the firm fails, the owners' investment may be wiped out, and bankruptcy may result. Only then, with the owners' equity completely dissipated, would the creditors suffer financial losses. Logically, then, the bearers of the greatest risks, the owners, are also given the ultimate control of the enterprise. The brunt of the loss resulting from an erroneous decision must be borne first by the initial risk-bearers, the owners or common stockholders; they cannot create losses for the other contributors without bringing a fair measure of ruin to themselves. The owners of a firm can be expected to behave rationally and, in protecting their position, shield the creditors from inordinate hazards.[5] In compensation for taking the most exposed risk position, the owners receive the best return if the operations are profitable.

Because it is accounting that classifies the amounts in the various legal and formal risk categories, the student of finance must have some acquaintance with operational accounting. The study of corporate finance draws upon both economics and accounting. Accounting provides the data needed to depict the relations of the capital and

[4]Although risk, income, and control can be divided in almost any manner in the various stages of single proprietorship, general partnership, and limited partnerships, combined with various short-run creditors, mortgage holders, and term loans, the corporation formalizes and illustrates these relationships better than the other business organizations. For this reason most discussion of financial structure uses examples from the corporate form. The flexibility of the corporation as a financing device is practically unlimited.

[5]The owners' position of control and greatest risk might be compared to a captain of a ship. In case disaster forces abandoning the vessel, the captain is traditionally the last person overboard. This tradition probably evolved as a result of the captain's control over operations, for the vulnerability of his position (the greatest risk-taker) ensures that he will direct the ship in the most responsible manner unless, of course, he is irrationally bent on self-destruction.

liability side of the balance sheet (the financial structure of the firm); it also provides the data necessary to derive the cash flow—the gradual conversion of assets to cash and back again.

Profitability and Risk

Risk and uncertainty pervade the field of finance. Assuming stockholder wealth maximization as a first approximation of the producer's goal, and developing marginal analysis as a tool, economics can provide useful insights into the organization of society and business decision-making. However, the typical financial decision is made under varying degrees of uncertainty, and the problem is to change these uncertainties, as far as possible, into objective probabilities, and arrive at the proper charge or premium for risk.[6] Uncertainty is moved toward more stable probability only as more and more individual but similar events are averaged together. This means that financial decision-making tends to deal in averages and probabilities.

In practice, financial decisions are basically choices between potential return and risk. Consider the structure of the firm's asset holdings. The greater the liquidity of the assets, the less is the risk of loss. The larger the percentage of assets the firm borrows, the greater is the risk of loss. Greater use of borrowed funds may raise the net return to the owners, but only at the cost of increasing the uncertainty of the firm's survival. The problem of the composition of asset holdings is characteristically posed as liquidity vs. earnings. The problem of choosing sources of funds is a question of maximizing earnings per share and the stock price. The choice between a conservative financial structure and an aggressive financial structure influences returns to the common shareholders.

Investors hold stocks as a major part of their wealth. We pay particular attention to the investment theories of Harry Markowitz and William F. ("Bill") Sharpe who shared the Nobel Prize in Economic Sciences in 1991. We develop the Markowitz concept of the efficient frontier, where expected return is maximized for a given level of risk, or risk is minimized for a given level of return. Extensive use is made of the Sharpe-Lintner-Mossin Capital Asset Pricing Model (CAPM) and its implications for stock valuation and portfolio construction and Management.

The most rational choice is not necessarily in favor of minimum risk. An extremely stable enterprise, which shows a minimum rate of return, may not be as valuable as one, which earned at a high rate for some years even if eventually its assets should dwindle or be dissipated through losses. The most stable firm would hold no assets but cash and it would have a capital structure of all common stock. It would also show no profit.

[6]Economic theorists allow for risk as a cost. But the problem is more often recognized than discussed.

Areas Covered in This Book

We conceive of the corporation as the major institution for capital formation in our economy. It is also a method for formalizing the division of risk-bearing, control, and income among the various contributors of funds.

The book develops the legal and organizational framework of the corporation from a functional point of view. The discussion of the major accounting statements provides an understanding of the source of financial data. A description and analysis of the capital markets and the instruments of long-term financing follow. Throughout we are interested in setting up the interplay of risk, control, and the cost of funds.

Although policy considerations (both internal and external) are introduced early, their discussion dominates the last half of the book. This text addresses the income statements, balance sheets, and ratios topics found in many great traditional texts such as Buchanan (1940), Dewing (1953), Schwartz (1962), Van Horne (2002), and Weston and Copeland (1986). The authors explore the factors governing the choice between return and risk in the various areas of financial decision-making and discuss the area of choice in the deployment investment in working assets and the retention of funds for emergencies or improved opportunities. The text deals with the analysis of expansion programs, analyzing their effect on cash flow, in the determination of whether the resultant risk is justified by the projected return. It should explore the models depicting the optimum capital structure, i.e., the composition of borrowed and equity funds, which would maximize the value of the ownership shares. Lastly, the authors discuss the circumstances under which a firm would be disbanded, liquidated, or reorganized. We discuss the financial decisions of firms, dividend and stock buyback strategies, liquidity strategies, leverage policies, merger and acquisition strategies from the perspective of stock price maximization. Robichek and Myers (1965) dealt with financial optimality in their outstanding monograph. Quantitative modeling in Mao (1969), Pogue and Bussard (1972), Vander Weide and Maier (1985), and Mansfield, Allen, Doherty, and Weigelt (2002) enhanced corporate planning and management. Mao (1969), Weston and Copeland (1986), Van Horne (2002), Brealey and Myers (2003), Brealey, Myers, and Allen (2006), and Guerard Jr. and Schwartz (2007) continue to address optimality. The Brealey, Myers, and Allen and Guerard and Schwartz are integrated with Compustat databases. We continue to use real-world business data in this text, particularly in financial statement and ratio analysis, dividend, stock buyback, and corporate export analysis, domestic and global multi-factor portfolio construction and management, merger and acquisition analysis, and socially responsible investing policies. Guerard Jr, Markowitz, and Xu (2014) addresses financial decision-making optimality. Chandler (1977), Carleton and McInnes (1982), and Ackoff (1999) specifically noted the dynamic nature of the corporation.

The text deals with the corporation as a social and economic institution. It covers the effects of the corporation operations on the economy: capital formation, economic stability and growth. It deals with the problem of the evolving relations between management, stockholders, and society. Lastly, the text carries a discussion of the different views of the effects of the large corporation in a competitive economy.

References

Ackoff, R. L. (1999). *Re-creating the corporation*. Oxford: Oxford University Press.

Brealey, R. A., & Myers, S. C. (2003). *Principles of corporate finance* (7th ed.). New York: McGraw-Hill/Irwin. Chapter 1.

Brealey, R. A., Myers, S. C., & Allen, F. (2006). In 8th edition (Ed.), *Principles of corporate finance*. New York: McGraw-Hill/Irwin. Chapter 1.

Buchanan, N. S. (1940). *The economics of corporate enterprise*. New York: Holt, Rinehart and Winston. Chapter 2.

Carleton, W. T., & McInnes, J. M. (1982). Theory, models, and implementation in financial management. *Management Science, 28*, 957–978.

Chandler Jr., A. D. (1977). *The visible hand: The managerial revolution in American business*. The Belknap Press of Harvard University Press: Cambridge. Chapter 13.

Dewing, A. S. (1953). *The financial policy of corporations* (5th ed.). New York: Ronald Press.

Guerard Jr., J. B., Markowitz, H. M., & Xu, G. (2014). The role of effective corporate decisions in the Creation of Efficient Portfolios. *IBM Journal of Research and Development, 58*(6), 1–11.

Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Guerard Jr., J. B., & Vaught, H. T. (1989). *The handbook of financial modeling*. Chicago: Probus. Chapters 1, 2, 4.

Mansfield, E. W., Allen, B., Doherty, N. A., & Weigelt, K. (2002). *Managerial economics* (5th ed.). New York: W.W. Norton and Company.

Mao, J. C. T. (1969). *Quantitative analysis of financial decisions*. New York: The Macmillan Company. Chapters 13, 14.

Meltzer, A. H. (2012). *Why capitalism?* Oxford: Oxford University Press.

Pogue, G. A., & Bussard, R. N. (1972). A linear programming model for short-term financial planning under uncertainty. *Sloan Management Review, 13*, 69–99.

Robichek, A. A., & Myers, S. C. (1965). *Optimal financing decisions*. Englewood Cliffs, NJ: Prentice-Hall.

Schwartz, E. (1962). *Corporate finance*. New York: St. Martin's Press. Chapter 13.

Van Horne, J. C. (2002). *Financial management & policy* (12th ed.). Upper Saddle River, NJ: Prentice Hall.

Vander Weide, J. H., & Maier, S. F. (1985). *Managing corporate liquidity: An introduction to working capital management*. New York: John Wiley & Sons. Chapter 10.

Weston, J. F., & Copeland, T. E. (1986). *Managerial finance* (8th ed.). Fort Worth: The Dryden Press. Chapters 8, 12.

Wright, D. M. (1951). *Capitalism*. New York: McGraw-Hill Book Company, Inc.

The Corporation and Other Forms of Business Organization



In Guerard Jr. and Schwartz (2007), we reported that in 2000 there were somewhat more than 25 million nonfarm business firms in the United States. About 5.045 million of these were corporations of all classes; the other 2.058 million were partnerships, and 17.805 million were nonfarm proprietorships. Corporations make the greatest profits, issue the most capital, and undertake the most investment in the US economy.[1]

Although the corporation is not the most numerous form of business organization, by far the greatest economic activity is carried on under the auspices of the corporate form. As Table 1 shows, some 66% of the gross national product originated in the business sector flows through the corporate sector in 2000.[2] The corporate form of business organization is dominant in the American economic scene.

In 2019, using the latest online *US Statistical Abstract of the United States*, for the year 2012, there were 5.73 million firms, with 115.9 million employees, $5414.6 million of payroll, and $32,637.8 million estimated receipts. Approximately 3.45 million firms had enterprise receipts of less than $500,000. The firms with receipts of less than $500K, employed 9.8 million employees, paid payrolls of approximately $190 million, and had estimated receipts of $6589.4 million. The 22,392 firms with enterprise receipts exceeding $100 million employed 55.9 million persons, paid payrolls of $3188.1 million, and had estimated receipts of $2359 million. The firms with receipts exceed $100 million employed 48.23% of the workers, paid 58.9% of the payrolls, and earned 71.6% of the estimated receipts.[3] The American economy is dominated in employment, payrolls, and receipts, by 23,932 firms. In this book, we concentrate on corporations because they employ the most people, raise the most capital, and generate the most profits. American business is the study

[1]Estimated from Statistical Abstract of the United States, 2003, US Department of Commerce, Bureau of Census, p. 495, and Survey of Current Business, 2001, US Department of Commerce.

[2]Ibid.

[3]Statistical Abstract of the United States, 2019, US Department of Commerce, Bureau of Census.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_2

Table 1 National income for 2000, originating in the business sector (in billions of dollars)

Legal form of organization	Amount	Percent of total
Corporations	$928	65.7
Partnerships	269	19.1
Nonfarm proprietorships	215	15.2
	$1412	1.000

Source: US Department of Commerce, US Census Bureau, *Statistical Abstract of the United States*, 2003, p. 495

Table 2 Number of returns and business receipts, 2000

	Returns (1000s)	Business receipts ($ Billions)
Corporations		
Under $25,000	5045	4
$500,000–$999,999	582	407
>$1,000,000	946	18,738
Partnerships		
Under $25,000	1105	5
$500,000–$999,999	82	86
>$1,000,000	137	2143
Nonfarm proprietorships		
Under $25,000	11,997	80
$500,000–$999,000	190	123
>$1,000,000	92	227

Source: US Department of Commerce, US Census Bureau, *Statistical Abstract of the United States*, 2003, p. 495

of corporations. The overriding advantage of the corporation, the limited liability of the shareholder, has made it the dominant form of business enterprise in the world in economic terms. The corporate form has the ability to amass capital from many sources and put it to productive use. The study of corporate finance has been addressed since the early days of business schools, when Professor Arthur S. Dewing published *The Financial Policy of Corporation* in 1934 (Arthur S. Dewing of Harvard Dead, 1971). We have the fifth edition of text (Dewing, 1953). Dewing taught at Harvard Business School from 1911 to 1933. The Dewing text was significantly reduced in length for undergraduate classes in the works of Bosland (1949), Gerstenberg (1959), Guthman and Dougall (1955), and Husband and Dockeray (1955). Graduate finance texts follow the Dewing tradition, and several outstanding examples are Brealey and Myers (2003) and Van Horne (2002). The authors are biased in their regard of Guerard and Schwartz (2007), an applied financial textbook.

Much of this book is concerned with the complex relationships and the nexus of institutions through which firms (corporations) can raise funds to finance growth and operations. The reader is reminded that management seeks to maximize stockholder wealth in its financial decisions. This chapter is mainly concerned with the institutional and legal arrangements that have enabled the corporation to be more effective

than other forms of business organization in performing this function on a large scale. Accordingly, in the next section of this chapter, we examine the major forms of business firms. We will examine the legal formality of the organization, the personal liability of the owners, the responsibility for management, and the tax status of the major business forms (Table 2).

The Sole or Single Proprietorship

The greatest numbers of firms are single proprietorships. The organization of the single proprietorship involves little legal formality. The owner and the business firm he owns are legally one. No special legal permission is required by the state to set up a sole proprietorship.[4] The proprietor has legal title to the assets of his business; she or he personally also assumes all debts. If in the course of operations, the assets of the business fail to satisfy all of the business liabilities, a proprietor's personal wealth or holdings may be used to help cover the claims of business creditors. Moreover, conversely, the net business assets are subject to the unfulfilled claims of personal creditors. This constitutes the basic rule of "unlimited liability for all debts whether personal or business." This rule actually strengthens the relative credit position of the single proprietor because the proprietor's personal wealth acts as a sort of a second guarantee for the safety of the business debts; it also is the major drawback of this business form, because a failing venture may cost an individual not only the funds directly risked in the business but the rest of the moneys, assets, or wealth he may have reserved for personal use.[5]

The single proprietor is responsible for running and managing his own business, but he may hire managers and agents as he sees fit. The single proprietor is taxed at the applicable individual income tax rate for all the net income arising from his business. After making proper arrangements with his creditors, the owner may dissolve his business any time he desires.

The single proprietorship has the advantages of simplicity, flexibility, and direct responsibility in management. It is limited, however, in its sources of ownership capital (i.e., risk capital) and in its ability to attract specialized managerial talent.

[4]Some kinds of enterprise may require a license. A license, however, is necessary only in certain activities—usually occupations where special skills or sanitary or hygienic considerations are involved—and not to the form of business organization.

[5]In case of bankruptcy, the rule of "marshaling the assets" is followed. Business assets are used first to satisfy business debt and personal assets satisfy personal debt first. Thus, if the ratio of debt to assets differs in the two categories, the rate of settlement for the different classes of creditors are not the same.

The Partnership

A partnership is an agreement by two or more individuals to own and run a business jointly. The agreement can be oral, but in most cases it is in writing, to prevent possible subsequent disputes. In some instances, it has even been created (constructed) by the courts where individuals have so acted as to lead others to believe that a partnership existed. The usual clauses in a written agreement are fairly well standardized.

Forming a partnership does not require any specific permission from the state. If the partners wish, however, they are generally entitled to file a copy of the agreement at the courthouse, which thus serves as a reliable neutral depository.

Each partner may bind the others to contracts incurred in the normal operations of the business.[6] Thus the partnership has often been defined as a contract of mutual agency. The legal profession repeatedly advises that one should have confidence in the reliability and judgment of the other party before entering a partnership.

The ordinary partner (strictly defined as a general partner unless the general partner is an LLC, a limited liability corporation) has unlimited liability for the partnership's business debts.[7] Thus if the partnership fails and the firm's assets fail to cover its liabilities, the creditors may seek to recoup their losses from the partners' private assets. Moreover, it is not the duty of the creditors to apportion losses among the partners. They may seek compensation for their claims where they can find it, regardless of any loss-sharing agreement among the partners. If one partner's personal assets are greater, or simply more available, he may well suffer disproportionate losses. A partner who loses more than his agreed share may have a counterclaim against the other partners,[8] but, of course, collection under these circumstances may be delayed considerably.

It is generally held that a rich man should be careful about entering a risky venture in partnership with a poorer individual. On the other hand, although their unlimited liability entails some financial dangers to the partners, it gives the partnership a stronger credit base than would a corporation of the same size.

No matter what formula has been set up for sharing returns out of the company, each partner will be taxed at the appropriate individual income tax rate for all the income realized or imputed to him out of the operations of the firm. Thus, if a partner receives a salary for his services to the partnership, interest on loans to the

[6]For example, one partner may, without the express consent of the others sell part of the normal inventory to a third party. The contract would hold, even if afterward the other partners should disapprove of the price or any other term of the sale. But a contract for the sale of major equipment might not be considered binding without the consent of the other partners if such a sale appeared to be outside the normal operations of the business.

[7]There is a variant of the partnership known as the limited partnership where one or more partners may limit their liability to their investment in the firm. There still must be at least one general partner.

[8]Unless stipulated otherwise, losses are shared on the same basis as profits. If there is no express agreement, profits and losses are shared equally by the partners.

partnership, and a share of the remaining pro forma profits,[9] he will pay tax on all of these items. He must pay tax on his total share of partnership profits, whether he draws them out for personal use or reinvests them in the business. The partnership as a firm files a Form 1065 Information return and distributes K-1s to the partners. Its report is made purely for information purposes, in order that 100% of the partnership income can be imputed to the various partners.

However, provision of the federal income tax laws allows a partnership that satisfies certain conditions to elect to be taxed as if it were a corporation. In this case, retained earnings may (for the time) bear a lower rate of tax than the applicable personal tax bracket of the partners.

Because the partnership rests on a foundation of mutual trust, its dissolution is made very easy. If no specific provision is written into the agreement, any partner can call for dissolution, usually with a required notice of, say, 30 days. Then a procedure called an "accounting" takes place; the assets are liquidated, and each partner is paid his ownership share (equity) according to the agreement. Since the assets upon liquidation are unlikely to bring in their "going concern" value, dissolution usually entails an economic loss. Often the partnership agreement sets a definite liquidation amount to be paid to a partner who requests dissolution so that the whole going concern need not be liquidated.

Since the partnership is a contract of close personal relationship, a partner's interest in the firm cannot be easily transferred or sold to a third party although the partner's interest can be assigned on death. A new party cannot enter the partnership without the consent of all the partners. If with the consent of the other partners, one partner sells out his interest in favor of someone else, the old partnership is actually legally dissolved, and a new firm is formed.

Again, since the partnership is constructed on the basis of personal contracts, the death of any partner dissolves the relationship. The heir of the deceased partner does not automatically become a member of the firm. If the surviving partners wish, they may let him in, actually forming a new partnership. Otherwise, the heir is entitled to dissolution and accounting. It is quite usual, however, for the payment to the heir to be preset in the partnership agreement. The difficulties that may occur because of the death of a partner give a partnership an insurable interest in the life of its principals. The insurance benefits may help in settling the partnership accounts.

The main advantages of the partnership form of business firm are flexibility in operation and ease of organization. More managerial talent can be assembled than is possible under the single proprietorship. The partnership can bring together more capital than is likely in the single proprietorship, and it may have a better credit standing relative to that of a small corporation because of the unlimited liability of the partners.

[9]Legally any payment of interest or salaries to the partners is not an expense but merely a way of distributing the partnership profits. However, these are generally subtracted on the accounting statements as an operating expense in order to obtain a pro forma profit figure.

On the other hand, the partnership may engender tensions between its principals. It may be unwieldy to operate if there are too many general partners since each partner is a general agent of the firm. This factor, together with the unlimited personal liability of the partners, tends to limit the total capital that can be raised. Because a partner may risk his personal fortune on the activities of the partnership, he generally wants a hand in the partnership affairs. To the drawbacks of the partnership, the difficulties of selling a partnership interest and the possible losses from a forced dissolution caused by either the death or withdrawal of one of the partners must be added.

Nevertheless, the partnership has proved serviceable where the total capital needs of the firm are not extremely large. The partnership is very common in smaller companies in the marketing field, in small manufacturing firms, in the professions, and in agriculture. The partnership form is very suitable where the personal skills and reliability of the partners is more important than capital in engendering income for the firm. (Thus doctors, lawyers, accountants, etc., have often been quite successful practicing as partners.) In the past, in those areas of financial activity where the bulk of the assets are carried in a liquid or highly marketable form and the problems of dissolution are not difficult (such as investment banking or security brokerage), the partnership was quite common.

The Limited Partnership

Given the versatility and imagination, with which persons direct business affairs and the ingenuity of the legal profession, it follows that there are many forms of business organization. Some of them are not very common today and are mainly of historical interest. The limited partnership, however, is an interesting and fairly common variant of the partnership form.

A limited partnership contains one or more general partners, one or more partners who are limited in their liability for business debts to the amount of their investment in the firm. The general partner or partners manage the firm and are subject to unlimited personal liability for the firm's debt. The limited partner, of course, receives some agreed-upon share of the profits. Sometimes called the "silent" or "sleeping partner," he cannot take an overt part in running the business, for if he does and the creditors come to believe he is an active partner, he may lose his limited status in the eyes of the law.

If a general partner wishes to retire from active management and become a silent partner, all interested creditors must be notified. Thereafter the limited partner cannot present himself in any way that could lead an "innocent third party" to believe he is a general partner.

The limited partnership is not widely applicable. The silent partner must rely heavily on the acumen and integrity of the general partners. He should, of course, keep himself informed of the progress of the business. If he is not satisfied with the performance of the enterprise, that is, the treatment he receives or the results

achieved with his capital, his sole legal course of action would be to ask for dissolution and to withdraw his capital. This may involve a loss for all concerned.[10]

The Corporation, Its Basic Characteristics

The corporation is a complex organization and takes many shapes. For the moment, we shall present a bare outline of the legal characteristics and organizational structure of the corporation.

The corporation is defined as a fictitious person created by the state. It can engage in certain defined activities, and in pursuing its purposes, it can obtain title to or dispose of property, enter into contracts, and engage agents to work and act for it. Of course, the corporation is not a human being and can act only through humans hired to work for it, yet the legal fiction of the "corporate person" is nevertheless a highly ingenious device. It enables the corporation (i.e., the properly state-sanctioned organization) to take responsibility and liability for legitimate activities which otherwise would be the final liability of the individuals in the organization. Unless something illegal has occurred, people dealing with the corporation must satisfy their claims against the corporation and may not look to those behind the company for ultimate settlement.

The limited liability of the owners is the single most important characteristic of the corporation. Limited liability simply means that in case of failure the owner or stockholder may lose what he has ventured in the firm, but even if the company cannot pay its creditors in full, the shareholder's personal assets cannot be endangered.[11] (The feature of limited liability is denoted in Great Britain by the abbreviation, LTD (limited), placed after the company's name). In France, it is denoted by the abbreviation, S.A., (Societé Anomie) the company of nameless ones. (The concept, of course, is that the owners cannot be named in a legal suit.) Because it possesses limited liability, the modem business corporation provides a method for many people to risk their funds in a distant organization, perhaps with the chance of gain if the business goes well, but with a definite limit of loss if the worst should happen. We will briefly trace the evolution of the modern corporation in chapter "Mergers and Acquisitions." It is hard to imagine how the large corporations with their funds amassed from many individuals could have developed without this provision.

Another important aspect of the corporate form is that its ownership shares are transferable. Unlike the partner, the owner of a share in a corporation has the inherent

[10]At that, the defensive weapon of calling for dissolution may be better than anything available to the minority stockholder in a small corporation.

[11]Unless, as sometimes happens in a closely held corporation, the major shareholder personally endorses the corporation's note in order to obtain additional credit, or he has failed to pay in the full par value of the stock. This last possibility is not too likely in practice.

right to sell or give away his holdings without the consent of the other shareholders,[12] the new holder acquires the same rights and privileges as the original owners. The characteristic of transferability is especially important to the development of modern free enterprise economies since it provides a high degree of capital mobility. Indeed, the ready transferability of corporate shares makes possible the whole intricate structure of capital markets.

The length of life of the corporate firm is independent of its stockholders. If a shareholder dies, his heirs take title to his shares, and the other surviving stockholders have to accept the new owners; there is no legal problem of "succession." The corporation virtually has "perpetual life." Although most corporation charters granted by the states run from 20 to 40 years and only a few are perpetual, most charters may be renewed with ease.

The corporation is a stable form of business organization because it can be dissolved only by a majority vote of the stockholders. In contrast, the partnership is dissolved on the death of a partner or on the request of any partner if he does not approve of the way the partnership is run or if he is displeased with his present or potential partners. In exchange for stability, the shareholder sacrifices the greater flexibility and the choice of associates inherent in the partnership. If the shareholder does not care for the other stockholders or if he thinks he can use his capital more productively elsewhere, his only remedy is to sell his shares for what he can get. The corporation shareholder cannot obtain dissolution of the firm merely by asking.

The single proprietor or the partners take direct responsibility for managing their firm. The owners of the corporation (i.e., the stockholders) have only an indirect role in the actual management of the company. The stockholders have the right to vote to elect a board of directors, who then appoint corporation officers, a President, Secretary, Treasurer, and various vice presidents, to run the business and make the day-to-day decisions for the firm. In a closely held company, where one man or a small closely connected group holds a majority of the shares, the major stockholders, the Board of Directors, and the management (i.e., the major officers) may all be the same people. In general, a widely held corporation is one where no single stockholder or close group of stockholders holds anywhere near a majority of the shares. Nevertheless, the management (though it usually holds only a small minority of the total shares) is the dominant voice in the corporation's affairs. In theory, the Board of Directors has strong powers. In actuality, the directors are nominated by the management and upon election reappoint the corporate officers.[13] The stockholders almost invariably vote for the management slate; usually, it is the only group

[12]Rights such as these can usually be voluntarily abridged. Thus a family-held corporation may require that a stockholder offer his shares to the corporation or other existing stockholders (at some fixed price) before he may sell them to someone else.

[13]In exercising independent choice in the appointment of officers, most boards resemble the electoral college since they are already pledged to a given set of candidates. In the rare instances of a contest for control, the management runs one slate of directors and the dissidents nominate an opposing slate.

running. The voting rights of the stockholders have become in practice a remote, ultimate power that might be used if the firm is badly mismanaged.[14]

The prevalence of management power over that of the shareholders has led to the development of agency-principal theory, which we discuss in chapter "Management-Stockholder Relations: Is Optimal Behavior All That Is Necessary?." Agency theory deals with the methods, the types of compensation, and other encouragement that the shareholders-owners (the principals) might devise to tie the operations of the management (the agents) to the basic interests of the shareholders.

Although the lack of influence the average stockholder has on the actual direction of the firm is often considered a drawback of the corporate form, it is not entirely a disadvantage. Most investors are not interested in the ordinary business decisions of the firm. They usually have other concerns, and they would consider it burdensome to devote any considerable time to the operation of the company. The large corporation develops its own management, which after time achieves certain autonomy, may develop a professional outlook, policies, traditions, and very likely a considerable devotion to the affairs and success of the company. Although the remuneration of management may be considerable, and even if at times it may take unfair advantage of its power, in the majority of corporations, the economic interests of the shareholders are well served by the corporate structure.

In regard to federal income taxation, the status of the corporation is quite unique. Since the corporation is a legal personality in its own right, it is taxed independently on its profits. The stockholders are further taxed at the individual tax rates for the profits they receive in the form of dividends or at the capital gains rate when reinvested earnings raise the value of shares and these shares are sold at a profit. Because profits have already been taxed once, corporations are described as being subject to "double taxation." In certain circumstances, however, the corporation form may actually offer some tax advantages. For example, if a firm whose owners are in a high-income tax bracket is in a profitable growth stage, it may retain a considerable proportion of its profits for reinvestment in the business. These may later be taxed at the lower capital gains rate. In any case, the other favorable aspects of the corporate form may decisively outweigh any extra tax burden.[15]

Chartering the Corporation

Since the corporation is a "creature" of the state, its formation requires a more complicated legal procedure than other forms of businesses. The legal "soul" of the corporate "personality" is the charter granted to it by the state.

[14]This is an old story. The gap between the theoretical right of the stockholders to run the firm through the election of the board of directors and their general helplessness in practice was labeled "the separation between ownership and control" by Berle and Means in *The Modern Corporation and Private Property*, Macmillan, 1933.

[15]Under the present tax laws, some closely held corporations which satisfy the requirements may elect to be taxed as partnerships. It depends on the individual situation whether this is advantageous.

The history of the corporation charter and of the relationship between the corporation and the state is a long one. The notion of some semiautonomous body functioning under a charter was not unknown in Roman law. Charters bestowing a grant of power by the sovereign to a subsidiary body existing under his control were used in the medieval times to define the relation between the crown and monasteries, certain guilds, universities, towns, etc. In England, the king later granted charters to various trading companies (actually independent merchants who formed a market) to engage in certain trades, deal in specific commodities, preempt certain geographical markets, or settle colonies. Lloyds of London, an association of insurance firms, is a survivor of these early-chartered companies. Some companies were financed with transferable stock and developed a definite resemblance to the modern corporation. The early charters almost always contained some type of monopoly privilege.

In time, British companies wishing to engage in regular commercial activities began to request charters from Parliament, who had begun to exercise this power on behalf of the Crown by about 1688. Each of these requests had to be passed upon separately. Action on these charters not only became a burden upon other legislative activities, but suspicion arose that favoritism existed in the granting of charters. In 1845, a general enabling act set up an administrative office to pass upon corporate charters. Any company fulfilling the standard requirements received a charter; the enabling act, moreover, redefined the concept of limited liability.

In the United States, developments ran parallel to those in England. The right to grant charters rested with the state legislatures. At first, each request for a corporate charter had to be passed on separately.[16] The politicking, favoritism, and sheer bribery that went into securing a charter became an overt scandal. In 1825 Connecticut passed the first inclusive general enabling act. In time all the states passed similar laws; they provided a procedure whereby the promoters of a business corporation who submitted their application in the proper form to a designated state official were automatically granted a charter.

Just how the grant of the charter defines the relation between the state and the corporation is still a matter of theoretical legal discussion. According to the "fiction theory" of the corporation that comes from the old common law, the corporate personality is a legal fiction of the state subordinate in every way to its creator. The "contract" theory, which has its roots in Roman law, holds that there was a vague but nevertheless existing antecedent body to the final corporation with which the state has made a binding contract by granting a charter. This theory would seem to limit severely the powers of the state to intervene in any way in the corporate structure. Yet in practice, the two theories have moved closer together. The corporation charter is almost inviolate and cannot be abrogated unless the parties behind the corporation have used it for fraudulent or criminal purposes. Under extreme

[16]Government corporations, e.g., municipalities, authorities, etc., still usually require individual action by the legislature to get a charter. The various federal corporations such as the Federal Reserve Banks and the Federal Deposit Insurance Corporation were chartered by separate acts of Congress.

provocation, the courts may pierce the "corporate veil" and hold the people maneuvering the corporation personally responsible for illegal actions, as we are currently seeing with many of the individuals associated with WorldCom, Enron, and Tyco.

A firm that plans to operate as a corporation can charter in any state; there is no requirement that its charter is in the state where its business is located. The procedure of obtaining a charter and the power it gives the corporation varies from state to state. By making chartering a matter, of course, the general enabling acts removed the competition of different groups within a state to secure a charter for their enterprise and possibly block the charter application of some rival. However, competition soon developed between the states to write easier enabling acts to induce corporations to charter with them. In their desire to obtain fees and annual franchise taxes (and bring business to local lawyers), some states reduced the difficulties in securing a charter, lowered fees, and, most importantly, widened the corporation privileges and powers allowed in their charters.[17] The striving by individual states for corporation chartering fees (which realistically have never amounted to much) by making incorporation easier and permitting wider corporate powers has been called "competitive laxity." One may wonder whether the United States would not have developed a simpler, more uniform, and possibly more responsible corporation managerial and administrative structure if the right to grant corporate charters had been reserved to the federal government in the Constitution.

Among the factors, a firm may consider before deciding upon the state in which it charters are differences in taxes such as chartering fees and the annual franchise taxes. The federal corporation profit tax does not differ no matter where the corporation is chartered; neither do the local property taxes, which are based on where the fixed assets of the company are located; nor the state corporation income taxes, which are set on where the firm does business. Chartering fees and annual franchise taxes are a minor fraction of the taxes a corporation is subject to and hardly the major influence in the choice of state in which to charter.[18] More likely the "liberality" of the charter provisions influence the choice of domicile (i.e., the state in which the charter is obtained); but even this was more crucial in the past than it is today. Most newly formed corporations obtain their charter in the state in which they intend to locate the major part of their business.

A single proprietor or partnerships whose principals have legal residence in the United States have the constitutional right to operate in any state. However, a corporation, although a legal personality, is not a citizen and has no inherent constitutional right to locate in states other than the state in which it is domiciled. We briefly trace the development of incorporation of holding companies and the creation of the modern corporation in chapter "Mergers and Acquisitions." In legal

[17]Many national corporations are chartered in Delaware, not so much because the fees are low but because Delaware was one of the first states to allow extensive holdings of the shares of other corporations. Thus Delaware became an ideal state to charter controlling and holding companies.

[18]Taxes may influence the location of the business, but that is a different matter from where it is chartered.

terminology, a corporation operating in the state in which it is chartered is called a domestic corporation, one operating in a state other than that in which it is chartered is a foreign corporation. A corporation of another nation is an alien corporation. Actually, a high degree of reciprocity prevails, and the states have not placed severe discriminatory restrictions on foreign corporations. A foreign corporation is generally required to pay an annual registration fee; the amount is usually equivalent to the annual franchise taxes paid by domestic corporations.

Administrative Organization

The overriding law applying to the corporation's government in any state is the state constitution, statutes passed on corporation matters, and court decisions that establish precedents. Where a corporation's charter or by-laws are in conflict with existing state laws, the charter or the by-laws must give way. Therefore, it is necessary to know that state law (or external law) in order to interpret properly the charter and the by-laws (or internal law) of the corporation.

The charter (or certificate of incorporation) is the corporation's right to legal existence, and it also states the general organization, authorized capitalization, the business and purposes of the company. Ordinarily, the charter does not make detailed provisions for running the affairs of the corporation. At the first meeting of the stockholders, after the charter is obtained, a set of by-laws is adopted, covering such topics as the election and remuneration of a board of directors, the establishment of the corporation offices and the definition of their powers and duties, the issuance and transfer of stock, methods of voting, and other matters pertaining to the management and administration of the company.

Although all corporations go through the formality of passing by-laws and electing directors, the practical importance of this procedure is determined by the structure of the ownership. By far the vast majority of corporations are closely held, closed, or family-held companies where a single individual, family, or closely related group holds all the shares. The closely held corporations are generally small, the principals are in close contact, and there is a minimum of formality in the procedures of administration. Much of the discussion that follows pertains to the widely held corporation.

After the board of directors have been elected (pursuant to the by-laws), it appoints the corporation officers. These might consist of a President, Secretary, Treasurer, Controller, and several vice presidents. The president is the chief executive officer (CEO) of the firm; he and the other officers direct, manage, and take responsibility for the regular operations of the business. In some firms, the chairman of the board of directors has a powerful position as overall strategist of the company; nevertheless, the president is usually in charge of the major internal affairs of the corporation. In some companies, basic administrative power may rest in an executive committee, consisting perhaps of the major officers and selected members of the board. Often the major officers are also on the Board of Directors. As we have

already indicated, in the going corporate concern, power to control the corporate affairs usually shifts from the stockholders through the Board of Directors finally to rest largely with the management.[19] The board is generally composed of the management's nominees, and the management's choice is voted down only if a dissenting group obtains enough stockholder votes to install a new management (and this seldom occurs).

A Board of Directors composed mostly of persons holding executive positions in the company is known as an inside board. A Board of Directors with some outside members is known as a mixed board. In Guerard and Schwartz (2007), we discussed the composition of the IBM Board of Directors. In its March 10, 2003, 10-K filing for the year ended December 31, 2002, IBM listed its Board of Directors. The Chairman of the Board, President, and Chief Executive Officer, is Samuel Palmisano. The executive officers of IBM are elected by the Board of Directors and serve until the next election (annually). Who is the Board of Directors of IBM? In 2002, the IBM Board had 15 members and included Kenneth Chenault, the Chairman and CEO of American Express Company; Nannerl Keohane, the President of Duke University; Sidney Taurel, the Chairman and CEO of Eli Lilly and Company; John Thompson, the Vice Chairman of the Board at IBM; Charles Vest, President of MIT; and Lockewijk C. VonWachem, Chairman of the Supervisory Board of the Royal Dutch Petroleum Company. The IBM Board of Directors is a mixed board. Currently, in December 2019, IBM has 24 members of its Board of Directors, including Virginia Rometty, Chairman, President, and CEO of IBM; Alex Gorsky, CEO of Johnson & Johnson; David Farr, CEO of Emerson Electric; Martha Pollack, President of Cornell University; and Shirley Jackson, President of Rensselaer Polytechnic Institute (RPI), and several retired CEOs, including Michael Eskew (UPS), Andrew Liveris (Dow Chemical), Joseph Swedish (Anthem), Sidney Taurel (Eli Lilly), Peter Voser (Royal Dutch Shell), and Frank Waddell (Northern Trust).[20] The IBM Board of Directors continues to be a mixed board.

The American Management Association has been in favor of outside directors; they feel such people may bring constructive, fresh, and critical insights to the board and provide a useful review of company and managerial policies.[21] The inside board is defended on grounds of efficiency; it is felt that the members of the board are able to act expeditiously if they are intimately acquainted with the problems they must deal with.

Contrary to popular impression, generally, directors as such are not extremely highly compensated, although many firms pay directors annual fees of $50,000, plus additional fees for committee meetings. There is a question when the directors are excessively complimented, $350,000 plus annually, e.g., Enron, whether they will

[19]Of course, all sorts of possibilities have to be allowed for. In some companies a dominant stockholder may be the real ultimate power, although he may elect not to be an officer or perhaps not even hold a seat on the board.

[20]https://www.ibm.com/investor/governance/board-of-directors.html

[21]On the other hand, outside directors are likely to have many interests, perhaps as officers of other corporations or serving on the boards of many other companies. They may be able to give no more than perfunctory attention to the matters at hand.

exercise any discretion on the corporate activities. Being on the board of a company is often considered an honorific position. (In most corporations a director does not initially have to be a stockholder of the firm.) Many prominent directors hold positions on a large number of boards. Under these circumstances, it is no wonder that the directors are often willing to accept the management's account of its stewardship and generally approve the programs that management proposes.

Just what are the responsibilities of the directors? The matter is not yet clearly decided in law. On a formal basis, the directors appoint the company officers approve executive compensation, approve dividend declarations and stock splits, and pass on major expansion plans or changes in company policies. The DuPont Board of Directors serves on various committees, including the Audit Committee, Compensation Committee, Corporate Governance Committee, Environmental Committee, and Strategic Direction Committee. In practice, the directors seldom initiate any policies but abide by management's suggestion. Moreover, matters of special interest are usually put to a special vote of the stockholders. The question is the degree of attention and care the directors owe to the affairs of the corporation. In the past, directors have often been cavalier in looking after the interests of the public stockholders. For some time, however, there has been a move in practice and in law regarding the directors as being in a quasi-fiduciary position. Holding a position of quasi-trust, the directors should exercise due care in protecting the equity of the stockholders. He should protect the stockholder against fraud and obvious malfeasance and make sure that he is fully informed on the corporation's condition and policies. A necessity for full disclosure holds with especial force when the director himself engages in a transaction with the corporation; it is important that the director not take advantage of his position to underpay the corporation or overpay himself or other interests he might represent.

Because they can be subject to stockholder suit, it can at times be difficult to recruit directors. To overcome this difficulty most corporations purchase liability insurance for their directors.

Major Rights of the Shareholders

The main value in owning stock in a publicly held corporation is the dividends and or capital gains that may be reasonably expected over time. The average shareholder depends on the innate fairness, reliability, skill, and goodwill of the management to give him a reasonable distribution of the gains accruing to a successful enterprise. Rights inhering to the stockholders protect them, to some degree, against arbitrary or capricious management. In practice, these rights give only negative protection; they may be a distant threat to some managements which might otherwise forget their obligations entirely.

The stockholders have the right to vote for the board of directors. Voting is done by share; each share equals one vote. An owner of 100 shares of IBM stock votes his or her opinion, although, with an average of 1720.4 million shares outstanding in

2003, and 885.6 million shares outstanding in 2019, the owner might not feel terribly important. The stockholders may vote by attending the annual meeting, or they may authorize someone else to vote their shares for them. Such authorization is called a proxy. The majority of proxies go to the management representatives. An innovation proposed some years back by students of corporation organization, and actually adopted by a few companies, is cumulative voting for the directors. Cumulative voting is a system of proportional representation. It allows a sufficiently large minority of the shareholders to concentrate their votes on one or more directors in order to achieve representation on the board. Under a straight voting system, where each position on the board is voted for separately, it is clearly possible for a bare majority to place every one of their candidates on the board, thus excluding any minority views entirely.

The proponents of cumulative voting argue that it is in the firm's interest to have a different point of view represented and that the minority directors will tend to check excesses of the majority. Opponents of cumulative voting point out that a dissident minority could get representation on the board merely to harass the normal operations of the company.

As has been pointed out, the shareholders almost always vote to retain the existing management. (This may be a fairly wise procedure; managerial talent is not necessarily in oversupply.) On occasion, however, a rebellion of stockholders (often dominated and financed by major shareholders who would like to be the new management) has succeeded in taking over a company. The stockholders generally have to be well aroused by extraordinarily poor management for this to occur. Nevertheless, although such rebellions are rare, the lesson is not entirely lost on other managements.

The shareholders are usually called upon to vote on any major change in the corporate structure. These changes may entail modifications or amendments of the by-laws, an increase in authorized shares, permission to float a convertible issue, a major expansion, a merger, a bonus or profit-sharing plan, a pension program, or stock options for the company executives. Generally, the shareholders approve the management's proposals.

The shareholder has the right to examine the company's books. This gives him some protection against possible fraud or manipulation of the accounts. It also provides him with the list of shareholders he needs to attempt a proxy drive to change the management. On the other hand, the right to examine the records can be put to unethical uses: to obtain a list of customers for a rival firm, to obtain a list of stockholders to solicit for other enterprises, to look for trade secrets, or otherwise harry the management. But the shareholder that wishes to examine the company's records at a reasonable time and place is allowed to do so.[22]

An important right of the stockholder is the right to institute a stockholder suit against the management on behalf of himself and the other stockholders. The right of

[22]Stories have arisen that some company's books are kept in cobwebby, cold vaults to discourage nosey stockholders. These stories are not wholly accurate.

the stockholders to sue on behalf of the company may be a powerful deterrent against errant management. Many times such suits are mere nuisances, started in hope the management will make some sort of settlement rather than fight through the courts; often, however, the results of the suits have been startling, uncovering fraud, manipulation, conflicts of interest, or pure stubborn mismanagement. If the stockholder wins his suit, he collects only court costs for himself; the damages are paid by the parties liable (the company executives and directors, as the case may be) to the corporation treasury.

In the vast majority of states, corporation stockholders enjoy the preemptive right or the right of privileged subscription. Existing shareholders are given the privilege of purchasing—in proportion to their holdings—the shares of a new issue of stock before these are offered on the general market. The preemptive right is said to give the stockholder the opportunity to preserve his proportionate interest in the corporation.[23] By exercising his or her rights, the existing shareholder can maintain the same proportionate claim to the firm's assets as he already has or maintains the same proportionate voting strength.[24]

Although often described in political-legal terminology the major function of the preemptive right is economic. It provides the shareholder with a compensating device if new stock is being floated at a price below the present going market price. (New shares are usually issued below the market price because it makes their flotation easier.) If the shareholder wishes, he can sell his rights, or he can exercise his rights and buy the new shares at the "offering" price. In either case, he loses no financial advantage to the newcomer shareholders.

The Advantages of the Corporate Form

When set against the possible machinations of the management, the protective rights of the shareholders hardly seems very strong. And it is true that innumerable cases can be cited where the management has taken advantage of the apathy, helplessness, or lack of knowledge of the stockholders to enrich themselves. On the other hand, such things have "also happened in government, politics, labor unions, social groups, and even religious organizations, and it is no more inherent in one [the corporation] than it is in another. Whatever may be the possibilities of unfair or unethical practices, it is probably true that the overwhelming majority of corporate managements are faithful to the trust they hold."[25]

---

[23]Both the preemptive right and general voting rights are commonly denied to preferred shareholders by charter provision. In a few states, it is possible to deprive the common stockholders of the preemptive right in the charter. The preemptive right on a particular issue can be waived by a vote of the shareholders.

[24]Rights are not given on the flotation of a security other than common stock unless the new issue is a convertible one (i.e., it can be changed into common stock at the option of the holder).

[25]Chelcie C. Bosland, *Corporate Finance and Regulation*, Ronald Press, 1949, p. 69.

In spite of its weaknesses, the corporate form gives many advantages to its stockholders (owners). If this were not so, it would be hard to account for the growth and economic dominance of the corporation over other forms of private business organization. Essentially, the corporate form permits the pooling of large amounts of capital from many sources (institutional and private) because it does not obligate the individual investor to be active in the affairs of the business nor subject him to worry over personal liability for business debts.

At most periods in our history, the publicly held corporation has been able to raise capital with relative ease because over time investment in a diversified portfolio of corporation shares has shown a comparatively good rate of return. As a corollary, in our economy characterized by the growth of industries requiring heavy capital investment, large-scale operations, and specialized professional managements, the corporation's ability to amass funds and to develop responsible administrative staff has made it the dominant business form.

References

Arthur S. Dewing of Harvard Dead. (1971, January 22). *The New York Times*.

Bosland, C. C. (1949). *Corporate finance and regulation*. New York: Ronald Press. Chapters 2, 3.

Brealey, R. A., & Myers, S. C. (2003). *Principles of corporate finance* (7th ed.). New York: McGraw-Hill/Irwin. Chapter 1.

Dewing, A. S. (1953). *The financial policy of corporations* (Vol. 1, 5th ed.). New York: Ronald Press. Chapters 1, 2, 3, 4.

Gerstenberg, C. W. (1959). Financial organization and management of business (4th rev. ed). Englewood Cliffs, NJ: Prentice-Hall. Chapters 1, 2, 3, 4.

Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Guthman, H. G., & Dougall, H. E. (1955). *Corporate financial policy* (3rd ed.). Englewood Cliffs, NJ: Prentice-Hall. Chapters 2, 3, 4.

Husband, W. H., & Dockeray, I. C. (1955). *Modern corporation finance*. Homeweed, IL: Richard D. Irwin. Chapters 1, 2, 3, 4.

Van Horne, J. C. (2002). *Financial management & policy* (12th ed.). Englewood Cliffs, NJ: Prentice-Hall. Chapter 1.

The Corporation Balance Sheet



This chapter on the balance sheet and the following one on the income statement are designed to serve two modest purposes: to acquaint the student with accounting and financial terminology and concepts used throughout the book and to explain the two important accounting statements on an uncomplicated level so that the student can appreciate and use some of the information presented by the accountants. It is not the purpose of these chapters to review all the procedures of accounting.

The Balance Sheet

The balance sheet is the financial picture of the firm at a precise point of time. On the left side of the balance sheet lists, categorizes, and sums the assets, the items of value the firm owns at a given point of time. The right side lists the debts and other payment obligations that the firm owes; these are the liabilities. The difference between the liabilities and the assets is the net worth, equity, or ownership capital. The liabilities plus the net worth of the firm must equal the sum of the firm's assets. The balance sheet then presents this equation:

$$\text{The sum of the assets} = \text{The sum of liabilities plus stockholder equity}$$

Although the balance sheet or position statement is a useful quantitative picture of a firm's financial position, it is not an exact reflection of the firm's economic worth. The balance sheet is constructed on the basis of formal rules and does not necessarily represent the market value of the firm as either a growing concern or liquidated (sold off) entirely. Moreover, the balance sheet represents the financial position' exactly at 12:00 midnight on the balance sheet date. The assets and liabilities shown are those accountants have ascertained to exist at that point in time. Since a business is always changing (buying a little here, selling something there, borrowing, lending, paying, and getting repaid), the balance sheet never represents more than an approximate

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_3

financial truth, even at the moment it is finally put together. The fact, however, that the balance sheet cannot tell us all we should like to know is not an indictment of accounting. The accountant's prime functions are to keep legal claims straight, present his data as consistently as possible, and stay as close as possible to objectively determined costs. The financial statements, a compromise between conflicting functions and demands made on the accountant, are the data very often used in making business decisions.

Several points are provided by the balance sheet equation, which holds that total assets = total liabilities + stockholder equity (or total debits = total credits). Therefore, if the firm increases its total assets (buys more goods, acquires new equipment, increases the money owed it by extending more sales credit), it follows that either the liability or stockholder equity (ownership accounts) must also increase to balance the rise in assets. The firm may increase the amount it owes its suppliers, borrow from the banks, float a mortgage bond issue, or increase its net worth by floating additional common stock or retaining additional earnings in the business. The issuance of debt increases the firm's interest expense, which it must pay to remain in business, but debt issuance does not possibly dilute the ownership and voting interests of management, inherent in an equity issuance. The problem of acquiring additional assets and the related question of financing the additional assets is discussed chapter "Investing in Assets: Theory of Investment Decision-Making." That is, management should finance the expansion of assets by pursuing the debt or equity strategy that maximizes the stock price of the firm.

Assets

The assets which the nonfinancial firm may acquire or own are usually broken down into three major categories: current assets, fixed assets, and other assets. The current assets and the fixed assets are usually much larger than the other assets.

Current Assets The current assets consist of cash, items that in the normal course of business will be turned into cash within a short time (the accountants use the rough measure of 1 year), and prepaid items that will be used up in the operation of the business within the year. The three largest accounts making up the current assets are usually in cash, receivables, and inventories.

Cash Cash is the sum of the cash on hand, held out to make changes and provide petty cash funds, and the deposits in the bank. IOUs, advances to employees, etc. are not counted in the cash account.

Receivables The receivables are amounts due to the firm from customers who have bought on credit. They are often segregated into accounts receivable and notes receivable. An account receivable is the usual way credit is given in American business practice. It simply means that the buyer of the goods is charged for his

purchases on the books of the seller—no other "formal" legal evidence is prepared. This type of credit is often called "book" or "ledger" credit.

If a note receivable is used, the purchaser of the goods has signed a promissory note in favor of the seller. A note, except in certain lines of business where they are customary, is generally required only of customers with weaker credit ratings or those who are already overdue on their accounts.

An account called reserve for bad debts or allowance for doubtful accounts is generally subtracted from the receivables. This is called a valuation reserve; it is an attempt to estimate the amount of receivables that may turn out to be uncollectible. The receivables minus this reserve, the net receivables, is counted as an asset on the balance sheet.

The receivables do not normally contain advances or loans that are not part of the normal business or sales operations of the firm. Loans to officers or employees or advances to subsidiaries are generally included in the other assets. Also, except for financial firms—banks and finance companies—such items as accrued interest receivable are usually not included with the other receivables.

Inventories Inventories are items making up the finished stock in trade of the business and the raw materials which in a manufacturing firm will in due course be turned into the finished products. In a mercantile or distributing firm, the inventory consists basically of "finished goods," that is, items which the company does not have to process further. In manufacturing companies, the inventory divides into three categories, raw materials, work in process, and finished goods. If we consider the current assets from the "flow of funds" aspect, that is, how close they are to being turned into cash, cash will be listed first, of course, next receivables (representing generally sales made but not yet collected), and then inventory. Obviously, for the going concern, finished goods are more current or liquid than work in process, and work in process more so than raw materials. The relative composition of the inventory can become a matter of importance—sometimes unfortunately overlooked—in analyzing the current credit position of a manufacturing firm. An inventory of 286 or 386 personal computers (PCs) are of little value to a computer firm.

A problem in presenting inventory values on the balance sheet is to keep separate the amount properly ascribed to supplies. Supplies are not part of the normal stock in trade nor are they processed directly into finished goods. Stationery and stenographic supplies, coal used for heat or power, and wrapping material used in a store are obviously supplies. It is difficult, however, to classify packaging items that are a distinctive part of the final product in a manufacturing concern, or the coal pile of a steel plant. In general, an item that is an integral part of the final product is part of the raw material inventory, whereas items used in corollary functions are supplies. Supplies are usually placed with the miscellaneous current assets; like the prepaid expenses, they represent expenditures made currently which save outlays in the future.

Valuation of the inventory is an additional problem with which the accountants must wrestle. The usual rule of valuation is "cost or market, whichever is lower."

This rule gives a conservative value to the inventory. The inventory is marked down in value if prices have declined since the items were purchased, but if prices have risen items are valued at their cost to the firm.[1]

Firms may now make a choice between the rule of "first-in, first-out" (FIFO) and "last-in, first-out" (LIFO) as methods of inventory valuation. Under FIFO (which most firms still use), it is considered (whether physically true or not) that sales have been made of the older items and that the items most recently manufactured or purchased compose the inventory. Conversely, if LIFO is used to value the inventory, then the new items coming in are considered to enter the cost of goods sold, and the cost of the older stock sets the value of the inventory. Under the first method, FIFO, the value given the inventory on the balance sheet is meaningful, but the cost-of-goods-sold figure used on the income statement may not truly represent current economic costs if price levels have been changing rapidly. Under FIFO the accounting figure for cost-of-goods-sold tends to lag behind price level changes, so that reported accounting profits are large on an upturn and decrease rapidly (or turn into reported losses) on a downturn in prices.[2] On the other hand, LIFO reduces the lag in accounting for cost-of-goods-sold when price level changes, thus modifying the swing of reported accounting profits during the trade cycle. The LIFO method of inventory valuation, however, tends to develop an inventory figure on the balance sheet that may not be at all representative of any current cost or price levels. The asset value of the inventory may become more and more fictitious or meaningless as time passes. In fact, more large companies surveyed annually by the American Institute of CPAs (AICPA) had switched to LIFO by 1981 (Horngren, 1984). In 1994, corresponding AICPA survey had 351 large firms using LIFO, with 186 firms using LIFO for 50% or more of its inventories.[3] The larger the firm's inventory balance, the greater is the potential tax savings for LIFO companies. Moreover, in defense of FIFO, any distortion it produces on the profit and loss statement is not very great for firms that turn over their inventory rapidly, i.e., for firms whose stock is replaced rapidly in relation to their sales. In fact, one of the earlier studies of market efficiency by Sunder found no significant excess returns for firms switching from FIFO to LIFO in inflationary times (from the date of the change), whereas firms switching from LIFO to FIFO experienced a drop in excess returns over 8%.

Miscellaneous Current Assets Other current assets besides cash, receivables, and inventories are accruals, prepaid expenses, and . Accrued items are amounts which the firm has earned over the accounting period but which are not yet collectible or legally due. For example, a firm may have earned interest on a note receivable given to it in the past even though the note is not yet due. The proportionate amount of interest earned on the note from the time it was issued to the date of the balance sheet

[1] Essentially unrealized profits are ignored but unrealized losses affect the accounting results.

[2] White, Sondhi, and Fried (1997), pp. 274–275.

[3] White et al. (1997), pp. 289–290.

is called accrued interest and under modern accounting procedures is brought on to the books as an asset.

Prepaid expenses are amounts the company has paid in advance for services still to be rendered. The company may have paid part of its rent in advance or paid in on an advertising campaign yet to get underway. Until the service is rendered, the prepayment is properly considered an asset (i.e., something of value due to the firm). When the service is rendered, the proportionate share of the prepaid item is charged off as an expense.

are holdings of highly marketable and liquid securities representing the investment of temporary excess cash balances. If these are to be classified as a current asset, the firm must intend eventually to use these funds in current operations. If, however, the securities are to be sold to finance the purchase of fixed assets or to cover some long-term obligation, such holdings are more correctly grouped with the other assets or miscellaneous assets.

Fixed Assets The fixed assets and the current assets are the two important asset classes. The fixed assets are items from which the funds invested are recovered over a relatively longer period than those invested in the current assets. Long-term receivables are included in fixed assets. The fixed assets are also called capital assets, capital equipment, or the fixed plant and equipment of the firm. The annual balance sheet usually lists a Property, Plant and Equipment-Net figure, composed of Property, Plant and Equipment-Gross, less accumulated depreciation. Almost all fixed assets except land are depreciable. Mineral deposits or reserves and standing timber are considered depletable and are important for mining and petroleum firms. In determining the value of a fixed asset, we must remember that their economic life is not unlimited, that eventually they will wear out or otherwise prove economically useless in their present employment.

The accounting reports allow for the loss of value on fixed assets through the passage of time by setting up a reserve for depreciation, allowance for depreciation, or accumulated depreciation account. Every fiscal period a previously determined amount is set up as the current charge for depreciation and is subtracted as an expense on the income statement. The matching credit is placed in the allowance for depreciation account, where it accumulates along with the entries from previous periods until (1) the allowance for depreciation equals the depreciable value (original cost less estimated scrap value) of the asset or (2) the asset is sold, lost, or destroyed. On the balance sheet, the allowance for depreciation constitutes a valuation account or reserve; the historically accumulated depreciation is subtracted from the original acquisition cost of the fixed assets, and the balance, called net fixed assets, is added into the sum of the total assets.

The problem of making adequate allowance for depreciation and determining the periodic depreciation charge has caused considerable difficulty for the accountants. A commonly used depreciation method presented to the public on the reported books is the “straight-line” method. This technique is quite simple (accounting, among other reasons, for its popularity). The probable useful life of the asset is estimated; the estimated scrap value of the asset is deducted from its original cost in order to

obtain its depreciable value; the depreciable value divided by the estimated life gives the yearly depreciation charge. The depreciation charge deducted as an expense remains the same year after year though the net book value of the asset is constantly reduced. For example, assume a warehouse has an estimated life of 20 years, an original cost of $225,000,000, and a $25,000,000 estimated scrap value. The yearly charge will be $10,000,000 per annum or 5% of its depreciable value. At the end of 10 years, the accumulated allowance for depreciation will be $100,000,000, and the net asset value of the warehouse will be carried at $125,000,000.

Although the straight-line method is popular for the public accounting reports, a constant depreciation charge does not reflect the fact that for most fixed assets the loss in economic value is higher in the earlier periods of use. Because of this, the Bureau of Internal Revenue now allows firms to adopt alternative, accelerated depreciation policies (such as the Accelerated Cost Recovery System, ACRS, instituted in 1981 and modified by the Tax Reform Act of 1986). Most industrial equipment is classified as having a 7-year life in ACRS, which allows firms to depreciate annually 14.29%, 24.49%, 17.49%, 12.49%, 8.93%, 8.93%, 8.93%, and 4.45%, respectively, in years 1 through 8. The great tax advantage of the accelerated depreciation is that it allows the company to defer some of its income tax liabilities to the future, thus providing a greater current after-tax cash flow. Accelerated depreciation is a significant tax saving in present value terms. Currently, most firms fudge their reporting; they use accelerated depreciation for tax reporting and straight line on the reports presented to the public. The difference is captured in an account labeled deferred income taxes which represents the difference between the taxes paid under accelerated depreciation and what the tax would have been if straight depreciation had been used. Presumably, sometime in the future when the accelerated depreciation runs out, the tax charge will rise. However, given the many possible tax and economic changes, the amount of future taxes is uncertain.

Actually reporting would be better, closer to the economic truth, and nothing would be lost if companies used accelerated depreciation on their public books. The sophisticated investor would understand the compensation of increased after-tax cash flow for the decrease in current reported book earnings.

Depreciation allowances are generally based on the original acquisition cost of the fixed asset to the firm. Any subsequent change in the value of the fixed asset—for example, through price level changes—is generally not reflected in the value of the asset on the books nor in the allowable depreciation rate. The depreciation rate is set by the original purchase price and is not changed for the new price level that may exist currently.[4] Thus even if the funds released to the business by its depreciation allowances were actually segregated (which they are not) for the replacement of fixed assets, they would not prove adequate if the replacement or reproduction cost of these assets had gone up in the meantime.[5] Many authorities have argued for

[4]This constitutes "historical cost depreciation" in contrast to, say, "replacement cost depreciation," where allowance might be made for changes in current costs.

[5]On the other hand, a fall in price levels would enable a firm to increase its physical capacity through the reinvestment of depreciation allowances.

changes in the basic accounting system that would adjust depreciation rates to movements of the general price level. While there is much in favor of such a reform, many practical difficulties stand in the way. Moreover, many accountants and public finance experts argue that the accountants' job is to measure and keep track of explicit costs and explicit legal claims as best they can; to introduce cost changes that do not reflect an actual transaction might put all accounting on an estimate, guess, and valuation basis.

In any case, much of the argument for current cost depreciation is deflected when accelerated depreciation is used. Because a large part of the depreciation is taken "up front," future changes in the value of the asset are of much less importance.

Akin to the allowance for depreciation is the account called allowance for depletion which appears on the books of mining or extractive companies and other companies, such as lumber firms, engaged in processing natural resources. The accumulated depletion account represents the proportionate cost of the amount of ore, crude oil, etc. that has been removed since the company started operation. It is subtracted on the balance sheet from the original acquisition costs of the company's estimated mineral reserves or resources. For income tax purposes, however, most companies take a percentage depletion allowance. (The rate varies for different types of minerals.) The allowable percentage is applied to the market value of the ore or crude oil and is subtracted from income before computing taxes. Under the law annual percentage depletion can continue to be taken even if the accumulated depletion already equals the original cost of the oil or mineral reserves.

Other Assets Other assets consist of items such as permanent investments and the so-called intangible assets, i.e., goodwill, franchises, trademarks, patents, and copyrights, and deferred charges.

Liabilities and Stockholder Equity

The liabilities and stockholder equity section of the balance sheet shows the claims of owners and creditors against the asset values of the business. It presents the various sources from which the firm obtained the funds to purchase its assets and thereby conduct its business. The liabilities represent the claims of people who have lent money or extended credit to the firm; the ownership, capital, net worth, or equity accounts (these terms are interchangeable) represent the investment of the owners in the business.

This, the credit side of the balance sheet, is often called the financial section of the balance sheet or the firm's financial structure. It is especially important to the student of finance. Many of the items found here will be discussed briefly since they will be taken up in considerably more detail in other parts of this book.

Current Liabilities The current liabilities are those liabilities, claims, or debts that fall due within the year. Among the more common current liabilities are accounts payable, representing creditors' claims for goods or services, and notes payable or

trade acceptances payable, arising out of similar economic transactions. Owed to bankers, obligations to bankers, notes payable to the bank, bank loans payable, or similar accounts show the amounts owing to banks for money borrowed. Usually, these arise from short-term loans, but the amounts due within the year on installment or term loans are also a current liability. Similarly, any portion of the long-term debt, i.e., bonds, mortgages, etc., maturing during the year is also carried in the current liability section.

Accruals A common group of current liabilities, represent claims that have built up but are not yet due, such as accrued wages, interest payable, and accrued taxes.[6] An item that bulks large for many corporations today is the amount owing on the federal and state corporation profits taxes. It appears as accrued income tax, provision for federal income tax, or other similar titles.

Dividends on the common or preferred stock that have been declared but have not yet been paid are carried among the current liabilities as dividends payable.

The relation of current liabilities to current assets is useful in many types of financial analysis and is especially important in analyzing the short-run credit position of the firm. Thus, the current liabilities are divided into the current assets to obtain the current ratio, and the current liabilities are subtracted from current assets to obtain the firm's net working capital. The larger the current ratio and the larger the net working capital relative to its total operations, the greater is the comparative safety of the firm's short-run financial position. The methods commonly used to judge the safety of the current liability coverage or the adequacy of net working capital vary with the type of firm and industry and with the judgment and analytical ability of the analyst. This subject will be discussed more thoroughly in chapter "Financing Current Operations and Efficiency Ratio Analysis."

Long-Term Debt Under the classification of long-term debt, fixed liabilities, or funded debt is placed the amount the corporation owes on bond issues, mortgage notes, debentures, borrowings from insurance companies, or term loans from banks. The company may have obtained funds to acquire assets and invest in the business from these sources, and this section of the balance sheet shows the amounts still owing.

There are generally three distinctions between the long-term debt and the current liabilities. First, the items making up the long-term debt are usually more "formal"

[6]Sloan (1996) and Richardson, Sloan, Soliman, and Tuna (2001) have shown that accrual's provide information about earnings' quality. Inventory accruals, the primary source of asset accruals, and accounts payables, the primary source of liability accruals, convey information. Richardson et al. (2001) showed that the setting of accruals, asset accruals less liability accruals, is an important source of financial information. Nondiscretionary accruals, accruals associated with sales growth and the level of firm operating activities, convey information regarding earnings quality. Accrual information influences information attributable to efficiency. Sloan uses FASB 95 to define accruals as the difference between net income and cash flows from operating activities. Sloan (1996) found that stocks with extreme accruals have less persistent earnings and experience mean reversion in the next year's earnings.

than those in the current liability section. A written legal contract or indenture describes the obligation, contains provisions for repayment under different circumstances, and details various devices for protecting the creditors against default and contains other clauses or provisions which might work to the benefit of the debtor company. The long-term debt is also often composed of "securities," or printed certificates issued by the corporation standing for evidence of the ownership of the debt which may be freely traded or negotiated. The second important distinction is that the long-term debt will not mature for at least a year and usually for some time longer than that. Moreover, the current liabilities are generally composed of recurring items, whereas the long-term obligations are incurred only on occasion. Third, the majority of long-term obligations carry some interest charge, whereas most current liabilities do not.

Deferred Credits Somewhere between the liability and equity section of the balance sheet we often find a category headed deferred credits or perhaps deferred, prepaid, or unearned income. These show a source of funds or assets for which the firm has not as yet performed any service. For example, suppose a company received a cash prepayment for a job on which work is not yet completed. The deferred credit classification does not mean that the firm owes money for this payment but that it owes for completion of the project. Furthermore, if the firm has made this contract on a normal basis, some part of the prepayment will not be covered by services or goods but will revert to the firm as profit. As the contract progresses, the accountants will normally analyze the results to date and apply a proportionate part of the prepayment to expenses, another part to profits (if any), and last leave (among the deferred credits) only that proportion which represents the uncompleted part of the contract.

Among the deferred credits are any rent payments made in advance by the company's tenants. Quite often, an unamortized bond premium may appear. If the company at one time issued bonds bearing a coupon interest rate above that of the going market rate at the time, the bonds would have sold on the market at some premium above their face or stated value. This premium is in a sense, then, compensation to the company for their generous interest rate. The premium is amortized over time as a reduction against the stated interest payment, i.e., it becomes an offset against an expense item. The unamortized portion is carried as a deferred credit since it is a sort of partial prepayment of the company's obligation to pay interest.[7]

Provisions for risks and charges and deferred taxes also are in total liabilities.

Common Equity Common stock, capital surplus, revaluation reserves, retained earnings, and ESOP guarantees constitute the majority of common equity.[8] Those terms and other variants are used interchangeably; they mean approximately the same thing, and the student should learn to identify these terms so that he will not be

[7]This matter is taken up again in chapter "Long-Term Debt" on Bonds.

[8]They may be called the proprietorship account in a single proprietorship or the partner's equity in a partnership.

confused if one or the other is used. This section of the balance sheet contains the items making up the ownership claims against the business. It represents the original investments of the owners plus any earnings they have retained in the business or less any accumulated losses the business may have suffered. Retained earnings change every year by the net income of the firm, less the dividend payment. The firm's retained earnings are the primary source of the firm's cash flow and growth. Common equity is often referred to as the capital section of the balance sheet (Schwartz, 1962).

Preferred Stock The amount shown as preferred stock represents the par or stated value of the various types of preferred stock issued, sold, and outstanding. The class of preferred stock is usually identified by its stated yearly dividends, i.e., the $7 p'f'd., the $5 p'f'd., or perhaps in percentage terms the 4 1/2% or 5%. Although the creditors may classify or consider the preferred issues simply as another form of equity, these shares have a prior claim on dividends and usually in case of dissolution have a claim prior to the common stockholders on the assets. They therefore, from the viewpoint of the common shareholders, take on some of the aspects of a creditor claim.

Common Stock The common stock account shows the par value or stated value of the common stock issued, sold, or outstanding. It is often said that this account represents the amount that the stockholders originally put into the business, but this is not likely to be literally true and should be modified by our historical knowledge of the firm's financial affairs. In one sense the capital stock account may represent more than the "original" investment, since common stock may have been issued and sold periodically on the primary market as the firm raised funds to expand and to improve its equity base. In another sense, the capital stock account may represent less than the original investment, for if in time the value of the firm's stock went over par, or over its stated value, and new issues were sold at a higher price, the difference is classified as capital or paid-in surplus. However, the new purchasing stockholders, at least, might well consider this amount part of their original investment.

Capital Surplus The capital surplus accounts, paid-in surplus, donated surplus, premium on capital stock, or perhaps investment in excess of par value of capital stock represents funds or assets given to the business on behalf of the ownership interests. These funds or assets do not arise out of the "normal" operations, of the firm but out of certain financial transactions. For example, a capital surplus would arise if someone, perhaps but not necessarily a stockholder, were to donate the firm some assets without asking for stock or other legal obligation in return. Most commonly capital surplus arises when a firm floats an issue of its stock at more than par value. After a company has been in operation for a time, the market value of the stock is more likely than not higher than the par value. The amount the company obtains in excess of the par value is classified as paid in surplus or premium on stock issued, etc. A premium on the issue price of either preferred or common stock is considered capital surplus.

Retained Earnings Retained earnings or earnings surplus shows the amount that the firm has reinvested in the business out of earnings that could otherwise have been

paid out in common stock dividends. These surpluses differ from capital surplus in that it arises out of accumulated retained earnings and not out of financial transactions.[9] If the firm's operations over time show accumulated losses rather than earnings, there is, of course, no earned surplus account but an accumulated deficit, which is subtracted from the other capital accounts on balance sheet. (One year's unsuccessful operation may not create a deficit on the balance sheet, since the losses of the current period may be more than covered by previously accumulated surplus.) The earned surplus accounts are basically derived from this equation: earnings minus losses minus dividends equal earned surplus. For the account to be negative, accumulated losses and dividends over time have to exceed the amounts earned.[10] The retained earnings-to-total assets ratio is a component of the Altman Z bankruptcy prediction model shown in chapter "Financing Current Operations and Efficiency Ratio Analysis."

Classified among the earned surplus accounts are the so-called appropriated surplus accounts, or surplus reserves. These accounts bear such titles as reserve for contingencies, reserve for plant expansion, or reserve for sinking fund. They are a source of much confusion; they do not represent cash or particular asset any more than the earned surplus account out of which they are derived represents cash or any distinguishable particular asset. These reserves are set up on the books by debiting earned surplus and crediting the reserve; i.e., the unappropriated earned surplus account is reduced, and a surplus reserve account is created by an equal amount. The purpose of these accounts is to warn the reader of the balance sheet that certain events are likely to occur that will reduce either the liquidity of the firm or both the liquidity and the asset holdings of the firm. For example, if the firm is embarked on a program of plant expansion and plans to use some of its present cash or marketable securities to finance the expansion, then the completion of the building program will reduce liquidity but not total assets. If, however, the firm faces a contract renegotiation with the possibility that it may have to relinquish some of its funds, the firm may lose cash without obtaining any other asset to offset it. If the management of the firm has plans for or is worried about losing some of its liquid assets, obviously it cannot distribute them as cash dividends. This is what the appropriated surplus accounts are supposed to show.

The question arises whether the existence of the appropriated surplus accounts means that the rest of the surplus—the unappropriated earned surplus—is available, for dividends. Unfortunately, the answer is no. The unappropriated earned surplus may have accumulated over the years, and the funds it represents may have long since gone into physical plant, inventory, or other operating assets. Although legally possible, it may not be financially possible to pay out any large percentage of the

[9]In bank balance sheets, the equivalent to the unappropriated earned surplus account is entitled undivided profits. The bank's surplus account consists of capital surplus plus the retained earnings that are considered permanently committed to the business.

[10]To add to the beginner's confusion, current operating losses are often called operating deficits, or just deficits.

"free" surplus without crippling the operations of the company. As long as the company maintains the level of its operations, for all practical purposes the earned surplus may be as permanently committed to the business as the rest of the capital accounts. In fact, cash is the source of dividends or stock buy backs, not the surplus account.

Book Value of Common Stock

The book value of the common stock is not directly indicated on the balance sheet, but it is readily derived from the balance sheet data. The book value is the net asset value of a share of common stock as presented by accounting convention on the balance sheet. To obtain this figure we subtract the total liabilities from total assets shown on the balance sheet, subtract the voluntary liquidation value of the preferred stock plus any accumulated dividends, and divide the remainder by the number of common shares outstanding. Alternately, the book value per share equals the stated or par value of the common shares issued and outstanding, plus all the capital surplus, earned surplus, and surplus reserve accounts, less any liquidation premium or accrued dividends on the preferred shares, divided by the number of common shares outstanding. The book value of common stock did not warrant much analysis in the original Graham and Dodd (1934), or even as late as Graham, Dodd, and Cottle (1962), but is a major component in the stock valuation analysis of Fama and French (1992, 1995) and Chan, Hamao, Lakonishok (1991), Chan, Lakonishok, and Hamao (1993), Lakonishok, Shliefer, and Vishay (1994), and Penman (2001). Most modern analysis links book value of return on equities and stock excess returns (see Bernstein, 1988; Brealey & Myers, 2003; Palepu, Healy, & Bernard, 2000; Penman, 2001; Weston & Copeland, 1986; White, Sondhi, & Fried, 1997).

Preferred stock is not included in book value either as a sum or as part of the divisor. If the term book value is used, it is usually understood as referring to the book value of the common stock, since the concept of book value of the preferred is not important or useful. Except in rare instances, the preferred stockholders are not conceived of as having any ownership or interest in the surplus accounts; it is the common shares pro rata equity in the surplus account which lends meaning to the concept of book value.

Calculating and understanding the concept of book value is not difficult; it shows the net equity per share of the common stock. The book value of a share of stock, however, reflects only the information, which is given formally by the accounting data on the balance sheet. Although book value has some significance in indicating the worth of a share, it cannot give the earning power per share of stock, its market value, its value for control, or its probable future value. Book value is only one of many financial benchmarks.

If a company owns a minor part of another firm, it can be represented by a Minority Interest item. The book value of the subsidiary company's minority stock (i.e., those shares of stock the patent company does not own) will be placed on the balance sheet midway between the liability and capital sections, since this account,

usually entitled interest of minority shareholders in consolidated subsidiaries, is somewhat of a hybrid.

The reader has been introduced to the component accounts of the firm's assets, total liabilities, and equity capital. A simple numerical example illustrates the accounts in a balance sheet.

Current assets		Current liabilities	
Cash	25	Accounts payable	115
Marketable securities	75	Short-term debt	110
Inventory	150	Total current liabilities	175
Total current assets	250	Long-term debt	525
Net fixed assets	750	Stockholder equity	300
Total assets	1000	Total liabilities and equity	1000

The simple firm has assets of $1000 (or $100 MM, if the reader prefers to think of larger numbers). The firm has current assets, cash, marketable securities, and inventory, of $250. The firm's current assets exceed its current liabilities, such that the immediate obligations of the firm could be paid, if payment would be requested, and its current assets, including inventory, could be liquidated at its book value. The bulk of the firm's assets are its net fixed assets, property, plant, and equipment, net of accumulated depreciation. One can think of the firm's fixed assets as including the factory where products are produced. The life of the fixed assets greatly exceeds the life of current assets, which is 1 year. Most firms with substantial fixed assets, as a percent of total assets, will seek to finance these assets by issuing long-term liabilities. The matching of asset and liability lives is often extremely important, particularly if the firm's creditors demand immediate payment. Management of the simple firm has issued debt of $525 and contributed (only) $300 of equity. That is, the firm has secured financing of its longer-lived assets by issuing debt (bonds, with a fixed maturity, or repayment date) than by issuing stock. Management maintains control of the firm's assets and profitability, as long as interest is paid on its debt and debt principle can be repaid. The concept of debt and equity financing is addressed in chapter "Debt, Equity, the Optimal Financial Structure, and the Cost of Funds." In the world of business, the management of many firms prefer to use debt financing, rather than equity financing, to pay for expansion of the firms' assets. Rates of return on equity investment are normally higher when the firm finances its asset growth using debt capital.

There are over 25 million firms in the US economy, as the reader will remember from chapter "The Corporation and Other Forms of Business Organization." At the time of our first edition, we urged the reader to find balance sheet information from many sources in the business world. Public libraries normally carry the Standard & Poor's *Stock Market Guide* or its *Stock Reports*, the Value Line *Investment Survey*, or Mergent's (formerly Moody's) *Industrial Manual*, which contain balance sheet information on as many as 3000 firms. Now, one can access this data on the Internet, at MSN/Money. In order that the reader may follow the discussion more readily, this chapter illustrates the debt and equity financing of assets with the presentation of the

balance sheets of three large corporations: International Business Machines (IBM), one of the world's largest computer and software firms; Boeing (BA), a leading aircraft manufacturer and defense contractor; and Dominion Energy (D), the holding company for Virginia Electric & Power Company, VEPCO, a utility firm. IBM is predominately a services and software provider, not the hardware firm of the first edition. Software and services account for the majority of IBM operating profits. VEPCO, an electric utility firm, provides electric power for Virginia and, to a lesser area, North Carolina. We show balances sheets for these firms in Tables 1, 2 and 3, respectively, for the 2015–2018 period, as they appear on MSN. We also show a modified set of balance sheet ratios for the three companies in Tables 4, 5 and 6 for selected years during the 1963–2017 period, drawing data from the WRDS Compustat database.

The reader may be concerned with an emphasis on manufacturing, or industrial firms. We illustrate the balance sheet of Dominion Resources, D, a diversified utility holding company principally in Virginia in Table 6. In 2004, Dominion Resources has a total debt-total assets ratio of 0.518, higher than its sector average of 0.451. Larger debt ratios imply greater risk in our economy, as we will see in chapter "Debt, Equity, the Optimal Financial Structure, and the Cost of Funds." The data of these three corporations can be found on the Wharton Research Data Services (WRDS) database. The WRDS data is as-reported data, not restated for a particular time, as is often the case with financial data; restated data makes the history of a particular firm consistent over time for events such as a merger. The investor can make rational financial decisions when he uses data known at that moment in time. Events, such as mergers and divestments, can create vastly different pictures of the firm. We show the ratio of current assets-to-total assets (CATA); current liabilities-to-total assets (CLTA); total debt-to-total assets (TDTA), defined as current liabilities plus long-term debt; and stockholder equity-to-total assets (SEQTA) for selected years during the 1950–2018 period for the three firms in Tables 4, 5 and 6.[11] The reader sees the vast differences among IBM, Boeing, and Dominion Resources with respect to their respective capital structures. The stockholder equity-to-total assets ratios are very different for the three firms; Dominion Resources has a larger stockholder equity to total assets ratio relative to the corresponding IBM and Boeing ratios during much of the period. It is interesting that the three firms have very similar stockholder equity to total assets ratios in 2003 and 2007. However, post-2007 and the Global Finance Crisis, Dominion has issued far less debt than Boeing and IBM. The stockholder equity-total assets ratio for Dominion Energy is far greater than the corresponding ratio post-2007. Dominion Resources has used far more long-term debt, relative to its current liabilities, than IBM or DuPont, to finance its assets. The greater use of long term debt may produce greater interest expenses and lower profitability for Dominion Resources, as we will see in chapters "The Annual Operating Statements:

[11]Because we use current liabilities and long-term debt in the TDTA calculation, and not total liabilities, the TDTA and SEQTA may not sum to 100%. Current liabilities and long-term debt represent the vast majority of total liabilities.

Table 1 IBM balance sheet

Period ending	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Current Assets				
Cash and Cash Equivalents	11,379,000	11,972,000	7,826,000	7,686,000
Short-Term Investments	618,000	608,000	701,000	508,000
Net Receivables	8,596,000	10,380,000	10,239,000	9,534,000
Inventory	1,682,000	1,583,000	1,553,000	1,551,000
Other Current Assets	2,902,000	2,266,000	532,000	293,000
Total Current Assets	**49,146,000**	**49,735,000**	**43,889,000**	**42,504,000**
Long-Term Investments	439,000	581,000	659,000	1,131,000
Property Plant and Equipment	10,792,000	11,116,000	10,830,000	10,727,000
Goodwill	36,265,000	36,788,000	36,199,000	32,021,000
Intangible Assets	3,087,000	3,742,000	4,688,000	3,487,000
Accumulated Amortization	–	–	–	–
Other Assets	14,505,000	13,844,000	12,184,000	10,612,000
Deferred Long-Term Asset Charges	5,216,000	4,862,000	5,224,000	4,822,000
Total Assets	**123,382,000**	**125,356,000**	**117,470,000**	**110,495,000**
Current Liabilities				
Accounts Payable	6,558,000	6,451,000	6,209,000	6,028,000
Short/Current Long-Term Debt	7,055,000	5,215,000	6,239,000	5,271,000
Other Current Liabilities	14,680,000	16,451,000	14,693,000	14,281,000
Total Current Liabilities	**38,227,000**	**37,363,000**	**36,275,000**	**34,269,000**
Long-Term Debt	35,681,000	39,871,000	34,663,000	33,431,000
Other Liabilities	32,544,000	30,397,000	28,140,000	28,371,000
Deferred Long-Term Liability Charges	2,676,000	2,136,000	1,496,000	1,626,000
Minority Interest	134,000	131,000	146,000	162,000
Negative Goodwill	–	–	–	–
Total Liabilities	**106,452,000**	**107,631,000**	**99,078,000**	**96,071,000**
Stockholders' Equity				
Misc. Stocks Options Warrants	–	–	–	–
Redeemable Preferred Stock	–	–	–	–
Preferred Stock	–	–	–	–
Common Stock	55,151,000	54,566,000	53,935,000	53,262,000
Retained Earnings	159,206,000	153,126,000	152,759,000	146,124,000
Treasury Stock	−197,561,000	−190,098,000	−188,448,000	−185,124,000
Capital Surplus	–	–	–	–
Other Stockholder Equity	−29,490,000	−26,591,000	−29,398,000	−29,606,000
Total Stockholder Equity	16,796,000	17,594,000	18,246,000	14,262,000
Net Tangible Assets	**−22,556,000**	**−22,936,000**	**−22,641,000**	**−21,246,000**

Table 2 Boeing balance sheet

Period ending	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Current Assets				
Cash and Cash Equivalents	7,637,000	8,813,000	8,801,000	11,302,000
Short-Term Investments	927,000	1,179,000	1,228,000	750,000
Net Receivables	13,904,000	11,088,000	8,832,000	8,713,000
Inventory	62,567,000	61,388,000	43,199,000	47,257,000
Other Current Assets	2,335,000	2,417,000	–	–
Total Current Assets	**87,830,000**	**85,194,000**	**62,488,000**	**68,234,000**
Long-Term Investments	1,087,000	1,260,000	1,317,000	1,284,000
Property Plant and Equipment	12,645,000	12,672,000	12,807,000	12,076,000
Goodwill	7,840,000	5,559,000	5,324,000	5,126,000
Intangible Assets	3,429,000	2,573,000	2,540,000	2,657,000
Accumulated Amortization	–	–	–	–
Other Assets	2,110,000	2,348,000	1,748,000	1,673,000
Deferred Long-Term Asset Charges	284,000	321,000	332,000	265,000
Total Assets	**117,359,000**	**112,362,000**	**89,997,000**	**94,408,000**
Current Liabilities				
Accounts Payable	12,916,000	12,202,000	11,190,000	10,800,000
Short/Current Long-Term Debt	2,690,000	435,000	284,000	734,000
Other Current Liabilities	54,936,000	51,260,000	27,623,000	27,589,000
Total Current Liabilities	**81,590,000**	**74,648,000**	**50,134,000**	**50,412,000**
Long-Term Debt	8,670,000	8,159,000	6,804,000	6,875,000
Other Liabilities	24,702,000	26,219,000	29,418,000	28,869,000
Deferred Long-Term Liability Charges	–	–	–	–
Minority Interest	71,000	57,000	60,000	62,000
Negative Goodwill	–	–	–	–
Total Liabilities	**116,949,000**	**110,649,000**	**89,120,000**	**88,011,000**
Stockholders' Equity				
Misc. Stocks Options Warrants	–	–	–	–
Redeemable Preferred Stock	–	–	–	–
Preferred Stock	–	–	–	–
Common Stock	5,061,000	5,061,000	5,061,000	5,061,000
Retained Earnings	55,941,000	49,618,000	40,714,000	38,756,000
Treasury Stock	−67,431,000	−59,827,000	−49,720,000	−42,316,000
Capital Surplus	6,768,000	6,804,000	4,762,000	4,834,000
Other Stockholder Equity	−15,083,000	−16,373,000	−13,623,000	−12,748,000
Total Stockholder Equity	339,000	1,656,000	817,000	6,335,000
Net Tangible Assets	**−10,930,000**	**−6,476,000**	**−7,047,000**	**−1,448,000**

Table 3 Dominion Energy balance sheet

Period ending	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Current Assets				
Cash and Cash Equivalents	268,000	120,000	261,000	607,000
Short-Term Investments	–	–	–	–
Net Receivables	2,080,000	1,786,000	1,706,000	1,369,000
Inventory	1,418,000	1,477,000	1,524,000	1,348,000
Other Current Assets	1,116,000	685,000	553,000	664,000
Total Current Assets	**5,161,000**	**4,334,000**	**4,248,000**	**4,189,000**
Long-Term Investments	1,626,000	1,882,000	1,866,000	1,612,000
Property Plant and Equipment	54,560,000	53,758,000	49,964,000	41,554,000
Goodwill	6,410,000	6,405,000	6,399,000	3,294,000
Intangible Assets	670,000	685,000	618,000	570,000
Accumulated Amortization	–	–	–	–
Other Assets	9,487,000	9,521,000	8,515,000	7,429,000
Deferred Long-Term Asset Charges	–	–	–	–
Total Assets	**77,914,000**	**76,585,000**	**71,610,000**	**58,648,000**
Current Liabilities				
Accounts Payable	914,000	875,000	1,000,000	726,000
Short/Current Long-Term Debt	3,650,000	3,131,000	1,742,000	1,989,000
Other Current Liabilities	1,804,000	1,476,000	1,372,000	1,363,000
Total Current Liabilities	**7,647,000**	**9,636,000**	**8,115,000**	**8,119,000**
Long-Term Debt	31,260,000	30,980,000	30,251,000	23,487,000
Other Liabilities	16,959,000	16,599,000	16,404,000	13,440,000
Deferred Long-Term Liability Charges	2,676,000	2,480,000	2,473,000	1,865,000
Minority Interest	1,941,000	2,228,000	2,235,000	938,000
Negative Goodwill	–	–	–	–
Total Liabilities	**55,866,000**	**57,215,000**	**54,770,000**	**45,046,000**
Stockholders' Equity				
Misc. Stocks Options Warrants	–	–	–	–
Redeemable Preferred Stock	–	–	–	–
Preferred Stock	–	–	–	–
Common Stock	12,588,000	9,865,000	8,550,000	6,680,000
Retained Earnings	9,219,000	7,936,000	6,854,000	6,458,000
Treasury Stock	−1,700,000	−659,000	−799,000	−474,000
Capital Surplus	–	–	–	–
Other Stockholder Equity	−1,700,000	−659,000	−799,000	−474,000
Total Stockholder Equity	20,107,000	17,142,000	14,605,000	12,664,000
Net Tangible Assets	**13,027,000**	**10,052,000**	**7,588,000**	**8,800,000**

Table 4 IBM. WRDS Database, selected years, 1963–2017

Year	CATA	CLTA	TDTA	SEQTA
1963	47.8%	15.0%	32.9%	67.1%
1970	39.9	21.9	28.7	69.6
1980	37.2	24.5	32.3	61.7
1985	49.6	21.7	29.2	60.8
1990	44.4	28.8	42.5	48.9
1995	50.6	39.4	52.0	27.8
2000	49.9	41.2	62.0	23.1
2004	43.0	36.5	50.0	27.3
2007	44.2	36.8	56.9	23.5
2010	42.4	35.8	55.1	20.3
2015	38.5	31.0	61.3	12.9
2017	39.7	29.8	61.6	14.0

CATA current assets-to-total assets, *CLTA* current liabilities-to-total assets, *TDTA* total debt-to-total assets, *SEQTA* stockholder equity-to-total assets

Table 5 Boeing Company. WRDS Database, selected years, 1963–2017

Year	CATA	CLTA	TDTA	SEQTA
1963	78.5%	43.3%	60.0%	41.8%
1970	67.5	42.4	66.2	30.9
1980	69.1	52.7	54.0	39.0
1985	73.2	47.8	47.9	47.2
1990	60.1	48.8	51.0	47.8
1995	59.6	33.6	44.2	44.8
2000	37.7	43.5	61.5	26.2
2004	27.9	38.6	58.8	20.9
2007	46.2	53.5	66.1	15.3
2010	59.2	51.6	68.4	4.0
2015	72.3	53.4	62.0	6.7
2017	70.6	60.9	71.5	0.4

CATA current assets-to-total assets, *CLTA* current liabilities-to-total assets, *TDTA* total debt-to-total assets, *SEQTA* stockholder equity-to-total assets

The Income Statement and Cash Flow Statement" and "Financing Current Operations and Efficiency Ratio Analysis."

The FactSet Research Investment System reports up to 20 years of data for a firm, as the data is known at a particular point in time, as we see in Table 7 for IBM; Table 8 for BA; and Table 9 for D. One must note in Tables 7, 8, and 9 that IBM and Boeing have incurred unfunded pension liabilities, Provision for Risks and Charges, such that Boeing's unfunded pension liabilities exceed its long-term debt. IBM's unfunded pension liabilities, Provision for Risks and Charges, almost exceed its long-term debt. Only Dominion maintains its unfunded pension liabilities such that the charges are substantially less than its long-term debt. Stockholder equity in IBM

Table 6 Dominion Resources. WRDS Database, selected years, 1963–2017

Year	CATA	CLTA	TDTA	SEQTA
1963	4.9	4.8	52.0	31.9
1970	4.0	6.0	56.9	31.5
1980	7.9	8.5	53.0	28.7
1985	8.5	12.8	46.5	31.6
1990	7.1	8.5	48.5	33.0
1995	7.9	10.0	44.2	34.1
2000	20.0	25.9	61.6	23.8
2004	15.7	17.7	51.8	25.1
2007	17.0	19.8	53.6	24.7
2010	12.6	13.5	50.3	28.6
2015	7.1	13.8	54.0	21.5
2017	5.6	12.5	53.0	22.4

CATA current assets-to-total assets, *CLTA* current liabilities-to-total assets, *TDTA* total debt-to-total assets, *SEQTA* stockholder equity-to-total assets

and Boeing is being greatly reduced by stockholder buybacks. The authors prefer more data to less. It is better to user balance sheet data of 20 years to build financial models than only 5–10 years (for statistical significance). We demonstrate the use of fundamental data in portfolio construction in chapters "Risk and Return of Equity and the Capital Asset Pricing Model" and "Multifactor Risk Models and Portfolio Construction and Management."

Summary and Conclusions

The corporate balance sheet allows the reader and investor to view the firm at a particular point in time. The reader sees the composition of the firm's capital structure, as well as the composition of current and fixed assets. Management's decisions on capital investment, dividend policy, and reliance on external financing can alter the presentation of corporate balance sheets and several financial ratios that many investors calculate to assess the firm's financial health over time. The balance sheet numbers are hard to analyze without using them in the returns on assets, sales, and net income, as in chapter "The Annual Operating Statements: The Income Statement and Cash Flow Statement," or in the Altman Z bankruptcy ratios, as in chapter "Financing Current Operations and Efficiency Ratio Analysis."

Table 7 International Business Machines Corporation. IBM 459200101 2005973 NYSE Common stock

	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Days in the Year	365	366	365	365	365	366	365	365	365	366	365	365	365	366	365	365	365	366	365	365
Asset																				
Cash & Short-Term Investments	5.83	3.72	6.39	5.98	7.65	10.57	13.69	10.66	16.15	12.91	13.97	11.65	11.92	11.13	11.07	8.48	8.19	8.53	12.58	12.22
Short-Term Receivables	27.62	30.73	27.02	27.36	28.92	28.14	24.43	26.85	28.79	27.56	26.79	28.23	29.56	30.58	31.84	31.83	28.55	29.25	31.63	30.98
Inventories	4.87	4.77	4.30	3.15	2.94	3.32	2.84	2.81	2.66	2.70	2.49	2.45	2.60	2.29	2.31	2.10	1.55	1.55	1.58	1.68
Other Current Assets	4.84	4.67	4.75	5.17	5.15	4.95	4.71	4.35	5.58	5.84	5.67	5.79	6.85	5.44	6.14	7.01	4.205	4.563	3.942	1.96
Total Current Assets	43.16	43.88	42.46	41.65	44.66	46.97	45.66	44.66	53.18	49.00	48.94	48.12	50.93	49.43	51.35	49.42	42.50	43.89	49.74	46.85
Net Property, Plant & Equip.	17.59	16.71	16.50	14.44	14.69	15.18	13.76	14.44	15.08	14.31	14.17	14.10	13.88	14.00	13.82	10.77	10.73	10.83	11.12	10.79
Total Investments & Advances	10.13	9.31	8.06	8.20	1.69	1.17	1.19	1.62	1.94	1.58	1.16	1.24	1.01	0.93	0.71	1.28	1.18	0.70	0.88	0.57
Long-Term Note Receivable	5.07	5.71	5.42	5.29	11.10	10.95	9.63	10.07	11.60	11.18	11.26	11.11	10.98	13.02	12.93	11.62	10.53	9.44	10.01	9.51
Intangible Assets	1.71	1.63	2.24	5.51	8.65	10.23	11.10	15.06	16.39	21.10	22.70	28.62	29.61	33.03	35.06	33.66	35.51	40.89	40.53	39.35
Deferred Tax Assets	2.65	2.97	2.40	4.31	4.29	3.02	1.83	3.88	1.51	7.27	4.20	3.22	3.50	3.97	3.05	4.81	4.82	5.22	4.86	5.22
Other Assets	7.19	8.14	11.22	17.08	19.38	21.66	22.58	13.51	20.73	5.08	6.61	7.05	6.52	4.83	9.31	5.98	5.23	6.51	8.23	11.10
Total Assets	**87.50**	**88.35**	**88.31**	**96.48**	**104.46**	**109.18**	**105.75**	**103.23**	**120.43**	**109.52**	**109.02**	**113.45**	**116.43**	**119.21**	**126.22**	**117.53**	**110.50**	**117.47**	**125.36**	**123.38**
Liabilities & Shareholders' Equity																				
ST Debt & Curr. Portion LT Debt	14.23	10.21	11.19	6.03	6.65	8.10	7.22	8.90	12.24	11.24	4.17	6.78	8.46	9.18	6.86	5.73	6.46	7.51	6.99	10.21
Accounts Payable	6.40	8.19	7.05	7.63	8.46	9.44	7.35	7.96	8.05	7.01	7.44	7.80	8.52	7.95	7.46	6.86	6.03	6.21	6.45	6.56
Income Tax Payable	4.79	4.83	4.64	5.48	5.48	4.73	–	–	–	–	–	4.22	3.31	4.95	4.63	5.08	2.85	3.24	4.22	3.05
Other Current Liabilities	14.16	13.18	12.24	15.08	17.04	17.53	20.59	23.23	24.02	24.19	24.40	21.76	21.83	21.54	21.20	21.92	18.93	19.32	19.71	18.42
Total Current Liabilities	39.58	36.41	35.12	34.22	37.62	39.80	35.15	40.09	44.31	42.44	36.00	40.56	42.12	43.63	40.15	39.60	34.27	36.28	37.36	38.23
Long-Term Debt	14.12	18.37	15.96	19.99	16.99	14.83	15.43	13.78	23.04	22.69	21.93	21.85	22.86	24.09	32.86	35.07	33.43	34.66	39.84	35.61
Provision for Risks & Charges	6.39	7.71	8.75	13.79	14.83	16.45	15.85	15.65	18.03	25.20	21.71	21.55	24.49	25.25	23.27	25.11	22.95	22.98	24.24	24.25
Deferred Tax Liabilities	1.35	1.62	1.59	1.45	1.83	1.88	1.62	0.67	1.06	0.27	0.47	0.38	0.55	0.45	1.74	0.29	0.25	0.42	0.55	3.70
Other Liabilities	5.54	3.61	3.27	4.26	5.32	6.49	4.61	4.54	5.52	5.47	6.15	5.95	6.18	6.82	5.27	5.45	5.18	4.75	5.65	4.67
Total Liabilities	**66.98**	**67.73**	**64.70**	**73.70**	**76.59**	**79.44**	**72.65**	**74.73**	**91.96**	**96.06**	86.27	90.28	96.20	100.23	103.29	105.52	96.07	99.08	107.63	106.45
Preferred Stock (Carrying Value)	0.25	0.25	0.00	0.00	0.00	0.00	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Common Equity	20.26	20.38	23.61	22.78	27.86	29.75	33.10	28.51	28.47	13.47	22.64	23.05	20.14	18.86	22.79	11.87	14.26	18.25	17.59	16.80
Total Shareholders' Equity	20.51	20.62	23.61	22.78	27.86	29.75	33.10	28.51	28.47	13.47	22.64	23.05	20.14	18.86	22.79	11.87	14.26	18.25	17.59	16.80
Accumulated Minority Interest	0.00	0.00	0.00	0.00	0.00	0.00	–	0.00	0.00	0.00	0.12	0.13	0.10	0.12	0.14	0.15	0.16	0.15	0.13	0.13
Total Equity	**20.51**	**20.62**	**23.61**	**22.78**	**27.86**	**29.75**	**33.10**	**28.51**	**28.47**	**13.47**	**22.76**	**23.17**	**20.24**	**18.98**	**22.93**	**12.02**	**14.42**	**18.39**	**17.73**	**16.93**
Total Liabilities & Shareholders' Equity	**87.50**	**88.35**	**88.31**	**96.48**	**104.46**	**109.18**	**105.75**	**103.23**	**120.43**	**109.52**	**109.02**	**113.45**	**116.43**	**119.21**	**126.22**	**117.53**	**110.50**	**117.47**	**125.36**	**123.38**
Per Share Values																				
Book Value per Share	11.23	11.56	13.70	13.23	16.44	18.08	21.03	18.92	20.55	10.06	17.34	18.77	17.31	16.88	21.62	11.98	14.77	19.29	19.08	18.82
Tangible Book Value per Share	10.28	10.63	12.40	10.03	11.34	11.86	13.97	8.93	8.72	–5.70	–0.05	–4.54	–8.14	–12.69	–11.63	–22.00	–22.00	–23.94	–24.87	–25.27

Source: FactSet Fundamentals

All figures in millions of US dollar except per share items

Table 8 Boeing Company. BA 097023105 2108601 NYSE Common stock

	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Days in the Year	365	366	365	365	365	366	365	365	365	366	365	365	365	366	365	365	365	366	365	365
Asset																				
Cash & Short-Term Investments	3.45	1.01	0.63	2.33	4.63	3.52	5.97	6.39	9.31	3.28	11.22	10.52	11.27	13.56	15.26	13.09	12.05	10.03	9.99	8.56
Short-Term Receivables	4.25	5.92	6.21	5.01	4.71	4.65	5.61	5.66	6.07	6.03	6.15	5.71	6.27	5.97	6.89	7.92	8.93	9.26	10.83	14.36
Inventories	6.54	6.79	6.92	6.18	5.34	4.25	7.94	8.11	9.56	15.61	16.93	24.32	32.24	37.75	42.91	46.76	47.26	43.20	44.34	62.57
Other Current Assets	1.47	2.14	2.44	3.33	2.57	2.68	2.45	2.84	2.34	1.05	0.97	0.03	0.03	0.03	0.01	0.02	–	–	–	2.34
Total Current Assets	15.71	15.86	16.21	16.86	17.26	15.10	21.97	22.98	27.28	25.96	35.28	40.57	49.81	57.31	65.07	67.79	68.23	62.49	65.16	87.83
Net Property, Plant & Equip.	8.25	8.81	8.46	8.77	8.43	8.44	8.42	7.68	8.27	8.76	8.78	8.93	9.31	9.66	10.22	11.01	12.08	12.81	12.67	12.65
Total Investments & Advances	0.16	0.27	7.21	8.80	10.02	11.71	2.85	4.09	4.11	1.33	1.03	1.11	1.04	1.18	1.20	1.15	1.28	1.32	1.26	1.09
Long-Term Note Receivable	5.21	5.96	2.41	2.61	2.66	1.80	9.64	8.52	6.78	5.86	5.47	4.40	4.65	4.41	3.98	3.73	3.85	4.27	3.23	2.66
Intangible Assets	2.23	5.21	6.44	3.89	2.95	2.90	2.80	4.75	5.17	6.33	7.20	7.92	7.99	8.15	8.10	7.99	7.78	7.86	8.13	11.27
Deferred Tax Assets	–	–	–	2.27	1.24	0.15	0.14	1.05	0.20	4.11	3.06	4.03	5.89	6.75	2.94	6.58	0.27	0.33	0.34	0.28
Other Assets	4.59	5.84	7.62	9.15	10.47	13.86	14.24	2.74	7.18	1.42	1.24	1.61	1.29	1.44	1.14	0.96	0.91	0.92	1.53	1.58
Total Assets	**36.15**	**41.97**	**48.34**	**52.34**	**53.04**	**53.96**	**60.06**	**51.79**	**58.99**	**53.78**	**62.05**	**68.57**	**79.99**	**88.90**	**92.66**	**99.20**	**94.41**	**90.00**	**92.33**	**117.36**
Liabilities & Shareholders' Equity																				
ST Debt & Curr. Portion LT Debt	0.75	1.23	1.40	1.81	1.14	1.32	1.19	1.38	0.76	0.56	0.71	0.95	2.35	1.44	1.56	0.93	1.23	0.38	1.34	3.19
Accounts Payable	4.91	5.04	4.79	4.43	3.82	4.56	5.12	5.64	5.71	5.87	7.10	7.72	8.41	9.39	9.50	10.67	10.80	11.19	12.20	12.92
Income Tax Payable	0.42	1.56	0.91	1.13	0.28	0.52	0.56	0.67	0.25	0.04	0.18	0.61	2.78	4.49	6.27	8.60	0.26	0.09	0.38	0.49
Other Current Liabilities	7.58	10.46	13.39	12.43	13.21	14.43	21.32	22.01	24.81	24.45	24.90	26.13	27.74	29.67	34.16	36.52	38.12	38.47	42.35	65.00
Total Current Liabilities	13.66	18.29	20.49	19.81	18.45	20.84	28.19	29.70	31.54	30.93	32.88	35.40	41.27	44.98	51.49	56.72	50.41	50.13	56.27	81.59
Long-Term Debt	5.98	7.57	10.87	12.59	13.30	10.88	9.54	8.16	7.46	6.95	12.22	11.47	10.02	8.97	8.07	8.14	8.73	9.57	9.78	10.66
Provision for Risks & Charges	4.88	5.15	5.37	11.71	12.37	9.13	8.94	8.81	8.16	15.71	13.36	17.83	24.06	27.18	17.00	23.98	24.40	25.86	22.02	19.91
Deferred Tax Liabilities	–	–	–	–	–	1.09	2.07	–	1.19	–	–	–	–	–	–	–	2.39	1.34	1.84	1.74
Total Liabilities	**24.69**	**30.95**	**37.52**	**44.65**	**44.90**	**42.68**	**49.00**	**47.06**	**49.98**	**55.07**	**59.83**	**65.70**	**76.38**	**82.93**	**77.67**	**90.41**	**88.01**	**89.12**	**91.92**	**116.95**
Common Equity	11.46	11.02	10.83	7.70	8.14	11.29	11.06	4.74	9.00	–1.29	2.13	2.77	3.52	5.87	14.88	8.67	6.34	0.82	0.36	0.34
Total Shareholders' Equity	11.46	11.02	10.83	7.70	8.14	11.29	11.06	4.74	9.00	–1.29	2.13	2.77	3.52	5.87	14.88	8.67	6.34	0.82	0.36	0.34
Accumulated Minority Interest	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.10	0.09	0.10	0.12	0.13	0.06	0.06	0.06	0.07
Total Equity	**11.46**	**11.02**	**10.83**	**7.70**	**8.14**	**11.29**	**11.06**	**4.74**	**9.00**	**–1.29**	**2.23**	**2.86**	**3.61**	**5.97**	**15.00**	**8.79**	**6.40**	**0.88**	**0.41**	**0.41**
Total Liabilities & Shareholders' Equity	**36.15**	**41.97**	**48.34**	**52.34**	**53.04**	**53.96**	**60.06**	**51.79**	**58.99**	**53.78**	**62.05**	**68.57**	**79.99**	**88.90**	**92.66**	**99.20**	**94.41**	**90.00**	**92.33**	**117.36**
Per Share Values																				
Book Value per Share	13.16	13.18	13.57	9.62	10.17	14.23	13.82	6.01	11.72	–1.78	2.82	3.76	4.72	7.76	19.90	12.26	9.50	1.32	0.60	0.60
Tangible Book Value per Share	10.60	6.94	5.49	4.76	6.49	10.57	10.32	–0.01	4.99	–10.50	–6.70	–7.00	–6.01	–3.02	9.07	0.96	–2.17	–11.42	–13.16	–19.26

Source: FactSet Fundamentals
All figures in millions of US dollar except per share items

Table 9 Dominion Energy Inc. D 25746U109 2542049 NYSE Common stock

	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Days in the year	365	366	365	365	365	366	365	365	365	366	365	365	365	366	365	365	365	366	365	365
Asset																				
Cash and Short-Term Investments	280	635	730	374	126	389	3575	1731	1044	1563	1176	801	807	766	1003	854	862	401	185	391
Short-Term Receivables	933	2358	1996	3054	3919	2905	3561	2753	2356	2559	2180	2246	2035	1717	1836	1633	1369	1706	1786	2080
Inventories	254	327	577	637	870	893	1167	1101	1045	1166	1185	1163	1348	1259	1176	1410	1348	1524	1477	1418
Other Current Assets	725	2546	2100	2760	2266	2935	1826	2513	2211	2373	2276	1190	1240	1398	1925	1718	612	617	886	1272
Total Current Assets	2192	5866	5403	6825	7181	7122	10,129	8098	6656	7661	6817	5400	5430	5140	5940	5615	4191	4248	4334	5161
Net Property, Plant and Equip.	10,764	14,849	18,364	20,257	25,850	26,716	28,940	29,382	21,352	23,274	25,592	26,713	29,670	30,773	32,628	36,270	41,554	49,964	53,758	54,560
Total Investments and Advances	2183	1926	3054	3174	3087	3873	5416	4467	4068	3257	3492	4151	3985	4191	5102	5561	5774	6343	6964	6560
Long-Term Note Receivable	2034	676	106	0	0	0	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Intangible Assets	132	3502	4527	4614	4656	4696	4917	4926	4094	4215	4047	3783	3778	3666	3646	3614	3864	7017	7090	7080
Deferred Tax Assets	109	241	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Other Assets	385	2347	2915	3039	3412	3039	3258	2396	2953	3646	2606	2770	2751	3068	2780	3267	3414	4038	4439	4553
Total Assets	**17,799**	**29,407**	**34,369**	**37,909**	**44,186**	**45,446**	**52,660**	**49,269**	**39,123**	**42,053**	**42,554**	**42,817**	**45,614**	**46,838**	**50,096**	**54,327**	**58,797**	**71,610**	**76,585**	**77,914**
Liabilities and Shareholders' Equity																				
ST Debt and Curr. Portion LT Debt	1406	3573	3213	3318	2704	1941	3948	4810	3234	2474	2432	1883	3293	4635	3446	4150	5335	4864	6376	4031
Accounts Payable	711	1736	1776	2310	2712	1984	2756	2142	1734	1499	1401	1562	1250	1137	1168	952	726	1000	875	914
Income Tax Payable	89	316	144	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Other Current Liabilities	793	1967	2345	2815	3451	4131	7776	4277	2778	3821	3000	2328	2419	1991	2380	2096	2059	2251	2385	2702
Total Current Liabilities	2999	7592	7478	8443	8867	8056	14,480	11,229	7746	7794	6833	5773	6962	7763	6994	7198	8120	8115	9636	7647

Long-Term Debt	6936	10,101	13,251	13,457	15,776	15,507	14,653	14,791	13,235	14,956	15,481	15,758	17,394	16,851	19,330	21,805	23,616	30,231	30,948	31,144
Provision for Risks and Charges	–	0	0	0	0	0	2249	1930	1722	3327	2865	2342	2345	3472	1965	2929	3086	4348	4329	4578
Deferred Tax Liabilities	1897	2879	–	4209	4563	5499	4984	5858	4253	4137	4244	4708	5216	5800	7114	7444	7414	8602	4523	5116
Other Liabilities	222	948	4888	1330	4185	4701	5640	2268	2476	1505	1689	1982	1937	2070	2794	2994	2959	3474	7779	7381
Total Liabilities	**12,054**	**21,520**	**25,617**	**27,439**	**33,391**	**33,763**	**42,006**	**36,076**	**29,432**	**31,719**	**31,112**	**30,563**	**33,854**	**35,956**	**38,197**	**42,370**	**45,195**	**54,770**	**57,215**	**55,866**
Preferred Stock (Carrying Value)	894	894	384	257	257	257	257	257	257	257	257	257	257	257	257	0	0	0	0	0
Common Equity	4752	6992	8368	10,213	10,538	11,426	10,397	12,913	9406	10,077	11,185	11,997	11,446	10,568	11,642	11,555	12,664	14,605	17,142	20,107
Total Share-holders' Equity	5646	7886	8752	10,470	10,795	11,683	10,654	13,170	9663	10,334	11,442	12,254	11,703	10,825	11,899	11,555	12,664	14,605	17,142	20,107
Accumulated Minority Interest	99	1	0	0	0	0	0	23	28	0	0	0	57	57	0	402	938	2235	2228	1941
Total Equity	**5745**	**7887**	**8752**	**10,470**	**10,795**	**11,683**	**10,654**	**13,193**	**9691**	**10,334**	**11,442**	**12,254**	**11,760**	**10,882**	**11,899**	**11,957**	**13,602**	**16,840**	**19,370**	**22,048**
Total Liabilities & Shareholders' Equity	**17,799**	**29,407**	**34,369**	**37,909**	**44,186**	**45,446**	**52,660**	**49,269**	**39,123**	**42,053**	**42,554**	**42,817**	**45,614**	**46,838**	**50,096**	**54,327**	**58,797**	**71,610**	**76,585**	**77,914**
Per Share Values																				
Book Value per Share	12.75	14.22	15.81	16.58	16.21	16.80	14.98	18.50	16.30	17.28	18.67	20.65	20.08	18.35	20.04	19.75	21.25	23.26	26.58	29.53
Tangible Book Value per Share	12.40	7.10	7.26	9.09	9.05	9.90	7.90	11.44	9.21	10.05	11.92	14.14	13.45	11.98	13.76	13.57	14.77	12.08	15.58	19.13

Source: FactSet Fundamentals
All figures in millions of US dollar except per share items

References

Bernstein, L. (1988). *Financial statement analysis: Theory, application, and interpretation* (4th ed.). New York: Irwin. Chapters 4, 5, 6, 7, 8.

Brealey, R. A., & Myers, S. C. (2003). *Principles of corporate finance* (7th ed.). New York: McGraw-Hill/Irwin. Chapter 31.

Chan, L. K. C., Hamao, Y., & Lakonishok, J. (1991). Fundamentals and stock returns in Japan. *Journal of Finance, 46*, 1739–1764.

Chan, L. K. C., Lakonishok, J., & Hamao, Y. (1993). Can fundamentals predict Japanese stock returns. *Financial Analysts Journal*. https://doi.org/10.2469/faj.v49.n4.63

Dewing, A. S. (1953). *The financial policy of corporations* (5th ed.). New York: Ronald Press.

Fama, E. F., & French, K. R. (1992). Cross-sectional variation in expected stock returns. *Journal of Finance, 47*, 427–465.

Fama, E. F., & French, K. R. (1995). Size and the book-to-market factors in earnings and returns. *Journal of Finance, 50*, 131–155.

Graham, B., & Dodd, D. (1934). *Security analysis: Principles and technique* (1st ed.). New York: McGraw-Hill Book Company.

Graham, B., Dodd, D., & Cottle, S. (1962). *Security analysis: Principles and technique* (4th ed.). New York: McGraw-Hill Book Company.

Horngren, C. T. (1984). *Introduction to financial accounting* (2nd ed.). Upper Saddle River, NJ: Prentice-Hall. Chapter 7.

Lakonishok, J., Shliefer, A., & Vishay, R. W. (1994). Contrarian investment, extrapolation and risk. *Journal of Finance*, 49, 1541–1578.

Palepu, K. G., Healy, P. M., & Bernard, V. L. (2000). *Business analysis & valuation* (2nd ed.). Cincinnati, OH: South-Western College Publishing. Chapters 4, 5, 8.

Penman, S. H. (2001). *Financial statement analysis and security analysis*. McGraw-Hill/Irwin, New York. Chapter 7.

Richardson, S. A., Sloan, R. G., Soliman, M. T., & Tuna, I. (2001) *Information in accruals about the quality of earnings*. University of Michigan Working Paper.

Schwartz, E. (1962). *Corporate finance*. New York: St. Martin's Press. Chapter 3.

Sloan, R. G. (1996). Do stock prices reflect information in accruals and cash flows about future earnings? *The Accounting Review, 71*, 289–315.

Weston, J. F., & Copeland, T. E. (1986). *Managerial finance* (8th ed.). Chicago: The Dryden Press. Chapters 2, 11, 26.

White, G. I., Sondhi, A. C., & Fried, D. (1997). *The analysis and use of financial statements* (2nd ed.). New York: John Wiley & Sons. Chapters 2, 4.

The Annual Operating Statements: The Income Statement and Cash Flow Statement



The balance sheet of a company and the income statement are related. The balance sheet is an accounting snapshot at a point of time. The income, profit and loss, or operating statement is a condensation of the firm's operating experiences over a given period of time.[1] It depicts certain changes that have occurred between the last balance sheet and the present one. The balance sheet (position statement) and the income statement may be reconciled through the retained earnings, or earned surplus, account. If this reconciliation is presented formally, it becomes the surplus statement.

The income statement is very important. The balance sheet depicts how much assets historically have been invested in a firm; the operating statement indicates how successful (whether by efficiency, daring, or chance) the company has been in making a return on the assets committed to it. In this volume, we regard net income, the bottom line of the income statement, as the most important result of a company's operations. The net income, and the resulting earnings per share, calculated as net income divided by the number of common stock shares outstanding, are the statistics that analysts seek to forecast and use in stock valuation models of the firm.

[1]Operating statements for internal control can be made up for any feasible time period. Most large firms present quarterly statements for their investors, although the annual results carry the most weight and go into the record books of the financial services, such as *Mergent*'s (formerly *Moody*'s), *Standard & Poor's*, etc., and are available on line at MSN/Money, as we will illustrate in this chapter.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_4

Form and Content of the Income Statement

The operating statement takes different forms, depending on the intended audience. The detailed breakdown made for the operating management is usually not presented in the annual report to the general stockholders. Furthermore, the format and the order of items on the report differ according to the tastes and traditions of the managements of different firms.

Financial services that gather data on corporations for the investment community use a similar format for all firms to make it easy to compare companies. The form used by the services breaks out most of the important variables that are interesting for investment analysis.

The following discussion explains the major items appearing in the suggested statement and gives something of their significance.

1. *Sales or Revenues*. The sales account shows the total gross revenue received by the firm during the period. It includes sales for cash and for credit, whether or not they were collected at the end of the period. The sales figure should be net of allowances made to the buyers for spoiled or poor quality goods or returned shipments.
2. *Direct Operating Cost, the Cost of Goods sold, or Cost of Revenue*. The direct operating costs are the amounts spent for material, labor on the goods sold, plus the other costs expended during the period such as selling, administrative, and advertising costs. Items such as the expense for fuel, power, light, local taxes other than those on income, and telephone charges are included in this figure. Although under certain circumstances it is helpful to have separate figures on the cost of goods sold and other costs, this breakdown is generally interesting only to the operating management. The direct operating costs do not in any case, include such noncash charges as the depreciation of fixed assets, depletion, or the amortization of franchises or patents.
3. *Regular Nonoperating Income.* This account includes interest income, dividends on investments, and similar items. Income from major subsidiaries should be consolidated on the reported income statement, even if this is not done for tax purposes. Thus this account does not include the dividends from dominated subsidiary companies. Irregular income, such as that which might occur from the sale of an operating asset at a profit, are presented near the foot of the statement after the results of regular operations are reported.
4. *Earnings Before Depreciation, Interest, and Taxes.* This figure represents the gross return on the company's operation. It is the amount by which the revenues exceed the "variable" costs, or costs of goods sold. Out of this sum come the funds to satisfy various claimants to a return from the firm and internal funds that can be used to reduce debt or buy new assets according to the company's position after fixed claims are met and the distribution to the owners is decided.
5. *Depreciation (and Other Noncash Charges)*. Noncash charges are such items as depletion, depreciation, or amortization of franchises, patents, etc. (these charges

were discussed in chapter "The Corporation Balance Sheet"). Depreciation is usually by far the most important of these items.

The-depreciation account is the estimated capital (i.e., fixed assets) used up during the year. The depreciation charge is based on the cost—not the present value—of the assets and the schedule of depreciation charges on an asset once set initially cannot be varied except under special circumstances. The level of depreciation charges is important in setting the amount of corporation profit tax due. It is important in reminding the management that not all the returns coming in are income, some must be considered a return of capital, and dividends policies should be set accordingly.

The annual depreciation charges do not, however, set the amount of fixed assets that will be replaced or new fixed assets purchased. This decision is based on the forecasted future profitability of the replacement or of the new assets, as the reader will see in chapter "Investing in Assets: Theory of Investment Decision-Making." If investment in new fixed assets appear to generate an acceptable rate of return, then available internal funds or other sources of funds will be found. If capital expenditures exceed internally generated cash flows, then new debt or equity must be raised to finance the investment. If the new investment in fixed assets does not exceed the amount of depreciation, then the extra funds can be used for something else,[2] such as retiring long-term debt, repurchasing equity, or paying larger dividends.

6. *Earnings Before Interest and Taxes (EBIT)*. The EBIT represents the income of the firm after the book charge for depreciation has been made. After this figure is determined, the effect of many past financial decisions comes into play. The amount of interest that will be paid is based on the amount of interest-bearing debt the firm has incurred, and this influences the profits tax. The amount available for the common stockholders is obtained after dividends on the preferred stock is subtracted. These figures are influenced by decisions on alternate methods of financing the firm. The pros and cons of these decisions make up a large part of the subject of corporation finance.
7. *Interest*. This item represents the interest paid by the company on debt. It is reduced by the amortization of any premium on bonds payable, and increased by the amortization of bond discounts. Interest expense is deductible before calculating the corporation income tax.
8. *Corporation Profits Tax*. The current corporate income tax schedule for tax years beginning after December 31, 1992, were:

[2]To refer to depreciation charges as a source of funds is the common shorthand usage, which to be frank is somewhat inaccurate. Strictly speaking, only the firm's operations provide funds. If the firm's revenues did not exceed its direct expenses, there would be no funds flow for the period. However, since the depreciation charge is added to retained earnings for the period to obtain the total of reinvested internally generated funds, the custom has grown of referring to depreciation as a source of funds.

Taxable income	Tax rate (%)	Minus $ = $ tax
$0–$50,000	15	Minus $0 = $ tax
$50,001–$75,000	25	Minus $5000 = $ tax
$75,001–$100,000	34	Minus $11,750 = $ tax
$100,001–$335,300	38	Minus $16,750 = $ tax
$335,301–$10,000,000	34	Minus $0 = $ tax
$10,000,001–$15,000,000	35	Minus $100,000 = $ tax
$15,000,001–$18,333,333	38	Minus $550,000 = $ tax
>$18,333,334	35	Minus $0 = $ tax

Source: Small Business, Quickfinder Handbook. www.quickfinder.com

Net income will be substantially higher for firms in the future because the US corporate income tax rate was reduced in H.R. 1, the Tax Cuts and Jobs Act of December 20, 2017, from 35% to 21%, for incomes exceeding 10 million dollars and because firms can now deduct all costs of new investment in the year of service, rather than depreciating them over multiple time periods.

9. *Earnings After Taxes, or Net Income*. The earnings after taxes as depicted on our model statement is the amount earned on the total equity of the corporation from regular sources. It is not the dividends paid on the owners' investment nor is it the amount earned on the common stock equity.

10 and 11. *Nonrecurring Losses or Gains (after taxes).* Nonrecurring losses or gains arise out of transactions such as the sale of fixed assets (buildings, land, or equipment), often associated with discontinued operations or the sale of subsidiaries or investments in securities. Losses can also occur because of natural disasters (floods or fires) or because of liabilities on lawsuits. In any case these gains or losses do not arise out of the normal operations of the business. These items are given separately because they are special or nonrecurring. If a firm sells a plant or subsidiary at a profit, the earnings for a given year are raised, but the earnings generated by the subsidiary will no longer be available in the future. The tax on nonrecurring gains will probably not be at the regular 35% rate, for if the transaction is classified as a capital gain, the maximum rate is 28%. Whether or not nonrecurring losses are fully deductible for tax purposes depends on the circumstances. It is suggested that when nonrecurring gains or losses occur, they be entered net of taxes (a loss would be reduced if there were regular income tax which could be used as an offset) after the regular part of the income report. Details should be provided in footnotes. In our illustrative statement, there are no nonrecurrent items.

12. *Dividends on Preferred Shares*. The dividends declared on the preferred shares are subtracted from net profits to obtain the earnings available to the common. Although the preferred dividends are not a legally fixed obligation of the firm, they represent a claim senior to any return on common shares, and there can be no calculation of earnings on the common shareholders' investment until they are accounted for. The preferred dividends are a prior charge from the view of the true residual owners of the firm—the common shareholders. Any accrued passed

preferred dividends should be cited in a footnote to the statement.[3]

13. *Earnings Available to Common Stock.* This amount represents the accounting profit or earnings accruing to the shareholders of the business after all prior deductions have been made. The term "accounting profits or earnings" is used deliberately. The accounting profits and the "true" or "economic" profits of the firm can differ considerably. (Economic profits is the amount which remains after all imputed costs—such as interest on all the funds employed no matter what the source—are subtracted.) However, the economists differ on their definition of economic profits, and none of these definitions are easily implemented in practice. Thus, the reported accounting profits serve as a useful available measure of the firm's success. Most stock valuation models, such as those presented in chapters "The Equity of the Corporation: Common and Preferred Stock" and "Risk and Return of Equity and the Capital Asset Pricing Model," rely upon reported earnings. Moreover, under the discipline of the accounting formalities, profits are reported on a sufficiently consistent basis to enable them to be used in the determination of important legal obligations and privileges. But even within the accounting rules, there exist legitimate alternative methods of reporting certain expenses and charges, which can cause considerable variation in the operating results of any year.[4]
14. *Dividends on Common Stock*. Dividends are shown here as a charge on earnings. In most jurisdictions, however, legally they constitute a charge against surplus, not current earnings, and may be declared as long as there is a sufficient credit balance in the surplus account, even if there are no current profits. Firms may elect to do this. As a practical matter, however, the dividend policies of most firms are conditioned by their current earnings position and not by their retained surplus account, and so from the point of view of functional relationships, the order of accounting presented seems quite correct.[5] Common stock dividends are a voluntary distribution of the profits of the firm and not a legal obligation. Their declaration does not reduce the profits of the firm. Thus, dividends are deducted after the earnings on the common are calculated. Moreover, the profits tax liability of the firm is not affected by either the payment of preferred dividends or common dividends.
15. *Addition to Earned Surplus for the Year*. What is left after all dividends are subtracted from the reported profits are the retained earnings, reinvested earnings, or net addition to surplus, for the year. If expenses exceed revenues, there would be instead an operating loss or deficit for the year. The retained earnings for the period depend on the level of profit and the dividend policy of the company.

[3]The relationship between common and preferred stock is explained more fully in chapter "The Equity of the Corporation: Common and Preferred Stock."

[4]See the discussion in chapter "The Corporation Balance Sheet" on depreciation and inventory valuation.

[5]For a fuller discussion of dividends and their relationship to earnings and the value of common stock, see chapter "The Equity of the Corporation: Common and Preferred Stock."

These in turn are influenced by factors such as the stability and amount of the company's cash flow, the firm's growth prospects, and its need for equity funds either to acquire additional assets or to repay debt.

16. *Earnings per Share and Cash Flow per Share*. The amount earned and paid in dividends on the individual stockholder's share is of more direct importance to him than the total amount earned by the firm. The earnings and trend of dividends on the individual shares in the long run establish their value in the market.

    The earnings per share are obtained by dividing the total earnings available to the common stock by the number of shares of stock outstanding. Often the earnings per share are shown once, reflecting the regular, recurring income and, again, including extraordinary income items. An additional figure, not always available but often useful, is the cash flow per share. It includes earnings available to the common shareholders plus noncash charges divided by the number of shares. This figure shows the gross funds available per share of stock which may be used to repay debt, acquire assets, and pay dividends. An interesting possibility is to subtract required amortization of debt from the cash flow per share and arrive at the figure of "free" or "disposable" cash flow per share. This figure might prove useful in comparing two firms where earnings are similar, but one firm is required to make payments on the principal of its debt.

    A simple income statement (in $MM) illustrates its components, as was done for the balance sheet in chapter "The Corporation Balance Sheet."

Sales	$722.50
– Cost of goods sold	470
– Depreciation	95
Gross income	157.50
– General, selling expenses	72.50
EBIT	85
– Interest paid	40
Earnings before taxes	45
– Taxes	9.45
Earnings after taxes	35.55

    The firm produced sales of $722.5 million by selling its products, and after subtracting its cost of goods sold and depreciation and amortization charges, gross income of $157.5 million is produced.[6] One subtracts general, selling, and administrative expense from gross income, to create Earnings before Interest and Taxes, EBIT ($85.0 MM for the simple firm). Interest expenses of $40 million must be paid to the firm's bondholders, such that Earnings before Taxes are $45 million. The firm pays taxes of $9.45 million, leaving Earnings after Taxes, or Net

[6]The previous tax rate, pre-2017, produced taxes of approximately $15.8 MM, producing earnings after taxes of $29.2 MM. The increased net income, $6.35 MM enhances the firm valuation by $63.5 MM, if the firm is valued by analysts with a price-earnings multiple of 10. Christmas came early for the management of many companies, and stock holders, in 2017!

Income, of $35.55 million. A net income figure is often used by analysts to value the firm. The net income number by itself may not as useful to an investor unless net income is represented as a percent of sales, total assets, or equity. The ratio of net income relative to its total assets, referred to as a firm's ROA, or return on assets, allows the investor to compare the profitability of the firm relative to all firms, or firms within an industry or sector.[7] Comparisons can be made regarding net income relative to a firm's sales, or return on sales, ROS, and equity, referred to as the return on equity, ROE. One prefers to see higher values of ROA, ROS, and ROE. We will calculate these ratios in chapter "Financing Current Operations and Efficiency Ratio Analysis."

Retained Earnings vs. Dividends

Corporate profits in the year earned are taxed by the federal government at a rate of 21%; and any distribution of dividends are taxed as additional income to the recipient at the appropriate personal income tax bracket rate.[8] This is what constitutes the "double taxation of corporation income."

Although the disparity between the corporation profits rate and the personal income tax rate, and the fact that they are applied separately, may be economically detrimental to some holders of corporation shares, it may benefit others. For individuals in very high income brackets, the 21% corporation profits tax may be lower than their personal tax rate. If the corporation retains some profits and pays dividends out of these at a later date—for example, at retirement, when the individual's tax bracket may have fallen—there can be a net savings of overall tax. More usual is the use of retained earnings to generate more earnings (and potentially higher dividends) so that, for example, the stock can be sold at a capital gain equivalent to the retained earnings. Retained earnings provide financing for the firm's investment projects, as we will see in chapters "The Equity of the Corporation: Common and Preferred Stock" and "Investing in Assets: Theory of Investment Decision-Making." Firms pay dividends to provide cash for its investors. Stock valuation is often extremely dependent upon dividends.

---

[7]The return on assets may be expressed as the ratio of net income to total assets, or the ratio of net income to average total assets. Accountants often prefer to express the ROA as earnings before interest and taxes (EBIT) divided by average total assets (White, Sondhi, & Fried, 1997), p. 147. Alternatively, White, Sondhi, and Fried present the ratio of net income plus after-tax interest cost, relative to average total assets (p. 147). These accountants use average sales and stockholder equity in the ROS and ROE calculations. We use the year-end total assets, sales, and stockholder equity in this text.

[8]That is, if the recipient of the income is an individual and not a nontaxable institution or another corporation. An 85% dividend received credit is given to corporations for dividends received from another corporation which is non-consolidated. (To consolidate for tax purposes, the parent must own a minimum of 80% of the subsidiary's shares.) The other 15% is taxed at the usual corporate income tax rate.

Income Statement in the World of Business: IBM, Boeing, and Dominion Energy

Income statements can be found for investors and researchers. A source of the most recent 4 years (2015–2018) of income statement data can be found on MSN, Money. We will continue with the examples of IBM, Boeing, and Dominion Energy that we used in chapter "The Corporation Balance Sheet." Table 1 is a modified income statement for IBM, for the 2015–2018 period, which might be highly for various financial analyses. Revenues, gross profit, operating income, net income, and basic and fully diluted earnings per share (EPS) are reported in millions of dollars. Similar income statements are shown for Boeing and Dominion Energy in Tables 2 and 3. We will stress net income and basic and fully diluted EPS as investment variables for stock selection and stock valuation. In Table 1, IBM's revenues and net income have fallen during the 2015–2018 time period from $81.74 billion to $479.59 billion and $13.19 to $8.73 billion, respectively. It would seem that IBM is in a virtually no-growth environment, if not declining growth environment, hardly the case that Guerard and Schwartz (2007) reported and discussed in our first edition. IBM had revenues of $79.59 billion in 2018, experienced costs of goods sold of $42.66 billion, producing gross profits of $36.94 billion. Research and development expenditures of $5.38 billion and general, selling, and administrative expenditures of 20.33 billion produced operating income of $12.23 billion. Subtracting other expenses of $0.89 billion produces an earnings before interest and Taxes, EBIT, figure of $12.23 billion. The EBIT is associated by investors as an "ordinary income" figure of the company. Interest expenses of $723 million produces an earnings before taxes, EBT, figure of $11.34 billion in 2018. Taxes of $2.62 billion, averaging approximately 23% of EBT, produces net income of $8.72 billion from recurring operations and approximately $8.73 billion of net income (the $8.728 billion figure is reported in the IBM Annual Report for 2018 in its Consolidated Statement of Earning; see page 70).

Boeing increased its revenues during the 2015–2018 time period from $96.11 billion to $101.13 billion and substantially increased its net income from $5.18 billion to $10.46 billion (see Table 2). This substantially increase in net income sent its stock price soaring! Boeing still spent $3.3 billion on research and development expenditures.

Dominion Energy increased its revenues and net income during the 2015–2018 time period from $11.68 billion to 13.37 billion and from $1.90 to $2.45 billion, respectively (see Table 3).

What is important about revenues and net income? Firms that produce higher revenue growth and earnings per share, EPS, growth that expected by analysts produce higher stock returns than lower growth firms. It is important to grow revenues and EPS, but more important to stockholders who seek to outperform the market that these growth rates exceed expectations, as we will see in chapter "The Equity of the Corporation: Common and Preferred Stock." Furthermore, the as-reported, "bottom-line" net income data for the three firms will be shown relative

Table 1 IBM income statement

	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Revenue				
Total Revenue	79,591,000	79,139,000	79,919,000	81,741,000
Cost of Revenue	42,655,000	42,196,000	41,402,000	41,057,000
Gross Profit	**36,936,000**	**36,943,000**	**38,517,000**	**40,684,000**
Operating Expenses				
Research Development	5,379,000	5,590,000	5,726,000	5,247,000
Selling General and Administrative	20,325,000	20,900,000	20,225,000	19,589,000
Nonrecurring	–	–	–	–
Others	−998,000	−1,445,000	−1,604,000	−669,000
Total Operating Expenses	67,361,000	67,241,000	65,749,000	65,224,000
Operating Income or Loss	**12,230,000**	**11,898,000**	**14,170,000**	**16,517,000**
Income from Continuing Operations				
Total Other Income/Expenses Net	−888,000	−498,000	−1,840,000	−572,000
Earnings Before Interest and Taxes	12,230,000	11,898,000	14,170,000	16,517,000
Interest Expense	−723,000	−615,000	−630,000	−468,000
Income Before Tax	11,342,000	11,400,000	12,330,000	15,945,000
Income Tax Expense	2,619,000	5,642,000	449,000	2,581,000
Minority Interest	134,000	131,000	146,000	162,000
Net Income From Continuing Ops	**8,723,000**	**5,758,000**	**11,881,000**	**13,364,000**
Nonrecurring Events				
Discontinued Operations	5000	−5000	−9000	−174,000
Extraordinary Items	–	–	–	–
Effect of Accounting Changes	–	–	–	–
Other Items	–	–	–	–
Net Income				
Net Income	**8,728,000**	**5,753,000**	**11,872,000**	**13,190,000**
Preferred Stock and Other Adjustments	–	–	–	–
Net Income Applicable to Common Shares	**8,728,000**	**5,753,000**	**11,872,000**	**13,190,000**

to the firm's total assets (ROA), sales (ROS), and equity (ROE) in the coming chapter. The ROA ratio illustrates the effective use of the firm's assets. That is, a higher ROA for the firm relative to a set of comparable firms, its industry, or sector or the economy shows that the firm's assets are employed to generate higher net income and a more effective use of assets. A higher relative ROS indicates that for a given sales volume, the firm generates higher net income than comparable firms. The ROE ratio helps quantify the firm's profitability relative to its use of leverage. Chapter "Financing Current Operations and Efficiency Ratio Analysis" makes extensive use of the ROA, ROS, and ROE ratios.

The online reader sees 4 years of income statements on MSN, its Money tab. Is 4 years of data enough to make an intellect assessment of the company's financial conditions? Probably not. Let us now show 20-year income statements for IBM,

Table 2 Boeing income statement

	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Revenue				
Total Revenue	101,127,000	94,005,000	93,496,000	96,114,000
Cost of Revenue	81,490,000	76,612,000	79,026,000	82,088,000
Gross Profit	**19,637,000**	**17,393,000**	**14,470,000**	**14,026,000**
Operating Expenses				
Research Development	3,269,000	3,179,000	3,391,000	3,331,000
Selling General and Administrative	4,525,000	4,101,000	4,091,000	3,525,000
Nonrecurring	–	–	–	–
Others	–	–	–	–
Total Operating Expenses	89,284,000	83,892,000	86,508,000	88,944,000
Operating Income or Loss	**11,843,000**	**10,113,000**	**6,988,000**	**7,170,000**
Income from Continuing Operations				
Total Other Income/Expenses Net	−239,000	−6000	−1,205,000	−15,000
Earnings Before Interest and Taxes	11,843,000	10,113,000	6,988,000	7,170,000
Interest Expense	−475,000	−360,000	−306,000	−275,000
Income Before Tax	11,604,000	10,107,000	5,783,000	7,155,000
Income Tax Expense	1,144,000	1,649,000	749,000	1,979,000
Minority Interest	71,000	57,000	60,000	62,000
Net Income from Continuing Ops	**10,460,000**	**8,458,000**	**5,034,000**	**5,176,000**
Nonrecurring Events				
Discontinued Operations	–	–	–	–
Extraordinary Items	–	–	–	–
Effect of Accounting Changes	–	–	–	–
Other Items	–	–	–	–
Net Income				
Net Income	**10,460,000**	**8,458,000**	**5,034,000**	**5,176,000**
Preferred Stock and Other Adjustments	–	–	–	–
Net Income Applicable to Common Shares	**10,460,000**	**8,458,000**	**5,034,000**	**5,176,000**

Boeing, and Dominion using the FactSet database service. We show the 20-year income statements in Tables 4, 5 and 6. IBM revenues, during the 1999–2018 time period, fell from $87.54 billion to $79.59 billion. Net income rose from $7.71 to $8.73 billion. Earnings per share, EPS, rose from $4.91 to $10.46. The reader may seem confused at this time; IBM's revenues fell, but IBM's net income and EPS rose. Is not EPS defined as net income divided by common stock shares outstanding? Indeed, it is. IBM EPS rose substantially because its number of common stock shares outstanding fell from 1.87 MM to 0.87 MM. IBM stock repurchases drove its EPS and stock price up during the 1999–2018 time period. See Table 4 for FactSet IBM income statements, 1999–2018. During the 1999–2018 time period, Boeing revenues increased from $57.99 billion to $101.11 billion and net income rose from

Table 3 Dominion energy income statement

Period Ending	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Revenue				
Total Revenue	13,366,000	12,586,000	11,737,000	11,683,000
Cost of Revenue	6,879,000	6,182,000	5,851,000	6,102,000
Gross Profit	**6,487,000**	**6,404,000**	**5,886,000**	**5,581,000**
Operating Expenses				
Research Development	–	–	–	–
Selling General and Administrative	−469,000	−478,000	−456,000	−456,000
Nonrecurring	–	–	–	–
Others	703,000	668,000	596,000	551,000
Total Operating Expenses	9,113,000	8,277,000	7,550,000	8,048,000
Operating Income or Loss	**4,253,000**	**4,309,000**	**4,187,000**	**3,635,000**
Income from Continuing Operations				
Total Other Income/Expenses Net	−1,124,000	−1,219,000	−1,320,000	−807,000
Earnings Before Interest and Taxes	4,253,000	4,309,000	4,187,000	3,635,000
Interest Expense	−1,493,000	−1,205,000	−994,000	−904,000
Income Before Tax	3,129,000	3,090,000	2,867,000	2,828,000
Income Tax Expense	580,000	−30,000	655,000	905,000
Minority Interest	1,941,000	2,228,000	2,235,000	938,000
Net Income from Continuing Ops	**2,549,000**	**3,120,000**	**2,212,000**	**1,923,000**
Nonrecurring Events				
Discontinued Operations	–	–	–	–
Extraordinary Items	–	–	–	–
Effect of Accounting Changes	–	–	–	–
Other Items	–	–	–	–
Net Income				
Net Income	**2,447,000**	**2,999,000**	**2,123,000**	**1,899,000**
Preferred Stock and Other Adjustments	–	–	–	–
Net Income Applicable to Common Shares	**2,447,000**	**2,999,000**	**2,123,000**	**1,899,000**

$2.31 to $10.46 billion, respectively. Boeing EPS rose from $2.49 to $17.85. The reader is referred to Table 5 for FactSet Boeing income statements, 1999–2018. During the 1999–2018 time period, Dominion Energy revenues increased from $5.52 to $13.46 billion; net income rose from $0.68 to $2.45 billion; and Dominion EPS rose from $1.41 to $4.17, respectively. The reader is referred to Table 6, for Dominion Energy income statements, 1999–2018. For most firms, as net income rises, so does its EPS. The growth in net income and EPS may not be proportional, because of stock repurchases.

Table 4 IBM income statements, 1999–2018. International Business Machines Corporation. IBM 459200101 2005973 NYSE Common stock

	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Sales	87,548	88,396	85,866	81,186	89,131	96,293	91,134	91,424	98,786	103,630	95,758	99,870	106,916	104,507	99,751	92,793	81,741	79,919	79,139	79,591
Cost of Goods Sold (COGS) incl. D&A	55,441	55,972	54,084	51,405	56,581	60,767	55,065	53,762	57,690	58,597	52,678	54,495	57,306	54,936	51,942	47,060	41,927	42,533	43,988	43,548
Gross Income	**32,107**	**32,424**	**31,782**	**29,781**	**32,550**	**35,526**	**36,069**	**37,662**	**41,096**	**45,033**	**43,080**	**45,375**	**49,610**	**49,571**	**47,809**	**45,733**	**39,814**	**37,386**	**35,151**	**36,043**
SG&A Expense	22,436	20,790	22,487	20,838	22,037	23,899	24,928	25,444	27,262	28,389	25,907	26,719	28,824	28,582	28,230	26,626	24,641	25,063	24,925	23,244
EBIT (Operating Income)	**9671**	**11,634**	**9295**	**8943**	**10,513**	**11,627**	**11,141**	**12,218**	**13,834**	**16,644**	**17,173**	**18,656**	**20,786**	**20,989**	**19,579**	**19,107**	**15,173**	**12,323**	**10,226**	**12,799**
Nonoperating Income — Net	4614	617	2301	1139	1019	1212	2526	1794	1966	1423	1627	1506	511	2254	1489	3437	902	1,874	1,722	–50
Interest Expense	727	717	238	145	145	139	220	278	611	673	498	471	510	569	469	575	574	777	673	686
Unusual Expense —Net	–	–	405	2413	513	672	1221	417	700	679	159	–41	–222	761	1,068	1977	–452	1,074	–142	704
Income Taxes	4,045	3441	3230	2190	3261	3580	4232	3901	4071	4381	4713	4890	5148	5298	3041	4235	2581	449	5,642	2,619
Consolidated Net Income	**7,712**	**8093**	**7723**	**334**	**7,613**	**8448**	**7994**	**9416**	**10,418**	**12,334**	**13,430**	**14,842**	**15,861**	**16,615**	**16,490**	**15,757**	**13,372**	**11,897**	**5,775**	**8,740**
Minority Interest	0	0	0	0	0	0	0	0	0	0	5	9	6	11	7	6	8	16	17	17
Net Income	**7,712**	**8093**	**7723**	**5334**	**7613**	**8448**	**7994**	**9416**	**10,418**	**12,334**	**13,425**	**14,833**	**15,855**	**16,604**	**16,483**	**15,751**	**13,364**	**11,881**	**5,758**	**8,723**
Discontinued Operations	0	0	0	–	–	–18	–24	76	0	0	0	0	0	0	0	997	–58	–9	–6	5
Preferred Dividends	20	20	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Net Income available to Common	7,692	8073	7713	5334	7613	8430	7970	9492	10,418	12,334	13,425	14,833	15,855	16,604	16,483	16,748	13,306	11,872	5,752	8,728
Per Share																				
EPS (recurring)	4.12	4.44	4.51	4.05	4.54	5.21	5.42	6.30	7.52	9.27	10.09	11.50	12.93	14.83	15.62	17.95	13.22	13.17	6.03	10.06
EPS (diluted)	4.12	4.44	4.35	3.07	4.34	4.93	4.90	6.11	7.18	8.93	10.01	11.52	13.06	14.37	14.94	16.58	13.54	12.38	6.13	9.52
Earnings Persistence	95.26	87.54	93.45	91.74	91.19	90.25	91.37	84.97	84.90	78.01	93.95	82.65	94.64	84.77	78.76	84.60	86.83	82.68	90.28	91.37
Dividends per Share	0.47	0.51	0.55	0.59	0.63	0.70	0.78	1.10	1.50	1.90	2.15	2.50	2.90	3.30	3.70	4.25	5.00	5.50	5.90	6.21
EBITDA																				
EBITDA	16,256	16,629	14,115	13,322	15,214	16,542	16,329	17,201	19,035	22,094	22,167	23,487	25,601	25,665	24,257	23,599	19,028	16,704	14,767	17,279

Source: FactSet Fundamentals
All figures in millions of U.S. Dollar except per share items

Table 5 Boeing income statements, 1999–2018. Boeing Company. BA 097023105 2108601 NYSE Common stock

	DEC '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Sales	57.99	51.32	58.20	54.07	50.49	52.46	54.85	61.53	66.39	60.91	68.28	64.31	68.74	81.70	86.62	90.76	96.11	94.57	93.39	101.13
Cost of Goods Sold (COGS) incl. D&A	51.05	43.71	48.78	45.50	43.86	44.68	46.21	50.46	53.43	50.39	56.54	51.84	55.87	68.64	73.27	76.75	82.09	80.79	76.07	81.49
Gross Income	**6.94**	**7.61**	**9.42**	**8.57**	**6.62**	**7.78**	**8.63**	**11.07**	**12.96**	**10.52**	**11.74**	**12.46**	**12.87**	**13.05**	**13.36**	**14.01**	**14.03**	**13.78**	**17.33**	**19.64**
SG&A Expense	3.39	3.78	4.33	4.17	4.42	5.54	6.43	7.43	7.38	6.85	9.87	7.77	7.33	7.02	7.03	6.81	6.86	8.24	7.27	7.84
Other Operating Expense	0.21	0.32	0.38	0.45	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
EBIT (Operating Income)	**3.35**	**3.52**	**4.72**	**3.95**	**1.75**	**2.25**	**2.20**	**3.64**	**5.58**	**3.67**	**1.87**	**4.70**	**5.54**	**6.04**	**6.33**	**7.20**	**7.17**	**5.54**	**10.05**	**11.80**
Nonoperating Income—Net	0.23	0.48	0.43	–0.04	0.49	0.40	0.97	0.36	0.78	0.52	0.34	0.53	0.41	0.39	0.32	0.44	0.41	0.39	0.45	0.28
Interest Expense	0.43	0.45	0.65	0.73	0.80	0.69	0.29	0.24	0.20	0.20	0.34	0.52	0.50	0.46	0.39	0.33	0.28	0.31	0.36	0.48
Unusual Expense—Net	–	0.56	0.94	0.00	0.89	0.00	0.06	0.57	0.05	0.00	0.14	0.20	0.06	0.05	0.03	0.16	0.15	0.06	0.10	0.00
Income Taxes	1.02	0.87	0.74	0.86	–0.17	0.14	0.26	0.99	2.06	1.34	0.40	1.20	1.38	2.01	1.65	1.69	1.98	0.67	1.85	1.14
Other After Tax Adjustments	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	–0.01	–0.01	–0.01	0.00	0.00	–0.01	–0.01
Consolidated Net Income	2.31	2.13	2.83	2.32	0.72	1.82	2.56	2.21	4.06	2.65	1.34	3.31	4.01	3.90	4.58	5.44	5.17	4.89	8.19	10.45
Net Income	**2.31**	**2.13**	**2.83**	**2.32**	**0.72**	**1.82**	**2.56**	**2.21**	**4.06**	**2.65**	**1.34**	**3.31**	**4.01**	**3.90**	**4.58**	**5.44**	**5.17**	**4.89**	**8.19**	**10.45**
Discontinued Operations	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Net Income available to Common	2.31	2.13	2.83	2.32	0.72	1.83	2.56	2.21	4.06	2.65	1.34	3.31	4.01	3.90	4.58	5.44	5.17	4.89	8.19	10.45
Per Share																				
EPS (recurring)	2.49	2.89	4.20	2.87	1.66	2.25	3.24	3.34	5.30	3.64	2.01	4.64	5.38	5.16	5.99	7.54	7.59	7.67	13.54	17.85
EPS (diluted)	2.49	2.44	3.41	2.87	0.89	2.25	3.19	2.84	5.26	3.64	1.87	4.45	5.33	5.11	5.97	7.38	7.44	7.61	13.43	17.85
Earnings Persistence	81.01	75.80	78.90	84.52	83.90	87.74	75.70	67.54	83.50	62.47	77.29	58.45	74.67	74.06	59.31	57.06	72.26	47.16	36.18	60.91
Dividends per Share	0.42	0.59	0.68	0.68	0.68	0.77	1.00	1.20	1.40	1.60	1.68	1.68	1.68	1.76	1.94	2.92	3.64	4.36	5.68	6.84
EBITDA																				
EBITDA	4.99	5.00	6.47	5.45	3.20	3.76	3.70	5.19	7.07	5.16	3.54	6.43	7.20	7.85	8.17	9.10	9.00	7.45	12.12	13.92

Source: FactSet Fundamentals
All figures in billions of U.S. Dollar except per share items

Table 6 Dominion energy income statements, 1999–2018. Dominion Energy Inc. D 25746U109 2542049 NYSE Common stock

	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Sales	5520	9260	10,558	10,218	12,078	13,972	18,041	16,482	15,674	16,290	14,059	14,598	14,226	12,905	13,178	12,566	11,480	11,407	12,505	13,456
Cost of Goods Sold (COGS) incl. D&A	3888	4775	5335	4714	6133	7988	11,962	9282	8336	8866	8031	7548	7409	6406	6725	6402	4985	4427	4911	5595
Gross Income	**1632**	**4485**	**5223**	**5504**	**5945**	**5984**	**6079**	**7200**	**7338**	**7424**	**6028**	**7050**	**6817**	**6499**	**6453**	**6164**	**6495**	**6980**	**7594**	**7861**
SG&A Expense	–	–	–	–	2260	–	3023	3280	3542	3257	334 0	3171	3301	2851	2384	2924	2595	3064	2,875	3458
Other Operating Expense	304	2496	2641	2603	476	3267	582	575	552	499	491	532	554	571	563	542	551	596	668	703
EBIT (Operating Income)	**1328**	**1989**	**2582**	**2901**	**3209**	**2717**	**2474**	**3345**	**3244**	**3668**	**2197**	**3347**	**2962**	**3077**	**3506**	**2698**	**3349**	**3320**	**4051**	**3700**
Nonoperating Income—Net	91	95	184	103	106	186	168	174	3196	–100	1095	3184	245	389	175	304	403	549	215	1277
Interest Expense	507	958	997	945	975	939	991	1030	941	853	890	904	856	897	862	1177	892	979	1153	1445
Unusual Expense—Net	–	460	757	16	794	–	35	–	1005	–	486	590	180	2,072	115	47	32	23	23	403
Income Taxes	259	183	370	681	597	700	582	920	1783	879	612	2057	745	146	892	452	905	655	–30	580
Consolidated Net Income	**636**	**483**	**642**	**1362**	**949**	**1264**	**1034**	**1569**	**2711**	**1836**	**1304**	**2980**	**1426**	**351**	**1812**	**1326**	**1923**	**2212**	**3120**	**2549**
Minority Interest	18	2	0	0	0	0	0	6	6	0	17	17	18	27	23	16	24	89	121	102
Net Income	**618**	**481**	**642**	**1362**	**949**	**1264**	**1034**	**1563**	**2705**	**1836**	**1287**	**2963**	**1408**	**324**	**1789**	**1310**	**1899**	**2123**	**2999**	**2447**
Discontinued Operations	0	0	0	0	–642	–15	5	–183	–	8	–2	–15	0	–22	–92	0	0	0	0	0
Preferred Dividends	67	66	98	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Net Income available to Common	551	415	544	1362	307	1249	1039	1380	2697	1834	1287	2948	1408	302	1697	1310	1899	2123	2999	2447
Per Share																				
EPS (recurring)	1.41	1.56	2.13	2.43	1.35	1.91	1.54	1.96	5.19	3.16	2.74	5.70	2.67	3.05	3.07	2.30	3.24	3.47	4.74	4.17
EPS (diluted)	1.41	0.88	1.08	2.41	0.49	1.91	1.51	1.96	4.12	3.16	2.17	5.00	2.45	0.53	2.93	2.24	3.20	3.44	4.72	3.74
Earnings Persistence	88.47	74.80	81.86	84.02	86.25	94.40	85.12	83.18	75.82	86.50	87.93	91.56	86.72	98.40	86.45	84.93	83.54	77.74	85.26	97.20
Dividends per Share	1.29	1.29	1.29	1.29	1.29	1.30	1.34	1.38	1.46	1.58	1.75	1.83	1.97	2.11	2.25	2.40	2.59	2.80	3.04	3.34
EBITDA																				
EBITDA	2126	3257	3904	4280	4543	4150	4012	5084	4777	4859	3516	4605	4257	4520	4896	4258	5018	5169	6253	5980

Source: FactSet Fundamentals
All figures in millions of U.S. Dollar except per share items

Annual Cash Flow Statement

The income statement provides a picture of the firm's operations during the past year. The "bottom line" of the income statement is the firm's net income or after-tax profits. The firm's sources of cash flow from its operations are positive net income, depreciation and other noncash expenses, net decreases in its current assets and net increases in its current liabilities (funds from operating activities), decreases in its long-term (fixed) assets (funds from investing), and issuing new debt or equity (funds from financing). The firm uses its cash flow to pay dividends, engage in capital expenditures, pursue research and development (R&D) expenditures, repurchase debt and equity, decrease its current liabilities, or increase its current and/or fixed assets. The authors personally believe that the cash flow statement is the most important financial statement because it reports net income, dividends paid, stock repurchases, and debt issued, among other items. The firm's sources of funds must equal its uses of funds.[9] Stockholders prefer to see its firms cash flow derived from profits, not depreciation, because depreciation is an expense that serves to provide the firm with cash flow to replenish its capital investment.

IBM net income and depreciation, $4.48 billion in 2018, provided the vast majority of its $15.25 billion in total cash flow from operations; see Table 7. IBM spent its operating funds on capital expenditures, $3.40 billion, dividends, $5.67 billion. The primary difference between cash flow from operations, $15.25 billion, and financing activities cash flow, $-10.47 billion, was the $4.4 billion spent on common stock repurchases.

During the 1999–2018 time period, IBM produced net income totaling $213.9 billion and depreciation expenses of $97.2 billion. The reader is referred to Table 8. IBM spent its cash flow generated on dividends, $56.3 billion; capital expenditures, $97.1 billion; and common stock repurchases, $151 billion, respectively.

In 2018, Boeing net income and depreciation, $10.46 and 2.11 billion, respectively, provided the vast majority of its $15.32 billion in total cash flow from operations; see Table 9. Boeing spent its operating funds to pursue capital expenditures, $1.72 billion, and pay dividends, $3.9 billion, and increased its liabilities, $2.6 billion, and net borrowings, $1.37 billion.

---

[9] Drtina and Largay (1985) examined cash flow reporting with regard to "cash flow from operations" (CFO). Problems can develop because of ambiguity in terms of the definition of "operations," the measurement of the current position of long-term leases (noted in chapter "Long-Term Debt"), diversity in reporting practices, and reclassification of current and noncurrent accounts. Krishnan and Largay (2000) reported that past direct method cash flows offer better predictions of future operating cash flows than indirect method cash flows information. The direct method of presenting cash flows OCF for time t is predicted to be:

$$OCF_t = \int (CSHRD_{t-1}, CSHPD_{t-1}, INTRD_{t-1}, INTPD_{t-1}, TXPD_{t-1})$$

where

$CSHRD_{t-1}$ = Cash received from customers at time t − 1
$CSHPD_{t-1}$ = Cash paid to suppliers and employees at time t − 1
$INTRD_{t-1}$ = Interest received at time t − 1
$INTPD_{t-1}$ = Interest paid at time t − 1
$TXPD_{t-1}$ = Taxes paid at time t − 1

Table 7 IBM cash flow

Period Ending	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Net Income	**8,728,000**	**5,753,000**	**11,872,000**	**13,190,000**
Operating Activities, Cash Flows Provided by or Used in				
Depreciation	4,480,000	4,541,000	4,381,000	3,855,000
Adjustments to Net Income	1,485,000	−383,000	−526,000	2,406,000
Changes in Accounts Receivables	1,006,000	1,297,000	712,000	812,000
Changes in Liabilities	126,000	47,000	197,000	81,000
Changes in Inventories	−127,000	18,000	−14,000	133,000
Changes in Other Operating Activities	−451,000	5,451,000	462,000	−3,222,000
Total Cash Flow from Operating Activities	**15,247,000**	**16,724,000**	**17,084,000**	**17,255,000**
Investing Activities, Cash Flows Provided by or Used in				
Capital Expenditures	−3,395,000	−3,229,000	−3,567,000	−3,579,000
Investments	−554,000	−1,039,000	−161,000	−231,000
Other Cash Flows from Investing Activities	−1000	−1000	−1000	1000
Total Cash Flows from Investing Activities	**−4,913,000**	**−7,081,000**	**−10,928,000**	**−8,159,000**
Financing Activities, Cash Flows Provided by or Used in				
Dividends Paid	−5,666,000	−5,506,000	−5,256,000	−4,897,000
Sale Purchase of Stock	–	–	–	–
Net Borrowings	−301,000	3,447,000	2,763,000	19,000
Other Cash Flows from Financing Activities	112,000	174,000	204,000	−1000
Total Cash Flows from Financing Activities	**−10,469,000**	**−6,418,000**	**−5,917,000**	**−9,413,000**
Effect of Exchange Rate Changes	−495,000	937,000	−51,000	−473,000
Change In Cash and Cash Equivalents	**−630,000**	**4,161,000**	**188,000**	**−790,000**

During the 1999–2018 time period, Boeing produced net income totaling $75 billion and depreciation expenses of $33.9 billion. The reader is referred to Table 10. Boeing spent its cash flow generated on dividends, $55.3 billion; capital expenditures, $34.1 billion; and common stock repurchases, $3.6 billion, respectively. The reader notices the dramatic impact on IBM expenditure activities on capital expenditures and common stock repurchases relative to Boeing.

In 2018, Dominion Energy produced net income and depreciation, $2.45 and 2.28 billion, respectively, which provided the vast majority of its $4.77 billion in total cash flow from operations; see Table 11. Dominion Energy spent its operating funds on capital expenditures, $4.41 billion, and dividends, $2.19 billion, and decreased its net borrowings $2.21 billion, respectively. The decrease in net borrowings was a $2.21 billion which was a common stock repurchase.

Table 8 IBM cash flow statements 1999–2018. International Business Machines Corporation. IBM 459200101 2005973 NYSE Common stock

Operating Activities	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08
Net Income / Starting Line	7712	8093	7723	5334	7613	8448	7934	9492	10418	12334
Depreciation, Depletion & Amortization	6585	4995	4820	4379	4916	4915	5188	4983	5201	5450
Deferred Taxes & Investment Tax Credit	−713	29	658	−67	1126	2081	2185	1724	740	1900
Other Funds	−4791	−792	88	343	−225	−483	−449	599	624	321
Funds from Operations	8793	12325	13289	9989	13430	14961	14858	16798	16983	20005
Changes in Working Capital	1318	−3051	976	3077	1139	362	56	−1779	−889	−1193
Net Operating Cash Flow	10111	9274	14265	13066	14569	15323	14914	15019	16094	18812
Investing Activities										
Capital Expenditures	−6423	−6181	−6315	−5350	−4974	−5056	−4634	−5166	−5505	−4887
Net Assets from Acquisitions	0	0	-916	−3158	−1836	−1738	−1482	−3799	−1009	−6313
Sale of Fixed Assets & Businesses	1207	1619	1165	2008	1136	1336	2039	430	847	421
Purchase/Sale of Investments	−1333	314	−40	−397	552	112	−346	−3014	992	1510
Other Funds	4880	0	0	0	−162	0	-40	−12	−5	−16
Net Investing Cash Flow	−1669	−4248	−6106	−6897	−5284	−5346	−4463	−11561	−4680	−9285
Financing Activities										
Cash Dividends Paid	−879	−929	−966	−1005	−1085	−1174	−1250	−1683	−2147	−2585
Change in Capital Stock	−6645	−6073	−3906	−3087	−3232	−5418	0	0	−18828	−10578
Issuance/Reduction of Debt, Net	−1101	643	−437	−3173	−3481	−1027	609	−122	12,112	−2444
Other Funds	0	0	0	0	0	0	−6506	−6399	4123	3773
Net Financing Cash Flow	−8625	−6359	−5309	−7265	−7798	−7619	−7147	−8204	−4740	−11834
Exchange Rate Effect	−149.0	−147.0	−83.0	148.0	421.0	405.0	−789.0	200.0	295.0	57.0
Miscellaneous Funds	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Net Change in Cash	−332	−1480	2767	−948	1908	2763	2515	−4546	6969	−2250
Free Cash Flow	**4152**	**3658**	**8605**	**8313**	**10176**	**10955**	**11072**	**10657**	**11464**	**14641**
Free Cash Flow per Share	2	2	5	5	6	6	7	7	8	11
Free Cash Flow Yield (%)	2.06	2.37	4.02	6.20	6.25	6.50	8.28	7.06	7.31	12.59

Operating Activities	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Net Income / Starting Line	13425	14833	15855	16604	16483	12022	13190	11872	5753	8728
Depreciation, Depletion & Amortization	4994	4831	4815	4676	4678	4492	3855	4381	4541	4480
Deferred Taxes & Investment Tax Credit	1773	1294	1212	797	−1610	−237	1387	−1132	−931	853
Other Funds	163	−172	355	−41	378	2357	1020	606	548	633
Funds from Operations	20355	20786	22237	22036	19929	18634	19452	15727	9911	14694
Changes in Working Capital	418	−1237	−2391	−2450	−2444	−1766	−2444	1231	6813	553
Net Operating Cash Flow	20773	19549	19846	19586	17485	16868	17008	16958	16724	15247
Investing Activities										
Capital Expenditures	−4077	−4754	−4667	−4717	−4140	−4183	−4151	−4150	−3773	−3964
Net Assets from Acquisitions	−1194	−5922	−1811	−3722	−3056	−656	−3349	−5679	−496	−139
Sale of Fixed Assets & Businesses	730	825	622	1009	669	2761	−31	−30	255	248
Purchase/Sale of Investments	−2005	1748	1751	−967	265	155	−231	−225	−1054	−554
Other Funds	−183	−404	−291	−607	−1064	−1078	−397	−892	−2028	−504
Net Investing Cash Flow	−6729	−8507	−4396	−9004	−7326	−3001	−8159	−10976	−7096	−4913
Financing Activities										
Cash Dividends Paid	−2860	−3177	−3473	−3773	−4058	−4265	−4897	−5256	−5506	−5666
Change in Capital Stock	−7429	−15375	−15046	−11995	−12785	−13679	−4609	−3502	−4340	−4443
Issuance/Reduction of Debt, Net	−7463	2350	2370	2252	6960	1783	19	2763	3447	−301
Other Funds	3052	3773	2453	1540	0	709	321	204	−19	−59
Net Financing Cash Flow	−14700	−12429	−13696	−11976	−9883	−15452	−9166	−5791	−6418	−10469
Exchange Rate Effect	98.0	−135.0	−493.0	−116.0	28.0	−655.0	−473.0	−51.0	936.0	−495.0
Miscellaneous Funds	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Net Change in Cash	−558	−1522	1261	−1510	304	−2240	−790	140	4146	−630
Free Cash Flow	**17326**	**15364**	**15738**	**15504**	**13862**	**13128**	**13429**	**13391**	**13495**	**11852**
Free Cash Flow per Share	13	12	13	13	13	13	14	14	14	13
Free Cash Flow Yield (%)	9.87	8.13	7.05	7.01	6.70	8.10	9.93	8.41	9.38	11.38

Source: FactSet Fundamentals
All figures in millions of US dollar except per share items

During the 1999–2018 time period, Dominion Energy produced net income totaling $30.0 billion and depreciation expenses of $29.8 billion. The reader is referred to Table 12. Dominion Energy net income and depreciation were approximately during the 1999–2018 time period. Dominion spent its cash flow generated on capital expenditures, $75.5 billion, dividends $22.5.2 billion, and common stock repurchases, $6.2 billion, and increased its long-term debt $22.6 billion, respectively.

IBM, Boeing, and Dominion Energy had different mixtures of net income and depreciation its income and cash flow statements. The firm spent the funds in

Table 9 Boeing cash flow

Period Ending	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Net Income	**10,460,000**	**8,458,000**	**5,034,000**	**5,176,000**
Operating Activities, Cash Flows Provided by or Used in				
Depreciation	2,114,000	2,047,000	1,889,000	1,833,000
Adjustments to Net Income	464,000	589,000	651,000	559,000
Changes in Accounts Receivables	−2,621,000	−2,440,000	1,118,000	−1,069,000
Changes in Liabilities	2,638,000	4,830,000	−740,000	954,000
Changes in Inventories	568,000	−1,403,000	4,004,000	−1,110,000
Changes in Other Operating Activities	1,879,000	609,000	−726,000	2,543,000
Total Cash Flow from Operating Activities	**15,322,000**	**13,346,000**	**10,496,000**	**9,363,000**
Investing Activities, Cash Flows Provided by or Used in				
Capital Expenditures	−1,722,000	−1,739,000	−2,613,000	−2,450,000
Investments	291,000	38,000	−513,000	554,000
Other Cash flows from Investing Activities	−11,000	6000	7000	39,000
Total Cash Flows from Investing Activities	**−4,621,000**	**−2,058,000**	**−3,378,000**	**−1,846,000**
Financing Activities, Cash Flows Provided by or Used in				
Dividends Paid	−3,946,000	−3,417,000	−2,756,000	−2,490,000
Sale Purchase of Stock	–	–	–	–
Net Borrowings	1,365,000	1,124,000	−34,000	861,000
Other Cash Flows from Financing Activities	35,000	35,000	−24,000	157,000
Total Cash Flows from Financing Activities	**−11,722,000**	**−11,350,000**	**−9,587,000**	**−7,920,000**
Effect of Exchange Rate Changes	−53,000	80,000	−33,000	−28,000
Change in Cash and Cash Equivalents	**−1,074,000**	**18,000**	**−2,502,000**	**−431,000**

different mixtures of capital expenditures, research and development expenditures, dividends, common stock repurchases, and long-term debt issuances.

What should the reader take away from income statements and cash flow statements? First, 2 years of annual report data is rarely sufficient to assess a company and its capacity to generate revenues and net income. Second, one should pay particular attention to the financial highlights, to see what management believes are the company's most meaningful sources and uses of funds, as reported by management (see Table 13).

Third, the reader must be aware of alternative sources of financial data, such as MSN, FactSet, and Compustat.

Table 10 Boeing cash flow statements 1999–2018. Boeing Company. BA 097023105 2108601 NYSE Common stock

Operating Activities	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08
Net Income / Starting Line	2.31	2.13	2.83	2.32	0.72	1.83	2.57	2.22	4.07	2.67
Depreciation, Depletion & Amortization	1.65	1.48	1.75	1.50	1.45	1.51	1.50	1.55	1.49	1.49
Deferred Taxes & Investment Tax Credit	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Other Funds	0.19	0.85	0.40	0.45	1.69	1.60	1.80	1.57	1.37	0.37
Funds from Operations	4.61	4.46	4.98	4.26	3.85	4.94	5.87	5.33	6.93	4.53
Extraordinaries	0.00	0.00	0.00	0.00	0.00	0.00	0.01	–0.01	–0.03	–0.03
Changes in Working Capital	1.61	1.48	–1.16	0.11	0.03	–1.48	1.12	2.19	2.68	–4.90
Net Operating Cash Flow	6.22	5.94	3.81	4.38	3.88	3.46	7.00	7.50	9.58	–0.40
Investing Activities										
Capital Expenditures	–3.63	–0.93	–1.07	–1.00	–0.74	–0.98	–1.55	–1.68	–1.73	–1.85
Net Assets from Acquisitions	0.00	–5.73	–0.02	–0.02	0.29	–0.03	–0.17	–1.85	–0.08	–0.96
Sale of Fixed Assets & Businesses	2.20	0.17	0.15	0.16	0.19	0.19	1.76	0.35	0.06	0.03
Purchase/Sale of Investments	0.00	0.00	0.00	–0.37	0.15	–2.74	–0.14	0.04	–1.89	4.67
Other Funds	0.00	–1.14	–3.78	–2.19	–0.95	2.19	0.00	–0.03	–0.18	0.00
Net Investing Cash Flow	–1.43	–7.63	–4.71	–3.42	–1.06	–1.37	–0.10	–3.19	–3.82	1.89
Financing Activities										
Cash Dividends Paid	–0.54	–0.50	–0.58	–0.57	–0.57	–0.65	–0.82	–0.96	–1.10	–1.19
Change in Capital Stock	–2.84	–2.22	–2.34	0.07	0.03	–0.66	–2.53	–1.40	–2.57	–2.89
Issuance/Reduction of Debt, Net	–0.24	2.07	3.44	1.25	0.02	–2.21	–1.38	–1.68	–1.37	–0.73
Other Funds	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.40	0.14	–0.39
Net Financing Cash Flow	–3.62	–0.66	0.52	0.75	–0.52	–3.52	–4.66	–3.65	–4.88	–5.20
Exchange Rate Effect	–	–	–	–	–	–	–0.037	0.038	0.046	–0.059
Net Change in Cash	1.17	–2.34	–0.38	1.70	2.30	–1.43	2.21	0.71	0.92	–3.77
Free Cash Flow	**4.99**	**5.01**	**2.75**	**3.37**	**3.14**	**2.48**	**5.45**	**5.82**	**7.85**	**–2.08**
Free Cash Flow per Share	5.39	5.75	3.31	4.17	3.88	3.05	6.79	7.39	10.17	–2.85
Free Cash Flow Yield (%)	13.00	8.71	8.54	12.65	9.21	5.89	9.67	8.31	11.62	–6.67

Operating Activities	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Net Income / Starting Line	1.31	3.31	4.02	3.90	4.59	5.45	5.18	4.90	8.20	10.46
Depreciation, Depletion & Amortization	1.67	1.73	1.66	1.81	1.84	1.91	1.83	1.91	2.07	2.11
Deferred Taxes & Investment Tax Credit	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Other Funds	0.68	0.95	0.49	0.91	0.67	0.61	0.56	0.65	0.58	0.46
Funds from Operations	3.66	5.98	6.17	6.62	7.10	7.96	7.57	7.45	10.85	13.04
Extraordinaries	0.04	0.01	–0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Changes in Working Capital	1.91	–3.03	–2.14	0.88	1.08	0.90	1.80	3.05	2.50	2.28
Net Operating Cash Flow	5.60	2.95	4.02	7.51	8.18	8.86	9.36	10.50	13.34	15.32
Investing Activities										
Capital Expenditures	–1.19	–1.13	–1.71	–1.71	–2.24	–2.24	–2.45	–2.61	–1.87	–1.79
Net Assets from Acquisitions	–0.68	–0.93	–0.04	–0.12	–0.03	–0.18	–0.03	–0.30	–0.32	–3.23
Sale of Fixed Assets & Businesses	0.03	0.06	0.09	0.10	0.05	0.03	0.04	0.04	0.09	0.16
Purchase/Sale of Investments	–1.59	–3.12	3.96	–2.02	–2.94	4.80	0.55	–0.51	0.04	0.29
Other Funds	–0.41	0.29	0.07	0.00	0.00	0.03	0.04	0.00	0.00	–0.01
Net Investing Cash Flow	–3.83	–4.83	2.37	–3.76	–5.15	2.46	–1.85	–3.38	–2.06	–4.59
Financing Activities										
Cash Dividends Paid	–1.22	–1.25	–1.24	–1.32	–1.47	–2.12	–2.49	–2.76	–3.42	–3.95
Change in Capital Stock	–0.04	0.09	0.11	0.12	–1.70	–5.66	–6.35	–6.68	–8.93	–8.92
Issuance/Reduction of Debt, Net	5.41	–0.65	–0.13	–2.02	–0.86	–0.64	0.86	–0.03	1.12	1.37
Other Funds	–0.02	–0.15	–0.44	–0.26	–0.22	–0.17	0.06	–0.12	–0.13	–0.26
Net Financing Cash Flow	4.13	–1.96	–1.70	–3.48	–4.25	–8.58	–7.92	–9.59	–11.35	–11.76
Exchange Rate Effect	0.044	–0.015	–0.002	0.018	–0.029	–0.087	–0.028	–0.033	0.080	–0.053
Net Change in Cash	5.95	–3.86	4.69	0.29	–1.25	2.65	–0.43	–2.50	0.01	–1.07
Free Cash Flow	**4.42**	**1.83**	**2.31**	**5.81**	**6.08**	**6.62**	**6.91**	**7.89**	**11.61**	**13.60**
Free Cash Flow per Share	6.19	2.45	3.07	7.62	7.92	8.99	9.95	12.27	19.02	23.23
Free Cash Flow Yield (%)	11.44	3.76	4.18	10.12	5.80	6.92	6.88	7.88	6.45	7.20

Source: FactSet Fundamentals
All figures in billions of US dollar except per share items

US Firm Cash Flow Analysis, 1971–2018

In this chapter, we specifically examined the income statement and cash flow statements of IBM, Boeing, and Dominion Energy. IBM, Boeing, and Dominion Energy had different mixtures of net income and depreciation in its income and cash flow statements. The firm spent the funds in different mixtures of dividends,

Table 11 Dominion energy cash flow

Period Ending	12/31/2018	12/31/2017	12/31/2016	12/31/2015
Net Income	**2,447,000**	**2,999,000**	**2,123,000**	**1,899,000**
Operating Activities, Cash Flows Provided by or Used in				
Depreciation	2,280,000	2,202,000	1,849,000	1,669,000
Adjustments to Net Income	287,000	75,000	665,000	812,000
Changes in Accounts Receivables	−110,000	−103,000	−286,000	294,000
Changes in Liabilities	67,000	−89,000	97,000	−199,000
Changes in Inventories	−29,000	15,000	1000	−26,000
Changes in Other Operating Activities	−169,000	−597,000	−298,000	26,000
Total Cash Flow from Operating Activities	**4,773,000**	**4,502,000**	**4,151,000**	**4,475,000**
Investing Activities, Cash Flows Provided by or Used in				
Capital Expenditures	−4,405,000	−5,909,000	−6,125,000	−5,993,000
Investments	−392,000	−95,000	−196,000	−35,000
Other Cash flows from Investing Activities	−103,000	−76,000	1000	−57,000
Total Cash Flows from Investing Activities	**−2,358,000**	**−5,942,000**	**−10,691,000**	**−6,503,000**
Financing Activities, Cash Flows Provided by or Used in				
Dividends Paid	−2,185,000	−1,931,000	−1,727,000	−1,536,000
Sale Purchase of Stock	–	–	–	–
Net Borrowings	−2,211,000	2,201,000	4,858,000	3,004,000
Other Cash Flows from Financing Activities	−274,000	−269,000	947,000	63,000
Total Cash Flows from Financing Activities	**−2,209,000**	**1,303,000**	**6,230,000**	**2,317,000**
Effect of Exchange Rate Changes	–	–	–	–
Change In Cash and Cash Equivalents	**206,000**	**−137,000**	**−310,000**	**289,000**

common stock repurchases, and long-term debt issuances. How might these data look for all firms? Let us examine the net income, dividends, equity issued, equity repurchased, long-term debt issued, and long-term debt repurchased for all US firms listed on US stock exchanges, 1971–2017 (see Table 14). The database consists of over 429,900 firm-years of data. The source of our data is the Compusat database. US firms totaled $34.78 billion of net income, ni; paid dividends, div, of $16.87 billion; repurchased common stock, prstkc, of $12.71 billion, issued common stock of $11.24 billion; issued long-term debt, dltis, of 12.15 billion; and repurchased long-term debt, dltr, of $10.84 billion. Firms produced increases in net income, paid larger divides, created net stock buy backs, and issued more long-term debt. These financial policies fueled equity markets. Firms paid 48% of their net income to stockholders as dividends; repurchased almost as much equity as firms paid dividends; increased net debt (long-term debt issued exceeded long-term debt repurchased); and decreased net equity (common stock issued was less than common stock repurchased). We will address dividends and stock buybacks in chapter "The Equity of the Corporation: Common and Preferred Stock."

Table 12 Dominion cash flow statements, 1999–2018. Dominion Energy Inc. D 25746U109 2542049 NYSE Common stock

Operating Activities	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08
Net Income / Starting Line	551	415	642	1362	307	1264	1033	1380	2539	1834
Depreciation, Depletion & Amortization	798	1268	1322	1379	1334	1433	1538	1739	1533	1191
Deferred Taxes & Investment Tax Credit	64	22	201	714	452	554	64	510	–1285	269
Other Funds	–91	–12	43	8	714	–65	411	183	–3054	342
Funds from Operations	1322	1693	2208	3463	2807	3186	3046	3812	–267	3636
Changes in Working Capital	–67	–350	195	–1015	–452	–347	–423	193	21	–977
Net Operating Cash Flow	1255	1343	2403	2448	2355	2839	2623	4005	–246	2659
Investing Activities										
Capital Expenditures	–961	–1738	–2168	–2828	–3438	–2750	–3358	–4052	–3972	–3554
Net Assets from Acquisitions	–167	–2779	–2215	–410	0	0	–877	–91	0	–21
Sale of Fixed Assets & Businesses	180	836	141	0	305	729	595	393	14,228	343
Purchase/Sale of Investments	–170	–238	–74	54	135	–24	–100	–86	–70	–337
Other Funds	–424	1322	123	–776	–375	789	380	342	6	79
Net Investing Cash Flow	–1542	–2597	–4193	–3960	–3373	–1256	–3360	–3494	10,192	–3490
Financing Activities										
Cash Dividends Paid	–493	–615	–649	–723	–825	–861	–923	–970	–931	–916
Change in Capital Stock	–372	–1109	992	2219	990	839	388	–61	–5542	240
Issuance/Reduction of Debt, Net	1050	3115	1552	1766	730	–1285	1108	530	–3352	1309
Other Funds	–44	–57	21	487	–42	–13	–51	–14	24	–18
Net Financing Cash Flow	141	1334	1916	1317	853	–1320	522	–515	–9801	615
Net Change in Cash	–146.0	80.0	126.0	–195.0	–165.0	263.0	–215.0	–4.0	145.0	–216.0
Free Cash Flow	**294.0**	**–395.0**	**235.0**	**–380.0**	**–1083.0**	**89.0**	**–735.0**	**–47.0**	**–4218.0**	**–895.0**
Free Cash Flow per Share	0.8	–0.8	0.5	–0.7	–1.7	0.1	–1.1	–0.1	–6.4	–1.5
Free Cash Flow Yield (%)	3.91	–2.50	1.55	–2.45	–5.32	0.40	–2.76	–0.16	–13.57	–4.30

Operating Activities	Dec '09	Dec '10	Dec '11	Dec '12	Dec '13	Dec '14	Dec '15	Dec '16	Dec '17	Dec '18
Net Income / Starting Line	1304	2825	1426	275	1720	1326	1923	2212	3120	2549
Depreciation, Depletion & Amortization	1319	1258	1288	1443	1390	1560	1669	1849	2202	2280
Deferred Taxes & Investment Tax Credit	–494	682	756	246	737	449	854	725	–3	517
Other Funds	1126	–3121	79	1747	–208	446	–66	–173	–11	–332
Funds from Operations	3255	1644	3549	3711	3639	3781	4380	4613	5308	5014
Changes in Working Capital	531	181	–566	426	–206	–342	95	–486	–759	–241
Net Operating Cash Flow	3786	1825	2983	4137	3433	3439	4475	4127	4549	4773
Investing Activities										
Capital Expenditures	–3837	–3422	–3652	–4145	–4104	–5551	–5575	–6125	–5909	–4405
Net Assets from Acquisitions	0	–	0	–	–923	0	–915	–4381	0	0
Sale of Fixed Assets & Businesses	0	4,191	0	115	625	272	0	0	0	–
Purchase/Sale of Investments	–76	–435	188	72	8	–6	14	–82	–251	1897
Other Funds	218	85	143	118	13	104	–27	–115	167	150
Net Investing Cash Flow	–3695	419	–3321	–3840	–4381	–5181	–6503	–10,703	–5993	–2358
Financing Activities										
Cash Dividends Paid	–1056	–1093	–1146	–1225	–1319	–1409	–1536	–1727	–1931	–2185
Change in Capital Stock	456	–826	–563	265	278	338	786	3124	1320	2465
Issuance/Reduction of Debt, Net	513	–311	2112	823	2,147	2,940	3,004	4858	2201	–2211
Other Funds	–25	–2	–25	–14	–90	–125	63	–25	–287	–278
Net Financing Cash Flow	–112	–2232	378	–151	1016	1744	2317	6230	1303	–2209
Net Change in Cash	–21.0	12.0	40.0	146.0	68.0	2.0	289.0	–346.0	–141.0	206.0
Free Cash Flow	**–51.0**	**–1597.0**	**–669.0**	**–8.0**	**–671.0**	**–2112.0**	**–1100.0**	**–1998.0**	**–1360.0**	**368.0**
Free Cash Flow per Share	–0.1	–2.7	–1.2	0.0	–1.2	–3.6	–1.9	–3.2	–2.1	0.6
Free Cash Flow Yield (%)	–0.22	–6.34	–2.19	–0.03	–1.79	–4.70	–2.74	–4.23	–2.64	0.79

Source: FactSet Fundamentals
All figures in millions of US dollar except per share items

Table 13 IBM financial highlights. International Business Machines Corporation and Subsidiary Companies

($ in millions except per share amounts)				
For the year ended December 31:	**2018**		**2017**	
Revenue	$79,591		$79,139	
Net Income	$8728	[a]	$5753	[a]
Income from continuing operations	$8723	[a]	$5758	[a]
Operating (non-GAAP) earnings[b]	$12,657		$12,807	[c]
Earnings per share of common stock-continuing operations				
Assuming dilution	$9.51	[a]	$6.14	[a]
Basic	$9.56	[a]	$6.17	[a]
Diluted operating (non-GAAP)[b]	$13.81		$13.66	[c]
Net cash provided by operating activities	$15,247		$16,724	
Capital expenditures, net	$3716		$3312	
Share repurchases	$4443		$4340	
Cash dividends paid on common stock	$5666		$5506	
Per share of common stock	$6.21		$5.90	
At December 31:	**2018**		**2017**	
Cash, cash equivalents, restricted cash and marketable securities	$12,222		$12,842	[d]
Total assets	$123,382		$125,356	
Working capital	$10,918		$12,373	
Total debt	$45,812		$46,824	
Total equity	$16,929		$17,725	
Common shares outstanding (in millions)	892		922	
Stock price per common share	$113.67		$153.42	
($ in millions)				

(continued)

Table 13 (continued)

For the year ended December 31:	2018	2017		2016	
Cash flows from operating activities					
Net Income	$8728	$5753		$11,872	
Adjustments to reconcile net income to cash provided by operating activities					
Depreciation	3127	3021		2837	
Amortization of intangibles	1353	1520		1544	
Stock-based compensation	510	534		544	
Deferred taxes	853	(931)		(1132)	
Net (gain)/loss on asset sales and other	123	14		62	
Change in operating assets and liabilities, net of acquisitions/divestitures					
Receivables (including financing receivables)	1006	1297		712	
Retirement related	1368	1014		54	
Inventories	(127)	18		(14)	
Other assets/other liabilities	(1819)	4437		408	
Accounts Payable	126	47		197	
Net cash provided by operating activities	15,247	16,724		17,084	
Cash flows from investing activities					
Payments for property, plant, and equipment	(3395)	(3229)		(3567)	
Proceeds from disposition of property, plant, and equipment	248	460		424	
Investment in software	(569)	(544)		(583)	
Purchases of marketable securities and other investments	(7.041)	(4949)	e	(5853)	e
Proceeds from disposition of marketable securities and other investments	6487	3910		5692	
Nonoperating finance receivables—net	(503)	(2028)		(891)	
Acquisition of businesses, net of cash acquired	(139)	(496)		(5,696)	e
Divestiture of businesses, net of cash transferred	–	(205)		(454)	
Net cash used in investing activities	(4913)	(7081)	e	(10,928)	e
Cash flows from financing activities					
Proceeds from new debt	6891	9643		9132	
Payments to settle debt	(8533)	(6816)		(6395)	
Short-term borrowings/(payments) less than 90 days—net	1341	620		26	
Common stock repurchases	(4443)	(4340)		(3502)	
Common stock repurchases for tax withholdings	(171)	(193)		(126)	
Financing—others	111	175		204	
Cash dividends paid	(5666)	(5506)		(5256)	
Net cash used in financing activities	(10,469)	(6418)		(5917	
Effect of exchange rate changes on cash, cash equivalents and restricted cash	(495)	937		(51)	
Net change in cash, cash equivalents, and restricted cash	(630)	4161	e	188	e
Cash, cash equivalents, and restricted cash at January 1	12,234	8073	e	7885	e

(continued)

Table 13 (continued)

Cash, cash equivalents, and restricted cash at December 31	11,604	12,234		8073	[e]
Supplemental data					
Income taxes paid—net of refunds received	$1745	$1597		$1078	
Interest paid on debt	$1423	$1208		$1158	

Amounts may not add due to rounding
The accompanying notes on pages 76 through 146 are an integral part of the financial statements
[a]Includes charges of $2.0 billion in 2018 and $5.5 billion in 2017 associated with US tax reform
[b]See page 41 for reconciliation of net income to operating earnings
[c]Recast to reflect adoption of the FASB guidance on presentation of net periodic pension and nonpension postretirement benefit costs
[d]Recast to reflect adoption of the FASB guidance on restricted cash
[e]Recast to reflect adoption of the FASB guidance on restricted cash

Table 14 WRDS cash flow 1971–2017, $MM

Variable	Sum	N
ni	34,782,762.14	429,905
div	16,866,503.44	429,905
prstkc	12,707,868.42	429,905
sstk	11,237,229.15	429,905
dltis	121,486,032	429,905
dltr	108,391,696	429,905
NetEquity	−1,470,639.27	429,905
Netdebt	13,094,336.29	429,905

Summary

The authors introduced the reader to operating statements of companies, the annual income statement and cash flow statements. We specifically examined IBM, Boeing, and Dominion Energy, firms introduced to the reader in chapter "The Corporation Balance Sheet." The reader learned about revenues, gross and net income, dividends, common stock issuance and repurchases, and long-term debt issuances and repurchases. We will use these data for common stock valuation in chapter "The Equity of the Corporation: Common and Preferred Stock." In this next chapter, we introduce the reader to ratio and concepts of firm efficiency using the data reported in chapters "The Corporation Balance Sheet" and this chapter.

References

Dtrina, R. E., & Largay III, J. A. (1985). Pitfalls in calculating cash flow from operations. *The Accounting Review, 40*, 314–326.

Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Krishnan, G. V., & Largay III, J. A. (2000). The predictive ability of direct method cash flow information. *Journal of Business Finance and Accounting, 27*, 215–245.

White, G. I., Sondhi, A. C., & Fried, D. (1997). *The analysis and use of financial statements* (2nd ed.). New York: John Wiley & Sons, Inc. Chapters 2, 3.

Financing Current Operations and Efficiency Ratio Analysis



Current financing encompasses managing and utilizing current assets, and incurring and repaying current debt. The current assets of a firm differ from fixed assets; these differences are not abrupt but represent a continuum. The current assets (cash, receivables, inventory, etc.) support the short-run operations of the business. Current assets are what the classical economists called "circulating capital." Within the current asset grouping, however, some items remain in the firm's possession longer than others. The purpose of this chapter is to focus the reader's attention to the efficient management of current assets and current liabilities. The use of ratios introduced in this chapter help analysts determine the financial health of corporations.

Working Capital Concepts

The reader was introduced to the firm's annual cash flow statement in the previous chapter. This chapter integrates current asset management, sources and uses of funds, and ratio analysis. In financial terminology, the total current assets, primarily cash, and other items likely to be turned into cash within the year, is called the gross working capital of the firm. The current assets of the firm, less the current liabilities, are the net working capital of the firm. When the term working capital is used, net is usually intended.

The net working capital of the firm is an important aspect of its financial strength. It represents the current assets, not offset by current creditor claims, that can be used to meet unexpected expenditures, absorb irregularities in the firm's cash flow, and cushion any short-run interruption, seasonality, or lumpiness in the flow of funds through the firm.[1] In addition, adequate working capital allows the firm to make

[1] Although only a fraction of net working capital is likely to be in cash, an adequate amount of net working capital assures a substantial short-term cash flow and serves as a good base for short-term credit.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_5

especially opportune purchases and to take advantage of immediate expansion possibilities. Net working capital is not to be confused with cash, but it functions similarly as a sort of generalized short-term working balance for the firm.

How much net working capital a firm should carry depends on many factors. Generally, the net working capital a firm carries depends on its sales volume, its needs for gross circulating capital relative to its sales volume, and the stability of its operations. A firm can increase its rate of profit by economizing on the use of net working capital, but this has the effect of increasing its short-term financial risk. A firm can minimize its net working capital requirements, however, if any of the following conditions exist: (1) its flow of cash receipts tends to be stable and predictable; (2) its current assets tend to be marketable, stable in value, or short-term and liquid; and (3) the amount of long-term debt is small for the type of firm.

On the other hand, net working capital needs are higher for firms with the following characteristics: (1) those with erratic or irregular cash flows or cash demands; (2) those carrying slow-moving inventories which do not have active trading markets or whose market prices tend to fluctuate; and (3) those firms that by the nature of their business extend considerable credit to dubious risks.

Quantitative Working Capital Models: Cash Management

There are several well-known quantitative working capital models. The first model, developed by Baumol in 1952, optimized the firm's cash balance by minimizing total costs per year of maintaining cash balances. Total costs, TC, were composed of transactions costs and an opportunity cost of maintaining additional cash balances:

$$TC = t\ \left(\frac{Cu}{C}\right) + i\ \left(\frac{C}{2}\right) \tag{1}$$

where

Cu = cash usage for the year
t = transaction costs per sale or purchase of marketable securities
i = annual interest rate on marketable securities
C = cash balance

One minimizes the costs of selling and buying securities and holding cash on which one could earn interest. Thus, to minimize one's total costs, one takes the derivative of TC with respect to its cash balance and sets the derivative to zero.[2] The optimal cash level, C*, is

[2] $dTC/dQ = i/2 - tCu/Q^2 = 0$,
$i/2 = tCu/Q^2$

$$Q = \sqrt{2\ tCu/i}$$

$$C^* = \sqrt{\frac{2tCu}{i}} \tag{2}$$

The optimal cash balance is positively associated with total cash usage and negatively associated with the interest rate, in the Baumol formulation. If a firm uses $100 million of cash in a given year, the interest rate is 5%, and its transactions cost is $50, then its optimal cash balance is

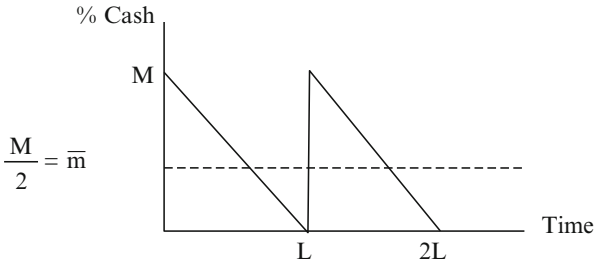
$$C^* = \sqrt{\frac{2(\$50)\$100,000,000}{.05}}$$

$$C^* = \sqrt{\frac{10,000,000,000}{.05}}$$

$$C^* = \$447,213.60$$

The optimal cash balance for the firm should be $447,213.60.

A second cash management model is the Miller-Orr (1966) control limit model that built upon the Baumol formulation. The firm earns interest on its savings deposits and has a noninterest-bearing cash balance. The firm transfers M dollars from its savings deposit into its cash account. The average cash balance is $\frac{M}{2}$, or m. Cash usage to a minimum acceptable level, zero, occurs every L days. Cash transfers occur every M/m days. In the Miller-Orr framework,



Miller and Orr defined ν as the annual interest rate and γ as the transaction cost. The Baumol formulation could be written as

$$M^* = \left(\frac{2\gamma m}{\nu}\right)^{.5} \tag{3}$$

The optimal period of M* transfers is

$$L^* = \left(\frac{2\gamma}{m^*}\right)^{.5} \tag{4}$$

If p is the probability that the cash balance will increase by m dollars, then 1 − p, or q, is the probability of a cash decrease. The expected costs of maintaining the cash balance over a T day planning horizon is

$$E(dc) = \gamma \frac{E(N)}{T} + \nu E(M) \tag{5}$$

where

E(dc) = expected daily costs
E(N) = expected number of cash transfers
E(M) = expected (average) daily cash balance

Miller and Orr further assumed that a firm maintained an upper bound on its cash balance of h, a balance level of z, and its transfer of lump-sum cash would be (h − z) dollars. Miller and Orr converted the expected value of the cash duration, Z(h − z), into time (days by dividing by t, the number of operating days. The expected cash duration is:

$$= \frac{Z^1\left(h^1 - Z^1\right)}{m^2 t} \tag{6}$$

where

$Z^1 = Z - m$
$h^1 = h - m$

The mean cash balance is $\frac{h+Z}{3}$.

The cash transfer, Z, is h − Z.[3]

Miller and Orr found that the optimal cash transfer was a positive function of cash transfer cost, daily cash balances, and an inverse function of the daily interest rate:

$$z^* = \left(\frac{3\gamma m^2 t}{4\upsilon}\right)^{.333} \tag{10}$$

$$Z^* = 2z^*$$

and
$h^* = 3z^*$.

The Miller-Orr model allows the corporate treasurer to invest excess cash when cash balances reach the upper control limit, h, returning the cash balance to its mean cash balance.

[3]Thus Miller and Orr minimized total costs

$$\min\ \varepsilon\ (dc) = \frac{\gamma m^2 t}{zZ} + \frac{\upsilon(Z+2z)}{3} \quad (7)$$

$$\frac{\partial \varepsilon(dc)}{\partial z} = -\frac{\gamma m^2 t}{Z^2 z} + \frac{2\upsilon}{3} = 0 \quad (8)$$

$$\frac{\partial \varepsilon(dc)}{\partial Z} = -\frac{\gamma m^2 t}{Z^2 z} + \frac{\gamma}{3} = 0 \quad (9)$$

Sources of Net Working Capital

Since net working capital is the current assets minus the current debt, net working capital (by definition) must be supplied by equity and/or long-term debt. Thus, net working capital is financed from permanent or semipermanent sources.

Working capital is not a static item, however, but expands and contracts with the level of the firm's operations. (Gross working capital is likely to show much wider proportionate changes.) Only on occasion does an established firm go out to the capital markets for additional working capital; normally, working capital is increased or depleted by internal operations of the business.

The main financial items affecting the level of working capital fall within two categories. Those that increase net working capital funds are (1) net operating profits; (2) current depreciation charges insofar as they are covered by revenues over variable costs; (3) sales of fixed or other assets for current funds; and (4) issuing new bonds or stocks if the funds obtained are not all used to acquire fixed assets. Financial events that decrease net working capital funds are (1) operating losses; (2) payment of dividends; (3) purchase of fixed or other noncurrent assets; (4) repayment of long-term debt; and (5) buybacks, i.e., repurchasing the corporation's own shares with available funds.

Working Capital Management and Efficiency Ratio Analysis

Ratio analysis is an alternative to the flow of funds method of working capital analysis, although the two can be used to supplement each other. Ratio analysis is older and possibly the more popular approach than the flow of funds method of management and is the most readily available to credit managers of other companies, or other outsiders. A person within the firm sometimes finds other analytical tools more useful.

Ratio analysis consists of studying ratio or percentage relationships of meaningful financial data. The results are compared (1) with standard ratios—i.e., the averages of similar firms, (2) with the firm's ratios in previous years, or (3) with some implicit standards existing in the mind of the analyst. In the hands of a skilled practitioner, both "external analysis" (comparisons to standard ratios) and internal analysis (i.e., trends and relationships of the ratios within the company) can be revealing.

Innumerable ratios can be developed, since the financial accounts can be placed in almost unlimited combinations. For most purposes, however, about 13 popular ratios suffice for whatever can be learned from this method about the firm's current financial position.[4] In many cases, only 6–10 of these ratios are needed for an understanding of the problem. If special areas seem to warrant additional attention, it is not difficult to develop other ratios.

Guerard Jr. and Schwartz (2007) calculated 13 ratios that were the most generally used in financial analysis. The first 6 are most relevant for current analysis. The

[4] As additional ratios are used, one soon discovers that the same information is being presented in a different form.

remaining 7 reveal more general relationships, many involving profitability, using net income or net profits.

Current Analysis Ratios

1. Current ratio, CR
2. Acid test, AT
3. Sales/receivables, SR
4. Sales/inventory, SI
5. Sales/net working capital, SNWK

General Analysis Ratios

6. The financial structure ratios—total debt/assets, TDA
7. Sales/total assets, SA
8. Net profit/total assets, ROA
9. Net profit/tangible net worth, ROTNW
10. Net profit/equity, ROE
11. EBIT/interest, TIE

Composite Firm Relative Valuation Ratios

12. DuPont analysis, DuPontA
13. Altman—Z model, NewZ

These 13 ratios are still used by many analysts; however, we will restrict ourselves to a small set of ratios with more demonstrated predictive power with regard to stock prices within the WRDS universe.

1. Current ratio, CR
2. Total debt ratio, TDA
3. Sales/total assets, SA
4. Net profit/equity, ROE
5. Net profit/total assets, ROA
6. Net profit/total sales, ROS
7. Altman—Z model, NewZ

Current Ratio The current ratio is obtained by dividing the current liabilities into the current assets. It indicates how many times current liabilities are covered by gross working capital. The higher the current ratio, the more conservative the current financial position of the firm. A two-to-one ratio is a rule-of-thumb benchmark indicating a minimum level of the working capital position. Other circumstances must always be considered; no financial analysis can proceed rigidly. A ratio below two does not necessarily make the firm unsafe, nor does a current ratio well over two ensure financial soundness. Much depends upon the collectibility and time structures of the firm's receivables and the type and quantity of inventory the firm carries. Public often have a current ratio of one to one or below.[5] In an electric utility

[5]This means that these companies, in effect, carry no net working capital.

company, for example, the low current ratio is possible because of its minimum inventory requirements and the stability of its revenues and cash flow.[6]

[6]For completeness, we will relegate several widely used ratios to footnotes.

Acid Test. The acid test, or quick ratio, is obtained by dividing current liabilities into the firm's net receivables and cash. This ratio highlights the firm's short-term liquidity position. The rule-of-thumb measure of a satisfactory acid test ratio is one to one. From an obverse point of view, the acid test ratio tends to indicate the amount of inventory in the working capital position of the firm. For example, if the current ratio is 3 to 1 and the acid test is only 0.85 to 1, the inventory account probably constitutes a heavy proportion of the current assets.

Sales to Receivables or the Receivable Turnover Ratio. This ratio is obtained by dividing credit sales by the outstanding trade accounts and trade notes receivable and indicates the collectibility and current condition of the receivables. The higher the sales to receivables ratio, the more current are the receivables. A variant of this ratio is to divide the turnover rate into 360 (representing the approximate number of days in the year). The resulting figure gives the number of days it takes to collect an average account. This figure can be compared to the usual terms (or allowable credit time) granted by the firm to ascertain whether the average account is collected in a period close to the credit terms.

Sales to Inventory: Approximate Inventory Turnover. This ratio is obtained by dividing the inventory into the sales figure. The result is useful in analyzing how rapidly the firm's inventory is sold. A slow turnover—relative to the type of business or its own previous performance—may indicate that the firm is overstocked, or that the inventory contains too many old or out-of-style items, or that the management is speculating in inventory. Again, as in the case of the receivable turnover, the inventory turnover figure can be divided into 360 days to get an average of how many days it takes for a given dollar amount of merchandise to be turned into an equivalent amount of sales.

Many analysts prefer to reserve the term "inventory turnover" for the ratio of inventory divided into the cost of goods sold. The ratio then indicates the true physical turnover of the inventory. Since the sales figure contains the gross markup (profit) over cost, the sales over inventory ratio overstates the actual physical turnover of the goods. The higher the customary gross margin, or markup over cost in the sales figure, the better the inventory into sales ratio appears in comparison to the true turnover ratio.[6] Unfortunately, the figure for the cost of goods sold is not always as available as the amount of sales. Thus, the standard ratios are more often based on sales. Ratio analysis, in any case, is not an exact science but is based on historical or intra-industry comparisons. The ratio serves its purpose as long as it depicts a logical relationship and comparisons using it are made on a consistent basis.

Sales/Net Working Capital: Working Capital Turnover. The net working capital turnover is obtained by dividing net working capital into the annual sales. This ratio is double-edged; a high ratio can indicate either efficiency or risk. A low turnover may indicate managerial inefficiency in moving goods and collecting receivables, or it may indicate excessively conservative management—a tendency to hold redundant idle funds or a failure to use a reasonable amount of available current credit. The other edge of the ratio appears if the turnover is too high in contrast with the industry norm. It may not necessarily indicate efficiency but a tendency to take on undesirable levels of risk. An especially high net working capital turnover can indicate overtrading on current account—an attempt to carry a heavy volume of business on an inadequate current capital base. Such speculative striving on the part of the management can be dangerous to both owners and creditors.

It should be obvious that the ratios are not to be used singly but in a composite manner to fill out a financial portrait of the company. Thus, the position of the net working capital turnover can be checked against the other current operating ratios. For example, a firm with a high net working capital turnover and ordinary inventory and receivable turnovers whose current ratio is tight is likely to be overtrading. A firm with a low net working capital turnover, a normal cash cycle, and a very high acid test ratio may be holding excessive idle funds. These are only two possibilities. An experienced analyst may be able to rough out normal relationships in his head while scanning the financial figures. His instinct may lead him quickly to any items that are out of line and suggest the few ratios necessary to highlight the potential trouble spot.

Sales to Total Assets The ratio of sales to total assets indicates how intensively the total assets are used in production. A low sales to total assets ratio in comparison with similar firms or with previous periods gives some indication of idle capacity, i.e., excess assets compared to the level of operations.

Inter-industry comparisons of this ratio are not very useful. A wholesale distributor, for example, with no processing costs, a small margin, and a large turnover of goods shows a relatively high volume of sales to total assets. A better ratio to measure the basic concept of the rate of utilization of capital would be value-added to total asset—i.e., something approaching a capital coefficient. Unfortunately, possibly because of statistical difficulties, value-added ratios are not commonly used in financial analysis.

Net Profit on Stockholder Equity This ratio is obtained by dividing the net profit by the common stock equity of the corporation.[7] It gives the rate of return made by the owners of the corporation on the book value of their investment. If this ratio is compared to the rate of return on total assets to determine whether financial leverage operated favorably for the firm, earnings after interest but before taxes (EBT) should be the numerator. Then if the earnings return on assets is properly calculated on the EBIT basis, these two ratios would be properly comparable on a financial basis. For some comparisons of profitability, therefore, it may be better to use the ratio of officers' salaries plus profits before taxes over the net worth as the basis of comparison.

Net Profit to Total Assets (Preferred Ratio: EBIT to Total Assets) This ratio is intended to relate the return of the firm to its total investment, i.e., the total assets it has available. It has some use, but it is subject to the criticism that the relationship presented is not the most logical one and that it does not present sufficient new information. The net profit figure has already been reduced by taxes and the cost of external funds (i.e., interest); to relate this figure to total assets is an illogical relating of a net concept (net profits) with a gross concept (total assets). Moreover, the ratio does not give much independent information if the net profit on owners' equity is to be calculated too, as it usually is. Obviously, the net profit rate of return (or rate of loss) on total assets is always less than that on the owners' equity. The difference depends upon the relative amount of total leverage. The larger the percentage of ownership capital in the financial structure, the lower will be any difference.

If we are to get any clear concept of how productively the management has utilized the total assets, the preferable ratio is the EBIT to total assets. In addition, a comparison of this ratio to rate of profit on net worth indicates how well the firm has adjusted its financial mix. Comparison of these ratios (allowing for the fact that one is an after-tax return) indicates something about the profitability—but not much about the risk of the firm's use of leverage.

[7] Intangible assets, such as good will, should be subtracted from the book value of the common equity.

Financial Ratios and the Perceived Financial Health of Firms

Financial analysis often combines the information of several ratios to gain insight into a picture of the firm's health. In this chapter, we examine two composite measures of the firm's health, the "DuPont system" rate of return, dating back to the early twentieth century, and the Altman Z bankruptcy prediction model, created in 1968. The DuPont system, or measure, takes its net operating income divided by sales and multiplies it by the ratio of sales to investment, producing a return on investment, ROI:

$$\frac{\text{NOI}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Investment}} = \text{ROI} \tag{11}$$

Stockholders should invest in firms with higher ROIs, and management could seek to maximize the DuPont ROI to maximize its stock price. The DuPont analysis uses information inherent in its return on sales and sales turnover ratios.

Pierre DuPont and Donald Brown, a DuPont employee, developed the DuPont return on investment relationship to access the firm's financial performance. General Electric calculated profitability by dividing earnings by sales (or costs). However, this calculation ignored the magnitude of invested capital. In 1903, Pierre DuPont created a new general ledger account for "permanent investment," where capital expenditures were charged at cost. The DuPont Corporation executive committee was presented with monthly sales, income, and return on invested capital on the firm's 13 products in 1904 (Chandler Jr, 1977). Donald Brown contributed to the DuPont analysis pointing out that as sales volume rose, the return of invested capital rose, even if prices remained constant. Brown's "turnover" analysis was defined as sales divided by total investment. The multiplication of turnover by the ratio of earnings to sales produced the DuPont return on invested capital, still in use by the DuPont Corporation and most American firms. Total investment includes working capital, cash, inventories, and accounts receivable, and permanent investment, bonds, preferred stock, and stocks. The DuPont return on invested capital combined and consolidated financial, capital, and cost accounting. The DuPont return on total investment helped DuPont develop many modern management procedures for creating operating and capital budgeting and making short-run and long-run financial forecasts.

A second composite model is the Altman Z model, which is useful to identify potential bankrupt firms. The Altman Z score used five primary ratios in its initial 1968 version:

$$Z = .012\ X_1 + .014\ X_2 + .033\ X_3 + .006\ X_4 + .999\ X_5 \tag{12}$$

where

$$X_1 = \frac{\text{Current Assets} - \text{Current Liabilities}}{\text{Total Assets}}$$

$$X_2 = \frac{\text{Retained Earnings}}{\text{Total Assets}}$$

$$X_3 = \frac{\text{EBIT}}{\text{Total Assets}}$$

$$X_4 = \frac{\text{Market Value of Equity}}{\text{Book Value of Debt}}$$

$$X_5 = \frac{\text{Sales}}{\text{Total Assets}}$$

The Altman Z score used a liquidity, past profitability, (present) profitability, leverage, and sales turnover ratios to produce a single score. An Altman Z score of less than 2.67 implied that the firm was not healthy. An Altman Z score exceeding 2.67 implied financial health. The Altman Z score successfully predicted impending bankruptcy for 32 of 33 firms (97%) in the year prior to bankruptcy, for his initial sample. The model correctly predicted 31 of 33 (94%) non-bankrupt firms in this sample for the year prior to bankruptcy. The Altman Z model has been reported since the early 1980s in corporate financial textbooks. Guerard and Vaught (1989), Van Horne (2002), Guerard and Schwartz (2007), and Markowitz and Guerard (2019) have reported its results.

Altman modified his equation in 2000 to become

$$Z = .717X_1 + .847X_2 + 3.107X_3 + .420X_4 + .998X_5 \tag{13}$$

where X_4 is now book value of equity relative to its book value of debt. The new critical level is 2.0.

The ratios of companies can be compared to all firms, companies in a sector, or broad segments of the economy, or specific industry groups. IBM maintains fewer liquid assets, as measured by its current ratio, than most companies in the WRDS universe. IBM uses greater leverage than the WRDS universe. IBM's greater leverage and profitability, as shown in Table 1, produced substantially higher ROE and DuPont ratio values than most US companies. In 2017, the IBM Altman Z is 1.865, exceeding the WRDS universe median value of 1.724.

Table 1 WRDS IBM efficiency ratios 1988–2017

Obs	tic	Fyear	sale	ni	re	CR	SA	TDA	NewZ
1	IBM	1988	59,681	5806	33,103	2.03273	0.81713	0.35468	2.44218
2	IBM	1989	62,710	3758	32,175	1.65323	0.80673	0.41841	2.10135
3	IBM	1990	69,018	6020	36,500	1.53980	0.78816	0.42503	2.11721
4	IBM	1991	64,792	−2827	30,506	1.21845	0.70066	0.50669	1.40198
5	IBM	1992	64,523	−4965	21,086	1.08044	0.74417	0.57194	1.12797
6	IBM	1993	62,716	−8101	11,667	1.18256	0.77319	0.59664	0.87580
7	IBM	1994	64,052	3021	15,024	1.41443	0.78988	0.51515	1.46224
8	IBM	1995	71,940	4178	14,723	1.28574	0.89598	0.51945	1.60290
9	IBM	1996	75,947	5429	13,758	1.19691	0.93609	0.54075	1.66594
10	IBM	1997	78,508	6093	11,909	1.20626	0.96330	0.57919	1.67776
11	IBM	1998	81,667	6328	11,052	1.15024	0.94851	0.60784	1.58207
12	IBM	1999	87,548	7712	18,039	1.09038	1.00061	0.61377	1.73106
13	IBM	2000	88,396	8093	23,489	1.20530	1.00053	0.62001	1.84788
14	IBM	2001	85,866	7723	29,480	1.20906	0.97229	0.57842	1.86827
15	IBM	2002	81,186	3579	28,137	1.20556	0.84145	0.56523	1.49533
16	IBM	2003	89,131	7583	35,629	1.18728	0.85328	0.52544	1.69846
17	IBM	2004	96,293	8430	42,464	1.18021	0.88194	0.50032	1.79382
18	IBM	2005	91,134	7934	42,718	1.29896	0.86180	0.47828	1.86239
19	IBM	2006	91,424	9492	43,531	1.11397	0.88560	0.52183	1.87241
20	IBM	2007	98,786	10,418	57,226	1.20011	0.82027	0.55923	1.82442
21	IBM	2008	103,630	12,334	48,507	1.15480	0.94619	0.59461	1.90094
22	IBM	2009	95,758	13,425	62,070	1.35923	0.87834	0.53140	2.06720
23	IBM	2010	99,871	14,833	73,789	1.18623	0.88029	0.55008	2.15256
24	IBM	2011	106,916	15,855	82,972	1.20903	0.91826	0.55809	2.26707
25	IBM	2012	104,507	16,604	91,882	1.13313	0.87664	0.56800	2.22159
26	IBM	2013	99,751	16,483	108,440	1.27883	0.79028	0.57842	2.24075
27	IBM	2014	92,793	12,022	109,918	1.24803	0.78951	0.63535	2.20372
28	IBM	2015	81,741	13,190	116,517	1.24030	0.73977	0.61267	2.24732
29	IBM	2016	79,920	11,872	123,361	1.20987	0.68034	0.60381	2.09744
30	IBM	2017	79,139	5753	126,534	1.33113	0.63131	0.61585	1.86534

IBM did not declare bankruptcy in 1993, when its Altman Z ratio collapsed below 1.0, to 0.876. The Altman Z data and comparable profitability ratio data for Boeing and Dominion Energy are reported in the Appendix (Table 2).

The Time Series of Ratios in the United States, 1962–2017

Is there a consistent pattern of movement in financial ratios over the 1962–2017 period? Yes. For all firms listed on the WRDS database, firms have substantially lowered their liquidity over the 55-year period; the median current ratio falling from

Table 2 Boeing profitability ratio, 1999–2018. Boeing Company. BA 097023105 2108601 NYSE Common stock

	Dec '99	Dec '00	Dec '01	Dec '02	Dec '03	Dec '04	Dec '05	Dec '06	Dec '07	Dec '08
Gross Margin	**11.97**	**14.83**	**16.19**	**15.85**	**13.12**	**14.84**	**15.74**	**17.99**	**19.52**	**17.28**
Gross Margin (Benchmark)	42.66	45.19	39.67	42.23	43.24	44.00	43.92	44.29	43.98	44.41
SG&A to Sales	5.84	7.36	7.43	7.72	8.75	10.55	11.73	12.07	11.12	11.25
Operating Margin	**5.77**	**6.85**	**8.11**	**7.31**	**3.46**	**4.28**	**4.01**	**5.92**	**8.41**	**6.03**
Operating Margin (Benchmark)	17.44	16.72	8.78	12.93	13.90	15.87	17.27	16.70	17.25	17.31
Pretax Margin	**5.73**	**5.84**	**6.12**	**5.88**	**1.09**	**3.74**	**5.14**	**5.19**	**9.22**	**6.56**
Pretax Margin (Benchmark)	18.24	22.36	0.58	7.14	13.70	15.58	17.69	17.66	17.35	13.84
Net Margin	**3.98**	**4.15**	**4.86**	**4.29**	**1.42**	**3.47**	**4.67**	**3.59**	**6.11**	**4.36**
Net Margin (Benchmark)	12.57	15.42	5.89	9.55	10.93	12.11	13.89	14.06	13.55	12.68
Free Cash Flow Margin	8.60	9.76	4.72	6.24	6.22	4.73	9.94	9.46	11.83	−3.41
Return on Assets (%)	**6.38**	**5.45**	**6.26**	**4.61**	**1.36**	**3.40**	**4.49**	**3.94**	**7.33**	**4.71**
Return on Assets (Benchmark)	10.85	11.60	4.29	6.54	7.63	9.07	11.17	11.10	10.57	10.05
Return on Equity (%)	**19.42**	**18.93**	**25.87**	**25.04**	**9.07**	**18.74**	**22.93**	**27.93**	**59.06**	**68.85**
Return on Equity (Benchmark)	21.76	20.96	8.01	12.49	14.02	16.40	20.40	21.43	22.18	22.82
Return on Common Equity	19.42	18.93	25.87	25.04	9.07	18.74	22.93	27.93	59.06	68.85
Return on Total Capital	17.87	18.50	21.99	17.48	7.82	9.75	9.72	20.19	35.43	31.32
Return on Invested Capital	12.88	11.81	14.03	11.05	3.44	8.35	11.98	13.17	27.65	24.00
Cash Flow Return on Invested Capital	34.71	32.98	18.94	20.85	18.60	15.86	32.74	44.78	65.30	−3.63
	Dec '09	**Dec '10**	**Dec '11**	**Dec '12**	**Dec '13**	**Dec '14**	**Dec '15**	**Dec '16**	**Dec '17**	**Dec '18**
Gross Margin	**17.20**	**19.38**	**18.72**	**15.98**	**15.42**	**15.44**	**14.59**	**14.57**	**18.55**	**19.42**
Gross Margin (Benchmark)	43.90	46.67	46.70	46.71	46.52	46.99	45.64	46.22	47.24	48.39
SG&A to Sales	14.45	12.08	10.66	8.59	8.11	7.51	7.13	8.72	7.79	7.75
Operating Margin	**2.74**	**7.31**	**8.06**	**7.39**	**7.31**	**7.93**	**7.46**	**5.86**	**10.76**	**11.67**
Operating Margin (Benchmark)	16.35	21.48	22.42	22.64	21.91	22.75	21.58	21.11	22.69	24.97
Pretax Margin	**2.54**	**7.01**	**7.85**	**7.23**	**7.19**	**7.86**	**7.44**	**5.89**	**10.76**	**11.47**

Pretax Margin (Benchmark)	12.19	21.48	21.59	19.12	21.52	22.16	20.26	20.00	21.84	23.46
Net Margin	**1.96**	**5.15**	**5.84**	**4.77**	**5.29**	**5.99**	**5.38**	**5.17**	**8.77**	**10.34**
Net Margin (Benchmark)	11.15	16.82	17.30	16.91	17.44	17.92	17.13	16.61	18.21	20.50
Free Cash Flow Margin	6.47	2.84	3.36	7.11	7.02	7.30	7.19	8.34	12.43	13.45
Return on Assets (%)	**2.31**	**5.07**	**5.40**	**4.61**	**5.04**	**5.67**	**5.34**	**5.31**	**8.98**	**9.97**
Return on Assets (Benchmark)	7.71	12.06	12.65	11.81	11.49	11.61	10.83	8.99	9.39	12.08
Return on Equity (%)	**320.14**	**135.31**	**127.72**	**83.03**	**44.15**	**46.22**	**68.96**	**136.80**	**1397.78**	**3012.39**
Return on Equity (Benchmark)	16.54	24.81	25.46	23.86	23.44	25.78	25.52	22.98	24.42	33.39
Return on Common Equity	320.14	135.31	127.72	83.03	44.15	46.22	68.96	136.80	1397.78	3012.39
Return on Total Capital	17.51	30.88	35.46	37.33	30.86	33.87	41.90	40.74	89.93	91.53
Return on Invested Capital	13.35	23.17	28.89	27.46	24.24	27.37	32.46	38.44	79.83	98.93
Cash Flow Return on Invested Capital	56.02	20.65	28.97	52.92	43.29	44.57	58.76	82.51	130.05	145.01

Source: FactSet Fundamentals

Table 3 WRDS Universe median values, 1962–2017, selected years

Fyear	CR	SA	TDA	DuPontA	ROE	New Altman Z
1962	2.477	1.273	0.384	0.068	0.102	
1970	2.065	1.816	0.449	0.063	0.098	
1975	1.879	1.262	0.487	0.062	0.102	
1980	1.765	1.224	0.486	0.078	0.125	
1985	1.870	0.996	0.452	0.056	0.093	
1990	1.578	0.957	0.460	0.047	0.084	2.097
1995	1.688	0.859	0.409	0.053	0.088	2.183
2000	1.630	0.629	0.402	0.031	0.065	1.765
2005	1.755	0.573	0.357	0.050	0.099	1.705
2010	1.797	0.500	0.338	0.039	0.066	1.635
2015	1.620	0.362	0.378	0.029	0.056	1.313
2017	1.724	0.353	0.375	0.063	0.044	1.389

2.477 in 1962 to 1.724 in 2017. The median debt-to-assets ratio has fallen slightly, from 0.384 in 1962 to 0.375 in 2017. Sales efficiency has fallen from 1.273 in 1962 to 0.353 in 2017.

The median on equity has fallen from 17% in 1962 to 4.4% in 2017. The falling liquidity, sales efficiency, and return on equity have driven the median Altman (new) Z from 2.097 in 1988 to 1.389 in 2017. The reader will not be surprised to see bankruptcies rise during the 1963–2017 period, as we will show in chapter "Liquidation, Failure, Bankruptcy, and Reorganization."

Profitability Ratios in Industry

Boeing has increased its operating income to sales (gross margin) during the 1988–2018 time period (see Table 2); however, its gross margin has been less than the benchmark used in Table 3 the stocks in the S&P 500. In fact, Boeing's net margin, net income to sales, has been substantially less than the S&P 500. The fact that Boeing has delivered higher return on equity, higher return on invested capital, and a higher (New) Altman Z score is the result of much higher use of debt. Boeing, like IBM, has used debt to achieve extremely high ROE values.

Limitations of Ratio Analysis

Although ratio analysis is an extremely versatile tool, applying it can be dangerous if its limitations are not understood. Ratios differ among industries, as we have shown. Ratio analysis may be useless if the analyst does not have a feeling for the normal differences among different industries. Furthermore, if the firm has misled the

financial community, such as WorldCom and Enron, in its financial disclosures, then ratio calculation and analysis using incorrect data cannot be correct. We will examine bankruptcy in chapter "Liquidation, Failure, Bankruptcy, and Reorganization" and calculate the WorldCom in 2001.

The user of ratios should be aware of seasonal factors. Usually, according to the results of the ratios, a firm appears in its best financial health at the seasonal low point of its annual operations. Inventory and collectibles are at minimum levels; funds normally invested in these assets have been released and are generally applied to reduce current obligations. These developments tend to improve almost all the working capital ratios. Most firms draw up their annual statements as of December 31. For many industries the end of the normal calendar year is as good a point as any to present the financial statements. In industries whose seasonal low point is not at December 31, some firms set the closing of their books at the end of their "natural" year. The analyst must be aware that the statement date often presents a better financial picture of the firm than that which prevails during the operating year. He must also be careful in making comparisons to allow for differing seasonal patterns among firms and industries and to take into account the use of possible different statement dates.

An attempt to solve the problem of seasonal or other variation in the statements has been made by taking such items as inventory or accounts receivable as an average between two statement figures. As in so many attempts to be extremely accurate, the slight improvement in results does not seem worth the additional effort. The figures may be difficult to obtain, and if the figures used to obtain the average come out of about the same seasonal position, the averages still do not represent the intra-operational relation of the accounts. Moreover, standard ratios used for comparison are not generally based on averages. Only where there has been a drastic change in the level of the firm's activity does the use of averages seem justified, and in such cases, ratios may not be the most fruitful method of analysis.

Working Capital Analysis and Granting Credit

The various tools of working capital analysis are useful to the internal management in forecasting the need for short-term funds and in assuring the safety and short-run liquidity of the firm. Investors may, of course, also be interested in the working capital relationships. Of all the techniques of short-term financial analysis, however, ratios are most commonly applied as a guide to decisions on the granting of credit. The most consistent users of ratio analysis are likely to be found in the credit departments of suppliers and in the loan departments of commercial banks.[8]

[8]Ratio analysis and related techniques are often called credit analysis.

When used in granting credit, ratio analysis helps classify the credit customer into different grades of risk. The grantor of credit desires some inkling of the chance of loss—the percentage of the accounts he may be unable to collect on each class of customer of different financial strength.[9] An experienced and perceptive credit analyst can roughly grade various credit applicants. The credit department might roughly predict that 0.5% of the total credit sales made over the year to firms having certain financial and business characteristics are likely to be uncollected. For another group of customers, perhaps the loss rate in uncollected accounts might amount to 15% of the annual sales: presumably there are all sorts of classes or grades in between.[10]

The credit department has to decide which grades of risks may be extended credit. Credit losses can be minimized by selling only to the financially strongest customers, but this would cause a loss of sales volume. The credit department's objective is to minimize losses, but management wishes also to maximize business volume. The problem can be brought near solution by the marginal approach. Credit is refused to that class of firms where the potential rate of loss is too large. All other applicants are extended normal credit terms. It is not necessary to grade the intramarginal credit risks. The credit department needs only two basic working grades. Their major task is at the margin where it must be decided whether a customer's financial position is too shaky to extend him credit profitably.

Where should the credit line be drawn? The amount of risk a creditor can take profitably depends largely on the operating and financial position of the credit-granting firm itself. A firm with a strong net working capital position may be able, from a financial point of view, to extend credit to poorer accounts than a firm with a narrower working capital base.[11] The supplier with a strong financial cushion can absorb occasional losses on credit sales, whereas a weaker supplier may fear that a bunching of credit losses would push his own firm close to the brink.

Pure economic analysis reveals that the firm should sell up to that grade of credit risk where the probable amount of loss on uncollected accounts just equals the difference between the sales price and the firm's marginal cost. Of course, in actual operations most firms have no precise measure of marginal costs but have adopted a variety of devices which, perhaps unconsciously, roughly approximate the results obtainable by the marginal theory.

---

[9]The banks, which have a lower margin for risk, obviously have to restrict their credit more severely than suppliers. The following discussion applies mainly to trade credit.

[10]If the loss experience in a given credit grade is to be fairly predictable, the classes have to be made quite large. Very fine grades might be useful for the purpose of finding the exact credit cutoff point. But they decrease the predictability of the results in each class.

[11]In some cases where immediate financial risks are high but the possibilities of good profits exist in the buying firm's activities, a supplier in a strong financial position has on occasion purchased some of its customer's common stock as well as sold to it on credit. This gives the supplier an opportunity to obtain possible compensating gains to offset the credit risk. The use of convertible securities is an analogous situation.

From the view of applied economics, a firm having open or extra capacity may profitably sell on credit up to that grade or class of risks where the chance of loss equals the margin of profit over the estimated variable costs at the going rate of production. For example, if at the current rate of output the variable costs on a product were 60% of the selling price and the credit loss in selling to a particular grade of customer was estimated at 30% of the sales, sales to this group of customers would cover out-of-pocket costs and return an additional 10% to cover fixed costs and possibly profits. To be precise, the seller could actually grant credit to the marginal customers up to the exact point where the predicted rate of credit losses (e.g., 40%) equaled the margin of profit over the variable costs at the given level of output.[12] However, credit acceptance tightens severely when the firm is operating at or near capacity.

Ideally, the credit managers' decision is mainly concerned with those cases where the financial characteristics of the customer (as determined by ratio analysis, etc.) are such as to make him appear to be close to the marginal credit risk. All firms which appear obviously stronger are extended credit, and weaker firms are denied credit terms. At the managerial level, there is often some tension between the sales manager whose goal is to maximize sales and the credit manager who wishes to minimize credit losses. As one observer noted, "If the credit manager and the sales manager do not get into an occasional fight, one or the other is not doing his job."

A Summary of Ratio Analysis

The financial community has long calculated ratios to assess the liquidity, profitability, leverage, and efficiency of firms. Ratio calculations and analysis summarize financial information reported on the balance sheet and income statement. The investor can often easily assess the financial health of a firm by calculating the ratios introduced in this chapter. Ratio analysis can be extremely useful in screening potentially poorer performing stocks, identifying problem firms, if not, bankrupt firms, using the Altman Z-score model.

[12]No net profit might be made out of selling to this group, but we presume there would be profits on the sales to all the firms that were intramarginal as far as credit loss probabilities are concerned.

Appendix

WRDS efficiency ratios 1988–2017												
Obs	tic	Fyear	sale	ni	re	CR	SA	TDA	NewZ	ROA	ROE	ROS
1	IBM	1988	59,681	5806	33,103	2.03273	0.81713	0.35468	2.44218	0.07949	0.14695	0.09728
2	IBM	1990	69,018	6020	36,500	1.53980	0.78816	0.42503	2.11721	0.06875	0.14055	0.08722
3	IBM	1993	62,716	−8101	11,667	1.18256	0.77319	0.59664	0.87580	−0.09987	−0.41043	−0.12917
4	IBM	1995	71,940	4178	14,723	1.28574	0.89598	0.51945	1.60290	0.05204	0.18633	0.05808
5	IBM	2000	88,396	8093	23,489	1.20530	1.00053	0.62001	1.84788	0.09160	0.39241	0.09155
6	IBM	2005	91,134	7934	42,718	1.29896	0.86180	0.47828	1.86239	0.07503	0.23971	0.08706
7	IBM	2010	99,871	14,833	73,789	1.18623	0.88029	0.55008	2.15256	0.13074	0.64363	0.14852
8	IBM	2015	81,741	13,190	116,517	1.24030	0.73977	0.61267	2.24732	0.11937	0.92484	0.16136

WRDS efficiency ratios 1988–2017												
Obs	tic	Fyear	sale	ni	re	CR	SA	TDA	NewZ	ROA	ROE	ROS
1	BA	1988	16,962	614	4137	1.27681	1.34534	0.55171	2.52851	0.048699	0.11362	0.036199
2	BA	1990	27,595	1385	4840	1.22967	1.89123	0.51011	3.06315	0.094922	0.19862	0.050190
3	BA	1995	19,515	393	7746	1.77721	0.88311	0.44162	1.86405	0.017784	0.03970	0.020138
4	BA	2000	51,321	2128	12,081	0.86741	1.22111	0.61521	1.82600	0.050633	0.19310	0.041465
5	BA	2005	54,845	2572	15,498	0.77934	0.91320	0.62816	1.33095	0.042825	0.23257	0.046896
6	BA	2010	64,306	3307	11,026	1.14626	0.93788	0.68356	1.35012	0.048232	1.19559	0.051426
7	BA	2015	96,114	5176	26,008	1.35353	1.01807	0.62645	1.66545	0.054826	0.81705	0.053853

WRDS efficiency ratios 1988–2017												
Obs	tic	Fyear	sale	ni	re	CR	SA	TDA	NewZ	ROA	ROE	ROS
1	D	1988	3344.1	491	955.5	0.96104	0.32523	0.48857	1.02682	0.047752	0.12399	0.14683
2	D	1990	3532.5	503.9	1129.6	0.84024	0.32140	0.48467	1.08209	0.045847	0.11500	0.14265
3	D	1995	4651.7	469.1	1420.9	0.78662	0.33458	0.44170	1.00236	0.033740	0.08637	0.10084
4	D	2000	9260	436	997	0.77266	0.31552	0.61599	0.65574	0.014856	0.05813	0.04708
5	D	2005	18,041	1049	−1014	0.69952	0.34259	0.55323	0.56875	0.019920	0.09846	0.05815
6	D	2010	15,197	2808	6088	0.93539	0.35493	0.50286	1.05089	0.065581	0.22915	0.18477
7	D	2015	11,683	1899	5984	0.51613	0.19870	0.53976	0.56156	0.032298	0.14995	0.16254

References

Altman, E. I. (1968). Financial ratios, discriminate analysis and the prediction of corporate bankruptcy. *Journal of Finance, 23*, 589–609.

Altman, E. I. (2000). Predicting financial of companies: revisiting the Z-Score and Zeta Models. New York University working paper.

Baumol, W. J. (1952). The transactions demand for cash: An inventory theoretic approach. *The Quarterly Journal of Economics, 65*, 545–556.

Chandler Jr., A. D. (1977). *The visible hand: The managerial revolution in American business*. Cambridge: The Belknap Press of Harvard University Press. Chapter 13.

Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Guerard Jr., J. B., & Vaught, H. T. (1989). *The handbook of financial modeling*. Chicago, IL: Probus. Chapters 1, 2, 4.

Markowitz, H. M. & Guerard, J. B., Jr. (2019). *The existence and persistence of financial anomalies*. Presented at the Institute of Quantitative Research in Finance, La Jolla, CA.

Miller, M. H., & Orr, D. (1966). A model of the demand for money by firms. *The Quarterly Journal of Economics, 80*, 413–435.

Van Horne, J. C. (2002). *Financial management & policy* (12th ed.). Upper Saddle River, NJ: Prentice Hall.

Financing Current Operations and the Cash Budget



A firm with increasing sales volume needs increased current assets to service the new level of activity. Given normal inventory turnover, higher sales necessitate a higher level of stocks. Similarly, greater sales levels enlarge the average amount of receivables the firm carries since additions to the accounts come in faster than old accounts are collected. The corporation officers responsible for working capital management must decide how to finance the required increase in current assets.

A temporary or seasonal rise in activity may be financed with short-term or "temporary" sources. The additional gross working capital required can be acquired by an increase in suppliers' credit, negotiating a bank loan, or selling open market commercial paper. The use of short-term financing neither adds to nor detracts from the absolute amount of a firm's net working capital. The working capital ratios, however, decline, indicating an increase of risk in the firm's current position, as we discussed in the previous chapter. Table 1 illustrates the effect on the working capital position of a firm when an increase in inventory is financed with short-term debt, specifically with a bank loan. The current and acid test ratios decline, and the net working capital turnover rises to what might be a dangerous level. The working capital position of the company would be considered rather risky if it maintained these proportions in its current accounts for any length of time.

However short-term financing is comparatively cheap and flexible and may be used with relative safety under any of the following conditions:

1. The growth in sales is temporary or seasonal.
   Under these circumstances, at the end of the peak period, sales decline, with a resulting decline in the level of inventory required. As sales slacken, more old receivables are collected than is added in new accounts. The funds released from inventory and receivables can be used to repay current liabilities and restore the firm to its original (or, if there are profits, a better) working capital position. Nevertheless, even if the need for more current assets is expected to be temporary, using much additional short-term financing is dangerous if the starting level of net working capital is inadequate.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_6

Table 1 Effect of a bank loan on the working capital ratios

Working capital position before bank loan			
Cash	$50,000	Accounts Payable	$100,000
Receivables	100		
Inventory	100,000		
Total current assets	$250,000	Total current liabilities	$100,000
Going level of sales		$1,500,000	
Net working capital		= $150,000	
N.W.C. Turnover		= 10	
Current ratio		= 2.5	
Acid test ratio		= 1.5	
Working capital position after bank loan			
Cash	$50,000	Accounts payable	$100,000
Receivables	100	Bank loan	100,000
Inventory	200,000		
Total current assets	$350,000	Total current liabilities	$200,000
Going level of sales		$2,000,000	
Net working capital		= $150,000	
N.W.C. Turnover		= 13.33	
Current ratio		= 1.75	
Acid test ratio		= .75	

2. The firm holds substantial net working capital.
   If the firm has a strong working capital position, it may be able to use some additional short-term financing with impunity. The resulting changes in the working capital ratios would merely bring them down to the standard or "normal" range. Within limits, the firm can safely finance an increase in operations without using any new long-term financing.
3. The operating margin over variable costs is high.
   A good profit margin over out-of-pocket costs may permit the firm to maintain its working capital from the operating profits on the increased sales. Whether this can be done depends on the comparative rate of fund inflow, the rate at which new assets are required, and the path of the rising sales curve. If the latter shows a short rise and then plateaus, the high profits may soon restore the working capital position that existed before the rise. If, however, there is no period of consolidation, profits alone may not be sufficient to furnish new net working capital at the desired rate. Relying on profits during a period of growth as the source of net working capital constrains the firm to careful budgeting of new fixed asset acquisitions. Moreover, such a policy is likely to keep dividend payouts niggardly.

Paradoxically, a firm finds the most pressure against its current financial position during periods of rising activity (and periods of sharp losses). In periods of mildly declining activity, the funds released by the decrease in current assets can be used to repay current debt. As long as the current ratio exceeds 1:1, an equal decrease in

current assets and current liabilities improves the working capital ratios. Thus, unless significant losses are experienced in realizing the current assets, the firm develops increased operating liquidity.[1]

On the other hand, the conditions for the financial ruin of a firm are often set during the upgrade. If financial planning is lax and if the management fails to provide adequate working capital, the company's current position can be weakened as its sales increase. A company whose current and acid test ratios are low and whose volume of business is high relative to its net working capital is "overtrading." A relatively minor economic reverse—the failure of a large account receivable or a drop in inventory values—can bring the firm to financial disaster. The reader is reminded of the Altman Z score calculated in the previous chapter and its predictive power in identifying potential distressed firms.

Sources of Short-Term Financing

Almost all firms make some use of short-term financing. It is comparatively cheap and relatively available. It may be obtained quickly and conveniently; some kinds of short-term financing are supplied practically automatically. (Among these are trade credit and expense and tax accruals.) The major types of short-term credit are trade credit, bank credit, open market commercial paper, finance companies, and factors. The order in which these are listed is largely based on the prevalence and volume of the type of credit as used by the nonfinancial firm. Trade credit is part of almost every firm's financial structure because credit is customarily given on almost every commercial sale. Bank credit is available to sound firms of all sizes and is used frequently. The use of commercial paper is restricted to large firms of good standing. Commercial paper is one of the credit instruments traded on the "money market."[2] The total volume of credit supplied by the finance companies and factors is quite large, but although its use is spreading, it is not widely used outside a few industries.

[1] In the early stages of a sales downturn, the firm may accumulate unintended inventory. The firm will cut current purchases to restore the inventory to a more "normal" relationship to sales. Adequate net working capital is important to bring the firm through just such transitional situations.

[2] In the broadest sense, the money market may be defined as the broad complex of institutions extending, dealing in, and using short-term money credit. It refers both to the dealers in short-term credit obligations and the debt instruments and types of loans customary on the market. The operations of the money market set the structure and level of the short-term interest rates. The center of its activities in the United States is New York City, although its eventual scope is nationwide and even international. Among the major suppliers of funds to the money market are the banks, insurance companies, other financial institutions, and corporations having temporary excess funds. The important rates set on the money market are the rates on short-term treasury bills, commercial paper, bankers' acceptances, prime commercial loans, and margin loans to brokers on security collateral. The rates on the capital and money markets are related.

Trade Credit

Trade credit, also called suppliers' credit, open book account, or ledger credit, develops because most commercial purchases need not be paid for immediately. During the period the purchasing firm has the goods but has not yet made payment, some part of its operating assets are being financed by its "creditor." On the books of the purchaser, the balance of not-yet-paid suppliers' bills is carried as accounts payable.

The length of time the purchaser has before payment and the discount, if any, for prompt or early payment is called the "terms." Credit terms vary according to the customs prevailing in different industries or lines of trade. One type of term, which is growing more common, is the single statement or end-of-the-month billing terms. A popular single-statement term is customarily written "2/10 e.o.m. n/30." All purchases made during the current month appear on the statement of the first day of the following month. If the purchaser pays the bill within 10 days after the end of the month (e.o.m.) in which he made the purchase, he may take 2% off the face of the bill as a cash discount or reward for prompt payment. If the purchaser fails to pay during the first 10 days of the month, the gross amount is due at the end of the month (i.e., net/30).

Single-statement terms save the seller booking costs and billing problems since he can consolidate the billing for sales during a month on a single statement. The cash discounts are actually less a reward for prompt payment than a penalty to the slow payer for the worry and extra risk put on the seller.

Open book credit is a cheap source of funds if the bills are paid within the 10-day end-of-the-month discount period. A purchaser who buys at the beginning of the month has 40 days in which to pay his bill and still receive his 2% discount. (As a matter of fact, the buyer may have upward of 45 days of free credit, since a purchase made after the 25th of the month usually does not appear on the statements until the end of the following month.) If the discount credit period passes, however, the purchaser should consider the loss of the discount a cost of receiving credit for only the additional 20 days from the tenth until the end of the month. This loss of the 2% discount comes to a high rate when reckoned on an annual basis. The payment of 2% for use of funds for 20 days is equivalent to an annual simple interest charge of 36%.[3]

A firm that foresees difficulty in meeting the cash discount period on its payables should, if it can, arrange a bank loan or alternate source of credit. (The cost of a bank loan is hardly likely to be more than 6–10% per annum, and perhaps the cost of factoring may run 12%. The cost in lost cash discounts may amount to 36% per annum or even more in some lines of trade.) However, some firms do not consider the lost cash discount as amounting to the equivalent of as much as 36% a year because in practice they delay payment beyond the whole term period. Of course, the firm is breaking the credit terms and is considered a delinquent or slow payer.

[3]That is, 20 days goes in 360 days 18 times ($18 \times 2\% = 36\%$).

Creditors may forgive such behavior if it happens infrequently. But frequent abuse of credit terms becomes known in the trade and the habitually delinquent firm may find its credit curtailed. It will not receive the best service, and some suppliers will not deal with it at all except on a cash basis.

Bank Credit

Types of Bank Loans Although commercial banks place funds in government and corporation bonds, purchase mortgages, and make personal loans, their prime function—one they are uniquely equipped to perform—is to make commercial loans.[4] The classical business or commercial loan is a short-term (30–90-day) grant of credit to a business unit to enable it to cover temporary gross working capital needs. A forecasted or planned event—i.e., a seasonal tapering off in activity that releases funds from inventory and receivables, or the development of long-term financing—will provide the firm with the money to repay the loan.

Although it is general bank policy to assume that all commercial loans will be paid out of the business' cash-generating capacity, there are two classes of loans, secured and unsecured. The unsecured loan is granted merely on the general credit of the firm; the secured loan is backed by a lien against a firm asset, by the pledge of the credit of some person in addition to that of the business, or by a lien against some personal assets of those connected with the business.

One type of secured loan is based on a lien against inventory. Inventories constitute good collateral for loans when the goods are homogeneous in nature and unspecialized and where there is a going market (preferably organized) and the market price of the goods is relatively stable. Sometimes banks make loans against warehouse receipts. The inventory is held in a public warehouse, and the goods may not be withdrawn from the warehouse by the firm without the counter-signature of the properly designated bank official. As any part of the goods is withdrawn, a proportionate amount of the original loan is repaid.

Some bank loans are secured by a lien against the firm's outstanding receivables. If the firm fails to repay its loan, the bank may try to satisfy its claim by collecting the accounts receivable for itself. This is, of course, a last resort; before granting the loan, the bank tries to ascertain whether the borrowing firm can repay out of its regular operations. The asset pledged as security is only an additional safety feature which may help to reduce the bank's losses if the loan is not repaid as expected. The bank, however, may set the terms of the loan so the company will have to pay back part of the loan as the pledged accounts are paid. This keeps the proportion of bank debt to the value of the assets pledged constant over the life of the loan.

[4]As of December 31, 1960, commercial loans totaled $42.4 billion and constituted 80% of aggregate earning assets of all commercial banks in the United States. As of December 2002, commercial loans totaled $959.8 billion, and $2351.5 billion, as of December 2019. Source: Federal Reserve Economic data. www.fred.stlouisfed.org. October 21, 2020.

Many business loans are backed by the pledge of personal assets of the owners as collateral. The assets are usually liquid, that is, marketable securities, cash values of insurance policies, savings accounts, etc. In a personal proprietorship these personal assets of the owners may furnish eventual backing for business debts, but only after the personal creditors are satisfied; moreover, all the general business creditors may look to the net personal assets. However, the direct pledge of personal assets as collateral gives the bank the first claim against these assets. The personal holdings of corporation owners are not normally within the reach of the business creditors. In a smaller company, this is not always an advantage, for it can restrict the amount of credit available to the firm. Often, then, the principal officers or major stockholders of a closely held corporation may pledge personal securities, savings, or other assets to help obtain a bank loan for their company.

Some loans are made on a cosigner basis. Some party other than the one primarily liable assumes contingent liability on the debt or guarantees the loan in case payment is not made as agreed. Friends and business acquaintances of the owner sometimes cosign a promissory note along with the proprietor of a small business. A supplier may cosign to help a new but growing customer establish bank credit. A parent company may sign or endorse the note of a subsidiary. In a family or closely held corporation, the principal officer may sign personally as well as in his capacity of agent for the corporation. In so doing, he waives his limited liability and pledges his personal credit at least for the amount of the particular debt. In general, as far as the bank is concerned, the effect of a cosigner loan is to bring in the credit and resources of another to help buttress the inadequate position of the original applicant for the loan.

A firm usually prefers to receive an unsecured loan rather than any of the varieties of secured loans. Receiving an unsecured loan is an implicit acknowledgment of a high credit rating. Moreover, although a secured loan does not change the overall situation of the other creditors, as shown by ratio analysis, from what would exist if an unsecured loan had been obtained, it does segregate a group of assets for the prior benefit of the secured creditor in the event financial difficulties should occur. Less desirable assets are thus left for the general creditors, and the rate of loss they could suffer may be higher than that of the secured creditors. If the general trade creditors know that a solid asset is pledged for a bank loan, they may be cautious in extending additional credit to the firm.

The foregoing analysis does not imply that incurring a bank loan normally weakens the general creditors' position. Ordinarily, the proceeds of the bank loan are used to reduce accounts payable or accruals. The remaining claims are in no worse position than before since the total debt is the same—only some of it has been transformed into bank debt.

Cost of Bank Credit The cost of bank credit varies with the size and credit standing of the firm and according to the general level and the shape of the market structure of the interest rate. The prime rate is the lowest commercial loan borrowing rate and is set by the banks in the large cities as the charge for their biggest and soundest customers. The average interest cost is lower on a large loan than on a smaller one,

because administrative, bookkeeping, investigating costs, etc. are more or less fixed and do not vary with the size of the individual loan. Also, credit risk tends to be higher for smaller concerns. For a given class of customers, the commercial loan rate is generally lower than the capital market rate (effective bond interest rate), because the bank loan rate belongs to the class of short-term interest rates and in a normal market the interest rate curve is lower at the shorter maturities.[5] The commercial loan rate tends to be low, of course, during periods of slack demand for funds and when the monetary authorities are pursuing an easy money policy; the rate is higher during times of credit stringency or tight money.

Not only is the commercial loan rate usually a notch below the effective capital market rate, but the use of bank borrowing in lieu of a bond or debenture issue avoids flotation costs, the legal and administrative costs of drawing up the indenture, and the administrative costs (if the issue requires it) of registering with SEC. Furthermore, a bank loan may be more closely tailored to the firm's requirements in contrast to a capital market issue; a bond issue, for example, is often floated ahead of the company's need for funds, thus requiring the firm to pay interest on the debt in the meantime. A seasonal inflow of funds can be used to extinguish bank debt and reduce the interest costs when the borrowed capital is not needed. If, however, the debt is a debenture or bond, the firm is not likely to make retirements out of seasonal funds, and interest payments continue. Of course, the net interest cost is somewhat reduced by the earnings on the short-term investments the firm can make in the interim.

On the other hand, although long-term capital may be costlier, its advantages over bank credit lie in its semipermanence—the lack of continuous current pressure on the firm gives management the opportunity to forecast and plan further into the future. Thus, many firms may prefer to fund (i.e., use long-term sources) as much of their current debt as possible, relying on bank credit for perhaps a small part of the peak of their seasonal needs. Such firms may have excess funds offseason, but capital costs can be reduced by placing seasonally redundant funds in such eminently secure short-term money market investments as treasury bills, tax certificates, marketable commercial paper, etc.

The interest cost on a commercial loan is usually computed on a simple interest basis for the life of the loan. Thus, if the interest rate is 5% per annum, the interest charge on a $100,000 loan for 60 days will be $833.33 (i.e., 5/6% of $100,000 for 60 days equals 5% interest on an annual basis). Where the amount of loan outstanding varies from day to day, the charge may be set up on a per-diem basis so that the lender pays only for the funds he actually uses. In making most loans, however, the bank discounts the note (i.e., the negotiable legal evidence of debt) and thereby earns a rate slightly above that of simple interest. When a note is discounted, the bank subtracts the interest for the term of the loan in advance and extends to the borrower the difference between the face amount of the note and the interest. In other words, the borrowers have the use of slightly less than the face value of their loans, and the

[5] See chapter "Long-Term Debt" for a discussion of this phenomenon.

bank retains the discounts which makes the interest charge minutely higher. The extra amount brought in by discounting rather than straight interest charges is relatively small.

Another device used by banks to vary the effective interest charge is the compensating balance. Traditionally the compensating balance has been set at 20%, which means that a successful applicant for bank accommodation is requested to keep 20% of his loan on deposit at the bank over the life of the loan. Although this requirement has been often cited as a safety measure, its actual effect is to raise the rate of interest on the usable portion of the loan.[6] To the extent the compensating balance exceeds the firm's normal cash balance, the effective interest cost on the loan exceeds the nominal rate. The use of the compensating balance is no longer a fixed custom. During times of easy money when the banks have considerable free reserves, no compensating balance is requested of prime borrowers. At other times, the percentage of compensating deposit required varies with conditions on the money market and the general credit standing of the borrower. Although a compensating balance is no longer an invariable condition, it is generally expected that the borrower will keep his normal working balance at the bank that accommodates his credit needs. Usually, a bank tries to meet the credit needs of its own depositors before financing other applicants.

Line of Credit The term line of credit is used in three different but related meanings. Suppliers may set a line of credit or credit limit for individual customers. This is the amount of outstanding trade credit they are willing to allow before cutting off further sales or reviewing the account. If the customer knows what the amount is, it is called an explicit line of credit; an implicit credit limit exists when the amount set by the suppliers' credit manager is not made known to the customer.

An open or general line of credit exists when a large corporation makes an agreement with its banks for a specified amount which the company may borrow at any time during a specified period, perhaps a year. The line of credit is subject to review and renewal at the end of the period. The firm is charged interest at the going commercial loan rate only for the time any loan is outstanding and only for the actual amount of the line of credit used. The firm usually pays an additional fee for the whole line of credit over the year—whether used or unused—since the bank obviously has to restrict its investment policies to have the standby credit available.

A seasonal or temporary line of credit with a bank provides a company with funds for its peak needs. The firm has only to pay interest on the funds it actually borrows, but the bank is morally committed to have the funds available. The amount of a seasonal line of credit is generally determined by drawing up a cash budget (i.e., a short-term cash forecast of the firm's operations). This not only enables the negotiators to project the firm's likely maximum cash needs; it also provides a quantitative

[6]On a loan granted with a compensating balance, the bank is able to keep some proportion of its reserves otherwise committed to the loan. This savings in reserves can be lent to another borrower with a subsequent increase in the bank's earning power.

framework to ascertain how the firm will develop the free funds to repay the bank debt after the operational peak is over.

Borrowing in Anticipation of Permanent Financing Bank credit is a simple and quick method of raising funds. Thus, it is often used as a temporary method of financing until the firm determines how much permanent financing it needs, decides upon the best type of security to float, or organizes the floating of an issue. Thus, a utility firm may use bank financing to begin an addition to plant and, after the final cost is determined, float new bonds to retire the bank debt. A firm with access to the capital markets may be caught in a sudden wave of expansion and finance its increased activity with bank borrowings until it can organize a long-term security issue. The bank may lend money to recognized companies that already have issues traded on the secondary markets, with the full knowledge that the debt is to be retired by the issue of additional stocks or bonds.[7] In some instances, the security issue may be an afterthought. Both parties may have thought the debt would be paid out of the cash flow and underestimated the time period for which the new financing was required. When it becomes apparent that the firm cannot clear up its bank debt without curtailing profitable opportunities and the banks are reluctant to supply long-term capital, the management will seek other financing to repay the bank. If the bank debt is repaid out of the proceeds of a bond issue, the whole process is called funding the debt.

The sequence of bank credit to permanent financing may work quite well for the firm already established on the capital markets. The pattern also describes a major reason why a firm may venture out with its first public issue. But for many firms, the link between the money markets and the capital market is broken. Because they are small, or new, or for other reasons, these firms cannot make the transition to the capital markets easily or cheaply. The company may have outgrown the resources of its owners, and it may have opportunities that the commercial bankers recognize, but which they are unwilling to finance because funds would be committed over too long a term.[8]

Other Forms of Short-Term Financing

Open Market Commercial Paper Open market commercial paper is the term for the short-term negotiable notes with maturities from about 1 to 9 months that strong corporations float on the money markets. Open market paper is moved onto the market by commercial paper dealers, who sell the issue to insurance companies'

[7]This seems to establish another advantage to having a publicly traded issue. The firm not only has access to the capital markets but has a better rating in the money market in addition.

[8]The small business investment companies growing out of the Small Business Act of 1959 were hopefully designed to fill this gap in our financial institutions by providing capital to intermediate-size firms.

pension funds, business corporations, and smaller commercial banks for the short-term investment of their liquid reserves.[9] It is an important fund-raising device of large commercial finance companies.[10] In total, the amount of finance company commercial paper floated is about double the amount floated by all other kinds of companies combined.

The rates on commercial paper are generally below the prime loan rate but somewhat above the prevailing rate on treasury bills of the same maturity. The advantages to the lender in buying commercial paper are its ready availability in convenient maturities and its minimal administrative detail and costs. The advantage to the borrower is that the wide distribution his paper may obtain in what is essentially a nationwide market frees him from relying on local bank resources and provides him with lower net interest rates. On the other hand, the use of commercial paper does not establish personal rapport with the bankers, who may sometimes advise and support the firm through temporary hard times. Neither does the use of commercial paper lend itself to easy change of terms and renewal as is sometimes possible with a bank loan. The total volume of commercial bank loans far exceeds the funds raised by issue of commercial paper, although the latter is by no means negligible.

Finance Companies and Factors The large commercial finance companies are the department stores of the financial world. Initially adjuncts or subsidiaries of the automobile companies, their major business still consists in discounting dealer sales paper and financing dealer automobile inventories. Branching out into other financing areas, the large companies usually have a personal finance subsidiary and an account receivable factoring subsidiary. They finance major furniture and appliance dealers both for inventory and on installment receivables, and they may do some direct sales financing themselves. In many activities, the finance companies compete with the commercial banks. Although their charges may be higher, the finance companies often benefit from closer contact with the dealers' specialization that enables them to act efficiently and quickly and the willingness to take more risk than the bank. A large percentage of the funds which the finance houses dispense are raised through the banking system—either by the establishment of lines of credit with large banks or by the sale of open market paper, a large percentage of which is picked up by the banks outside of New York.

Factoring is an important source of financing in the textile and garment trades and also appears in leather goods, sporting goods, and lumber trades, among other industries. Many factoring companies are today subsidiaries of the major finance

[9]The spread received by the dealers for floating the notes is very low, but since the issues are large and the issuing firms have the highest credit standing, marketing problems are not difficult.

[10]In 1960 about $3.5 billion of finance company commercial paper outstanding was placed with insurance companies, corporations, or other large buyers without passing through the dealers and the open market. By December 2002, the commercial paper market had grown to $1321.5 billion, of which $1167.5 billion was finance and $154 billion was nonfinance. Source: US Federal Reserve (www.federalreserve.gov)

companies. Factoring arrangements are based on an advance of funds by the factor against an assignment of trade receivables. The factor's relation with the company is governed by an underlying agreement which can be varied to meet different circumstances. Generally, the factoring agreement is not a one-shot deal but a continuing arrangement; the company's debt to the factor is decreased as the assigned receivables are collected, but new debt is created as the receivables arising out of new sales are surrendered to the factor for immediate funds. Usually, the factor provides cash up to about 75% of the firm's accounts receivable. In certain cases, however, the firm may be given the privilege of assigning up to 85% of its accounts if it pays a higher financing charge for the last 10%.

Factoring arrangements are either recourse or nonrecourse. In a nonrecourse factoring agreement, the factor buys the accounts outright and cannot collect from the selling company for any losses on defaulted accounts. Of course, the factor advances only about 75% of the face of the accounts, and this gives him some margin to cover losses.[11] Moreover, in nonrecourse factoring, the factor has the right to reject accounts that do not meet his credit standards. In factoring with recourse, the factor does not take the loss on bad accounts; the borrowing firm makes good on all noncollected receivables. The financing costs are generally lower on the recourse agreement.

Factoring agreements are also classified as being on a notification or non-notification basis. In a non-notification arrangement, the firm's customers are not necessarily told of the factoring agreement, and they continue to send their payments to the office of the selling firm. Here the receipts are segregated, and the proper share turned over to the factor. Borrowing companies generally prefer a non-notification basis. When the accounts are on a notification basis, the customers are instructed to send their payments directly to the factor. The factor, of course, applies these receipts against the selling firm's indebtedness. Under a notification, nonrecourse basis, the factor often in effect operates the selling firm's credit department. He makes the initial credit analysis on a sale, takes charge of collections, and follows up on bad accounts.

Factoring charges are usually based on the firm's daily balance. Depending on the level of the general interest rate, the specific factoring agreement, and the credit standing of the selling firm and its customers, factoring rates may run from something below 1–3% per month. These relatively high charges, however, compare favorably with the customary cash discounts running as high as 8/10 n/60 in some trades where factoring is common. Factoring costs, of course, run higher than the bank loan rates; but a factor takes risks the bank will not, and many of the services a factor performs are not available at the banks.

Factors are heavily used in industries where entry of new firms has been comparatively easy and where many companies are short of working capital. Thus, although the use of factoring does not weaken the financial position of the firm, its

[11] When the account is collected in full the 25% coverage belongs to the seller; it is either returned to him or used to reduce his balance at the factors.

customary use by companies with inadequate initial capital has sometimes led to the mistaken notion that factoring is a "last resort" type of financing. In spite of this psychological drawback, factoring is a useful and convenient method of speeding up the cash cycle for many business firms.

The Cash Budget

Short-term borrowing is generally arranged on the anticipation that the normal operations of the business will create enough "cash throw-off" to retire the debt. The cash budget is a device for making the assumptions behind this anticipation explicit. The cash budget is a short-term forecast of the firm's cash operations. It shows the items that will absorb funds as activity rises, and it indicates how cash will be made available to retire current obligations after the peak need is over.[12]

Since the cash budget estimates the firm's short-term needs for cash, it is extremely useful in arranging bank loans and seasonal lines of credit. However, our purpose in presenting the cash budget here transcends its practical value as a tool of financial planning. The cash budget serves as a clear expository device depicting the absorption and release of funds as the pulse of business operations surges upward and then slackens.

The following information or estimates, broken down by the relevant time periods, are needed to construct a cash budget:

1. A forecast of sales, broken down into the probable proportions of cash and credit sales.
2. An estimate of purchases coupled with information on customary credit terms.
3. The estimation and construction of a function relating the volume of cash collections to the amount of outstanding receivables.
4. An estimate of out-of-pocket charges (or cash operating expenses) for the period ahead.

For illustration suppose we construct a cash budget to apply to the following problem: you are the treasurer of Goldman Dress Manufacturing, Inc. You are about to negotiate a credit limit at the bank for your seasonal operating needs.

Schedule A Estimated purchases		Schedule B Estimated sales	
1st month	150,000	1st month	100,000
2nd month	200,000	2nd month	150,000
3rd month	25,000	3rd month	250,000
4th month	25,000	4th month	200,000

[12]If the cash budget indicates that the firm will require funds for a longer term, the management should, if it can, arrange more permanent financing.

Table 2 Cash budget of Goldman Dress Manufacturing, Inc.

	1st Month	2nd Month	3rd Month	4th Month	5th Month
Accounts receivable, beginning of the month	50,000	110,000	172,000	284,000	256,880
Gross collections for the month (80% of the above)	40,000	88,000	137,600	227,520	205,500
	10,000	22,000	34,400	56,880	51,380
Sales (estimated per schedule B)	100,000	150,000	250,000	200,000	n.a.
Accounts receivable—end of the month	110,000	172,000	284,400	256,880	n.a.
Cash receipts: Net cash collection after discounts (98.2% of the gross collections)	39,280	86,416	135,123	223,425	201,805
Estimated cash disbursements: Payment of trade accounts—less discounts	36,800	138,000	184,000	23,000	23,000
Variable expenses (est. for mo. 5)	20,000	30,000	50,000	40,000	30,000
Administrative expenses	10,000	10,000	10,000	10,000	10,000
Total disbursements	66,800	178,000	244,000	73,000	63,000
Net cash gain or (outflow) for the month	(27,520)	(91,584)	(108,877)	150,425	138,805
Starting cash 1st month	10,000-				
Cumulative cash position (deficit)	(17,520)	(109,104)	(217,981)[a]	(67,556)	71,249

[a]Peak forecasted cash needs. Probable line of credit around $230,000

Schedule A Estimated purchases	Schedule B Estimated sales
(Terms on purchase 8/10 e.o.m.)	(Terms of sales 2/10 e.o.m. n/30)
Some significant accounts at the start of the season:	
Cash	$10,000
Accounts receivable	50,000
Accounts payable	40,000

It is estimated that about 80% of the accounts receivables of the preceding month are collected each month. About 90% of the collections obtain the cash discount:

Variable expenses are 20% of each month's sales.
Staff salaries and other administrative expenses run $10,000 per month.
Depreciation on equipment is $5000 per month.

Table 2 depicts the construction of a cash budget based on the information given on the Goldman Dress Manufacturing Company. Although all the figures on the budget appear solid, it is important to remember that the whole schedule is built on a forecast of short-term sales, an estimated purchase schedule related to these sales, and a cash collection estimate based on past experience of the relation of cash

payments to the receivable balances. If any of these factors deviate from expectations, the budget will be more or less imprecise. Of course, a budget is not sacred and is subject to revision to conform to developing facts. It is a guide to intelligent action, not a fixed concept.

According to the Goldman Dress Manufacturing, Inc. budget, the peak need for cash will reach approximately $218,000. This estimate is based on the assumption that payments for the purchases and receipts from the receivables will come in about the same time during the month. For some companies payments are customarily concentrated at the beginning of the month and receipts come in evenly throughout the month. It is possible to have an intra-monthly peak which, though it exists but a short time, must be covered. In such a case, the time periods upon which the budget is constructed should be made shorter than a month so that the peak need for cash is clearly indicated.[13]

The information given in the Goldman Manufacturing Company example included depreciation charges of $5000 per month. The reader might note that this item is extraneous and was not used in constructing the cash budget.

Although data were not specifically given for the fifth month, the major cash flow items affecting the fifth month were implicitly contained in the information presented. The estimated cash flow for the fifth month is particularly interesting, for it shows the company well out of temporary debt with cash holdings of $71,000—approximately $60,000 more than at the start of the season. This does not necessarily mean that the company will make a profit for the year's operation, or even for the season. Since there are other expenses or incomes which may be noncash in nature, the cash flow for any short-term period seldom corresponds to the usual definition of profit or loss. However, in constructing a budget even for so simplified a case as the Goldman Company, we can observe how the slackening activity led to a net cash inflow more than sufficient to repay the indebtedness built up at the height of the seasonal operations. Thus we have an illustration of the argument presented at the beginning of this chapter.

If a cash budget shows that the firm's increasing activity is long run so that the repayment of short-term borrowing might prove difficult, the management may consider raising longer-term capital-funding the short-term debt. Of course, it must determine whether moving to this new, more permanent level of activity is profitable. A firm does not have to increase its activities if there is no economic gain in it. If the firm decides to use more permanent financing, the factors of cost, risk, and the existing financial structure will determine the type of issue.

The cash budget problem has been presented here in isolation. Using the information given, we determined that the Goldman Company should apply for a seasonal line of credit with a maximum (allowing for possible minor miscalculations) of about $230,000. In the context of our little example, the other circumstances of the Goldman Manufacturing Company's financial position can be assumed to

[13] A budget can be made to cover the first third of every month and then the last two-thirds. This is useful, if there is a peak need around the tenth of the month.

support this conclusion as the best course of action. On the other hand, suppose the financial structure of the Goldman Company was already risky. Then the firm should try to arrange additional long-term financing carry a bigger net working capital balance and reduce the amount of its necessary seasonal short-term borrowing. If this procedure were to lead to an unduly heavy cash balance in the off-season, the excess funds can earn a return (albeit low) by investment in the money market.[14]

The cash budget—like other tools of financial analysis—is not an end in itself but merely a guide to understanding. Comprehending the financial problems of a firm may require the use of more than one method of analysis. When, however, these methods overlap or duplicate the same information, the simplest tool adequate to the problem is preferable.

The controller wishes to maintain a minimum cash balance of $200,000. Since his collections and the bulk of his payments come about the same time of the month, he foresees no intra-monthly difficulties.

Construct a cash budget for the controller so that he can estimate his financing needs for the coming period.

We have the following significant information about Firm A:

	1956	1957
Current assets	3,500,000	3,250,000
Current liabilities	1,800,000	800,000
Fixed liabilities at 4%	200,000	1,400,000
Net worth (40,000's)	4,000,000	4,300,000
Sales	17,000,000	17,300,000
Profit after taxes	600,000	593,000

Explain what has happened to the financial position of this firm. Very importantly, what changes were made in its "current" capital position? Has the firm's general financial position improved? Use figures. Discuss.

[14]Many large corporations are suppliers of funds to the money market, investing temporarily idle cash in commercial paper, bankers' acceptances, etc.—but above all in short-term US Treasury notes and bills. In the past, short-term corporate investment funds often went into "call loans" to stock brokers. In 1929, an exceptionally large amount of these corporation funds helped support the excesses of the stock market by supplying cash for margin loans to speculators. Member banks were prohibited from placing "call loans" for nonbank lenders in 1933.

Capital and New Issue Markets



The essential advantage of the corporation over other forms of business organization lies in its ability to bring together large amounts of capital. This is especially true of those corporations, which are *publicly held* or *widely held*.[1] The small *family-held*, *closely held*, or *closed* corporation has basically the same financial sources as other business forms, i.e., trade credit, the banks, and the resources of family and friends willing to invest in the business. A widely held corporation, although it will also finance through trade creditors and banks, has the additional ability to tap the flow of investment funds available in the security markets, the market for stocks and bonds. The security market, alternately called the investment, financial, or capital market, comprises the group of institutions through which old corporate securities are traded and new issues of securities are floated. These markets have provided a significant part of the capital that has nourished the growth of corporate enterprise. Thus, to understand the birth of new corporations and the growth of established firms, it is necessary to understand the functions of the security markets.

The capital market is divided by function into two aspects: the secondary or old issues market and the primary or new issues market. The secondary market makes it convenient for an owner of securities—stocks or bonds—to sell them to someone else. The primary market is concerned with floating new issues, with raising funds for existing corporate expansion or the promotion and creation of new publicly held firms. In this chapter, we report the continued extensive debt issuance in the U.S. economy relative to new stock issues. The increased leverage of corporation is great when the economy is booming, but can present major problems with an economic downturn.

[1]English practice distinguishes between "public corporations" (basically our widely held corporations) and "private corporations."

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_7

The Secondary Markets

The secondary markets consist of the organized stock exchanges—the so-called listed exchanges—and the group of security dealers and traders composing the over-the-counter market organized as the National Association of Security Dealers Automated Quotation (i.e., NASDAQ). Stock exchanges have changed since Guerard and Schwartz (2007). In the United States the organized exchanges include the famous New York Stock Exchange, NYSE, the much smaller American Stock Exchange (AMEX, also in New York City), and the various smaller regional exchanges. NYSE American, formerly known as the AMEX, and more recently as NYSE MKT, is an American stock exchange situated in New York City.[2] NYSE Euronext acquired AMEX on October 1, 2008, with AMEX integrated with the Alternext European small-cap exchange and renamed the NYSE Alternext US. In March 2009, NYSE Alternext US was changed to NYSE Amex Equities. On May 10, 2012, NYSE Amex Equities changed its name to NYSE MKT LLC. Following the SEC approval of competing stock exchange IEX in 2016, NYSE MKT rebranded as NYSE American and introduced a 350-μs delay in trading, referred to as a "speed bump," which is also present on the IEX.

In 2019, some 2400 companies were listed on the NYSE with a market capitalization of $22.9 trillion. The NYSE domestic companies met the following criteria for listing:[3]

1. Total number of shareholders of 2200 or 2000 round-lot holders
2. 1,100,000 shares of stock outstanding
3. Market value of public companies of at least $200,000,000
4. An affiliated company standard in which the stockholders (1) requirement is met and market capitalization of entity exceeds $500,000,000
5. Aggregate pre-tax earnings over last 3 fiscal years of $10,000,000 as:

   (a) Most recent fiscal years pre-tax earnings of $2,500,000 and previous 2 fiscal years pre-tax earnings of $2,000,000 (each)
   (b) Aggregate last 3 fiscal years operating cash flow of $25,000,000, if global market value exceeds $500,000,000 and revenues exceed $100,000,000 in last 12 months
   (c) Average global market capitalization exceeds $1,000,000,000 and last fiscal year revenues exceed $100,000,000

The number of companies listed for trading on the regional exchanges is considerably less. Listed companies are considered to have an active following or market. However, the securities of any company which develops a "following" may be traded at some time or other. Thus the over-the-counter market may deal

[2] AMEX was previously a mutual organization, owned by its members. In December 2002, some 796 companies were traded on the American Stock Exchange.

[3] Source: www.nyse.com. Rule 102.01C

sporadically with the securities of more than 70,000 companies. The shares of most major well-known companies are listed on the New York Stock Exchange, but some prominent newer companies are traded on the NASDAQ which has grown enormously in recent years including such notable firms as Microsoft and Apple Computers. The relative out-performance of the NASDAQ during the 1985–2000 period gave rise to its new Tower at Times square and coverage on finance news. The world's largest electronic exchange lists over 3600 companies, and its informal over-the-counter market handles the largest volume of transactions in both government and corporation bonds.

The organized stock exchanges are strictly secondary markets: they do not float or sell any new issues. The over-the-counter market straddles both functions; the firms on it handle the trading of previously issued securities and also take part in the sales effort on new issues. The function of the organized exchanges and the NASDAQ should not be misunderstood. The moneys invested for the purchase of existing stocks do not go to the underlying company. They pass from the individual buyers to the individual sellers. The stock exchange does not set the price of securities. The "market," an abstraction for the freely competing buyers and sellers bidding for and offering various securities, sets the price on each individual security.

A letter published in the *Wall Street Journal* some years ago defines the contribution of the secondary market to the economy:

> A stock exchange is a second-hand market; that is, it enables those who have "old" securities to sell them to others. Such change of ownership obviously has no effect on the amount of capital assembled by the company whose securities are traded. The great economic significance of the stock exchanges in this context is that if such a splendidly functioning second-hand market were not available for disposing of securities once purchased, there would be a greater difficulty than there is in selling new issues and, thus, in accumulating large aggregates of capital.[4]

The secondary market furnishes the desirable quality of liquidity or marketability to investments in the stocks and bonds of traded corporations. The market will accept a somewhat lower rate of return on liquid or marketable investments over those that have no market. The cost of capital is lower for corporations that are established on the secondary markets. Equivalently we can say that the issuing corporation can acquire funds at a lower cost if its securities have access to the secondary market.

The secondary market provides some other useful function in a modern economy. By setting the levels of prices on already issued securities, it indicates the yields, interest rates, and price earnings levels that must be placed on new securities to float them successfully on the primary markets. Thus the secondary market acts as an indicator or measuring gauge for the costs of new capital funds. It also helps to ration capital. Since a relatively higher price is set on the securities of industries whose prospects seem promising and a lower price placed on the securities of industries

[4]Quoted from a letter by Leverett Lyon appearing in the *Wall Street Journal*, March 20, 1955.

whose future seems less bright, the secondary markets help direct the flow of capital funds toward the more favored areas of the economy.[5]

The secondary market facilitates the mobility of funds. It harnesses the resources of investors who have somewhat more aversion to risk in support of the primary market. This occurs because some of the overflow of funds out of the secondary market helps to furnish money for the new issue (primary) market.[6] An idealized schematic explanation of this statement would take three steps:

1. Primary market. The venturesome investor puts his funds into a new enterprise.
2. The over-the-counter market (part of the secondary market). The new company having become fairly well established, our risk-taking investor sells its securities. He takes the funds released by the sale and searches for another new venture.
3. The organized exchange. After some seasoning on the over-the-counter market, the firm's securities are listed on an organized exchange. The securities are gradually taken by relatively more conservative investors. The funds from these sales are placed in the securities of newer and smaller companies traded over the counter. As the securities are passed up the ladder to somewhat more conservative investors, funds filter down to the new issue market. The secondary market enables the providers of true risk capital to withdraw their funds from successful ventures and seek out new enterprises.

The secondary market performs its functions well when it is in a more or less "normal" range. When the market is in a speculative fever or seized in a wave of excessive pessimism, its malfunctioning probably adds an extra disequilibrating factor to the aggregate problem.

The Primary Market

The primary market is the complex of investment bankers, over-the-counter houses, and security dealers who combine to sell new issues to individual investors and institutions.

Although it is smaller in dollar volume than the trading markets, the primary market from an aggregate economic view is more important. On the secondary market, if one investor sells his securities to another, no new funds necessarily enter the economy, no new business assets appear, the underlying company does

[5] In a price or market economy, the process of allocation starts when increased demand raises profits or rates of return in a given industry or segment of the economy, or decreased demand lowers profits. However, the security markets are quick to recognize such portents and mark prices up or down accordingly.

[6] If *net* new funds are invested in the secondary markets, some portions of old portfolios—securities—will have been liquidated, turned into cash. Since it is unlikely that all of these funds will be spent on consumption, a large portion must be reinvested in the primary market, i.e., net new securities.

not grow, nor does the productive capacity of the economy increase. But when a new issue is sold on the primary market, the company making the sale raises new funds from the investors. These funds may be used to pay off old debt, but more likely they will be used to acquire new assets: new inventory, plant, or equipment. So used, these funds reenter the income stream, increasing the gross national product. Moreover, these new assets constitute real capital formation, increasing the productive capacity of the firm and the economy. The primary market is a part of the machinery through which aggregate savings are tapped to create net new investment and economic growth. If the savings of the economy do not reenter the income stream at the proper rate, the rate of economic growth will be retarded.

The investment banking firms and the new issues department of the brokerage firms are an integral part of the primary market. The new issue firms help devise the legal content of the new issue, agree on the interest rate or price that can be set on the securities, and then takeover the sale of the issue on the market. (The investment firm that finds the issue and guides it through the primary market is called the originating house.) When a security firm floats an issue of securities on a firm commitment or underwriting basis, it purchases the whole issue directly from the corporation before it is placed for sale on the market. US firms have moved from using rights issues to using the firm commitment method of selling seasoned public offerings (SPOs) during the 1950–2003 period, and 60% of IPOs are sold using the firm commitment method (Eckbo & Masulis, 1995). The funds to enable the investment bankers to carry the securities (their inventory) until they are sold are generally obtained from the commercial bankers. The price the new issue firms pay to the issuing corporation is somewhat less than the buying price set for the market. This difference, the charge, or "spread" compensates the security firms for their advice, their selling services, and the risks they assume.

There are some eight to ten investment firms who handle most of the originating and underwriting business. Such firms as Goldman Sachs, Merrill Lynch, and Morgan Stanley are among the largest underwriters. Although these firms are rivals, they will cooperate and form groups or syndicates of various sizes to float large issues. Many possible combinations of bankers and dealers and organizational structures can be used in floating an issue of securities; there are, however, three recognizable functions performed for most security issues: originating, underwriting, and selling. (On a small issue, one or all of these functions may be combined.)

The Originating House

The originating house advises the issuing firm on the features, terms, and prices of the security to be floated. It sets the spread it will take for floating the issue. It undertakes to get together the underwriting group. The spread is divided among the underwriters and sellers. In addition to the originating house, a relatively small fee for being the sponsor of the whole issue obtains its pro rata share of the spread, usually a major allotment, for its participation in the underwriting. (The largest share

of the spread usually goes to the selling group since they generally put in the most effort and time.) The originating house also receives its proportion of the spread on the amount of the securities that they hold out to sell themselves. Thus the competition to originate issues rests on the advantages in functioning and obtaining returns on three levels, originating, underwriting, and selling.

The Underwriting Group

The underwriting group, of which the originating house is a part, agrees to purchase the total issue at a fixed price. Each house assumes a part of the issue. The members of the underwriting group are on a firm commitment basis. Thus they may take a loss if the issue moves slowly or if the price at which the market will take the securities drops below the spread; in short, they risk a loss if they are not able to sell all of their quota at the price agreed upon with the issuing corporation.

The Selling Group

The selling group is composed of the underwriters plus other firms invited to help sell the issue. Usually a concerted drive is made to sell the issue (close the books) as quickly as possible since the risk to the underwriters increases as the selling period lengthens. Generally, the selling group is on a "best effort" basis. This means they get a commission on what they sell; they are not, however, forced to buy the unsold part of their quota, but may return it to the underwriters. Of course, a firm in the selling group that fails to place its quota too often is likely to find itself uninvited to participate on later flotations.

Other Aspects of Investment Banking

Best Effort vs. Firm Commitment or Underwriting Basis

Sometimes the securities of a new, lesser-known, or weaker firm are floated on a "best effort" basis. The investment banker has no commitment to purchase the issue at a fixed price. He does his best to sell the issue, and he gets a commission on all he sells. In contrast to the underwriting agreement, the investment bankers take no risk of loss since they may return any unsold securities to the issuing corporation.

Initial Public Offerings (IPOs)

An interesting aspect of investment banking is the market for IPOs, the initial public offering of the shares of new companies. In the 1990s, the flotation of IPOs in the "Hi-Tech" industries was a source of considerable income for the investment banking firms. IPOs have been sold on a firm commitment or underwriting basis rather than floated on a "best effort" basis during the 1980–2003 period. The investment bankers gave no commitment to purchase the issue at an agreed upon fixed price.

Profits to the security firms on a successful IPO could be quite large. The spread between the market offering price and the return to the issuing companies could be as high as 20% (Ibbotson & Jaffe, 1975; Ritter, 1984; Chen & Ritter, 2000). In the boom period for new issues, it has been apparently common practice for the IPO shares to be underpriced. There is a pronounced underpricing differential between IPOs with an offering price below $3.00 and those with an offering price exceeding $3.00. Ibbotson, Sindelar, and Ritter (1994) report that IPOs with offering prices less than $3.00 produced returns of 42.8% for the 1975–1984 period, whereas the corresponding IPOs with offering prices exceeding $3.00 produced an average return of 8.6%. In the trading hours shortly after the issue was placed on the market, the market price might rise 50–100% over the original offer price. It seemed that the managers of the corporation were not opposed to these phenomena. Although the funds going directly to the corporation were somewhat less than they might be, the pro forma wealth in their own holdings of shares rose with the market. In the 3 years following IPOs during the 1975–1984 period, Ritter (1991) found that IPOs substantially underperformed matched firms on a risk-adjusted basis. (A profitable investor should only be a short-term holder in IPOs.) A recent study of more than 1100 IPOs, raising between $20 and $80 million, launched during the 1995–1998 period found that 90% of the issuers paid gross spreads on underwriting discounts of exactly 7% (Chen & Ritter, 2000). The IPO gross spreads in many developed nations were approximately one-half the US spreads, calling into question the competitiveness of the US IPO market. The largest five investment banks underwrote 60% of the deals during the 1999–2000 period, and the largest 15 banks accounted for 95% of the deals (Fang, 2005). Fang found evidence that the larger, more reputable banks had more stringent underwriting standards, particularly for junk bonds, and produced higher net proceeds for their clients, despite charging higher fees.

New corporations start out by selling common shares. A corporation is formed by stockholders, the owners of the firm. Most corporations are relatively small, are held privately, and remain so throughout their economic life. However, an interesting juncture in a firm's life is when a decision is made to "go public." The firm engages in an initial public offering (IPO). It sells shares on the market to outside stockholders. In the promotion of a new venture, the inside shareholders may decide on an IPO to finance the enterprise. They find an investment banker willing to form a syndicate to float shares to the general public. A new public issue requires a published prospectus presenting a detailed study of the proposed new business,

which must obtain the approval of the SEC. The prospectus contains estimates of potential revenues, etc. and projects the necessary initial investment. The prospectus must specifically acknowledge that all details are estimates and that the whole proposal is subject to risk. Of course, a completely new enterprise has no past record of revenues and profits.

Expansion of a Privately Held Firm into a Public Corporation

A new business generally starts out as a proprietorship or a partnership. If it does take the corporate form, the initial investors are usually a group of relatives, associates, or friends. Such a new corporation is classified as closely or family held. If the firm has growth or expansion plans, it may invite the public to participate in the firm by selling shares through a flotation of stock on the market.[7] If the publicly held shares exceed that of the internal group, it is now a publicly held company.

The flotation of new stock may secure equity capital in a larger amount and at a more rapid rate than the retained earnings of the firm or the private resources of the initial owners can or are willing to supply. If capital is required to take advantage of a profitable opportunity to expand, a sale of new common stock not only supplies new equity funds, but by buttressing the equity base, it enables the firm to borrow additional debt on better terms. Alternatively, a firm that has financed past expansion with debt capital might wish to float a stock issue to secure additional risk-bearing equity capital in order to place its financial affairs on a firmer footing.

The owners of a firm may go to the market with a capital stock flotation for reasons other than that raising of additional funds. Having some part of the stock traded on the market establishes a going price. This makes it easier to adjust the ownership value of different individual owners and to settle estates. The original owners may no longer be interested in the direct management of the company or in having so much invested in one company. A public flotation of new stock brings outside funds and participation into the company. By allowing a market to develop for the company's shares, the original holders can dispose of some of their present shares and diversify their holdings. A company with shares on the market can also make more appealing offers in the way of stock option or stock bonus plans to attract managerial talent. In the initial flotation of a closely held company, shares are often sold both for the company and for the account of the original owners. This serves the dual purpose of increasing the corporation's equity base and decreasing the participation of the original shareholders.

[7] If the firm is legally a single proprietorship or partnership, it must, of course, first organize as a corporation before it can sell shares.

Floating the stock of a privately held company that wishes to go public is not as difficult as selling the stock of an entirely new venture. The SEC prospectus can show a record of past earnings and perhaps even dividends on which to base the market price of the shares. Potential investors can compare the firm's past performance and growth possibilities with that of similar companies. The price for the new issue—based on such criteria as price earnings ratios, dividend yields, financial structure, earnings growth per share, and asset structure—can be set competitively with other stocks already on the market.

The Problem of Control

A firm contemplating a market flotation may consider the problem of control. Fear of losing the advantages of controlling the management of the company has prevented some firms from selling stock to the public even though from a purely financial point of view it would be profitable to both the original and the new shareholders. On the other hand if the original ownership group has lost interest in the direct management of the firm, the problem arises of how to obtain and maintain loyal and competent management. In normal times, there is a shortage of good managerial and executive talent, rather than the reverse.

Going public creates marketable shares which can be used to help solve the agency-principal problem where the principals, the owners (stockholders) of the firm, need to set up a reward or compensation structure which will induce the managers (the agents) to operate as fully as possible in the owner's interests. (In simpler times, the problem was resolved by offering the manager the owner's daughter and/or a share in the business.) In the corporate world, part of the agent's compensation can be in the form of grants of shares or stock options. Presumably if the managers hold shares, they will operate to maximize the share value, thus increasing their own wealth and that of the rest of the stockholders. In actuality, the solution has not always worked that well, especially in cases where the holding period for the managerial shares was very short and the agents operated to push up the short run price of the stock at the cost of the long-run health of the company and a poor outcome for the longer-term holders of the stock.

If there is a strong desire to maintain control, the holding group may issue only a minority of the common stock. (The company is then classed as closely or family controlled rather than closed or family held.) Close control, however, often breaks down as with the passage of time as the ties among the original founding group become tenuous. At this point, a strong outside interest may link up with some faction or descendants of the original holders and acquire control.

Promotion of a Subsidiary by Parent Corporations

A firm venturing into a new field that differs from its main line of business may find it strategic to do so by forming a subsidiary. The parent company lends organizational and managerial talent to the new company and helps in the financing, both with credit and by purchasing all or a significant part of the new company's shares. If the parent company owns all of the subsidiary company's common stock, the subsidiary is called 100% owned or wholly owned. Alternatively some of the subsidiary's shares may be floated on the market and the firm organized with some public participation.[8]

Why should a company go through the trouble of promoting a subsidiary rather than simply creating a division of the main firm? The reasons are legal, organizational, and financial. From the legal point of view, the subsidiary may be able to obtain a broader charter or operate in a different jurisdiction or country than the parent company. From the organizational point of view, the subsidiary allows for decentralization of administration. The parent company can endow the management of the subsidiary with some independent responsibility for the economic success of the new venture. Moreover by organizing a separate subsidiary, the firm is able to limit its financial responsibility in the new venture. If all goes well, the parent firm benefits almost as much from the subsidiary's prosperity as if it were a division or branch. In the event of losses or operating difficulties, the parent company can minimize its losses by cutting the subsidiary loose. At worst, the parent company is not likely to lose more than its original investment unless it has endorsed all or part of the subsidiary's debt.

When the parent company organizes a subsidiary with "public participation," it purchases or acquires a large part of the new subsidiary's stock, while the rest of the shares are sold to the public. This limits the investment at risk of the parent company, but still gives the new subsidiary an adequate equity base. Moreover, if the new company starts out with some public ownership of its stock, its shares will develop investor acceptance, and it will be easier to float new stock to finance future expansion plans. A political advantage in having public participation lies in reducing the onus for demonstrating "monopolistic expansion tendencies." If the subsidiary is located in a foreign country, it is often useful to have a group of native stockholders interested in the success of the venture.

The subsidiary as a new legal entity or firm might be considered a new venture. However, it is easier for a subsidiary to carry out a successful public flotation of its stock than it would be for an entirely new company. A firm partly financed by a parent company possesses considerable advantages; probably it has a seasoned management, and its assets are likely to be in working order from the beginning.

[8]None of these arrangements need be perpetual. Companies have sold off all or part of their investment in a subsidiary or "spun off" the shares to their own stockholders. In other instances, a parent company has purchased the public's holding in a subsidiary.

In short, the parent company may bestow considerable operating and credit facilities on its offspring.

Formation of a Joint Subsidiary by Two or More Parent Companies

Sometimes two or more parent companies form a new venture because the new area of operation falls somewhere between the usual operations of both companies. Each company limits its overall financial participation in the new company, but each may contribute technical or economic know-how in the area it knows best.

Of special interest is the formation of joint ventures linking firms in different countries. Using one country's firm's processes or products, the other country's firm may furnish local manufacturing facilities and/or knowledge of local markets. For example, if a foreign firm locates a joint subsidiary in the United States, the American parent may furnish the production facilities or knowledge of distribution channels, while the foreign parent brings a new patent, products, or processes.

Of course, any of these joint subsidiaries may involve public participation. If the subsidiary is successful and grows, sometimes one or both of the parent companies may divest itself of its holdings either by selling its shares in the market or by "spinning them off," i.e., giving its holdings in the subsidiary to its own shareholders. This may be done if there is an antitrust problem or if the combined management becomes too unwieldy. In a sense the parent corporation performs the role of the "classical promoter." Another possibility is an eventual reshuffling of interests between the parent companies so that one buys out the interest of the other and makes the offspring into a wholly owned subsidiary.

The SEC and the Flotation of New Issues

From 1932 to 1934, the Senate Committee on Banking and Currency, whose chief counsel was Ferdinand Pecora, held a series of investigations on the operations of the securities business in the 1920s. These hearings disclosed considerable evidence that many areas of the security market operations fell short of normal notions of honesty and that in the promotion of many stock and bond issues, there had been a considerable disregard of the interests of the investors. No doubt the reaction to the hearings was partially colored by the attitudes of the Depression decade to the prodigal and profligate 1920s; nevertheless, the legislation that resulted was well considered and constructive. More recently, the market plunge of 2002 brought about a demand and rules for increased surveillance.

The provisions of the Securities Act of 1933 as amended require the preregistration of all public issues over a set dollar amount that are not the obligations of the

federal, state, or local governments or their instrumentalities.[9] A registration statement containing exhaustive details on the new issue—reports by accountants and engineers, a description of the company's properties and business, a description of the securities offered for sale, past or projected earnings, liabilities incurred and assets acquired, the names of the promoters and their interest in the firm, the names of the officers, and the names and addresses of the major stockholders—must be filed with the SEC. The registration statement approximates the size of a "small telephone book." The SEC puts in a "stop order," holding up the issue, if any item on the registration statement appears questionable. If the corporation fails to amend the statement, the commission may refuse to grant permission to float the issue. Nevertheless, the commission does not vouch for the accuracy of every item in the registration statement, and the law specifically forbids issues to state or imply that the SEC has approved the issue. Since the registration statement is very lengthy, the issuing corporation draws up a prospectus, a booklet of about 10–20 pages summarizing the most pertinent data in the registration statement. The prospectus, which must be given to all prospective purchasers, is full of information useful to a thoughtful investor. The Securities Act of 1933 prohibited deceit, misrepresentations, and other fraud in the sale of securities, but could not guarantee the information provided was accurate.[10]

Although fines and imprisonment may be enforced against violators of the commission's orders, the heart of the enforcement of the Securities Act lies in the potential individual liability which exists against the (1) corporation, (2) its chief officers, (3) all of its directors, and major stockholders (who may control directors), (4) underwriters and sellers, and (5) all experts whose statements are included, for any omission of material fact or misstatement appearing on either the prospectus or registration statement. Any purchaser of the issue may bring civil suit for damages against any or all of these groups if he suffers a loss and an inaccurate statement appears on the prospectus or registration statement.

The registration statement has been criticized for its length and cost, since very few investors need (or indeed are willing) to study and evaluate any more information than that presented in the prospectus. The statement is put on file, however, and prospective investors may request photostatic copies. The institutional investors (banks, insurance companies, etc.), who constitute the backbone of the bond market, make good use of the details on the registration statement, and perhaps the mutual funds, the managers of pension funds, the brokerage houses, and very large and sophisticated individual investors may find it valuable to study the quantity of data available in the registration statement. Because these groups constitute a sort of leadership among investors, the results of their evaluations can affect the acceptance

[9]Of course, the exemption of issues from federal preregistration, which are under the minimum amount, does not excuse the promoters of such securities (nor, as a matter of fact, the promoters of registered securities) for violations of pertinent state statutes nor from the penalties of the general antifraud provisions of the common law.

[10]www.sec.gov/about/

of the issue by the capital markets. Of course if they behave badly, the result may be a speculative bubble which could upset the normal operations of the market for a considerable time.

Risky, foolish, or ill-conceived companies may still legally sell their securities so long as they reveal the truth in their registration statement and prospectus. (A prospectus, for example, may reveal that an overwhelming proportion of the company's stock is going to the organizers for assets of nebulous value, whereas the outside investors are expected to furnish all the cash and shoulder almost all the risk for a minority share of the prospective gains. Nevertheless, all the facts are revealed, and the issue is perfectly legal.) The Securities Act of 1933 as amended has been called the "Truth in Securities Act."

The reforms brought about by the Securities Act and the power granted to the SEC have entailed some economic costs. The expenses of drawing up the registration statement and prospectus may weigh heavily on the issues of small companies. The fear of the many-sided penalties may preclude the promotion of some risky but justifiable ventures. Nevertheless the greater the reliance investors can repose in the facts offered in the new issue market increases their willingness to risk their capital. Investor confidence is an important part of the dynamic of a relatively free enterprise system. Without it, a large source of risk capital dries up, and the formation of new enterprise increases in productive capacity, and economic growth can slow to a deleterious level.

Secondary Flotations

Investment bankers on occasion underwrite the sale of a large block of securities for an estate, a large individual owner, or an endowed institution. These secondary flotations or secondary offerings differ from a regular primary issue in that the proceeds of the sale do not go to the corporation but to the original holders of the securities. Selling a large block of securities through the regular market channels might depress the price; moreover, if the securities are sold slowly over time, the sellers assume the risk of changes in the market price in the interim. Often, then, the underwriting function, the marketing facilities, and the skills of the investment bankers are used to sell the securities at a determined price at an immediate time.

Issuing Securities Through Rights

Unless the shareholders waive their rights, an established corporation must allow its common stockholder a privileged subscription position on new common stock issues. When a company issues new stock under the conditions of privileged subscription or pre-emptive rights, the present common shareholders are given the

first opportunity to buy the new shares in proportion to their present holdings.[11] If an issue of new stock is made through privileged subscription, the new shares are usually priced under the going market price. The privilege of buying the new shares at this bargain price attaches to each outstanding share. It is represented by a negotiable "right" which can be either exercised or sold before the subscription date to be used by someone else to purchase the new shares. The intrinsic value of a right on the market can be obtained through the use of a simple formula, namely,

$$\text{VR} = \frac{\text{MPOS} - \text{PSP}}{\text{N} + 1}$$

where VR is the value of the right, MPOS is the market price of the existing shares, PSP is the privileged subscription price, and N is the number of old shares required to buy one new share.

For example, suppose a firm with 6,000,000 outstanding shares, floats a new issue of 600,000 shares giving pre-emptive rights to the existing stockholders. The going price of the stock on the market is $36.60, and the subscription price is $30.00:

$$\text{Value of a Right} = \frac{\$36.60 - \$30.00}{10 + 1} = \frac{\$6.60}{11} = \$0.60$$

The intrinsic value of a right is $0.60, and it requires ten of these rights plus $30 to buy a new share. If a shareholder sells a right he will get $0.60 for it, representing a compensation for the dilution of his equity, and the market price of his stock drops to $30.00, since the stock is now quoted *ex-rights*. Conversely, a prospective new shareholder can buy ten rights (10 × $0.60) and use them to buy a new share, which, including the payment for the rights, costs him $36.00 ($0.60 less than the present price); however, the new share is also ex-rights. If the original holder exercises his rights himself, he acquires one new share at $30 for every ten old shares, valued at $36.60, he presently holds. The average value of the 11 shares works out exactly to $36.00. Because the value of a right is considered a return of capital, the stockholder who sells his rights need not pay personal income tax on the proceeds; however, the sale is used to reduce his original acquisition costs so that if he later sells his securities at a profit, his taxable capital gains will be that much larger.

Because the market value of rights moves up or down proportionately more than the value of the underlying shares, generally rights tend to sell at a slight premium. Since they have a more volatile price movement, they give the speculator a "greater kick" for his investment, and there are those who will pay for the privilege of gambling.

Although a company could manage to sell a new issue of stock by itself by pricing the new shares below the market and using rights, most firms prefer to enlist the aid of an investment banker to ensure the success of the issue. In fact, the use of rights

[11] Preemptive rights are not confined to issues of common stock but may be extended to the issue of other types of securities if these are *convertible* to common stock.

costs substantially less (about one-third) than the cost of using an investment banker (Smith, 1977). The investment banker makes a "standby pledge" by which he is obligated to purchase all shares not subscribed to by the rights holders. The period between the issue of the rights and the subscription deadline may be as long as 30 days. It is possible in this time for the market price of the stock to drop below its subscription price; the banker would then suffer a loss under his standby commitment. Ordinarily, however, since the new issue has been floated below market, all rights (except those of some few stockholders who may not be fully informed or wait too long or are negligent) are exercised. Under these circumstances picking up the few neglected rights furnishes an additional source of income to the investment bankers. In addition to the standby agreement, the underwriting banker on a privileged subscription usually administers and records the exercise of the rights and the accounting for the new shares. The costs of rights financing are often approximately 30% of the cost of using an underwriter.

Stock Tenders

Technically, a tender is a pledge of the willingness of a holder to sell his securities in response to an open bid by a corporation or an individual to buy outstanding securities at a designated price. In financial language, however, the whole transaction—bid and response—is likely to be called a tender. Sometimes a firm may bid for bond tenders when it wishes to pick up bonds for retirement. The stock tender, however, is probably more common and certainly more interesting.

Stock tenders may be thought of as a reverse flotation. On the completion of a successful tender request, shares formerly held in the market are concentrated with one holder or retired. A bid for tenders may be used for the following purposes:

1. By a corporation to retire some of its own shares
2. By one group of shareholders to buyout dissident shareholders
3. By a corporation to acquire another corporation
4. By a corporation to extend its holdings in a subsidiary, possibly with the view of acquiring enough shares to enable it to consolidate its income statement for tax purposes
5. By an individual or group of individuals to acquire enough shares to obtain control of a corporation

During the past 50 years, most tender offers have been merger-related.

The group making the offer to buy will designate a transfer agent—usually a large bank—to register the shares tendered for sale according to the terms of the offer. The price at which the stock will be purchased is fixed somewhat above the market price. Once the offer is made, however, the market price of the stock will not fall below the tender price as long as the tender is still open. A date is set for the termination of the tender. Most requests for tenders run about 2 weeks to a month. A tender offer may be open, in which case all stock tendered before the termination date will be

purchased. The terms may, however, set an upper limit; e.g., only the first 100,000 shares submitted will be picked up. When a definite number of shares are required for the purchasing group's purpose, there may be an all or nothing tender. The terms will state, for example, that a minimum of 100,000 shares must be submitted or none of the shares will be bought. An all or nothing tender offer can be combined, of course, with an upper limit. The tender price can be stated in cash, in an exchange of securities, or in a combination of both.

The purposes of a request for tenders could be accomplished by purchasing the desired shares on the open market. By using the tender device, however, the price paid per share can be limited. Perhaps some of these shares could have been bought for less on the open market, but the purchasing activity in the shares might raise their price so that the average cost of the shares becomes greater than they would be if purchased by tender.

One interesting use of a request for the submission of tenders is the retirement of some of its outstanding common shares by the corporation itself. Such a tender is akin to a partial liquidation and might be employed if the corporation has redundant liquid funds which are not contributing much to its earnings. The tender accomplishes a reduction in the firm's ownership capital with a subsequent increase in the earnings of the remaining shares. Perhaps a partial liquidating dividend would be the fairer way to accomplish the same purpose. However, the buyback or tender limits the personal tax to the capital gains rate on the shares.

Costs of Floating an Issue

The major cost of a security is its servicing charge—the amount of interest or the earnings yield the firm must make over time to entice the market to buy the issue. The actual costs of floating the issue—the legal expenses, fees, and the spread of the investment banker—are usually a relatively minor consideration and only rarely affects the type or timing of the issue.

In general, the costs of flotation depend on the expected risks and difficulties of selling the securities. This is affected by the type of underwriting agreement, the class of security, the size of the issue, and the security market seasoning and reputation of the issuing company. Equity flotation costs are considerably greater than debt flotation costs. Straight debt costs less to issue than convertible debt; see Table 1.

1. Issues of greater price stability cost less to float than issues, which are by nature somewhat more volatile. Debt securities (bonds and debentures) are relatively inexpensive to float compared to stocks, and preferred stocks are floated for a cheaper percentage than common shares.
2. Issues of seasoned, well-known companies can be floated for a relatively lower spread than those of new or more speculative companies.

Table 1 Direct costs as a percentage of gross proceeds for equity (IPOs and SEOs) and straight and convertible bonds offered by domestic operating companies, 1990–1994

	Equity						Bonds					
	IPOs			SEOs			Convertible bonds			Straight bonds		
Proceeds (millions of dollars)	GS[a]	E[b]	TDC[c]	GS	E	TDC	GS	E	TDC	GS	E	TDC
$2–9.99	9.05%	7.91%	16.96%	7.72%	5.56%	13.28%	6.07%	2.68%	8.75%	2.07%	2.32%	4.39%
10–19.99	7.24	4.39	11.63	6.23	2.49	8.72	5.48	3.18	8.66	1.36	1.40	2.76
20–39.99	7.01	2.69	9.70	5.60	1.33	6.93	4.16	1.95	6.11	1.54	0.88	2.42
40–59.99	6.96	1.76	8.72	5.05	0.82	5.87	3.26	1.04	4.30	0.72	0.60	1.32
60–79.99	6.74	1.46	8.20	4.57	0.61	5.18	2.64	0.59	3.23	1.76	0.58	2.34
80–99.99	6.47	1.44	7.91	4.25	0.48	4.73	2.43	0.61	3.04	1.55	0.61	2.16
100–199.99	6.03	1.03	7.06	3.85	0.37	4.22	2.34	0.43	2.76	1.77	0.54	2.31
200–499.99	5.67	0.86	6.53	3.26	0.21	3.47	1.99	0.19	2.18	1.79	0.40	2.19
500 and up	5.21	0.51	5.72	3.03	0.12	3.15	2.00	0.09	2.09	1.39	0.25	1.64
Average	7.31%	3.69%	11.00%	5.44%	1.67%	7.11%	2.92%	0.87%	3.79%	1.62%	0.62%	2.24

Source: Reprinted with permission from the *Journal of Financial Research*, Vol. 19, No. 1 (Spring 1996), pp. 59–74, "The Costs of Raising Capital," by Inmoo Lee, Scott Lochhead, Jay Ritter, and Quanshui Zhao

[a]GS—gross spreads as a percentage of total proceeds, including management fee, underwriting fee, and selling concession

[b]E—other direct expenses as a percentage of total proceeds, including management fee, underwriting fee, and selling concession

[c]TDC—total direct costs as a percentage of total proceeds (total direct costs are the sum of gross spreads and other direct expenses)

3. An issue of common shares through the use of rights and a standby arrangement with the investment bankers will cost less than an underwritten flotation price with rights waived.
4. Since the bankers incur certain costs, such as advertising, which do not vary proportionately with the size of the issue, larger issues are generally handled at lower percentage underwriting and selling fees than smaller issues of the same quality.
5. The split between the selling fees, underwriting fees, and the originating house or managing house charges depends on the expected risks and difficulties in selling the issue. Although the underwriters take the risks, the sellers may have to incur more expenses. Their share of the spread must be large enough to encourage them to sell the issue.

Regulation of the Capital Markets

Under the New Deal administration of the 1930s, Congress passed a number of laws designed to eliminate questionable practices and to enforce a higher level of business morality in the investment markets. Over time as conditions developed additional acts and regulations have been added to the mix.

Securities Act of 1933

As previously discussed, the Securities Act of 1933 mandates new issues to file full information, registration statements, and prospectuses, with the SEC, as was stated earlier in this chapter. The SEC is composed of five commissioners, appointed by the President, confirmed by the Senate, and whose term is 5 years. No more than three commissioners can belong to the same political party, reducing potential partisanship. The President designates one of the Commissioners as Chairman, the top executive in the SEC. President Franklin D. Roosevelt appointed Mr. Joseph P. Kennedy as the first SEC Chairman. The current chairman is Mr. William Donaldson, and Cynthia Glassman, Harvey Goldschmid, Paul Atkins, and Roel Campos. The Commissioners interpret Federal Securities laws, amend existing rules, propose new rules in response to changing market conditions, and enforce rules and laws. The SEC is composed of four divisions: (1) Corporate Finance, which reviews registration statement's for newly offered securities as well as 10-K (annual) and 10-Q (quarterly) filings, annual reports and proxies, and merger and acquisition filings; (2) Market Regulation, which establishes and maintains standard for fair, orderly, and efficient markets, including monitoring the financial integrity program for broker-dealers, reviewing self-regulatory organizations (SROs), and surveilling markets; (3) Investment Management, overseeing and regulating the

$15 trillion investment management industry, including mutual funds and investment advisors; and (4) Enforcement, which investigates possible violations of security laws. Enforcement obtains evidence of possible violations. The most common SEC investigations include insider trading, misrepresentation of information about securities, manipulating market prices of securities, sale of securities without proper registration, and stealing customer's funds or securities. The Act had the objective of enforcing full and truthful disclosure of all pertinent facts to prospective investors on the primary market.

The Securities Exchange Act of 1934

The Securities Act of 1933 was followed by the Securities Exchange Act of 1934, which established the Securities Exchange Commission, the SEC. The SEC was empowered to register, regulate, and oversee brokerage firms, transfer agents, clearing agencies, as well as the self-regulatory organizations (SROs), which include the NYSE and AMEX. The Securities Exchange Act of 1934 required annual (and other periodic) reports of firms with assets exceeding $10 MM, and more than 500 owners. The Securities Exchange Act governs proxy solicitations, tender offers involving direct purchase, or a tender offer of more than 5% of a company's securities. Fraudulent insider trading was prohibited by the Securities Exchange Act of 1934. The existence of the SEC has not prevented all losses on questionable securities. Unscrupulous promoters and security salesmen may still take advantage of investors simply by risking the fines, prison terms, and civil suits which false statements may entail. The SEC does not guarantee the profitability of any venture. An investor making a bad choice will lose money on his purchase of securities.

Trading on the exchange has been regulated in the following respects:

1. The Federal Reserve Board is given authority to set initial margin requirements (the amount required of the purchaser in cash as a percentage of the market price of stock) and to regulate the extension of credit on the market.[12] (Low margin requirements create a weak market. If the stocks start to drop, the brokers may strengthen the downtrend by selling the margined shares in order to protect their loans.)
2. Brokers' indebtedness to all persons may be regulated and such indebtedness may not exceed 20 times the net capital of the firm.
3. Brokers may not lend customers' securities without their consent. The lending of securities is used to effectuate a short sale. A short sale is made by someone who expects the price to go down. He borrows the stock and sells it at the going market price. When (or if) the price falls, he repurchases the stock (covers his sale) and

[12]In the feverish era of the 1920s, initial margin went as low as 10%.

returns it to the lender. The difference between the repurchase price and the price at which the shares were originally sold is the short sellers' profit.
4. The manipulation of stock prices by wash sales, excessive activity, and misleading statements is made unlawful.
5. Options (puts, calls, etc.) and short selling are regulated by the commission. Short sales can only be made on the "uptick"—the stock must have moved up a fraction of a trading point since its last regular sale. (This rule handicaps a "bear raid," where a speculative group hammers down the price of a stock by continuously offering more shares as it moves downward.) We discuss option pricing and strategies in chapter "Options."
6. The commission was authorized to supervise the activities of the "specialists," members of the exchange who act as a broker on certain transactions and in other instances "create a market" (act as dealers) on company stocks assigned to them by the exchange. The specialist is a critical part of the stock exchange's machinery; it is doubtful if a manipulative "pool" could be successful without enlisting the sub rosa aid of the specialist.

Banking Act (Glass-Steagall Act) of 1933

One of the clauses of the Banking Act of 1933 was a requirement that deposit bankers divest themselves of investment banking affiliates. (Commercial banks were prohibited from floating or dealing in any securities except federal and state and local government obligations.) Many of the investment banking affiliates were organized as subsidiaries by the big city commercial banks in order to participate in the profitable security issue business of the 1920s. The history of these investment banking affiliates did not in the end reflect very favorably on the acumen a financial judgment of their organizers and management.[13]

The security affiliates of commercial banking firms represented a potential conflict of interest; they violated the general legal concept of the "arm's-length transaction" which is enforced wherever fiduciary (or quasi-fiduciary) relationships are involved. Possible conflicts of interests between the commercial bank, its depositors, and its investment banking subsidiary could arise in the following instances:

1. If a depositor asked his banker for advice on investments, the banker might be tempted to recommend an issue being floated by the security affiliate. Indeed, the list of depositors of the commercial bank often served as a prime source of prospects for the affiliate's sales.
2. When the commercial bank bought securities for its portfolio, it might choose an inferior issue floated by the affiliate. (And the high quality of the bank's investment is one aspect of the safety of its deposits.)

[13]See F.A. Bradford, *Money and Banking*, Longmans, Green, 1949, p. 800.

3. If the bank felt that one of its borrowers was stretching out his repayment, they might have their affiliate float an issue for him, the proceeds to repay the loan. Here the commercial bank would get rid of a potentially embarrassing asset by shifting it to the outside investors. This is a subtle violation of the implicit investment banking code—that the investment banker lends his name to the general soundness of the issue considering its description and class.

Glass-Steagall Act Amended

In 1999 the Gram-Leach-Bliley Act (GLB) severely amended the original Glass-Steagall Act. Large financial conglomerates, deregulated as financial holding companies (FHL), are allowed to have an investment banking subsidiaries as long as a "fire wall" exists between the activities of investment banking (new issues) and the regular banking or other activities of the conglomerate. The result has been a merger of all types of banks, stock brokerage firms, and insurance companies offering a department store of financial services. Each sector is supposed to be overseen by its own traditional functional regulator.

Retail Brokerage Houses

Of course the problem of conflicts of interest between different activities in the financial industry is endemic. A recent area of concern has developed in the brokerage industry where the brokerage firms also had large investment banking operations. Here a conflict of interest arises when the retail broker advises his clients to purchase a company's new issues unwritten by his firm.

Public Utility Act of 1935

The main purpose of this Act is breaking up and regulating holding companies in the gas and electric utility industries. In addition, however, the Public Utility Act of 1935 gave the Securities and Exchange Commission considerable powers over the financial affairs of the holding companies and operating subsidiaries.[14] The SEC must approve the terms and types of new utility issues, and it sets broad bounds on permissible financial structures for the companies.

[14]This authority over financial affairs should not be confused with the rate-regulating function given to the various state Public Utility Commissions and to the Federal Power Commission for interstate business.

The SEC, in its authority under the 1935 Act, encouraged the use of competitive bidding in floating utility company securities. In 1941 it imposed a competitive bidding requirement on all public utility issues. (The ICC has required competitive bidding on the issue of railroad equipment trust certificates since 1926, and it imposed this requirement on all railroad securities in 1944. Municipal issues are almost always floated under competitive bids.)

Most corporation issues are still floated on a negotiated basis. The originating house advises the company on the features and terms of the issue and negotiates the spread and fees. Under competitive bidding, the issuing firm prepares all the terms of the issue except the coupon rate (stated interest rate) on bonds and the price. The investment bankers then submit bids citing the stated interest rate they will put on the issue, the price to the issuing firm, and the price to the market. The investment banker whose bid shows the lowest effective interest lost to the issuer (function of the rate and the price) wins the privilege of floating the issue and managing the underwriting syndicate. On a competitive bid issue, the firm performing the function corresponding to the originating house is called the managing firm.

Competitive bidding was pushed by some Midwestern investment bankers who felt that eastern bankers had established relationships with the corporations which gave them an unfair share of the new issue market, but their complaints found sympathetic ears at the government commissions. It was argued that the connections developed by the negotiated type of offering lead to investment banker domination of some firms. (The corporation became more interested in deals which furnished securities to be marketed than in company operations.) The most pertinent argument was that competitive bidding would lower flotation costs.

The old-line investment bankers originally opposed competitive bidding. They argued that it deprived the issuing firm of the private advice, market knowledge, and competent guidance of the investment banker in selecting the most advantageous type of security and setting up the best possible terms and provisions before the issue of the security. There may be some force to this argument when applied to industrial issues. However, the features of the issues commonly subject to competitive bidding—utilities, municipals, and equipment trust certificates—are today actually quite uniform and standardized. There may not be any pressing need for the special expertise of the investment banker.

The Maloney Amendment, 1938

This amendment to the Securities Exchange Act gave the SEC some control over the secondary market activities of the over-the-counter dealers. (The primary market activities of these dealers fall under the general purview of the Securities Act of 1933.) The Maloney Amendment provides a system of self-regulation for the over-the-counter traders promulgated and administered by the National Association of Security Dealers. The Securities Exchange Commission passes on the rules, supervises their administration, and reviews the disciplinary decisions of the Association.

The Investment Company Act of 1940[15]

This Act, largely unchanged to this day, provides for the federal registration of both the closed-end and open-end investment companies. (Open-end companies—mutual funds—stand ready to redeem their own shares at the pro rata underlying portfolio value. An investor in a closed-end company must sell his shares on the market.) Investment companies are financial intermediaries; they place the funds given to them by their own investors into securities of operating companies. People invest with them because they may provide better diversification and better portfolio management than the investor could achieve himself.

Although investment companies had a long history of successful operations dating from Scotland in the 1870s, they fell into disrepute in the United States in the 1930s. The Investment Company Act provided that companies who announced their investment policy clearly and adhered to it could register with the commission. Registration gave the company status in the eyes of the investor and removed some of the onus of the 1929 debacle. Among other things, a registered company must practice proper diversification of its portfolio, limit the issue of its prior claim securities and the use of margin trading, and renounce any holding company ambitions. If a registered company pays out 90% of its income and capital gains to its own shareholders, it need pay no federal income tax. The effects of the Investment Company Act must be considered salutary. The investment companies, especially the mutual funds, have experienced vigorous growth over the last 60 years and brought a healthy flow of savings into the capital markets.

Sarbanes-Oxley Act of 2002

President Bush signed the Sarbanes-Oxley Act of 2002, on July 30, 2002, which enhanced corporate responsibility and financial disclosure. The Act created a "Public Company Accounting Oversight Board" (PCAOB), to oversee activities the auditing process.[16] The SEC adopted requiring mutual fund CEOs and CFOs verification of stockholder reports, disclosure of codes of ethics, audit independence, and listing standards for audit committees. The audit independence ruing prohibited auditors from providing certain non-audit services related to conflict of interests standards. Exchanges and national security associations were prohibited from listing securities of any security not in compliance with certain audit committee requirements. Disclosure of the firm's adoption of a code of ethics was required, as was the

[15]This act follows many of the provisions suggested by Henry Simon in his pamphlet "A Positive Program for Laissez Faire," University of Chicago, 1935, reprinted in *Economic Policy for a Free Society*, University of Chicago Press, 1948.

[16]www.sec.gov/about

presence of one "financial expert" serving on the audit committee.[17] Investment companies were encouraged by rule amendments to disclosure more balanced information to potential investors with respect to past performance. In December 204, the SEC established an advisory committee to examine the impact of the Sarbanes-Oxley Act on smaller public companies. The advisory committee, known as the Securities and Exchange Commission Advisory Committee on Smaller Public Companies, will concentrate on (1) the frameworks for internal control over financial reporting for smaller companies;(2) corporate disclosure and reporting requirements for smaller public firms, including differential requirements based on market capitalizations; and (3) accounting standards applicable to smaller firms.[18] There could well be a cost to the public by the Sarbanes-Oxley Act. In 2002, some 67 firms delisted their common stock, whereas in 2003, 198 firms delisted common stock, or "went dark." The costs of compliance may be too high for many firms.[19]

The Capital Market as a Source of Funds

Financial institutions—the financial intermediaries—far exceed individual suppliers of funds. However the institutions are not an autonomous source of funds; they merely reinvest the moneys of their individual policy holders and savers. The amount of funds shown supplied by individuals on their own account appears to be relatively small. This is a strategic item, however, because although the data is not available, it is generally presumed that this is the largest source of financing for new enterprise.

In Table 2 we have the breakdown of corporation use of the capital market according to type of issues. The low interest rates that have prevailed since the 1930s and the income tax deductibility of interest seem to have encouraged debt issues over stock. A larger proportion of debt is retired every year than common stock and that on a net basis the issuance of debt has been five times the issuance of equity since 1971. Moreover, the use of equity is understated because the largest part of equity finance arises from the retention of earnings rather than new issues. In general, Table 2 shows a sharp increase in corporate use of the capital markets since 1975. This has been a true "growth industry."

In the late 1930s and the early 1940s, financial writers were predicting the demise of the capital markets as a source of long-term financing. Firms were doing most of their long-term financing internally through depreciation and retained earnings and

[17]SEC Annual Report, 2003, pp. 44–52.

[18]www.sec.gov/news/press/2004-174

[19]C. Leuz, "Why Firms go Dark: Causes and Economic Consequences of Voluntary SEC Deregulations," working paper, The Wharton School, University of Pennsylvania. See also R.E. Verrecchia) and C. Leuz "The Economic Consequences of Increased Disclosure." *Journal of Accounting Research* (2000).

Table 2 Gross proceeds of new securities of all corporations offered for sale for selected years 1935–2017 (in millions of dollars)

Calendar year	Gross proceeds from issues of bonds, debentures, and notes	Gross proceeds from issues of common stock
1935	$2224	$22
1940	2386	108
1945	4855	397
1950	4919	811
1955	7420	2185
1960	8081	1664
1971	37,809	10,146
1975	97,088	18,819
1980	179,633	44,254
1985	357,436	83,347
1990	696,799	53,944
1995	1,197,158	97,636
2000	2,748,933	392,684
2003	2,645,478	172,294
2010	6,038,081	541,186
2015	7,911,335	511,987
2017	8,899,657	454,323

Source: Securities and Exchange Commission, 27th Annual Report, Compustat Database

were divorcing themselves from any reliance on external financing, i.e., the capital markets. Even as they wrote, however, there were indications of a recovery in the capital markets.

Table 3 depicts the relative importance of internal financing and external financing sources. Internal financing looms very large in financing firm growth. The capital markets allow firms to raise capital to support capital expenditures and generate future profits. Nevertheless firms generate most of their sources of funds from retained earnings, net income less dividends. The capital markets, as we will see in chapter "The Equity of the Corporation: Common and Preferred Stock," evaluate the effectiveness of the firm's retained earnings policy by establishing a price-earnings multiple for the security. A higher price-earnings multiple often shows confidence in management, among other things. Nevertheless, although depreciation is counted as one of the sources of funds, it is part of the gross savings of the economy, not net. Over time, depreciation tends to approximate capital consumption. Fixed assets must be replaced eventually. Corporations earned net income of $37,005.19 million during the 1950–2017 period. The corporations paid dividends of $27,505.91 million. Amortization and depreciation charges totaled $37,681.25 million during the corresponding period. Cash flow, net income plus depreciation, totaled approximately $74.8 billion. Corporations raised $90,765.43 million of new debt issues and $10,617 million of new equity issues; however, if one calculates net debt and equity issues, defined as new debt issues less debt repurchased, and new equity issues less equity repurchased, one finds that nets debt issues totaled $26,466.74 million during the 1950–2017 period, whereas net equity issues were $−6341.6 million. Thus, new debt issues exceeded new equity issues by an 8.5:1 ratio. Moreover, the internal generated cash flow totaled $52.30 billion, whereas net debt and equity issues totaled

Table 3 Financing of US corporations for selected years, 1971–2017 ($B)

Sources	**1971**	**1975**	**1980**	**1985**
Internal	76.2	127.1	260.9	363.1
Retained profits	28.6	51.1	112.7	109.3
Depreciation	47.6	76.0	148.2	253.8
External	29.3	64.3	130.1	165.4
Net debt	17.2	47.1	94.5	137.8
Net equity	8.2	17.2	35.6	27.6
Sources	**1990**	**1995**	**2000**	**2003**
Internal	504.9	916.1	1557.0	1477.4
Retained profits	120.4	312.8	526.7	601.7
Depreciation	384.5	603.3	1030.3	875.7
External	146.1	349.0	1203.6	497.9
Net debt	148.8	324.9	1055.6	525.1
Net equity	−2.7	24.1	148.0	−27.2
Sources	**2005**	**2007**	**2010**	**2017**
Internal	2038.1	2394.5	2648.2	2933.8
Retained profits	929.1	1084.5	1209.7	1135.2
Depreciation	1108.0	1310.0	1439.5	1798.6
External	979.0	2287.6	−1125.0	622.3
Net debt	739.4	1817.5	−1105.9	254.0
Net equity	240.6	470.1	−20.1	368.3

Source: Standard & Poor's, Compustat
Net debt = Debt issued − debt repurchased
Net equity = equity issued − equity repurchased

Table 4 Industry statistics: public offerings and mutual fund assets

	1980	1985	1990	1995	1998
Public offerings: IPOs registered with SEC	1.3	90.0	50.0	122.0	257.0
Public offerings (excluding rivate placements)	58.0	133.0	309.0	705.0	1819.0
Managed funds: assets under management by investment companies	235.0	591.0	1350.0	2879.0	5180.0
Mutual funds	135.0	495.0	1065.0	2812.0	5525.0

Source: SEC, 1999 Annual Performance Report

$14.56 billion. Internal financing is the greatest source of corporate financing. Internal funds, defined as retained profits and depreciation, exceed external funds, defined as net debt issues plus net equity issues, during much of the 1950–2017 period. Depreciation has become the largest source of internal funds; see Table 3. Depreciation has exceeded retained profits during the 1971–2018 period. Net debt issues dominate net equity issues. External financing rose relative to internal financing during the late 1990s and early 2000s, but still fell short relative to internal financing because of depreciation. The capital market, especially in recent years, is a significant supplier of savings for net capital formation. The relative insignificance of IPOs is shown in Table 4, where IPOs, public offerings, and mutual fund assets

Table 5 Post-IPO performances in 2019

Post-IPO stock performance for tech companies				
Company	IPO date	Offer price	Price as of Dec 13, 2019	Price return
Lyft, Inc.	Mar 28, 2019	$72.00	$46.75	(35.1%)
Peloton Interactive, Inc.	Sep 25, 2019	29.00	31.53	8.7%
Pinterest, Inc.	Apr 17, 2019	19.00	17.45	(8.2%)
Slack Technologies, Inc.	Jun 19, 2019	26.00	21.39	(17.7%)
Uber Technologies, Inc.	May 09, 2019	45.00	28.36	(36.7%)

are shown. Mutual fund assets dwarf IPOs and are 2–3 times the amount of public offerings during the 1980–1998 period (Table 5).

The Debate on the Optimal Organization of the Capital Market

In the European continental capital markets and in Japan, the various financial functions are quite integrated. Thus it is quite common for the commercial banks to sell insurance to their clients and perform the brokerage function, buying and selling shares and bonds for their customers. Until recently, in the United States and Great Britain, there has been a tendency to keep various financial institutions independent so that, for example, the transactions between a bank and an insurance company would be at arm's length. The growth of holding company financial companies owning insurance companies and brokerage company subsidiaries and brokerage companies offering a variety of banking and insurance options have begun to blur the boundaries.

Putting activities under one roof may increase "transaction efficiency." It can lower transaction costs. However, as shown in the recent case of Merrill Lynch's financial advisors lying in bed with its new issue branch and promoting risky new issues to their customers, a source of abuse can arise when the presumed neutral financial advisor and the seller wear the same uniform. The advantages of transaction efficiency can soon evaporate if the external investor begins to doubt that he is getting the benefit of an "arm's-length transaction" and operating in an unbiased market. A loss of investor confidence raises the risk premium on all financial sources and taxes the economy across the board.

Capital Markets and Long-Term Economic Growth

The capital market is an institution of tremendous importance. It may be enlightening to sketch its influence on three levels: how it affects investor, corporation management, and the economy as a whole. From the viewpoint of the investor, the

prices shown by the securities market is the considered judgment of his peers on the success and potential success of the company in which he has put his funds. The attitude he will have toward the management, his willingness to put more money in the company or similar companies will be affected by the verdict of the investment market.

The securities markets play a significant role in the managerial and financial decisions of the large publicly held corporations. The security markets affect the stockholder's attitude toward the corporation and its management. The verdict of the securities market reflects what sophisticated investors think of the success and potential of the company. From the operational viewpoint of the corporation's financial management, the security or capital funds markets are a source of long-term capital. Through the machinery of the capital markets and the investment bankers, the corporation can obtain funds from long-term debt issues (bonds or debentures) or by increasing its equity (floating more stock). The financial management of the corporation must understand these markets and the relations between the primary and secondary markets if it is to make careful, rational decisions on problems of long-term financing.

From the view of the economy as a whole, the activities of the investment markets play a major role in capital formation and economic growth. The tradition of investing savings in securities is quite strong in economically developed countries; in other parts of the world, savings are too often placed in precious metals, jewelry, or land rather than invested in productive enterprises.

The investment markets are in nature international as well as national. The securities of many large foreign corporations are listed on the American exchanges, and efficient machinery exists for buying foreign securities from foreign exchanges or co-listings. The vast apparatus of the financial market serves an essential service in a free enterprise economy of allocating a large portion of new capital formation among various firms, countries, organizations, and governments.

References

Chen, H. C., & Ritter, J. R. (2000). The seven percent solution. *Journal of Finance*, *55*, 1105–1131.

Eckbo, B. E., & Masulis, R. W. (1995). Seasoned equity offerings: A survey. In R. A. Jarrow, V. Maksimovic, & W. T. Ziemba (Eds.), *Handbooks in operations research and management: Finance* (Vol. 9, pp. 1017–1072). Amsterdam: North Holland.

Fang, L. H. (2005). Investment bank reputation and price and quality of underwriting services. *Journal of Finance*, *60*, 2729–2762.

Guerard, J., & Schwartz, E. (2007). *Quantitative corporation finance*. New York: Kluwer Academic Publishers.

Ibbotson, R., Sindelar, J., & Ritter, J. (1994). The market's problem with the pricing of initial public offerings. *Journal of Applied Corporate Finance, 6*, 66–74.

Ibbotson, R. G., & Jaffe, J. F. (1975). 'Hot issue' markets. *Journal of Finance, 30*, 1027–1042.
Ritter, J. R. (1984). The 'hot issue' market of 1980. *Journal of Business, 57*, 215–240.
Ritter, J. R. (1991). The long-run performance of initial public offerings. *Journal of Finance, 46*, 3–27.
Smith, C. (1977). Alternative methods for raising capital: Rights versus underwritten offerings. *Journal of Financial Economics, 5*, 273–307.

The Equity of the Corporation: Common and Preferred Stock



This chapter deals mainly with the financial function of stockholding, i.e., the supplying of risk capital and the expected rewards thereof. The shareholders' expected return is the supply cost of equity capital. Only if the firm is able to give its shareholders at the minimum the "normal" rate of return on risk capital can the company be considered an economic success. Thus, a large part of the discussion is centered on the behavior of the investment markets. This follows from the assumption that the major objective of financial management is to maximize the long-run value of the common stock. If management is to develop financial strategies aimed at maximizing the long-run value of the common stock, it must understand the rationale of the investment markets. It is this market that measures relative risk and provides approximations of the rates of return on different classes of risk capital. The purpose of this chapter is to discuss why investors own common stock. It is well known that stock returns have been high relative to bond returns during the 1926–2017 time period. We discuss common stock valuation models in this chapter.

Common Stock

Common Stock as Risk Capital The basic function of the shareholder is to furnish risk capital. We briefly discussed capitalism and the individual in chapter "Introduction: Capital Formation, Risk, and the Corporation." The individual in this chapter is a stockholder, who is compensated for bearing risk. All the creditors and investors in the firm take some risk, but because the claims of the common shareholders are subordinated to those of all the others, the common stock absorbs the initial brunt of any downturn.

The owner of the common shares of the corporation receives no definite rate of return. The corporation owes no contractual obligation to pay any fixed amount to its owners. The return to the shareholders depends on how much is left after bank debts, bonds, and other obligations which have been acquired at a fixed rate have been

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_8

serviced. As long as the earnings of a business are subject to external variation, at least one class of capital suppliers must be content with a variable return. A firm could not promise fixed returns to the short-term creditors and bond owners if there were no equity holders willing to take whatever earnings are left or to absorb losses. The equity holders bear the initial risks of operation; the debt holders' return is not usually endangered until the earnings of the shareholders shrink or turn into losses.

The investments of the creditors are safe until the equity can no longer absorb the loss of asset value. The equity of a corporation is a shield protecting the debt holders. Other things being equal, the proportion of ownership capital to total assets affects both the safety margin of earnings before the fixed interest charges and the asset safety margin of the debt holders.

Shareholders bear the initial economic risks of the firm, but they are rewarded with the residual earnings left after all operating expenses, financial charges, and taxes are deducted; in a successful firm, the residual earnings exceed the fixed returns promised to the creditors. The reader is referred to the income statements of IBM, Dominion Energy, and Boeing discussed in chapter "The Annual Operating Statements: The Income Statement and Cash Flow Statement." The net income of a corporation is most often expressed as a percent of sales, total assets, or stockholder equity, as the reader saw in chapters "The Annual Operating Statements: The Income Statement and Cash Flow Statement" and "Financing Current Operations and Efficiency Ratio Analysis." The earnings of the common stockholders are calculated only after the expenses, interest charges, and profit taxes are deducted. Though the shareholders' reward comes last, they have one great advantage; their return is not contractually fixed or limited. As legal owners of the company entitled to all the earnings left after the creditors and the preferred shareholders receive their fixed return, they receive the rewards of successful operation and company growth. If sales increase, if profitable new products are introduced, or cost-saving techniques are successfully applied, the increased returns redound mainly to the benefit of the stockholders. Those who take the larger end of the risk do not necessarily lose. When both dividends and appreciation of capital are taken into account, a "long-run" investment in a diversified portfolio of common stocks has generally proven more profitable than investment in any other type of security.[1] The Center for Research in Security Prices, CRSP, database was developed in 1960 to measure the returns of common stocks relative to corporate and government bonds. The empirical evidence of CRSP stocks, presented in Fisher and Lorie (1964), Fisher and Lorie (1970), and Lorie, Dodd, and Kimpton (1985) reported great stock returns relative to bonds. The annual Ibbottson and Sinquefield books continue to report higher stock returns.

Of course, the superior performance of common shares in the past is no guarantee of gains in the future. The long-run nature of the test periods (12–20 years or even 54 years as in the WRDS database) is no comfort to the short-term speculator or to those who wish or need an assured amount of money at a given time, and the

[1] See Chelcie C. Bosland, *The Common Stock Theory of Investment: Its Development and Significance*, Ronald Press, 1937.

selection of an improperly diversified or over-priced portfolio can surely lead to losses. Nevertheless, the data indicates that in general those who have risked funds in the ownership of corporation shares have not gone unrewarded.

Rewards to Common Shareholders

The returns to the outside shareholder of a corporation fall into three categories: dividends, earnings, and capital appreciation on the market price of the stock. These categories are not independent; the emergence of capital appreciation depends heavily on the future course of earnings. The two explicit measures of return are the dividend yield, current dividends divided by the market price, and the P/E ratio, current earnings divided into the market price (the reciprocal is the earnings rate).[2] Implicit in the valuation of shares is g, the expected growth rate of earnings.

Buybacks The two cash returns that shareholders may expect to receive from their holdings are buybacks and dividends. Buybacks are a program where the corporation periodically uses a proportion of its cash flow earnings to buy back its shares from the open market. A buyback reduces the amount the firm reinvests in the business and is a full or partial substitute for dividends. Currently, the cash amount of buybacks exceeds the amount of dividends. Under past tax policy where dividends were fully taxable at the personal level but capital gains if any were capped at 20%, buybacks had a tax advantage. (Hopefully, an eventual sensible tax reform would allow a full or partial deduction of dividends at the corporate level. This would do much to restore dividends as the main cash payout program of the corporations.)

Buybacks and dividends seem at first glance to be different creatures. However, at the internal corporate level, the two have the same effect; they reduce the amount of cash available within the corporation and the proportion of equity in the financial structure. For the shareholder, receiving a dividend gives him cash, and all else remaining equal, the value of his shareholdings remains the same. In case of a buyback, the "universal shareholder" receives cash for a portion of his shares and now holds fewer shares. However, the divisor (the number of shares) against total earnings has been reduced, earnings per share increase, and thus the price of each of the remaining shares should rise. Fewer shares times a higher price per share, all else being equal, the value of the portfolio should stay the same, just as in the case of dividend payouts.

Equity repurchases (buybacks) are generally motivated by several factors. First, equity repurchases, as contrast with dividends, serve to increase the earnings per share and thus lead to a possible increase in the stock price. Equity buybacks

[2]Price/earnings ratios are given as multipliers. They are the number of times the going market price of the stock exceeds annual earnings per share. The price/earnings ratio equals $\frac{\text{market price}}{\text{annual earnings}}$. Current dividend yield, on the other hand, is obtained by dividing annual dividends by the market price. It is therefore stated as a percentage. No special reason can be adduced as to why these two relationships are customarily presented in these different manners. It does, however, help to prevent unnecessary confusion on the security markets.

increase the financial leverage of the firm and are often used in mergers and acquisitions and stock option plans. Buybacks may be viewed as flexible dividends by investors who do not want cash from the sale (Bierman, 2001a, b). Equity repurchases may be a good use of corporate cash and can be viewed as an investment. Buybacks enhance the value of the book value of the remaining shares, particularly if the shares are repurchased at a price less than the current book value. Bierman (2001a, b) holds that the primary advantages to share buybacks are:

1. The tax shield that occurs when the firm repurchases stock. A portion of the transaction is regarded as a return to stockholders capital and not taxed, generating increased after-tax cash flow, whereas the increase in stock prices generates capital gains, taxed at 20%, as opposed to dividends taxed as ordinary income, at a rate of 35%.
2. The lower transaction costs for investors who do not sell and increase their "investment" in the firm.
3. A higher stock price than paying an equivalent dividend.
4. Hiding dilution effects of issuing stock options.
5. Supporting the stock price.
6. Provides a signal that management believes the price is too low.
7. Often increases earnings per share.

Of course, one important difference remains. Dividends go willy nilly to all the shareholders; during a buyback, some shareholders may sell varying proportions of their holdings, and others may hold on to all their shares.

Dividends If a firm is making some earnings, has cash resources, and is reasonably well established, the management often endeavors to set up some sort of dividend policy. Generally, dividends are paid out of current earnings and constitute an interim sharing of the profits of the enterprise. The proportion of earnings distributed as dividends is the payout rate.

The board of directors is not legally obligated to declare dividends. Failing to pay dividends on stock is not an act of bankruptcy, as is the failure or inability to pay the interest on debt. This flexibility of return—the ability to forego, postpone, or cut dividends—makes capital stock the safety element in a firm's financial structure.

For a purely dividend-paying company, the expected flow of dividends sets the basic value of a company's shares. If a stock has a current dividend of $2 and a forecasted growth rate of 3.0% per annum and the market desires an overall rate of return of 9%, the stock would have a market price of about $33.33 $\left(\frac{\$2.00}{(.09-.03)} = \$33.33\right)$ and a current yield of 6.0%.

Various aspects of risk raise the yield rate on common shares and reduce the relative market price. On the other hand, the prospect of a growing stream of earnings and dividends raises the market price and reduces the present yield. The expectation of a growth in earnings and dividends will lower current dividend yields below the interest yield on sound bonds.

Thus the expression

$$\$1 + \frac{1}{(1.08)} + \frac{1}{(1.08)^2} + \frac{1}{(1.08)^3} \quad \cdots \quad \frac{1}{(1.08)^n}$$

carried to a large number of times eventually sums to $12.50, but the unlimited progression of

$$\$1 + \frac{1(1.08)}{(1.08)} + \frac{1(1.08)^2}{(1.08)^2} + \frac{1(1.08)^3}{(1.08)^3} \quad \cdots \quad \frac{(1.08)^{\infty}}{(1.08)^{\infty}}$$

has no finite number. Presumably, a stock whose dividend is expected to grow at a rate equal to or above the going yield rate for an indefinitely long period could fetch an infinitely high price on the market. But long-run dividends based on earnings cannot grow above the yield rate forever. Nevertheless, prospective growth will raise the stock price and the P/E ratio and lower present yields.

The Corporate Sector: A Net Exporter of Funds

There is a "balanced economic growth model" which carries the assumption that the amount of income from capital equals the total of aggregate savings and real investment and that perforce the rate of return on capital and the growth rate of the economy are equal. However, the cash flow generated by the operating assets exceeds the amount of possible net real investment, and it necessarily follows that at the end of the fiscal period, the overall corporate sector will have more funds than it can desirably reinvest in the business. In short, the data shows that if in any given year the growth rate is 4.0% and the aggregate after-tax earnings on the real value of *corporate equity* is 8.0%, then the corporate sector one way or another will distribute about 50% of its equity earnings or profits to the rest of the economy. The outflow of cash in the form of dividends, interest paid on debt, buybacks, and repayment of debt has substantially exceeded new funds raised on the capital markets, primarily raised by issuing debt. The reader is referred to Table 1.

In 2003, the corporate sector exported over $350 billion of funds and over $1453 billion by 2014, see Table 2. The surplus of funds over any possible reasonable capital investment policy is the rationale behind the cash buyback of shares and the payment of dividends. In short, perforce, the corporate sector is a net exporter of funds to the rest of the economy.[3]

Earnings The investor's cash returns, buybacks, and dividends are a fruit of the company's earnings. The level of current and projected earnings impacts dividend policy and potential buybacks. A forecast of dividend levels can be obtained by studying past payout rates and attempting to predict future earnings levels. Dividend policies show a high correlation of dividends with earnings, although as earnings rise there is usually some lag before former payout levels are resumed. Buybacks are more difficult to predict. In the past, before the prevalence of buybacks, if a company had a "normal" dividend payout policy, the prospective investor would not be far wrong if he concentrated on the trend of earnings in evaluating the worth of a share of stock.

[3]See E. Schwartz and R.J. Aronson, "The Corporate Sector: A Net Exporter of Funds," *Southern Economic Journal*, October 1966.

Table 1 The corporate sector net exports of funds 1971–2000 ($MM)

	Year						
Fund sources/uses	1971	1975	1980	1985	1990	1995	2000
Debt issued	$35,762.8	84,427.6	144,809.8	299,454.4	625,378.4	1,156,219.5	2,204,155.7
Equity issued	9262.5	14,611.9	37,132.9	76,050.7	51,664.4	118,036.6	360,740.4
Debt repurchased	19,907.5	45,473.0	75,759.6	118,685.9	485,998.4	838,614.1	1,546,809.7
Equity repurchased	1585.3	1501.3	7959.1	51,989.7	52,183.7	88,825.8	230,042.4
Net debt	15,855.2	38,954.6	69,050.2	111,798.5	139,380.0	317,605.4	657,346.0
Net equity	7667.2	13,110.9	29,173.8	24,061.1	−519.4	29,210.8	130,698.0
Dividends	23,743.6	33,438.9	67,759.3	100,397.8	163,604.3	228,352.1	303,767.2
Interest paid	20,331.2	44,028.2	97,911.8	161,184.8	338,105.4	357,941.9	634,214.4
Net exports	20,542.3	25,401.9	67,447.1	56,753.1	362,849.0	239,477.8	149,938.3

Source: WRDS Database

The amount left after common dividends are paid represents retained earnings or earnings reinvested in the firm. Retained earnings add to the equity of the common shareholders, and the amount of correlated funds can be used to finance additions to the operating assets of the corporation or to retire debt.

Additional assets, properly employed, should add to the future earnings of the corporation. An increase in assets financed by ownership capital as opposed to debt improves the credit standing of the firm and enables it to acquire debt funds at a relatively lower rate. Funds represented in earnings should increase the future profits of the shareholders and eventually result in buybacks or higher dividends. It is not the increment in the book value of the shares but a hoped-for sequence of increased earnings that makes retained earnings of value to the shareholder.

Capital Appreciation If the shareholder sells his shares at a higher price than he paid for them, the difference is called capital appreciation or capital gains. The reverse can be true and the sale of securities may show a capital loss. However, a long-term investment in a portfolio of common shares almost always outperforms a comparable investment in other financial assets. Essentially as one might expect, there is a net return for bearing risk.

Capital appreciation develops from growth in earnings or from a decline in the basic discount rate or a favorable reevaluation of the company's risk.

If the market reevaluates a stock favorably, assigning it plus factors for future growth, increased dividends and buybacks, and greater stability all at once, the stock can show a rapid rise in price in a brief span of time. However, the market for common stock is notoriously volatile, and a wave of optimism as to the future of the economy and the share of corporate earnings can send the general level of stock prices unsustainably high. Widespread pessimism can have the opposite effect.

Corporate Exports

Corporations seek to maximize stockholder wealth. Brealey, Myers, and Allen (2006) stated that the goal of corporate finance is to implement corporate decisions to enhance stockholder wealth. We assume that the market mechanism produces the most effective measurement of corporate strategies. Schwartz and Aronson (1966) held that the rate of return on capital and the growth rate of the economy are equal. If corporate cash flows exceeded real investment opportunities, then the firm should distribute the surplus funds in the form of dividends, interest paid on debt, buybacks, and repayment of debt. Schwartz and Aronson reported that dividends and interest paid substantially exceeded new funds raised on the capital markets. Schwartz and Aronson referred to the excess of corporate cash flows exported as "corporate exports." Guerard and Schwartz (2007) updated the analysis and confirmed the dramatic growth in corporate funds exported during the 1971–2005 period using a universe of publically traded firms in the United States with total assets exceeding $200 million. Dividends paid and interest paid rose substantially, as did stock repurchases (Table 1).

We specifically look at dividend, debt and equity issuance and buyback strategies, and the implied leverage decisions of firms. Fama and Babick (1968) reported that dividend policies show a high correlation of dividends with earnings, although as earnings rise there is usually some lag before former payout levels are resumed. Buybacks are more difficult to predict. In the past, before the prevalence of buybacks, if a company had a "normal," or consistent, dividend payout policy, the prospective investor would not be far wrong if he concentrated on the trend of earnings in evaluating the worth of a share of stock. A consistent dividend policy means the company pays reasonably consistent ratio of dividends divided by earnings. Equity repurchases, whether privately negotiated or via a tender offer, generally specify the number of shares the firm seeks to repurchase, the tender price at which it will repurchase shares, and the expiration date of the tender offer, which the firm may extend. Dann (1981) examined 143 cash tender offers to repurchase equity during the 1962–1976 period, made by 122 different firms, and reported a 22.46% tender offer premium, relative to the previous day of the announcement (20.85% relative to the 1-month period before the announcement).

Large excess returns are associated with share buybacks, particularly for stocks in the 1970s. Firms repurchase equity to enhance stockholder wealth. There were marginal debt effects, and approximately 95% of the enhanced value accrued to stockholders. Lakonishok and Vermaelen (1990) reported excess returns to repurchases continuing through 1986, but at a (slightly) diminished rate.[4] The level of current and projected earnings impacts dividend policy and potential buybacks (see Bierman, 2001a, b, 2010).[5] Deutsche Bank reports modest long-term buyback premia. A recently published study, Fu and Haung (2016), studied 14,309 stock repurchases over the 1984–2012 period and reported that long-run abnormal returns from stock repurchases for the 2000–2012 period became negative. We find no such results with corporate exports. Moreover, we find no such results over the 2003–2014 time period with global firms' stock buybacks.

Debt is issued to finance capital expenditures and dividend payment.[6] Retained earnings increase the equity of the common shareholders and represent the majority

[4]Lakonishok and Vermalen reported premiums of 21.79%, 24.09%, and 18.54% on tender offers during the 1962–1986, 1962–1979, and 1980–1986 periods, respectively. They also reported cumulative abnormal returns of 12.54%, 14.58%, and 9.78% to non-tendering stockholders during the corresponding periods. Smaller firms produced the highest abnormal returns.

[5]A forecast of dividend levels can be obtained by studying past payout rates and attempting to predict future earnings levels. The amount left after common dividends are paid represents retained earnings or earnings reinvested in the firm. Firms target a target debt ratio by choosing between debt and equity repurchases rather than choosing between debt and equity issuances (see Bierman, 2001a, b). Repurchases and dividends are motivated by contracting future investment opportunities (see Bierman, 2010).

[6]Dhrymes and Kurz (1967) proposed an explicit link among these dividend, investment, and debt issuance decisions and econometrically implemented using sample consists of 181 industrial and commercial firms for which a continuous record exists over the period 1947–1960. Guerard, Bean, and Andrews (1987) updated the Dhrymes and Kurz analysis and reported (1) a strong interdependence is evident between the investment and dividend decisions and (2) a strong interdependence is evident between the investment and new debt financing decisions.

of funds which can be used to finance additions to the operating assets of the corporation or to retire debt. An increase in assets financed by ownership capital as opposed to debt improves the credit standing of the firm and enables it to acquire debt funds at a relatively lower rate. Debt and internally generated funds are the primary sources of aggregate investment outlays.

New debt issues exceed new equity issues by a multiple exceeding eight times, a result consistent with Dhrymes and Kurz (1967) and Guerard, Bean, and Andrews (1987). New debt issued is identified with the financing capital expenditures in the simultaneous equation modeling. Funds represented in earnings should increase the future profits of the shareholders and eventually result in buybacks or higher dividends. It is not the increment in the book value of the shares but a hoped-for sequence of increased earnings that makes retained earnings of value to the shareholder.

Let's define a variable to designate the net corporate export of funds, CE, of the corporate sector, as defined in Guerard and Schwartz (2007) and Guerard et al. (2014):

$$\begin{aligned} \text{CE} = {} & \text{Dividends Paid} + \text{Interest Paid} + \text{Net Equity Repurchased} \\ & - \text{Net Debt Issued} \end{aligned} \tag{1}$$

In 1972, the corporate sector exported over $32 billion of funds and, by 2014, the corporate sector funds exported to grown to over $1453 billion (see Table 2). The surplus of funds over any possible reasonable capital investment policy is the rationale behind the cash buyback of shares and the payment of dividends. In short, the US corporate sector continues to be a net exporter of funds to the rest of the economy and has risen almost consistently throughout the 1972–2014 period. The steady growth of dividends paid and equity repurchased led to a great growth in corporate exports.[7] IBM's corporate exports have risen substantially with its

[7] An analysis of the CE components reveals that interest paid has risen faster than dividends paid during the 1971–2014 time period. Net debt issues have risen at an undiminished rate, with the notable exception of 2001–2005 and 2009. Stock repurchases rose substantially following the crash of October 1987. Net equity repurchases increased substantially in the 2002–2006 time period and fell dramatically with the financial crisis. To illustrate the corporate fund generation process, Guerard (2010) showed the ten largest and smallest corporate exporter firms in 1983; the largest corporate exports firms that included AT&T, IBM, and several of the large oil companies dominated positive corporate exports in 1983 as they paid large dividends and interest and generally repurchased more debt than was issued (which made a great deal of sense given the level of interest rates in 1983). A similar process occurred in 2006 as Microsoft, Pfizer, and the oil companies dominated the largest corporate exporting firm (IBM fell to only the 24th largest exporter in 2006). Dividends paid exceeded equity repurchased of the Compustat firms from 1971 to 2014, although equity repurchases have risen relatively to dividends since 1982. We examine the 1971–2014 period because Compustat does not maintain debt and equity issuance, and repurchases, prior to 1971. Guerard et al. (2014) created mean-variance efficient portfolios and tilt the portfolios to purchase stocks with the largest corporate exports. We believe, and will show, that buying stocks on a forecasted earnings acceleration variable basis can be complemented with buying companies that generate positive and corporate exports. The portfolios produce statistically significant outperformance relative to the universe benchmark. We report that portfolios with the largest corporate export stocks produce higher returns, relative to risk, when the stocks that are not identified as undervalued by a stock selection model are excluded from the portfolios.

Table 2 Corporate export components, $MM. All Compustat companies and IBM, selected years

Year	Number of companies	Long-term debt issued	Long-term debt repurchased	Stock dividends	Equity repurchased	Equity issued	Net new debt	Corporate exports
1972	4508	35,435.7	20,728.3	30,444.9	2113.7	8679.8	14,707.4	32,123.1
1975	6757	93,832.6	47,912.0	41,668.1	15,238.0	18,180.1	459,320.6	28,349.7
1980	6889	173,022.6	82,885.2	82,523.2	8295.3	42,367.2	90,137.4	76,272.3
1987	9225	437,062.9	329,071.9	154,007.9	71,264.0	102,673.0	107,991.1	250,345.3
1990	9571	666,371.5	512,349.7	192,196.8	55,208.9	57,453.2	154,021.1	397,666.3
1995	12,491	1,231,890.1	898,656.4	263,986.6	96,141.0	132,137.5	333,236.6	277,587.0
2000	12,092	2,724,008.9	1,721,730.0	367,292.3	247,253.7	411,075.7	1002,281.5	22,446.5
2008	10,884	6,085,805.7	5,138,659.0	776,022.6	703,572.4	891,732.3	947,147.0	954,624.8
2010	11,088	6,036,302.0	7,142,278.4	727,427.8	521,110.3	541,200.9	−1,105,976.4	26,009,941.2
2014	10,921	6,703,998.7	6,000,589.7	979,835.3	824,651.0	497,677.5	703,409.0	1,453,053.5
1972	IBM	201.6	98.3	626.2	0.0	270.6	103.3	37.4
1975	IBM	29.7	870.4	968.9	0.0	284.5	−40.7	787.4
1980	IBM	604.0	94.0	484.0	484.0	422.0	510.0	1885.0
1987	IBM	408.0	719.0	2654.0	1425.0	133.0	−311.0	4874.0
1990	IBM	4676.0	4184.0	2774.0	491.0	135.0	1592.0	2874.0
1995	IBM	6636.0	9460.0	634.0	10.0	0.0	−4110.0	4346.0
2000	IBM	9604.0	756.0	929.0	6093.0	0.0	2043.0	5696.0
2008	IBM	4363.0	3522.0	1250.0	6506.0	3774.0	841.0	7676.0
2010	IBM	8055.0	6522.0	3177.0	15,376.0	3774.0	1533.0	14,174.0
2014	IBM	8180.0	4644.0	4265.0	13,679.0	709.0	3536.0	14,729.0

Source: WRDS Database

dividend payments, great reduction in common stock due to equity repurchases. Equity repurchases have substantially exceeded dividends for IBM in the post-2000 time period.

Definitions of the Value of Common Shares

Par Value Par value is basically a legal concept that sets the minimum initial price at which the company's shares are sold at the time of organization. (Par value can be changed through reorganization, recapitalization, or stock splits.) Any initial sale of stock at a discount from par presumably involves a legal obligation which the creditors can press against the holder of the stock should the company ever fail. In some jurisdictions, the issue of stock at a discount from par is illegal; in any case, it is extremely rare. (This is why almost all stock certificates bear the notation "fully paid and nonassessable.") The dollar amount of stocks sold at par comprises the capital stock account. In most jurisdictions, no dividends can be declared unless there is a positive balance in the surplus account sufficient to cover the dividend. If the books show no surplus, no dividends can be declared under this rule even though there might be considerable current earnings. In short the general rule "prohibits dividends out of capital."

Historically, par value was designed to serve a twofold purpose. It set up a sum of paid-in capital to act as a trust fund for the creditors of the corporation in lieu of the unlimited personal liability of the owners. It was also to act as assurance to a subscriber for the original shares that other shareholders were purchasing their shares at an equal price. In practice par value has served neither of these two functions very well. Credit ratings are more likely to be based on earnings potential, safety and liquidity of assets, and total financial structure than the presumed last line of defense shown by the capital stock account. Furthermore, any attempt to favor stockholders by giving them dividends at a time when the firm appeared unable to meet its credit obligations would constitute an act of "involuntary bankruptcy" and could be halted by invoking bankruptcy proceedings. Both aspects of par value, including the assurance that all are venturing equally in the new company, are considerably vitiated by the fact that the directors have wide discretion in valuing noncash assets exchanged for common stock. Essentially, the investor in a new venture must estimate and study all of the potential rewards and risks and not rely upon the fictitious safety of par value.

Because of the mistaken impressions par value has on occasion given to the investor and because of its disadvantages in the initial issue, there has been a large growth in the employment of no-par value stock. With an issue of no-par stock, the directors have some discretion as to how much is to be allocated to the capital stock account and how much to the paid-in surplus account. A great advantage of no par is that if the initial operations of the company do not go well, additional equity capital may be raised by selling new shares at a lower price than the original issue. With par value shares, this might be difficult to accomplish because it would be illegal to sell the second flotation at a discount from par. However, there have been some tax

difficulties with no-par shares; in New York State, for example, these shares have been given a constructive value of $100 par for purposes of calculating the transfer tax on their sale. Mainly because of this problem, the most popular issue today, replacing no-par stock, is low par stock. These shares may have par values such as $1.00, $3.00, or $5.00 and are usually floated at a price considerably higher than their par value. This stock has most of the advantages of no-par stock without being subject to possible extra burdens under the franchise or transfer tax.

Book Value The book value is the net balance sheet asset value per share. It is obtained by subtracting liabilities and preferred stock from the accounting value of the total assets and dividing the remainder by the number of common shares outstanding. Book value may be used in certain financial comparisons. It is sometimes useful to see which of a set of companies earns a greater percentage on its book value. Because book value is increased with additions to surplus (i.e., retained earnings), tracing the course of book value over time shows how much net internal financing the company has used. (This historical analysis becomes difficult if any stock dividends have taken place.) The usefulness of book value for financial analysis is reduced by the time changes in price levels and variations in depreciation policies, which result in a wide difference between the balance sheet accounting value of the company's assets and their current economic value.

When the market is pessimistic or depressed, stocks may tend to sell below book value; in a buoyant or optimistic market, shares are generally priced above book value. Growth companies, which show a strong upward trend in earnings, may sell at a considerable multiple of their book value. The statistical association between equity holding period returns and book value of stockholder equity will be calculated in chapter "Risk and Return of Equity and the Capital Asset Pricing Model."

Market Value The relative market price of a corporation's shares is an index of the firm's success. To the so-called value investor, the market price of the stock relative to its prospective yield is the vital datum.

Market price is set by the investors' projection of the firm's potential earnings discounted by the going rate of return for the type of firm. The market price relationship is usually summed up in the going price/earnings ratio. A relatively high P/E ratio indicates that the market has hopes for growing earnings from the firm, and a low P/E ratio suggests fear for the future. The price/earnings ratio sometimes rises with a drop in corporation earnings which the investors believe to be temporary.[8]

Intrinsic or "Normal" Value Intrinsic value is an idealized concept. It is the price a stock would sell at if the market was supposedly not subject to psychological aberrations.[9] Under the method given by Graham and Dodd, intrinsic value is

[8]Thus if a stock had annual earnings of $6 and sold for $60, its P/E was 10 times; if earnings should drop to $5 and the stock price drops to $55, still the P/E ratio would have risen to 11 times.

[9]For the basic statement of this concept, see Graham Dodd, *Security Analysis*, McGraw-Hill, 1951, pp: 410–411.

approximated by studying the past level of earnings, earnings trends, and the pattern of dividends. (Buybacks did not appear in the Graham and Dodd model.) A capitalization rate based on standard levels modified for various qualitative factors affecting the particular firm is applied to a weighted combination of earnings and dividends. The result is an intrinsic or normal value.

Graham and Dodd's basic formula was

$$\text{Intrinsic Value} = \text{M}\left(\frac{\text{E}}{3} + \text{D}\right) \pm \text{asset factor.}$$

M represents a "normal capitalization multiplier" which ran from 8 to 15 for ordinary industrials—with lower multipliers given for highly leveraged and volatile companies and the higher multipliers reserved for stable and blue-chip companies. E is the projected earnings per share, generally based on the average earnings of the last 5 years corrected for any pronounced trend. The figure for earnings is divided by 3, and then D, the projected dividend, is added to the multicand. The amount for D is generally based on the dividends of the last 5 years. When a company's payout rate approximated 66%, M approaches the pure earnings multiplier. The asset correction is used to reduce the intrinsic value if it exceeds book value by too large an amount; a plus correction is made if intrinsic value based on earnings falls below the rule-of-thumb "liquidating value" per share.

Graham and Dodd admitted the difficulty of applying the formula to very new companies, to high-risk situations, to highly leveraged companies, and to strong growth companies.

According to the intrinsic value theorists, share prices at any time may swing well above or below intrinsic value, but they tend to return to it over the long run. Although long-term economic growth and stability have raised expectations and raised the P/E multipliers above those proposed by Graham and Dodd and although buybacks have vitiated the emphasis on dividends, there is still a sense in which the intrinsic value concept influences managerial decisions. The determination of the cost of equity financing cannot be based on current market data unless the company's shares are selling within some "normal range." A growth firm whose stock is selling at a relatively high price cannot use the explicit current market rate of return as its cost of equity funds, but must develop an implicit cost based on some normative valuation.

Liquidation Value If all the assets of a company were sold off, all creditors paid in full, and the preferred stockholders paid the voluntary liquidation price, the remainder per share is the liquidation value. In practice, the exact liquidation value of the shares cannot be known unless this process (which unfortunately is not reversible) was carried out. Rough estimates of the liquidation value can be made by experienced appraisers estimating the realizable value of the assets. The accuracy of such estimates varies considerably for different types of assets. In general, the receipts from current assets and land can be estimated more accurately than the amount which might be recovered from plant and equipment.

A crude index of liquidation value has been developed by the investment analysts. This measure is called "net current assets" per share and is obtained by subtracting all liabilities and preferred stock from the current assets and dividing the remainder by the common shares. No value is given to plant and equipment, but on the other hand, it is assumed that no loss will be incurred in the liquidation of the current assets. Net current asset value is in a sense a minimal liquidation value.

When the shares of a company sell for any considerable time below their liquidation value, it indicates either extremely poor management or that the capital of the business is employed at a rate of return lower than that obtainable in other industries.

The company may be ripe for seizure by a new management group. The new managers may try to raise the rate of return, merge with another company, or engage in a partial or complete liquidation of the firm. From an overall economic viewpoint, the demise of such a company is not distressing; presumably the resources released could be more productively employed elsewhere.

Stock Splits and Stock Dividends A stock split is a financial maneuver by which the book value of each share is reduced but the holders of outstanding stock receive certificates for new shares prorated on their present holdings. After a stock split of 2 for 1, a holder of 100 shares of a corporation holds 200 shares; after a split of 3 for 2, she holds 150 shares. The book value of the shares is reduced proportionately by the split, as are the earnings on the new shares. Nothing on the balance sheet changes; the surplus and capital stock accounts have the same totals, but in the information attached to the balance sheet, the number of outstanding shares is increased. A stock split gives the share owners nothing of value that they did not have previously. Nevertheless, the investment market tends to react favorably to a split. Rumors of an impending split generally send the price of the stock upward. A potential favorable effect of the stock split may be to bring the price of the shares down into a more marketable range. The reduced base price draws increased trading interest and adds a slight increment to the market price. Another presumed favorable aspect of a split is the "announcement effect," the assumption that the split implies an optimistic management view of the prospects of the firm.

A stock dividend is the distribution of new shares to the present stockholders, reducing the surplus account and crediting the capital stock account on the corporation's books.[10] Its effects are similar to those of a stock split, although the accounting formalities differ slightly. A stock dividend runs only from about 2% to 10% of the shares in contrast to the much larger division of the stock split. In a stock dividend, the par value of the shares is left unchanged. However, a larger number of shares are now divided into the same total equity, and the book value of the stock is lower. The earnings per share is also reduced proportionately. The stock dividend brings the stockholder nothing of objective value except as it may raise the effective cash dividend rate. The usual stock dividend is under 10%, and generally the cash dividend rate on the shares remains unchanged. Thus a 10% stock dividend is usually equivalent to a 10% rise in the cash dividend and probably has a positive effect on the value of the investor's portfolio.

[10]In rather old-fashioned parlance, a stock dividend was described as "capitalizing earnings."

Federal tax laws do not consider a stock split or a stock dividend personal income. The stock split or stock dividend is used to reduce the acquisition cost of the shares, so that in case of any partial liquidation of the portfolio, capital gains or losses may be properly calculated.

Stock Prices and Dividends: An Example

Stock values are determined, to a great extent, by the expectations of earnings and dividends growth. Equity valuation models developed by Williams (1937), Gordon and Shapiro (1956), Gordon (1959, 1962) and Kenan (1970) capitalized dividends. Let us examine the case of Dominion Resources, D, the holding company of Virginia Electric and Power Company (VEPCO), as of October 2000. The current price of D was $53.25. The respective dividend and earnings per share data were provided in Table 3.

The reader notes that Dominion Resources maintained its dividend per share during the 1995–1999 period, despite the volatility of its earnings per share. How can one determine the fair market value of Dominion Resources stock, P? Let us use the Williams/Gordon discounted dividends stock valuation equation:

$$P = \frac{E(DPS)}{k - g}$$

where

E(DPS) = dividend per share
k = required rate of return
g = growth rate

Gordon assumed in his model that dividends and earnings growth is equal. This is not a reasonable assumption for Dominion Resources during the 1990–1999 period:

$$g_{eps} = \sqrt[9]{\frac{eps_{1999}}{eps_{1990}}} = \sqrt[9]{\frac{\$2.81}{\$2.92}} = -.004$$

$$g_{DPS} = \sqrt[9]{\frac{DPS_{1999}}{DPS_{1990}}} = \sqrt[9]{\frac{\$2.58}{\$2.32}} = .012$$

Table 3 Stock data for dominion energy

	1999	1998	1997	1996	1995	1994	1990
Dividends	$2.58	2.58	2.58	2.58	2.58	2.55	2.32
Earnings	$2.81	2.75	2.15	2.65	2.45	2.81	2.92
Payout ratio	.92	.94	1.20	.97	1.05	.91	.78

Source: Standard & Poor's Stock Guide. 2005

The 9-year compounded earnings per share growth is −0.4%, whereas dividends grew at a 1.2%. One will use the 9-year historic dividend per share growth to price the stock.

What should one use for the required rate of return? We can use the required rate of return, k, from the Capital Asset Pricing Model, which we develop in chapter "Risk and Return of Equity and the Capital Asset Pricing Model." The required rate of return on equity is an opportunity cost that the investor incurs by purchasing the stock. That is, the stock must produce a return that is consistent with its level of risk. The return of a stock is determined by its dividend payments and its stock price appreciation. Investors must be compensated for bearing risk, and risk is measured by the stock's beta, its measure of systematic risk. The S&P 500 beta for D is 0.20, a defensive asset in that if the market is expected to rise by 10%, then Dominion Resources stock return is expected to rise by only 2%. We will estimate a beta in chapter "Risk and Return of Equity and the Capital Asset Pricing Model."

The 90-day Treasury bill yield is 3.5% in October 2001, and the expected market equity premium of stocks relative to Treasury bills historically is 8.8%. The required return on return for Dominion Resources is 5.26%:

$$k = R_F + [E(R_M) - R_F]\,\beta;$$

$$k = .035 + (.088)(.20) = .0526$$

The market's required rate of return is 5.26%; we should purchase Dominion Resources if we expect the stock to earn more than 5.26%. An alternative strategy, yielding the same answer, is to price Dominion Resources shares and compare its fair market value to its current stock price:

$$P = \frac{\$2.58(1.012)}{.0526 - .012} = \$64.29$$

In October 2001, one could have determined a fair market value for Dominion Resources of $64.29, far exceeding its current stock price of $53.25. One should purchase Dominion Resources.[11]

The current date the authors write this chapter is August 23, 2004. The current 3-month Treasury bill yield is 1.41%, the current Dominion Resources beta is 0.23, and the 10-year growth in earnings per share for Dominion Resources is

[11]One of the authors used Dominion Resources as an example of security valuation for an Advanced Financial Management class at Rutgers University in October 2001. Dominion Resources has outperformed the market, the S&P 500, by approximately 3% over almost 3 years; the relative outperformance reaching 15% in October 2003. One might smirk at such a return; however, one must remember that the beta of Dominion Resources is approximately 0.20, creating a more reasonable market-adjusted excess return. Dominion Resources has cumulatively underperformed the market in 1 month of the 36 months (which one of my former Penn students noticed with an interesting e-mail).

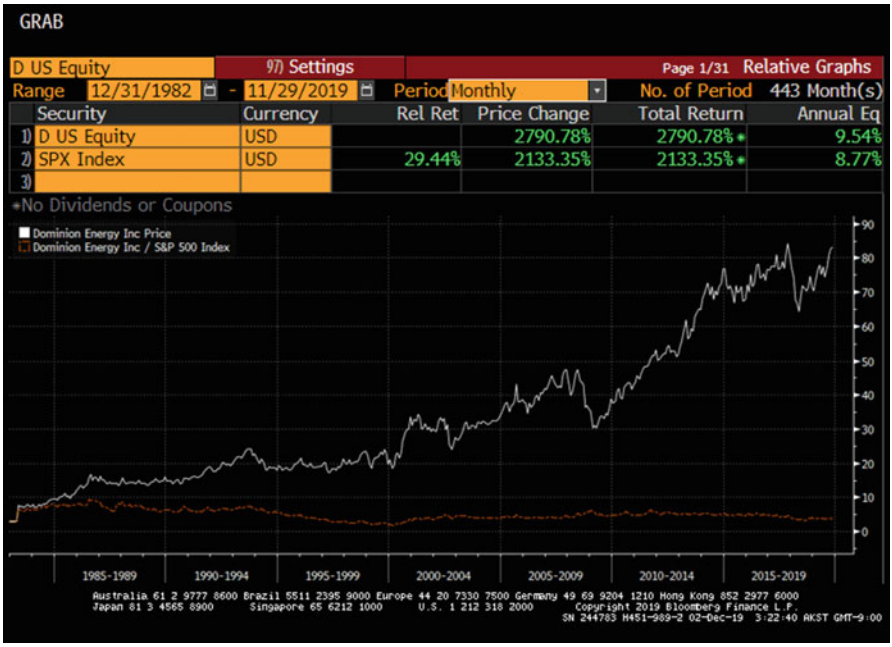
0.65% (the ninth root of $2.98 in 2003 divided by $2.81 in 1994), and Dominion Resources pays a $2.58 dividend (still). The current fundamental value of Dominion Resources is

$$k = .014 + .088\ (.23) = .032$$

$$P = \frac{\$2.58(1.0066)}{.032 - .0065} \quad = \quad \$100$$

The current price of Dominion Resources is $64.58.[12] One should purchase Dominion Resources as it is substantially undervalued.

How has D performed since 1982, when purchased, as of December 2019?



Dominion Resources, now Dominion Energy, has outperformed the S&P 500 by over 80 basis points, annually, with a beta of about 0.26, one-quarter the risk of the market! An investment of $600 for 50 shares now exceeds $64,000. Have you any Grey Poupon? But of course. Not all stock transactions are profitable, but in many cases, portfolio returns are often realized by the great appreciation of 2–3 stocks out of 10–15.

[12]On August 10, 2006, a final version of Guerard and Schwartz (2007) was completed. The price of Dominion Resources was $78.90. D, with a current beta of 0.44, had outperformed the market (S&P 500 Index) by over 500 basis points during the past 2 years.

One could calculate dividend discount models for all exchange traded stocks in the United States (and globally) and rank the stocks relative to their stock price deviation from fair market value. Alternatively, one could use a multifactor stock selection model to quantify the purchase decision of stocks.[13] The authors prefer the latter method that is developed in chapter "Risk and Return of Equity and the Capital Asset Pricing Model."

Noncash-Paying Growth Shares

The value of any corporation stock is its current or its potential future cash throw-off either in the form of dividends, cash buybacks, or purchase by another company. At any point of time, there are shares on the market that currently pay no dividend and show no buyback policy but still command a positive price. The value of such shares is based on an expectation or forecast that sometime in the future the shares will yield a cash throw-off. If the shares are those of a *super-growth company*, one whose periodic earnings grow at a rate in excess of the normal capitalization rate, the current market price will be quite high.

Super growth cannot be infinite; sometime in the future, the firm's growth rate will slow to some norm, the company will begin to throw off cash either in the form of buybacks or dividends, and the price of the shares will be evaluated on the basis of the earnings reached at this time. The current market price of the shares is the present value of this future price. An example of the valuation of a noncash-paying, fictional growth company, Crampton Inc., is presented below. The formula for the valuation of the shares follows:

$$\text{Current price} \approx \frac{E(1+g)^n \times \text{M}}{(1+i)^n}$$

where

E equals the current earnings
g is the expected average super-growth rate
M is a "normal" market multiplier
n is the forecasted period of super growth
i is a risk-adjusted desired rate of return

The data for the fictional Crampton Corporation:

- Crampton's current earnings per share is $1.00
- The current growth rate of earnings is 30% per annum[14]

[13]See Bierman (2010) for an interesting analysis of stock buybacks and dividends.

[14]To make calculations easier, we have set up an average growth rate. In fact, we are likely to have a curvilinear growth path. This does not vitiate the basic concept.

- The forecasted growth period is 20 years
- The risk-adjusted rate of return is 9.0%

The final value of the shares is obtained thusly:

- At the end of 20 years, earnings compounded at a growth rate of 30% p.a. = $190
- The future price of the normalized shares with earnings capitalized at 9% (M = 11.11) equals $2111
- The present value of a Crampton share, $2111, discounted at 9% for 20 years equals $376

Under this supposed scenario, the P/E ratio could be as high as 376 times current earnings. Of course, financial disaster follows if the growth period is shorter than 20 years and/or the growth rate does not reach 20%. On the other hand, much money can be made if the reverse is true or if the investor jumps in at a lower price before the market recognizes Crampton's potential.

An argument sometimes presented against any valuation of long-term deferred returns is that an investor would be irrational to buy an investment whose cash payout is beyond her expected life or planning horizon. The argument does not hold as long as a market exists containing sufficient purchasers with differing and/or longer horizons. As time passes and the payoff period becomes closer, the price of the investment moves continuously upward. The analogy can be made to that of a forest, which is growing at a rate in excess of the financial rate and will not be harvested for 50 years. One may buy the forest at a current price given the knowledge that there will be other buyers in the interim willing to buy and hold at all intervals before the harvest.

Valuing a Dividend-Paying Growth Stock

A growth company may present the investor with a stream of dividends and earnings growing in excess of the normal yield rate. A solution of the problem is presented for the fictional Shoreham Corporation. In evaluating Shoreham shares, it is necessary to obtain the present value of both the stream of cash dividends and the present value of the future price of the stock.

Illustration of the Valuation of a Dividend-Paying: Growth Stock—Shoreham Inc.

A. Data

Current Earnings = $5.00
Current Dividend = $2.00

B. Assumptions

1. Earnings per share will grow at the rate of 20.0% per annum.
2. The risk-adjusted desired rate of return is 10% per annum.
3. The expected growth period, n, is 10 years.
4. At the end of 10 years, Crampton shares will have a "normal valuation" at a P/E of 11 times current earnings.
5. Dividends will be 40% of annual earnings during the growth period and will necessarily grow at 20% per annum.

C. The basic valuation equation is

$$PV = \frac{D(1+g)}{(1+i)} + \ \ldots + \frac{D(1+g)^{10}}{(1+i)^{10}} + \frac{11 \times (E)(1+g)^{10}}{(1+i)^{n}}$$

or the value of the increasing dividend stream for 10 years, plus the present value of the price of the shares at the end of the super-growth period 10 years from date.

D. 1. The value of the dividends is the value of an annuity, $s_{\overline{n}|}{}^{e}$
where e equals g − i (20–10%) or 10.0% and n = 10 years.

Thus, $s_{\overline{10}|}{}^{10\%}$	=	15.937
	×	$2.00
The PV of the dividends	≈	$31.87

2. The value of the future share price is obtained by multiplying the future earnings by 11 and bringing this price back to present value.
Future earnings equal $E(1 + g)^n$ or $\$5(1.20)^{10}$

$(1.20)^{10}$	=	6.192
	×	$5.00
The amount of future earning	=	$30.96
Times the future P/E factor	×	11
The future price of the shares	≈	$340.56
Discounted at the present value factor of $1/(1.10)^{10}$	×	.3855
The present value of the future price of the shares	=	$131.29

The current value of Crampton shares equals the pv of the dividends plus the pv of the future price of the shares

pv of dividends	$31.87
pv of future price	131.29
Current value	$163.16

E. Note that on the basis of the current value, the

P/E ratio, \$163.26/\$5	=	32.7
Or earnings to price \$5/163.26	=	3.06%
The current dividend yield, \$2/163.26	=	1.2%

The growth stock has a high current price-to-earnings multiple and low current yield.

Super Growth Cannot Be Infinite

An investment that forecasts earnings growth for an infinite period in excess of the market's normal rate of return would have an infinite value.[15] However, in actuality, a super-growth period cannot be infinite, and therefore a growth investment has a defined value.

Three constraining factors prevent the growth period from running for an indefinite length. First, it is apparent that no option can keep generating a return that grows forever at a rate that exceeds the growth rate of the economy. A super-growth enterprise that grows at a rate exceeding that of the GDP (Gross Domestic Product) becomes a larger and larger part of the GDP; it becomes a bigger and bigger proportion of the total economy. Taking this proposition to the logical extreme, when the super-growth enterprise becomes the total GDP, it cannot grow faster than the GDP. More realistically, we may note that as the enterprise becomes a larger part of the economy, its rate of growth is constrained by the limits of the system in which it operates.

The second limiting factor involves the relation of the level of the interest rate to available super-growth projects. Theoretically a super-growth investment of infinite duration has an infinite value. However, if there were a prevalence of super-growth options, the demand for capital would take a quantum leap. But savings, the supply of capital, is not infinitely elastic. Presumably in any society, people must consume some resources, eat, clothe themselves, and find shelter. They cannot save all income. The price for savings, i.e., the interest rate, must rise, and equilibrium is reached when the interest rate became higher than the rate of growth. Last, we may take a look at the actual financial side of a super-growth process. Presumably, a super-growth situation requires an increase in net capital per period which is in excess of the investment in the previous period and which requires that in each period the return on the total capital continues at a rate in excess of the cost of capital.

[15]If the growth rate exceeds the prevailing discount rate and is of unlimited length, the theoretical present value of the investment option is infinite. This result at the extreme has led to much philosophical discussion in a variant form of the "St. Petersburg paradox" as to whether any value can be infinite.

Although given some initial superior advantage in technology or product, an enterprise might enjoy super growth for a considerable number of years; such growth is not likely to go on indefinitely on an increasing capital base. When in the particular situation the enterprise begins to evidence a declining marginal return to capital, the period of super-growth must begin to slow and halt.

The Paradox of the Low Current Return on Growth Options

The existing current return on a growth stock appears meager on the basis of market valuation. Relative to the price, current earnings are very low.[16] The explanation is that the discounting of a projected rising stream of return at a normal desired market rate of return produces a very high present value. It is not the existing low earnings but the high current price that makes them seem so.

Would purchasers of the shares at the current high price lose return if they do not hold the stock to the end of the growth period? Not so. The current earnings yield is low, but the price of the stock over time will rise toward its future price, resulting in capital gains over each holding period. The periodic capital gains yields the normal rate of return. If actual events, expectations, and market discount rates do not change, every owner for every holding period will obtain the same overall rate of return.

Risk and Returns to Growth Investments

The volatility of the market price growth stock shares is higher than that for standard shares. Not only must an estimate of the rate of growth be used in the evaluation, but the number of years of forecasted growth, n, is a highly strategic variable. The market price is highly sensitive to n, and a change in the forecast number of the remaining years of super growth can bring about wide swings in the price of the shares. Investors who come in early, before the market recognizes the facts of potential growth, may do very well indeed. Even later investors may garner well above a normal return if, after a period, the forecast of the growth period inherent in the price of the shares is extended. On the other hand, if the expected length of the growth period is shortened, the price will drop sharply.

A bit of analytical advice to potential purchasers of growth shares goes as follows. Given the current market price of the shares and given an estimate of the percentage rate of growth and a reasonable risk-adjusted desired rate of return (discount rate), the number of years of growth necessary to justify the investment will fall out of the equation. If the number of years of sustained growth implicit in the current market

[16] A similar phenomenon also holds for real estate in a growth area where the current short-term rents appear to be low relative to the market price of the properties.

price seems reasonable, the investment may be worthwhile. If the required years of growth seem too long, the investment is dangerous.

The Cost of Capital to a Growth Firm

Because the P/E ratio on a growth share can go very high, some analysts have argued that the cost of capital to a growth company is very low and that, for example in the Shoreham case, the company could invest in projects that at the margin yielded a return on equity of a mere 3.06%. This is preposterous. The marginal investment would not cover the normal cost of capital. If this strategy were followed, the market would soon note that the relative returns generated by the company were falling and the Shoreham share price would plunge downward.

The cost of capital to a growth firm is the same as that for normal enterprises in its class Real capital investments at the margin must yield the norm of the market rate of return. However, the returns made on the intramarginal projects of the growth company are well above the normal rate, so that the **average rate of return** is significantly above the cutoff point. This excess rate of return is sustained because increases in product demand and/or decreases in production costs are continuous over the length of the growth period.

The value of a growth stock is raised by factors outside the financial system. The rise in the value of the stock does not lower the cost of capital.

The Cost of Common Stock Financing: The Norm

Assuming the firm is operating in the range of an acceptable financial structure, the cost of common equity capital is the "normal" earnings yield for shares of the given industry. The earnings yield is the inverse of the price/earnings ratio corrected for projected growth or decline. A stock with average earnings of $4, an estimated growth rate of 3.0%, and a "normal" price of $48 has a P/E ratio of 15 and a current yield of 6%. Given the projected growth rate, the minimum rate of return after taxes that the company must make on new equity capital is 9.0%. Investing new equity funds in projects earning less than 9.0% lowers the return for both new and old shareholders. In a growth situation where the investors believe earnings are likely to increase for some indefinite period, the ratio of current earnings may be very low. Here the cost of common stock financing must be imputed from some normal concept.[17]

[17]It may be of interest that Adam Smith wrote that the normal rate of profit on ownership capital would be about double the interest rate on sound commercial loans. A. Smith, *The Wealth of Nations*, Modern Library, E. Cannan, Ed. 1937, Ch. IX, p. 97.

The cost of any one source of financing cannot be determined singly. Its interactions with other types of financing must be considered. An increase in equity capital decreases the firm's financial risk and thus lowers the costs of all sources of funds. On the other hand, equity in itself is the most expensive source of funds. Determining the best position between the reduction of risk and the cost of equity funds is the crux of the problem of reaching the "optimum" financial structure. The cost of capital to the firm is the weighted sum of the costs of each source of funds in the optimum mix.[18] However, the basic rule is that unless it is extremely desirable to reduce financial risk, new equity financing should not be used if the effect is to lower the expected earnings per share.

The management must consider financial risk and the cost of funds if it is to succeed in its basic obligation of maximizing the long-run economic position of the common shareholders. The management must not overuse debt financing. Maximizing short-run earnings is not optimum if the shareholders are subject to inordinate financial risk. In making investment decisions, the firm must consider the alternate returns that stockholders could get on their funds elsewhere. The costs of financing new assets must not exceed the projected return on these assets, and neither should the shareholders be subjected to undue risks. These considerations apply to those decisions involving the management of assets and financial decisions, which are under the control of the management. A decline in the value of shares because of unfavorable developments in the general external economy cannot be blamed on management.

Preferred Stock

Preferred stock is a form of equity or ownership capital. Its claims to a return out of earnings or to a share of the assets in case of liquidation come after the creditors but before those of the common shares. From the viewpoint of the common shareholder, the preferred stock issue performs a function resembling a form of debt—albeit a debt subordinated to the rest of the liabilities. However, the preferred has one basic equity feature: its dividend is not an inflexible claim; it can be postponed or passed without the firm having committed an act of bankruptcy. The equity nature of the preferred shares is recognized in the current corporation income tax law; preferred dividends are not deducted before calculating net income for tax purposes.

Features of Preferred Stock

Priority on a Stated Dividend The dividend on a preferred is stated on a fixed basis. The dividends on preferred stocks must be paid before any dividends on common

[18] See chapter "Debt, Equity, the Optimal Financial Structure, and the Cost of Funds."

can be paid. A corporation has to be in a rather tight financial position before it passes the preferred dividend. The fixity of the preferred stock dividend eliminates any significant gain for the ordinary run of preferred stock from an increase in the prosperity and earnings of the corporation but sets a basic value in the ordinary run of affairs.

A preferred stock has the features of a perpetual annuity. The market price is obtained by dividing the stated dividend by the going rate of return. Thus if the market rate of return on a $5 dividend preferred is 6.0%, the stock sells at approximately $83.33. The market price rises or falls with the level of the interest rate.

Cumulative Dividends Most preferred shares have a cumulative dividend feature. If dividends are passed, they accumulate, and no dividends can be paid on common stock until all arrearages are paid up. The cumulative features deter management from passing the preferred dividend.

A preferred stock carrying passed dividends may sell below par as long as the possibility of the arrearages being paid appears remote. If the economic position of a firm takes a turn for the better and payments of some amounts of the passed dividends are imminent, the price of the preferred stock will rise to somewhere between its long-run investment value plus a discounted value of the accrued past dividends.

Preference to Assets Preferred stock has a priority over the common in case of liquidation. In case of dissolution, the funds from the sale of the assets go first to the creditors and then to the preferred shareholders, and only then is any remainder distributed to the common stockholders. If the funds are insufficient to pay the liquidation value of the preferred stock, the common shares receive nothing. However, in involuntary liquidation, through failure or bankruptcy, most of the creditors are probably not paid in full, and both the preferred and the common stock are worthless.

The preferred stock generally has two values: one for "voluntary" liquidation and one for "involuntary" liquidation. Liquidation is involuntary in the case of bankruptcy. The involuntary liquidation value of the preferred generally consists of the par value of the preferred shares plus any accumulated dividends. Voluntary liquidation occurs when the stockholders themselves vote to go out of business. Usually the voluntary liquidation value of the preferred has a premium over par (e.g., a $100 par preferred may have a voluntary liquidation value of $120).

Most companies attach a call option on a preferred issue. This enables the company to buy back (i.e., call) its preferred stock from the holders at a fixed maximum price. The call price is usually set at a premium over par. The voluntary liquidation price and the call price are usually the same.

Nonvoting The preferred shares are generally nonvoting on ordinary matters of corporate policy but usually have voting rights on matters directly affecting their own position. Many preferred stocks are given the full right to vote on corporation matters if preferred dividends have been passed.

Rational for Preferred Stock Financing

If the going dividend yield on preferred is 6% and the tax rate is 35%, the pretax cost of preferred stock financing is 9.23%. The cost of preferred stock financing is relatively high compared to that of tax-sheltered debt. On the other hand, as a subordinated form of financing—bearing some of the risks of equity—it helps to support the debt structure. Thus if the firm can net 10% on its equity capital after interest and taxes (about 15.4% before a 35% tax), an issue of a 6% preferred can be to the advantage of the common shareholders. The 4% earned over the preferred dividend becomes part of the return of the common shareholders. *The preferred is a form of leverage within the equity section of the financial structure*. Nevertheless, the corporate income tax makes preferred stock financing expensive compared to debt. At present, preferred stock is not very common on the new issue market.

Convertible Preferred

A number of preferred issues have a convertible feature. At the option of the holder, the preferred shares may be turned in for common stock at an agreed ratio or price.[19] The conversion option may be open indefinitely, or it may expire after a stated date.

Charter provisions and indentures are generally drawn up to protect the holder of a convertible security against dilution of the value of the convertible privilege. The conversion ratio is adjusted for any common stock splits or dividends. For example, a conversion ratio of 2 for 1 will be raised to 4 for 1 if the common shares are split 2 for 1.

Rationale of Convertible Preferred The logic of issuing convertible preferred rests on the possibility of the firm obtaining, for the time being, equity funds at a lower cost than that desired on the common equity. The conversion feature induces investors to supply the firm with equity capital at a reasonable rate. The convertible feature gives the purchasers the advantages of a hedged bet; they have some of the safety of return given by the preference feature of the shares, and the conversion feature allows them to share in the gains if the earnings and market value of the common shares rises.

Convertible preferred shares are sometimes used to make merger terms more attractive. The stockholders of the firm being wooed into the merger are given convertible preferred shares of the acquiring firm in return for the common stockholdings in their present company. The new group of stockholders has some

[19]Since a convertible issue, whether preferred stock, debenture, etc., can be turned into common stock, possibly changing the future sharing in earnings, assets, and voting strength, preferred subscription privileges (the preemptive right) are often given to the common stockholders on the initial flotation of any convertible security.

protection against a decline in the fortunes of the acquiring company plus a chance to share in its possible success.

The advantages to one side on a convertible issue are of some disadvantage to the other. If the enterprise is successful, the potential gain to the common is diluted as the preferred shareholders turn their shares into common. The rise in common share earnings is delayed, and there is a drag on the market price of the common until all the preferred is converted. On the other hand, the initial flotation of the converted preferred gives the firm equity funds relatively cheaply, and in the interim before conversion, the amount earned on these funds over and above the preferred dividend benefits the common shareholders. Perhaps the increased equity funds acquired at a strategic time enable the corporation to obtain additional debt funds at better terms, and, of course, the common stockholders benefit from all returns in excess of the cost of the financing mix. Thus the decision to float a convertible issue rests on a favorable weighing of present benefits against a potential future detriment.

Effects of Conversion Features on the Market Price The convertible feature on a security is an option to buy the common stock at a fixed price, and the option fetches a price in the market. The more optimistic the market, the higher the price of the option. The amount at which the market prices the option can be inferred from the difference between the market price of the preferred and its pure investment or the value, where the investment value is the price the stock would sell for as a pure preferred stock and the conversion value is the amount the preferred is worth if turned into common stock at the going price.

Table 4 gives some examples of possible relationships between these various prices. The preferred illustrated in Table 4 has a dividend of $5.50 annually and a conversion rate of 2 for 1 on the common. If the normal yield for a straight preferred stock were 5.5%, the stock would have an investment value of $100. In example 1 in Table 4, the conversion value of the preferred is $80, but the market will pay $110 for the preferred; the option price is $10. However, not only is the $10 option premium at risk if the common should fall significantly, but in the interim, the conversion option price reduces the current yield on the preferred from 5½% to 5%.

Table 4 Possible market prices on a $5.50 dividend convertible preferred stock (convertible into common at a rate of 2 for 1) (assume a normal preferred stock yield of 5½%)

	Market price common	Investment value preferred	Conversion value preferred	Market price preferred	Premium for conversion option
1. Preferred has no positive conversion value	$40	$100	$80	$110	$10
2. Preferred has a slight positive conversion value	60	100	120	130	10
3. Preferred has a considerable conversion value	80	100	160	162	2

The price of the preferred in example 2 illustrates an optimistic market. The premium for the convertible option is $10 over the conversion price; a current purchaser of the preferred could not convert except at a loss. Although the option price is still only $10, the amount put at risk over the investment value is $30 per share, and the current yield is down to 4.23%. In example 3, the conversion value of the preferred is $160, but the option premium has dropped to $2. In a rational market, the conversion option price goes down if the conversion value rises high enough. A purchaser of the preferred would have $62 at risk. He no longer has any significant comparative safety as against a direct investment in the common. Moreover, the yield on the current price of the preferred is now only 3.4%. In short, as the conversion value rises, the advantage of the hedged position diminishes.

Call Prices and Forced Conversion If the convertible preferred has a call feature, the firm may force conversion by calling in the preferred. If, for example, the preferred is convertible to common 1 for 1, the market price of the common is $70, and the call price of the preferred is $60, an investor has no choice but to change his preferred into common if the preferred is called.

The reasons for a forced conversion vary. The preferred may have a restrictive covenant—such as limitation on bank borrowing or senior debt—which put difficulties in the way of new financing. Or the company might wish to simplify its capital structure preliminary to raising funds on a new security issue. There does not seem to be any obvious economic gain in forcing conversion, for if the common has risen so high as to make this operation possible, then the return on the common must be quite good and the risk in carrying the preferred cannot be very large.

Protective Features on Preferred Shares

The indentures on preferred issues may carry features designed to give additional protection to the preferred stockholders. These provisions are similar to the protective clauses found on debt issues. There may be provisions relating to the maintenance of minimum working capital ratios before common stock dividends can be paid. There may be limitations on the amount of debt the firm may carry. Minimum working capital ratios and financial structure ratios may be required before additional senior debt can be incurred.

A preferred issue may have a sinking fund provision. The sinking fund is used to retire some fixed portion of the preferred stock annually. By reducing the amount of the preferred stock outstanding in a systematic manner, the position of remaining shares is strengthened. The sinking fund provision usually allows the firm to buy the required amount of preferred shares either at the market price or at a predetermined sinking fund call price.[20]

[20]The sinking fund call price is usually lower than the general call price.

Floating New Common Equity Issues

Very seldom does an ongoing, mature company issue major amounts of additional capital stock. Usually, any increase in equity capital is provided by retained earnings. New equity may be required in a new company when the inflow of retained earnings does not provide a sufficient equity base to finance the growth of the firm. Or firm may issue new equity when special investment opportunity opens up or the firm wishes to finance the acquisition of another company. The firm could finance the expansion solely with debt, but if the proportion of debt to equity grows too large, the firm's financial risk increases. Thus a large package of new debt finance often includes an accompanying stock issue.

Rights As a matter of precaution, to make sure the flotation is successful, the announced price on a new issue of shares is usually set below the market price of the existing stock. To compensate the old shareholders for the potential dilution in the value of their holdings, they are given rights, i.e., options to purchase the new issue at the discounted market price. A market develops for the rights, and the existing shareholders may exercise the rights to purchase additional shares or sell the rights on the market.[21] In either case, they are compensated for allowing in new shares at a reduced price. In case where there are major shareholders who exercise some degree of control, exercise of the rights allows them to maintain their proportional share of the voting common.

Advantage of New Share Financing

The main advantage of new capital stock financing is that the shares require no fixed rate of return and there is no maturity date on the principle. Although the new shareholders will share equally with existing shareholders in the company's success, they do not constitute a legal liability if the company does not do as well as expected. If the firm experiences temporary economic setbacks, it need not pay anything to the shareholders until conditions improve. For a new firm, which may suffer operating losses before establishing a profitable level of operations, a heavy equity base if obtainable increases the chance of surviving the lean times. If the firm relied on debt-financing, then the failure to pay its interest obligations could force the firm into bankruptcy.

[21]Because the rights need not be exercised immediately and provide an option to buy at a fixed price, they generally develop a speculative premium over their current exercise value. Of course, the rights will go to their net value when the rights expire.

If the corporation is to justify new common stock financing, it must earn the going rate of return on equity on the new issue. If it does not reach this rate of return, the value of its present stockholders will be drawn down. However, additional equity financing reduces internal risk and lowers the cost of additional debt financing and the overall cost of funds.

References

Bierman Jr., H. (2001a). *Increasing shareholder value: Distribution policy, a corporate finance challenge*. Boston: Kluwer Academic Publishers.

Bierman Jr., H. (2001b). Valuation of stocks with extraordinary growth prospects. *Journal of Investing, 10*, 23–26.

Bierman Jr., H. (2010). My favorite two corporate finance puzzles. In J. R. Aronson, H. Parmet, & R. L. J. Thornton (Eds.), *Variations in economic analysis: Essays in honor of Eli Schwartz*. New York: Springer.

Brealey, R. A., Myers, S. C., & Allen, F. (2006). *Principles of corporate finance* (8th ed.). New York: McGraw-Hill/Irwin.

Dann, L. (1981). Common stock repurchases: An analysis of returns to bondholders & stockholders. *Journal of Financial Economics, 9*, 115–138.

Dhrymes, P. J., & Kurz, M. (1967). Investment, dividends, and external finance behavior of firms. In R. Ferber (Ed.), *Determinants of investment behavior*. New York: Columbia University Press.

Fama, E. F., & Babiak, H. (1968). Dividend policy: An empirical analysis. *Journal of the American Statistical Association, 63*, 1132–1161.

Fisher, L., & Lorie, J. H. (1964). Rates of returns on investments in common stocks. *Journal of Business, 37*, 1–24.

Fisher, L., & Lorie, J. H. (1970). Some studies of variability of returns on investment in common stocks. *Journal of Business, 43*, 99–134.

Fu, F., & Haung, S. (2016). The persistence of long-run abnormal returns following stock repurchases & offerings. *Management Science, 62*, 964–984.

Gordon, M. J., & Shapiro, E. (1956). Capital equipment analysis: The required rate of profit. *Management Science, 3*(1), 102–110.

Gordon, M. J. (1959). Dividends, earnings, and stock prices. *Review of Economics and Statistics, 41*, 96–105.

Gordon, M. J. (1962). The savings investment, and valuation of a corporation. *Review of Economics and Statistics, 44*, 37–51.

Guerard Jr., J. B. (2010). The corporation as a net exporter of funds: Additional evidence. In R. J. Aronson, H. L. Parmet, & R. J. Thornton (Eds.), *Variations in economic analysis: Essays in honor of Eli Schwartz* (p. 2010). New York: Springer.

Guerard Jr., J. B., Bean, A. S., & Andrews, S. (1987). R&D management and corporate financial policy. *Management Science, 3*, 419–1427.

Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Guerard Jr., J. B., Markowitz, H. M., & Xu, G. (2014). The role of effective corporate decisions in the creation of efficient portfolios. *IBM Journal of Research & Development, 58*(4), Paper 11.

Kenan, M. (1970). Models of Equity Valuation. The great SERM bubble. *Journal of Finance, 25,* 243–273.

Lakonishok, J., & Vermaelen, T. (1990). Anomalous price behavior around repurchase tender offers. *The Journal of Finance, 45*, 455–477.

Lorie, J. H., Dodd, P., & Kimpton, M. H. (1985). *The stock market: Theories & evidence* (2nd ed.). Homewood, IL: Richard D. Irwin, Inc.

Schwartz, E., & Aronson, R. J. (1966). The corporate sector: A net exporter of funds. *Southern Economic Journal, 33*, 252–257.

Williams, J. B. (1937). *The theory of investment value*. Cambridge, MA: Harvard University Press.

Long-Term Debt



The use of debt in capital structure has been analyzed since Schwartz (1962) and re-visited in Guerard (2010) in the Springer volume to honor Schwartz. Long-term debt is the term given to those obligations the firm does not have to pay for at least a year. They are also called funded debt or fixed liabilities. Items that may be classed as long-term debt are bonds, debentures, term loans, or, in small firms, mortgages on buildings. The portion of the long-term debt due within the current year is carried in the current liability section of the balance sheet. Firms in the US issue far more debt than equity shares. In most years during the 1963–2003 period, firms have issued six to eight times more debt than equity. The reader is referred to Myers (1977) and Brealey, Myers and Allen (2006). (Of course, most increase in equity is through retained earnings.) One of the most important financial decisions of forms is its debt decision. Internal financing accounts for most of the funds for asset expansion, see Brealey and Myers (2003). Firms issue primarily debt to provide funds to undertake investment, or asset expansion. Issuing debt raises capital for firm growth and expansion without possibly lessening current stockholder control. Corporations issue debt and then retire the previously issued long-term debt when interest rates fall and newer debt can be issued with lower interest charges. The floatation costs are less on debt than on equity, and the cost of debt is less than the expected shareholder return on equity.

Bonds

A bond is a formal long-term promissory note. It is a promise made by the company to pay a fixed sum of money at a determinable future date with a stipulated interest rate. A bond issue is composed of a number of notes, which may be owned by different holders. Since bonds are negotiable instruments, they are readily transferred from one investor to another, and bonafide holders for value are given the usual protections of the negotiable instrument laws against counterclaims.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_9

The major provisions of a bond issue are listed in the bond indenture. This legal document describes the agreements and pledges made between the corporation and the bondholders (creditors). The indenture is usually quite long, and it is impractical to attach a copy to each bond certificate. Each bond, however, carries a reference to the indenture, and the bond owner is presumed to have access to the indenture; at least he can learn of its major features in the original prospectus, written up at the initial issue of the security.

Because in most cases it would be impractical for an individual bond owner to enforce his rights under the indenture, and because it would be a matter of some difficulty for the company to deal with each bondholder separately, the actual enforcement of the indenture is left to a trustee.[1] It is the trustee's duty to ensure that the firm lives up to the provisions in the indenture protecting the bondholder. Because Congressional hearings in the 1930s indicated that some trustees had not always been fully vigilant in enforcing the indenture provisions, and even that some trustees had interests in the corporation in conflict with the interests of the bondholders, the Trust Indenture Act was passed in 1940. This law establishes a high standard of conduct for the trustees and prohibits individuals or firms who might have conflicts of interest with the bondholders from acting as trustees on the indenture.

Mortgage Agreement The usual bond issue is generally backed by a mortgage, basically a lien given to the creditor against some of the firm's assets. This lien gives the creditors a provisional title against the assets which can be exercised if the debtor does not pay his notes as agreed. If the lien is not paid, the mortgaged property may be seized and auctioned off to satisfy the debt. If the property brings more than the mortgage at the sale, the difference goes back to the owner; if the property is not sold for an amount sufficient to satisfy the debt in full, the unpaid portion remains as an obligation of or a general claim against the debtor. The mortgage is given by a company as a protective feature of the indenture to provide additional security for all the owners of the bond issue. A mortgage usually contains a clause requiring minimum maintenance standards on the property and provides for the compulsory retirement of a proportionate part of the outstanding debt if some part of the underlying property is disposed of.

Many bond mortgages contain an "after-acquired" clause. This clause requires that all new property coming under the ownership of the firm fall under the lien of the existing mortgage. If the indenture is closed, i.e., that is no new debt can be issued under the cover of the existing mortgage, it could mean that on the purchase of new assets the firm could offer prospective new lenders only a second mortgage, i.e., only a subsidiary lien on the new properties. In certain instances, these circumstances have put an unfortunate rigidity on the corporation's financial plans. However, in most cases where the mortgage is closed, the lien is restricted to definite property and does not flow over (at least as a prior lien) to other assets.

In most existing bond issues where the after-acquired clause is part of the indenture, the mortgage is conditionally open. Additional debt can be created

[1]The trustee is compensated with an annual fee. The business of being a trustee for major corporation issues is concentrated among the larger commercial banks.

under the existing mortgage up to 60%, for example, of the amount of new assets acquired by the firm. The corporation thus has considerable flexibility in financing new property without diluting the underlying protection of the old bondholders.

The importance of mortgage provisions to the security of the bondholders has been exaggerated in the past. Earnings and cash flow are more relevant to the safety of the interest payment and the ultimate repayment of the bonds than the pledge of property. The property itself is not likely to have much dollar value unless it can generate earnings; it must show economic usefulness in our competitive society if it is to bring any cash on the market. On the other hand, there has been a recent tendency to depreciate any value to the bondholders in a mortgage provision. But a mortgage claim has considerable worth in the event of ultimate financial disaster. In bankruptcy, the mortgage lien and its priority have tremendous importance in determining the amount and value of the securities the bondholders will get if the firm is reorganized or the amount of cash they will receive in settlement of their claim if the bankrupt firm is liquidated. Because the mortgage pledge is generally placed mainly against fixed assets, it provides additional security to the bondholders even within the "going concern" concept. Because depreciation is a charge against income, but not an out-of-pocket expense, a firm having a considerable amount of depreciable fixed assets has a relatively larger cash flow than a firm with similar earnings but relatively little fixed assets.[2] The inflow of cash is one criterion of a firm's ability to carry debt; a firm with earnings and substantial fixed assets is obviously in a safer position to incur funded debt.

Subordinate Mortgages Bonds are sometimes secured by a second or third mortgage if a prior mortgage already exists on the property. In case of default, the pledged assets are used first to satisfy the first mortgage holder's claims, and any remaining value goes to the second mortgage. The holders of any part of an unsatisfied mortgage claim become general creditors.

If the prior bond issue is small compared to the total value of the assets pledged, a secondary lien may actually have considerable strength. If a property valued at $1,000,000 carried a first mortgage of $200,000 and a second mortgage of $300,000, the position of the second mortgage bonds might be better than a first mortgage of, let us say, $700,000 on a property of similar value. Bonds backed by a secondary lien, however, are seldom called second mortgage bonds but tend to bear titles like "Improvement Bonds, 1990," "General Consolidated and Refunding Bonds, 1995," etc. Many railroad bond issues have extremely complicated mortgage positions. They may have first mortgage claims over one section of trackage, secondary liens over another, and tertiary claims over still another part of the company's property. These complexities tend to add difficulties to the problem of reorganization in case of failure.

Other Liens Many bond issues, besides holding mortgages against the firm's fixed assets, may be given a lien against stocks or bonds the borrowing firm carries as

[2]The depreciation charges, although they reduce reported accounting earnings, are not a current out-of-pocket expense. See chapters "The Corporation Balance Sheet" and "The Annual Operating Statements: The Income Statement and Cash Flow Statement".

investments. (A bond issue backed mainly by a contingent claim against such securities is a collateral trust bond.) The pledge of securities does not always add safety to a bond issue. The stocks and bonds are often those of firms or enterprises whose economic position is closely tied to the borrowing company; if the borrowing corporation falls on hard times, the issues of satellite companies may not be too valuable.

Net Working Capital Maintenance Requirements Some indentures require a firm to keep its net working capital above a stipulated proportion of the funded debt. If the firm suffers losses, this proportion may be difficult to maintain. If interest payments and the other indenture provisions are met, however, the trustees often ask the bondholders to relax the working capital requirement temporarily, rather than force failure on the firm. Nevertheless, although working capital maintenance provisions may add little to the debt holder's security in a downturn, in normal periods they warn the management of the dangers of drawing down working capital with excessive dividends or by overinvesting in fixed assets.

Dividend Restrictions Dividend restrictions help safeguard the bond issue by preserving the debt-to-equity ratio in the firm's financial structure. Usually, dividends are restricted to earnings made after the bond issue, perhaps plus some arbitrary amount. Dividends may be limited to a fraction (e.g., two-thirds of all subsequent earnings) or prohibited entirely for a definite period of time. The general effect of these provisions is to keep assets within the firm unless there are operating losses.

Sinking Funds Most bond indentures provide for a periodic repayment of at least a substantial proportion of the bonds before maturity. The required amount of bonds can be repurchased on the market (at market price) or called in by the use of the sinking fund call option. The call option gives the sinking fund trustees the right to buy back a given portion of bonds at a previously fixed price. The periodic retirement of part of the issue strengthens the position of the remaining outstanding bonds.

Restrictions on Creating New Debt If a firm incurs new obligations at too heavy a rate (especially if the new creditors have the same or higher priority than the present creditors), the position of the existing debt holders is eroded. Thus restrictions are generally placed on the amount and types of new issues. Yet the firm must have some flexibility in its financial operations; if restrictions are unreasonable, it may turn to a more liberal competitive source to obtain its funds. Therefore most indentures do not prohibit the firm from borrowing new funds outright but attempt to set bounds of safety before additional debt can be floated. Most indentures have a number of conditions, all of which must be satisfied simultaneously.

In first mortgage bonds, one likely condition is to permit the issue of new bonds up to some percentage of new fixed assets acquired. New bonds are also allowed to replace any part of the issue which may have been retired outside of the required sinking fund. Other common restrictions require the firm to have met certain minimum interest charge coverages over perhaps the last 5 years before new fixed

debt obligations can be created. Often, especially for industrial firms, net working capital has to bear a given percentage to the total proposed funded debt.

Indentures on junior debt—claims of lower priority—may limit the total bank debt that can be created and restrict the amount and kind of senior debt that the firm may incur.

Other Types of Long-Term Debt

Debentures A debenture is a bond unsecured by a lien or mortgage against the firm's assets. It is a general obligation of the corporation and holds the same position as any other general creditor. A debenture issued after other secured long-term debt is already outstanding is weakened by the priority of claim of the senior issue and has to bear a higher yield in order to find acceptance in the capital markets. Commonly, however, the debentures constitute the sole long-term debt issue of industrial firms; if the company is strong, the debenture's financial position may be quite secure, and the interest rate comparatively low.

A type of debenture that has become quite popular in financing commercial finance companies is the subordinated debenture. A major clause of the indenture provides that the claims of the debenture rank behind those of bank loans the company is carrying or may incur. A subordinated debenture issue adds to the safety of the bankers' loans since it provides additional risk-bearing funds in the firm's financial structure.[3] It serves the same function as preferred stock. From the view of the company, the subordinated debentures allow the firm to negotiate more bank borrowing without putting new equity capital immediately into the firm. Moreover, the interest paid on the debentures is deductible for tax purposes, whereas the dividends on a preferred stock issue are not.

Term Loans A company may negotiate a loan with a bank, a group of banks, or an insurance company that may last from 1 to perhaps 15 years, usually 3 to 10 years. These loans, amortized periodically, are designated term loans; whereas the ordinary commercial loan is hardly ever made for more than 90–120 days. The term loan is often used to help finance the purchase of equipment and is customarily secured by a chattel mortgage[4] against new machinery, trucks, etc. They were quite popular in the immediate postwar period when banks were diligently seeking additional ways to put their funds to work. The development of the term loan has provided the medium-sized business with one source of comparatively available, reasonably priced intermediate-term funds.

[3]To provide a little "sweetening" to attract the investor, the subordinated debentures are often convertible.

[4]A chattel mortgage is placed against personality, movable or personal property, as distinguished from realty, i.e., land and building.

Private Placements When a company sells or "places" all of a given issue to a single buyer, e.g., an insurance company, a large bank, or a group of insurance companies or banks, rather than sell its securities to the investors at large in the capital market, the transaction is called a private placement.

Equipment Trust Certificates The equipment trust certificate has been used for some time in financing railroad rolling stock (freight and passenger cars, and locomotives) and now appears in the financing of airline equipment. The basic legal provision of the equipment trust certificates indenture leaves the legal title to the newly purchased aircraft in the possession of the trustee on the issue. The trustee (usually a bank) leases the equipment to the airline. The airline company puts up about 20% of the cost of the new airplanes and agrees to pay an annual or semiannual rent sufficient to cover the interest on the outstanding trust certificates and retire the principal of the certificates, usually in 15 years. The certificates, which have been floated to obtain the money for the equipment, are issued in series so that some part comes due every year; when the last issue is retired (that is, when the last rental payment is made), the airline exercises its option to buy, and the title to the planes is released to the airline.

The holders of equipment trust certificates are not legally creditors of the corporation but participants in a leasing agreement. Legally their position is that of beneficial owners (through a trust arrangement) of an identifiable lot of aircraft. This actually makes their credit position quite strong. Should the airline fail to meet its rental payments (i.e., the principal and interest due on the certificates), the trustee on behalf of the certificate holders can take possession of the planes and sell them to another line. There have been very few cases of default on equipment trust certificates; even in case of bankruptcy and reorganization, the trustee in bankruptcy will keep up the rental payments rather than lose the use of the aircraft, without which the operation of the airline becomes impossible. Equipment trust certificates tend to bear comparatively low-interest rates, Because of their rather special position, which gives them considerable safety.

Long-Term Lease

Some firms acquire fixed assets by renting them rather than purchasing and financing them. The lessee (user) makes periodic payments, monthly, quarterly, or semiannual, payments, to the lessor (owner) for the exclusive rights to use the asset. A long-term lease lasts from 10 to 20 years. One form of long-term lease is sale-and-leaseback. A firm sells a building or plant to another company, sometimes a subsidiary or more frequently an insurance company, to obtain immediate funds. It then commits itself to renting the property on a long-term basis. Typically lease payments are made in advance. Leases may be operating or financial. An operating lease is a short-term obligation, usually for a period less than the economic life of the asset, and is cancelable. Financial leases are longer-termed obligations, approximating the

economic life of the asset, and are non-cancelable. Sale-and-leaseback arrangements usually allow for repurchase at a fixed price after the rental term is over.

Traditionally leases were not disclosed and were referred to as "off-balance sheet" financing. Financial Accounting Standards Board (FASB) Statement No. 13, adopted in 1981, requires capitalization on the balance sheet of leases, known as capital leases, meeting any of the following criteria:

1. The lease transfers the title to the lessee by the end of the lease period.
2. The lease contains an option to purchase the asset at a bargain price.
3. The lease period is equal to or exceeds 75% of the economic life of the asset.
4. The present value of the minimum lease payments equal to exceeds 90% of the fair value of the leased property, at the beginning of the lease.

The absence of all four conditions creates an operating lease than is cancelable and not capitalized. The capitalized value of a capital lease is the present value of the minimum lease payments over the lease period. The discount rate used to determine the present value of the lease payments is the lessee's cost of borrowing or the lessor's implicit interest rate. Operating leases can be disclosed in financial footnotes. Five years of non-cancelable operating leases must be disclosed, annually, as well as the total future minimum lease payments.[5]

A capital lease must be amortized over the lease payment. The lease is amortized as the firm depreciates its assets, and the annual interest implied in the lease payment is treated as an expense. The interest expense reduces net income. In an operating lease, only the lease payment itself is deductible as an expense.

Logically, it would seem that long-term leasing or outright purchase of an asset should be considered alternative forms of financing. Renting requires little immediate outlay of funds, but the future cash flow from the project is reduced by the rental payments. The rental payments must contain all sorts of implicit capital costs since the owner of the asset will expect to earn a fair return on his funds. Obviously, the implicit interest rate on a long-term lease will be higher than the cost of "loan" capital to the firm. Whether renting or purchasing assets is better (for those firms which have easy access to the capital markets) depends upon which method shows a higher rate of return.[6] If no special hindering factors exist, purchase rather than rental

[5]In 2003, the S&P 500 Index firms disclosed $482 billion of off-balance sheet operating lease commitments in their footnotes. J. Weil, "Lease Accounting: Accounting Still has an Impact," *Wall Street Journal*, September 22, 2004, A5–A6.

The operating lease commitments were approximately 8% of reported debt of the S&P firms. For many firms, operating lease obligations are much higher. When UAL, the parent firm of United Airlines files for bankruptcy in December 2002, its assets were $ 24.5 billion, its liabilities were $ 22.2 billion, and its non-cancelable operating lease commitments were $24.5 billion (for aircraft). US Airways had substantial operating obligations at the time of its bankruptcy.

[6]One must discount the difference between the forecasted after-tax cash flow for the project under a purchase plan and under a rental arrangement, in order to obtain the implicit rate of return on the additional funds required under an outright purchase plan. In general, the rate of return is obtained mathematically by finding the discount factor which equates the cash flow plus any remainder value of the assets with the original investment.

is economically justified if the firm can make a going rate of return on the extra funds put into the purchase.

Rental saves the firm from the eventual burden of holding obsolescent equipment. This argument may be true where the rate of technical improvement is rapid, the equipment mobile, and the rental period short. In this case, after the lease is over, the company can obtain new equipment if the old proves unserviceable or better designs are available; however, one must assume that owner of the equipment will require high enough rental payments—on the average—to cover the possibilities of rapid obsolescence. In the usual long-term rental contract on real property, which often runs for 20 or 25 years and where the rent payment amortizes the whole cost of the property over this period, potential losses from obsolescence or changes in the economic value of the location can hardly be shifted from the renter to the owner. As a matter of fact, if the property declines in use value, a renter may be in a more difficult position because he finds it harder to adapt the property to new uses than would an owner.

Renting saves taxes. The rental payments in capital leases include both a charge for interest on the funds invested in the property plus the amortization or repayment of the principal amount of these funds. The whole of the rent is deductible for tax purposes; on a bond issue, only the interest is a deductible expense and repayment of the principal is not. This argument oversimplifies the problem of taxes, however, for if the property is owned by the firm depreciation charges may be deducted before calculating taxable income. The tax advantages, if any, can be found by calculating the rate of return under ownership or rental using the net cash flow after taxes. In many cases, the rental program offers no net tax savings.

A post-inflationary situation can encourage sale-and-leaseback arrangements for the tax advantage. At the end of the period, many companies may have assets of considerable market value which have been written off on the corporation's books so that no more depreciation charges are allowable. The possibilities of a sale-and-leaseback arrangement for tax savings now arises. The firm sells its property to an insurance company (sometimes to its own special real estate subsidiary) and agrees to rent the property back at a rate which would cover the going interest rate plus repayment of the value of the property. The firm pays, out of the receipts for the sale of the property, the capital gains tax on the profits of the sale, but it saves out of future taxes on income the amortization on the principal value.[7]

In general, the sale-and-leaseback arrangement is worthwhile if the present value of the savings to be obtained out of future taxes exceeds the amount of the immediate capital gains tax. In a bankruptcy, the lessor's position may be better than a traditional creditor, as the lessor owns the asset and can retrieve it upon the lessee default.

[7]These savings accrue because no deduction would be left for depreciation were the firm to keep the title to the property.

Lease Accounting: Recent Changes

In 2016, FASB introduced a new standard that will require organizations that lease assets—referred to as "lessees"—to recognize on the balance sheet the assets and liabilities for the rights and obligations created by those leases.

Under the new guidance, a lessee will be required to recognize assets and liabilities for leases with lease terms of more than 12 months. Consistent with Generally Accepted Accounting Principles (GAAP), the recognition, measurement, and presentation of expenses and cash flows arising from a lease by a lessee primarily will depend on its classification as a finance or operating lease.

However, unlike current GAAP—which requires only capital leases to be recognized on the balance sheet—the new ASU will require both types of leases to be recognized on the balance sheet. The ASU also will require disclosures to help investors and other financial statement users better understand the amount, timing, and uncertainty of cash flows arising from leases. These disclosures include qualitative and quantitative requirements, providing additional information about the amounts recorded in the financial statements.

The accounting by organizations that own the assets leased by the lessee—also known as lessor accounting—will remain largely unchanged from current GAAP. However, the ASU contains some targeted improvements that are intended to align, where necessary, lessor accounting with the lessee accounting model and with the updated revenue recognition guidance issued in 2014.

Off-balance sheet accounting of operating leases can skew various financial ratios positively and can mask underlying contractual liabilities for a firm that are not recognized on the balance sheet. The latest guidance on FASB is a step in the right direction that helps to level the playing field and remove the arbitrariness that was inherent in the balance sheet treatment of operating and capital leases.

The Cost of Debt Capital

The explicit cost of a long-term debt issue is the interest rate the firm pays on the net funds obtained by the bond issue. But there are costs other than the explicit interest rate which a firm must consider before floating a debt issue. Debt increases the financial risk of the firm and thus raises the cost of equity capital. Further, every additional amount of debt has its effect on the interest rate that the firm will eventually have to pay on its overall debt and thus may have a marginal cost measurably higher than the actual rate at which the bonds are floated.

A firm also incurs intangible "managerial" costs on a bond or debt issue. It must meet cash drains periodically as the issue is amortized, and the indenture on the bond issue usually carries restrictive clauses that may at some time hamper managerial freedom or flexibility.

Of course, some of the problems and costs of incurring long-term debt exist with other types of debt financing. For example, a firm may have the choice of using more short-term debt (economizing on net working capital) or using additional long-term debt. Long-term debt may have a higher explicit interest cost, but heavy short-term debt brings with it more implicit costs and difficulties. If open-book creditors are not paid on time, the costs of lost cash discounts can come to an astounding rate.[8] Heavy short-term debt may force the firm to forego immediately profitable buying opportunities or hurriedly to arrange financing at inopportune times. Nor is the notion that short-term debt is passable but fixed debt is risky or dangerous a valid one. Long-term debt has its risks, but trying continually to balance and meet current indebtedness is riskier. If current debt is defined as those obligations due within the year, it is a truism that a firm fails by being unable to meet current obligations, not by being unable to meet its funded debt.

The intangible costs of all debt issues are extremely important; however, the going interest rate (the explicit cost) is an index of certain aspects of these intangible costs. Furthermore, the interest rate is an observable, measurable phenomenon, whereas the implicit costs are not.

The interest rate a firm has to pay on any debt issue is conditioned by four major factors: (1) the basic level of prevailing market interest rates; (2) the length to maturity of the issue; (3) the particular type of issue, its provisions, and its security position relative to other financing the firm is already carrying or may incur; (4) the general credit standing of the firm.

How can one measure the credit quality of a firm or a firm's particular issue? The marketplace relies upon several bond rating services to assess risk. Mergent's and Standard & Poor's have long histories of rating debt. Standard & Poor's Corporation uses the following debt ratings:

Rating	Financial strength or condition
AAA	Extremely strong to repay debt and interest
AA	Strong capacity to repay principal and interest
A	Strong capacity to serve debt but more volatile than AA debt
BBB	Adequate debt-serving capacity but more volatile than A debt
BB	Bonds have long-term uncertainty but adequate short-term debt-serving capacity
B	Larger volatility than BB Bonds
CCC	Current vulnerability to default and dependent on favorable economic conditions to service debt
CC	Debt subordinated to CCC senior debt
C	Same as CC
D	Debt is in default

[8]Suppose a firm is given customary terms of 2/10 n/30 e.o.m. This means the firm may deduct 2% from the monthly statement if it pays within the first 10 days of the month. If it misses this period, the gross amount is due 30 days from the statement date. The 2% discount lost if the firm does not pay within 10 days of the monthly statement amounts to 36% on an annual basis.

All these factors are significant, although some of them may be objectively determined and others are subjective. Standard & Poor's uses the following criteria to partially determine debt ratings:

Adjusted key industrial financial ratios. US industrial long-term debt

3-year (1998–2000) medians	AAA	AA	A	BBB	BB	B	CCC
EBIT int. cov. (x)	21.4	10.1	6.1	3.7	2.1	0.8	0.1
EBITDA int. cov. (X)	26.5	12.9	9.1	5.8	3.4	1.8	1.3
Free oper. cash flow/total debt (%)	84.2	25.2	15	8.5	2.6	−3.2	−12.9
FFO/total debt (%)	128.8	55.4	43.2	30.8	18.8	7.8	1.6
Return on capital (%)	34.9	21.7	19.4	13.6	11.6	6.6	1.0
Operating income/sales (%)	27	22.1	18.6	15.4	15.9	11.9	11.9
Long-term debt/capital (%)	13.3	28.2	33.9	42.5	57.2	69.7	68.8
Total debt/capital (incl. STD) (%)	22.9	37.7	42.5	48.2	62.6	74.8	87.7
Companies	8	29	136	218	273	281	22

Source: www.standardandpoors.com
Data for earlier years and in greater detail are available by subscribing to Standard & Poor's CreditStats

The general level and time structure of the interest rate can be determined by ascertaining the going market yield on various bonds currently traded. The market yield—which is the important quantity—is not the same as the coupon or stated interest rate of the bond. The true yield on a bond is obtained by considering (1) the contractual or stated rate on the bond, (2) the remaining life to maturity of the bond, and (3) the current market price of the bond. The purchasers or investors in a bond obtain two items of value: they are to receive the face or par value of the bond at maturity, and they are to receive the stated interest rate annually or semiannually over the life of the instrument. The price they pay is the present value of these two features; the yield they obtain is that rate which equates the present value of the two items with the going market price. This is represented by the following equation:

$$\text{Market Price} = \frac{CR}{(1+i)} + \frac{CR}{(1+i)^2} + \frac{CR}{(1+i)^3} \cdots + \frac{CR}{(1+i)^n} + \frac{FV}{(1+n)^n}$$

CR is the contractual or coupon rate, FV equals the face value of the bond to be paid at maturity, n is the number of years to maturity, and i is the unknown yield. Although the formula appears complex, it is not difficult to use. Furthermore, every investment house, knowledgeable investor, or broker has a computer program table that quickly reveals the yield if the market price is known or gives the price which should be paid if the buyer wishes to stipulate the yield.

Bond Prices on the Market Since the maturity date and the coupon or stated interest rate are fixed at the time of the initial flotation, the market price of the bond moves in response to the changing level of the interest rate on the market. If the current market

interest rate is higher than the fixed rate on a bond, the bond sells below par value or at a discount. If the going interest rate is below the coupon rate, the bond sells at a premium. The general bond market moves inversely to the interest rate; when the rate is falling, bond prices move upward, and when the yield rate is rising bond prices drop. A "strong" market signifies a lowering of interest rates and a "weak" market means the opposite.

The general level of interest rates is determined by the supply and demand for investment funds on the capital and money markets. The bond market, however, is not the model of a perfectly free market because the interest rate is influenced by government monetary and fiscal policy as implemented by the Federal Reserve System and the Treasury. The bond market is also a speculative market; the market interest rate is influenced by forecasts or expectations as to the future level of the interest rate.

Supplying the capital funds market are the net purchasers of bonds—insurance companies, investment companies, banks, trust funds, pension funds, and individual investors. The demand for funds comes from the net issuers of bonds—state and municipal governments, the federal government and its agencies, foreign governments, international agencies, and private corporations. The market can be analyzed in terms of the supply and demand for funds or in terms of the supply and demand for securities. When the demand for funds increases at a faster rate than supply, the demanders of funds (borrowers of capital) have to pay higher interest rates. Or, conversely, if the supply of bonds on the market increases relative to the demand for debt securities, bond prices fall, i.e., the interest rate rises. Of course, the opposite of these statements holds, and during a period of easy money, when the supply of funds seeking investment is relatively greater than the demand for capital, the interest rate falls, and bond prices rise.

Level and Structure of the Interest Rates

Table 1 presents average interest rates prevailing on high-grade securities for the 1960–2018 time period. Interest rates, on even the best securities, show considerable movement. The general movement of interest rates reported in Table 1 show dramatic downward movement in the post-Gobal Financial Cisis time period, 2009–2018. Treasury bonds are considered the safest of all investments and therefore their interest rate is lower than other bonds of comparative maturity and tax status. The yield curve shows the relationship between the time to maturity and the yield to maturity, for a given issuer (or risk of issuer). That is, as time to maturity increases, what happens to the yield to maturity of debt? From the view of default risk, it is usually felt that short-term issues (of equal quality) are safer, from the investor's point of view, than long-term issues.

The Liquidity Preference Theory of the Term Structure If any credit risk is involved at all, the short-term issue is safer because the borrower's position at repayment time is more easily forecast. Hicks (1937) referred to the additional

Table 1 Selected interest rates in the U.S., in selected years

	U.S. Treasury securities					Corporate bonds		High-grade muni. bonds	Home mortgage rates	Prime rate charged	Federal funds
	Bills		Constant maturities								
	3-month	6-month	3-year	10-year	30-year	Aaa	Baa				
1960	2.93	3.25	3.98	4.12	–	4.41	5.19	3.73	–	4.82	3.53
1965	3.95	4.06	4.22	4.28	–	4.49	4.87	3.27	5.81	4.54	4.04
1970	6.46	6.56	7.29	7.35	–	8.04	9.11	6.51	8.45	7.91	5.95
1975	5.84	6.12	7.49	7.99	–	8.83	10.61	6.89	9.00	7.86	6.25
1980	11.51	11.37	11.55	11.46	11.27	11.94	13.67	8.51	12.66	15.27	11.77
1985	7.48	7.66	9.64	10.62	10.79	11.37	12.72	9.18	11.55	9.93	7.69
1990	7.51	7.47	8.26	8.55	8.61	9.32	10.36	7.25	10.05	10.01	6.98
1995	5.51	5.59	6.25	6.57	6.88	7.59	8.20	5.95	7.87	8.83	5.21
2000	5.85	5.92	6.22	6.03	5.94	7.62	8.36	5.77	7.52	9.23	5.73
2001	3.45	3.39	4.09	5.02	5.49	7.08	7.95	5.19	7.00	6.91	3.40
2002	1.62	1.69	3.10	4.61	–	6.49	7.80	5.05	6.43	4.67	1.17
2003	1.02	1.06	2.10	4.01		5.67	6.77	4.73	5.80	4.12	2.12
2004	1.38	1.57	2.78	4.27	–	5.63	6.39	4.63	5.77	4.34	1.35
2005	3.16	3.40	3.93	4.29	–	5.24	6.06	4.29	5.94	6.19	3.22
2007	4.41	4.48	4.35	4.63	4.84	5.56	6.48	4.42	6.41	8.05	5.02
2008	1.48	1.71	2.24	3.66	4.28	5.63	7.45	4.80	6.05	5.09	1.92
2009	0.16	0.29	1.43	3.26	4.08	5.31	7.30	4.64	5.14	3.25	0.16
2010	0.14	0.20	1.11	3.22	4.25	4.94	6.04	4.16	4.80	3.25	0.18
2012	0.09	0.13	0.38	1.80	2.92	3.67	4.94	3.14	3.69	3.25	0.14

(continued)

Table 1 (continued)

	U.S. Treasury securities					Corporate bonds		High-grade muni. bonds	Home mortgage rates	Prime rate charged	Federal funds
	Bills		Constant maturities								
	3-month	6-month	3-year	10-year	30-year	Aaa	Baa				
2015	0.06	0.17	1.02	2.14	2.84	3.89	5.00	3.48	4.01	3.26	0.13
2017	0.94	1.05	1.58	2.33	2.89	3.74	4.44	3.36	3.97	4.10	1.00
2018	1.94	2.10	2.63	2.91	3.11	3.93	4.80	3.53	4.53	4.91	1.83

Source: Economic Report of the President, Table B-42, March 2019 p. 684

yield on longer-term instruments as a "liquidity premium." Furthermore, short-term debt instruments are less subject to the interest rate risk. If the basic interest rate rises, the near maturity date of the short-term issue prevents any significant capital loss. In the same circumstance, an investor in long-term bonds may take a considerable drop in price if he wishes to sell them on the market. Because by and large, the investor risks less on short-term issues, in a normal market the short-term rate is generally below the long-term rate.

The Pure Expectations Theory of the Term Structure The pure expectations theory of term structure of interest rates holds that the expected one-period return on investments is the same, regardless of the maturity of the security. The rate of interest, or yield, on a bond, is its current, or spot, rate. A forward rate is the interest rate that will prevail in the future. If the current (spot) 2-year rate is R_2, and the current 3-year rate is R_3, then there is a 1-year interest rate, 2 years from now, denoted r_1 that equates the 2-year and 3-year bond yields.

$$(1 + R_3)^3 = \left(1 + R_2^2\right)^2 (1 + r_1)$$

$$1 + r_1 = \frac{(1 + R_3)^3}{(1 + R_2)^2}$$

$$r_1 = \frac{(1 + R_3)^3}{(1 + R_2)^2} - 1$$

If the current 2-year bond rate is 4%, and the current 3-year bond rate is 5%, then the 1-year bond rate, 2 years from now, should be

$$r_1 = \frac{(1.05)^3}{(1.04)^2} - 1$$

$$r_1 = \frac{1.1576}{1.0816} - 1 = 7.03$$

If the 1-year bond rate 2 years from now exceeds 7.03%, then the investor should have purchased the 2-year bond and reinvested the proceeds, rather than purchase the current 3-year bond. It is obvious to the reader that an infinite number of expectations are incorporated into the yield curve. When the short-term rate rises above the long-term rate, it may be because of an inflationary condition which is expected to decline in the future.[9] A general forecast that the long-term rate is

[9] Although the short term rate went to 22.0% as against 13% on long-term treasuries, subsequently as the rate of inflation fell, the interest rates declined, and the long-term bonds proved to be the better investment.

relatively high and likely to decline will lead borrowers to postpone long-term borrowing and finance short term for the time being.[10]

The Market Segmentation Theory of the Term Structure Life insurance companies are renowned for being able to forecast when their clients will die and thus can manage their cash flow, leading them to be one of the larger purchasers of corporate bonds and being able to operate in the longer time to maturity instruments. There is a "market segmentation" theory of the term structure of interest rates which holds that particular financial intermediaries, matching their asset and liability lives, operate in different regions of the yield curve.

Knowledge of the factors affecting the general level of the rates and the time structure of the rates is important to both the investor and the corporate financial management. If the management desires to save on costs, it needs some insight into the structure and level of the interest rate and the factors likely to affect its future course. The timing of an issue is an important consideration. By anticipating long-run needs for funds, management can advance issue dates to take advantage of prevailing low rates; by financing short term, they can defer an issue if the long-term rates appear cyclically high.

In addition, since the maturity date of issue is a variable feature up to the time it is floated, management should try to ascertain the best maturity for an issue, considering both present and future costs.

When a firm decides whether it can make profitable use of long-term borrowed capital, the question arises of exactly how high an interest rate it has to pay. As the previous discussion indicated, the going interest rate and the length of the maturity of the issue have a great deal to do with the effective interest rate the firm has to offer in order to float its bonds. Also to be considered are the general credit standing of the firm, the type of issue, and the financial structure the firm already carries. If the firm is considered sound, with a good history of repayment on past obligations, having a good market position for its product and relatively stable demand, its interest cost tends to be low. The greater the equity the firm has relative to its debt, the lower is the interest rate it has to pay. A heavy amount of debt in proportion to the firm's ownership capital (residual risk-bearing capital) calls for a higher interest rate on any given issue.

Setting the Rate on a New Issue A firm's investment bankers can evaluate a prospective issue and indicate the approximate interest rate it will have to bear. Many issues are already being traded on the secondary market, and some are likely to resemble the position of the proposed issue. A fairly precise estimate of the necessary interest rate can be achieved by comparing the new security to those with similar features already on the market.

[10]For a period in the 1970s, the short-term rate went considerably above the long-term rate during a period of inflation. See Macaulay (1938) and Malkiel (1962a, 1962b) for further analysis of expectations and the term structure of interest rates.

The commission to be paid the investment bankers for underwriting and selling the issue is another cost entering the calculation. These commissions, or flotation costs, are relatively low on high-grade debt issues. Flotation costs are brought into perspective by comparing the effective yield which the investors will obtain (as determined by the market price they pay) and the effective interest rate the company pays (as determined by the net proceeds of the issue). The difference between these two rates represents the flotation costs.

Often a new issue is not floated at par but at a small discount or a small premium. The coupon rate on bonds is customarily set at round percentages or fractions, such as 4%, 4½%, or 5%, occasionally at quarters, e.g., 4¾% or 5¼%; hardly ever are any fractions below ¼% used. Effective yields on the market, however, move over a far more continuous range than the coupon rates. Thus, if the effective rate on the market is 4.68% for a given class of 30-year bonds and a firm floats a new issue with a coupon rate of 4¾%, the new issue will be priced out at 101½, a premium of 1½ points, making the effective rate 4.68%. Similarly, if a new issue bore a 5% rate and the going market rate was 5.10%, it would be issued at a discount; it would sell for 99 5/8. On the issuing company's books, bond discounts are carried as deferred debits; bond premiums are carried halfway between the liability and equity section as a deferred credit. Each interest payment period, a proportionate amount of the discount or premium is amortized and applied against the current charge for interest expense. As the discount is amortized it increases the current charge for interest expense, whereas the amortization of a bond premium reduces the amount charged for interest expense. Although bond premium and discount complicate the accounting, there is no particular advantage or disadvantage to issuing bonds at a discount or premium; in most issues the coupon rate is set as close to the market as possible in order to float the issue near par. If the firm sets the coupon rate above the market, it can obtain more funds than the face value of the issue, but it pays for these funds through the higher than necessary coupon rate. On the other hand, a firm can set the coupon rate below market; it saves on the rate but does not obtain the full face value of the issue. In any case, the effective rate the firm pays on the net proceeds it receives from an issue is the true explicit cost of these funds.

The Call Feature on Bonds

About 95% of corporate bond issues outstanding today carry a call feature. A provision in the bond indenture enables the company to call, i.e., retire or pay back, some or all of the bonds before maturity at the company's option. The firm exercises its option by paying a penalty or call premium of a few points above the face value of the bonds. The general call premium is usually set somewhat higher than the sinking fund call and declines over the life of the issue so that it reaches zero near the maturity date.

The call option gives the borrowing firm an opportunity to repay its obligations before maturity, should any of the following develop: (1) The corporation obtains a heavy inflow of funds, and the best use of these funds at the moment appears to be a

repayment of debt in order to save the interest charges. (2) New financing seems desirable (possibly in connection with an acquisition or merger), and the old debt contains some onerous or restrictive provisions interfering with the new plans. (3) Interest rates in the market drop, and the company can save on interest costs by calling the old issue and replacing it with a new one at a lower effective rate.

Replacing one issue with another, refunding before maturity, has many interesting facets. In favor of the operation is the lower effective interest rate obtainable; opposed to it is the call premium, other costs of retiring the old issue, and flotation costs on the new issue. The reader is referred to Schwartz (1967), Bierman (1972), and Van Horne (2002a) for more complete bond refunding analysis. However, the profitability of the maneuver can be ascertained by studying the market. The price the old issue would sell at, if there were no call, can be calculated by applying going market yields to the issue and obtaining the estimated free market price.[11] The problem can be illustrated by an example:

Amount of bond issue	$10,000,000
Coupon rate	5¼%
Years to maturity	25 years
Call price at this time	104
Present market interest rate on similar bonds	4½%
Additional expenses of calling old bonds and floating new issue	$250,000

If there were no call feature, the old bonds would sell on the market for approximately 111¼ giving a yield to maturity of 4.5%. The net present value of the interest savings on a refunding operation can be obtained by subtracting the call price and the additional flotation expenses from the implicit market price, thus:

Implicit market price (old bonds) of the whole issue		$11,125,000
Less: call price	$10,400,000	
Additional expenses	250,000	10,650,000
Net present value of annual interest savings over the life of the issue		$475,000

In the problem above, the annual difference between the interest charge on the old bonds at 5¼% and the interest rate obtainable on a new issue at 4½% comes to $75,000 ($525,000 minus $450,000) for 25 years. Against this saving must be set the call premium of $400,000 and various additional flotation and retirement expenses of $250,000. The solution illustrated essentially brings all the items to a present

[11]This cannot be obtained for the bond directly from the market because the effect of a drop in interest rates combined with a call price may force the bond investors into a locked-in position. They cannot sell at the call price or lower without losing on the interest rate; potential buyers cannot purchase above the call price because of the obvious danger of having the bonds called in by the company at the lower price. Any sales recorded during such a period are not likely to be normal market sales.

value basis and gives the immediate value of the savings obtainable on a refunding operation.

The call feature operates essentially for the benefit of the borrowing firms because they can retire the bonds when it is to their advantage. For the investors in the bonds, the call feature may deprive them of capital appreciation on their bonds or of receiving the benefit of the higher contractual interest rate if the market interest rate falls. On the other hand, when easy money prevails and the long-term interest rate is low, investors are not too concerned about giving the corporation a relatively flexible call feature. With the interest rate already at a secular low, it does not seem that any further drop would be sufficient to make a refunding worthwhile. When interest rates are at a "normal" level bond investors are more wary of allowing the company too lax a call feature. The call penalty or premium may be set a bit higher, perhaps the initial coupon rate may be set a fraction of a point higher, and/or the bonds may be made noncallable or nonredeemable for a certain number of years after issue.

Convertible Bonds and Bonds with Warrants Attached

A convertible bond or debenture may at the option of the holder be turned into the common stock of the company at some agreed ratio or rate of exchange. Sometimes the convertible feature involves a fixed ratio; a $500 debenture may be turned into ten shares of stock. More commonly, the debentures may be turned into common shares for a fixed amount of face value plus an additional set amount of cash. For example, $50 of face value of the debentures plus $20 in cash can be turned in for one share of common stock. Often the cash premium increases over time to compensate for the likely rise in the book value of the common shares. The agreement on the conversion privilege allows for a compensating adjustment in the rate of conversion in case of any stock splits, stock dividends, etc.

A warrant differs from a conversion feature in that the warrant can be detached from the bond and exercised separately. A warrant entitles its holder to a share of stock upon surrender of the warrant plus, usually, a set amount of money.[12] Some warrants run indefinitely, whereas others expire after a given term. Again, as with the convertible, the cash requirement may be increased at a set rate over time. Since the warrant can be detached and exercised separately, the original holder can sell his warrant and the subsequent holder can exercise the warrants, hold them, or trade them as he sees fit. A regular trading market may develop for the warrants.

Whereas the call feature operates for the benefit of the corporation, a conversion feature is to the advantage of the security holder. If the company's fortunes are fair,

[12]Warrants sometimes appear after reorganizations, when they are given to the stockholders allowing them the chance to get back into the company if the reorganized company should do better than its failed predecessor.

the owners of a convertible security can hold on to their bonds and obtain the contracted income and repayment of the principal at maturity. If the firm does well, the owner of a convertible security can choose to change the bond into common stock, sharing its increased dividends and appreciation. (Of course, this does not preclude the possibility of a secondary purchaser of a convertible security paying a relatively high premium for the conversion feature and suffering a decline in the value of the investment if the underlying stock drops on the market.)

The market price of a convertible security is subject to forces in addition to those affecting the ordinary bond or debenture. If the conversion feature is not immediately valuable, the bonds will sell close to their basic investment value, i.e., their value as a normal credit instrument. But if the market price of the underlying stock is high enough, the market price of the convertible bonds is determined basically by the price level of the shares. In no case will the bonds sell below their conversion value. As a matter of fact, in an optimistic atmosphere, market forces may push the price above the conversion value, so that a premium exists for the conversion privilege beyond its explicit value.

Some issues are both callable and convertible. If the conversion value is high enough, such bonds furnish an exception to the rule that a security cannot sell above its call price. Here the rule that an issue never sells below its conversion value has more force. If a purchaser buys a bond above its redemption price but at its conversion value into common, he has nothing to fear from a call. If the company calls the bond, he can escape loss by turning the bond into its equivalent in common stock. If he wants his investment back, he can sell the shares. Or he can find someone willing to buy the bonds and make the conversion; there exists a going market.

These relationships make possible an operation known as forcing conversion. This is done simply by calling an issue when the market price (due to the conversion value) is above the call price. Since no one takes a loss deliberately, no securities will be surrendered to the company; almost all will be turned into stock. The only rationale for forcing a conversion would seem to be a desire to get rid of an issue that in one way or another is troublesome to the arrangement of future financing.

Since a convertible issue may be eventually turned into common shares, the common stockholders are generally given the preemptive right on new convertible issues. This right is especially important where the price on new securities is below their immediate conversion value.[13]

The disadvantage of the convertible security rests on the potential dilution of the per share earnings of the common stock when and if the senior issue is converted. On the other hand, there are occasions when the convertible is the easiest and cheapest issue to float and is most readily accepted by the market. For some issues, the risk position is such that at the given interest rate the convertible feature is necessary to make the issue attractive to outside investors.[14] Convertible debentures often appeal

[13]American Telephone and Telegraph Company for years raised equity funds by issuing rights to the stockholders for convertible debentures priced below their conversion value.

[14]This is probably true of the subordinated debentures—a popular security of the large finance companies, which is made more acceptable to the market by its convertible feature.

to institutional investors, who are generally limited to a small amount of stock in their portfolio. A convertible debenture may be on the legally approved list for insurance company portfolios and still give the investment department a chance to try their hand at making a little capital appreciation.

Sometimes the convertible security is initially floated at a price below its value in underlying shares; basically, the company is floating an equity issue. But instead of coming in all at once, the equity funds flow in more conveniently over time. The company gains on the lower financing costs and saves the tax on the interest for the period investors hold the bonds before deciding to convert.

The use of a convertible debt issue seems most logical in the financing of an untried company. The outside investor may want more safety than is provided by common shares and yet desire some gain, if the company is successful, as compensation for the extra risk of investing in a newer or smaller company. The use of convertible debentures, therefore, as the prime financial device of the Small Business Investment Companies in providing capital for smaller firms (falling under the special small business classifications) appears eminently appropriate.[15]

Some explanations given for the use of convertible securities are not always valid. Thus it is suggested that the convertible security is a good method of raising delayed equity funds when the current market for equities (common stocks) is depressed. The convertible security brings in immediate funds; when the stock market recovers sufficiently, the debentures are turned into shares. Thus the company avoids floating new shares at a very low price and obtains equity funds on better terms after the market rises. The issue of the convertible should be regarded simply as a delayed issue of common stock.

The fallacy in the foregoing explanation lies in the notion that business activities necessarily proceed in a smooth upward trend. But if earnings take a few sharp dips, then the in service chases on the debt issue, convertible, or otherwise may prove a strain on the financial resources of the firm. Under such circumstances, obviously, the debt holders do not convert. It is precisely the initial risk-bearing characteristic of equity financing, that is, the non-fixity of its' return in good or bad times that makes it an important part of the financial mix. The convertible issue (as long as it is outstanding) provides none of the cushion against business fluctuations which constitutes the uniquely valuable aspect of common stock financing. Thus the convertible issue will do the company no good if its estimate of future business developments is wrong. Moreover, if the company were certain that its shares were undervalued, it would do better to raise present funds with a callable debt issue. When the company's shares rise in price later, new stock can be floated, and the debt retired. There is no additional risk in this procedure over the flotation of a convertible, and if the company's fortunes improve, the present shareholders do not have to sacrifice part of their earnings gain or capital appreciation to the holders of the convertibles.

[15]The Small Business Investment Companies are the outgrowth of the Federal Small Business Act of 1958. A major objective of the Act is to help make more long-term capital available for smaller firms.

The Advantages and Disadvantages of Long-Term Debt

To the conservative investor, a bond represents a relatively safe investment of his money at a stated rate of return. True, there have been some defaults on bond issues with subsequent losses to the bondholders, but by and large a diversified portfolio of bonds in the past has earned enough to cover any loss in principal and still give its owner a positive rate of return. (Substantial evidence indicates that a diversified mix of secondary bonds over time has outperformed prime bonds.[16]) Because of their relative stability, bonds are a desirable component of the investment portfolio of insurance companies, pension funds, and other financial intermediaries, whose own obligations to policyholders, savers, pensioners, and annuitants are generally on a fixed or contractual basis. These financial intermediaries have very substantial funds to invest and are an important part of the capital market.

The desirability of bonds as investment media for large segments of the financial market has enabled corporations to obtain funds on debt issues at comparatively low out-of-pocket costs. This and the fact that bond interest charges are deductible before computing the corporate profit tax are the main advantage of using debt financing for the firm. On the other hand, the use of debt financing binds the firm to the obligation of periodically meeting fixed interest charges and to the repayment of the principal as agreed. These obligations add certain rigidity to the financial operations and increase the financial risk of the firm.

Malkiel's Bond Theorems

Let us look at several aspects of the pricing of long-term debt, as seen with 10-year bonds. Let us assume that the current 10-year bond yield is 5%. Let us further assume that our bond pays an annual (once a year) interest payment. A bond with a coupon rate of 5% would sell for $1000, its par value. If we purchased a 5% bond today, and interest rates fell to 4% tomorrow, then we would own a bond with a higher yield than the current market bond selling at par, and our bond would sell for a premium, a price exceeding its face, or par, value! How can we price us a bond? Remember that a bond pays interest, and its interest payment is constant during the life of the bond. That is, the interest payment is an annuity, a fixed dollar payment for a specified time period. We can price a bond by calculating the present value of the bond's interest payments plus the present value of the return of the bond principal. In the case of our 10-year 5% coupon bond, with a 4% yield, the bond price is

[16]See W. Bradford Hickman, *Corporate Bond Quality and Investor Experience*, Princeton University Press, 1958.

$$B_P = \$50\,(7.7217) + \$1000\,(.6139) = \$1081.15$$

The bond price rises to $1081.15, an 8.1% increase. If the interest rate rose to 6%, then our 10-year 5% bond would not yield a rate equal to the current yield of bonds and would sell for a discount, a price less than its par value.

$$B_P = \$50\,(7.3601) + \$1000\,(.5584) = \$926.41$$

Note that the value of the bond falls to $926.41, a loss of 7.36% if the interest rate rises to 6%. The reader notes that if the interest rate rises 1%, then the bond price falls 7.36%, whereas a 1% decrease in the interest rate creates an 8.12% bond price increase. We can establish two of Malkiel's bond theorems:

1. A rise in the interest rate causes bond price to fall, whereas a decrease in the interest rate causes bond prices to rise.
2. The bond price appreciation when interest rates fall is greater than the bond price depreciation when interest rates rise.

Let us now create a 3 × 3 bond price matrix to establish several additional bond theorems:

Coupon rates

		4.0	5.0	6.0
Yields (interest rates)	4.0	$1000.00	$50 (8.1109) + $1000 (.6756) = $1081.15	$60 (8.1109) + $1000 (.6756) = $1162.25
	5.0	$40 (7.7217) + $1000 (−6139) = $922.77	$1000.00	$60 (7.7217) + $1000 (.6139) = $1077.20
	6.0	$40 (7.3601) + $1000 (.5584) = $852.80	$50 (7.3601) + $1000 (.5584) = $926.41	$1000.00
OR				
		4.0	5.0	6.0
	4.0	$1000.00	$1081.15	$1162.25
	5.0	$922.77	$1000.00	$1077.20
	6.0	$852.80	$926.41	$1000.00

If the initial interest rate was 5%, and interest rates fell to 4%, which can be caused by a Federal Reserve (the "Fed") Federal Funds rate cut, then a 4% coupon bond rises in value from $922.77 to $1000, an increase of 8.37%. The corresponding appreciations for the 5% and 6% coupon bonds are 8.12 and ($1000 to $1081.15) and 7.90% ($1077.20 to $1162.25), respectively. When interest rates fall, and bond prices rise, an investor's return is higher in lower coupon bonds. If the Fed raised interest rates from 5% to 6%, the 4% coupon bond price falls to $852.80, a decline of 7.58%. The corresponding depreciation for 5% and 6% bonds are 7.36 ($1000 to $926.41) and 7.17 ($1077.20 to $1000)%, respectively. An investor's loss is minimized in higher coupon bonds when interest rates rise. It appears that the lower

coupon bonds experience the highest appreciation when interest rates fall and the highest depreciation when interest rates rise. A third bond theorem by Malkiel holds that the lower the coupon rate, the greater the change in bond prices, for a given change in the interest rate. The relationship between coupon rates and changes in interest rates is an elasticity concept.[17] The ultimate interest rate play is to purchase zero-coupon bonds if interest rates are expected to decline. The fourth bond theorem holds that the longer the time to maturity of the bond, the greater the bond price volatility for a given change in the interest rate. See Malkiel (1962a).

If the initial interest rate was 5%, and interest rates fell to 4%, then a 5% bond would sell for $922.77, a loss of 7.72%. A 4% coupon bond would fall in price from $1081.15 at an interest rate of 5% to $1000 or par. The 4% coupon bond declines 8.12% of its value. A corresponding 6% coupon bond falls from a price of $926.41, at an interest rate of 5%, to $852.80, at an interest rate of 4%. The 6% coupon bond falls 7.58% when interest rates rise. If interest rates rise from 5% to 6%, the 4% coupon bond rises to $1162.25, an increase of 7.50%, while a 5% coupon bond rises to $1077.20, a 7.72% increase, and the 6% coupon bond rises from $926.41 to $1000, an increase of 7.94%.

Retirement of Debt

Corporations find it profitable at times to add to their debt financing. Other circumstances may call for the reduction of debt. An actual reduction of the aggregate amount of debt financing should, however, be distinguished from merely shifting its distribution or form. A firm may retire particular debt instruments or issues, yet the total amount of debt financing remains constant. Moreover, if ownership funds and debt financing are substitutable for each other within limits, the range in which various types of liability financing, i.e., bank borrowing, open-book credit, and funded debt, may be substituted for each other is far wider. It is easy to detect the ordinary refunding operation either at or before maturity when one issue is directly replaced by another; but if a firm gradually accumulates funds from many credit

[17] The elasticity concept is proportional to the duration concept. Duration measures the sensitivity of bond prices to changes in the yield curve. The yield curve can slope and shift in different scenarios. The duration of a bond may be found by solving the equation:

$\frac{dP}{P} = -D_i dr$

where

P = bond price,
dP = change in bond price,
dr = change in interest rates,
D = duration.

$D_i = \frac{-\frac{dP}{P}}{dr}$.

sources to repay other debt instruments, what appears to be a retirement of debt is actually a mere shifting within the debt structure.

For purposes of this discussion, debt reduction is defined as the retirement of a debt issue with no significant increase in other debt forms to replace it. Discovering when a firm has retired debt under this definition is more difficult than it sounds. If the balance sheet of a firm is compared immediately before and after a legal or accounting retirement of a debt issue, then, of course, there always appears to be a reduction of total debt. To be useful, however, balance sheet comparisons and application of funds analysis should be made over a relevant time period or periods. This time period need not necessarily be the regular or fiscal period. The period selected for review can be less or more than a year, depending upon what sort of information one hopes to uncover. At any rate, in order to measure debt reduction, a sufficient time span should be allowed in order to observe if there has been a build-up of floating debt on the way to the absorption of a funded issue.

There are really only two reasons for a decrease in total debt, as distinguished from a redistribution of debt, the firm is moving toward a more conservative capital structure and it is moving toward a smaller scale of operations; or it could perhaps be both. The capital structure can be modified in a conservative direction (i.e., toward less risk and less potential equity income) by processes other than reducing debt. If a firm's expansion is financed with a larger proportion of equity funds than existed in the original capital structure, the resulting capital structure will be more conservative. If the analyst concerns himself only with funded debt and capital stock as constituting the capital structure, it is possible to overlook another method by which the firm's financial structure can be made more conservative. A firm that floats bonds to provide net working capital—i.e., to reduce its dependence on short-term debt-generally lessens its short- or intermediate-term financial risk. The firm has either decreased its current liabilities—funded its current debt—or added to its current assets, in any case improving its current ratio or working capital position. That a firm reduces risk by this type of action is widely acknowledged, yet it might be missed in practice by financial analysts who concentrate on the bond-to-stock relationships. The reduction of debt is not a necessary condition for a less risky capital structure since this can be accomplished by other means. Nor is it a sufficient condition, since if the equity contribution decreases at a more rapid rate than debt, the result will be greater risk than before.

The equity funds may be reduced by losses or withdrawals even while the firm is repaying debt. In this case, of course, the scale of operations is necessarily decreased. If, however, debt is reduced while net worth is held intact or decreased by only a moderate amount in comparison to the decrease in debt, there is both a reduction in the scale of operations and a more conservative capital structure.

Toward a More Conservative Capital Structure The motives underlying the shifts in capital structure entailed by a reduction in debt vary. A move toward a more conservative capital structure is probably a sign of the advancing maturity or solidity of the firm. Many newly established firms commence operations with a high financing risk. This may be the only way they can get started. If the firm begins to

operate successfully, management acts to strengthen its capital structure, possibly by funding short-term liabilities (since in new business sources of short-term capital are relatively less scarce than long-term funds), perhaps by accumulating net worth at a greater rate than liabilities, or perhaps by actually retiring some debt. As the firm approaches maturity, and its rate of growth slows down, the risk attendant upon the capital structure of the intermediate, still moderately growing firm may seem too great for a conservative management, and debt may be scaled down still further. Perhaps in older firms, management might judge that the only major use of retained earnings is to repay liabilities. In this case, the reduction of debt may be carried to a point where the capital structure is pushed past its optimum position on the conservative side.[18]

Decreasing the Level of Operations A decrease in the level of output takes place whenever the firm faces a seasonal, cyclical, or secular decline in demand or possibly a decline in its competitive position. A decrease in production also occurs if the firm's marginal costs rise without a compensating increase in demand. The change in the firm's financial structure corresponding to the decline in output insofar as the management can control these changes on the downturn is largely affected by the situation the management thought it faced. A seasonal decline in demand is traditionally taken up by the repayment of current liabilities. This is a task of working capital management—the control and financial management of the current cash flow. A cyclical or temporary decline in the firm's position probably finds the management retiring debt while attempting to hold the ownership capital constant except as it might be impaired by unavoidable losses. At junctions such as this flexibility in the capital structure is important. The firm would like to be able to manage an orderly retirement of debt, enabling it to maintain its credit and capital base intact. Thus its survival chances would be high, and it could resume expansion on borrowed funds when the economic climate once again became propitious.

If the decline in the scale of the firm's operations is expected to be secular or permanent, the company will move toward a smaller capital structure. The mixture of debt and equity in the new financial structure would be appropriate for the new size of operations, given risks and costs, and both liabilities and net worth would be reduced in proportions resulting in a balance sheet tailored to the new situation. Management occasionally resists the movement toward a smaller balance sheet. The result is likely to be a redundant asset structure. The firm is over-conservatively financed, and it holds far too much cash and reserves in relation to its real output. This situation, however, is an aberration and not a rational development of managerial responsibility. Though it may exist for a while, it will hardly last indefinitely. Such a firm is ripe for a merger offer or a proxy fight.

A reduction in the absolute amount of debt is, in general, more probable during a decline in the rate of expansion—possibly a contraction in output and an actual decrease in earnings—than during periods when the firm appears most prosperous.

[18]One may be tempted to draw an analogy on the financial side to Alfred Marshall's categories of the young and vigorous, the mature, and, finally, the old or senile firm.

To the unsophisticated, it may seem an anomaly that debt can be reduced when earnings are declining. But obligations are not necessarily paid with earnings; they are paid with cash. These are related but are not identical. Thus there may be an automatic increase in available cash (even with a decline in earnings) if accounts receivable are collected and new credit accounts are not created at the same rate or if inventory is sold off and not restored to its previous level. Quite simply, a reduction in gross working capital assets releases cash funds, and these may be applied against debt if the level of operations is declining.[19]

Similarly, the noncash charges against earnings, such as depreciation, depletion, or amortization, represent a cash flow which is independent of net reported accounting earnings and even, in a sense, of current losses. As long as current operating losses do not exceed such noncash charges as depreciation and there is no build-up of other nonliquid assets, these charges represent a net source of funds to the firm. In periods of contracting activity a logical use of these funds would be the reduction of outstanding debt.

Methods for Retiring Specific Issues A whole system and complex of clauses can be written into the bond indenture for the purpose of setting debt retirement terms. There are four basic methods, however, by which specific debt issues can be repaid; and all debt retirement plans consist of one or a combination of these four methods:

1. Repurchase on open market
2. Repayment at maturity
3. Repayment of the bond issue by periodic payments
   (a) Sinking fund bonds
   (b) Serial bonds
4. (a) Callable bonds permitting retirement before maturity in whole or part.
   (b) Callable convertible bonds which may permit forced conversion

Most marketable bond issues today have become fairly standard. They call for a definite maturity date, averaging about 30 years before the bulk of the issue becomes due. They usually have a partial sinking fund provision requiring the company to retire about 2% of the original issue yearly, leaving about 40% of the debt to be met at maturity. The sinking fund trustees may purchase the yearly retirement quota at market or use a sinking fund call provision (paying a slight premium over par), whichever is cheaper. Most bonds contain a call provision enabling the issuer to call back the bonds in whole or part before maturity by paying a fixed call premium over the par value. This general call premium is higher than the one that prevails on bonds bought back under the sinking fund provision: however, both kinds of call premiums are highest near the date of issue and decrease in some regular manner over the life of the bonds. On occasion, bonds have been issued which are not redeemable under a general call for, say, the first 5 years. This enables the lenders to enjoy at least a few

[19]Of course, if the firm suffers steep losses in the liquidation of current assets, it may experience a stringency of cash.

years of good interest rates if the bonds were floated in a period of relatively high rates.

Repurchase on the Open Market A basic tool of the management used to retire debt, unless the issue has been privately placed, is repurchasing the debt on the market. If the company wishes to pay the going market price, there is no reason why it shouldn't buy back its own bonds or notes and thus effectively reduce its debt. This method is the only way a perpetual bond can be retired, but it could be used on all other issues. The disadvantage of market retirement arises if the company has made a bad bargain or sold bond at an interest rate that is now appreciably above the market rate. It would have to pay a heavy premium in the form of an increased market price to be released from its contract of debt.

The ability to repurchase on the open market can be valuable, and the loss of this ability is a chief drawback to private placement. Of course, if the firm presents an attractive proposition to the insurance companies or other financial institutions holding its bonds, these institutions would surely accept it. They would be able to use the funds released just as gainfully in an alternative investment. A repurchase of debt from private holders would have to be negotiated, however, and a price approximating the market price agreed upon. Purchase from the open market involves no such negotiation, and the amount to be bought at any particular time can be readily varied. A possible objection to a firm's repurchasing its debt on the open market is that the technique could be used to favor one set of creditors over the others. A shaky firm might use funds to buy back a junior issue, releasing these holders from risk while increasing the risk to the holders of the senior issue. That this is a possibility is attested by the attempt to guard against it by inserting minimum working capital maintenance requirements in most bond indentures. This abuse of purchasing on the open market would take place only if the management had questionable motives. In most cases, the management would not buy the junior issues if the firm's survival position were dubious, because the increased remaining risk to the senior securities would be even more intensified for the common stock.

Repayment or Refunding at Maturity The due date, or date of maturity, is the stated time the corporation promises to pay a specific debt instrument. The maturity date permits the borrower and the suppliers of funds to reassess their positions, and by terminating the old debt contract, it allows them to negotiate a new loan under current market conditions. Thus the maturity date protects both the borrower and the lenders if later developments should make the initial debt terms disadvantageous to either party. If credit conditions have moved in its favor, the firm can get out of debt at par, without paying a premium. On the other hand, the maturity date assures the security owners of some positive action. The issue must be paid at maturity, as agreed, or else the bondholders begin action to enforce their claim. Thus the maturity date enables the bondholders, as well as the corporation, to limit the losses arising from an incorrect forecast of the general market interest rate or the specific risk rate for the firm.

Since a longer term to maturity on a bond lengthens the time during which the lenders cannot renegotiate, it increases the risk and interest charges. The due date is

an important item upon which the firm and the investors must agree. The specific terms of an issue, including the length of maturity, are part of the bargain between the demands of the market as represented by the investment bankers and the concessions given by the borrower.

In most modern bond issues, a large part of the debt is retired before maturity by the sinking fund or by partial calls. Even so, the amount of debt to be repaid at maturity is usually too great to be paid back with the cash flow of a given year. If the company is intent on repaying the maturing debt, it must accumulate cash some time in advance. These funds can be invested in short-term liquid assets, but, usually, the best return on the incoming funds is obtained if they are employed in buying back the issue. Sometimes the firm incurs other debt along the way to provide the funds to pay the bulk of the maturing issue. If so, what appears to be retirement of debt is actually a refunding of the issue once removed.

A financial operation known as refunding at maturity occurs when the firm either puts a new issue on the market to raise funds to repay the old or attempts an exchange of new bonds with the old bondholders. The latter, sometimes called refunding through exchange, generally indicates financial weakness in the firm. The old bondholders, however, often find it wiser to accept the new securities rather than call for repayment. Perhaps the firm has been meeting interest payments so far, but rather than force the firm to the market place to search for someone else to refinance the maturing debt, the present bondholders accept the offer of new securities, usually with some improvement in the terms. Accepting the exchange offer may be safer for the lenders than attempting to force payment and finding themselves in the midst of a reorganization.

Gradual Reduction of Debt Issue by Sinking Fund Purchase Originally the sinking fund was thought of as assets set aside by the company for the repayment of the total bond issue at maturity. Today most sinking fund provisions retire only part of the issue, generally around 40–60%. Under modern sinking fund provisions, the trustees are directed to buy back a portion of the issue each year at the sinking fund call price or at the market price, whichever is lower. Instead of a formal device to retire the whole debt issue, the sinking fund effects a gradual reduction in the bonds outstanding and supports the market by giving the holders a probability that some of their holdings will be only short term.

In its earlier conception, the sinking fund consisted of assets, earmarked so that confusing them with other assets was impossible. The amount set aside annually was either fixed or contingent upon earnings. Eventually, it was to total the amount of the issue. A sinking fund reserve to balance the accumulation of assets in the fund was usually set up on the equity side of the balance sheet. This sinking fund reserve "appropriated" a portion of the surplus. Supposedly it served as a reminder to forestall too large a distribution of earnings by an overly liberal dividend policy. The sinking fund assets were far more important in repaying the debt than the so-called sinking fund reserve. A corporation showing a large sinking fund surplus reserve would find it difficult to redeem its bonds if it neglected to accumulate the

sinking fund or suffered losses in the sinking fund assets. A surplus reserve cannot repay debt; only assets can do so.

Although the old fashioned sinking fund is still used occasionally on municipal bond issues[20] it is rarely used for corporate issues. The basic disadvantage of an asset sinking fund is the difficulty of finding safe investments with a suitable yield. The sinking fund trustee had to minimize risk in investing his funds. Capital appreciation was not a prime consideration since the fund was to pay off a fixed sum of money. On the other hand, a loss in the sinking fund principal subjected the corporation to considerable hazard. If safety was the major requirement, obtaining a reasonable sinking fund income sometimes posed a dilemma. Suppose that safe investments for the sinking fund had a market yield of only 3%, whereas the market rate for the borrowing corporation was 5%. The real cost of the corporation's borrowed funds exceeded the nominal cost. Assume that the corporation had outstanding $1,000,000 in 20-year sinking fund bonds, issued at par, with an interest rate of 5%. The yearly interest cost is $50,000. At the end of the tenth year, the sinking fund totals $500,000 and earns 3%. During the 11th year, the firm's net interest cost is $35,000, i.e., $50,000 less the $15,000 interest income earned on the sinking fund. The firm's net borrowed funds are only $500,000–$1,000,000 outstanding on the debt less the $500,000 tied up in sinking fund assets. Thus the real cost to this firm for the use of funds is $35,000 over $500,000 or 7%. Considering all the problems of sinking fund asset selection, the best investment for a bond sinking fund is the retirement of the bond issue itself. Such an investment eliminates all risks, including the risk of market fluctuations, present even in prime government bonds. Moreover, the interest cost saved by the sinking fund and the effective market rate on the bond issue are very close.

Thus the present type of sinking fund developed. The bond indentures carry provisions instructing the trustee to purchase a certain percentage of the issue (perhaps a minimum of 2–5%) annually. Actual retirement is accomplished by purchasing the bonds in the open market or by call.[21] The specific bonds to be called are determined by lot, so that no buyer knows at the time of his purchase the exact date he may be asked to surrender his bonds.

Reduction of Debt Through Serial Issues An alternative to the modern sinking fund as a method of providing periodic retirement of bonded debt is the serial bond issue.

Under a serial bond issue, a definite schedule of maturity dates is established. In effect, there is a simultaneous issue of bonds with varying due dates. The corporation

[20]Sinking funds may be advantageous on municipal issues if the yield on safe taxable federal bonds is higher than the rate on the tax exempt local bonds. The sinking fund trustees should be careful, however, to choose the maturities of the sinking fund investments to coincide with the due dates of their own bonds if they wish to avoid the possibility of capital losses on the sinking fund assets.

[21]Bonds may also be retired by tender, where bondholders are invited to submit the price at which they are willing to sell their bonds back to the company. The company picks up the bonds tendered at the lowest prices.

must pay each annual or semiannual fraction of the issue as it comes due. It usually has the choice of retiring a greater amount than the annual installment through the use of the call privilege. Given a plan of repayment so definite, the average interest charges on a serial issue is usually lower than under other systems of repayment.

Serial bonds, although prevalent as state or municipal issues, are not common corporation issues today. They are, however, the basic design of the equipment trust certificate. In general, the most common form for commercial bond issues is the partial sinking fund callable issue.

In order to give a serial bond issue some of the flexibility of the callable sinking fund bond issue, the serial bond issue also often contains a call feature. If the call privilege is exercised, the indenture frequently provides that the longest maturities be called first. This is in marked contrast to sinking fund bond indentures, which provide for call by lot.

An advantage of the sinking fund issue, for management, is that failing to make the annual sinking fund payments may be less serious than failing to pay a maturing serial bond installment. For one thing, the bondholders often may not know of defaults in sinking fund payments for some time after the event. Second, as long as interest payments are met, the bond trustees may not wish to take action other than registering a protest. Of course, the failure to make the regular sinking fund installments is a breach of contract, and the trustee may appeal to a court of equity to enforce payment of the installment. Such a course of action, however, could conceivably lead to the appointment of a receiver. The bondholders may hesitate to apply so drastic a remedy. The dilemma of the bondholders can be conceived as accruing to the benefit of the issuing corporation, giving it a flexibility not possible under a serial bond indenture. This argument can be overstated; obviously, no rational organization enters into a contract with the notion of possibly breaking it. By and large, the sinking fund requirement has to be considered as fixed an obligation as the serial bond maturities.

Debt Retirement and the Call Privilege The call feature on a bond considerably increases managerial control over the issue; it facilitates a program of gradual debt reduction or perhaps refunding before maturity. Should there be a downward shift in market rates or a strengthening in the firm's credit standing, the call price places an effective ceiling on the premium that the company has to pay to retire its bonds.

If either the market interest rate or the specific risk rate applicable to the firm declines, then, other things being equal, a firm's securities rise in the market. However, were the call option an original term of the issue, the firm need not concern itself with what the market price would have otherwise been. It can pay the call price or the market price, whichever is lower. On the other hand, if there were no call provisions, a firm seeking to reduce its indebtedness would have no choice but to pay the market rate.

Forced Conversion If a bond is both convertible and callable, the management may elect to reduce the risk of its capital structure by forcing conversion. If the market price of the common stock rises above the conversion rate, a convertible bond may command a market price higher than the call price. The management is then in a

position to force the conversion from bonds to common stock. If the corporation issues a call for redemption, the security holders must convert or lose the difference between the market price of the bond and its call price. Under these circumstances, the firm can reduce its debt obligation and increase its equity capital at the same time.[22] The position of the original shareholders, however, will in a sense deteriorate, since the new stockholders pay for their shares with the surrender of debt whose value, as such, is below the going market price of the company's shares. On the other hand, the original stockholders are presumed to have gained on the initial issue of the convertible securities at a low-interest rate or have been recompensed with salable rights. Furthermore, the management's call only hastens the conversion which, under the assumed circumstances, is bound eventually to take place.

Reference

Bierman, H. (1972). The bond refunding decision. *Financial Management*, 22–29.
Brealey, R. A., & Myers, S. C. (2003). *Principles of corporate finance* (7th ed.). Boston, MA: McGraw-Hill/Irwin. Chapter 26.
Guerard Jr., J. B. (2010). The corporate sector as a net exporter of funds. In J. R. Aronson, H. L. Parmet, & R. J. Thorton (Eds.), *Variations in economic analysis: essays in Honor of Eli Schwartz*. New York: Springer.
Hicks, J. R. (1937). *Value and capital*. London: Oxford University Press.
Macaulay, F. R. (1938). *Some theoretical problems suggested by the movements of interest rates*. New York: Columbia University Press.
Malkiel, B. G. (1962a). Expectations, bond prices, and the term structure of interest rates. *Quarterly Journal of Economics, 76*, 197–218.
Malkiel, B. G. (1962b). The term structure of interest rates. *American Economic Review, 54*, 532–543.
Myers, S. (1977). Determinants of corporate borrowing. *Journal of Financial Economics, 5*, 147–176.
Schwartz, E. (1962). *Corporate finance*. New York: St. Martin's Press, Chapter 9.
Schwartz, E. (1967). The refunding decision. *Journal of Business, 40*, 448–449.
Van Horne, J. C. (2002a). *Financial management & policy* (12th ed.). Upper Saddle River: Prentice Hall. Chapter 18.
Van Horne, J. C. (2002b). *Financial market rates and flows* (6th ed.). New York: Prentice-Hall, Inc..

[22] In rare cases the conversion must be accompanied by a payment of money. The company not only retires the debt issue, it increases equity capital and strengthens its working capital position.

Debt, Equity, the Optimal Financial Structure, and the Cost of Funds



Traditionally, the capital structure of a firm has been defined as the book value of its common stock, its preferred stock, and its bonds, or fixed liabilities. If the goal of corporate finance is to maximize the price of common stock, then the question must be asked if there is an optional capital structure that maximizes the stock price. Schwartz (1959, 1962) long advocated that optimal capital structure exist as leverage increases the tenders risk and increases the borrower's risk. Optimal capital structures exist in different industries because of different industry risks. These items are considered to be the "permanent" financing of the firm. The special importance given to them, however, may lead to error in financial analysis. Thus, a company which has only common shares in its capital structure is often described as conservatively or safely financed. But if, for example, the firm has considerable trade debt outstanding, owes on a bank loan, or is tied up with long-run rental contracts, it may not be "safely" financed.

Although distinguishing between current liabilities and longer-term financing is convenient in some analyses, the degree of difference between current and funded debt is often grossly exaggerated. The so-called permanent financing is not unalterable; bonds can be retired, reduced, or increased, so can common stock, as we noted in chapters "The Corporation Balance Sheet" and "Financing Current Operations and Efficiency Ratio Analysis," stock; and the book value of the total common stock equity may also be varied. On the other hand, no operating firm is likely to function without some amount of current liabilities; thus, some current debt is permanent to the financial structure. Thus, it would be better to consider a firm's capital or financial structure as consisting of all the items on the credit side of the balance sheet representing the equity and all the liability accounts.

An important general tool of financial structure analysis is the ratio of total debt to total assets. Of course, in a detailed financial analysis, the relationships and ratios among the items on the credit side of the balance sheet and among liability groupings and certain assets are significant and useful; but the usual financing analysis may be misleading when only the fixed debt is employed in depicting the capital structure of the firm. In this chapter, we discuss optimal capital structure. We constrast Miller and Modigliani and the Schwartz theories of optimal capital structure, the

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_10

implications for the return an equity, and why capital structure is a major corporate finance decision.

A Most Misleading Relationship

Perhaps the most misleading ratio used as index of credit quality is the ratio of debt to the market value of the shares. Too often the market value of the stock is an ephemeral number based on a notion of future growth and hope for the future. If another recommended index is used, i.e., debt service into cash flow, the potential operating coverage of the debt has already been captured. Because cash flow and the market value of the shares are related, the relation between debt and the market value of the stock is not an independent variable.

Definition of Leverage: Profits and Financial Risk

An important concept in understanding the relationships in the financial structure of the firm is the ancient idea of "trading on the equity" now long going under the current term "leverage."[1] Leverage is the amount of outside funds (debt) the owners use in proportion to their own contributions to the financing of the firm. The use of debt is called leverage because these funds, acquired at a priority of repayment and given a priority of return, widen the potential swing of both gains and losses to the ownership shares. Any earnings on the assets acquired by borrowing in excess of the rate that has to be paid to the creditors belongs to the owners and increases their net rate of return; however, if the earnings on the assets acquired with borrowed funds falls below the contracted rate, or if there are overall losses, the negative difference sharply reduces the rate of return or increases the loss on the equity. But as long as the marginal assets employed in the firm earn more than the cost of the borrowed funds, it will be profitable to use leverage, with the proviso that the financial risk of the firm is not thereby inordinately increased. The degree of leverage in a firm's capital structure is measured by noting how much the rate of return on equity would change with any change in the average rate of return on the total assets. The greater the proportion of outside funds to ownership capital, the more emphatic is the leverage effect.

Some financial analysts apparently recognize leverage only if the outside funds are acquired under a definite contract and the suppliers of these funds are paid a fixed positive rate of return. Leverage is thus limited to the use of bonds, preferred stock, or long-term bank loans. Under this concept, many banks, for example, are not considered as leveraged, since they often have no bonds or preferred stock outstanding in their capital structure. Nevertheless, authorities in the field of money and banking note

[1] Called "gearing" in England.

Table 1 Effect of leverage using current debt. Woodrow Corporation

Total liabilities and capital		
Current liabilities		$300,000
Surplus	$100,000	
Capital stock	100,000	
Total common stock equity		200,000
Total capital and liabilities		$500,000
Rates of return at different levels of earnings on total assets		
Rate of return (EBIT)[a] on total assets	Total profit (EBIT) (or loss)	Rate of return (EBT)[b] on equity (ownership capital)
10%	$50,000	25%
5%	25,000	12½%
2%	10,000	5%
-0-	-0-	-0-
(3%) Loss	(15,000) Loss	(7½%) Loss
(5%) Loss	(25,000) Loss	(12½%) Loss

[a]*EBIT* earnings before interest and taxes
[b]*EBT* earnings before taxes

the "highly leveraged aspect" of the typical bank's capital structure and the small percentage of equity in comparison to the total deposits or liabilities carried.

A broad definition of leverage covers the relationship between all the prior claim securities and obligations to the ownership capital. Trade accounts and other current liabilities are included in this concept of leverage. These obligations have priority over the ownership shares; they must be paid at least a zero rate of return. This seems a paradox until we remember that ours is a profit and loss economy. Shareholders may earn a negative rate of return, the owners may absorb losses, but liability claims are not written down unless there is a failure or reorganization. The zero return placed on current liabilities is thus, in a sense, a fixed return, and it accordingly widens the possibilities for gains and losses on the ownership investment just as does any other fund borrowed from outside sources.

Illustrations of Leverage: Return and Risk

The effect of "zero" cost liabilities on the possible dispersion of return for the owners is demonstrated by the simplified example presented in Table 1. The Woodrow firm is financed solely with current liabilities and ownership capital. Whatever the rate of return on total assets, which by accounting definition is equal to the total liabilities and capital, the return on the ownership capital is two and a half times as large. Thus, if the Woodrow Corporation earns 10% of its total assets, it would earn a before-taxes return of 25% on its owners' investment. If, however, the firm should experience even a small overall loss, the rate of loss to be absorbed by the equity capital is again two and

Table 2 Effect of leverage using considerable fixed debt. Meredith Company

Liabilities and capital				
Current liabilities				$10,000,000
First mortgage bonds (4's)				40,000,000
Debentures (4½'s)				15,000,000
Capital stock 00 common (1,000,000's)			$20,000,000	200,000
Surplus			15,000,000	35,000,000
Total capital and liabilities				$100,000,000
Effect of leverage on the rate of return under variations in the profitability of the assets				
Rate of return (EBIT) on total assets	Total earnings before interest and taxes	Interest charges	Earnings before taxes (loss)	Rate of return on common stock equity before taxes
10%	$10,000,000	$2,275,000	$7,725,000	22.1%
8	8,000,000	2,275,000	5,725,000	16.3
6	6,000,000	2,275,000	3,725,000	10.6
4	4,000,000	2,275,000	1,725,000	4.9
2	2,000,000	2,275,000	(275,000) Loss	(0.8%)
0	-0-	2,275,000	(2,275,000) Loss	(6.5%)

a half times as large as the overall rate. Thus, if the Woodrow Corporation lost 3% of its total assets, the loss on the ownership funds would be 7½%.

The use of current debt is a cheap method of financing; carried too far, it may become quite risky. Current liabilities constrict the firm's net working capital position. Although current liabilities carry a minimum interest (charge) if any, the principal amount is continually coming due. From this point of view, fixed debt, when it can be obtained on favorable terms, is a safer component of leverage than current debt. The interest charges on long-term debt reduce the profits derived from successful leverage and increase the possibilities of loss in case of a downturn, but at least the repayment of the principal of the debt is delayed into the future. Thus, the firm has a chance to recover its financial position before the due date.

Table 2 illustrates the possible behavior of the rate of earnings on the stockholders' investment as the rate of earnings on the total assets changes, in a firm which has a considerable component of long-term debt leverage. For the Meredith Company, the rate of return on the common stock equity changes by approximately 2.85 percentage points for each percentage point change in the earnings on the total assets. If the company earned 10% on its total assets, the earnings on the common shareholders' equity would be 22.1%; at a 2% return on total assets, however, the company shareholders suffer a loss. The Meredith Company must gross 2.21½% on its total assets to break even.

An interesting method to calculate the break-even point of debt and equity financing is to express earnings before interest and taxes, EBIT, as a function of sales. If one calculates the earnings per share (EPS) of the debt option, incurring higher interest costs and lower earnings, and equates the EPS debt option with the EPS equity option, keeping interest expenses and earnings constant, but issuing

more shares, one can find the break-even sales level for the debt and equity decisions. For example, let us assume that EBIT equals 15% of sales. Interest charges are currently $10 million; there are 5,000,000 shares of stock outstanding, and the firm seeks to raise $100 million of new funds. If the firm's stock price is $100, then the firm must issue 1 million new shares of stock. If the current cost of debt is 8%, then the firm must pay an additional $8 million of interest expense. The break-even equation for indifference between debt and equity financing is

$$\text{eps Equity} = (.15 * \textit{Sales} - \$10\text{ MM})/(5\text{ MM} + 1\text{MM}) =$$

$$\text{eps Debt} = (.15 * \text{Sales} - (10\text{MM} + 8\text{MM}))/5\text{ MM}$$

$$(.15 * \text{Sales} - 10\text{ MM})/6\text{MM} = (.15 * \text{Sales} - 23\text{MM})/5\text{MM}$$

$$\text{Sales, Break-even} = \$586.67\text{ MM}.$$

If sales are expected to be less than $586.67 million, then the firm should issue equity; if sales are expected to exceed $586.67 million, then the firm should issue debt. Management should seek to maximize the earnings per share, EPS, of the firm to maximize stockholder wealth.

Leverage is profitable if the rate of earnings on total assets is higher than the going rate of interest on the debt. Of course, the risk to the stockholders of loss and failure in case of a downturn must always be considered. It is generally felt that to finance safely with leverage, the stability of the earnings, or better the cash flow, is more important than its level. The reader is invited to follow the Lerner and Carleton (1966) derivation of a return on equity and the issue of leverage. The operating return on assets, ROA, or R is the ratio of the firm's EBIT to total assets. The firm pays interest on its liabilities, L, with a coupon rate of r:

$$\text{EBIT} = \text{R}\,(\text{Total Assets})$$

$$\text{Operating Income} = \text{EBIT} = \text{R}\,(\text{Liabilities} + \text{Equity}) = \text{R}\,(\text{L} + \text{E})$$

$$\text{Less Interest Paid} = -\text{I} = \text{r}\,(\text{Liabilities}) = \text{rL}$$

$$\text{Earnings before Taxes} = \text{EBT} = \text{R}\,(\text{L} + \text{E}) - \text{rL}$$

$$\text{Taxes Paid} = -\text{Taxes} = \text{t}[\text{R}(\text{L} + \text{E}) - \text{rL}]$$

$$\text{Earnings after Taxes} = \text{EAT} = (1 - \text{t})[\text{R}(\text{L} + \text{E}) - \text{rL}]$$

The return on equity is given by earnings after taxes divided by equity and is a positive function of the liabilities-to-equity ratio:

$$\begin{aligned}\text{ROE} = \frac{\text{EAT}}{\text{E}} &= \frac{(1 - \text{t})\ \ [\text{R}(\text{L} + \text{E}) - \text{rL}]}{\text{E}} \\ &= \frac{(1 - \text{t})\quad [\text{RL} + \text{RE} - \text{rL}]}{\text{E}} \\ &= (1 - \text{t})\left[\text{R} + (\text{R} - \text{r})\frac{\text{L}}{\text{E}}\right]\end{aligned}$$

Thus, as long as the return on asset exceeds the cost of debt, then the return on equity rises linearly with leverage. The linearity of the ROE with leverage was established in Modigliani and Miller (1958), Proposition II. Leverage is extremely important to the firm's stockholders. The choice of capital structure must be made with management's perception of the return on assets and its expected cost of debt.

Surrogate Evidence on the Development of "Optimum" Financial Structure

The basic types of firms have tended to develop characteristic financial structures. These regularities in the typical financial structures imply that certain similar forces operate on all the firms in an identifiable class. Occasionally, an individual firm, because of special characteristics in its financial history or the predilections of its management, may diverge noticeably from the typical structure for a firm in the given industry. Nevertheless, a financial specialist can almost invariably determine whether she is reading the balance sheet of an electric utility, a railroad, a bank, a mercantile, or an industrial firm. Although one utility may differ in its financial structure ratios from another, these differences are usually small compared to the differences from a typical industrial or rail. The persistence of these differences seems to indicate that there is an "optimum" firm financial structure for various industries.

An optimum financial structure maximizes the long-run market value of the firm's common stock. This is not the same as asserting that the optimum capital structure maximizes profit or earnings per share. For both the earnings per share and the risk-adjusted rate at which the market capitalizes, these earnings must be considered. The amount of financial risk or leverage a firm carries helps set the capitalization rate of the shares. If a firm's financial structure carries too much debt, i.e., borrower's risk, the market may set a lower price for the shares than it would give for similar shares with smaller earnings but a more "conservative" financial structure.

Under "normal" conditions, a company can add to its rate of profit by adding more debt to its financial structure. However, leverage increases the potential loss and lowers the survival rate in case of an economic downturn. In setting the value of the shares, investors consider both the financial risk of the firm's capital structure and the intensity of the economic or operating hazards typically faced by the industry. The investing market will accept the sacrifice of some financial safety for increased earnings, as long as other operating hazards are not too high, but it will discount the returns of a firm whose total risk seems too large. The hypothesis that the corporation management attempts to adjust to both economic risk and financial risk to obtain the best possible position for their shareholders may explain why companies in different industries have tended to develop typical financial structures.

If the theory of the optimum capital structure is correct, every firm has a preferred combination of risk and return.[2] Suppose three firms, A, B, and C, are engaged in similar operations:

Earnings per share	$2.00	$2.50	$3.00
Market price	24.00	25.00	24.00
Price/earnings ratio	12×	10×	8×

Firm A carries very little leverage or financial risk, Firm B has some financial risk, and Firm C is highly leveraged. The earnings per share are for a "normal year," and the differences in earnings are due solely to the different financial structures. If these conditions prevailed, then according to the revealed preferences, of the investors on the market, Firm A is too conservatively financed, Firm B has the optimum capital structure, and Firm C is "overleveraged."

The optimum capital structure not only maximizes the value of the shares, but it also implies the minimization of capital costs. The explicit cost of debt is cheaper than equity. As a moderate amount of debt is added to the financial structure, the cost of the financial mix decreases. However, as the proportion of debt in the total financial structure rises, the investors' appraisal of the quality of the debt falls. The cost of borrowed funds rises. In addition, as the internal risk of the firm increases, the price/earnings ratio falls, and the cost of equity funds rises. In an overleveraged company, the cost of the total financial mix is higher than the norm. For the ideal financial structure, the cost of capital mix is at a minimum.

The Pure Theory of the Optimal Financial Structure[3]

The pure theory of the optimal capital structure, as expressed by Schwartz (1959), is based on the assumption that the firm is a semi-monopsonistic demander of funds from the capital market. By discriminating against the suppliers of funds through employing varying debt instruments and judiciously balancing the total of financial risk and external risks, the firm can achieve an optimum financial structure, reducing total financing costs and maximizing the value of its shares. The parameters constituting the environment in which the firm exists are as follows:

[2] A fuller theoretical discussion of the optimum financial structure may be found in E. Schwartz, "Theory of the Capital Structure of the Firm," *Journal of Finance*, March 1959. The arithmetical example follows the general pattern of Graham and Dodd, *Security Analysis*, McGraw-Hill, 1951, Chapter 37.

[3] E. Schwartz, Op.Cit., *Journal of Finance*, March, 1959.

1. The individual firm is confronted by two types of risks. One type we might call the "external risk," the other type the "internal" or "financial risk."[4]
2. The external risks are a composite of the stability of earnings, or cash flow of the firm, and the liquidity, safety, and marketability of the assets typically held by the firm. The level of external risk is in large part dictated by the nature of the industry in which the firm is engaged and is not subject to any great extent to the control of the financial decision-makers.
3. Internal risk is the financial risk of the firm's capital structure. It is set by the types of liabilities (short term or funded) that the firm carries and the total amounts of the liabilities in proportion to the firm's equity capital. The factors constituting the firm's capital or financial structure can be varied considerably by the financial management.
4. The two types of risks together are the sum of the hazards to which the owners and the creditors of the firm may be subjected. The external risks are a parameter given by the nature of the industry; these external risks are borne in mind by both borrowers and lenders and influence the optimum financial risk that different types of firms are likely to carry.

The optimum capital structure for any widely held company is one which maximizes the long-run market value per share of the common stock. This is not quite the same as asserting that the optimum capital structure is one which will maximize profit or earnings per share. For both the earnings per share of stock and the rate at which they are capitalized must be considered. The amount of financial risk that a firm carries helps set the capitalization rate. If a firm's financial structure carries too much borrowers' risk, the market may set a lower price for the shares that it would give for similar shares with perhaps somewhat smaller earnings but less financial risk.

The ability of the firm to set up an optimum capital structure implies the ability to discriminate against suppliers of funds, investors, individuals, or financial institutions, with different preferences for income and aversions to risk. Discrimination on one level leads to complex financial structures. It means that by raising funds through securities and contracts with varying return and security provisions, the firm could lower its total financial costs. On a broader macro level, varying preferences for return and risk implies that by a judicious mix of overall debt (financial risk) and equity, the firm could maximize the value of its shares (minimize the cost of capital), i.e., achieve an optimum capital structure.

The theoretical trade-off for a given firm between the rate of return on ownership capital (equity), the degree of financial risk (debt), and the market preference yielding the maximum price for the shares is illustrated in Fig. 1. The financial risk factor is indicated indirectly in Fig. 1. It is shown on the horizontal axis, inversely related to the proportion of equity (capital stock) in the capital structure. Thus, as the amount of equity capital increases in a particular firm's capital structure,

[4]Financial risk was divided into borrower's risk and lender's risk by Keynes in The General Theory, pp. 144–145.

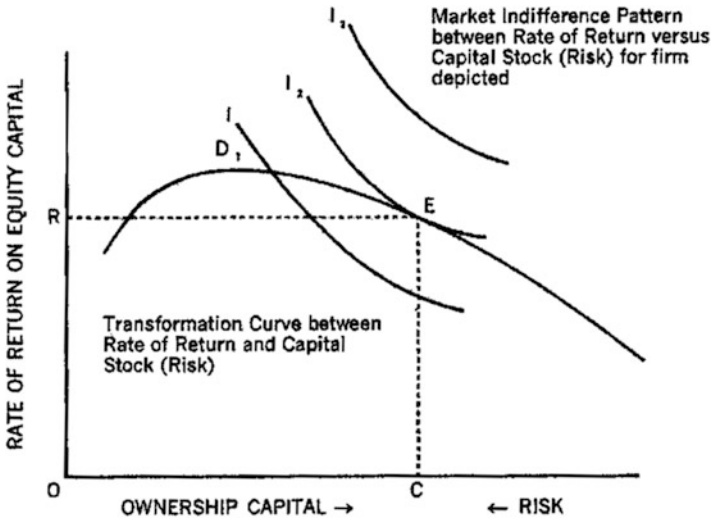


Fig. 1 Formal solution of a firm's optimum capita structure

the debt-equity ratio and the degree of financial risk decrease. The conventional rate of return on the equity is depicted on the vertical axis. Because of the pro forma profitability of leverage, the rate of return on the equity falls as the proportional amount of share financing increases although volatility and financial risk decrease.

The transformation curve, D, gives the average rate of return for shares and degree of risk for a financial structure containing varying amounts of ownership capital. Sur-imposed in the figure are investor's indifference curves showing the investors substitution rate between earnings and the degree of risk for a firm of this type. Each indifference curve represents a given constant stock price. The tangency point E indicates the financial structure and the trade-off between risk and return which will fetch the highest price for the shares on the market. It is the point where the earnings and the risk-adjusted discount rate yield' the highest amount.

The tangency point E indicates the optimum amount of equity capital and rate of return on equity capital for the firm. In brief, the conditions of the optimum are

$$\frac{\text{Marginal sacrifice in earnings}}{\text{Marginal decrease in risk}\quad(\text{investor's choice})} = \frac{\text{Marginal decrease in earnings}}{\text{Marginal increase in ownership}\quad(\text{decrease risk})\ (\text{in the financial structure of the firm})}$$

The letter R indicates the rate of earnings on equity investment and OC the optimum amount of equity capital for the firm, setting up the market capitalization rate and expected earnings that maximize the value of the shares. Schwartz (1959, 1962) finds an optimal capital structure for corporations.

The optimum capital structure varies for firms in different industries because the typical asset structures and the stability of earnings which determine inherent risks vary for different types of production. The theoretical solution of the optimum capital structure is made in a very formal manner since it must give consideration to many variables—increasing lender's risk and increasing borrower's risk, the interest rate structure, the forecasted earnings function, and the possibility of discriminating against the market supply of outside capital.

Modigliani and Miller: Constant Capital Costs

Contrary to traditional views of the relative costs of debt, equity, and the weighted cost of capital found in Schwartz (1959), Professors Franco Modigliani and Merton Miller (M&M 1958) posited a model where in a nontax world, for a firm of a given risk class, capital costs are constant regardless of the financial risk.[5] There is no optimum financial structure, until taxes are introduced (see Modigliani & Miller, 1963).

In the M&M model, the trade-off between financial risk and the cost of funds is unitary; if more debt is added to the financial mix, the cost of debt rises and the desired rate of return on equity rises, so that the weighted cost of the financial mix remains constant. Let us briefly recount the three propositions of M&M in their seminal presentation of the cost of capital and valuation.[6] First, M&M hold for many firms in the same line of business, the cost of capital is a constant, ρ_o. The constant is determined by dividing the expected return per share by the stock price; i.e., the cost of capital is determined by dividing net operating earnings of the firm by its total market value. Hence, the M&M hypothesis is often referred to as "the net operating income" approach.[7] Thus, first, the average cost of capital is independent of capital structure. Second, the expected cost of equity rises linearly with the debt-to-equity ratio. The earlier Lerner and Carleton derivation is a variation of the M&M Proposition II. M&M argued that the firm must earn a return on investments exceeding ρ_o. M&M's Proposition III holds that if the firm earns at least ρ_o on its investments, the project(s) is acceptable regardless of the securities issued to finance the investment. M&M presented empirical evidence in their 1958 study, using the 40 firm electric utility study of Allen (1954) and the 42 firm oil company sample of Smith (1955). Both Allen and Smith provided data on the average values of debt and preferred stocks and market values of securities, such that M&M could calculate the debt-to-

[5]Franco Modigliani and Merton H. Miller, "The Cost of Capital, Corporation Finance and the Theory of Investment," *American Economic Review*, June 1958.

[6]M&M discussed their initial proposition of capital structure irrelevance in terms of a class of homogeneous firms, firms that are perfect substitutes of each other, such as firms in the same industry.

[7]J.C. Van Horne, *Financial Management & Policy*, Prentice Hall, 2002, 12th edition, pp. 255–260.

total value of securities ratio, d. M&M regressed the net returns, x, defined as the sum of interest, preferred dividends, and net income, as a function of ratio d. The Allen electric utility sample covered 1947–1948, and the Smith sample of oil companies was for year 1953. The M&M regressions were

$$\text{Electric Utilities} \quad \begin{matrix} x = 5.3 + .006d \\ (\text{s.e.}) \quad (.008) \end{matrix}$$

$$\text{Oils} \quad \begin{matrix} x = 8.5 + .006d \\ (.024) \end{matrix}$$

M&M held that the regression results supported their Proposition I. The calculated t-statistics, obtained by the ratio of the regression slope, b, divided by its standard error (in parenthesis), should be 1.96 (or 1.645 at the 10% level), to be statistically significant. The calculated t-statistic on the electric utility sample is 0.75, far less than 1.645. The calculated t-statistic for the oil sample is 0.25. Thus, there is no statistically significance between net returns and the debt-to-asset ratio in the initial M&M study. We will take a detailed look at hypothesis testing in chapter "Investing in Assets: Theory of Investment Decision-Making." M&M used the Allen and Smith samples to test their Proposition II. M&M regressed ROEs, defined by dividing net income by equity, as a function of the debt-to-equity ratio, b:

$$\text{Electric Utilities} = \quad \begin{matrix} \text{ROE} = 6.6 + .017h \\ (\text{s.e.}) \quad (.004) \end{matrix}$$

$$\text{Oil Companies} = \quad \begin{matrix} \text{ROE} = 8.9 + .051h \\ (.012) \end{matrix}$$

The estimated t-statistics of the electric utilities and oil company sample of 4.25 and 4.35, respectively, rejected the null hypothesis of no association between ROE and the debt-to-equity ratio. Thus, support is found for Proposition M&M II, that the cost of equity rises linearly with the debt-to-equity ratio.

A graphic depiction of the M&M model is shown in Fig. 2. The substitution rate between the risks and the costs of the elements of the financial mix is linear so that the average cost of capital is constant.

In an equation presenting the M&M hypothesis which is in a linear form, the function for the overall cost of capital appears as follows:

$$k_o/A = D\left(i_j + D/_A \cdot p\right) + E\left(e_j + D/_A \cdot p\right)$$

where

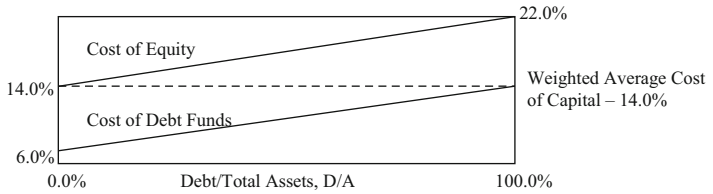


Fig. 2 A graphic depiction of the M&M hypothesis

Table 3 Overall cost of funds in a simplified M&M model, where the unleveraged cost of equity is 14%, the initial cost of debt is 6%, and the financial risk premium is 8%

	A Total assets	D Debt	E Equity	k_i Cost of debt at the given financial mix (%)	K_e Cost of equity at the given financial mix (%)	Total cost of debt financing	Total cost of equity financing	K_0 Weighted cost of total capital at the given financial mix, $k_i \times D + k_e \times E$ (%)
1.	100	20	80	7.6	15.6	152	1248	14.0
2.	100	50	50	10.0	18.0	500	900	14.0
3.	100	90	10	13.2	21.2	1188	212	14.0

k_o/A = the overall cost of capital
D = the amount of debt the firm carries
i_j = the rate on the very first incremental amount of debt, the pure borrowing rate
$D/_A$ = equals the percentage of debt to total assets, the degree of leverage
A = total assets equals the sum of D and E
E = the amount of equity capital
e_j = the return required on a non-leveraged share of stock for an industry of this risk class
p = the sum of the borrower's and lender's risk premium, set equal to the difference between i_j and e_j
$k_i = (i_j + D/A \cdot p)$; it is the overall market interest rate on debt at the given degree of leverage
$k_e = (e_j + D/A \cdot p)$; it is the market's desired return on equity at the given level of leverage

Showing how this works in a simple arithmetical example, let us set:

i_j at 6.0%
ej at 14.0%
and p becomes 8.0%

The result is the same as that in the graph in Fig. 2. No matter what the degree of leverage, the nontax cost of funds is constant at 14.0%. Moreover in Table 3, when

the resultant change in the costs of the other components of the financial mix is considered, the marginal cost of capital is constant of 14.0%, again no matter what type of financing is employed.[8]

The simplified version of the M&M model points out its difficulties. As the debt ratio reaches 90% of the financial structure, the least economic blip can bring about failure and bankruptcy. The costs of restructuring after bankruptcy are substantial. If the firm's capital structure is 100% equity, issuing debt increases the return per share without substantially increasing risk. Of course, Professor's Modigliani and Miller did not present their hypothesis as a pure mathematical abstraction. They presented devices by which the outside investors could offset any anomalies in the financial structure of firms having marketable shares.

(a) Arbitrage—M&M argued that the external investors could engage in arbitrage to create an increased return on shares that were underleveraged. They could increase risk but raise the expected rate of return by employing debt to buy the shares.

    The counter-argument is that this is not a perfect offsetting tool. There are additional transaction costs, and external debt is more dangerous to maintain under varying returns and asset values and more dangerous in terms of legal liability to the external or individual investor than the internal debt of the firm.

(b) Hedging—External investors could reduce the risk on a firm which was over-leveraged by purchasing both the shares and the bonds (debt) of the firm. This might reduce the expected return, but it would provide a minimum fail-safe position (i.e., the final value of the bonds) in case of failure.

    The counter-argument involves the problem of transaction costs and the fact that the offset in holding the bonds is far from perfect in case of failure and bankruptcy. The legal and transaction costs of going through bankruptcy and reorganization are likely to erode a large part of the value of the firm.

The major contribution of the M&M model is to show that each type of financing, debt or equity, effects changes in the costs of the other. Nevertheless, when the costs of failure, bankruptcy, reorganization, and various transaction costs are considered, it is clear that the trade-off is not likely to be perfect. Perhaps we can sum up by noting that the central difficulty of the pure M&M model is the problem of the asymmetry of information. Should not a judicious financial management knowing the environmental conditions of their firm do a better job of setting up the financial structure than the outside investor? And if this is so, have we not arrived at the notion of an optimal capital structure?

[8]Modern financial management uses the beta of a security to determine the firm's cost of equity. The reader will see the beta estimation in chapter "Risk and Return of Equity and the Capital Asset Pricing Model," see Hamada (1971, 1972). The estimated beta is produced by regression a security monthly returns as a function of the corresponding market returns. Because the vast majority of securities have issued long-term debt, the estimated betas involve leverage. The beta of a security rises with debt. M&M II is consistent with estimated betas and costs of equity.

Finally, the empirical evidence that financial structures are not random, but appear to be significantly different for varying classes of firms, points in the direction of the existence of optimal capital structure.[9]

Nevertheless, M&M made an important contribution. Firstly, it is clear that the bounds of the optimal capital structure are much broader than theory would suggest. Secondly, the M&M hypothesis sharpens the argument or more clearly points out the tax advantage (the tax deductibility of interest) of debt under our current corporation income tax laws.

M&M recognized the cost of capital implications of interest deductibility in their original 1958 study. M&M held that the interest deductibility feature of corporate taxation leads to a decreasing cost of capital as the debt ratio rises. By 1963, M&M formulated the before-tax earnings yield, the ratio of expected earnings before interest and taxes, $\overline{x}$, to the market value of the firm, $\widehat{v}$, as

$$\frac{\overline{x}}{v} = \frac{\rho^T}{1-t}\left[1 - t \ \ \frac{\cdot D}{V}\right].$$

Due to the tax advantage, the cost of capital of the firm decreases with leverage, and the value of the firm will rise with the use of debt. The reader is referred to Solomon (1959, 1963a, 1963b) for presentations of the early theories of capital structure and Harris and Raviv (1991) for a more modern summary of the cost of capital. Miller (1977) maintains that tax rates create optimal capital structures using higher leverage.

The Optimal Capital Structure and the M&M Hypothesis

The difference between optimal capital structure theory and the M&M hypothesis can be exaggerated. Both models emphasize the point that the use of one class of financing has rebound effects on the costs of the rest of the financial structure. In the optimal model, the overall cost of capital at any given time is constant within the range of the optimal capital structure. Debt or equity financing or some combination may be used for any particular project, as long as the financial mix is kept within an optimal range. Nevertheless, because every type of financing has interactions with the other sources of financing, the return on a project is not to be compared to the direct cost of its mode of financing but to the overall cost of the financial mix.

In the M&M model, the interaction between different types of financing is complete, so there is no optimal financial structure. Thus, the firm's overall cost of capital at any point of time is constant at the proper financial mix, or it is constant regardless of the mix. Quite importantly, both of these views are in opposition to the

[9]E. Schwartz and J.R. Aronson, "Some Surrogate Evidence in Support of the Concept of Optimal Financial Structures," *Journal of Finance*, March 1967.

sequential cost models, in which the cost of capital depends on the financing which is being used currently so that the cost is lowest when the firm uses retained earnings, rises for outside borrowing, and becomes still higher when borrowing capacity is strained and additional funds depend on the flotation of new shares. In short, in making real investment decisions, Schwartz and M&M agree that the appropriate discount variable is not the immediate financial source but the overall cost of capital.[10]

Empirical Factors Influencing Financial Structures

The two main external factors influencing the financial structure of a firm are the composition of its assets and the stability of its cash flow. Financial firms, such as banks and insurance companies, are prime examples of enterprises where the liquidity and marketability of their assets enable them to carry a high proportion of liabilities. Of course, in this instance, the firm's selection of assets for safety, marketability, and liquidity may be predetermined by the heavy volume of the firms' contingent or short-term liabilities, rather than the other way around. Nevertheless, a firm with safe marketable or short-term assets can finance these assets with a high proportion of debt with relatively matching maturities. Thus, marketing firms carry short-term inventories, and creditable short-term accounts receivable can safely carry a relatively high proportion of short-term debt.

The stability of cash flow is influential in shaping the financial structure. The cash flow is the amount of free funds the firm can utilize over a short-run period. Cash flow and accounting profits or earnings may differ considerably. Cash flow is less than earnings, for example, by any increases in costs incurred on work in process; cash flow exceeds reported earnings by the extent of depreciation, depletion, and other book or noncash changes—i.e., noncash charges representing the using up of assets acquired in the past. Although for internal control and budgeting purposes detailed analyses are made of the components of the cash flow, the rough rule of thumb for measuring the cash flow is reported earnings for the period plus depreciation, depletion, and any other noncash charges.

In calculating the leverage a firm might reasonably carry, the financial decision-makers must not only estimate the average level of the cash flow over time but the likelihood and extent of deviations from the norm. Where fluctuations from the average are not expected to be either deep or sustained, the firm may safely carry a high percentage of debt.

The inclusion of depreciation charges in the cash flow helps explain why firms with a good proportion of fixed assets may also carry more long-term debt. The fact that firms having a considerable fixed plant usually float bonds is not related to any physical attribute of the fixed plant; it is not dependent on any presumed safety that

[10]Schwartz has observed that the cost of capital is probably more invariant with respect to the debt-to-total assets ratio than he thought in 1959 and observed in 1966.

bricks and mortar bring to the bond mortgage. The affinity of fixed assets and long-term debt rests on the fact that the cash flow of firms holding considerable fixed assets must exceed their reported earnings. The depreciation charges taken against the fixed assets act as an extra cushion, which, added to the accounting net earnings, may help the firm meet its interest and principal obligations. A firm may show zero accounting profits after depreciation, yet have a positive flow of cash. As long as reported losses do not exceed depreciation and depletion charges, some cash flow will be available to pay debt obligations. In other words, some cash is always generated as long as operating revenues are greater than out-of-pocket operating costs, no matter what the depreciation charges may be.

Measures for Approximating Financial Risk

There are no precise methods for measuring the degree of risk in a firm's financing. In practice, however, there are some rough measures of risk that can be computed and then compared to certain traditional "rules of thumb" or "benchmarks." These have been established from experience, and not surprisingly, econometric studies have shown that weakness in these measures is quite predictive of financial failure or bankruptcy.

For current liabilities, the best measures of risk are the various working capital ratios. Chapter "Financing Current Operations and Efficiency Ratio Analysis" focuses on credit analysis and risk using these ratios. For long-term debt, introduced in the previous chapter, the best measures of financial risk (or safety) are the various "times interest earned" ratios and the balance sheet financing structure percentages. The most insightful presentation of the financing structure shows the major financing categories as a percentage of total assets. A growing misleading financial structure ratio is the percentage of debt to the market value of the stock! In an overexuberant market, the price of the shares will balloon, encouraging the incurring of debt levels which cannot be borne in more reasonable time.

Dividing the annual interest charges into the annual pretax earnings gives the interest earned coverage. Other things being equal, the larger the "times interest earned" ratio, the greater the safety of the funded debt. Of course, the coverage the financial analysts deem adequate is larger for an industrial, with its greater likelihood of fluctuating cash flow, than for the base service electric utility company. The rule-of-thumb minimum requirement for the highest grade industrial bonds is an interest coverage averaging eight times over the last 7 years, whereas only three times was required for the standard utility bonds.

The measure of earnings generally used in constructing the "times interest earned" ratio is EBIT, earnings before interest and taxes, or gross operating income. The earnings of a firm before interest and taxes represent the difference between costs and revenue; following economic theory, this is the variable the firm attempts to

maximize. EBIT is the firm's overall economic return, given the best efforts of the marketing people and those in charge of production costs. Net profit, the earnings of the shareholders, is derived from EBIT, but it is also a function of the financial structure, how much goes to bondholders and how much remains for the shareholder.

EBIT is the proper measure of earnings flow adequate to support a given debt structure. Interest is a deductible expense before income or profit taxes are computed. Net after-tax profit is an understatement of the firm's ability to pay its fixed charges. Thus, a firm with interest charges of, say, $200,000, reporting a zero income (no profits, no losses), will have covered its interest charges at least once. Assuming a 35% average tax rate, the same firm reporting a net after taxes income of $100,000 will have covered its fixed charges about 1.76 times.

The Nemo Corporation has the following financial structure:

Financial structure			
			Percent
Current liabilities—trade		$170,000	22
Bank loan, 4½%		50	
Long-term debt (5% interest)		200,000	20
Preferred stock (6% dividend)		30,000	3
Common stock	$200,000		
Surplus	350,000	550,000	55
Total capital and liabilities		$1,000,000	100

and the following income statement:

Income statement	
Revenues	$1,500,000
Operating expenses	1,300,000
	200,000
Depreciation	50,000
EBIT	150,000
Interest expense	12,250
Earnings before taxes	137,750
Taxes (assume 50% approximate rate)	68,875
Earnings after taxes	68,875
Preferred stock dividend	1800
Earnings available for common	67,075

$20,000 of fixed debt must be retired for sinking fund every year

Using Nemo Company's figures, we can present examples of various measures of the times charges covered under various definitions and approaches. All of the methods are valid; they vary, but they show different useful relationships. The first example is the coverage ratio most commonly used and the one which is usually indicated in the term "times interest or times fixed charges earned."

1. Times fixed charges earned (usual basis)

$$\frac{\text{EBIT}}{\text{Interest Charges}} = \frac{150,000}{12,250} = 12.3$$

All interest charges are divided into the EBIT. Preferred stock dividends are not included in the fixed charges, since they are not a legal liability of the company and do not have to be met in any given year. Although the sinking fund contribution is an annual obligation, it is not considered an expense or interest charge but a repayment of debt and thus is not normally included in the fixed charges. Nor is it deducted before computing normal accounting profits or income for tax purposes. In railroad financial analysis, contractual rental payments on leased lines are often included in the fixed charges. In analyzing other types of firms, rental obligations are generally not included in the fixed charges. A firm that rents a large part of its fixed assets may appear much more conservatively financed than a similar firm that has title to its fixed assets and has financed them by borrowing on mortgage bonds. Thus, many financial analysts suggest that, at least for internal analysis, some proportion, perhaps a third, of annual rental payments be added to the fixed charges.

2. Times prior charges earned (includes preferred dividends)

$$\frac{\text{EBIT}}{\text{Interest} + \text{Preferred Dividends} + \text{Taxes on Preferred Dividends}} = \frac{150,000}{12,250 + 1800 + 1800} = \frac{150,000}{15,850} = 9.5$$

This measure (times prior charges earned) includes the coverage on the preferred stock. Although it is a comprehensive measure, it is not commonly used. Since the preferred stock dividends are not deductible before income taxes are calculated, the taxes applicable to these dividends must be added back before the coverage is computed. In the example presented, $15,850 is the minimum the firm must earn, before interest and taxes, in order to cover interest, taxes, and preferred stock dividends. Nevertheless, the preferred dividends are not fixed charges since their payment is at the discretion of the directors, and a decision not to pay is not considered a failure to meet contractual obligations.

3. Times interest earned on a cash flow basis

$$\frac{\text{EBIT} + \text{Depreciation}}{\text{Interest Charges}} = \frac{200,000}{12,250} = 16.3$$

The measurement of times interest earned on a cash flow basis is a gauge of the short-term ability of the firm to meet its interest charges. The annual depreciation allowance is not an actual expenditure in any given year, but a bookkeeping charge to approximate the firm's average annual capital consumption. Funds from both earnings and depreciation—as long as they haven't been shifted into nonliquid assets—are available to pay charges on the debt, at least on a short-term basis.

4. Times debt service charges earned on a cash flow basis

$$\frac{\text{EBIT} + \text{Depreciation}}{\text{Interest Charges} + \text{Debt Retirement}} = \frac{200,000}{12,250 + 20,000} = \frac{200,000}{32,250} = 6.2$$

This ratio points out the significant relationship between the cash flow and all the service charges on the debt. Service charges here include interest and the annual obligation to repay part of the principal.[11] By relating debt service charges to the cash flow, we can achieve some estimate of the company's short-run ability to meet its debt obligations. In our example, no tax component is added to the sinking fund charge, although repayment of debt is not deductible before computing profit taxes. However, the funds available from depreciation, which are also nontaxable, more than cover the sinking fund requirement. If the sinking fund requirement exceeds the annual depreciation, it would be necessary to add income taxes to the excess sinking fund requirements before computing the coverage. If the ratio had been calculated as "times debt service charges covered on an earnings basis," the taxes applying to the sinking fund installment would have to be taken into account. Thus, if a firm has to meet a $300,000 sinking fund installment and the income tax rate is approximately 50%, then the firm has to earn $600,000 before taxes if we are to consider the sinking fund as being paid out of earnings. However, relating the sinking fund payment and the interest charges to the estimated cash flow is the more relevant procedure.

It is apparent that various debt and prior charge coverages can be calculated in many ways. The stability of the coverage is generally more important than the absolute level of the coverage. Thus, the ratios are often calculated for 5 or 10 years back; the average coverage for the period is determined and compared to the coverage of a few of the poorest years. A firm showing minimal fluctuations (i.e., one that is probably cyclically resistant) can be allowed a lower average coverage than the one whose EBIT shows strong fluctuations and whose coverage of fixed charges in the poorer years shows a wide deviation from its average coverage.

Outside Financing Capacity

Until now, we have discussed the theoretical foundations of optimal capital structure and seminal role of M&M theorems These discussions are predicated on the notion that a firm that is willing to pay a sufficiently high-interest rate has theoretically

[11]This is, of course, the concept generally used in local public finance. Since governments do not usually measure their success in terms of maximum money returns and minimum money costs, the commercial accounting that distinguishes between expenses and repayment of principal is not too important. A prime tool of municipal financial analysis is the level of debt service charges related to regular yearly revenues; this relation is some indication of the issuing area's economic ability to carry its debt.

unbounded borrowing capacity. In this section, we introduce the notion of credit rationing that questions the validity of this central assumption and discusses some practical ramifications of its failure. We refer the reader to Tirole (2006) for more advanced analytical discussion on this topic and limit ourselves to key insights.

Bester and Hellwig (1987) argued that a would-be borrower is said to be *rationed* if he cannot obtain the loan that he wants even though he is willing to pay the interest that lenders are asking, perhaps even a higher interest. In practice, such credit rationing seems to be commonplace: some borrowers are constrained by fixed lines of credit which they must not exceed under any circumstances; others are refused loans altogether. As far as one can tell, these rationing phenomena are more than temporary dislocations in otherwise efficient lending markets. Indeed, they seem to inhere in the very nature of the loan market.

Economists have studied credit rationing for several years and broadly attribute its existence to two key factors, namely, moral hazard and information asymmetry. Both of these factors pivot on a simple observation: an interest rate increase has no impact on the borrower in the event of bankruptcy as long as the borrower is protected by limited liability. Moral hazard explanation is that the reduced stake may demotivate the borrower and prompt him to engage in suboptimal capital allocation by favoring projects with higher personal benefits as compared to projects with higher net present value (NPV) for the firm and, in extreme cases, engage in outright fraud. Adverse selection explanation is that, in a situation where the lenders cannot directly tell good and bad borrowers apart, higher interest rates tend to attract low-quality borrowers, for low-quality borrowers are less affected by a rise in the interest rate than high-quality borrowers. Lenders may want to keep interest rates low in order to face a better sample of borrowers.

Credit rationing affects different firms differently depending on the strength of their balance sheet, nature of the underlying business, uncertainty of the future cash flows, and the ability of the lender to discern if the borrower used the funds for the overall benefit of the firm or for personal benefit. Next, we provide a simple example to illustrate circumstances that lead to credit rationing and effect of company fundamentals in alleviating or aggravating the resulting problems.

Consider an entrepreneur who has a project that requires fixed investment I with the following characteristics. If undertaken, the project either succeeds, that is, yields income $R > 0$, or fails and yields zero income. The probability of success is denoted by p. Note that p depends on both exogenous (macro factors, market conditions, competition, etc.) and endogenous (entrepreneur motivation) factors.

The entrepreneur initially has "assets" $A < I$. In order to undertake the said project, the entrepreneur needs to borrow I − A from lenders. Notably, once the entrepreneur has secured loan for his project, he can "behave" or "misbehave." Behaving in this context corresponds to efficient deployment of capital so as to maximize the chances of success for the project; we denote by pH the probability of success for the project when the entrepreneur "behaves." Misbehaving, on the other hand, corresponds to a scenario wherein the entrepreneur takes personal benefit B from the raised capital and operates in a manner that reduces the probability of

success for the project; we denote by *pL (<pH)* the probability of success for the project when the entrepreneur "misbehaves."

Among other things, the loan agreement also determines how the lender and entrepreneur will share the income from the project. Let RL denote the share of the lenders in the project profits, and let RB = R − RL denote the share of the entrepreneur. Note that expected profit for the lender in the "behave" scenario is given by pH*RL − (I − A). Assuming that the lending supply market is perfectly competitive, we can assume without loss of generality that the lender's profit is equal to zero, i.e. pH*RL = (I − A).

From the lender's perspective, it is imperative that the entrepreneur "behaves" and allocates the loaned capital in an efficient manner; in other words, the lender will enter into an agreement only if it can establish without doubt that the entrepreneur has sufficient monetary incentives to "behave." The entrepreneur's perspective is a little more complex. The entrepreneur has different incomes under the "behave" and "misbehave" scenarios as discussed below.

- "Behave": entrepreneur income = pH * RB
- "Misbehave": entrepreneur income = pL * RB + B

Consequently, a necessary and sufficient condition for the entrepreneur to behave is

$$\mathrm{pH} * \mathrm{RB} >= \mathrm{pL} * \mathrm{RB} + \mathrm{B}\,[\text{Viability Condition}]$$

Note that lenders will be unwilling to lend capital to the entrepreneur even for projects with positive NPV if the above condition is violated, resulting in credit rationing. We illustrate this statement through a numerical example.

Consider a hypothetical scenario when I = 100, A = 50, B = 20, pH = 70%, pL = 50%, and R = 150. In this case, the entrepreneur needs to raise I − A = 50 units of capital from the lender for the project. The project produces income of R = 150 units when successful and zero units in case of failure. Furthermore, the probability of success for the project under the scenario that the entrepreneur behaves (misbehaves) is 70% (50%). Note that the project has a positive NPV under the "behave" scenario computed as pH*R − I = 5 > 0.

As discussed earlier, we can compute the lender's share of the income by solving the equation, pH*RL = (I − A) yielding RL = 71.43 and RB = R − RL = 78.57. Next, we evaluate the income of the entrepreneur under the two scenarios:

- "Behave": entrepreneur income = pH * RB = 55
- "Misbehave": entrepreneur income = pL * RB + B = 95

Given that the entrepreneur has a higher income under the "misbehave" scenario, no lender will be willing to fund this positive NPV project, resulting in credit rationing. Note that credit rationing in this particular instance results from a credibility gap. If the entrepreneur can convince the lender that he will "behave" once the funds have been received, the lender will be willing to fund the project. However, prima facie the high private benefits (B = 20) and high level of investment (I − A = 50) make it difficult for the entrepreneur to formulate a credible story.

Let's revisit the viability condition which can be restated as follows:

$$pH * RB >= pL * RB + B$$

$$(D.p) * RB >= B \,(\text{where } D.p = pH - pL)$$

$$RB >= B/D.p$$

$$R - RL >= B/D.p \,(\text{since } RB + RL = R)$$

$$R - (I - A)/pH >= B/D.p$$

$$A >= I - pH * (R - B/D.p).$$

The final inequality has the following interpretation. An entrepreneur can obtain loan for a project only if the initial assets he has access to, namely, A, are greater than or equal to the threshold amount A(min) = I − pH* (R − B/D.p). Indeed, in the numerical example presented above, A(min) = 65 < A = 50.

Factors that increase A(min) − A entrench the credit rationing phenomenon, making it difficult for the entrepreneur to raise capital. On the other hand, factors that reduce A(min) − A mitigate credit rationing. We examine a few such factors and study their marginal effect on the credit rationing phenomenon.

Increasing the initial amount of fixed assets, namely A, reduces A(min) − A and can go a long way in mitigating credit rationing problems. Stated differently, ceteris paribus firms with strong balance sheets that have adequate amount of liquid assets (cash and cash equivalents) are less likely to be exposed to credit rationing problems.

Reducing the private benefit, namely B, that the entrepreneur can potentially derive by misbehaving reduces A(min) − A and can help avoid credit rationing problems. This can be arranged by carefully drafted loan agreement that earmarks the funds for project milestones and contractually disallow misuse of funds for entrepreneur personal use. Alternatively, the lender can provide fixed assets that have utility only in the context of the project under consideration, thereby further reducing the private benefit that can accrue to the entrepreneur in the possibility he decides to misbehave.

Assuming that the differential, D.p = pH − pL, remains constant, increasing the probability of success pH reduces A(min) − A and can be helpful in avoiding credit rationing problems. In other words, if the entrepreneur can demonstrate factors that have increased the likelihood of success for the project that can reduce the barriers to procuring credit. For example, successful completion of a critical R&D project or abnormal success of an initial limited launch of a product can provide evidence for higher probability of success for the project. Similarly, increasing values of R reduce A(min) − A and can be assist in alleviating credit rationing problems. In other words, reduced uncertainty (higher pH) and/or higher magnitude (high R) of future cash flows associated with a project reduces the likelihood of credit rationing.

References

Bester, H., & Hellwig, M. (1987). Moral hazard and equilibrium credit rationing: an overview of the issues. In G. Bamberg, K. Spremann (Eds.), *Agency theory, information, and incentives*. Springer, Berlin, Heidelberg.

Hamada, R. S. (1971). Investment decision with a general equilibrium mean-variance approach. *Quarterly Journal of Economics, 85*, 667–683.

Hamada, R. S. (1972). The effect of the firm's capital structure on the systematic risk of common stocks. *Journal of Finance, 27*, 435–452.

Harris, M., & Raviv, A. (1991). The theory of the optimal capital structure. *Journal of Finance, 48*, 297–356.

Lerner, E. M., & Carleton, W. T. (1966). *A theory of financial analysis*. New York: Harcourt, Brace & World.

Miller, M. H. (1977). Debt and taxes. *Journal of Finance, 32*, 261–276.

Modigliani, F., & Miller, M. (1958). The cost of capital, corporate finance, and the theory of investment. *The American Economic Review, 48*, 261–297.

Modigliani, F., & Miller, M. (1963). Corporate income taxes and the cost of capital: A correction. *American Economic Review, 53*, 433–443.

Schwartz, E. (1959). Theory of the capital structure of the firm. *Journal of Finance, 14*(1), 18–39.

Schwartz, E. (1962). *Corporate finance*. New York: St. Martin's Press. Chapter 10.

Solomon, E. (1959). *The management of corporate capital*. New York: The Free Press.

Solomon, E. (1963a). Leverage and the cost of capital. *Journal of Finance, 18*(2), 273–279.

Solomon, E. (1963b). *The theory of financial management*. New York: Columbia University Press.

Tirole, J. (2006). *The theory of corporate finance*. Princeton, NJ: Princeton University Press.

Investing in Assets: Theory of Investment Decision-Making



Capital budgeting, or investment decision, depends heavily on forecasts of the cash inflow and a correct calculation of the firm's cost of capital.[1] Given the cost of capital, i.e., the appropriate discount rate, and a reasonable forecast of the inflows, the determination of a worthwhile capital investment is straightforward. An investment is desirable when the present value of the estimated net inflow of benefits (or net cash inflow for pure financial investments) over time, discounted at the cost of capital, exceeds or equals the initial outlay on the project. If the project's present value of expected cash flow meets these criteria, it is potentially "profitable" or economically desirable; its yield equals or exceeds the appropriate discount rate. On a formal level, it does not appear too difficult to carry out the theoretical criteria. The stream of the forecasted net future cash flows must be quantified; each year's return must be discounted to obtain its present value. The sum of the present values is compared to the total investment outlay on the project; if the sum of the present values exceeds this outlay, the project should be accepted.[2]

The formula for obtaining the net present value of a project runs in this form:

$$PV = \frac{CF_1}{(1+i)} + \frac{CF_2}{(1+i)^2} + \frac{CF_n}{(1+i)^n} + \frac{S_n}{(1+i)^n} \tag{1}$$

$$\text{NPV} = \text{PV} - \text{I}.$$

[1]One need consider the case of Joel Dean's degree of necessity, in which projects must be undertaken regardless of their economic benefit (profit). Dean (1954) suggested that replacing a shop destroyed by fire could be an example of the degree of necessity, and he further suggested that you could not quantify the degree of necessity.

[2]The discounted cash flow approach has been widely accepted since Dean (1954).

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_11

Table 1 Net present value of capital project

Years	Investment cost of project	Estimated net annual inflows	Discount factor (cost of capital = 12%) $\frac{1}{(1.12)^n}$	Present value of inflows
0	$4,000,000	0	0	0
1		$1,000,000	.8929	$892,857.14
2		1,500,000	.7972	1,195,800.00
3		2,000,000	.7118	1,423,600.00
4		1,500,000	.6355	635,500.00
5		1,000,000	.5674	283,713.43
6[a]		500,000	.5674	283,713.43
Total	**$4,000,000**			**$4,431,470.57**

[a]Period 6 includes a return of $250,000 and $250,000 scrap value

PV is the present value of the net cash flow stream (CF_1, CF_2, etc.) over time to n years, S_n is the scrap value or the remaining value of the project at the end of its economic life at year n, and i is the applicable discount rate or cost of capital. NPV equals the net present value, the present value of the benefit stream minus I, the full investment cost of the project.

If there is a cost for removing the project at the end of its economic life, then S (the scrap value) is negative. If the stream of returns is constant, their present value can be obtained by the summarization annuity formula: $PV = CF[1 - (1 + i)^{-n}]/i$.

Table 1 provides an example of the mechanics of the capital evaluation problem. The project illustrated would be accepted because the present value of the estimated stream of net returns is $4,431,470.57, which is $431,470.57 above the project's initial cost of $4,000,000.

Thus, the projected rate of return on the project is higher than the 12.0% discount rate, the estimated cost of capital.

Net Present Value and the Internal Rate of Return

There are two standard criteria for selecting desirable projects, but sometimes they may yield conflicting rankings.

1. Net present value. Net present value is obtained by subtracting the initial outlays from the gross present value of the benefits discounted at the given cost of capital. A project is acceptable if the NPV is positive.
2. The internal rate of return. The internal rate of return is the rate that brings the present value of the cash flows into equality with the initial outlay. The equation for the internal rate of return is formally similar to that for present value.

$$I = \frac{CF}{(1+r)} + \frac{CF_1}{(1+r)^1} + \quad \ldots \quad + \frac{CF_n}{(1+r)^n} \tag{2}$$

I (the initial investment) is a given factor and one solves for r, the internal rate of return, i.e., the rate of discount, that brings the present value of the benefits equal to the outlay, I. If the internal rate of return (IRR) exceeds the cost of capital, the project is economically feasible.

In the vast majority of capital investment projects, both criteria give the proper signal as to whether a single project is acceptable. If a project's net present value is positive, it necessarily follows that its rate of return also exceeds the company's cost of capital. The project's positive net present value increases the firm's net income and cash flow and increases its stock price, ceteris paribus. However a selection conflict may arise when in comparing mutually exclusive projects, one project has a higher internal rate of return and one shows a higher net present value. If two projects are mutually exclusive, then one project can be accepted.

A sound capital budgeting takes into account increments to net working capital, additions to current assets in excess of current liabilities, and annual incremental capital expenditures. Let us assume that we can introduce a new machine that costs $30 MM, which will produce a new product, producing sales in its initial year of production of $17.5 MM, and a 5% annual sales growth in sales during the 8-year life of the asset. These are several costs associated with the machine. Fixed costs, costs incurred regardless of production, will total $1.0 MM, and variable costs, such as materials, are estimated to be 65% of sales. Depreciation will be charged as allowed during a 7-year life by the Accelerated Cost Recovery System, ACRS, reported in chapter "The Corporation Balance Sheet." Earnings before interest and taxes, EBIT, are found by Sales-Variable Costs-Fixed Costs-Depreciation. Taxes are paid at the incremental tax rate of 35%. The change in net working capital, dNWK, are $85,000 in year 1 and 10% of the change in sales, thereafter, and annual capital expenditures, CE, are $950,000 in year 1 and 15% of the change in sales, thereafter. The project operating cash flow is the machine's net income plus depreciation, and the machine's cash flow is its operating cash flow less its change in net working capital and its capital expenditures. The machine (project) discounted cash flow is discounted at a 10% cost of capital. The machine's present value, NPV, is $27.58 MM, which is less than its $30 MM cost.

	Year							
	1	2	3	4	5	6	7	8
Sales	17,500,000	18,375,000	19,293,750	20,258,438	21,271,359	22,334,927	23,451,674	24,624,257
Variable costs	11,375,000	11,943,750	12,540,938	13,167,984	13,826,384	14,517,703	15,243,588	16,005,767
Fixed costs	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000
Depreciation	4,287,000	7,347,000	5,247,000	3,747,000	2,679,000	2,679,000	2,679,000	1,335,000
EBIT	838,000	(1,915,750)	505,813	2,343,453	3,765,976	4,138,225	4,529,086	6,283,490
Taxes	293,300	(670,513)	177,034	820,209	1,318,092	1,448,379	1,585,180	2,199,222
Net income	544,700	(1,245,238)	328,778	1,523,245	2,447,884	2,689,846	2,943,906	4,084,269
dNWK	85,000	87,500	91,875	96,469	101,292	106,357	111,675	117,258
CE	95,000	131,250	137,813	144,703	151,938	159,535	167,512	175,888
Project Op. CF	4,831,700	6,101,763	5,575,778	5,270,245	5,126,884	5,368,846	5,622,906	5,419,269
Project CF	4,651,700	5,883,013	5,346,091	5,029,073	4,873,654	5,102,954	5,343,719	5,126,123
Disct CF	4,228,860	4,855,839	4,016,518	3,434,857	3,026,052	2,880,618	2,742,397	2,391,336
PV(CF)	27,576,476							
Costs	30,000,000							
NPV (CF)	(2,423,524)							

The machine's net present value, its PV less the cost, is −2.42 MM. The new machine should not be purchased because its NPV is negative.

Mutually Exclusive Projects

A problem of selecting projects apparently exists when more projects pass the economic test than a fixed capital budget will allow and/or when projects exist that are mutually exclusive alternatives (i.e., only one is to be taken from among several possible projects because they serve the same function). The NPV and the IRR can give conflicting signals in these cases where a selection must be made out of mutually exclusive projects.

As will be shown later, the constraints imposed by a prefixed capital budget is the result of a wrongful management decision. Nevertheless, in the general situation, supposedly a conflict in the rankings of mutually exclusive projects poses a problem. However, in every case, NPV indicates the optimum choice.

Projects may show up with conflicting rankings if the following three (not necessarily exclusive) conditions exist:

1. The shapes of the inflows over time differ.
2. The investment sizes of the projects differ.
3. The duration of the inflows differ.

The first condition, i.e., the shape of the inflows, is illustrated in Table 2. Project A has a higher internal rate of return than B (11.3% as against 10.0%), but given the cost of funds rate of 6.0%, Project B's net present value is $113,500 as against $85,000 for A. When this type of conflict in ranking exists, the project with the higher internal rate of return has the higher earlier inflows: The rival project with the higher net present value (NPV) has a greater total inflow and relatively higher returns in later periods. If the decision between the projects is made on the basis of the rate of return, the earlier returns of A are acquired at the sacrifice of the greater returns from

Table 2 Comparison of projects when the shapes of the cash flows differ

Years	Outlays	Inflows	Discount rate 6%	PV of inflows
Project A				
1	$1,000,000	$700,000	.9434	$660,400
2	0	200,000	.8900	178,400
3	0	200,000	.8396	167,900
4	0	100,000	.7921	79,200
	$1,000,000	$1,200,000		$1,085,500
Project A has a net present value of $85,500. Its internal rate of return is calculated at 11.3%				
Project B				
1	$1,000,000	$100,000	.9434	$94,300
2	0	200,000	.8900	178,000
3	0	700,000	.8396	587,700
4	0	320,000	.7921	253,500
	$1,000,000	$1,320,000		$1,113,500
Project B has a net present value of $113,500. Its internal rate of return is calculated at 10.0%				

B offers $38,000 in additional net present value over Project A

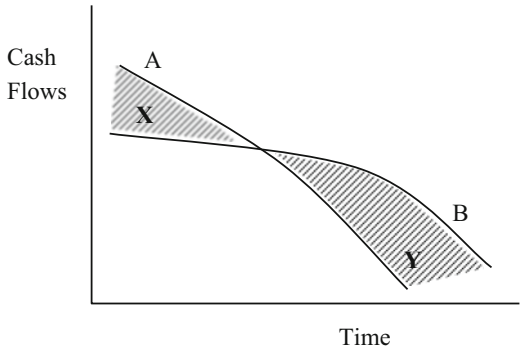


Fig. 1 Different shapes of cash flows of two investment options

B later on, and this trade-off is made at a higher rate than the appropriate cost-of-funds warrants.

The effect of the differing shapes of the cash flows (the size of the investment constant) may be further explained with the aid of the graph shown in Fig. 1.

Project A has the advantage of a greater earlier cash inflow, but Project B promises higher later cash flow.

Indeed, Project B offers higher total flow. The area X indicates the earlier differential inflow of A; the area Y depicts the later extra inflows of B. Absolute Y is greater than absolute X. (If the absolute earlier returns were greater, there would be no problem; A would always win under either criteria.)

If the discount rate was zero, benefits would not be discounted, and B would always win. (B wins at lower rates.) On the other hand, at some high enough positive discount rate, A will win; however, if the going cost of capital is used as the discount rate and B still shows the higher present value, it follows that the sum of the present values of the funds depicted in Y must be higher than the present value of X. Alternatively, if the funds depicted as X were to be invested to earn the cost of capital, they would not grow to equal the amount of Y.

In the case illustrated in Table 2, if the discount rate was 8.26%, the NPV of the two projects would come to the same amount—$47,661. However, if the discount rate goes past 8.26% (but is less than 11.7%), the present value of X will exceed Y. This involves no paradox; it simply means that when the cost of capital is high, earlier returns are preferred over later returns (because capital is relatively scarce and very productive), and the present value of Project A will exceed that of B.

Sometimes the argument is made that the comparison need not be made between Projects A and B but that a whole investment strategy should be considered. Suppose the argument goes that X funds could be used in other projects, the returns for which show a higher present value than Y. Should not A be undertaken in preference to B? The answer is no. B is still preferable to A; X can be replaced by market supplied funds, which will be priced at the firm's cost of capital. The original argument is fallacious because it is based on a misuse of the concept of alternative or opportunity costs. In fact, the present value criterion is correct because it uses an objective

measure in its calculations. The firm's cost of capital is a market-determined rate (modified by the firm's peculiarities), and it basically reflects the supply and demand for capital in the economy as a whole. Properly calculated, it is an opportunity cost; it is the rate the firm must pay to acquire additional new funds, and it is also the imputed value of the moneys to stockholders should the firm elect to use the funds to repay some of its stockholders.

Difference in Project Size

Condition 2, in which the sizes of the two projects differ, is illustrated in Table 3. Here, the smaller project, A, has a greater internal rate of return, but the larger project, B, has a higher net present value. This means that the size increment in Project B earns benefits at a rate higher than the cost of capital, even though there is an "averaging down" of the rate of return. A simple proof is to conceive of the bigger project in two parts, one of which (i.e., B − A) may be considered an increment to the smaller project. Since the incremental project has a positive NPV, it is a desirable or profitable addition. Here again, the criterion of net present value gives an unambiguous correct answer.

A simple intuitive example comparing rates of return to maximizing wealth is to consider two **exclusive** options, K and L. Option K requires a payment of $100 and will pay an assured return of $500 at the end of the year. Option L costs $1000 and returns $2000 at the end of year. The return on K is 500%, on L 100%. For any normal rate of interest and financing availability, L is far preferable. The extra return of $1500 for an additional outlay of $900 makes L a more valuable choice than K.

One rating criterion that has often been suggested is the benefit/cost ratio, i.e., where the present value of the benefits divided by the cost of the project. However, as shown in Table 3, where the benefit/cost ratio favors Project A, this ratio is

Table 3 Comparison of projects

Project	**A**	**B**
Cost	$1,000,000	$1,200,000
PV of inflows at 6%	1,500,000	1,770,000
Internal rate of return	12.0%	11.5%
Benefits/cost ratio	1.5×	1.48×
Net present value	$500,000	$570,000
Consider the difference between A and B as an independent project, B − A		
Project	**B − A**	
Cost	$200,000	
PV of inflows at 6%	270,000	
Internal rate of return	9.0%	
Benefits/cost ratio	1.35	
Net present value	$70,000	

misleading when the projects differ in size. In fact, the optimal project size is reached when dPV/dI = 1 or when the discounted increase in inflows equals the marginal increase in the cost of the project. The proper criteria may not maximize the benefit/cost ratio; it does maximize net present value or total wealth.

Differing Duration of the Inflows

The third condition, differing duration of the investment benefits, is a problem when the competing investment projects are renewable. If the projects are once-and-done activities, the one with the highest net present value is preferable. However, when the substitutable projects are renewable, not only may their internal rates of return differ but a net present value of, say, $100,000 for a stream of benefits lasting 5 years cannot be compared directly to the net present value of $120,000 for an alternate project of 7 years' duration.

The problem involves a comparison of different strategies carried out over time, not merely a comparison of the return on the initial investments. The evaluation of strategies may not be very difficult if it can be assumed that each project could be renewed at the end of its life at similar costs and benefits as the present project. In this case, the net present values of each project can be annualized—reconverted into an equivalent flat annual amount over the life of each project. This is done by dividing the net present values by the present value annuity formula at the appropriate rate to yield the uniform annual equivalent return (UAER). The UAER amounts then can be compared directly. Table 4 illustrates such a problem. Here, although the NPV of

Table 4 Comparison of projects when durations differ (using the method of annualization)

Year	Outlay	Returns	Disc. factor (at 10.0%)	Present value
Project A				
1	80,000	60,000	.909	54,540
2		50,000	.826	41,300
3		40,000	.751	30,040
	80,000			125,880
NPV = $125,880 − 80,000 = $45,880 Equivalent annual stream of return for 3 years at 10% = $18,450				
Project B				
1	$170,000	80,000	.909	72,720
2		70,000	.826	57,820
3		60,000	.751	45,060
4		50,000	.683	34,150
5		40,000	.621	24,840
	$170,000			$234,590
NPV = $234,590 − $170,000 = $64,590 Equivalent annual stream of return for 5 years at 10% = $17,040				

Project B is larger, the equivalent annuity of Project A (at 10.0%) is $18,450 per annum; the equivalent annuity of B is $17,040 per annum. Thus, a series of A projects is preferred to one of B projects since it results in a higher stream of net benefits over time.

It might be noted at this point that if the cost of capital (discount rate) should drop below 6.8%, Project B yields the greater annualized return and should be selected. This conforms to the basic rule that lower interest rates favor more capital-intensive and longer-run projects.

In the case in which the future renewal costs and benefits of each project may not be exact replicas of the original projects, annualization will not work. The solution must be obtained by comparing the net present value of a series of linked, shorter-lived projects to the net present value of an alternate series of longer projects, both ending at a reasonably common time. Thus, if capital Project A lasts 4 years and rival Project B lasts 6 years, a comparison of the net present value of three A-type projects with forecasted costs and benefits renewed at the end of 4 years and 8 years, respectively, should be made with the net present value of two linked type B projects also lasting a total of 12 years. Forecasting costs and returns accurately is difficult, but the calculation itself is not so hard. Thus, if

I = the project cost at each renewal period
CF = benefits for each time period
r = cost of capital (discount rate)
NPV = net present value.
then

$$1.\mathrm{NPV_A} = -\mathrm{I_{A_1}} + \frac{\mathrm{b_1}}{(1+\mathrm{r})} + \cdots + \frac{\mathrm{b_4}}{(1+\mathrm{r})^4} - \frac{\mathrm{I_{A4}}}{(1+\mathrm{r})^4} + \frac{\mathrm{b_8}}{(1+\mathrm{r})^8} + \cdots + \frac{\mathrm{b_8}}{(1+\mathrm{r})^8} - \frac{\mathrm{I_{A8}}}{(1+\mathrm{r})^8} + \frac{\mathrm{b_8}}{(1+\mathrm{r})^8} + \cdots + \frac{\mathrm{b_{12}}}{(1+\mathrm{r})^{12}}.$$

$$2.\mathrm{NPV_B} = -\mathrm{I_{B_1}} + \frac{\mathrm{CF_1}}{(1+\mathrm{r})} + \cdots + \frac{\mathrm{CF_8}}{(1+\mathrm{r})^8} - \frac{\mathrm{I_{B8}}}{(1+\mathrm{r})^8} + \frac{\mathrm{CF_7}}{(1+\mathrm{r})^7} + \cdots + \frac{\mathrm{CF_{12}}}{(1+\mathrm{r})^{12}}.$$

The forecasted costs of renewing each project when the time comes and estimates of the extended benefits must be made specific. Then NPV A may be compared with NPVB to see which is larger.

Lowest Annualized Total Costs

The annualization method comes into play in a variant problem where the benefits are a given and the question involves choosing the equipment or technique that shows the minimum cost. For example, suppose one piece of equipment has a lower initial cost but higher projected annual operating costs than an alternative which has a higher initial purchase price but lower projected ongoing operating and maintenance costs. The choice is made by taking the present value of the operating costs in each case and adding it to the purchase price. This total is then divided by the present value of the appropriate annuity factor to obtain the uniform annual equivalent cost (UAEC). The choice goes to the equipment showing the lowest UAEC.

The Irrational Fixed Capital Budget

As noted previously, in some circumstances, the decision-makers are forced to choose between projects because they are faced with a fixed capital investment budget. It should be noted before proceeding that an arbitrary fixed budget constraint is basically irrational. Presumably, an operating, viable corporation can acquire funds at an appropriate cost of capital, just as it can buy or hire any other productive factor. The proper capital budget is flexible and not fixed. It is set at the amount of profitable investment opportunities within the purview of the firm. Instead of equating the amount of funds to be made available to the opportunities for profitable real investment, a fixed capital budget arbitrarily rations the amount of capital investment. Such a budget can result in uneconomic behavior; it can force the substitution of less efficient projects or delay the implementation of worthwhile improvements. Any economically desirable project that cannot be undertaken because of a limited budget represents a loss of an opportunity to create net wealth and represents a loss of wealth.[3] In short, the planned outlays for capital projects, the capital budget, should not be fixed a priori; the so-called budget should be constructed after all projects, the costs and returns for which have been properly counted, are evaluated, and are passed upon. Presumably, if all projects are evaluated at the ongoing and obtainable cost of funds, the ideal procedure is to enlarge the budget to accommodate all worthwhile noncompeting projects.

[3] In a case of capital rationing, the profitability index is calculated and used. The profitability index is the project's net present value divided by its initial cost (Brealey & Myers, 2003).

Operating Practice and the Internal Rate of Return

In operating practice, it may be well to show decision-makers both the results of the net present value criteria and the internal rate of return in order to demonstrate that the return on a feasible project having a positive net present value is above the cost of capital. Nevertheless, the optimum decision must go to the project showing the higher net present value. The internal rate of return is a ratio; a higher ratio is not the proper choice when it forecloses a project that has a higher absolute addition of wealth over cost. Essentially, the objective of capital choice is to maximize total wealth or value rather than the rate of return.

Real Investments and the Cost of Funds

The problem central to a firm's long-run financial policy is the selection of investment projects—that is, the choice of business projects based on the likely return compared to the cost of the funds needed to finance them. The theoretical concept for determining the desirability or profitability of a project does not entail any difficulties. In application, however, it may involve fairly complicated calculations.

The relation between the firm's cost of funds and the return from the earnings opportunities open to it is the major determinant of the success of the company. If the company can consistently apply funds to its operations at a rate of return above the cost of the funds used, it is considered a growth firm, and the owners of the common equity will experience financial happiness. In any case, as long as the management of the corporation does not regularly invest funds at a return lower than their costs—or consistently overlook profitable investment opportunities—the management must be deemed a success.

Applying Investment Theory

The application of investment theory requires the decision-makers to insert numerical estimates into the equations:

$$PV = \frac{CF_1}{(1+i)} + \frac{CFn}{(1+i)^n} + \frac{Sn}{(1+i)^n} \tag{3}$$

$$\text{NPV} = \text{PV} - \text{I}$$

All the variables are subject to some degree of uncertainty in the calculation.

CFs forecasted periodic estimated benefits. In the case of a pure business operation, the usual problem involves the calculation of the net after-tax cash flow of the project. Sometimes a project, e.g., a recreation area for the employees, has no cash

inflow but is estimated to yield benefits in terms of increased employee morale and employee improved retention rates. CF for each period is an estimate of some reasonable amount that the firm might pay for these beneficial improvements. If the project is a business operation, then CF is the periodic net after-tax cash flow the investment is forecasted to yield. This involves a forecast of the gross cash revenues minus the estimated out-of-pocket operating costs. These costs do not include economic depreciation which is implicit in the fact that n, the life of the project, is finite. The amount of depreciation allowable under the tax code is used to calculate the amount of cash flow subject to tax. The amount of tax, not the depreciation, is then subtracted to obtain the after-tax cash flow. The project should bear a share of managerial costs insofar as management costs increase with the volume of operations and presumably decrease with a cutback.

S is the forecasted scrap or withdrawal value of the project at the end of its economic life. S can include current or working capital no longer required when the operation ceases. S need not be a positive amount. It can be a negative amount if there are to be clean up or demolition costs; on the other hand, there may be recoverable land value.

The Estimated Economic Life of the Investment in the Project The decision-maker needs to make a forecast as to when machinery or buildings will wear out and maintenance costs become excessive or when obsolesce could require the substitution of new technology. However, because of the effect of discounting, mistakes in estimating n are inconsequential when the forecast for n is a long period.

The appropriate discount rate used to discount the inflows in order to obtain PV, the present value of the project is depicted as i. For a project falling within the normal purview of the firm, the ideal discount rate is the weighted after-tax cost of the components making up the optimal financial structure for the firm. (An after-tax cost is employed because CF, the cash inflow, has been calculated on an after-tax basis.) Various adjustments are suggested where the level of risk differs from that of the usual operation.

I is the total investment outlay on the project. The investment on the project includes all the machinery, fixtures, and construction costs. It should include the imputed value of land costs. I should include working capital assets, cash needs, and working inventory that will be required by the project. If the project requires time to complete, a periodic charge at the cost of capital rate should be added to the investment cost of project. If projected initial operating returns are negative and show operating losses before the project gets "online," these start-up costs should be added to the total investment.

The investment, I, is subtracted from PV, the present value of the inflows, to obtain NPV, the net present value of the project. If NPV is zero, the project is expected to generate a normal rate of return, one that will satisfy all the investors in the firm. If NPV is positive, the return is better than the norm; extra profits rebound to the shareholders. If investment, I, exceeds PV, then the project does not cover the costs of capital.

Illustration of an investment decision, ABC, Inc. financial structure

	Normal financial structure percent of total assets (%)	Pretax cost current market	After-tax cost (tax rate 35%) (%)	Weighted after-tax cost of K_0 (%)
Current liability	30	5.0%	3.25	0.98
L.T. debt	30	8.33%	5.42	1.63
Equity	40	15.0% (desired pretax rate of return)	10.50	4.20
				Total 6.81

The project requires an outlay of $300,000 in fixed assets and $200,000 in gross working capital (i.e., current assets). That is, investment, I equals $500,000.[4] $180,000 of the current assets will be available to the firm after the project ends in 4 years. The fixed assets will be worthless. The tax rate is 35%.

Year	Before tax cash flow in excess of operating costs	ACRS depreciation expenses	Effective taxes (35%)	After-tax cash flow	Discount factor (k = .0681)	P.V.
1	$120,000	$99,990	42,000	$177,990	.9363	$166,652
2	280,000	133,350	98,000	315,350	.8767	276,467
3	300,000	44,430	105,000	239,430	.8209	196,549
4	100,000	22,230	22,230	100,000	.7686	76,860
4a	180,000[a]			180,000	.7686	138,348
	Total PV					$854,876

[a]Return of working capital

$$\text{NPV} = \$854,876 \text{ minus } \$500,000 \text{ or } \$354,876.$$

The project is worthwhile because its NPV exceeds zero.

Myers (1974) has suggested that the adjusted present value model (APV) be considered relative to the weighted average cost of capital (WACC). In the APV approach, project cash flows are broken down into unleveraged, operating cash flow and cash flow associated with financing the project. Different discount rates can be used to reflect the present value of the tax shield on debt used to finance the project. The reader is reminded that we discussed this tax shield on debt in chapter "Debt, Equity, the Optimal Financial Structure, and the Cost of Funds."

[4]The ACRS depreciation charges for a 3-year asset are 0.3363, 0.4445, 0.1481, and 0.0741, respectively.

CFO Practice

In their survey of 392 chief financial officers (CFOs), Graham and Harvey (2001) found that approximately 76% of the CFOs almost always used internal rate of return and net present value calculations in project or acquisition decisions. These decisions are well-grounded in the financial decisions previously discussed. Graham and Harvey (2001) also found that approximately 57% of the CFOs surveyed used payback period and hurdle rate calculations. The payback period and hurdle rate calculations were the third and fourth methods, respectively, of analyzing projects. Adjusted present value and the profitability index, the latter being optimal for capital rationing, or a fixed capital budget, were the least used techniques. Let us discuss the payback period. The payback period is the ratio of the cost of capital investment divided by its annual earnings before depreciation (Gordon, 1955). The payback period gives equal weight to the cash flow before the payback period, or "cutoff date," and ignores all cash flow after the cutoff date (Brealey & Myers, 2003).[5]

[5]Gordon (1955) revealed that the payback period is proportional to the reciprocal of the project's rate of profit.

I = Cost of the project
E = Earnings or savings before depreciation of the project
n = Number of years of the project
s = Scrap value at time t = n
k = project's expected rate of profit

$$C = \frac{E_1}{1+k} + \frac{E_2}{(1+k)^2} + \cdots + \frac{E_k}{a+k} + \frac{E_n+S_n}{(1+k)^n} \quad (4)$$

Gordon found if one ignored the scrap value, that

$$C = \sum_{t=1}^{n} \frac{S}{(a+k)^t}$$

$$= \frac{S}{k} - \frac{S}{k}\left(1+\frac{1}{(a+k)}\right)^n \quad (5)$$

where Eq. (5) is the present value of an annuity

$$k = \frac{E}{C} - \frac{E}{C}\left(\frac{1}{1+k}\right)^n \quad (6)$$

As the life of the project becomes very large, Eq. (6) becomes

$$k = \frac{E}{C}$$

and the rate of profit is the reciprocal of the payback period. Furthermore, if the project life is infinite, then the rate of profit is the reciprocal of the payback period. The reader immediately recognizes the Gordon payback period analysis is consistent with the Gordon valuation of stock model shown in chapter "The Equity of the Corporation: Common and Preferred Stock." If we buy a stock, we purchase a stream of expected future dividends. The stock price, P_{CS}, or P_0, is determined by the current dividends per share, D_0, and the growth rate of future dividends, g.

$P_0 = \frac{D_0}{d-g}$

where $k = \frac{D_0}{P_0+g}$.

The hurdle rate is a minimum acceptable rate of return established by management. It is generally higher than the firm's cost of capital (Lanzilotti, 1958; Lerner & Carleton, 1966).

Current Costs of the "Optimum" Financial Mix

For normal projects falling within the regular operations of the firm, the basic discount rate use the cost of the financial mix which appears optimum for the type of firm. This involves some qualitative judgments as to the long-run reaction of the financial and investment market to financial risk. The analyst must estimate the amount of debt of all types the firm can safely and profitably carry, given the typical nature of its operations and of its asset structure. An important feature of this approach is its attempt to take into account the "full cost" of capital, including costs which may not be immediately explicit. If, for example, the firm already has a large equity base, it may be able to borrow at a relatively low rate to finance new operations. To assume that the cost of capital for the new projects is merely the external interest rate is to fail to account properly for the implicit costs of the "extra" equity capital already existing in the firm. Although a new project might be financed by debt to bring the firm's financial structure into balance, the firm must consider the full imputed costs of its capital mix. A given project is not burdened or credited with just the cost of its particular financial source but with the costs of the overall "optimum" capital costs for the firm.

Adjusting the Capital Mix and Costs for Individual Project

A rather subtle but perhaps useful concept in getting at the cost of capital for a particular project is to consider it, if it is large enough, as a financial entity. Potential projects differ not only in their forecasted rate of return but also in the certainties of the forecast and the possibilities of asset losses. Thus risky projects implicitly require heavier admixtures of relatively expensive equity capital (in a non-M&M world), whereas less risky ventures can be financed more heavily with cheaper debt. This approach furnishes one method of dealing with risk; the more risky projects are discounted at a higher rate before they are accepted.

If in actuality as the firm expands it could adjust its financing to the type of projects it undertakes, the capital structure will change over time. Such a policy

The rate of profits earned by purchasing stock is its current dividend yield, $\frac{D_0}{P_0}$, plus the rate at which dividends are expected to grow, as long as $k > g$ (Gordon & Shapiro, 1956).

slowly but continually moves the firm's financial structure into line with the type of enterprise, product, operations, and assets it currently maintains.

Closing or Continuing Operations

A difficulty may arise in calculating the desirability of additional investment in projects or areas already in operation. In some circumstances, outlays for the purpose of preventing a fall in earnings may be profitable. The difference in the forecasted flow of earnings with the project and without the project is the implied earnings of the investment. If, however, the mistake is made of setting the investment at just the cost of the additional assets, the rate of return will be grossly exaggerated. Properly, the amount of the investment is the sum of the new assets plus the liquidation or withdrawal value of the assets presently used in the operation. For example, a firm may contemplate additional outlays in a sector of its operations, not with the hope of increased returns but to prevent deterioration in its competitive position. The value of the losses forestalled may appear high compared to the cost of the additional assets. Yet the firm has another alternative, to withdraw entirely from this area of its activities. The salvage value of the existing investment plus the cost of the new assets must be compared to the present value of the flow of funds forecasted from the operation.[6] Only if the return is large enough on this basis should the operation be continued.[7]

On the other hand, the recoverable value of assets already committed to a given line of production is usually quite low; this furnishes one explanation why firms often continue to invest in activities whose return is stable or declining. (The return on the sum of the new repair and maintenance assets plus the low liquidation value of the old assets may be relatively good.) Thus economically it may be difficult for a firm to withdraw from a going operation.

[6]If a firm does not make the proper cost-of-investment determination, it could develop an unprogressive policy in regard to equipment replacement. The firm could repair its equipment long past an economic optimum because the fairly large total earnings on a repaired piece of equipment is credited just to the costs of repair. The economic investment in a reconstruction job, however, is the repair costs plus the salvage value of the unrepaired equipment. Furthermore, the forecasted cash flow should subtract the estimate of maintenance costs that may still have to be made in the future. When the problem is thus properly formulated, it may be that a replacement will show a higher rate of return than a repair job.

[7]In a sense, the existence of the whole firm may be periodically measured against the same criterion: how does the withdrawal value of the firm's assets compare to the discounted value of its projected cash flow.

Stability of Forecasts: Risk of the Investment

The level of the cash flow, CF, an investment is expected to bring in over the years is an estimate or forecast. In some cases, the net cash flow can be estimated with more confidence than in others. For example, the cash flow generated by a cost savings project usually can be estimated with more certainty than a project directed to expanding output. The projected return over out-of-pocket cost is likely to be more accurate on the expansion of the output of an old product than on the introduction of a new item, and the funds flow can be forecasted more reliably for a product with a stable demand than for one subject to high style or income change volatility or price uncertainty.

For different types of projects, the estimated profitability on forecasts is more problematical in some instances than others. The dispersion of the probability outcome of the rates of return for different kinds of company investments is one aspect of the varying level of risk of different projects. Thus, suppose Project A is likely to return something between 10% and 16% over its life with an expected mean return of 13%, project B has the same expected return of 13%, but the range of probable outcomes is between −2% and 28%. Project A is less risky and, other things, being equal, is preferred; possibly another way of putting it is that it can be financed with a riskier—i.e., less costly—financial mix.

Another aspect of risk is the fixity or specialization of the assets that must be committed to the project. A project requiring an investment mostly in current assets can be salvaged (if it seems desirable) with less loss than a project requiring mainly an investment in new or remodeled plant. Similarly, an investment in automotive equipment can be withdrawn with less loss than the funds put into fixtures or into a specialized piece of machinery. To make intelligent decisions on investment of the corporation's funds, the management should have not only the estimates on the rate of return but also the possible range of the rate of return and estimates of the maximum possible loss.

The Theory of Dealing with Risk

For the typical project, the analysis usually presents a single-point forecast or estimates of the future cash flows and the projected expenditures and then applies the given cost of funds, to decide whether the project is financially worthwhile. For a large number of usual business investment decisions, the element of uncertainty is not a major factor because problems that are likely to arise in the usual course of operations are included in the forecasts, and the use of a normal cost of funds as the discount rate seems reasonable.

Theory still leaves considerable difficulty providing solutions to problems which involve large factors of risk or uncertainty in the forecast of future benefits, in the stream of estimated future costs, or in the estimate of the full cost of the total outlay

required to implement the project. In general, the extant literature on how to handle inordinate risk leaves us with a considerable degree of uncertainty.

Risk-Adjusted Discount Rate

For some projects for which the range and duration of future cash flows are difficult to forecast, it has been suggested that an extra premium sufficient to offset the disutility of the hazards inherent in the project be added to the normal discount rate. For example, if the decision-makers would be content in the usual case with an overall 10.0% return, they are asked to evaluate the project with a risk-adjusted rate of 20.0%. Thus, a higher-than-normal expected return is required for acceptance of a risky project. It has, however, been pointed out that the increased discount rate places a proportionately greater value on near returns and severely devalues later inflows. It clearly favors shorter-run projects over longer-run projects, even though the degree of uncertainty may be the same.

However, it could be argued that in circumstances where the level of future operating costs, future demand, or the potential development of competitive goods are hard to predict or when there is a great possibility of natural hazards or political instability, the length of time needed to recoup the investment is the essence of the risk. In these cases, for example, a project in a less then stable foreign country political environment, the risk-adjusted discount rate performs quite well.

Risk-Return Distribution

The risk versus the potential return on a potential investment is sometimes presented in a mathematical or graphical form in order to arrive at a decision. Figure 2 shows the risk-return profile of two projects, A and B. The probabilities of the different return outcomes are not scientific probabilities derived from frequency observations but a priori probabilities derived from the best estimates of experience in these types of activities. Project A has a higher expected return than B, but it has much greater variance and a much larger possibility of coming in as at a loss. The expected return for B is relatively low, but the range of possible outcomes is quite constrained, and there is only a slight probability of loss. If B's expected return is minimally above the risk-free rate of return, there should be no difficulty in finding takers. On the other hand, even though Project A has the possibility of being highly profitable, it appears to have a high level of risk. Whether it is implemented or not depends on the utility characteristics of the management, how strongly they evaluate high potential returns, and how willing and able they are to absorb any potential losses.

This discussion of variance is pertinent to the case in which the systematic risk of the projects is not much different than that of the rest of the economy (i.e., covariance is close to 1.0). However, according to portfolio theory, a project that has a large

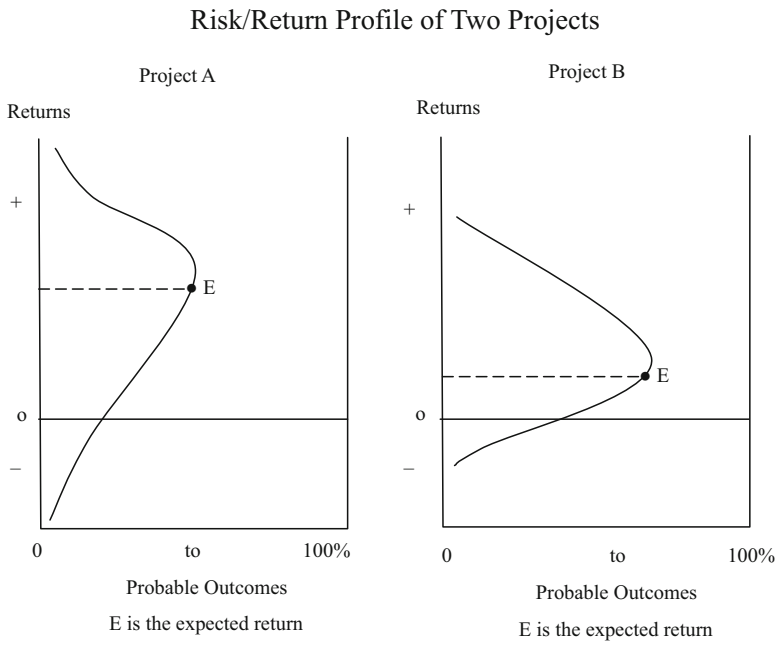


Fig. 2 Risk-return profile of two projects

variance which runs in an opposite direction than most of the rest of the economy serves the function of a hedge and may find willing takers. The cost of equity may be calculated using the required rate of return from the Capital Asset Pricing Model, which is introduced in chapter "Risk and Return of Equity and the Capital Asset Pricing Model."

Certainty Equivalence

Another suggested method of handling risk is to set up or forecast the range of each periodic benefit or return. The decision-maker is then asked to search his psyche (his utility function) and select a certain return for which he would be willing to trade the possible projected outcome. (This return is, of course, lower than the mean expected periodic returns on the project.) These certainty equivalent returns are then discounted by a nonrisk cost of capital to see whether the net present value calculated on this basis is profitable and the project worthwhile.

The certainty equivalence model furnishes some revealing insights on how individual utility preferences and risk aversion might enter into the investment decision. The inherent problem is that one model does not have an external market guide but relies on the decision-maker's psychological evaluation to frame the problem.

Maximum Loss and Reversibility

In dealing with risk, an important variable often left out of the discussion is the maximum possible loss. The maximum possible loss is related to the type of assets that must be committed to the project in case the investment needs to be liquidated. An investment that requires highly specialized, immobile assets (e.g., specific capital equipment and structures) which cannot be readily redirected to other uses will show a high maximum loss if operations prove unsuccessful. (Stated somewhat differently, investments in long-term, specialized assets are not readily reversible.) Outlays using skilled labor, research, development, and scientific testing are not likely to be recoverable if the outcome proves unusable. (However, the investment in training in skills may be turned to other uses in time.)

An investment with asset requirements that are mainly nonspecialized, standardized items, small unit flexible capital equipment or inventory that may be readily marketable (albeit with some loss) is not as risky in terms of termination costs. In order to make an informed decision, the decision-maker needs in addition to the range of possible outcomes, some estimate of the maximum loss in case the project is to be liquidated.

Gross Uncertainty

There are some ventures for which the gross payoff in case of success is simply tremendous. The potential returns are not truly calculable in percentages but in multiples of the original investment. Using interest or capital theory analysis in such a case is an exercise in pedantry. The present value of any successful outcome is simply off the board. Decision-making in these areas is a matter of weighing the odds; it is essentially a problem in game theory. The psychology of the decision-makers, their talents, and judgment shape the decision.

Market Risk

We will discuss the development of market, or systematic risk, in chapter "Risk and Return of Equity and the Capital Asset Pricing Model." Management can estimate the security beta to use in calculating a cost of equity and capital. We illustrate the calculation of costs of equity and capital using the Capital Asset Pricing Model beta in chapter "Mergers and Acquisitions" and value the estimated cash flow of a potential merger candidate. The discounted cash flow approach to merger valuation is an important example of the valuation technique employed in this chapter.

The Effect of Taxes on the Financial Structure

A corporate tax system that allows the deduction of interest on debt but taxes the return to the stockholders encourages the use of debt financing or leverage. An illustration of this effect, under the assumption of a defined optimal financial mix, is shown in Table 5. In a nontax world, the optimal financial mix would be at line 3 (60% equity, 40% debt), with the lowest weighted capital costs, 12.15% to be applied to *nontaxable* profits. Under a 34.0% tax rate on equity profits, the optimum financial mix shifts to line 5 (60% debt and 40% equity), showing a cost of capital 10.45% to be applied however to the after-tax cash flow. The example presented illustrates the existence of an optimal financial mix. Presumably in a theoretical frictionless, M&M world, the introduction of taxes would lead to a corner solution, with the lowest amount of equity possible as the optimum.

Costing the Components of the Financial Mix

In evaluating a project, the decision-maker will employ an overall cost of capital. But arriving at this figure requires an understanding and a casting of the components going into the financial mix. The financial includes all the sources used to acquire and support all the assets employed by the firm.

Cost of trade credit If the firm pays its trade creditors within the cash discount period, the cost of the credit is zero to the firm which obtains it.[8] There is a cost if the accounts are not paid within the allotted time, i.e., the loss of the discount. The supply of trade or book credit is not unlimited; the sellers ration it. In addition, the use of excessive book credit strains the debtor firm's liquidity position.

Cost of bank credit Bank credit may be needed only seasonally to help finance peak operations. Its cost can be charged in at its estimated average annual use multiplied by the going interest rate on bank loans. Since the rate of return on the project has been calculated after taxes and interest is tax-deductible, the cost of loan capital to the firm can be lowered by the reduction in the tax bill. If the tax rate is approximately 35%, then a bank loan at 5% costs the firm only 3.25% net.

Cost of long-term debt The cost of various long-term loans (bonds, debentures, subordinated debt, etc.) is averaged into the overall cost at their relative weights in the financial mix multiplied by their current market interest rates. The current rate of interest is the one that must be considered. The company may have bonds outstand-

[8]The cost is not zero to the economy as a whole, but the individual firm is not concerned with aggregate economic costs at this point.

Table 5 Effect of the corporate income tax on the optimum financial structure of the firm

Overall financial mix		Percent of each (%)	Pretax cost (investors' desired rate of return or current cost) (%)	Adjusted after-tax cost (assume 34% tax rate) (%)	Weighted cost of capital (assuming to corp. tax) (%)	After-tax weighted cost of capital (used to discount after-tax cash flows (%)
1.	Total debt	20.0	9.00	5.94	1.80	1.19
	Equity	80.0	13.25	13.25	10.60	10.60
	Total weighted cost				12.40	11.79
2.	Total debt	30.0	9.25	6.11	2.78	1.83
	Equity	70.0	13.50	13.50	9.45	9.45
	Total weighted cost				12.23	11.28
3.	Total debt	40.0	9.75	6.44	3.90	2.57
	Equity	60.0	13.75	13.75	8.25	8.25
	Total weighted cost				12.15[a]	10.82
4.	Total debt	50.0	10.75	7.10	5.38	3.55
	Equity	50.0	14.00	14.00	7.00	7.00
	Total weighted cost				12.38	10.55
5.	Total debt	60.0	12.00	7.92	7.20	4.75
	Equity	40.0	14.25	14.25	5.70	5.70
	Total weighted cost				12.90	10.45[b]
6.	Total debt	70.0	13.75	9.08	9.63	6.35
	Equity	30.0	16.78	16.78	5.03	5.03
	Total weighted cost				14.66	11.39

[a]Least costly financial mix without taxes
[b]Least costly financial mix after accounting for tax effect

ing bearing a 5% coupon rate, but if similar new bonds cannot be floated today at less than 7%, the cost of debt capital is 7%. The final after-tax cost of long-term debt capital is reduced by the equivalent amount of tax saving.

Cost of preferred stock Given an ongoing corporate profit tax, preferred stock is an expensive form of equity capital. It is generally used only because its subordinate position helps secure the safety of debt and thus lowers its cost. If preferred stock is employed as part of the capital mix, its cost to the firm is its current dividend yield. There is no tax saving on preferred since its dividends are not considered a cost but a division of profits and come out after taxes are calculated.

Cost of common stock The various uncertainties involved in the market evaluation of common shares make the costs of common equity funds the hardest to measure. A few guidelines help narrow the approximate range of these costs. The cost of common stock financing is based on its earnings yield and not its current dividend yield.

For normal stocks in a moderate market, the cost of equity capital may be approximated by the ratio $\frac{\text{earnings per share}}{\text{market price per share}}$ or the inverse of the traditional price/earnings ratio. The earnings used in the numerator must be a "normal" run of earnings. Neither the earnings per share of a poor year nor a boom year are appropriate.

The cost of equity funds is calculated on the market price of the shares. The amount earned on the book or asset value of the stock is of minor importance, since with passing of time the book value of the equity may diverge further and further from current economic realities and values.[9]

The earnings yield ratio is only a starting approximation of the cost of common stock financing. The current yield of a share on the market is influenced by the security markets' expectations and evaluation of the firm's future prospects. To get a usable cost of equity financing, the earnings yield has to be corrected for normal growth, potential recovery, forecasted decline, and super normal growth. In the case of normal sustainable growth, if the earnings yield on market is 6% (i.e., a P/E ratio of 16.6 times) and the sustainable growth rate of earnings is 4.0% per annum, the after-tax desired return is 10.0%. When the market expects a potential recovery from a depressed period, the P/E ratio may appear quite high, e.g., 40 times or an earnings yield of 2½%. Here, the current yield may be ignored. The cost of equity financing is imputed from the market, i.e., the normal average return for the general run of stocks in the broad industry class. Chapter "Risk and Return of Equity and the Capital Asset Pricing Model" further develops and estimates the return on equity by formally introducing the reader to the required rate of return on equity. The required rate of return on equity and the weighted average cost of capital are used to value a potential merger candidate in chapter "Mergers and Acquisitions."

[9]The amount made on the asset value of the equity—especially its growth or trend over time—may, however, be a fairly good index of the firm's economic success.

When the market expects a decline in firms' earnings, the current P/E ratio will be rather low, perhaps showing an earnings yield of 25%. Here again, after allowing for variant risk (i.e., the beta function), the cost of equity capital has to be imputed from that prevailing on the market.

In the case of super normal growth, the firm may possess technological or marketing advantages which enable it to obtain earnings growth well over the norm, e.g., 15–20% per annum. The P/E may be as high (or higher) than 50 times earnings or 2% earnings yield. Of course, such a growth rate cannot be sustained indefinitely; as someone has remarked, over time a firm with such a growth rate would soon constitute the whole economy. The 2% earnings yield is not the true cost of equity; the investors would soon be extremely dissatisfied if the firm invested in projects that returned a mere 2%. Once again the cost of equity has to be imputed from the long-run norm on the market.

A countercheck to the earnings yield is to consider the earnings yield relative to the long-run interest rate in a reasonably noninflationary market. A risk premium of perhaps 4–6% might be added to the rate on top-grade long-term debt to obtain the cost of equity capital.

Internal funds The net internal funds available to the firm for additional investment constitute the money inflow for the period in excess of the costs of operations less interest, taxes, preferred stock dividends, and required repayment of debt. Cash flow is a broad undifferentiated concept; nevertheless, for purposes of exposition, we should divide the internal funds to those derived from earnings and those derived from true economic depreciation.

The cost of retained earnings Historically the cost of using retained earnings to help finance a firm has been the subject of considerable debate. Some writings in corporation finance seemed to imply that retained earnings were costless; the use of retained earnings in lieu of other sources of funds was sometimes given almost "moral" approval. Internal financing supposedly avoided the use of debt and prevented the firm from having to deal with bankers or from having to appeal for new stockholders on the capital market. A careful reading of these arguments shows them to be mostly homiletical; they gave no guides as to the cost of using retained earnings in the firm and no aid in deciding how much of a given flow of earnings may be reasonably kept and how much should be paid out in dividends or used for stock buybacks.

A more scientific approach to the costing retained earnings is to impute to the retained funds their alternative or opportunity costs—that is, how much they could earn if they were paid out to the stockholders in dividends or in stock buybacks and reinvested in the best opportunities (of equal risk) available elsewhere. In effect, the firm should retain earnings only to the point where it can make a return on these funds equal to the normal market rate on equity funds. The cost of retained earnings should be imputed at the same rate as the normal return on shares of the same industry class. The general rule that retained earnings be given the imputed costs of equity is useful in solving the problem of capital cost, and it provides an answer to the problem of the amount of cash dividends or cash buybacks the firm should

disperse and how much earnings it might retain. All cash earnings which cannot be employed at the proper rate of profit should be returned to the shareholders.

For the average firm in the economy, there is no shortage of internal funds. Overall the corporate sector generates a greater cash inflow rate then the growth of the economy. The aggregate inflow of cash exceeds the amount of economically feasible real investment, and thus the corporate sector is a net exporter of cash to the rest of the economy.[10]

Other internal funds: depreciation, depletion, etc. As a sort of "shorthand" method of expression, depreciation, depletion, amortization of intangible assets, etc. are commonly listed as sources of funds. But deeper analysis makes it clear that an excess of *revenues* over out-of-pocket *operating* costs is only source of funds. Annual charges for depreciation and other such noncash charges are accounted as funds because reported accounting earnings are less than the total current funds flow. Depreciation, estimated amount of past capital dollar investment in fixed assets used up in the appropriate accounting period, is a charge against earnings but not against funds.

True economic depreciation based on the declining replacement value of the fixed assets represents a reemployment in the investment in capital. Funds arising from true economic depreciation carry the cost of the financial mix used to fund capital assets. In a theoretical sense, these funds might be returned to the original investors in the capital assets. The cost of keeping them in the business is what could be saved by not paying off an equivalent amount of debt and equity. Insofar as new projects are financed with funds arising from true depreciation, these must be given the imputed costs of the suitable financial mix. Under the current tax code, the allowable depreciation deduction exceeds the amount of true economic depreciation. Funds arising from this source must be considered under a different rubric then true economic depreciation. These funds constitute a sort of deferred equity fund granted to the company by the taxing authority. The proper charge for these funds, when retained in the business, is the imputed cost of equity funds.

Investments Under Negative Interest Rates and Hyperinflation

For better part of the past century, most developed markets have been characterized by positively sloped yield curve, positive interest rates, and "normal" rate of inflation (1–10%). However, there have been exceptions when the yield curve has gotten inverted, the interest rates have turned negative, or economies have suffered from hyperinflation (30–100% or more). While the inverted yield curve does not create

[10]"The Corporate Sector: A Net Exporter of Funds," J.R. Aronson and E. Schwartz, *Southern Economic Journal*, October 1966.

Table 6 Present value of $1 for various combinations of discount rates (−5%, 0%, and 5%) and durations (1, 2, . . ., 10 years)

Year 1	−5%	0%	5%
1	1.05	1.00	0.95
2	1.11	1.00	0.91
3	1.17	1.00	0.86
4	1.23	1.00	0.82
5	1.29	1.00	0.78
6	1.36	1.00	0.75
7	1.43	1.00	0.71
8	1.51	1.00	0.68
9	1.59	1.00	0.64
10	1.67	1.00	0.61

any technical issues with the material presented in this chapter, negative interest rates and hyperinflation do require special treatment as discussed below.

Negative Interest Rates It is important to note that the discount rate computation starts with the risk-free rates and then applies a series of usually positive adjustments such as beta adjustment for covariance with market returns, country/industry premiums, currency adjustment, and private equity or illiquidity discount. In the presence of these adjustments, the discount rate can turn positive even when the underlying risk-free rates are negative. However, in a few cases, the discount rate can still be negative, adjustments notwithstanding. We illustrate some of the pitfalls of using the NPV approach directly in such cases and offer guidance on how to proceed.

Table 6 reports the present value of $1 for various combinations of discount rates (−5%, 0%, and 5%) and durations (1, 2, . . ., 10 years).

As it is evident from Table 6, negative discount rates have an amplifying effect on the present value computation—present value of $1 for any duration is greater than $1. This creates a unique problem with respect to the computation of the terminal value; specifically, in the presence of negative discount rates, constant or positively growing stream of perpetual cash flows has infinite present value. In order to circumvent this problem, we need to understand and question the assumption of constant or positively growing perpetual cash flow stream in the presence of negative interest rates. Usually, negative interest rates arise in peculiar economic regimes that are also accompanied with deflationary tendencies, transient or short-term fiscal/monetary imbalances, or unusual geopolitical situations.

In such cases, an investor needs to determine if it is reasonable to assume negative interest rates in perpetuity and, if so, characterize the economic regime that will be concomitant with such a scenario. Among other things, perpetual negative interest rate will be accompanied with deflationary pressures and anemic growth that contradicts the assumption of constant or positive growing perpetual cash flow streams. Alternatively, one can conclude that negative interest rates are a short-term phenomenon and they will revert back to their "normal" (positive) behavior after a certain period of time, thereby resolving the issues with terminal value computation.

Hyperinflationary Regime In economics, hyperinflation refers to an excessive amount of high and accelerating inflation that quickly erodes the real value of local currency resulting in rapid and continuing rise in nominal prices of goods/services. Countries such as Brazil, Germany, Greece, Peru, the Philippines, etc. have experienced hyperinflation in the past century.

Performing NPV computation in a hyperinflationary suffers from three key challenges. First, it becomes extremely difficult to forecast future cash flows given the inherent uncertainty around supply and demand of goods/services. Even in situations where such a demand can be forecasted within reasonable economic precision in real terms, computing the nominal value of these cash flows becomes problematic due to unstable value of the local currency. Second, it is questionable whether the value of real assets (land, building, etc.) and present values of future cash flow streams are fungible in a hyperinflationary environment. This problem gets magnified while valuing firms and enterprises, one of the key practical applications of NPV methodology. Third, in multinational companies that do business in different parts of the world, hyperinflation presents a particularly unique conundrum, namely, how to combine the NPV of cash flows in a stable currency with those derived from a currency suffering from hyperinflation.

There are two ways of circumventing the hyperinflation problems in the context of NPV methods. First, one can move to performing all of the computations in real terms. While this approach does not relieve the investor from forecasting the supply and demands of goods/services, it does annul the need to forecast unstable inflation rates into the future. The second approach is to perform computations in a stable foreign currency that offers the best de facto stable proxy for the local currency.

Summary

A crucial problem in the management of a firm's capital is the selection of assets or projects whose estimated return will justify the cost of the funds invested in them. Although the details of the calculations vary, essentially projects are worthwhile when the cash inflow discounted at the appropriate cost of capital shows a present value equal or in excess of the original required investment. If the data underlying the calculation is correct, the project is estimated to have a return equal to or better than the cost of capital. Because the net cash flow on the project is a forecast and the required investment is an estimate, the actual profitability is also only a forecast; the project in operation may realize more or less.[11] Nevertheless, such an initial forecast

[11]It should also be recognized that the accounting results and the estimated rate of return may not necessarily coincide because accounting conventions on depreciation and calculation of profits may differ from the assumptions underlying the cash flow rate of return analysis. This needs not invalidate the usefulness of either the discounted cash flow techniques or of accounting procedures.

and calculation of expected value are essential steps in real asset investment decision-making.

If the management of a firm does its best in selecting assets and weighing their potential return against the cost of funds, then within the context of the prevailing economic conditions the best possible return will be maintained on the stockholders' investment without subjecting the firm to inordinate risk of failure.

Problems

1. A firm is considering making an investment decision. Given the following data, should the firm purchase the investment, given its 5-year-ahead cash flow valuation horizon? Please use straight line depreciation on the asset, and recover the working capital investment at the conclusion of the project.

Acquiring firm data:
Long-term debt is $20,000 MM.
Stockholder equity is 30,000 MM.
The current risk-free rate is 1.87%.
The relevant market risk premium is 8.8%.
Firm beta is 1.45 (value line).
Tax rate = 35%.
The AAA bond yield is 5.47%.

Project data:
Project cost: $32,000,000
Sales, 2004: $17,000,000
Variable costs = .70 × sales.
Fixed costs = $1500,000.
Tax rate = 35%.
Change in net working capital = .1 × change in sales.
Capital expenditures = .15 × change in sales.
Initial working capital investment = $500,000.
Initial capital expenditures = $600,000.

References

Brealey, R. A., & Myers, S. C. (2003). *Principles of corporate finance* (7th ed.). New York: McGraw-Hill/Irwin.

Dean, J. (1954). Measuring the productivity of capital. *Harvard Business Review, 32*, 120–130.

Gordon, M. J. (1955). The payback period and the rate of profit. *Journal of Business*.

Gordon, M. J., & Shapiro, E. (1956). Capital equipment analysis: The required rate of profit. *Management Science, 3*(1), 102–110.

Graham, J. R., & Harvey, C. R. (2001). The theory and practice of corporate finance: Evidence from the field. *Journal of Finance Economics, 60*, 187–243.

Lanzilotti, R. F. (1958). Pricing objectives in large companies. *American Economics Review, 48*, 921–940.

Lerner, E. M., & Carleton, W. T. (1966). *A theory of financial analysis*. New York: Harcourt, Brace, and World.

Myers, S. C. (1974). Interactions of corporate financing and investment decisions: Implications for capital budgeting. *Journal of Finance, 29*, 1–25.

Regression Analysis and Estimating Regression Models



A forecast is merely a prediction about the future values of data. Financial forecasts span a broad range of areas, and each of the forecasts is of interest to a number of people and departments in a firm. A sales manager may wish to forecast sales (either in units sold or revenues generated). This prediction is of interest to the operations (manufacturing) department in order to predict the materials and time needed to create the product. The corporate financial officer is interested in the amount of cash required to support the projected level of sales and how much available cash inflow he can eventually expect to pay financial costs, cover expansion programs, and provide cash payouts to investors. In short, good forecasting underlies the construction of an operational cash budget.[1]

There are a number of ways to make forecasts. One could consult tarot cards or be guided solely by one's intuition. Recent years have seen the explicit search for "megatrends" which, presumably, might continue into the future. However, regression analysis is the primarily techniques used in this chapter. Regression analysis is a preferred technique used to analyze quantitative data to make forecasts. More specifically, regression analysis seeks to find the "line of best fit" through the points. Basically, the regression line is drawn to best approximate the relationship between the two variables. Techniques for estimating the regression line (i.e., its intercept on the Y-axis and its slope) are the subject of this chapter.

Modern quantitative analysts use the regression techniques introduced in this chapter on a daily basis. One uses ordinary least squares and robust regression techniques to estimate stock betas, estimate stock selection model weights, and analyze the effectiveness of the leading economic indicators.

In simple regression analysis, one seeks to measure the statistical association between two variables, X and Y. Regression analysis is generally used to measure how changes in the independent variable, X, influence changes in the dependent

[1] See chapter "Financing Current Operations and the Cash Budget."

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_12

variable, Y. Regression analysis shows a statistical association or correlation among variables, rather than a causal relationship among variables.

Estimating an Ordinary Least Square Regression Line

The case of simple, linear, least squares regression may be written in the form

$$Y = \alpha + \beta X + \varepsilon \tag{1}$$

where Y, the dependent variable, is a linear function of X, the independent variable. The parameters α and β characterize the population regression line, and ε is the randomly distributed error term. The regression estimates of α and β will be derived from the principle of least squares. In applying least squares, the sum of the squared regression errors will be minimized; our regression errors equal the actual dependent variable minus the estimated value from the regression line. If Y represents the actual value and Y the estimated value, their difference is the error term, *e*. Least squares regression minimized the sum of the squared error terms. The simple regression line will yield an estimated value of Y, $\widehat{Y}$ by the use of the sample regression:

$$\widehat{Y} = a + bX + e \tag{2}$$

In the estimation Eq. (2), a is the least squares estimate of α, and b is the estimate of β. Thus, a and b are the regression constants that must be estimated. The least squares regression constants (or statistics) α and β are unbiased and efficient (smallest variance) estimators of α and β. The error term, e_i, is the difference between the actual and estimated dependent variable value for any given independent variable values, X_i:

$$e_i = \widehat{Y}_i - Y_i \tag{3}$$

The regression error term, e_i, is the least squares estimate of ε_i, the actual error term.[2]

To minimize the error terms, the least squares technique minimizes the sum of the squares error terms of the N observations

$$\sum_{i=1}^{N} e_i^2 \tag{4}$$

The error terms from the N observations will be minimized. Thus, least squares regression minimizes:

[2]The reader is referred to an excellent statistical reference, such as Irwin Miller and J.E. Freund, *Probability and Statistics for Engineers*, (Englewood Cliffs, NJ: Prentice-Hall, 1965).

$$\sum_{i=1}^{N} e_i^2 = \sum_{i=1}^{N} \left[Y_i - \widehat{Y}_i \right]^2 = \sum_{i=1}^{N} [Y_i - (\alpha + bX_i)]^2 \tag{5}$$

To assure that a minimum is reached, the partial derivatives of the squared error terms function

$$\sum_{i=1}^{N} = [Y_i - (\alpha + bX_i)]^2$$

will be taken with respect to *a* and *b*:

$$\begin{aligned}
\frac{\partial \sum_{i=1}^{N} e_i^2}{\partial a} &= 2 \sum_{i=1}^{N} (Y_i - a - bX_i)(-1) \\
&= -2 \left(\sum_{i=1}^{N} Y_i - \sum_{i=1}^{N} a - b \sum_{i=1}^{N} X_i \right) \\
\frac{\partial \sum_{i=1}^{N} e_i^2}{\partial b} &= 2 \sum_{i=1}^{N} (Y_i - a - bX_i)(-Xi) \\
&= -2 \left(\sum_{i=1}^{N} Y_i X_i - \sum_{i=1}^{N} X_i - b \sum_{i=1}^{N} X_l^2 \right)
\end{aligned}$$

The partial derivatives will then be set equal to zero:

$$\begin{aligned}
\frac{\partial \sum_{i=1}^{N} e_i^2}{\partial a} &= -2 \left(\sum_{i=1}^{N} Y_i - \sum_{i=1}^{N} a - b \sum_{i=1}^{N} X_i \right) = 0 \\
\frac{\partial \sum_{i=1}^{N} e_i^2}{\partial b} &= -2 \left(\sum_{i=1}^{N} YX_i - \sum_{i=1}^{N} X_l - b \sum_{i=1}^{N} X_l^2 \right) = 0
\end{aligned} \tag{6}$$

Rewriting these equations, one obtains the normal equations:

$$\begin{aligned}
\sum_{i=1}^{N} Y_i &= \sum_{i=1}^{N} a + b \sum_{i=1}^{N} X_i \\
\sum_{i=1}^{N} Y_i X_i &= a \sum_{i=1}^{N} X_i + b \sum_{i=1}^{N} X_l^2
\end{aligned} \tag{7}$$

Solving the normal equations simultaneously for a and b yields the least squares regression estimates:

$$\begin{aligned}\widehat{a} &= \frac{\left(\sum_{i=1}^{N} X_i^2\right)\left(\sum_{i=1}^{N} Y_i\right) - \left(\sum_{i=1}^{N} X_i Y_i\right)}{N\left(\sum_{i=1}^{N} X_i^2\right) - \left(\sum_{i=1}^{N} X_i\right)^2}, \\ \widehat{b} &= \frac{\left(\sum_{i=1}^{N} X_i Y_i\right) - \left(\sum_{i=1}^{N} X_i\right)\left(\sum_{i=1}^{N} Y_i\right)}{N\left(\sum_{i=1}^{N} X_i^2\right) - \left(\sum_{i=1}^{N} X_i\right)^2},\end{aligned} \tag{8}$$

An estimation of the regression line's coefficients and goodness of fit also can be found in terms of expressing the dependent and independent variables in terms of deviations from their means, their sample moments. The sample moments will be denoted by M:

$$\begin{aligned} M_{XX} &= \sum_{i=1}^{N} x_i^2 = \sum_{i=1}^{N} (x_i - \overline{x})^2 \\ &= N\sum_{i=1}^{N} X_i - \left(\sum_{i=1}^{N} X_i\right)^2 \end{aligned}$$

$$\begin{aligned} M_{XY} &= \sum_{i=1}^{N} x_i y_i = \sum_{i=1}^{N} (X_i - \overline{X})(Y_i - \overline{Y}) \\ &= N\sum_{i=1}^{N} X_i Y_i - \left(\sum_{i=1}^{N} X_i\right)\left(\sum_{i=1}^{N} Y_i\right) \end{aligned}$$

$$\begin{aligned} M_{YY} &= \sum_{i=1}^{N} y_i^2 = \sum_{i=1}^{N} (Y - \overline{Y})^2 \\ &= N\left(\sum_{i=1}^{N} Y_i^2\right) - \sum_{i=1}^{N} (Y_i)^2 \end{aligned}$$

The slope of the regression line, *b*, can be found by

$$\widehat{b} = \frac{M_{XY}}{M_{XX}} \tag{9}$$

$$\widehat{a} = \frac{\sum_{i=1}^{N} Y_i}{N} - b\frac{\sum_{i=1}^{N} X_i}{N} = \overline{y} - b\overline{X} \tag{10}$$

The standard error of the regression line can be found in terms of the sample moments:

$$\begin{aligned} S_e^2 &= \frac{M_{XX}(M_{YY}) - (M_{XY})^2}{N(N-2)M_{XX}} \\ S_e &= \sqrt{S_e^2} \end{aligned} \tag{11}$$

The major benefit in calculating the sample moments is that the correlation coefficient, r, and the coefficient of determination, r^2, can easily be found:

$$\begin{aligned} r &= \frac{M_{XY}}{(M_{XX})(M_{YY})} \\ R^2 &= (r)^2 \end{aligned} \tag{12}$$

The coefficient of determination, R^2, is the percentage of the variance of the dependent variable explained by the independent variable. The coefficient of determination cannot exceed 1 nor be less than zero. In the case of $R^2 = 0$, the regression line's Y = Y and no variation in the dependent variable is explained. If the dependent variable pattern continues as in the past, the model with time as the independent variable should be of good use in forecasting.

The firm can test whether the *a* and *b* coefficients are statistically different from zero, the generally accepted null hypothesis. A t-test is used to test the two null hypotheses:

$$H_{0_1} : a = 0$$

$$H_{A_1} : a \,\text{ne}\, 0$$

$$H_{0_2} : \beta = 0$$

$$H_{A_2} : \beta \text{ ne } 0$$

where ne denotes not equal.

The *HO* represents the null hypothesis while *HA* represents the alternative hypothesis. To reject the null hypothesis, the calculated t-value must exceed the critical t-value given in the t-tables in the appendix. The calculated t-values for *a* and *b* are found by

$$t_a = \frac{a - \alpha}{S_e}\sqrt{\frac{N(M_{XX})}{M_{XX} + \left(N\overline{X}\right)^2}}$$
$$t_b = \frac{b - \beta}{S_e}\sqrt{\frac{(M_{XX})}{N}} \tag{13}$$

The critical t-value, t_c, for the .05 level of significance with N – 2 degrees of freedom can be found in a t-table in any statistical econometric text. One has a statistically significant regression model if one can reject the null hypothesis of the estimated slope coefficient.

We can create 95% confidence intervals for a and b, where the limits of a and b are

$$a + ta/_2 S_e + \sqrt{\frac{\left(N\overline{X}\right)^2 + M_{XX}}{N(M_{XX})}}$$
$$b + ta/_2 S_e \sqrt{\frac{N}{M_{XX}}} \tag{14}$$

To test whether the model is a useful model, an F-test is performed where

$$H_o = \alpha = \beta = 0$$
$$H_A = \alpha \text{ ne } \beta \text{ ne } 0$$
$$F = \frac{\sum_{i=1}^{N} Y^2 \div 1 - \beta^2 \sum_{i=1}^{N} X_i^2}{\sum_{i=1}^{N} e^2 \div N - 2} \tag{15}$$

As the calculated F-value exceeds the critical F-value with (1, N − 2) degrees of freedom of 5.99 at the .05 level of significance, the null hypothesis must be rejected. The 95% confidence level limit of prediction can be found in terms of the dependent variable value:

$$(a + bX_0) + ta/_2 S_e \sqrt{\frac{N\left(X_0 - \overline{X}\right)^2}{1 + N + M_{XX}}} \tag{16}$$

Autocorrelation

An estimated regression equation is plagued by the first-order correlation of residuals. That is, the regression error terms are not white noise (random) as is assumed in the general linear model, but are serially correlated where

$$\varepsilon_t = \rho\varepsilon_{t=1} + U_t \quad t = 1, 2, \ldots N \tag{17}$$

ε_t = regression error term at time *t*.
ρ = first-order correlation coefficient.
U_t = normally and independently distributed random variable.

The serial correlation of error terms, known as autocorrelation, is a violation of a regression assumption and may be corrected by the application of the Cochrane-Orcutt procedure.[3] Autocorrelation produces unbiased (the expected value of parameter is the population parameter) but inefficient parameters. The variances of the parameters are biased (too low) among the set of linear unbiased estimators and the sample t and F-statistics are too large. The Cochrane-Orcutt (CORC) procedure was developed to produce the best linear unbiased estimators (BLUE) given the autocorrelation of regression residuals. The CORC procedure uses the information implicit in the first-order correlative of residuals to produce unbiased and efficient estimators:

$$\begin{aligned} Y_t &= \alpha + \beta X_t + \varepsilon_t \\ \widehat{\rho} &= \frac{\sum e_t, e_t - 1}{\sum e_t^2 - 1} \end{aligned} \tag{18}$$

The dependent and independent variables are transformed by the estimated rho, $\widehat{\rho}$, to obtain more efficient ordinary least squares estimates:

$$Y_t - \rho Y_{t-1} = \alpha(1 - \rho) + \beta(X_t - \rho X_{t-1}) + \mu t \tag{19}$$

The Cochrane-Orcutt procedure is an iterative procedure that can be repeated until the coefficients converge. One immediately recognizes that as ρ approaches unity the regression model approaches a first-difference model.

The Durbin-Watson, D-W, statistic was developed to test for the absence of autocorrelation:

$$H_o : \rho = 0$$

One generally tests for the presence of autocorrelation ($\rho = 0$) using the Durbin-Watson statistic:

[3]Cochrane D. and G.H. Orcutt. 1949. "Application of Least Squares Regression to Relationships Containing Autocorrelated Error Terms," *Journal of the American Statistical Association* 44: 32–61.

$$D - W = d = \frac{\sum_{t=2}^{N} (e_t = e_{t-1})^2}{\sum_{t=2}^{N} e_t^2} \tag{20}$$

The e represents the ordinary least squares regression residuals, one uses a two-tailed test examine the randomness of residuals. One rejects the null hypothesis of no statistically significant autocorrelation if

$$d < d_L \text{ or } d > 4 - d_u$$

where

d_L is the "lower" Durbin-Watson level
and
d_u is the "upper" Durbin-Watson level

The upper and lower level Durbin-Watson statistic levels are given in Johnston (1972). The Durbin-Watson statistic is used to test only for first-order correlation among residuals:

$$D = 2(1 - \rho) \tag{21}$$

If the first-order correlation of model residuals is zero, the Durbin-Watson statistic is 2. A very low value of the Durbin-Watson statistic, $d < d_L$, indicates positive autocorrelation between residuals and produces a regression model that is not statistically plagued by autocorrelation.

The inconclusive range for the estimated Durbin-Watson statistic is

$$d_L < d < d_u \text{ or } 4 - d_u < 4 - d_u.$$

One does not reject the null hypothesis of no autocorrelation of residuals if $d_u < d < 4 - d_u$.

One of the weaknesses of the Durbin-Watson test for serial correlation is that only first-order autocorrelation of residuals is examined; one should plot the correlation of residual with various time lags

$$\text{corr } (e_t, e_{t-k})$$

to identify higher-order correlations among residuals.

In a nonlinear least squares (NLLS) model, one seeks to estimate an equation in which the dependent variable increases by a constant growth rate rather than a constant amount.[4] The nonlinear regression equation is

[4]The reader is referred to C.T. Clark and L.L. Schkade, Statistical Analysis for Administrative Decisions (Cincinnati: South-Western Publishing Company, 1979) for an excellent treatment of this topic.

$$\begin{aligned} Y &= ab^x \\ \text{or } \log\ Y &= \log\ a + \log\ BX \end{aligned} \tag{22}$$

The normal equations are derived from minimizing the sum of the squared error terms (as in ordinary least squares) and may be written as

$$\begin{aligned} \Sigma(\log\ Y) &= N(\log\ a) + (\log\ b)\,\Sigma X \\ \Sigma(X\ \log\ Y) &= (\log\ a)\Sigma X + (\log\ b)\Sigma X^2 \end{aligned} \tag{23}$$

The solutions to the simplified NLLS estimation equation are

$$\begin{aligned} \log a &= \frac{\Sigma(\log Y)}{N} \\ \log b &= \frac{\Sigma(X\ \log\ Y)}{\Sigma X^2} \end{aligned} \tag{24}$$

Estimating Multiple Regression Lines

It may well be that several economic variables influence the variable that one is interested in forecasting. For example, the levels of the gross domestic product (GDP), personal disposable income, or price indices can assert influences on the firm. Multiple regression is an extremely easy statistical tool for researchers and management to employ due to the great proliferation of computer software such as SAS, used in chapter "Financing Current Operations and Efficiency Ratio Analysis." The general form of the two-independent variable multiple regression is

$$Y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + \varepsilon_t, \quad t = 1, \ldots, N \tag{25}$$

In matrix notation multiple regression can be written

$$Y = X\beta + \varepsilon \tag{26}$$

Multiple regression requires unbiasedness, the expected value of the error term is zero, and the X's are fixed and independent of the error term. The error term is an identically and independently distributed normal variable. Least squares estimation of the coefficients yields

$$\begin{aligned} \widehat{\beta} &= \left(\widehat{\beta}_1, \widehat{\beta}_2, \widehat{\beta}_3\right) \\ Y &= X\widehat{\beta} + e \end{aligned} \tag{27}$$

Multiple regression, using the least squared principle, minimizes the sum of the squared error terms:

$$\sum_{i=1}^{N} e_1^2 = e'e$$
$$\left(\mathrm{Y} - \mathrm{X}\widehat{\beta}\right)' \left(\mathrm{Y} - \mathrm{X}\widehat{\beta}\right) \tag{28}$$

To minimize the sum of the squared error terms, one takes the partial derivative of the squared errors with respect to $\widehat{\beta}$ and the partial derivative set equal to zero:

$$\partial \frac{(e'e)}{\partial \beta} = -2\mathrm{X}'\mathrm{Y} + 2\mathrm{X}'\mathrm{X}\widehat{\beta} = 0$$
$$\widehat{\beta} = (\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'\mathrm{Y} \tag{29}$$

Alternatively, one could solve the normal equations for the two-variable to determine the regression coefficients:

$$\Sigma \mathrm{Y} = \beta_1 \mathrm{N} + \widehat{\beta}_2 \Sigma \mathrm{X}_2 + \widehat{\beta}_3 \Sigma \mathrm{X}_3$$
$$\Sigma \mathrm{X}_2 \mathrm{Y} = \widehat{\beta}_1 \Sigma \mathrm{X}_2 + \widehat{\beta}_2 \mathrm{X}_2{}^2 + \widehat{\beta}_3 \Sigma \mathrm{X}_3{}^2 \tag{30}$$
$$\Sigma \mathrm{X}_3 \mathrm{Y} = \widehat{\beta}_1 \Sigma \mathrm{X}_3 + \widehat{\beta}_2 \Sigma \mathrm{X}_2 \mathrm{X}_3 + \widehat{\beta}_3 \Sigma \mathrm{X}_3{}^2$$

When we solved the normal equations, (7), to find the a and b that minimized the sum of our squared error terms in simple liner regression, and when we solved the two variable normal equations, Eq. (30) to find the multiple regression estimated parameters, we made several assumptions. First, we assumed that the error term is independently and identically distributed, i.e., a random variable with an expected value, or mean of zero, and a finite, and constant, standard deviation. The error term should not be a function of time, as we discussed with the Durbin-Watson statistic, Eq. (21), nor should the error term be a function of the size of the independent variable(s), a condition known as heteroscedasticity. One may plot the residuals as a function of the independent variable(s) to be certain that the residuals are independent of the independent variables. The error term should be a normally distributed variable. That is, the error terms should have an expected value of zero, and 67.6% of the observed error terms should fall within the mean value plus and minus one standard deviation of the error terms (the so-called Bell Curve or normal distribution). Ninety-five percent of the observations should fall within the plus or minus two standard deviation levels, the so-called 95% confidence interval. The presence of extreme, or influential, observations may distort estimated regression lines and the corresponding estimated residuals. Another problem in regression analysis is the assumed independence of the independent variables in Eq. (30). Significant correlations may produce estimated regression coefficients that are "unstable" and have the "incorrect" signs, conditions that we will observe in this chapter. Let us spend some time discussing two problems discussed in this section, the problems of

influential observations, commonly known as outliers, and the correlation among independent variables, known as multicollinearity.

There are several methods that one can use to identify influential observations or outliers. First, we can plot the residuals and 95% confidence intervals and examine how many observations have residuals falling outside these limits. One should expect no more than 5% of the observations to fall outside of these intervals. One may find that one or two observations may distort a regression estimate even if there are 100 observations in the database. The estimated residuals should be normally distributed, and the ratio of the residuals divided by their standard deviation, known as standardized residuals, should be a normal variable. We showed, in Eq. (28), that in multiple regression

$$\widehat{\beta} = (X'X)X'Y$$

The residuals of the multiple regression line is given by

$$e = Y' - \widehat{\beta}X$$

Influential Observations and Possible Outliers and the Application of Robust Regression

The standardized residual concept can be modified such that the reader can calculate a variation on that term to identify influential observations. If we delete observation i in a regression, we can measure the change in estimated regression coefficients and residuals. Belsley, Kuh, and Welsch (1980) showed that the estimated regression coefficients change by an amount, DFBETA, where

$$DFBETA_i = \frac{(X'X)^{-1}X'e_i}{1 - h_i} \tag{31}$$

where $h_i = X_i(X'X)^{-1}X_i'$.

The h_i or "hat" term is calculated by deleting observation i. The corresponding residual is known as the studentized residual, sr, and defined as

$$sr_i = \frac{e_i}{\widehat{\sigma}\sqrt{1 - h_i}} \tag{32}$$

where $\widehat{\sigma}$ is the estimated standard deviation of the residuals. A studentized residual that exceeds 2.0 indicates a potential influential observation (Belsley et al., 1980). Another distance measure is the Cook distance measure which modifies the studentized residual, to calculate a scaled residual, reported in SAS. As the researcher or modeler deletes observations, one needs to compare the original matrix

of the estimated residuals variance matrix. The COVRATIO calculation performs this calculation, where

$$COVRATIO = \frac{1}{\left[\frac{n-p-1}{n-p} + \frac{e_i^*}{(n-p)}\right]^p (1-h_i)} \tag{33}$$

where

n = number of observations
p = number of independent variables
e_i* = deleted observations

If the absolute value of the deleted observation >2, then the COVRATIO calculation approaches

$$1 - \frac{3p}{n} \tag{34}$$

A calculated COVRATIO that is larger than 3p/n indicates an influential observation. The DFBETA, studentized residual, CookD, and COVRATIO calculations may be performed within SAS. The identification of influential data is an important component of regression analysis. One may create variables for use in multiple regression that make use of the influential data, or outliers, to which they are commonly referred.

The modeler can identify outliers, or influential data, and rerun the ordinary least squares regressions on the re-weighted data, a process referred to as robust (ROB) regression. In ordinary least squares, OLS, all data is equally weighted. The weights are 1.0. In robust regression one weights the data universally with its OLS residual; i.e., the larger the residual, the smaller the weight of the observation in the robust regression. In robust regression, several weights may be used. We will report the Huber (1973), Beaton and Tukey (1974), and where we use the Tukey biweight function in the least trimmed squares (LTS) algorithm. 99%, discussed in Maronna, Martin, Yohai, and Salibian-Barrera (2019) weighting schemes in our analysis. In the Huber robust regression procedure, one uses the following calculation to weigh the data:

$$w_i = \left(1 - \left(\frac{|e_i|}{\sigma_i}\right)^2\right)^2 \tag{35}$$

where

e_i = residual I
σ_I = standard deviation of residual
w_i = weight of observation i

The intuition is that the larger the estimated residual, the smaller the weight. A second robust re-weighting scheme is calculated from the Beaton-Tukey bisquare criteria where

$$w_i = \left(1 - \left(\frac{\frac{|e_i|}{\sigma_e}}{4.685}\right)^2\right)^2, \quad if \quad \frac{|e_i|}{\sigma_e} > 4.685;$$
$$1, \quad if \quad \frac{|e_i|}{\sigma_e} < 4.685. \tag{36}$$

A second major problem is one of multicollinearity, the condition of correlations among the independent variables. If the independent variables are perfectly correlated in multiple regression, then the (X′X) matrix of Eq. (29) cannot be inverted, and the multiple regression coefficients have multiple solutions. In reality, highly correlated independent variables can produce unstable regression coefficients due to an unstable $(X'X)^{-1}$ matrix. Belsley et al. advocate the calculation of a condition number, which is the ratio of the largest latent root of the correlation matrix relative to the smallest latent root of the correlation matrix. A condition number exceeding 30.0 indicates severe multicollinearity. Readers are referred to Guerard, Xu, and Wang (2019) for more applications of SAS in regression analysis.

Efron, Hastie, Johnstone, and Tibshirani (2004) introduce LAR to the reader by discussing automatic model-building algorithms, including forward selection, all subsets, and back elimination. One can measure the goodness of fit in terms of predictive accuracy, but one uses a different manner, how well model subsets perform in terms of portfolio geometric means using out-of-sample variable weights. LAR is a variation of forward selection; that is, the technique selects the variable with the largest absolute correlation, x_{j1}, with the response variable, y, and performs simple linear regression of y on x_{j1}. The regression produces a residual vector orthogonal to x_{j1}, now considered to be the response, or dependent variable. One projects the other predictor variables orthogonally to x_{j1} and repeats the selection process. The application of K steps produces a set of predictor variables, x_{j1}, x_{j2}, x_{j3}, . . ., x_{jk}, to construct a K-parameter linear model. Hastie, Tibshirani, and Friedman (2016) state that forward selection is an aggressive fitting technique that can be overly greedy, eliminating at a second step useful prediction correlated with x_{j1}.

We will use robust regression and LAR techniques in chapter "Risk and Return of Equity and the Capital Asset Pricing Model" to model US stock returns in chapter "Risk and Return of Equity and the Capital Asset Pricing Model".

The Conference Board Composite Index of Leading Economic Indicators and Real US GDP Growth: A Regression Example

The composite indexes of leading (LEI), coincident, and lagging economic indicators produced by The Conference Board are summary statistics for the US economy. Wesley Clair Mitchell of Columbia University constructed the indicators in 1913 to serve as a barometer of economic activity. The leading indicator series was

developed to turn upward before aggregate economic activity increased and decrease before aggregate economic activity diminished. Historically, the cyclical turning points in the leading index have occurred before those in aggregate economic activity, cyclical turning points in the coincident index have occurred at about the same time as those in aggregate economic activity, and cyclical turning points in the lagging index generally have occurred after those in aggregate economic activity.

The Conference Board's components of the composite leading index for the year 2018 reflects the work and variables shown in Levanon, Manini, Ozyildirim, Schaitkin, and Tanchua (2015) which continued work of the Mitchell (1913, 1927, 1951), Burns and Mitchell (1946), Moore (1961), and Zarnowitz (1992). The Conference Board index of leading indicators, in November 2019, is composed of:

Average weekly hours, manufacturing.
Average weekly initial claims for unemployment insurance.
Manufacturers' new orders, consumer goods and materials.
ISM® Index of New Orders.
Manufacturers' new orders, nondefense capital goods excluding aircraft orders.
Building permits, new private housing units.
Stock prices, 500 common stocks.
Leading Credit Index™.
Interest rate spread, 10-year treasury bonds less federal funds.

The one change with the first edition of Guerard and Schwartz (2007) is the exclusion of the money supply. See Levanon, Manini, Ozyildirim, Schaitkin, and Tanchua (2015).

The Conference Board composite index of leading economic indicators, LEI, is an equally weighted index in which its components are standardized to produce constant variances. Details of the LEI can be found on The Conference Board website, www.conference-board.org, and the reader is referred to Zarnowitz (1992) for his seminal development of underlying economic assumption and theory of the LEI and business cycles. We transform both real GDP, the output of the economy, and LEI series to stationarity and constant variance by taking the logarithm of each series ad creating a first difference of the data, 1959Q2–2018Q2:

$$\begin{aligned} &\text{RGDP} = .007 + 0.128\ \text{LEI}, \text{R-Squared} = .010, \text{F} = 3.43 \\ &\quad (\text{t}) \quad (18.55) \quad (1.85) \end{aligned}$$

The growth of US real GDP is correlated with the contemporaneous growth of the US LEI. The regression coefficient on the LEI variable, 0.128, is statistically significant because the calculated t-value of 1.85 exceeds 1.645, the 10% critical level. One can reject the null hypothesis of no association between the growth rate of US GDP and the growth rate of the LEI. The reader notes, however, that we estimated the regression line with current or contemporaneous, values of the LEI series. There is no forecasting test of the regression relationship.

Table 1 OLS DLRGDP DLLEI

The REG procedure					
Model: MODEL1					
Dependent variable: DLRGDP					
Number of observations read					239
Number of observations used					237
Number of observations with missing values					2
Analysis of variance					
Source	**DF**	**Sum of squares**	**Mean square**	**F value**	**Pr > F**
Model	1	0.00008059	0.00008059	2.16	0.1432
Error	235	0.00878	0.00003735		
Corrected total	236	0.00886			

Root MSE	0.00611	**R-square**	0.0091
Dependent mean	0.00745	**Adj R-Sq**	0.0049
Coeff var	82.02672		

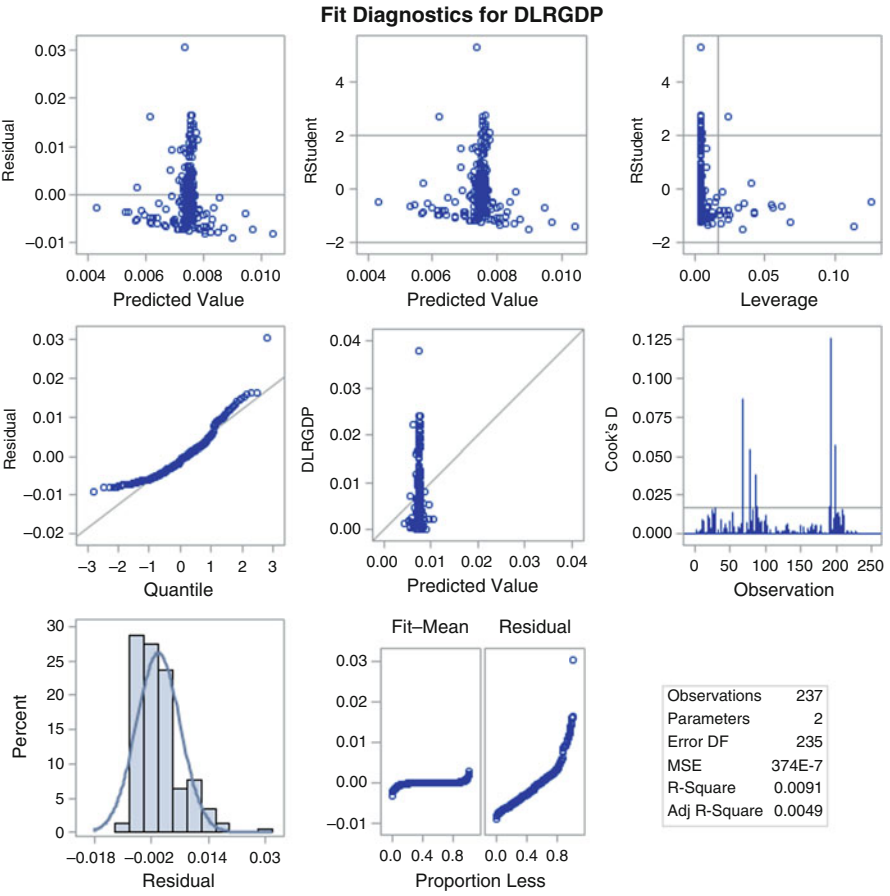
Parameter estimates

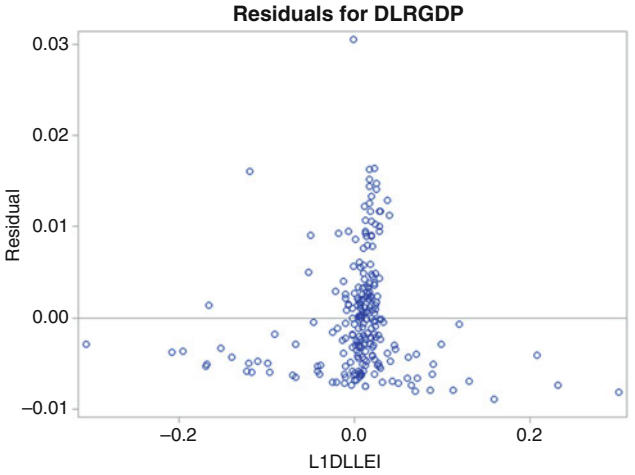
Variable	**DF**	**Parameter estimate**	**Standard error**	**t-value**	**Pr > |t|**
Intercept	**1**	0.00739	0.00039894	18.53	<0.0001
L1DLLEI	**1**	0.01006	0.00685	1.47	0.1432

Source: SAS, proc reg

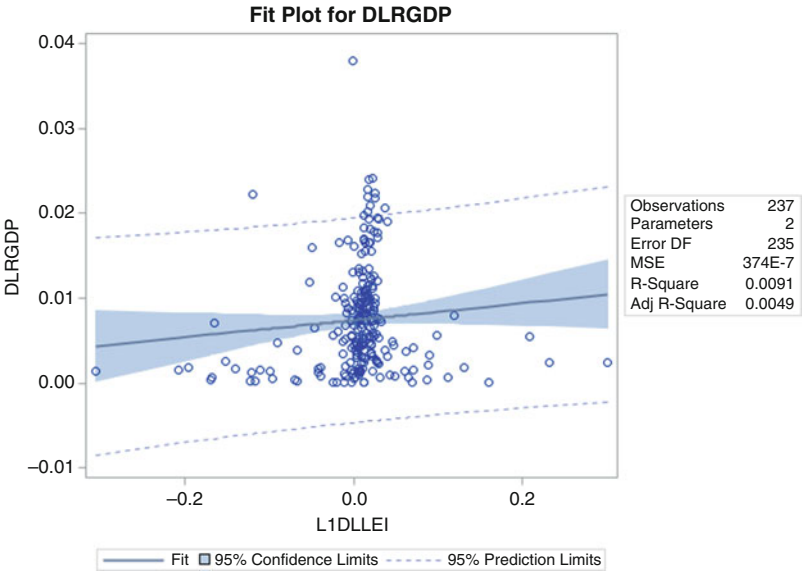
What we need to know is whether past growth rates in the LEI are statistically correlated with US real GDP growth.

The LEI series was developed to "forecast" future economic activity such that current growth of the LEI series should be associated with future US GDP growth rates. Alternatively, one can examine the regression association of the current values of real US GDP growth and previous or lagged values, of the LEI series. How many lags might be appropriate? Let us estimate regression lines using up to four lags of the US LEI series. If one estimates a simple regression model using the SAS software, proc reg as shown in Table 1, the first lag of the LEI series is not statistically significant at the 10% level, having an estimated t-value of 1.47, less the critical level of 1.645, for 30 observations, and statistical studies often use the 10% level as a minimum acceptable critical level.





Source: SAS, proc reg



Source: SAS

The residual plots associated with Table 1 suggest that regression outliers can be a large problem, distorting the statistical associate between US real GDP growth and the growth in the US LEI.

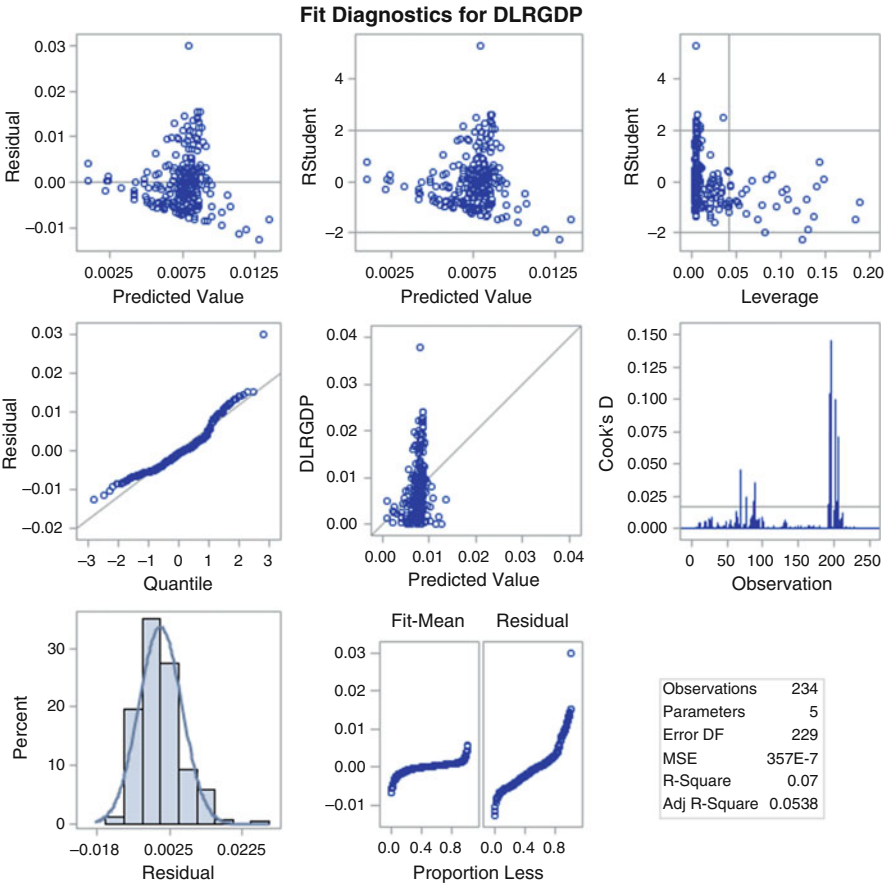
In Table 2, we present regression results for 1, 2, 3, and 4 lags of the LEI variable. The inclusion of two LEI lags in the estimated regression line produces an estimated multiple regression line is which only the first lag of LEI is statistically significant at the 10% level, having an estimated t-value of 1.67. In the regression analysis using three lags of the LEI series, the LEI variable is statistically significant at lags 1 and 3. The F-statistic indicates that the three lag LEI estimated model is statistically significant. As in Guerard and Schwartz (2007), we report four quarters of LEI lags, based on Guerard Jr. (2001).

In the regression analysis using four lags of the DL(LEI)series, LEI lags 1, 3, and 4 are statistically significant, 1959–2018Q2; see Tables 2 and 3. We include a regression diagnostic analysis of Table 3 as a Regression Appendix 3.

Table 2 SAS analysis of US real GDP and LEI lags

Time period: 1959–2018Q2						
Dependent variable: Real US GDP						
		Lags of LEI				
					Adjusted	
Constant	**L1DLLEI**	**L2DLLEI**	**L3DLLEI**	**L4DLLEI**	**R-square**	**F-statistic**
0.007	0.1000				0.005	2.16
(18.53)	(1.47)					
0.007	0.0117	0.0056			0.005	1.57
(18.33)	(1.67)	(0.94)				
0.007	0.0136	0.0130	0.0144		0.019	2.48
(18.01)	(1.94)	(1.43)	(2.05)			
0.007	0.0209	4.334	0.0209	0.0226	0.054	4.31
(17.39)	(2.86)	(0.21)	(2.89)	(3.09)		

() denotes T-statistic
Source: SAS, proc reg



Source: SAS, proc reg

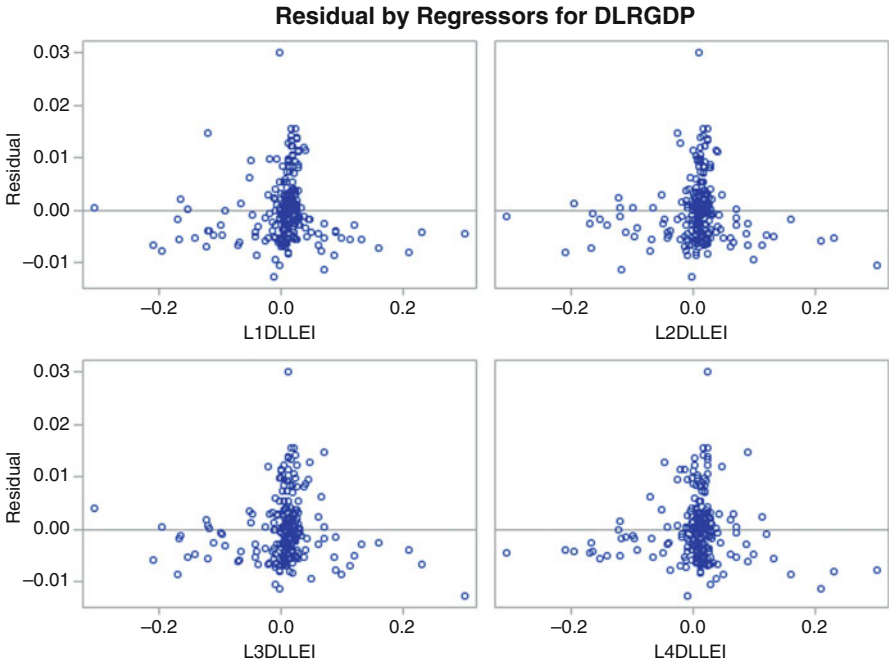
Table 3 OLS DLRGDP DLLEI

The REG procedure					
MODEL: MODEL1					
Dependent variable: DLRGDP					
Number of observations read					239
Number of observations used					234
Number of observations with missing values					5
Analysis of variance					
Source	**DF**	**Sum of squares**	**Mean square**	**F Value**	**Pr > F**
Model	4	0.00061550	0.00015388	4.31	0.0022
Error	229	0.00817	0.00003570		
Corrected total	233	0.00879			

Root MSE	0.00597	**R-square**	0.0700
Dependent mean	0.00749	**Adj R-Sq**	0.0538
Coeff var	79.72828		

Parameter estimates					
Variable	**DF**	**Parameter estimate**	**Standard error**	**t Value**	**Pr > |t|**
Intercept	**1**	0.00705	0.00040548	17.39	<.0001
L1DLLEI	**1**	0.02085	0.00730	2.86	0.0047
L2DLLEI	**1**	0.01491	0.00723	2.06	0.0403
L3DLLEI	**1**	0.02086	0.00723	2.89	0.0043
L4DLLEI	**1**	0.02259	0.00730	3.09	0.0022

Source: SAS, proc reg



We use the SAS robust regression (proc robustreg) procedure and the Tukey bisquare weighting scale in optimal influence function and finds that the adjusted R-square rises from 0.054 to 0.067; see Table 4.

Summary and Conclusions

This chapter develops the theory of ordinary least squares, influential observations, and possible outlier issues. We examined the regressive association between US real GDP and the leading economic indicators during the 1959–2018Q2 time period. We report that The Conference Board leading economic indicators are statistically significant. Moreover, robust regression enhances the estimated models. The reader sees the costs and benefits of regression modeling. We will use roust regression in chapter "Risk and Return of Equity and the Capital Asset Pricing Model," for estimating the determinants of cross-sectional stock returns in Japan and the United States and in chapter "Liquidation, Failure, Bankruptcy, and Reorganization" for non-US and global stock universes.

Table 4 Rob Reg RGDP LEI EFF = 0.99

The ROBUSTREG procedure	
Model information	
Data set	WORK.WIP
Dependent variable	DLRGDP
Number of independent variables	4
Number of observations	234
Missing values	5
Method	MM estimation
Number of observations read	239
Number of observations used	234
Missing values	5
Parameter information	
Parameter	**Effect**
Intercept	Intercept
L1DLLEI	L1DLLEI
L2DLLEI	L2DLLEI
L3DLLEI	L3DLLEI
L4DLLEI	L4DLLEI
Profile for the initial LTS estimate	
Total number of observations	234
Number of squares minimized	177
Number of coefficients	5
Highest possible breakdown value	0.2479

(continued)

Table 4 (continued)

MM profile							
Chi function							Tukey
K1							7.0410
Efficiency							0.9900
Parameter estimates							
Parameter	**DF**	**Estimate**	**Standard error**	**95% confidence limits**		**Chi-square**	**Pr > ChiSq**
Intercept	1	0.0068	0.0004	0.0060	0.0075	315.79	<0.0001
L1DLLEI	1	0.0206	0.0069	0.0071	0.0340	9.01	0.0027
L2DLLEI	1	0.0141	0.0068	0.0008	0.0275	4.32	0.0376
L3DLLEI	1	0.0201	0.0068	0.0067	0.0334	8.62	0.0033
L4DLLEI	1	0.0219	0.0069	0.0084	0.0354	10.13	0.0015
Scale	0	0.0055					

Diagnostics summary		
Observation type	**Proportion**	**Cutoff**
Outlier	0.0043	3.0000

Goodness-of-fit	
Statistic	**Value**
R-square	0.0670
AICR	250.8919
BICR	269.1247
Deviance	0.0073

Appendix 1: Least Angle Regression

Efron et al. (2004) introduce LAR to the reader by discussing automatic model-building algorithms, including forward selection, all subsets, and back elimination. One can measure the goodness of fit in terms of predictive accuracy, but we will use a different manner, how well model subsets perform in terms of portfolio geometric means using out-of-sample variable weights. LAR is a variation of forward selection; that is, the technique selects the variable with the largest absolute correlation, x_{j1}, with the response variable, y, and performs simple linear regression of y on x_{j1}. The regression produces a residual vector orthogonal to x_{j1}, now considered to be the response, or dependent variable. One projects the other predictor variables orthogonally to x_{j1} and repeats the selection process. The application of K steps produces a set of predictor variables, $x_{j1}, x_{j2}, x_{j3}, \ldots, x_{jk}$, to construct a K-parameter linear model. Hastie et al. (2016) state that forward selection is an aggressive fitting technique that can be overly greedy, eliminating at a second step useful prediction correlated with x_{j1}.

Forward stagewise regression is a cautious version of forward selection and is geometrically related to LASSO. All variables are standardized, to be zero mean and unit variance, or

$$\sum_{i=1}^{n} y_i = 0, \quad \sum_{i=1}^{n} x_{ij} = 0, \quad \sum_{i=1}^{n} x_{ij}^{z} = 1, \tag{37}$$

A prediction vector, $\widehat{\mu}$, can be written as an matrix form:

$$\widehat{\mu} = \sum_{j=1}^{m} x_j \widehat{\beta}_j = X\widehat{\beta} \tag{38}$$

The total squared error is

$$S\left(\widehat{\beta}\right) = \|y - \widehat{\mu}\|^2 = \sum_{i=1}^{n} (y_i - \mu_i)^2 \tag{39}$$

If $T\left(\widehat{\beta}\right)$ is the absoluter norm of $\widehat{\beta}$, then

$$T\left(\widehat{\beta}\right) = \sum_{j=1}^{m} \left|\widehat{\beta}_j\right| \tag{40}$$

The LASSO chooses $\widehat{\beta}$ by minimizing $S\left(\widehat{\beta}\right)$ subject to a bound on $T\left(\widehat{\beta}\right)$. The LASSO tends to shrink the OLS coefficients to zero, the shrinkage improves predictive accuracy, and LASSO is parsimonious, with only a subset of covariates that have non-zero values of $\widehat{\beta}_j$. Stagewise, or forward stagewise linear regression, is an iterative process that begins with $\widehat{\mu} = 0$ and builds the regression function in successive small steps. If $c(\widehat{\mu})$ is the vector of current correlations

$$\widehat{c} = c\left(\widehat{\mu}\right) = X^T(y - \widehat{\mu}) \tag{41}$$

Then the next step of the stagewise algorithm is taken in the direction of the greatest current correlation

$$\widehat{J} = \arg\max \left|\widehat{c}_j\right| \quad \text{and} \quad \widehat{\mu} \to \widehat{\mu} + \varepsilon \cdot \text{sign}\left(\widehat{c}_j\right) \cdot x_j. \tag{42}$$

The stagewise procedure does not lead to the "big" choice $\varepsilon = \left|\widehat{c}_j\right|$, the classic forward selection technique, but rather to a "small" choice, which is not overly greedy.

LAR is a version of the stagewise procedure to accelerate calculations. All coefficients start as equal to zero. LAR is similar to forward selection in that it finds the predictor most correlated with the response.

LAR next proceeds to find the predictor with as much correlation with the current residual. LAR finds in an equiangular direction between the first two predictors and a third predictor that is equally correlated with the "most correlated" set, along the "least angle direction." LAR creates a regression model, one covariate at a step, such that after K steps, only K of the $\widehat{\beta}_j$s are non-zero:

$$\widehat{\mu}_1 = \widehat{\mu}_0 + \widehat{\gamma}_1 x_1 \tag{43}$$

LAR uses $\widehat{\gamma}_1$ such that $\overline{y}_2 - \widehat{\mu}$ is equally correlated with x_1 and x_2. Efron et al. (46) demonstrate that $\overline{y}_2 - \widehat{\mu}_1$ bisects the angle between x_1 and x_2 such that $c_1(\widehat{\mu}_1) = c_2(\widehat{\mu}_1)$.

$$\widehat{\mu}_2 = \widehat{\mu}_1 + \widehat{\gamma}_2 u_2, \tag{44}$$

where u_2 is the unit vector lying along the bisector.

The LAR subsequent steps are taken along equiangular vectors. The covariate vectors are linearly independent.

LAR begins at $\widehat{\mu}_0 = 0$ with the stagewise procedure. Each step builds $\widehat{\mu}$. If the current $\widehat{\mu}_A$ is $\widehat{c} = X^T(y - \widehat{\mu}_A)$, then C is the current correlations vector. The active set A is the set of indices corresponding to

$$\widehat{c}_j = \max\left\{\left|\widehat{c}_j\right|\right\} \quad \text{and} \quad \text{A} = \left\{\left|j|\widehat{c}_j\right| = \widehat{c}\right\} \tag{45}$$

Let $s_j = \text{sign}\left\{\widehat{c}_j\right\}$ for jϵA. For A the subset of the indices {1, 2,. . ., m}, define the matrix $X_A = (\ldots, s_j x_j, \ldots)$, $g_A = X_A^T X_A$. Calculate $A_A = \left(1_A^T g_A^{-1} 1_A\right)^{-0.5}$.

Define equiangular vector

$$u_A = X_A w_A \tag{46}$$

where $w_A = A_A g_A^{-1} 1_A$. Compute u_A inner product vector a: $a \equiv X^T u_A$. The next LAR step update is

$$\widehat{\mu}_{A+} = \widehat{\mu}_A + \widehat{\gamma} u_A$$

where

$$\widehat{\gamma} = \min_{j \notin A}{}^{+} \left\{ \frac{\widehat{c} - \widehat{c}_j}{A_A - a_j}, \frac{\widehat{c} + \widehat{c}_j}{A_A + a_j} \right\} \tag{47}$$

Thus, $\widehat{\gamma}$ is the smallest positive value of γ such that the new index j joins the active set, A_+. The new maximum absolute correlation is $c_+ = \widehat{c} - \widehat{\gamma} A_A$.

Appendix 2: The US Leading Economic Indicators

Let us follow The Conference Board components and their definitions, as of November 29, 2019:

BCI-01 Average Weekly Hours, Manufacturing

The average hours worked per week by production workers in manufacturing industries tend to lead the business cycle because employers usually adjust work hours before increasing or decreasing their workforce.

BCI-05 Average Weekly Initial Claims for Unemployment Insurance

The number of new claims filed for unemployment insurance are typically more sensitive than either total employment or unemployment to overall business conditions, and this series tends to lead the business cycle. It is inverted when included in the leading index; the signs of the month-to-month changes are reversed, because initial claims increase when employment conditions worsen (i.e., layoffs rise and new hirings fall).

BCI-08 Manufacturers' New Orders, Consumer Goods and Materials (in 1982 $)

These goods are primarily used by consumers. The inflation-adjusted value of new orders leads actual production because new orders directly affect the level of both unfilled orders and inventories that firms monitor when making production

decisions. The Conference Board deflates the current dollar orders data using price indexes constructed from various sources at the industry level and a chain-weighted aggregate price index formula.

BCI-130 ISM New Order Index

This index reflects the levels of new orders from customers. As a diffusion index, its value reflects the number of participants reporting increased orders during the previous month compared to the number reporting decreased orders, and this series tends to lead the business cycle. When the index has a reading of greater than 50, it is an indication that orders have increased during the past month. This index, therefore, tends to lead the business cycle. ISM new orders are based on a monthly survey conducted by Institution for Supply Management (formerly known as National Association of Purchasing Management). The Conference Board takes normalized value of this index as a measure of its contribution to LEI.

BCI-33 Manufacturers' New Orders, Non-defense Capital Goods Excl. Aircraft (in 1982 $)

This index, combing with orders from aircraft (in inflation-adjusted dollars) are the producers' counterpart to BCI-08.

BCI-29 Building Permits, New Private Housing Units

The number of residential building permits issued is an indicator of construction activity, which typically leads most other types of economic production.

BCI-19 Stock Prices, 500 Common Stocks

The Standard & Poor's 500 stock index reflects the price movements of a broad selection of common stocks traded on the New York Stock Exchange. Increases (decreases) of the stock index can reflect both the general sentiments of investors and the movements of interest rates, which is usually another good indicator for future economic activity.

BCI-107 Leading Credit Index™

This index is consisted of six financial indicators: 2-year swap spread (real time), LIBOR 3-month less 3-month treasury bill yield spread (real time); debit balances at margin account at broker dealer (monthly); AAII Investors Sentiment Bullish (%) less Bearish (%) (weekly); Senior Loan Officers C&I loan survey, bank tightening credit to large and medium firms (quarterly); and Security Repurchases (quarterly) from the Total Finance-Liabilities section of Federal Reserve's flow of fund report. Because of these financial indicators' forward-looking content, LCI leads economic activities.

BCI-129 Interest Rate Spread, 10-Year Treasury Bonds Less Federal Funds

The spread or difference between long and short rates is often called the yield curve. This series is constructed using the 10-year treasury bond rate and the federal funds rate, an overnight interbank borrowing rate. It is felt to be an indicator of the stance of monetary policy and general financial conditions because it rises (falls) when short rates are relatively low (high). When it becomes negative (i.e., short rates are higher than long rates and the yield curve inverts), its record as an indicator of recessions is particularly strong.

BCI-125 Avg. Consumer Expectations for Business and Economic Conditions

This index reflects changes in consumer attitudes concerning future economic conditions and, therefore, is the only indicator in the leading index that is completely expectations-based. It is an equally weighted average of consumer expectations of business and economic conditions using two questions, Consumer Expectations for Economic Conditions 12-months ahead from Surveys of Consumers conducted by Reuters/University of Michigan, and Consumer Expectations for Business Conditions 6-months ahead from Consumer Confidence Survey by The Conference Board. Responses to the questions concerning various business and economic conditions are classified as positive, negative, or unchanged.

Regression Appendix 3

Regression Appendix 3: Regression Diagnostics of Table 3

OLS DLRGDP DLLEI

The REG procedure
Model: MODEL1
Dependent variable: DLRGDP

Output statistics

Obs	Residual	RStudent	Hat DiagH	Covratio	DFFITS	DFBETAS				
						Intercept	L1DLLEI	L2DLLEI	L3DLLEI	L4DLLEI
1	.	.	.	.	.	.	.	.	.	.
2	.	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.
6	−0.001723	−0.2888	0.0063	1.0268	−0.0231	−0.0156	−0.0087	−0.0002	−0.0056	−0.0112
7	−0.006392	−1.0728	0.0049	1.0016	−0.0754	−0.0642	−0.0083	−0.0217	0.0060	−0.0114
8	−0.002344	−0.3926	0.0053	1.0242	−0.0288	−0.0259	0.0091	0.0009	−0.0052	0.0063
9	−0.006978	−1.1718	0.0051	0.9970	−0.0841	−0.0752	0.0073	0.0235	0.0002	−0.0184
10	0.003192	0.5348	0.0050	1.0209	0.0381	0.0371	−0.0071	−0.0081	−0.0139	−0.0048
11	0.0114	1.9304	0.0059	0.9482	0.1490	0.1200	0.0599	−0.0027	−0.0131	−0.0257
12	0.0112	1.8859	0.0071	0.9528	0.1595	0.1001	0.0695	0.0875	0.0170	0.0167
13	0.008805	1.4841	0.0088	0.9828	0.1397	0.0699	0.0624	0.0693	0.0791	0.0293
14	−0.000485	−0.0814	0.0109	1.0332	−0.0085	−0.0034	−0.0041	−0.0041	−0.0043	−0.0051
15	0.003134	0.5260	0.0085	1.0247	0.0487	0.0241	0.0209	0.0220	0.0241	0.0247
16	−0.005292	−0.8883	0.0066	1.0113	−0.0722	−0.0457	−0.0137	−0.0263	−0.0313	−0.0327
17	0.002552	0.4276	0.0055	1.0237	0.0319	0.0235	0.0075	0.0038	0.0110	0.0128
18	0.002931	0.4912	0.0054	1.0223	0.0363	0.0271	0.0140	0.0081	0.0043	0.0127
19	0.0135	2.2902	0.0057	0.9175	0.1734	0.1244	0.0697	0.0672	0.0398	0.0248
20	−0.002046	−0.3430	0.0065	1.0261	−0.0277	−0.0174	−0.0103	−0.0124	−0.0114	−0.0081

21	0.0121	2.0495	0.0070	0.9395	0.1718	0.1009	0.0666	0.0662	0.0777	0.0749
22	0.002195	0.3679	0.0064	1.0257	0.0296	0.0187	0.0103	0.0108	0.0112	0.0132
23	0.006956	1.1687	0.0061	0.9981	0.0915	0.0604	0.0358	0.0301	0.0329	0.0346
24	−0.005506	−0.9242	0.0064	1.0096	−0.0740	−0.0471	−0.0314	−0.0290	−0.0247	−0.0279
25	0.0154	2.6142	0.0063	0.8873	0.2074	0.1341	0.0736	0.0860	0.0808	0.0698
26	0.003792	0.6360	0.0069	1.0202	0.0531	0.0314	0.0237	0.0196	0.0223	0.0221
27	0.0134	2.2642	0.0066	0.9206	0.1846	0.1139	0.0673	0.0795	0.0673	0.0774
28	0.0142	2.4164	0.0061	0.9063	0.1898	0.1249	0.0636	0.0647	0.0809	0.0676
29	0.0154	2.6149	0.0066	0.8876	0.2128	0.1318	0.0918	0.0729	0.0739	0.0949
30	−0.005134	−0.8615	0.0062	1.0120	−0.0682	−0.0442	−0.0267	−0.0283	−0.0231	−0.0236
31	−0.000168	−0.0282	0.0064	1.0287	−0.0023	−0.0014	−0.0009	−0.0009	−0.0009	−0.0008
32	0.0000709	0.0119	0.0056	1.0279	0.0009	0.0007	−0.0000	0.0003	0.0003	0.0003
33	0.001162	0.1946	0.0052	1.0266	0.0140	0.0122	−0.0021	−0.0025	0.0028	0.0028
34	−0.006646	−1.1157	0.0050	0.9997	−0.0790	−0.0741	0.0073	0.0194	0.0194	−0.0089
35	0.002568	0.4302	0.0051	1.0232	0.0307	0.0301	−0.0049	−0.0053	−0.0092	−0.0100
36	0.000102	0.0171	0.0046	1.0269	0.0012	0.0011	0.0002	−0.0000	−0.0001	−0.0002
37	0.0123	2.0854	0.0048	0.9345	0.1453	0.1221	0.0362	0.0414	0.0108	0.0072
38	0.008363	1.4065	0.0054	0.9842	0.1033	0.0773	0.0328	0.0327	0.0337	0.0162
39	−0.000587	−0.0983	0.0055	1.0276	−0.0073	−0.0053	−0.0016	−0.0024	−0.0024	−0.0025
40	−0.004282	−0.7177	0.0051	1.0159	−0.0515	−0.0399	−0.0122	−0.0095	−0.0164	−0.0158
41	0.007396	1.2424	0.0050	0.9932	0.0880	0.0698	0.0232	0.0193	0.0156	0.0276
42	−0.004745	−0.7954	0.0045	1.0126	−0.0536	−0.0474	−0.0030	−0.0102	−0.0093	−0.0057
43	−0.007912	−1.3296	0.0046	0.9880	−0.0905	−0.0780	−0.0159	−0.0067	−0.0184	−0.0177
44	−0.006044	−1.0175	0.0115	1.0109	−0.1098	−0.0782	0.0861	0.0169	0.0154	0.0174
45	−0.005447	−0.9165	0.0112	1.0149	−0.0976	−0.0680	0.0032	0.0741	0.0127	0.0104
46	−0.005672	−0.9556	0.0133	1.0154	−0.1111	−0.0773	0.0455	0.0129	0.0845	0.0245
47	−0.001680	−0.2824	0.0129	1.0336	−0.0322	−0.0208	−0.0040	0.0101	0.0015	0.0214

(continued)

(continued)

48	−0.003634	−0.6097	0.0074	1.0214	−0.0526	−0.0349	−0.0083	−0.0259	0.0104	−0.0127
49	0.009513	1.6073	0.0118	0.9776	0.1754	0.1188	−0.1119	−0.0208	0.0385	−0.0725
50	−0.002451	−0.4116	0.0099	1.0285	−0.0411	−0.0260	−0.0052	0.0214	0.0004	−0.0167
51	0.001280	0.2147	0.0084	1.0297	0.0198	0.0147	0.0025	0.0007	−0.0124	−0.0019
52	−0.004563	−0.7663	0.0083	1.0176	−0.0703	−0.0518	0.0033	−0.0104	−0.0032	0.0430
53	0.009859	1.6611	0.0056	0.9679	0.1247	0.0892	0.0323	0.0323	0.0485	0.0437
54	0.0139	2.3583	0.0061	0.9117	0.1850	0.1230	0.0820	0.0504	0.0500	0.0765
55	0.000936	0.1569	0.0063	1.0280	0.0125	0.0081	0.0054	0.0055	0.0034	0.0036
56	0.007999	1.3457	0.0067	0.9891	0.1102	0.0678	0.0406	0.0491	0.0489	0.0332
57	−0.001503	−0.2521	0.0084	1.0293	−0.0232	−0.0116	−0.0122	−0.0092	−0.0104	−0.0112
58	0.002096	0.3578	0.0422	1.0642	0.0751	0.0324	−0.0692	−0.0075	−0.0043	−0.0118
59	−0.000662	−0.1130	0.0434	1.0682	−0.0241	−0.0105	0.0055	0.0225	0.0027	0.0020
60	−0.001279	−0.2183	0.0431	1.0671	−0.0464	−0.0194	−0.0013	0.0091	0.0423	0.0037
61	−0.004271	−0.7299	0.0429	1.0555	−0.1545	−0.0629	0.0014	−0.0080	0.0280	0.1369
62	−0.005219	−0.8856	0.0281	1.0338	−0.1507	−0.0231	−0.1236	−0.0873	−0.0693	−0.0533
63	−0.009366	−1.5936	0.0259	0.9928	−0.2601	−0.0485	−0.0585	−0.2107	−0.1493	−0.1146
64	−0.008422	−1.4280	0.0213	0.9989	−0.2108	−0.0674	0.0296	−0.0115	−0.1611	−0.0932
65	−0.004626	−0.7867	0.0328	1.0426	−0.1450	−0.0650	0.1003	0.0621	0.0251	−0.0415
66	−0.004091	−0.6951	0.0319	1.0447	−0.1263	−0.0585	−0.0301	0.0925	0.0563	0.0229
67	−0.002459	−0.4182	0.0347	1.0548	−0.0793	−0.0279	−0.0263	−0.0304	0.0474	0.0212
68	−0.001361	−0.2313	0.0338	1.0566	−0.0432	−0.0150	0.0092	−0.0158	−0.0174	0.0255
69	0.0146	2.5141	0.0353	0.9242	0.4812	0.1668	−0.2956	−0.0965	0.1685	0.1721
70	0.000467	0.0792	0.0308	1.0544	0.0141	0.0063	−0.0003	−0.0112	−0.0042	0.0036
71	0.000643	0.1090	0.0290	1.0524	0.0188	0.0098	−0.0017	−0.0025	−0.0168	−0.0077
72	0.001585	0.2680	0.0246	1.0462	0.0425	0.0216	−0.0052	0.0005	−0.0033	−0.0370
73	0.003120	0.5231	0.0066	1.0227	0.0425	0.0265	0.0125	0.0165	0.0183	0.0182
74	0.0107	1.8028	0.0062	0.9582	0.1419	0.0933	0.0515	0.0385	0.0546	0.0607
75	0.009352	1.5753	0.0062	0.9743	0.1242	0.0814	0.0534	0.0440	0.0337	0.0490

76	−0.008270	−1.3908	0.0056	0.9854	−0.1046	−0.0747	−0.0280	−0.0419	−0.0357	−0.0259
77	−0.005148	−0.8635	0.0055	1.0112	−0.0644	−0.0468	−0.0137	−0.0164	−0.0256	−0.0216
78	0.0300	5.3214	0.0049	0.5720	0.3735	0.3118	0.0043	0.0439	0.0744	0.1187
79	0.001973	0.3304	0.0051	1.0249	0.0236	0.0191	0.0085	0.0004	0.0030	0.0052
80	−0.000460	−0.0770	0.0049	1.0270	−0.0054	−0.0046	−0.0005	−0.0018	0.0001	−0.0005
81	−0.003912	−0.6633	0.0281	1.0416	−0.1129	−0.0596	0.1027	0.0270	0.0067	0.0345
82	−0.001064	−0.1802	0.0272	1.0499	−0.0301	−0.0158	0.0075	0.0273	0.0070	0.0015
83	−0.005604	−0.9579	0.0415	1.0452	−0.1993	−0.0617	−0.1174	0.0039	0.1203	−0.0029
84	−0.004997	−0.8540	0.0422	1.0503	−0.1794	−0.0601	0.0241	−0.0979	0.0089	0.1146
85	−0.002671	−0.4520	0.0250	1.0437	−0.0724	−0.0202	−0.0020	−0.0072	−0.0643	−0.0132
86	−0.005472	−0.9416	0.0542	1.0599	−0.2253	−0.0784	0.1677	0.0532	0.0252	−0.0729
87	−0.007299	−1.2629	0.0620	1.0523	−0.3247	−0.0954	−0.1516	0.2158	0.0608	0.0136
88	−0.001759	−0.3107	0.1059	1.1408	−0.1069	−0.0361	0.0689	−0.0197	0.0679	0.0362
89	−0.002522	−0.4447	0.1023	1.1337	−0.1501	−0.0501	0.0434	0.0954	−0.0299	0.0951
90	−0.008574	−1.4955	0.0743	1.0515	−0.4236	−0.0755	−0.1773	−0.0078	0.2140	−0.2448
91	−0.004422	−0.7586	0.0501	1.0626	−0.1743	−0.0567	−0.0203	−0.0561	0.0172	0.1348
92	−0.005203	−0.8882	0.0396	1.0461	−0.1804	−0.0606	0.1264	−0.0380	−0.0717	0.0152
93	−0.002696	−0.4638	0.0567	1.0785	−0.1137	−0.0188	−0.0631	0.0463	−0.0347	−0.0598
94	−0.004734	−0.8128	0.0512	1.0618	−0.1888	−0.0673	0.0838	−0.0535	0.1199	−0.0033
95	−0.004872	−0.8368	0.0515	1.0612	−0.1950	−0.0729	0.0163	0.0929	−0.0510	0.1284
96	−0.000896	−0.1519	0.0302	1.0534	−0.0268	−0.0098	0.0050	−0.0030	0.0129	−0.0141
97	−0.001574	−0.2658	0.0219	1.0434	−0.0398	−0.0214	−0.0004	0.0166	−0.0003	0.0293
98	0.002811	0.4721	0.0103	1.0278	0.0483	0.0299	−0.0080	0.0150	−0.0202	0.0128
99	0.0129	2.1867	0.0098	0.9304	0.2171	0.1543	−0.0268	−0.0642	0.0519	−0.1185
100	0.0120	2.0355	0.0094	0.9430	0.1985	0.1044	0.1037	0.0406	−0.0155	0.1125
101	0.0114	1.9222	0.0074	0.9502	0.1657	0.1070	0.0601	0.0822	0.0288	−0.0257
102	0.008003	1.3483	0.0094	0.9916	0.1311	0.0603	0.0629	0.0668	0.0745	0.0453

(continued)

(continued)

103	0.000334	0.0561	0.0097	1.0321	0.0056	0.0025	0.0022	0.0027	0.0028	0.0032
104	−0.000644	−0.1080	0.0073	1.0293	−0.0092	−0.0053	−0.0031	−0.0031	−0.0045	−0.0046
105	0.001340	0.2244	0.0056	1.0267	0.0168	0.0123	0.0026	0.0041	0.0051	0.0073
106	0.000678	0.1136	0.0049	1.0269	0.0080	0.0065	0.0021	0.0008	0.0018	0.0021
107	0.007276	1.2219	0.0047	0.9940	0.0838	0.0714	0.0144	0.0201	0.0064	0.0166
108	−0.000501	−0.0839	0.0047	1.0267	−0.0057	−0.0049	−0.0011	−0.0009	−0.0014	−0.0004
109	0.001295	0.2168	0.0048	1.0259	0.0150	0.0125	0.0024	0.0031	0.0027	0.0039
110	−0.003410	−0.5711	0.0046	1.0195	−0.0386	−0.0335	−0.0065	−0.0050	−0.0074	−0.0062
111	0.001639	0.2744	0.0045	1.0251	0.0185	0.0161	0.0028	0.0030	0.0024	0.0036
112	−0.002481	−0.4154	0.0045	1.0228	−0.0279	−0.0246	−0.0043	−0.0039	−0.0044	−0.0034
113	−0.000406	−0.0680	0.0045	1.0266	−0.0046	−0.0040	−0.0005	−0.0007	−0.0006	−0.0007
114	0.002926	0.4901	0.0044	1.0213	0.0327	0.0293	0.0040	0.0034	0.0047	0.0043
115	0.000816	0.1365	0.0045	1.0263	0.0091	0.0081	0.0014	0.0012	0.0010	0.0014
116	0.009202	1.5483	0.0045	0.9744	0.1040	0.0918	0.0178	0.0160	0.0135	0.0120
117	−0.002764	−0.4630	0.0046	1.0220	−0.0314	−0.0270	−0.0059	−0.0060	−0.0053	−0.0048
118	0.005357	0.8983	0.0045	1.0088	0.0603	0.0546	−0.0026	0.0079	0.0091	0.0066
119	−0.002015	−0.3375	0.0046	1.0243	−0.0229	−0.0202	−0.0043	0.0005	−0.0033	−0.0040
120	0.005517	0.9252	0.0045	1.0077	0.0625	0.0557	0.0088	0.0110	−0.0018	0.0087
121	0.002458	0.4115	0.0045	1.0229	0.0276	0.0251	0.0020	0.0033	0.0045	−0.0013
122	−0.000238	−0.0399	0.0045	1.0267	−0.0027	−0.0024	−0.0002	−0.0003	−0.0004	−0.0005
123	−0.000111	−0.0186	0.0044	1.0266	−0.0012	−0.0012	0.0001	0.0000	−0.0001	−0.0001
124	−0.005145	−0.8628	0.0048	1.0105	−0.0600	−0.0584	0.0181	0.0129	0.0048	0.0038
125	−0.005643	−0.9466	0.0050	1.0073	−0.0671	−0.0658	0.0137	0.0226	0.0161	0.0085
126	−0.000200	−0.0338	0.0211	1.0441	−0.0050	−0.0031	0.0043	0.0017	0.0016	0.0019
127	−0.003233	−0.5453	0.0180	1.0341	−0.0739	−0.0444	−0.0006	0.0607	0.0203	0.0175
128	−0.003163	−0.5345	0.0217	1.0383	−0.0796	−0.0472	0.0369	0.0045	0.0636	0.0266
129	−0.003870	−0.6544	0.0226	1.0360	−0.0995	−0.0449	−0.0382	0.0241	−0.0091	0.0557

130	−0.006643	−1.1235	0.0193	1.0139	−0.1576	−0.0723	0.0694	−0.0769	0.0358	−0.0207
131	−0.007674	−1.3003	0.0214	1.0066	−0.1921	−0.0763	−0.0645	0.0701	−0.0975	0.0275
132	−0.006015	−1.0178	0.0214	1.0211	−0.1506	−0.0516	−0.0227	−0.0646	0.0457	−0.0889
133	0.006243	1.0568	0.0219	1.0198	0.1583	0.0880	−0.0956	−0.0336	0.0283	−0.1040
134	0.003054	0.5136	0.0124	1.0290	0.0577	0.0334	−0.0006	−0.0306	−0.0059	0.0277
135	0.003534	0.5936	0.0096	1.0241	0.0583	0.0448	−0.0020	−0.0115	−0.0423	−0.0166
136	0.003799	0.6379	0.0088	1.0221	0.0602	0.0455	−0.0034	0.0027	−0.0094	−0.0410
137	−0.006319	−1.0606	0.0049	1.0022	−0.0743	−0.0607	−0.0131	−0.0202	−0.0212	−0.0083
138	−0.002733	−0.4580	0.0061	1.0237	−0.0360	−0.0238	−0.0165	−0.0094	−0.0114	−0.0133
139	−0.003521	−0.5902	0.0057	1.0202	−0.0447	−0.0317	−0.0119	−0.0195	−0.0110	−0.0134
140	0.005365	0.9000	0.0053	1.0095	0.0655	0.0505	0.0084	0.0142	0.0273	0.0134
141	0.001325	0.2219	0.0056	1.0268	0.0166	0.0121	0.0052	0.0026	0.0039	0.0074
142	0.005409	0.9072	0.0049	1.0089	0.0640	0.0514	0.0191	0.0175	0.0082	0.0132
143	−0.002434	−0.4078	0.0055	1.0241	−0.0304	−0.0221	−0.0115	−0.0101	−0.0090	−0.0056
144	0.002897	0.4855	0.0061	1.0231	0.0380	0.0251	0.0133	0.0155	0.0134	0.0130
145	−0.005078	−0.8520	0.0064	1.0125	−0.0682	−0.0434	−0.0268	−0.0242	−0.0281	−0.0253
146	−0.005484	−0.9203	0.0060	1.0094	−0.0713	−0.0481	−0.0202	−0.0264	−0.0246	−0.0287
147	0.000490	0.0821	0.0050	1.0271	0.0058	0.0048	0.0000	0.0010	0.0018	0.0014
148	−0.000952	−0.1593	0.0047	1.0264	−0.0110	−0.0098	0.0004	0.0009	−0.0013	−0.0026
149	0.0000724	0.0121	0.0044	1.0267	0.0008	0.0008	−0.0000	−0.0001	−0.0001	0.0000
150	0.009821	1.6545	0.0055	0.9683	0.1232	0.1179	−0.0546	−0.0188	−0.0253	−0.0345
151	0.001620	0.2713	0.0051	1.0257	0.0194	0.0180	0.0017	−0.0067	−0.0016	−0.0020
152	0.002935	0.4918	0.0051	1.0220	0.0354	0.0311	0.0044	0.0055	−0.0105	−0.0004
153	−0.000994	−0.1665	0.0051	1.0268	−0.0120	−0.0105	−0.0001	−0.0016	−0.0019	0.0034
154	0.008305	1.3964	0.0050	0.9845	0.0994	0.0782	0.0249	0.0177	0.0247	0.0322
155	0.004291	0.7192	0.0051	1.0157	0.0512	0.0402	0.0167	0.0126	0.0092	0.0131
156	0.000254	0.0426	0.0056	1.0278	0.0032	0.0023	0.0013	0.0011	0.0008	0.0007

(continued)

(continued)

157	0.001553	0.2602	0.0058	1.0266	0.0200	0.0137	0.0059	0.0082	0.0072	0.0057
158	0.000912	0.1528	0.0055	1.0273	0.0114	0.0083	0.0020	0.0030	0.0045	0.0038
159	0.004282	0.7178	0.0051	1.0158	0.0515	0.0404	0.0105	0.0066	0.0126	0.0191
160	0.008373	1.4076	0.0044	0.9832	0.0937	0.0862	−0.0009	0.0087	0.0050	0.0134
161	0.002224	0.3724	0.0047	1.0238	0.0256	0.0248	−0.0074	−0.0035	0.0000	−0.0021
162	0.000415	0.0694	0.0047	1.0268	0.0048	0.0046	−0.0004	−0.0014	−0.0007	−0.0000
163	0.005940	0.9965	0.0047	1.0049	0.0687	0.0672	−0.0047	−0.0074	−0.0207	−0.0108
164	0.009580	1.6127	0.0046	0.9703	0.1099	0.1050	0.0034	−0.0009	−0.0076	−0.0271
165	−0.003958	−0.6631	0.0043	1.0167	−0.0438	−0.0410	−0.0005	−0.0052	−0.0021	−0.0002
166	0.0102	1.7193	0.0047	0.9629	0.1181	0.1006	0.0326	0.0108	0.0202	0.0155
167	−0.006491	−1.0894	0.0047	1.0006	−0.0746	−0.0641	−0.0076	−0.0200	−0.0064	−0.0122
168	−0.004015	−0.6729	0.0049	1.0171	−0.0474	−0.0434	0.0133	0.0015	−0.0081	0.0026
169	−0.004861	−0.8165	0.0085	1.0159	−0.0755	−0.0612	0.0490	0.0295	0.0113	0.0051
170	−0.004212	−0.7095	0.0148	1.0261	−0.0869	−0.0629	0.0569	0.0581	0.0365	0.0267
171	−0.005193	−0.8754	0.0153	1.0207	−0.1092	−0.0644	−0.0389	0.0490	0.0569	0.0229
172	−0.002784	−0.4731	0.0329	1.0517	−0.0873	−0.0448	0.0650	−0.0066	0.0375	0.0479
173	0.000476	0.0804	0.0230	1.0461	0.0123	0.0062	−0.0005	−0.0086	0.0027	−0.0041
174	−0.000639	−0.1078	0.0209	1.0437	−0.0158	−0.0077	0.0023	−0.0005	0.0109	−0.0044
175	−0.001201	−0.2026	0.0199	1.0419	−0.0289	−0.0169	0.0051	0.0085	0.0019	0.0243
176	−0.006292	−1.0566	0.0060	1.0034	−0.0818	−0.0623	−0.0119	−0.0176	0.0158	−0.0301
177	−0.001561	−0.2614	0.0054	1.0261	−0.0192	−0.0175	0.0040	0.0006	−0.0020	0.0073
178	0.000792	0.1325	0.0046	1.0265	0.0090	0.0081	0.0007	−0.0006	0.0006	0.0023
179	0.009725	1.6374	0.0046	0.9686	0.1116	0.1097	−0.0269	−0.0083	−0.0192	−0.0091
180	0.003911	0.6554	0.0048	1.0174	0.0453	0.0417	0.0085	−0.0072	−0.0008	−0.0039
181	−0.002683	−0.4496	0.0058	1.0235	−0.0342	−0.0255	−0.0146	−0.0110	0.0013	−0.0048
182	−0.000795	−0.1332	0.0069	1.0289	−0.0111	−0.0069	−0.0048	−0.0053	−0.0040	−0.0007
183	0.000307	0.0515	0.0088	1.0312	0.0049	0.0023	0.0023	0.0024	0.0025	0.0022

184	0.000894	0.1499	0.0085	1.0304	0.0139	0.0069	0.0057	0.0063	0.0070	0.0071
185	0.002214	0.3711	0.0071	1.0264	0.0315	0.0182	0.0109	0.0118	0.0142	0.0155
186	−0.004053	−0.6798	0.0065	1.0184	−0.0550	−0.0344	−0.0225	−0.0175	−0.0204	−0.0247
187	0.000635	0.1063	0.0054	1.0274	0.0078	0.0058	0.0015	0.0028	0.0023	0.0025
188	−0.001736	−0.2908	0.0049	1.0252	−0.0204	−0.0168	−0.0021	−0.0027	−0.0066	−0.0048
189	−0.003363	−0.5634	0.0046	1.0198	−0.0384	−0.0337	−0.0024	−0.0018	−0.0038	−0.0106
190	0.000439	0.0796	0.1491	1.2011	0.0333	0.0115	−0.0328	−0.0097	−0.0068	−0.0104
191	−0.001098	−0.1979	0.1412	1.1891	−0.0802	−0.0267	0.0173	0.0788	0.0222	0.0150
192	0.004129	0.7464	0.1443	1.1800	0.3065	0.1039	−0.0725	−0.0689	−0.3010	−0.0900
193	−0.004533	−0.8420	0.1893	1.2413	−0.4068	−0.0606	−0.2240	−0.0136	0.0120	0.2440
194	−0.0104	−1.8711	0.1309	1.0898	−0.7261	0.0008	−0.1970	−0.7117	−0.1786	−0.1879
195	−0.0127	−2.2912	0.1240	1.0413	−0.8621	−0.0217	−0.1059	−0.2000	−0.8427	−0.1836
196	−0.007697	−1.3900	0.1376	1.1363	−0.5554	−0.0714	0.2083	0.0307	−0.0560	−0.4114
197	0.001411	0.2470	0.0900	1.1217	0.0777	0.0332	−0.0432	−0.0718	−0.0290	−0.0247
198	0.000449	0.0784	0.0842	1.1159	0.0238	0.0099	−0.0047	−0.0127	−0.0221	−0.0083
199	−0.004097	−0.7255	0.1085	1.1334	−0.2531	−0.0526	−0.1432	−0.0225	0.0729	0.1369
200	−0.005375	−0.9372	0.0790	1.0887	−0.2745	−0.0272	−0.0456	−0.2531	−0.0742	0.0272
201	−0.006512	−1.1614	0.1180	1.1251	−0.4247	−0.0733	0.2465	0.0017	−0.2768	−0.0242
202	−0.008050	−1.4958	0.1842	1.1933	−0.7109	−0.0248	−0.4281	0.2194	−0.0793	−0.4940
203	−0.005687	−1.0008	0.0953	1.1053	−0.3248	−0.0725	−0.0043	−0.1749	0.2034	0.0387
204	−0.003921	−0.7022	0.1285	1.1602	−0.2697	−0.0726	0.1398	0.0284	−0.1019	0.1806
205	−0.0113	−1.9962	0.0832	1.0223	−0.6012	−0.0664	−0.2325	0.1623	−0.0728	−0.4971
206	0.000204	0.0352	0.0682	1.0969	0.0095	0.0041	−0.0075	−0.0009	−0.0062	−0.0036
207	−0.001775	−0.3061	0.0618	1.0872	−0.0786	−0.0333	0.0186	0.0606	0.0048	0.0500
208	−0.005338	−0.9107	0.0382	1.0436	−0.1816	−0.0775	0.0355	0.0182	0.1550	−0.0169
209	−0.005627	−0.9621	0.0421	1.0457	−0.2018	−0.0737	−0.0639	0.0252	0.0097	0.1472
210	−0.006517	−1.1089	0.0315	1.0273	−0.1999	−0.0315	−0.1222	−0.1639	−0.0314	−0.0764

(continued)

(continued)

211	−0.007004	−1.1986	0.0416	1.0335	−0.2496	−0.0766	0.1482	−0.0646	−0.1304	0.0373
212	0.002481	0.4232	0.0407	1.0613	0.0872	0.0255	−0.0052	−0.0500	0.0244	0.0476
213	0.001939	0.3287	0.0287	1.0499	0.0565	0.0272	−0.0045	−0.0172	−0.0472	0.0066
214	−2.713E-6	−0.000460	0.0309	1.0547	−0.0001	−0.0000	0.0000	0.0000	0.0000	0.0001
215	−0.005868	−0.9844	0.0047	1.0054	−0.0677	−0.0646	0.0027	−0.0001	0.0019	0.0197
216	−0.006363	−1.0677	0.0043	1.0013	−0.0705	−0.0673	0.0063	−0.0022	−0.0032	−0.0020
217	0.001369	0.2292	0.0043	1.0254	0.0152	0.0147	−0.0008	−0.0017	0.0002	0.0003
218	−0.006191	−1.0386	0.0043	1.0026	−0.0686	−0.0664	−0.0013	0.0037	0.0077	−0.0009
219	0.000449	0.0752	0.0044	1.0265	0.0050	0.0048	−0.0001	0.0001	−0.0003	−0.0006
220	−0.002079	−0.3481	0.0043	1.0238	−0.0228	−0.0219	−0.0000	−0.0006	−0.0009	0.0006
221	−0.005483	−0.9197	0.0050	1.0084	−0.0651	−0.0531	−0.0244	−0.0077	−0.0069	−0.0104
222	0.002458	0.4117	0.0050	1.0235	0.0292	0.0256	−0.0051	0.0081	0.0014	0.0001
223	0.003872	0.6491	0.0057	1.0186	0.0493	0.0371	0.0174	−0.0031	0.0163	0.0077
224	−0.003506	−0.5878	0.0063	1.0209	−0.0468	−0.0320	−0.0158	−0.0183	0.0010	−0.0176
225	0.000299	0.0500	0.0055	1.0278	0.0037	0.0029	0.0005	0.0010	0.0014	−0.0003
226	−0.000207	−0.0347	0.0058	1.0281	−0.0027	−0.0019	−0.0006	−0.0006	−0.0009	−0.0012
227	−0.005623	−0.9431	0.0048	1.0072	−0.0652	−0.0542	−0.0112	−0.0103	−0.0129	−0.0183
228	−0.006767	−1.1359	0.0044	0.9981	−0.0756	−0.0683	−0.0056	−0.0083	−0.0090	−0.0108
229	−0.003741	−0.6267	0.0043	1.0178	−0.0413	−0.0390	0.0012	−0.0009	−0.0031	−0.0028
230	−0.001733	−0.2901	0.0044	1.0247	−0.0193	−0.0188	0.0026	0.0017	0.0004	−0.0002
231	−0.002517	−0.4215	0.0044	1.0227	−0.0281	−0.0277	0.0023	0.0046	0.0030	0.0014
232	−0.002872	−0.4810	0.0044	1.0215	−0.0320	−0.0316	0.0016	0.0025	0.0052	0.0035
233	−0.002779	−0.4653	0.0044	1.0218	−0.0311	−0.0307	0.0041	0.0017	0.0026	0.0054
234	−0.000197	−0.0330	0.0044	1.0266	−0.0022	−0.0021	−0.0003	0.0001	−0.0000	−0.0000
235	−0.000662	−0.1108	0.0044	1.0264	−0.0074	−0.0068	−0.0008	−0.0011	0.0002	−0.0003
236	−0.002062	−0.3452	0.0045	1.0240	−0.0231	−0.0208	−0.0027	−0.0030	−0.0039	−0.0001

237	−0.002621	−0.4390	0.0049	1.0229	−0.0309	−0.0248	−0.0094	−0.0062	−0.0057	−0.0078
238	0.002089	0.3499	0.0050	1.0245	0.0248	0.0196	0.0070	0.0076	0.0051	0.0049
239	0.0000374	0.006262	0.0052	1.0275	0.0005	0.0003	0.0001	0.0001	0.0001	0.0001
Sum of residuals										0
Sum of squared residuals										0.00817
Predicted residual SS (PRESS)										0.00856

References

Beaton, A. E., & Tukey, J. W. (1974). The fitting of power series, meaning polynomials, illustrated on bank-spectroscopic data. *Technometrics, 16*, 147–185.

Belsley, D. A., Kuh, E., & Welsch, R. E. (1980). *Regression diagnostics: Identifying influential data and sources of collinearity*. New York: Wiley. Chapter 2.

Burns, A. F., & Mitchell, W. C. (1946). *Measuring business cycles*. New York: NBER.

Efron, B., Hastie, T., Johnstone, J., & Tibshirani, R. (2004). Least angle regression. *The Annals of Statistics, 32*, 407–499.

Guerard Jr., J. B. (2001). A note on the forecasting effectiveness of the U.S. leading economic indicators. *Indian Economic Review, 36*, 251–268.

Guerard Jr., J. B. (2004). The forecasting effectiveness of the U.S. leading economic indicators: Further evidence and initial G7 results. In P. Dua (Ed.), *Business cycles and economic growth: An analysis using leading indicators* (pp. 174–187). New York: Oxford University Press.

Guerard, J. B., Jr., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Guerard, J. B., Xu, G., & Wang, Z. (2019). Portfolio and Investment Analysis with SAS®: Financial Modelling Techniques for optimization. Cary, NK: SAS Press.

Hastie, T., Tibshirani, R., & Friedman, J. (2016). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed, 11th printing). New York: Springer.

Huber, P. J. (1973). Robust regression: Asymptotics, conjectures, and Monte Carlo. *Annals of Statistics, 1*, 799–782.

Johnston, J. (1972). *Econometric methods* (2nd ed.). New York: McGraw-Hill.

Levanon, G., Manini, J-C., Ozyildirim, A., Schaitkin, B., & Tanchua, J. (2015). Using financial indicators to predict turning points in the business cycle: the case of the leading economic index for the United States. *International Journal of Forecasting, 31*, 419–438.

Maronna, R. A., Martin, R. D., Yohai, V. J., & Salibian-Barrera, M. (2019). *Robust statistics: Theory and practice (with R)*. New York: Wiley.

Mitchell, W. C. (1913). *Business cycles*. New York: Burt Franklin reprint.

Mitchell, W. C. (1927). *Business cycles: the problem & its setting*. New York: National Bureau of Economic Research.

Mitchell, W. C. (1951). *What happens during business cycles: A progress report*. New York: NBER.

Moore, G. H. (1961). *Business cycle indicators*, 2 volumes. Princeton: Princeton University Press.

Zarnowitz, V. (1992). *Business cycles: Theory, history, indicators, and forecasting*. Chicago: University of Chicago Press.

Time Series Modeling and the Forecasting Effectiveness of the US Leading Economic Indicators



An important aspect of financial decision-making may depend on the forecasting effectiveness of the composite index of leading economic indicators, LEI. The leading indicators can be used as an input to a transfer function model of real gross domestic product, GDP. The previous chapter employed four quarterly lags of the LEI series to estimate regression models of association between current rates of growth of real US GDP and the composite index of leading economic indicators. This chapter asks the question as to whether changes in forecasted economic indexes help forecast changes in real economic growth. The transfer function model forecasts are compared to several naïve models in terms of testing which model produces the most accurate forecast of real GDP. No-change forecasts of real GDP and random walk with drift models may be useful forecasting benchmarks (Granger & Newbold, 1977; Mincer & Zarnowitz, 1969). Economists have constructed leading economic indicator series to serve as a business barometer of the changing US economy since the time of Wesley C. Mitchell (1913). We introduced the LEI in the previous chapter on regression modeling. Now, we seek to use time series and causality modeling with the LEI. The purpose of this study is to examine the time series forecasts of composite economic indexes produced by The Conference Board (TCB) and test the hypothesis that the leading indicators are useful as an input to a time series model to forecast real output in the United States.

Economic indicators are descriptive and anticipatory time series data used to analyze and forecast changing business conditions. Cyclical indicators are comprehensive series that are systemically related to the business cycle. Business cycles are recurrent sequences of expansions and contractions in aggregate economic activity. Coincident indicators have cyclical movements that approximately correspond with the overall business cycle expansions and contractions. Leading indicators reach their turning points before the corresponding business cycle turns. The lagging indicators reach their turning points after the corresponding turns in the business cycle.

In section "Basic Statistical Properties of Economic Series," we introduce the reader to time series modeling and forecasting. Section "ARMA Model Identification in Practice" of this chapter presents an introduction to the models that are estimated and tested in the analysis of the forecasting effectiveness of the leading

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_13

indicators. Section "Leading Economic Indicators (LEI) and Real GDP Analysis: The Statistical Evidence of the Initial Period, 1970–2002" presents the empirical evidence to support the time series models and reports how models adequately describe the data. Out-of-sample forecasting results are shown in section "Updating the Forecasting Effectiveness of Time Series Modeling: Studies of the US Real GDP, 1959Q2–2018Q3" for the United States and the G7 nations.[1] Section "Updating the Forecasting Effectiveness of Time Series Modeling: Studies of the US Real GDP, 1959Q2–2018Q3" of this chapter presents an introduction to SAS time series modeling and its outlier analysis. In section "Forecasting Effectiveness of Time Series Modeling Using Autometrics and SCA Model the US Unemployment Rate, 1959Q2–2018Q3," models that are estimated and tested in the analysis of the forecasting effectiveness of the leading indicators, using SCA and OxMetrics statistical analysis software systems, reflecting recent time series research in Tsay (2010), Clements and Hendry (1998), Doornik (2009), Doornik and Hendry (2013a, 2013b, 2015), Hendry and Doornik (2014), and Dhrymes (2017) are presented. The reader notes the recent increase in outstanding texts on time series modeling. This chapter was enhanced greatly from the Guerard and Schwartz (2007) edition.

Basic Statistical Properties of Economic Series

This chapter develops and forecasts models of economic time series in which we initially use only the past history of the series. The chapter later explores explanatory variables in the forecast models. The time series modeling approach of Box and Jenkins involves the identification, estimation, and forecasting of stationary (or series transformed to stationarity) series through the analysis of the series autocorrelation and partial autocorrelation functions.[2] The autocorrelation function examines the correlations of the current value of the economic times series and its previous k-lags. That is, one can measure the correlation of a daily series, of shares, or other assets, by calculating:

$$p_{jt} = a + bp_{jt-1} \tag{1}$$

where

p_{jt} = today's price of stock j;
p_{jt-1} = yesterday's price of stock j;
b is the correlation coefficient.

[1] Section "ARMA Model Identification in Practice" can be omitted with little loss of continuity with readers more interested in the application of time series models.

[2] This section draws heavily from Box and Jenkins, *Time Series Analysis*, Chapters 2 and 3. The time series analysis of Box and Jenkins was a great intellectual advance. The 1970s and 1980s were its high water mark period. The authors continue to use Box and Jenkins modeling in the SCA system, see Liu (2006).

In a daily shares price series, b is quite large, often approaching a value of 1.00. As the number of lags, or the previous number of periods increases, the correlation tends to fall. The decrease is usually very gradual.

The partial autocorrelation function examines the correlation between p_{jt} and p_{jt-2}, holding constant the association between p_{jt} and p_{jt-1}. If a series follows a random walk, the correlation between p_{jt} and p_{jt-1} is one, and the correlation between p_j and p_{jt-2}, holding constant the correlation of p_{jt} and p_{jt-1}, is zero. Random walk series are characterized by decaying autocorrelation functions and a partial autocorrelation function with a "spike" at lag one and zeros thereafter. Stationarity implies that the joint probability [p(Z)] distribution $P(Z_{t1}, Z_{t2})$ is the same for all times t, t_1, and t_2 where the observations are separated by a constant time interval. The autocovariance of a time series at some lag or interval, k, is defined to be the covariance between Z_t and Z_{t+k}:

$$\gamma_k = \mathrm{cov}[Z_t, Z_{t+k}] = \mathrm{E}[(Z_t - \mu)(Z_{t+k} - \mu)]. \tag{2}$$

One must standardize the autocovariance, as one standardizes the covariance in traditional regression analysis before one can quantify the statistically significant association between Z_t and Z_{t+k}. The autocorrelation of a time series is the standardization of the autocovariance of a time series relative to the variance of the time series, and the autocorrelation at lag k, ρ_k, is bounded between +1 and −1:

$$\begin{aligned} \rho_k &= \frac{E[(Z_t - \mu)(Z_{t+k} - \mu)]}{\sqrt{E\left[(Z_t - \mu)^2\right] E\left[(Z_{t+k} - \mu)^2\right]}} \\ &= \frac{E[(Z_t - \mu)(Z_{t+k} - \mu)]}{\sigma_Z^2} = \frac{r_k}{r_0} \end{aligned} \tag{3}$$

The autocorrelation function of the process, {ρk}, represents the plotting of r_k versus time, the lag of k. The autocorrelation function is symmetric about series and thus $\rho_k = \rho_{-k}$; thus, time series analysis normally examines only the positive segment of the autocorrelation function. One may also refer to the autocorrelation function as the correlogram. The statistical estimates of the autocorrelation function are calculated from a finite series of N observations, $Z_1, Z_2, Z_3, \ldots, Z_n$. The statistical estimate of the autocorrelation function at lag k, r_k, is found by

$$r_k = \frac{C_k}{C_0}$$

where

$$C_k = \frac{1}{N} \sum_{t=1}^{N-k} \left(Z_t - \overline{Z}\right)\left(Z_{t+k} - \overline{Z}\right), \quad k = 0, 1, 2, \ldots, K.$$

Ck is, of course, the statistical estimate of the autocovariance function at lag k. In identifying and estimating parameters in a time series model, one seeks to identify orders (lags) of the time series that are statistically different from zero. The

implication of testing whether an autocorrelation estimate is statistically different from zero leads one back to the t-tests used in regression analysis to examine the statistically significant association between variables. One must develop a standard error of the autocorrelation estimate such that a formal t-test can be performed to measure the statistical significance of the autocorrelation estimate. Such a standard error, S_e, estimate was found by Bartlett and, in large samples, is approximated by

$$\begin{aligned} Var[r_k] &\cong \frac{1}{N}, \quad \text{and} \\ S_e[r_k] &\cong \frac{1}{\sqrt{N}}. \end{aligned} \tag{4}$$

An autocorrelation estimate is considered statistically different from zero if it exceeds approximately twice its standard error.

A second statistical estimate useful in time series analysis is the partial autocorrelation estimate of coefficient j at lag k, ϕ_{kj}. The partial autocorrelations are found in the following manner:

$$\rho_j = \phi_{k1}\rho_{j-1} + \phi_{k2}\rho_{j-2} + \cdots + \phi_{k(k-1)}\rho_{jk-1} + \phi_{kk}\rho_{j-k} \quad j = 1, 2, \ldots, k$$

or

$$\begin{bmatrix} 1 & \rho_1 & \rho_2 & \cdots & \rho_{k-1} \\ \rho_1 & 1 & \rho_1 & \cdots & \rho_{k-2} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ \rho_{k-1} & \rho_{k-2} & \rho_{k-3} & \cdots & 1 \end{bmatrix} \begin{bmatrix} \phi_{k1} \\ \phi_{k2} \\ \vdots \\ \phi_{kk} \end{bmatrix} \begin{bmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_k \end{bmatrix}$$

The partial autocorrelation estimates may be found by solving the above equation systems for k = 1,2,3, ... k:

$$\phi_{11} = \rho_1$$

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = \frac{\begin{vmatrix} 1 & \rho_1 \\ \rho_2 & \rho_2 \end{vmatrix}}{\begin{vmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{vmatrix}}$$

$$\phi_{33} = \frac{\begin{vmatrix} 1 & \rho_1 & \rho_1 \\ \rho_1 & 1 & \rho_2 \\ \rho_2 & \rho_1 & \rho_3 \end{vmatrix}}{\begin{vmatrix} 1 & \rho_1 & \rho_2 \\ \rho_1 & 1 & \rho_1 \\ \rho_2 & \rho_1 & 1 \end{vmatrix}}$$

The partial autocorrelation function is estimated by expressing the current autocorrelation function estimates as a linear combination of previous orders of autocorrelation estimates:

$$\widehat{r}_1 = \widehat{\phi}_{k1^r j-1} + \widehat{\phi}_{k2^2 j-2} + \cdots + \widehat{\phi}_{k(k-1)^r j+k-1} + \widehat{\phi}_{kk^2 j-k} \quad j = 1, 2, \ldots, k.$$

The standard error of the partial autocorrelation function is approximately

$$Var\left[\widehat{\phi}_{kk}\right] \cong \frac{1}{N}, \quad \text{and}$$
$$S_e[\phi_{kk}] \cong \frac{1}{\sqrt{N}}.$$

The Autoregressive and Moving Average Processes

A stochastic process, or time series, can be repeated as the output resulting from a white noise input, α_t:[3]

$$\begin{aligned} \widetilde{Z}_t &= \alpha_t + \psi_1 \alpha_{t-1} + \psi_2 \alpha_{t-2} + \ldots \\ &= \alpha_t + \sum_{j=1}^{\infty} \psi_j a_{t-j}. \end{aligned} \tag{5}$$

The filter weight, ψ_j, transforms input into the output series. One normally expresses the output, $\widetilde{Z}_t$, as a deviation of the time series from its mean, μ, or origin:

$$\widetilde{Z}_t = Z_t - \mu.$$

The general linear process leads one to represent the output of a time series, $\widetilde{Z}_t$, as a function of the current and previous value of the white noise process, α_t, which may be represented as a series of shocks. The white noise process, α_t, is a series of random variables characterized by

$$\begin{aligned} &E[\alpha_t] \cong 0 \\ &Var[\alpha_t] = \sigma_\alpha^2 \\ &\gamma_k = E[\alpha_t \alpha_{t+k}] = \sigma_\alpha^2 \quad k \neq 0 \\ &0 \qquad\qquad\qquad\qquad k = 0. \end{aligned}$$

The autocorrelation function of a linear process may be given by

[3]Please see Box and Jenkins, *Time Series Analysis*, Chapter 3, for the most complete discussion of the ARMA (p,q) models. Granger and Newbold (1977), Granger (1989), and Hamilton (1994) are considered standard time series modeling texts.

$$\gamma_k = \sigma_\alpha^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+k}.$$

The backward shift operator, B, is defined as $BZ_t = Z_{t-1}$ and $B^jZ_t = Z_{t-}j$. The autocorrelation generating function may be written as

$$\gamma(B) = \sum_{k=-\infty}^{\infty} \gamma_k B^k$$

For stationarity, the ψ weights of a linear process must satisfy that $\psi(B)$ converges on or lies within the unit circle.

In an autoregressive, AR, model, the current value of the time series may be expressed as a linear combination of the previous values of the series and a random shock, α_t:

$$\widetilde{Z}_t = \phi_1 \widetilde{Z}_{t-1} + \phi_2 \widetilde{Z}_{t-2} + \cdots + \phi_p \widetilde{Z}_{t-p} + \alpha_t$$

The autoregressive operator of order P is given by

$$\begin{aligned} &\phi(B) = 1 - \phi_1 B^1 - \phi_2 B^2 - \cdots - \phi_p B^p \\ &\text{or} \\ &\phi(B)\widetilde{Z}_t = \alpha_t \end{aligned} \tag{6}$$

In an autoregressive model, the current value of the time series, $\widetilde{Z}_t$, is a function of previous values of the time series, $\widetilde{Z}_{t-1}$, $\widetilde{Z}_{t-2}$, ..., and is similar to a multiple regression model. An autoregressive model of order p implies that only the first p order weights are non-zero. In many economic time series, the relevant autoregressive order is one and the autoregressive process of order p, AR(p), is written as

$$\begin{aligned} &\widetilde{Z}_t = \phi_1 \widetilde{Z}_{t-1} + \alpha_t \\ &\text{or} \\ &(1 - \phi_1 B)\widetilde{Z}_t = \alpha_t \\ &\widetilde{Z}_t = \phi^{-1}(B)\alpha_t. \end{aligned}$$

The relevant stationarity condition is $|B| < 1$ implying that $|\phi_1| < 1$. The autocorrelation function of a stationary autoregressive process

$$\widetilde{Z}_t = \phi_1 \widetilde{Z}_{t-1} + \phi_2 \widetilde{Z}_{t-2} + \cdots + \phi_p \widetilde{Z}_{t-p} + \alpha_t$$

may be expressed by the difference equation

$$P_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \cdots + \phi_k \rho_{k-p} \quad k > 0$$

or expressed in terms of the Yule-Walker equation as

$$\begin{aligned}
\rho_1 &= \phi_1 + \phi_2 \rho_1 + \cdots + \phi_p \rho_{p-1} \\
\rho_2 &= \phi_1 \rho_1 + \phi_2 + \cdots + \phi_p \rho_{p-2} \\
\rho_p^{=} &= \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \cdots + \overline{\overline{\phi}}_p
\end{aligned}$$

For the first-order AR process, AR(1),

$$\rho_k = \phi_1 \rho_{k-1} = \overline{\overline{\phi}}_p.$$

The autocorrelation function decays exponentially to zero when ϕ_1 is positive and oscillates in sign and decays exponentially to zero when ϕ_1 is negative:

$$P_1 = \phi_1$$

and

$$\sigma_2 = \frac{\sigma_\alpha^2}{1 - \phi_1^2}.$$

The partial autocorrelation function cuts off after lag one in an AR(1) process. For a second-order AR process, AR(2),

$$\widetilde{Z}_t = \phi_1 \widetilde{Z}_{t-1} + \phi_2 \widetilde{Z}_{t-k} + \alpha_t$$

with roots

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 = 0$$

and, for stationarity, roots lying outside the unit circle, ϕ_1 and ϕ_2 must obey the following conditions:

$$\begin{aligned}
\phi_2 + \phi_1 &< 1 \\
\phi_2 - \phi_1 &< 1 \\
-1 < \phi_2 &< 1.
\end{aligned}$$

The autocorrelation function of an AR(2) model is

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} \tag{7}$$

The autocorrelation coefficients may be expressed in terms of the Yule-Walker equations as

$$\rho_1 = \phi_1 + \phi_2\rho_2$$
$$\rho_2 = \phi_1\rho_1 + \phi_2$$

which implies

$$\phi_1 = \frac{\rho_1(1-\rho_2)}{1-\rho_1^2}$$
$$\phi_2 = \frac{\rho_2(1-\rho_1^2)}{1-\rho_1^2}$$

and

$$\rho_1 = \frac{\phi_1}{1-\phi_2} \quad \text{and} \quad \rho_2 = \phi_2 + \frac{\phi_1^2}{1-\phi_2}.$$

For a stationary AR(2) process,

$$-1 < \phi_1 < 1$$
$$-1 < \rho_2 < 1$$
$$\rho_1^2 < \frac{1}{2}(\rho_2 + 1).$$

In an AR(2) process, the autocorrelation coefficients tail off after order two and the partial autocorrelation function cuts off after the second-order (lag).[4]

In a q-order moving average (MA) model, the current value of the series can be expressed as a linear combination of the current and previous shock variables:

$$\begin{aligned}\widetilde{Z}_t &= \alpha_1 - \theta_1\alpha_{t-1} - \cdots - \alpha_q\theta_{t-q} \\ &= \left(1 - \theta_1 B_1 - \cdots - \theta_q B_q\right)\alpha_t \\ &= \theta(B)\alpha_t\end{aligned}$$

The autocovariance function of a q-order moving average model is

[4] A stationary AR(p) process can be expressed as an infinite weighted sum of the previous shock variables:

$$\widetilde{Z}_t = \phi^{-1}(B)\alpha_t.$$

In an invertible time series, the current shock variable may be expressed as an infinite weighted sum of the previous values of the series:

$$\theta^{-1}(B)\widetilde{Z}_t = \alpha_t.$$

$$\gamma_k = E\left[\left(\alpha_t - \theta_1\alpha_{t-1} - \cdots - \theta_q\alpha_{t-q}\right)\left(\alpha_{t-k} - \theta_1\alpha_{t-k-1} - \cdots - \theta_q\alpha_{t-k-q}\right)\right]$$

The autocorrelation function, ρ_k, is

$$\rho_k = \begin{array}{ll} \dfrac{-\theta_k + \theta_1\theta_{k+1} + \cdots + \theta_{q-k}\theta_q}{1 + \theta_1^2 + \cdots + \theta_q^2} & k = 1, 2, \ldots, q \\ 0 & k > q \end{array}$$

The autocorrelation function of a MA(q) model cuts off, to zero, after lag q and its partial autocorrelation function tails off to zero after lag q. There are no restrictions on the moving average model parameters for stationarity; however, moving average parameters must be invertible. Invertibility implies that for the π weights of the linear filter transforming the input into the output series, the π weights lie outside the unit circle.

$$\begin{aligned} \pi(B) &= \psi^{-1}(B) \\ &= \sum_{j=0}^{a} \phi^j B^j \end{aligned}$$

In a first-order moving average model, MA(1),

$$\widehat{Z}_t = (1 - \theta_1 B)\alpha_t$$

and the invertibility condition is $|\theta_1| < 1$. The autocorrelation function of the MA (1) model is

$$\rho_k = \begin{array}{ll} \dfrac{-\theta_1}{1 + \theta_1^2} & k = 1 \\ & k > 2. \end{array}$$

The partial autocorrelation function of an MA(1) process tails off after lag one and its autocorrelation function cuts off after lag one.

In a second-order moving average model, MA(2),

$$\widetilde{Z}_t = \alpha_t - \theta_1\alpha_{t-1} - \theta_2\alpha_{t-2}$$

the invertibility conditions require

$$\theta_2 < \theta_1 < 1$$

$$\theta_2 - \theta_1 < 1$$

$$-1 < \theta_2 < 1$$

The autocorrelation function of the MA(2) is

$$\rho_1 = \frac{-\theta_1(1+\theta_2)}{1+\theta_1^2+\theta_1^2}$$

$$\rho_2 = \frac{-\theta_2}{1+\theta_1^2+\theta_1^2}$$

and

$$\rho_k = \theta \quad \text{for} \quad k > 3.$$

The partial autocorrelation function of an MA(2) tails off after lag two.

In many economic time series, it is necessary to employ a mixed autoregressive moving average (ARMA) model of the form

$$\begin{aligned} &\widetilde{Z}_t = \phi_1 \widetilde{Z}_{t-1} + \cdots + \phi_p \widetilde{Z}_{t-p} + \alpha_t - \theta_1 \alpha_{t-1} - \cdots - \theta_q \alpha_{t-q} \\ &\text{or} \\ &\left(1 - \phi_1 B - \phi_2 B^2 - \cdots - \theta_p B^p\right)\widetilde{Z}_t = \left(1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q\right)\alpha_t \end{aligned} \tag{8}$$

that may be more simply expressed as

$$\phi(B)\widetilde{Z}_t = \theta(B)\alpha_t$$

The autocorrelation function of the ARMA model is

$$\begin{aligned} &\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \cdots + \phi_p \rho_{k-p} \\ &\text{or} \\ &\phi(B)\rho_k = 0. \end{aligned}$$

The first-order autoregressive first-order moving average operator ARMA(1,1) process is written

$$\begin{aligned} &\widetilde{Z}_t - \phi_1 \widetilde{Z}_{t-1} = \alpha_t - \theta_1 \alpha_{t-1} \\ &\text{or} \\ &(1 - \phi_1)\widetilde{Z}_t = (1 - \theta 1 B)\alpha_t. \end{aligned}$$

The stationary condition is $-1 < \phi_1 < 1$ and the invertibility condition is $-1 < \phi_1 < 1$. The first two autocorrelations of the ARMA (1,1) model are

$$\rho_1 = \frac{(1 - \phi_1\theta_1)(\phi_1 + \theta_1)}{1 + \theta_1^2 - 2\phi_1\theta_1}$$

and

$$\rho_2 = \phi_1\rho_1$$

The partial autocorrelation function consists only of $\phi_{11} = \rho_1$ and has a damped exponential.

An integrated stochastic progress generates a time series if the series is made stationary by differencing (applying a time-invariant filter) the data. In an integrated process, the general form of the time series model is

$$\phi(B)(1 - B)^d X_t = \theta(B)\varepsilon_t \tag{9}$$

where $\phi(B)$ and $\theta(B)$ are the autoregressive and moving average polynominals in B of orders p and q, ε_t is a white noise error term, and d is an integer representing the order of the data differencing. In economic time series, a first difference of the data is normally performed.[5] The application of the differencing operator, d, produces a stationary ARMA(p,q) process. The autoregressive integrated moving average, ARMA, model is characterized by orders p, d, and q [ARIMA(p,d.q)]. Many economics series follow a random walk with drift; an ARMA (1,1) may be written as

$$\overline{V}^d X_t = X_t - X_{t-1} = \varepsilon_t + b\varepsilon_{t-1}. \tag{10}$$

An examination of the autocorrelation function estimates may lead one to investigate using a first-difference model when the autocorrelation function estimates decay slowly. In an integrated process, the corr(X_t, $X_{t-\tau}$) is approximately unity for small values of time, τ.

ARMA Model Identification in Practice

Time series specialists use many statistical tools to identify models; however, the sample autocorrelation and partial autocorrelation function estimates are particularly useful in modeling. Univariate time series modeling normally requires larger datasets than regression and exponential smoothing models. It has been suggested that at least 40–50 observations be used to obtain reliable estimates.[6] One normally calculates the sample autocorrelation and partial autocorrelation estimates for the

[5] Box and Jenkins, *Time Series Analysis*. Chapter 6; C.W.J. Granger and Paul Newbold, *Forecasting Economic Time Series*. Second Edition (New York: Academic Press, 1986), pp. 109–110, 115–117, 206.

[6] Granger and Newbo1d, *Forecasting Economic Time Series*. pp. 185–186.

raw time series and its first (and possibly second) differences. The failure of the autocorrelation function estimates of the raw data series to die out as large lags implies that a first difference is necessary. The autocorrelation function estimates of a MA(q) process should cut off after q. To test whether the autocorrelation estimates are statistically different from zero, one uses a t-test where the standard error of $\upsilon\tau$ is[7]

$$\mathrm{n}^{-1/2}\left[1+2\left(\rho_1^2+\rho_2^2+\cdots\rho_q^2\right)\right]^{1/2} \quad \text{for} \quad \tau > \mathrm{q}.$$

The partial autocorrelation function estimates of an AR(p) process cut off after lag p. A t-test is used to statistically examine whether the partial autocorrelations are statistically different from zero. The standard error of the partial autocorrelation estimates is approximately

$$\frac{1}{\sqrt{N}} \quad \text{for} \quad \mathrm{K} > p. \tag{11}$$

One can use the normality assumption of large samples in the t-tests of the autocorrelation and partial autocorrelation estimates. The identified parameters are generally considered statistically significant if the parameters exceed twice the standard errors.

The ARMA model parameters may be estimated using nonlinear least squares. Given the following ARMA framework generally pack-forecasts the initial parameter estimates and assumes that the shock terms are to be normally distributed:

$$\alpha_\mathrm{t} = \widetilde{W_t} - \phi_1\,\widetilde{W}_{t-1} - \phi_2\,\widetilde{W}_{t-2} - \cdots - \phi_\mathrm{p}\,\widetilde{W}_{t-p} + \theta_1\,\alpha_{t-1} + \cdots + \theta_\mathrm{q}\alpha_{t-q}$$

where

$$\mathrm{W_t} = \overline{V}^d \mathrm{Z_t} \quad \text{and} \quad \widetilde{W}_t = \mathrm{W_t} - \mu.$$

The minimization of the sum of squared errors with respect to the autoregressive and moving average parameter estimates produces starting values for the p order AR estimates and q order MA estimates:

$$\left.\frac{\partial e_t}{-\partial \phi_j}\right|_{\beta_0} = \mu_{\,j,t} \quad \text{and} \quad \left.\frac{\partial e_t}{-\partial \theta_i}\right|_{\beta_0} = X_{\,j,t}$$

[7] Box and Jenkins, *Time Series Analysis*. pp. 173–179.

It may be appropriate to transform a series of data such that the residuals of a fitted model have a constant variance or are normally distributed. The log transformation is such a data transformation that is often used in modeling economic time series. Box and Cox (1964) put forth a series of power transformations useful in modeling time series.[8] The data is transformed by choosing a value of λ that is suggested by the relationship between the series amplitude (which may be approximated by the range of subsets) and mean:[9]

$$X_t^{\lambda} = \frac{X_t^{\lambda} - 1}{\overline{X}^{\lambda-1}} \tag{12}$$

where X is the geometric mean of the series. One immediately recognizes that if $\lambda = 0$, the series is a logarithmic transformation. The log transformation is appropriate when there is a positive relationship between the amplitude and mean of the series. A $\lambda = 1$ implies that the raw data should be analyzed and there is no relationship between the series range and mean subsets. One generally selects the λ that minimizes the smallest residual sum of squares, although an unusual value of λ may make the model difficult to interpret. Some authors may suggest that only values of λ of −.5, 0, .5, and 1.0 be considered to ease in the model building process.[10]

Many time series, involving quarterly or monthly data, may be characterized by rather large seasonal components. The ARIMA model may be supplemented with seasonal autoregressive and moving average terms:

$$\begin{aligned}
&\left(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p\right)\left(1 - \phi_{1,s} B^s - \cdots - \phi_{p,s} B^p S^s\right)(1 - B)^d \\
&(1 - B^s)^{ds} X_t = \left(1 - \theta_1 B - \cdots - \theta_q B^q\right)\left(1 - \theta_{1,s} B^s - \cdots - \theta_{q,s} B^{q,s}\right)\alpha_t \\
&\text{or} \quad \theta_p(B)\Phi_p(B^s)\overline{V}^d \overline{V}_x^D Z_t = \theta_q(B)\theta_Q(B^s)\alpha_t
\end{aligned} \tag{13}$$

One recognizes seasonal components by an examination of the autocorrelation and partial autocorrelation function estimates. That is, the autocorrelation and partial autocorrelation function estimates should have significantly large values at lags 1 and 12 as well as smaller (but statistically significant) values at lag 13 for monthly data.[11] One seasonally differences the data (a 12th-order seasonal difference for monthly data and estimates the seasonal AR or MA parameters.) A random walk with drift model with a monthly component may be written as

[8]G.E. Box and D.R. Cox, "An Analysis of Transformations," *Journal of the Royal Statistical Society*, B 26 (1964), 211–243.

[9]G.M. Jenkins, "Practical Experience with Modeling and Forecasting Time Series," *Forecasting* (Amsterdam: North-Holland Publishing Company, 1979).

[10]Jenkins, *op. cit.,* pp. 135–138.

[11]Box and Jenkins, *Time Series Analysis*, pp. 305–308.

$$\overline{VV}_{12}Z_t = (1-B)\left(1-\theta B^{12}\right)\alpha_t \tag{14}$$

The multiplicative form of the (0,1,1) × (0,1,1)12 model has a moving average operator that may be written as

$$(1-\theta \mathrm{B})\left(1-\theta \mathrm{B}^{12}\right) = 1-\theta \mathrm{B}-\theta \mathrm{B}^{12}+\theta \mathrm{B}^{13}$$

The random walk with drift with the monthly seasonal adjustments is the basis of the "airline model" in honor of the analysis by Professors Box and Jenkins of total airline passengers during the 1949–1960 period.[12] The airline passenger data analysis employed the natural logarithmic transformation.

There are several tests and procedures that are available for checking the adequacy of fitted time series models. The most widely used test is the Box-Pierce test, where one examines the autocorrelation among residuals, α_t:

$$\widehat{\nu}_k = \frac{t = \sum_{k+1}^{n} \alpha_t \alpha_{t-k}}{\sum_{t=1}^{n} \alpha_t^2}, \quad k = 1, 2, \ldots$$

The test statistic, Q, should be X^2 distributed with (m-p-q) degrees of freedom.

$$Q = n\sum_{k=1}^{m} \widehat{\nu}_k^2.$$

The Ljung-Box statistic is a variation on the Box-Pierce statistic, and the Ljung-Box Q statistic tends to produce significant levels closer to the asymptotic levels than the Box-Pierce statistic for first-order moving average processes. The Ljung-Box statistic, the model adequacy check reported in the SAS system, can be written as

$$\mathrm{Q} = \mathrm{n}(\mathrm{n}+2)\sum_{k=1}^{m} (n=k)^{-1}\widehat{\nu}_k^2. \tag{15}$$

Residual plots are generally useful in examining model adequacy; such plots may identify outliers as we noted in the chapter. The normalized cumulative periodogram of residuals should be examined.

Granger and Newbold (1977) and McCracken (2000) use several criteria to evaluate the effectiveness of the forecasts with respect to the forecast errors. In this chapter, we use the root mean square error (RMSE) criteria. One seeks to

[12] Box and Jenkins, *op. cit.*

minimize the square root of the sum of the absolute value of the forecast errors squared. That is, we calculate the absolute value of the forecast error, square the error, sum the squared errors, divided by the number of forecast periods, and take the square root of the resulting calculation. Intuitively, one seeks to minimize the forecast errors. The absolute value of the forecast errors is important because if one calculated only a mean error, a 5% positive error could "cancel out" a 5% negative error. Thus, we minimize the out-of-sample forecast errors. We need a benchmark for forecast error evaluation. An accepted benchmark (Mincer & Zarnowitz, 1969) for forecast evaluation is a no-change, NoCH. A forecasting model should produce a lower root mean square error (RMSE) than the no-change model. If several models are tested, the lowest RMSE model is preferred.

In the world of business and statistics, one often speaks of autoregressive, moving average, and random walk with drift models, or processes, as we have just introduced.

It is well-known that the majority of economic series, including real gross national product (GDP) in the United States, follow a random walk with drift and are represented with autoregressive integrated moving average (ARIMA) model with a first-order moving average operator applied to the first difference of the data. The data is differenced to produce stationarity, where a process has a (finite) mean and variance that do not change over time, and the covariance between data points of two series depends upon the distance between the data points, not on the time itself. The RWD process, estimated with an ARIMA(0,1,1) model, is approximately equal to a first-order exponential smoothing model. The random walk with drift model has been supported by the work of Nelson and Plosser (1982).

Leading Economic Indicators (LEI) and Real GDP Analysis: The Statistical Evidence of the Initial Period, 1970–2002

We introduce the time series modeling process in this study because we will use The Conference Board US composite leading economic indicator as an input to a transfer function model of US real GDP, both series being first-differenced and log-transformed. The authors test the null hypothesis that there is no statistical association between changes in the logged leading economic indicators and changes in logged real GDP in the United States. A positive and statistically significant coefficient indicates that the leading indicator composite series is associated with rising real output and leads to the rejection of the null hypothesis.

Zarnowitz (1992) examined the determinants of real GDP, 1953–1982, using VAR models. In this analysis, we test the statistical significance of The Conference Board leading economic indicators (LEI) by adding the lags of the variable to an AR (1) model. Does the knowledge of the leading economic indicator help forecast future changes in GDP, and can past values of the GDP data predict the future

growth of GDP? In a recent study of univariate and time series model post-sample forecasting, Thomakos and Guerard (2004) compared RWD and transfer-function models with no-change forecasts using rolling one-period-ahead post-sample periods. Guerard Jr. (2001) found that the AR(1) and random walk with drift processes are adequate representations of the time series process of real GDP, given the lags of the autocorrelation and partial autocorrelation functions. Guerard Jr. (2001) reported the estimated cross-correlation functions between the G7 respective LEI and real GDP for the 1970–2000 period and found that the resulting transfer function models were statistically significant in forecasting real GDP in the G7 nations.

In this chapter, the authors report the estimated autocorrelation and partial autocorrelation functions of US real GDP, 1963–March 2002, shown in Table 1. Let us look at Table 1, the estimated autocorrelation and partial autocorrelations functions of real quarterly US GDP, March 1963–March 2002. The estimated autocorrelation function decays gradually, falling from 0.979 for a one period (quarter lag), 0.958 for a 2-quarter lag, to 0.584 for a 20-quarter lag, and 0.318 for a 36-quarter lag. The estimated partial autocorrelation function is characterized by the "spike" at a one-quarter lag. The first estimated partial autocorrelation is 0.979, and the second partial autocorrelation is −.005. The US real GDP series can be estimated as a random walk with drift series for the 1963–2002 period.

The estimated functions substantiate the estimation of the first-order moving average operator of the first-differenced, log-transformed US real GDP series, denoted RWD, shown in Table 2. Guerard Jr. (2001) used an autoregressive variation of the random walk with drift model as a forecasting benchmark. The residuals of the RWD model show few deviations from normality. The RWD is a statistically adequately fitted model.

We estimate the cross-correlation function of the LEI and real GDP for an initial 32-quarter estimation period, following Thomakos and Guerard (2004), and use the 1978–March 2002 period for initial US post-sample evaluation. Similar estimations are derived for real GDP series in France (FR), Germany (GY), and the United Kingdom (see Table 3). The leading economic indicators are statistically significantly associated with real GDP in the respective countries during the 1978–2002 period, as are shown in the respective GDP regressions in Table 3. The lag structures of the models were discussed in Guerard Jr. (2001), and we refer the reader to the initial modeling and forecasting analysis. The statistical significance of the transfer functions in Table 3 leads one to reject the null hypothesis of no statistical association changes in the leading economic indicators and changes in real GDP. The statistically significant lags in the cross-correlation functions show how past values of the LEI series are associated with the current values of the respective real GDP. That is, the LEI series lead their respective real GDP series and can be used as inputs to transfer function models of real GDP. The multiple regressions of the post-sample period are generally statistically significant at the 1% level, as shown by their respective F-statistics of the regressions. The exception to this result is the French real GDP estimate (see Table 3) that is significant at approximately the 5% level.

Table 1 Correlogram of USGDP

Date: 05/09/03 Time: 14:28
Sample: 1 158
Included observations: 157

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.979	0.979	153.31	0.000
		2	0.958	-0.005	301.09	0.000
		3	0.937	-0.010	443.41	0.000
		4	0.916	-0.024	580.18	0.000
		5	0.894	-0.024	711.33	0.000
		6	0.871	-0.022	836.82	0.000
		7	0.849	-0.015	956.70	0.000
		8	0.826	-0.021	1071.0	0.000
		9	0.804	0.003	1179.9	0.000
		10	0.781	-0.011	1283.6	0.000
		11	0.760	0.008	1382.4	0.000
		12	0.739	0.004	1476.5	0.000
		13	0.719	-0.005	1566.1	0.000
		14	0.699	-0.011	1651.3	0.000
		15	0.679	0.002	1732.4	0.000
		16	0.660	-0.004	1809.4	0.000
		17	0.640	-0.012	1882.4	0.000
		18	0.621	-0.003	1951.7	0.000
		19	0.602	-0.011	2017.4	0.000
		20	0.584	-0.006	2079.4	0.000
		21	0.566	0.004	2138.2	0.000
		22	0.548	-0.001	2193.7	0.000
		23	0.531	-0.004	2246.2	0.000
		24	0.514	-0.013	2295.7	0.000
		25	0.497	0.005	2342.4	0.000
		26	0.480	-0.010	2386.4	0.000
		27	0.464	-0.007	2427.7	0.000
		28	0.448	-0.013	2466.5	0.000
		29	0.431	-0.018	2502.7	0.000
		30	0.414	-0.016	2536.4	0.000
		31	0.397	-0.001	2567.7	0.000
		32	0.381	-0.020	2596.6	0.000
		33	0.365	0.008	2623.3	0.000
		34	0.349	-0.008	2648.0	0.000
		35	0.333	0.001	2670.7	0.000
		36	0.318	-0.013	2691.5	0.000

Thus, the estimation of the transfer function is statistically significant relative to simply using an AR(1) time series model.

Guerard and Schwartz (2007) concluded Chapter 13, time series analysis, with Table 3. We will update the LEI analysis for the 1959–2018 time period. We will update the transfer function modeling using two approaches to modern time series

Table 2 Random walk with drift time series model of real US GDP

Dependent variable: DLUSGDP				
Variable	**Coefficient**	**Std. error**	**t-statistic**	**Prob.**
C	0.008	0.0001	8.149	0.000
MA(1)	0.218	0.087	2.507	0.013
R-squared	0.061			
Adjusted R-squared	0.053			
S.E. of regression	0.0086	Akaike info criterion		−6.6575
Sum squared resid	0.0093	Schwarz criterion		−6.6129
Log likelihood	428.08	F-statistic		8.1570
Durbin-Watson stat	1.92	Prob(F-statistic)		0.0050

Table 3 Post-sample regression coefficients of the leading economic indicators, 1978–March 2002

Country	Const.	LEI (−1)	LEI (−2)	LEI (−3)	LEI (−4)	AR(1)	Adj. R-sq.	F-stat.
USA	0.005	0.337	0.060	0.141		0.053	0.283	10.400
(t)	7.200	4.800	0.890	2.130		0.480		
UK	0.005			0.214		−0.166	0.088	5.600
	7.500			2.610		−2.300		
Germany	0.004	0.242		0.211		−0.250	0.102	4.610
	5.750	2.610		2.370		−2.300		
France	0.004		0.140	0.133	−0.064	0.038	0.058	2.470
	7.960		1.930	1.870	−0.910	0.360		
Japan	0.005	0.217				−0.437	0.174	11.030
	5.860	2.900				−4.660		
Canada	0.008		0.306	0.036	−0.263	0.150	0.240	3.290
	4.880		2.340	0.270	−2.100	0.640		
Italy	0.004		0.132	−0.089	−0.009	−0.050	0.059	1.460
	4.670		2.260	−1.480	−1.490	−0.240		

modeling: the outlier analysis work of Tsay (1988, 1989, and 2010) and the general-to-specific (GUM) analysis of David Hendry and his Oxford colleagues. In our "Why a Second Edition" preface, we specifically mentioned the need for an updated time series chapter update.

Updating the Forecasting Effectiveness of Time Series Modeling: Studies of the US Real GDP, 1959Q2–2018Q3

The present (September 2016) ten components of **The Conference Board Leading Economic Index®** for the United States include:

1. Average weekly hours, manufacturing
2. Average weekly initial claims for unemployment insurance
3. Manufacturers' new orders, consumer goods and materials
4. ISM® Index of New Orders
5. Manufacturers' new orders, nondefense capital goods excluding aircraft orders
6. Building permits, new private housing units
7. Stock prices, 500 common stocks
8. Leading Credit Index™
9. Interest rate spread, 10-year Treasury bonds less federal funds
10. Average consumer expectations for business conditions

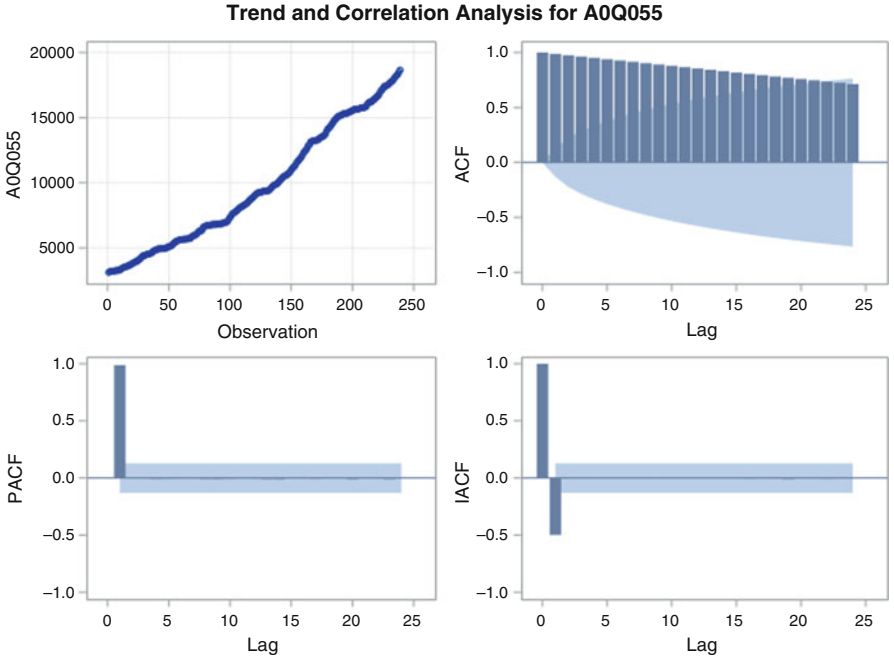
A database of the monthly leading economic indicators is merged with the quarterly FRED database for the real GDP growth previously analyzed. We test the hypothesis that the changes LEI lead real GDP growth. Dhrymes (2017) reports that real GDP growth is statistically associated with contemporaneous and one-through four-quarter lags in the LEI.

One can find, examine, and download real gross domestic product from the St. Louis Federal Reserve Bank database, FRED, as we do, The Conference Board Business Cycle database. The data is downloaded from the first quarter of 1959 to the second quarter of 2018. We estimate several variations of random walk with drift, RWD, an ARIMA(0,1,1), time series models in the spirit of Box and Jenkins and Granger, previously discussed. An ARIMA(1,1,0) model, RWD, is estimated in SAS and reported in Table 4. Let us first examine the correlogram (the autocorrelation lags) and partial autocorrelation lags of real GDP to examine the usefulness of first differencing the data, 1959–2018. We use SAS for an initial time estimate model estimate.

The ARIMA procedure

Name of variable = real GDP (A0Q055)	
Mean of working series	9694.133
Standard deviation	4635.121
Number of observations	239

Autocorrelation check for white noise									
To lag	**Chi-square**	**DF**	**Pr > ChiSq**	**Autocorrelations**					
6	1346.08	6	<.0001	0.988	0.976	0.964	0.952	0.940	0.928
12	2526.39	12	<.0001	0.915	0.903	0.891	0.879	0.867	0.855
18	3547.64	18	<.0001	0.843	0.831	0.818	0.806	0.794	0.783
24	4422.55	24	<.0001	0.771	0.759	0.748	0.736	0.724	0.713



Name of variable = A0Q055	
Period(s) of differencing	1
Mean of working series	65.30714
Standard deviation	52.58961
Number of observations	238
Observation(s) eliminated by differencing	1

Source: SAS

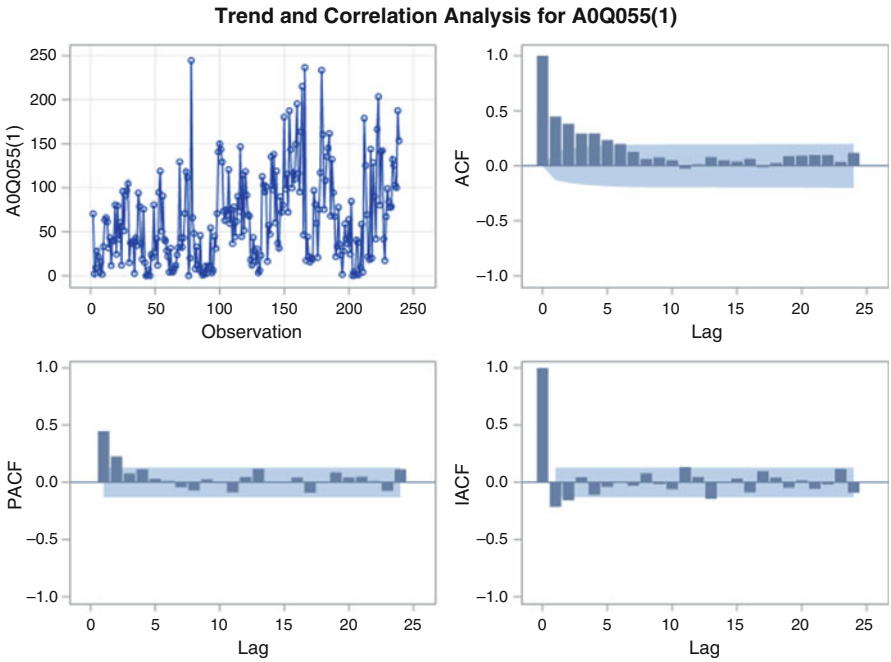
The graphic depiction of the gradual decline of the estimated partial autocorrelation function is consistent with the need to (first) difference for stationarity.

Autocorrelation check for white noise									
To lag	**Chi-square**	**DF**	**Pr > ChiSq**	**Autocorrelations**					
6	148.56	6	<.0001	0.448	0.381	0.293	0.294	0.235	0.199
12	155.76	12	<.0001	0.126	0.061	0.077	0.050	−0.026	0.013
18	159.63	18	<.0001	0.079	0.051	0.039	0.063	−0.012	0.025
24	172.97	24	<.0001	0.088	0.092	0.098	0.097	0.036	0.118

Table 4 ARIMA model identification and estimation of US real GDP, an ARIMA(0,1,1) estimation, 1959–2018Q2

Parameter	Estimate	Standarderror	t-value	ApproxPr > |t|	Lag
MU	65.41400	4.16845	15.69	<.0001	0
MA1,1	−0.31046	0.06193	−5.01	<.0001	1
Constant estimate					65.414
Variance estimate					2414.122
Std error estimate					49.13372
AIC					2531.21
SBC					2538.154
Number of residuals					238

Source: SAS



Source: SAS

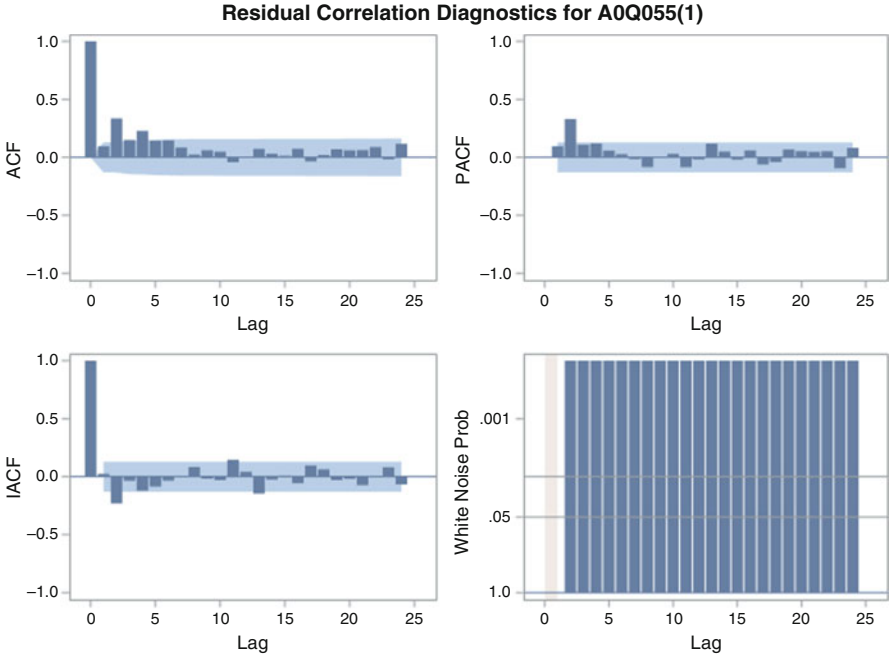
The graphic depiction of the gradual decline of the estimated partial autocorrelation function and the one-period "spike" in the estimated partial autocorrelation function are consistent with an estimated RWD model. Let us estimate an RWD model for US real GDP, 1959–2018Q2.

The estimated first-order moving average term, MA(1), is −.310 and is highly statistically significant, having a t-statistic of −5.01 (see Table 4).

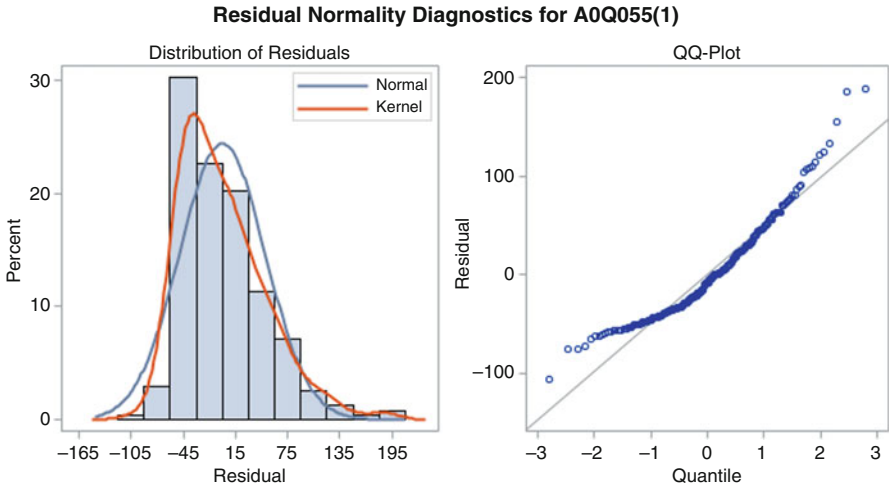
* AIC and SBC do not include log determinant

Correlations of parameter estimates		
Parameter	**MU**	**MA1,1**
MU	1.000	−0.002
MA1,1	−0.002	1.000

Autocorrelation check of residuals									
To lag	**Chi-square**	**DF**	**Pr > ChiSq**	**Autocorrelations**					
6	57.86	5	<.0001	0.096	0.336	0.150	0.228	0.143	0.146
12	61.78	11	<.0001	0.085	0.024	0.061	0.049	−0.043	0.004
18	65.25	17	<.0001	0.073	0.031	0.013	0.074	−0.036	0.020
24	74.31	23	<.0001	0.069	0.061	0.063	0.089	−0.018	0.117
30	75.94	29	<.0001	0.041	0.044	−0.001	0.026	0.014	−0.040
36	78.10	35	<.0001	−0.006	−0.045	0.071	0.008	−0.023	−0.004
42	85.51	41	<.0001	0.034	0.108	−0.008	0.101	0.029	0.046



The RWD residuals are not white noise because the chi-square values and probabilities indicate severe non-randomness.



Source: SAS

At the suggestion of victor zarnowitz, the authors estimated an ARIMA (1,1,0), a near random-walk model, to serve as a time series benchmark. The estimated first-order autoregressive term, AR(1), is 0.452 and is highly statistically significant, having a t-statistic of 7.75 (see Table 5).

Constant estimate	35.91417
Variance estimate	2223.82
Std error estimate	47.1574
AIC	2511.668
SBC	2518.613
Number of residuals	238

Source: SAS

Table 5 ARIMA model identification and estimation of US real GDP, an ARIMA(1,1,0) estimation, 1959–2018Q2

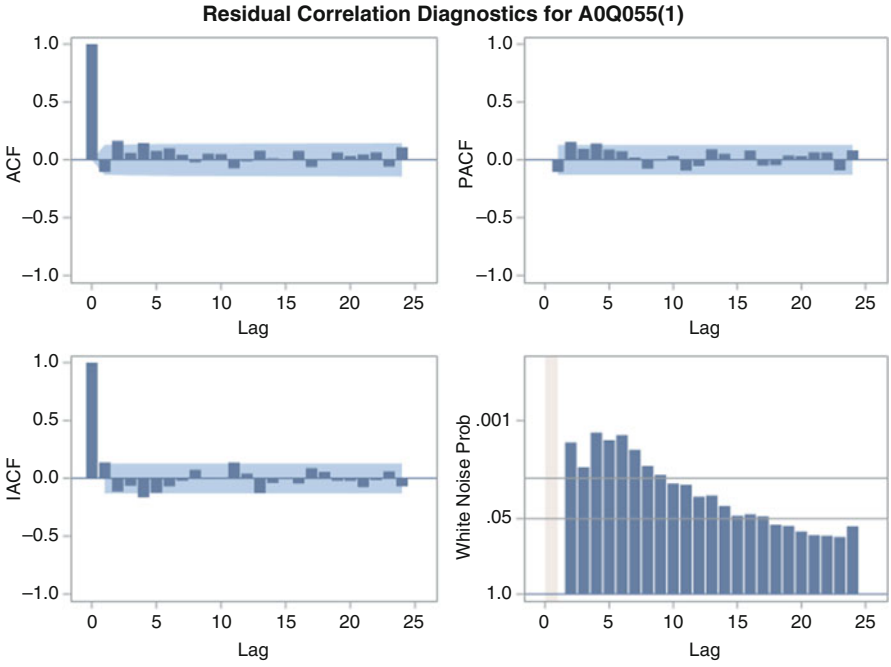
Model for variable A0Q055, real GDP					
Estimated mean				65.414	
Period(s) of differencing				1	
Moving average factors					
Factor 1:				1 + 0.31046 B**(1)	
Conditional least squares estimation					
Parameter	**Estimate**	**Standarderror**	**t-value**	**ApproxPr > |t|**	**Lag**
MU	65.64324	5.56026	11.81	<.0001	0
AR1,1	0.45289	0.05847	7.75	<.0001	1

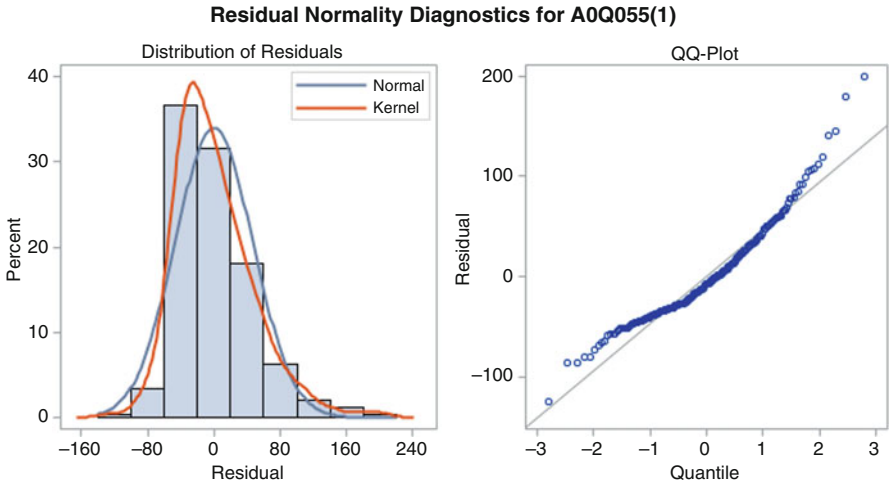
Source: SAS

Correlations of parameter estimates		
Parameter	**MU**	**AR1,1**
MU	1.000	0.013
AR1,1	0.013	1.000

Autocorrelation check of residuals									
To lag	**Chi-square**	**DF**	**Pr > ChiSq**	**Autocorrelations**					
6	19.17	5	0.0018	−0.106	0.164	0.060	0.145	0.077	0.100
12	22.51	11	0.0207	0.044	−0.022	0.053	0.050	−0.074	−0.013
18	26.67	17	0.0632	0.079	0.013	−0.001	0.077	−0.062	0.004
24	33.83	23	0.0676	0.064	0.034	0.047	0.065	−0.060	0.109
30	35.11	29	0.2009	0.025	0.021	−0.019	0.033	0.018	−0.043
36	38.98	35	0.2954	−0.009	−0.057	0.088	0.010	−0.044	−0.025
42	45.11	41	0.3041	0.025	0.104	−0.047	0.087	−0.000	0.010

The ARIMA(1,1,0) model residuals are not white noise because the chi-square values and probabilities indicate severe non-randomness.





Let us estimate an ARIMA(1,1,1) model for US real GDP, 1959–2018Q2.

The ARIMA(1,1,1) model residuals are white noise because the chi-square values and probabilities indicate randomness. Thus, the ARIMA(1,1,1) model is an adequate time series model of US real GDP, 1959–2018Q2 (see Table 6).

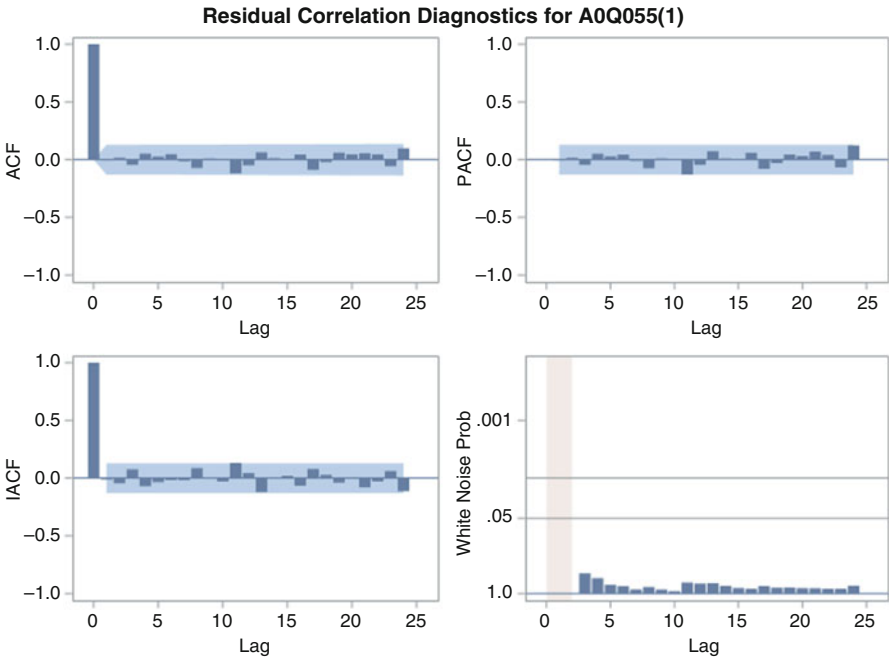


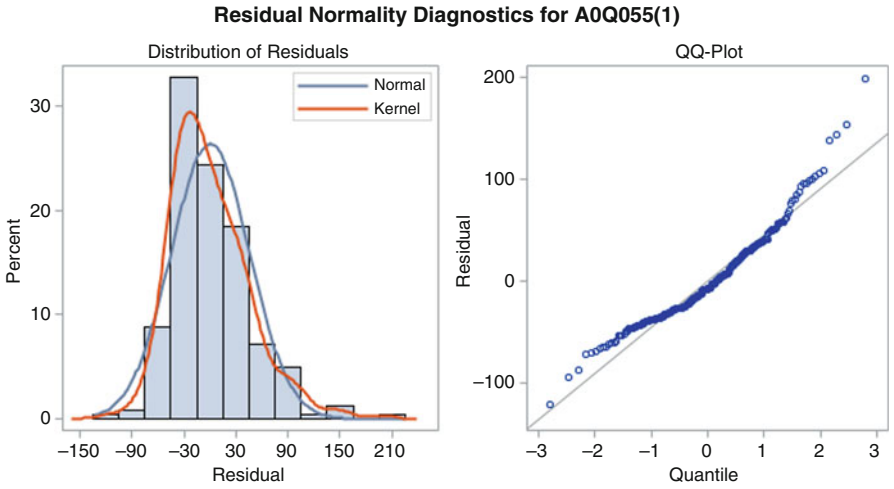
Table 6 ARIMA model identification and estimation of US real GDP, an ARIMA(1,1,1) estimation, 1959–2018Q2

Autoregressive factors					
1 − 0.45289 B**(1)					
Conditional least squares estimation					
Parameter	**Estimate**	**Standarderror**	**t-value**	**ApproxPr > |t|**	**Lag**
MU	65.63017	9.02460	7.27	<.0001	0
MA1,1	0.53013	0.09712	5.46	<.0001	1
AR1,1	0.85091	0.06171	13.79	<.0001	1
Constant estimate					9.784723
Variance estimate					2076.026
Std error estimate					45.56343
AIC					2496.29
SBC					2506.707
Number of residuals					238
The AR(1) and MA(1) terms are both highly statistically significant, with t-statistics exceeding 5					
*** AIC and SBC do not include log determinant**					

Correlations of parameter estimates			
Parameter	**MU**	**MA1,1**	**AR1,1**
MU	1.000	0.035	0.046
MA1,1	0.035	1.000	0.819
AR1,1	0.046	0.819	1.000

Autocorrelation check of residuals									
To lag	**Chi-square**	**DF**	**Pr > ChiSq**	**Autocorrelations**					
6	1.94	4	0.7470	−0.006	0.016	−0.045	0.052	0.026	0.047
12	7.55	10	0.6728	−0.014	−0.073	0.010	0.004	−0.120	−0.050
18	11.26	16	0.7933	0.063	0.013	−0.002	0.045	−0.088	−0.022
24	17.53	22	0.7332	0.060	0.046	0.055	0.045	−0.057	0.098
30	19.35	28	0.8869	0.036	0.005	−0.030	0.028	0.012	−0.059
36	25.59	34	0.8498	−0.044	−0.056	0.089	0.012	−0.075	−0.059
42	29.87	40	0.8788	0.022	0.096	−0.037	0.061	−0.012	0.001

Source: SAS



Model for variable A0Q055

The ARIMA(0,1,2) model is characterized by two highly statistically significant MA terms (see Table 7). The ARIMA(0,1,2) model residuals are white noise because the chi-square values and probabilities indicate randomness. Thus, the ARIMA (0,0,2) model is an adequate time series model of US real GDP, 1959–2018Q2.[13] The time series procedures in SAS reflect the research of Tsay (1988) and outlier analysis is reported in Table 8.

[13]A similar answer with R is reported in Guerard and Thomakos, *Financial Forecasting in the Chinese Stock Market*, Peking University Press, forthcoming.

Table 7 ARIMA model identification and estimation of US real GDP, an ARIMA(0,1,2) estimation, 1959–2018Q2

Estimated mean	65.63017
Period(s) of differencing	1
Autoregressive factors	
Factor 1:	1 − 0.85091 B**(1)
Moving average factors	
Factor 1:	1 − 0.53013 B**(1)

Conditional least squares estimation

Parameter	Estimate	Standarderror	t-value	ApproxPr > |t|	Lag
MU	65.56955	4.87686	13.45	<.0001	0
MA1,1	−0.34381	0.06325	−5.44	<.0001	1
MA1,2	−0.25024	0.06385	−3.92	0.0001	2

Constant estimate	65.56955
Variance estimate	2241.236
Std error estimate	47.34169
AIC	2514.514
SBC	2524.931
Number of residuals	238

*** AIC and SBC do not include log determinant**

Correlations of parameter estimates

Parameter	MU	MA1,1	MA1,2
MU	1.000	−0.000	−0.008
MA1,1	−0.000	1.000	0.276
MA1,2	−0.008	0.276	1.000

Autocorrelation check of residuals

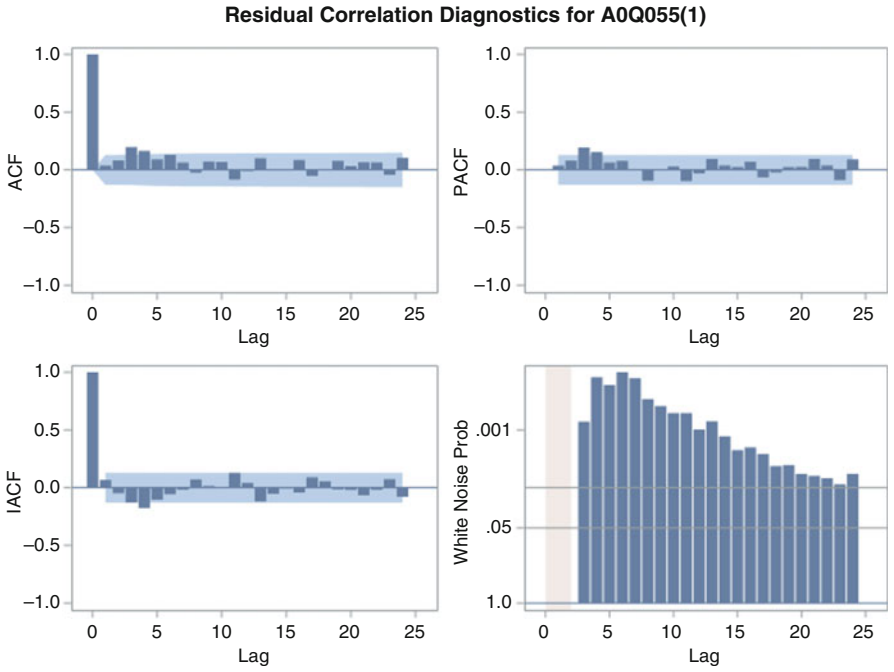
To lag	Chi-square	DF	Pr > ChiSq	Autocorrelations					
6	24.23	4	<.0001	0.037	0.082	0.198	0.164	0.091	0.130
12	29.64	10	0.0010	0.063	−0.025	0.072	0.068	−0.084	−0.011
18	34.81	16	0.0042	0.100	0.004	0.005	0.086	−0.053	0.002
24	42.30	22	0.0058	0.078	0.033	0.066	0.064	−0.044	0.104
30	43.93	28	0.0282	0.054	0.009	−0.011	0.039	0.015	−0.033
36	47.99	34	0.0564	−0.022	−0.051	0.084	0.027	−0.053	−0.031
42	54.15	40	0.0670	0.055	0.094	−0.033	0.090	0.017	0.006

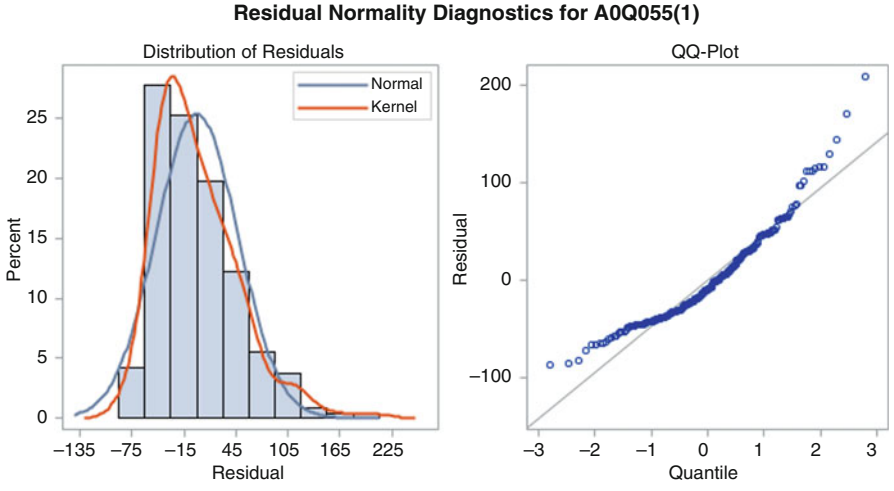
Source: SAS

Table 8 Outlier analysis of ARIMA(0,1,2) real GDP estimation

Model for variable A0Q055 (real GDP)				
Period(s) of differencing				1
Moving average factors				
Factor 1:				1 + 0.34381 B**(1) + 0.25024 B**(2)
Outlier detection summary				
Maximum number searched				5
Number found				5
Significance used				0.05
Outlier details				
Obs	**Type**	**Estimate**	**Chi-square**	**Approx prob>ChiSq**
78	Shift	203.43897	20.45	<.0001
165	Additive	−115.00980	17.24	<.0001
211	Additive	−77.16038	7.91	0.0049
217	Additive	76.97682	8.00	0.0047
179	Shift	122.57778	8.10	0.0044
Forecasts for variable A0Q055				
Obs	**Forecast**	**Std error**	**95% confidence limits**	
240	18,773.6252	47.3417	18,680.8372	18,866.4132
241	18,850.1466	79.3001	18,694.7213	19,005.5719
242	18,915.7161	109.4691	18,701.1607	19,130.2716
243	18,981.2857	132.9604	18,720.6881	19,241.8833
244	19,046.8553	152.8838	18,747.2086	19,346.5019
245	19,112.4248	170.4947	18,778.2614	19,446.5882
246	19,177.9944	186.4495	18,812.5601	19,543.4286
247	19,243.5639	201.1427	18,849.3314	19,637.7964
248	19,309.1335	214.8334	18,888.0678	19,730.1991
249	19,374.7030	227.7023	18,928.4146	19,820.9914
250	19,440.2726	239.8819	18,970.1126	19,910.4325
251	19,505.8421	251.4723	19,012.9655	19,998.7188

Source: SAS





The ARIMA(0, 1,2) model has five outliers that may be very useful to address (see Table 8).

Correlations of parameter estimates					
Variable parameter	**DLRGDPMA1,1**	**DLRGDPMA1,2**	**DLLEINUM1**	**DLLEIDEN1,1**	**DLLEIDEN1,2**
DLRGDPMA1,1	1.000	0.229	0.066	0.004	−0.008
DLRGDPMA1,2	0.229	1.000	−0.025	0.014	−0.012
DLLEINUM1	0.066	−0.025	1.000	0.283	−0.334
DLLEIDEN1,1	0.004	0.014	0.283	1.000	−0.998
DLLEIDEN1,2	−0.008	−0.012	−0.334	−0.998	1.000

Autocorrelation check of residuals									
To lag	**Chi-square**	**DF**	**Pr > ChiSq**	**Autocorrelations**					
6	10.44	4	0.0337	0.009	0.047	0.063	0.154	−0.067	0.095
12	19.30	10	0.0367	−0.027	−0.051	0.010	0.017	−0.160	−0.080
18	26.36	16	0.0492	−0.020	−0.060	−0.027	0.083	−0.128	−0.001
24	35.29	22	0.0361	−0.033	0.001	0.018	0.055	−0.132	0.111
30	37.48	28	0.1087	0.041	−0.004	−0.016	0.002	0.011	-0.078
36	43.70	34	0.1231	0.004	−0.099	0.012	−0.020	−0.097	−0.052
42	49.46	40	0.1452	0.074	0.001	−0.104	0.045	0.041	0.017

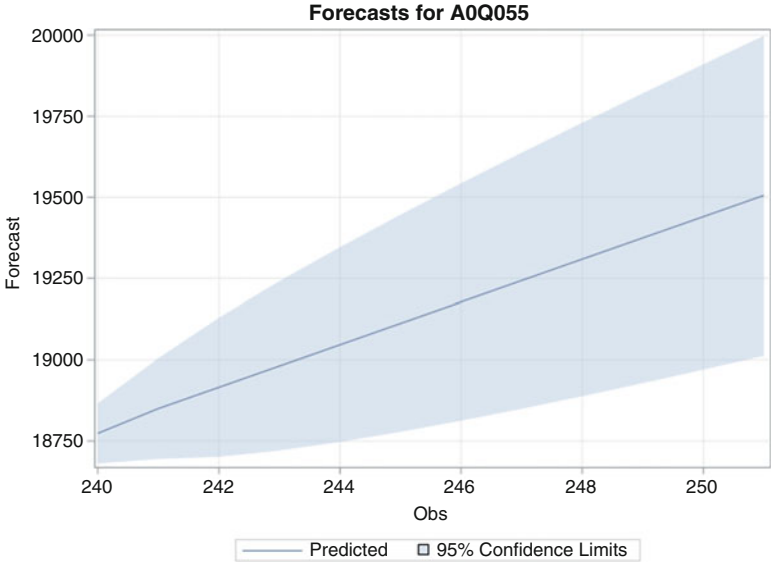


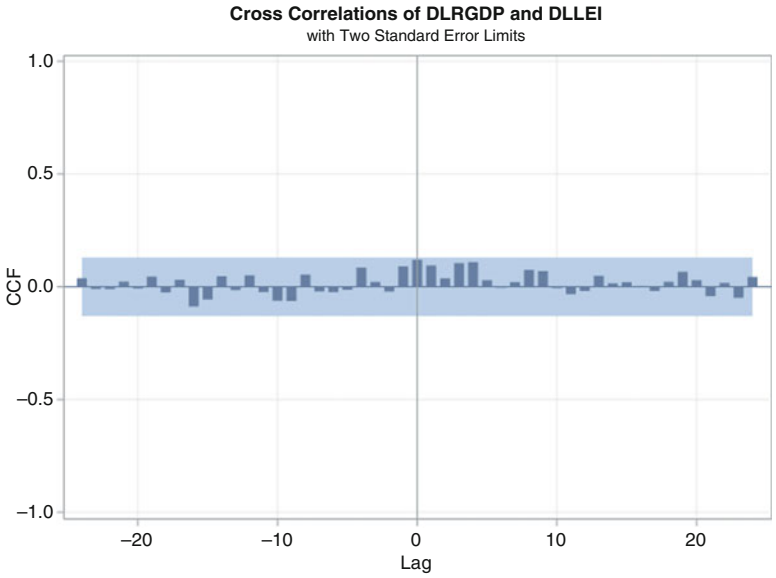
Table 9 ARIMA transfer function model estimation of US real GDP, an ARIMA(0,1,2) estimation, with LEI input, 1959–2018Q2

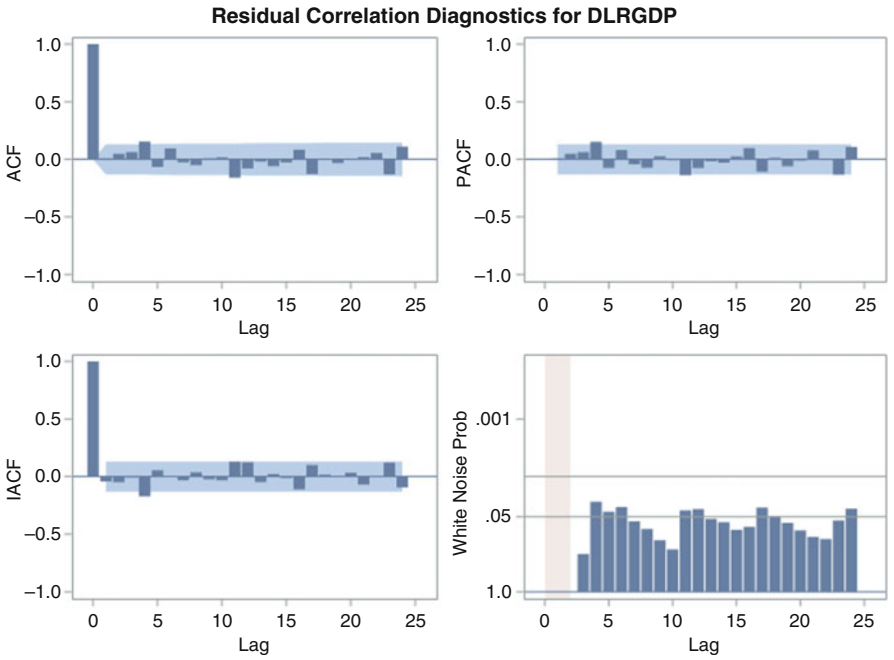
Conditional least squares estimation							
Parameter	**Estimate**	**Standarderror**	**t-value**	**ApproxPr > |t|**	**Lag**	**Variable**	**Shift**
MA1,1	−0.28626	0.06445	−4.44	<.0001	1	DLRGDP	0
MA1,2	−0.23133	0.06445	−3.59	0.0004	2	DLRGDP	0
NUM1	0.0053405	0.0019890	2.69	0.0078	0	DLLEI	0
DEN1,1	0.10224	0.07365	1.39	0.1664	1	DLLEI	0
DEN1,2	0.88204	0.07473	11.80	<.0001	4	DLLEI	0
Variance estimate							5.672E − 6
Std error estimate							0.002382
AIC							−2157.71
SBC							−2140.43
Number of residuals							234

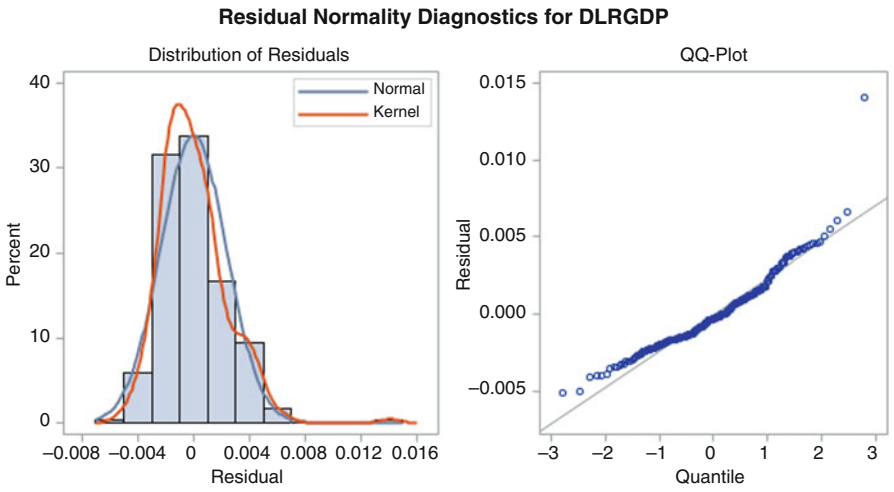
Source: SAS

Dhrymes (2017) and Guerard Jr. (2004) both estimated transfer function in which the US LEI was useful as an input to modeling real US GDP. We continue on that modeling effort. Let us estimate a transfer function of US real GDP with the US LEI as an input.

A transfer function estimation requires an estimated cross-correlogram (see Ashley, Granger, & Schmalensee, 1980; Granger & Newbold, 1977). The cross-correlation function identifies the possibility of lags 1 and 4 of the US LEI being significant in modeling real GDP.




Residual Correlation Diagnostics for DLRGDP
ACF
PACF
IACF
White Noise Prob
Lag
1.0
0.5
0.0
−0.5
−1.0
.001
.05
0
5
10
15
20
25


Residual Normality Diagnostics for DLRGDP
Distribution of Residuals
QQ-Plot
Normal
Kernel
Percent
Residual
Quantile
40
30
20
10
0
0.015
0.010
0.005
0.000
−0.005
−0.008
−0.004
0.004
0.008
0.012
0.016
−3
−2
−1
1
2
3

Model for variable DLRGDP	
Moving average factors	
Factor 1:	1 + 0.28626 B**(1) + 0.23133 B**(2)

Input number 1	
Input variable	DLLEI
Overall regression factor	0.005341

Denominator factors	
Factor 1:	1 − 0.10224 B**(1) − 0.88204 B**(4)

Outlier detection summary	
Maximum number searched	5
Number found	5
Significance used	0.05

Outlier details

Obs	Type	Estimate	Chi-square	Approx prob>ChiSq
78	Additive	0.01358	42.04	<.0001
69	Additive	0.0063587	9.51	0.0020
25	Additive	0.0060233	9.26	0.0023
166	Additive	0.0054298	7.52	0.0061
29	Additive	0.0048940	6.31	0.0120

Source: SAS

The SAS-estimated US real GDP transfer function model has outlier issues, and the residuals are not white noise. The updated analysis of real GDP during the 1959–2018 time period reports a statistically significant transfer function model and outliers. Let us turn to Autometrics to expand upon time series breaks in the next section and the SCA system to test for causality in US real GDP and the LEI.

Forecasting Effectiveness of Time Series Modeling Using Autometrics and SCA Model the US Unemployment Rate, 1959Q2–2018Q3

In this section, we report on monthly time series modeling and forecasting using the US leading economic indicator (LEI) as an input to forecast the US unemployment rate, 1959Q2 to 2018Q3. Montgomery, Zarnowitz, Tsay, and Tiao (MZTT, 1998) modeled the quarterly unemployment rate for the 1948–1993 period and reached several very interesting conclusions. Among the conclusions, the unemployment rate

contained no consistent trend, and in times of rising unemployment, the weekly unemployment insurance claims were a useful input–however unemployment claims were not useful over the entire 1948–1993 time period. MZTT suggested that although future models could build upon asymmetric modeling analysis, long-run models had to forecast stable, slowing declining periods of unemployment. These time series have been addressed before, but our results are more statistically significant using more recently developed time series modeling techniques and software.

We employ the automatic time series modeling and forecasting of Hendry and Doornik (2014) and Doornik and Hendry (2015) with its emphasis on saturation variables which is very relevant for modeling the MZTT unemployment rate data. Montgomery et al. (MZTT, 1998) modeled the US unemployment rate as a function of the weekly unemployment claims time series, 1948–1993. We use The Conference Board (TCB) database, 1959–2018, to retrieve the US unemployment rate, TCBUER, The Conference Board Index of Leading Economic Indicators, LEI or TCBLEI, and weekly unemployment claims data WkUNCL. A similar conclusion is found for the impact of the LEI and weekly unemployment claims series leading the unemployment rate series. We report statistically significant saturation variables in the US unemployment rate time series, 1959–1993 and 1959 to the present. We apply the Tsay (1988, 2010) methodology within the SCA system to address causality testing between LEI and the US unemployment rate and weekly unemployment claims and the US unemployment rate.

Automatic time series models have recently been discussed in Hendry (1986, 2000), Krolzig and Hendry (2001), Hendry and Krolzig (2005), Hendry and Nielsen (2007), Castle, Doornik, and Hendry (2013), and Hendry and Doornik (2014) and implemented in the Autometrics software.[14] Hendry sets the tone for automatic modeling by contrasting how statistically—based his PC-Give and Autometrics work in contrast to the "data mining" and "garbage in, garbage out" routines, citing their forecasting efficiency and performance. Automatic model selection is discussed by stating that the economic data generation process (DGP) is the joint density of all the variables in the economy, which is very complex, is evolving over time, and is of high dimensionality. Computer implementation of the automatic modeling procedure (1) forces previse specification of the process, (2) allows operational studies of alternative strategies, and (3) creates a tool, Autometrics, for empirical modeling. Because economists only analyze a small subset of possible variables, Hendry and his colleagues seek to reduce the analysis to a manageable size such that the local (LDGP) is the DGP of the selected variables. Empirical models of the LDGP satisfy

[14] Automatic time series modeling has advocated since the early days of Box and Jenkins (1970). Reilly (1980), with the Autobox system, pioneered early automatic time series model implementation. Tsay (1988) identified outliers, level shifts, and variance change models that were implemented in PC-SCA. SCA was used in modeling time series in MZTT (1998). Guerard (1990) used both Autobox and SCA systems to model the monthly NYSE stock volume series, January 1965 to December 1987. SCA correctly identified the introduction of the DOT (trading) system as a level shift in March 1976 and the market crash of October 1987 as an additive outlier. Autobox identified October 1987 as a pulse variable.

several criteria: a model is congruent when it matches the available evidence in all dimensions examined, such as homoscedastic errors and constant parameters; and encompassing, a model represents the set of feasible variable reductions and alternative models contain no additional information about the variables being modeled.

If one starts with a large number of predictors, or candidate explanatory variables, say n, then the general model can be written as

$$y_t = \sum_{i=1}^{n} \gamma_i Z_{it} + u_t \tag{16}$$

while the (conditional) data generating processes is assumed to be given by

$$y_t = \sum_{i=1}^{n} \beta_i Z_{(i),t} + \epsilon_t \tag{17}$$

where $\epsilon_t \cong IN\left(0, \sigma_e^2\right)$ for any $n < N$.

One must select the relevant regressors where $\beta_j \neq 0$ in (5). Hendry and his colleagues refer to Eq. (16) as the most general, statistical model that can be postulated, given the availability of data and previous empirical and theoretical research as the general unrestricted model (GUM). The Hendry general-to-specific modeling process is referred to as *Gets*. One seeks to identify all relevant variables, the relevant lag structure and cointegrating relations, forming near-orthogonal variables, Z. The general unrestricted model, GUM, with s lags of all variables can then be written as

$$y_t = \sum_{i=1}^{n} \sum_{j=0}^{s} \beta_{i,j} x_{i,t-j} + \sum_{i=1}^{n} \sum_{j=0}^{s} k_{\mathrm{i},j} z_{i,t-j} + \sum_{j=1}^{s} \theta_j y_{t-j} + \sum_{i=1}^{T} \delta_i 1_{\{i=t\}} + e_t \tag{18}$$

where $\varepsilon_t \tilde{\mathrm{I}}\mathrm{N}\left(0, \sigma_e^2\right)$. Furthermore, outliers and shifts for T observations can be modeled with saturation variables (see Doornik & Hendry, 2015; Hendry & Doornik, 2014, Chapters 7 and 14).

Automatic modeling seeks to eliminate irrelevant variables, variables with insignificant estimated coefficients, lag-length reductions, the nonlinearity of the principal components, and combinations of "small effects" represented by principal components and reduce saturation variables (for each observation).[15] One can consider the orthogonal regressor case in which one ranks the variables by their *t*-statistics, highest to lowest, and defines m to be the smallest, but statistically significant t-statistic, t_m^2, and discards all variables with *t*-statistics below the m

[15]Doornik and Hendry (2013a, 2013b) remind the reader that the data generation process (DGP) is impossible to model, and the best solution that one can achieve is estimate the models to reflect the local DGP, through reduction, described above. The Automatic *Gets* algorithm reduces GUM to nest LGDP, the locally relevant variables. Congruency, in which the LGDP has the same shape and size as the GUM, or models reflect the local DGP. See Hendry and Krolzig (2005) for original Gets modeling.

largest *t*-values. One must be reminded that every test statistic has a distribution occurring in different samples by the specification of Eqs. (16) and (17). We seek to select a model of the form

$$y_t = \sum_{r=1}^{m} \delta_r Z_{\{r\},t} + n_t \tag{19}$$

where $Z_{\{r\},\,t}$ is a subset of the initial N variables and that model may differ from either the one postulated in (16), (17), (18), or (19) depending on which variables remain at the end of the selection process.[16] One progresses from the general unrestricted model to the "final" model in (19) by establishing that model residuals are approximately normal, homoscedastic, and independent. Model reduction proceeds by tree searches of insignificant variables. The last, non-rejected model is referred to as the terminal equation. Selected model regressors have coefficients that are largely relative to their estimated standard errors; since the estimators obtained by the initial model (17) are unbiased, the selected estimators are upward biased conditional on retaining $Z_{(j),\,t}$. The unselected variables will have downward biased estimators. By omitting irrelevant variables, the selection model does not "overfit" the model, and the relevant (retained) variables have estimated standard errors close to those from fitting equation (19).

The automatic time series modeling program (PCGets) or (Autometrics) is efficient, but Hendry and Nielsen (2007) state that the largest selection bias can arise from strongly correlated regressors. Autometrics deals with outliers and breaks in its automatic time series modeling. The regression sum of squares, RSS, rises as the outlier criteria shrinks. Autometrics can apply impulse-indicator saturation variables IIS, step-indicator saturation variables (SIS), differenced IIS (DIIS), and trend saturation (TIS) to all marginal models, where there are significant indicators. The impulse indicator saturation variables are dummy variables for each observation. The step-indicator saturation (SIS) variables are generalized IIS variables with higher statistical power to detect location shifts; that is, SIS variables are step dummy variables. One can include outlier detection indicators (impulse-indicator saturation, IIS, and step-indicator saturation, SIS, variables) in the Autometrics analysis of the LEI component effectiveness estimates.[17]

We estimate GUM to Gets models for the US unemployment rate, 1959Q2 to 2018Q2, using *Autometrics*. We start with an AR(1) model, estimate saturation variables and models, and estimate transfer function models with LEI, and weekly unemployment claims, a component of the LEI, both time series reported monthly. We employ Autometrics on the raw, not differenced, data, on the unemployment rate

[16]In the selection process, one tests the null hypothesis that the parameter in front of a variable is zero. The relevant *t*-statistic from a two-sided test is used.

[17]The use of saturation variables avoids the issue of forcing a unit root to capture the shifts, leading to an upward biased estimate of the lagged dependent variable coefficient. The authors are indebted to Jenny Castle for her comments on the application of saturation variables, which she observes addresses this very well.

Table 10 Modeling TCBUER (the US unemployment rate) by OLS input is TCBLEI, The Conference Board leading economic indicator (LEI) time series, reported monthly

The dataset is C:\OxMetrics\USMZTT.csv					
The estimation sample is 13-716					
	Coefficient	Std. error	t-value	t-prob	Part.R^2
TCBUER_1	0.995288	0.004251	234.	0.0000	0.9874
Constant	0.0373680	0.02785	1.34	0.1801	0.0026
Trend	−3.01533e − 05	3.319e − 05	−0.908	0.3639	0.0012

Sigma	0.177814	RSS	22.1640691
R^2	0.987525	F(2,701) =	2.775e + 04 [0.000]**
Adj.R^2	0.987489	Log likelihood	218.391
no. of observations	704	No. of parameters	3
Mean(TCBUER)	6.01733	Se(TCBUER)	1.58975

AR 1-2 test	F(2,699) = 29.548 [0.0000]**
ARCH 1-1 test	F(1,702) = 50.651 [0.0000]**
Normality test	Chi^2(2) = 50.406 [0.0000]**
Hetero test	F(4,699) = 8.9991 [0.0000]**
Hetero-X test	F(5,698) = 7.8361 [0.0000]**
RESET23 test	F(2,699) = 0.11436 [0.8920]

series. We collect our results on Tables 10, 11, 12, 13, 14, 15, 16, and 17. In the table, we have four models, the plain AR(1) model, an AR(1) model with saturation variables only, and then the AR(1) model with saturation variables plus the weekly claims and the LEI components (separately and together). We report the search results of Autometrics in the tables and we can see immediately some very interesting results. First, note that a simple autoregression does not capture the cyclical variability of the level of unemployment as expected, with a low estimate of persistence and a relatively low R^2. The RSS is large, and there is autocorrelation, heteroscedasticity, and non-normality in the residuals. The reader is referred to Table 10.

If one adds the monthly US weekly unemployment claims, WkUNCL, to the basis US unemployment rate model, the weekly unemployment claims variable, with the one-period lag, is statistically significant with the expected positive coefficient. That is, the higher the WkUNCL, the higher should be the future US unemployment rate (see Table 11).

If one adds the US LEI to the basis US unemployment rate model, The Conference Board LEI variable, with the one-period lag, is statistically significant with the expected negative coefficient. That is, the higher the LEI, the lower should be the future US unemployment rate (see Table 12).

Table 11 Modeling TCBUER by OLS input is WKUNCL, The Conference Board weekly unemployment claims (WkUNCL) time series, reported monthly

The dataset is C:\OxMetrics\USMZTT.csv					
The estimation sample is 2–716					
	Coefficient	Std. error	t-value	t-prob	Part.R^2
TCBUER_1	0.925612	0.006266	148.	0.0000	0.9684
WKUNCL_1	0.00162429	0.0001165	13.9	0.0000	0.2147
Constant	−0.0780523	0.02592	−3.01	0.0027	0.0126
Trend	−0.000105084	2.963e − 05	−3.55	0.0004	0.0174

Sigma	0.158907	RSS	17.9538104
R^2	0.989924	F(3,711) =	2.328e + 04 [0.000]**
Adj.R^2	0.989881	Log likelihood	302.661
No. of observations	715	No. of parameters	4
Mean(TCBUER)	6.00783	Se(TCBUER)	1.57972

AR 1-2 test	F(2,709) = 8.2189 [0.0003]**
ARCH 1-1 test	F(1,713) = 15.909 [0.0001]**
Normality test	Chi^2(2) = 23.768 [0.0000]**
Hetero test	F(6,708) = 10.323 [0.0000]**
Hetero-X test	F(9,705) = 9.6441 [0.0000]**
RESET23 test	F(2,709) = 3.1805 [0.0422]*

Model saved to C:\OxMetrics\Model AR1 WkUnCL OLS.pdf
Model saved to C:\OxMetrics\Model AR1 WkUnCL OLS.pdf.gwg

Table 12 Modeling TCBUER (the US unemployment rate) by OLS input is TCBLEI, The Conference Board leading economic indicator (LEI) time series, reported monthly

The dataset is C:\OxMetrics\USMZTT.csv					
The estimation sample is 2-716					
	Coefficient	Std. error	t-value	t-prob	Part.R^2
TCBUER_1	0.953773	0.007205	132.	0.0000	0.9610
Constant	0.665178	0.09394	7.08	0.0000	0.0659
Trend	0.00131122	0.0001924	6.81	0.0000	0.0613
TCBLEI_1	−0.0128775	0.001835	−7.02	0.0000	0.0647

Sigma	0.17342	RSS	21.3829931
R^2	0.987999	F(3,711) =	1.951e + 04 [0.000]**
Adj.R^2	0.987949	Log likelihood	240.172
No. of observations	715	No. of parameters	4
Mean(TCBUER)	6.00783	Se(TCBUER)	1.57972

AR 1-2 test	F(2,709) = 27.425 [0.0000]**
ARCH 1-1 test	F(1,713) = 39.758 [0.0000]**
Normality test	Chi^2(2) = 51.144 [0.0000]**
Hetero test	F(6,708) = 10.181 [0.0000]**
Hetero-X test	F(9,705) = 7.0422 [0.0000]**
RESET23 test	F(2,709) = 0.094305 [0.9100]

Table 13 Modeling TCBIUER by OLS, input is the WkUNCL and estimation of impulse-indicator and step-indicator saturation (IIS and SIS) variables

The dataset is C:\OxMetrics\USMZTT.csv					
The estimation sample is 2-716					
	Coefficient	Std. error	t-value	t-prob	Part.R^2
TCBUER_1	0.881390	0.008094	109.	0.0000	0.9518
WKUNCL_1	0.00183639	0.0001438	12.8	0.0000	0.2138
I:2	0.240813	0.1292	1.86	0.0628	0.0058
I:12	−0.480995	0.1493	−3.22	0.0013	0.0170
I:15	0.818444	0.1218	6.72	0.0000	0.0700
I:22	0.601527	0.1218	4.94	0.0000	0.0391
I:24	0.510623	0.1218	4.19	0.0000	0.0285
I:35	−0.212370	0.1103	−1.93	0.0546	0.0061
I:44	0.359363	0.1102	3.26	0.0012	0.0174
I:51	−0.321908	0.1179	−2.73	0.0065	0.0123
I:67	−0.275309	0.1089	−2.53	0.0117	0.0105
I:75	−0.307656	0.1069	−2.88	0.0041	0.0136
I:143	0.230736	0.1084	2.13	0.0337	0.0075
I:145	−0.252188	0.1085	−2.32	0.0204	0.0089
I:170	0.311588	0.1218	2.56	0.0108	0.0108
I:189	0.297152	0.1084	2.74	0.0063	0.0124
I:217	−0.291614	0.1120	−2.60	0.0094	0.0112
I:225	−0.221523	0.1218	−1.82	0.0694	0.0055
I:245	−0.316518	0.1088	−2.91	0.0037	0.0139
I:253	0.178556	0.1088	1.64	0.1014	0.0045
I:271	−0.465201	0.1085	−4.29	0.0000	0.0297
I:295	−0.603408	0.1140	−5.29	0.0000	0.0446
I:308	0.254724	0.1492	1.71	0.0883	0.0048
I:311	−0.271060	0.1112	−2.44	0.0151	0.0098
I:326	0.556277	0.1071	5.19	0.0000	0.0430
I:352	−0.250130	0.1072	−2.33	0.0199	0.0090
I:388	−0.289402	0.1073	−2.70	0.0072	0.0120
I:406	−0.316579	0.1099	−2.88	0.0041	0.0136
I:650	−0.215791	0.1069	−2.02	0.0440	0.0067
I:659	−0.310312	0.1070	−2.90	0.0039	0.0138
I:664	−0.287188	0.1075	−2.67	0.0078	0.0117
I:669	−0.208480	0.1074	−1.94	0.0528	0.0062
I:671	0.165637	0.1074	1.54	0.1234	0.0040
I:673	0.155429	0.1073	1.45	0.1481	0.0035
I:689	−0.279744	0.1073	−2.61	0.0094	0.0112
I:695	−0.279118	0.1073	−2.60	0.0095	0.0111
I:714	0.236613	0.1070	2.21	0.0273	0.0081
S1:4	−0.226396	0.1055	−2.15	0.0323	0.0076
S1:6	−0.215630	0.08832	−2.44	0.0149	0.0098

(continued)

Table 13 (continued)

S1:11	0.253205	0.1155	2.19	0.0288	0.0079
S1:13	0.169301	0.1218	1.39	0.1651	0.0032
S1:17	−0.349326	0.08621	−4.05	0.0001	0.0266
S1:20	0.198498	0.08619	2.30	0.0216	0.0088
S1:25	−0.354503	0.1218	−2.91	0.0037	0.0139
S1:26	0.243768	0.1180	2.07	0.0392	0.0071
S1:30	−0.189167	0.1181	−1.60	0.1098	0.0043
S1:31	0.515134	0.1491	3.45	0.0006	0.0195
S1:32	−0.512973	0.1492	−3.44	0.0006	0.0193
S1:33	0.278189	0.1103	2.52	0.0119	0.0105
S1:46	−0.356975	0.1102	−3.24	0.0013	0.0172
S1:47	0.454868	0.1492	3.05	0.0024	0.0153
S1:48	−0.342325	0.1179	−2.90	0.0038	0.0139
S1:53	0.397301	0.1180	3.37	0.0008	0.0186
S1:54	−0.249761	0.1090	−2.29	0.0222	0.0087
S1:70	0.273581	0.1089	2.51	0.0123	0.0104
S1:71	−0.209676	0.1069	−1.96	0.0502	0.0064
S1:113	−0.201732	0.1068	−1.89	0.0593	0.0059
S1:114	0.206128	0.1083	1.90	0.0576	0.0060
S1:132	−0.168005	0.03599	−4.67	0.0000	0.0350
S1:152	0.227309	0.07873	2.89	0.0040	0.0137
S1:154	−0.174916	0.08066	−2.17	0.0305	0.0078
S1:166	0.214517	0.06810	3.15	0.0017	0.0163
S1:170	−0.176070	0.06582	−2.68	0.0077	0.0118
S1:190	−0.417491	0.07999	−5.22	0.0000	0.0434
S1:192	−0.248361	0.1294	−1.92	0.0554	0.0061
S1:193	0.861014	0.1492	5.77	0.0000	0.0526
S1:194	−0.395251	0.1220	−3.24	0.0013	0.0172
S1:197	0.364395	0.08668	4.20	0.0000	0.0286
S1:200	−0.185947	0.07185	−2.59	0.0099	0.0110
S1:208	0.216619	0.1119	1.94	0.0533	0.0062
S1:209	−0.309535	0.1119	−2.77	0.0059	0.0126
S1:218	0.284662	0.07141	3.99	0.0001	0.0258
S1:221	−0.376857	0.1218	−3.09	0.0021	0.0157
S1:222	0.435510	0.1492	2.92	0.0036	0.0140
S1:223	−0.280722	0.1218	−2.30	0.0215	0.0088
S1:227	0.420086	0.1218	3.45	0.0006	0.0194
S1:228	−0.262327	0.1140	−2.30	0.0217	0.0088
S1:234	−0.367101	0.1139	−3.22	0.0013	0.0170
S1:235	0.596186	0.1491	4.00	0.0001	0.0259
S1:236	−0.250443	0.1086	−2.31	0.0215	0.0088
S1:255	−0.426501	0.08010	−5.32	0.0000	0.0451
S1:257	0.620075	0.09154	6.77	0.0000	0.0710

(continued)

Table 13 (continued)

S1:261	−0.206983	0.09282	−2.23	0.0261	0.0082
S1:263	0.219842	0.1293	1.70	0.0896	0.0048
S1:264	−0.566253	0.1492	−3.80	0.0002	0.0235
S1:265	0.384812	0.1218	3.16	0.0017	0.0164
S1:268	−0.243726	0.06630	−3.68	0.0003	0.0220
S1:288	0.330798	0.1098	3.01	0.0027	0.0149
S1:289	−0.372434	0.1140	−3.27	0.0011	0.0175
S1:296	0.239001	0.05506	4.34	0.0000	0.0304
S1:306	−0.424210	0.1108	−3.83	0.0001	0.0238
S1:307	0.514367	0.1492	3.45	0.0006	0.0194
S1:309	−0.205576	0.1112	−1.85	0.0649	0.0057
S1:319	0.114504	0.03990	2.87	0.0043	0.0135
S1:353	0.176866	0.1072	1.65	0.0994	0.0045
S1:354	−0.195481	0.1127	−1.73	0.0835	0.0050
S1:361	0.207926	0.08456	2.46	0.0142	0.0100
S1:363	−0.200406	0.07684	−2.61	0.0093	0.0112
S1:405	−0.0851370	0.03715	−2.29	0.0223	0.0087
S1:418	0.140162	0.04035	3.47	0.0006	0.0197
S1:435	−0.397064	0.1087	−3.65	0.0003	0.0218
S1:436	0.462275	0.1092	4.23	0.0000	0.0290
S1:450	−0.243987	0.1092	−2.23	0.0258	0.0083
S1:451	0.565076	0.1492	3.79	0.0002	0.0234
S1:452	−0.337734	0.1082	−3.12	0.0019	0.0160
S1:471	0.392589	0.1082	3.63	0.0003	0.0215
S1:472	−0.364029	0.1078	−3.38	0.0008	0.0186
S1:494	0.130341	0.07791	1.67	0.0948	0.0046
S1:496	−0.150370	0.07556	−1.99	0.0470	0.0066
S1:592	−0.259811	0.02654	−9.79	0.0000	0.1377
S1:622	−0.470829	0.1083	−4.35	0.0000	0.0306
S1:623	0.814543	0.1493	5.46	0.0000	0.0473
S1:624	−0.338369	0.1072	−3.16	0.0017	0.0163
S1:663	0.129909	0.02959	4.39	0.0000	0.0311
S1:697	0.0878047	0.03083	2.85	0.0046	0.0133

Sigma	0.105457	RSS	6.67275007
Log likelihood	656.503		
No. of observations	715	No. of parameters	115
Mean(TCBUER)	6.00783	Se(TCBUER)	1.57972

AR 1-2 test	F(2,598) = 4.5368 [0.0111]*
ARCH 1-1 test	F(1,713) = 1.6353 [0.2014]
Normality test	Chi^2(2) = 0.98028 [0.6125]
Hetero test	F(52,597) = 0.55210 [0.9956]
RESET23 test	F(2,598) = 0.0093440 [0.9907]

Model saved to C:\OxMetrics\Model UE AR1 WkUnCL IIS SIS.pdf
Model saved to C:\OxMetrics\Model UE AR1 WkUnCL IIS SIS.pdf.gwg

Table 14 OxMetrics saturation variable summary of the modeling of the US unemployment rate with the weekly unemployment claims data time period: 1959Q2–2018Q2

Model	Regression sum of squares, RSS	Adjusted R-squared	Sigma
AR1	22.863	0.987	0.179
AR1 with WkUNCL	17.953	0.989	0.159
AR1 with WkUNCL and IIS variables	14.337	0.992	0.144
AR1 with WkUNCL and SIS variables	10.044	0.992	0.123
AR1 with WkUNCL and IIS, SIS variables	6.673	0.999	0.105
AR1 with WkUNCL and DIIS variables	14.762	0.992	0.142
AR1 with WkUNCL and TIIS variables	12.659	0.999	0.136
AR1 with WkUNCL and IIS, SIS, DIIS, TIIS variables	11.765	0.999	0.131

The residuals reported in Tables 11 and 12 are also plagued with autocorrelation, denoted as the AR 1-2 test , heteroscedasticity denoted as the Hetero test, and non-normality, denoted as the normality test.

Adding impulse-indicator saturation variables, IIS, and step-indicator saturation variables, SIS, to the basis unemployment rate model substantially reduces the regression sum of squares, RSS, from 22.163 to 6.673 (see Table 13). The autocorrelated residuals remain, but heteroscedasticity and non-normality disappear with the application of the IIS and SIS variables.

A summary of the US unemployment rate with its AR1 estimation and the WkUNCL variable and *OxMetrics* applications of several saturation variables is reported in Table 14, with significant RSS and sigma, the regression equation standard error.

Dhrymes (2017) reported that the use of longer (3 and up to 12 monthly) lags of TCBLEI and the WkUNCL variables, with IIS, SIS, DIIS, and TIS variables, produced models not plagued with autocorrelation, heteroscedasticity, and non-normality. We substantiate the Dhrymes analysis. The use of longer lags, up to 12, in the TCBLEI and the WkUNCL time series, reduce the RSS from 21.673 to 15.046, despite the facts that few of the lags were statistically significant (see Table 15).

The application of *Autometrics* produces statistically significant LEI 6-month and 12-month lags and a 1-month WkUNCL lag with IIS and SIS variables such that the regression standard error falls from 0.149 to 0.109 and the RSS falls from 15.076 to 7.310. These are substantial estimated error reductions, and heteroscedasticity and normalized residuals are reported (see Table 16).

The further application of differenced impulse-indicator saturation variables, DIIS, and trend saturation variables, TIS, removes autocorrelation in residuals and maintains heteroscedasticity and normalized residuals. We report validation of the MZTT analysis (see Table 17).

Table 15 Modeling TCBUER longer lags of LEI and WkUNCL time series

The dataset is C:\OxMetrics\USMZTT.csv
The estimation sample is 13-716

	Coefficient	Std. error	t-value	t-prob	Part.R^2
TCBUER_1	0.942029	0.01244	75.7	0.0000	0.8944
TCBLEI_1	−0.00955904	0.01919	−0.498	0.6186	0.0004
TCBLEI_2	−0.0186564	0.03073	−0.607	0.5440	0.0005
TCBLEI_3	0.00676126	0.03068	0.220	0.8256	0.0001
TCBLEI_4	0.0108450	0.03079	0.352	0.7247	0.0002
TCBLEI_5	0.0365468	0.03095	1.18	0.2380	0.0021
TCBLEI_6	−0.0518495	0.03089	−1.68	0.0937	0.0041
TCBLEI_7	0.0454777	0.03093	1.47	0.1419	0.0032
TCBLEI_8	−0.0238333	0.03084	−0.773	0.4400	0.0009
TCBLEI_9	−0.0183403	0.03075	−0.596	0.5510	0.0005
TCBLEI_10	−0.00558617	0.03083	−0.181	0.8563	0.0000
TCBLEI_11	0.00160296	0.03085	0.0520	0.9586	0.0000
TCBLEI_12	0.0199111	0.01950	1.02	0.3077	0.0015
WKUNCL_1	0.00209300	0.0003701	5.66	0.0000	0.0451
WKUNCL_2	2.57908e − 05	0.0005030	0.0513	0.9591	0.0000
WKUNCL_3	−0.000805880	0.0005044	−1.60	0.1106	0.0038
WKUNCL_4	−0.000369858	0.0005021	−0.737	0.4616	0.0008
WKUNCL_5	0.000517410	0.0005004	1.03	0.3015	0.0016
WKUNCL_6	−0.000467630	0.0005006	−0.934	0.3505	0.0013
WKUNCL_7	0.000511163	0.0005004	1.02	0.3073	0.0015
WKUNCL_8	−0.000882234	0.0005004	−1.76	0.0783	0.0046
WKUNCL_9	0.000888125	0.0005006	1.77	0.0765	0.0046
WKUNCL_10	−0.00121982	0.0005009	−2.44	0.0151	0.0087
WKUNCL_11	0.000615748	0.0005002	1.23	0.2188	0.0022
WKUNCL_12	−0.000126419	0.0003672	−0.344	0.7308	0.0002
Constant	0.306182	0.09283	3.30	0.0010	0.0158
Trend	0.000628407	0.0001990	3.16	0.0017	0.0145

Sigma	0.149077	RSS	15.0456405
R^2	0.991532	F(26,677) =	3049 [0.000]**
Adj.R^2	0.991206	Log likelihood	354.75
No. of observations	704	No. of parameters	27
Mean(TCBUER)	6.01733	Se(TCBUER)	1.58975

AR 1-2 test	F(2,675) = 23.299 [0.0000]**
ARCH 1-1 test	F(1,702) = 22.011 [0.0000]**
Normality test	Chi^2(2) = 17.461 [0.0002]**
Hetero test	F(52,651) = 2.3465 [0.0000]**
RESET23 test	F(2,675) = 0.82400 [0.4391]

Model saved to C:\OxMetrics\Model UE AR1 LEI12 WkUCL12 OLS.pdf
Model saved to C:\OxMetrics\Model UE AR1 LEI12 WkUCL12 OLS.pdf.gwg

Table 16 Modeling TCBUER by OLS longer lags of LEI and WkUNCL time series and estimation of saturation (IIS and SIS) variables

The dataset is C:\OxMetrics\USMZTT.csv
The estimation sample is 13–716

	Coefficient	Std. error	t-value	t-prob	Part.R^2
TCBUER_1	0.753143	0.01798	41.9	0.0000	0.7387
TCBLEI_6	−0.0256720	0.004169	−6.16	0.0000	0.0575
TCBLEI_12	−0.0118922	0.003331	−3.57	0.0004	0.0201
WKUNCL_1	0.00199255	0.0001807	11.0	0.0000	0.1638
Constant	3.14563	0.3769	8.35	0.0000	0.1008
Trend	0.00178787	0.0004887	3.66	0.0003	0.0211
I:14	−0.269182	0.1253	−2.15	0.0321	0.0074
I:15	0.654970	0.1255	5.22	0.0000	0.0420
I:22	0.470861	0.1173	4.01	0.0001	0.0253
I:24	0.455375	0.1174	3.88	0.0001	0.0237
I:32	−0.391650	0.1161	−3.37	0.0008	0.0180
I:75	−0.280408	0.1097	−2.56	0.0108	0.0104
I:193	0.779814	0.1539	5.07	0.0000	0.0397
I:235	0.315773	0.1108	2.85	0.0045	0.0129
I:245	−0.311096	0.1111	−2.80	0.0053	0.0125
I:307	0.307971	0.1102	2.80	0.0053	0.0124
I:326	0.532650	0.1096	4.86	0.0000	0.0367
I:388	−0.304081	0.1112	−2.73	0.0064	0.0119
I:406	−0.275779	0.1107	−2.49	0.0130	0.0099
I:450	−0.242743	0.1102	−2.20	0.0280	0.0077
I:452	−0.363923	0.1104	−3.30	0.0010	0.0172
I:624	−0.749718	0.1538	−4.87	0.0000	0.0368
I:649	0.304336	0.1105	2.75	0.0061	0.0121
I:652	0.217843	0.1101	1.98	0.0484	0.0063
I:664	−0.335184	0.1096	−3.06	0.0023	0.0148
I:666	−0.184159	0.1096	−1.68	0.0936	0.0045
I:672	−0.180063	0.1095	−1.65	0.1005	0.0043
I:689	−0.232399	0.1095	−2.12	0.0342	0.0072
I:695	−0.250072	0.1097	−2.28	0.0229	0.0083
I:714	0.250455	0.1105	2.27	0.0237	0.0082
S1:17	−0.193868	0.07731	−2.51	0.0124	0.0100
S1:25	−0.240667	0.06316	−3.81	0.0002	0.0228
S1:33	0.253074	0.05477	4.62	0.0000	0.0332
S1:43	−0.355830	0.1139	−3.12	0.0019	0.0155
S1:44	0.382163	0.1330	2.87	0.0042	0.0131
S1:46	−0.392613	0.1329	−2.95	0.0033	0.0139
S1:47	0.416369	0.1536	2.71	0.0069	0.0117
S1:48	−0.291738	0.1189	−2.45	0.0144	0.0096
S1:53	0.294606	0.1190	2.48	0.0136	0.0098
S1:54	−0.216635	0.1100	−1.97	0.0494	0.0062

(continued)

Table 16 (continued)

S1:113	−0.0764296	0.03185	−2.40	0.0167	0.0092
S1:132	−0.266956	0.04531	−5.89	0.0000	0.0529
S1:142	−0.229766	0.08489	−2.71	0.0070	0.0117
S1:144	0.212549	0.08448	2.52	0.0121	0.0101
S1:154	−0.0812795	0.04695	−1.73	0.0839	0.0048
S1:166	0.181605	0.07018	2.59	0.0099	0.0107
S1:169	−0.221101	0.06779	−3.26	0.0012	0.0168
S1:188	−0.278787	0.1118	−2.49	0.0129	0.0099
S1:189	0.260239	0.1535	1.70	0.0905	0.0046
S1:190	−0.422165	0.1334	−3.17	0.0016	0.0159
S1:192	0.461044	0.1348	3.42	0.0007	0.0185
S1:194	−0.398496	0.1256	−3.17	0.0016	0.0159
S1:197	0.328761	0.07383	4.45	0.0000	0.0309
S1:208	0.220266	0.1138	1.94	0.0534	0.0060
S1:209	−0.332219	0.1146	−2.90	0.0039	0.0133
S1:218	0.119115	0.05132	2.32	0.0206	0.0086
S1:227	0.287236	0.1144	2.51	0.0123	0.0100
S1:228	−0.221658	0.1113	−1.99	0.0468	0.0064
S1:255	−0.426285	0.08170	−5.22	0.0000	0.0420
S1:257	0.529592	0.08756	6.05	0.0000	0.0556
S1:264	−0.403154	0.1178	−3.42	0.0007	0.0185
S1:265	0.359727	0.1255	2.87	0.0043	0.0131
S1:268	−0.215883	0.09938	−2.17	0.0302	0.0075
S1:270	0.467421	0.1329	3.52	0.0005	0.0195
S1:271	−0.578624	0.1126	−5.14	0.0000	0.0408
S1:294	0.476863	0.1134	4.20	0.0000	0.0277
S1:295	−0.696051	0.1538	−4.53	0.0000	0.0319
S1:296	0.366225	0.1108	3.31	0.0010	0.0173
S1:352	−0.174898	0.1103	−1.59	0.1133	0.0040
S1:353	0.206328	0.1108	1.86	0.0631	0.0056
S1:378	−0.122316	0.03445	−3.55	0.0004	0.0199
S1:405	−0.0942329	0.03559	−2.65	0.0083	0.0112
S1:435	−0.404908	0.1110	−3.65	0.0003	0.0210
S1:436	0.365521	0.1105	3.31	0.0010	0.0173
S1:471	0.284758	0.1109	2.57	0.0105	0.0105
S1:472	−0.377882	0.1103	−3.43	0.0007	0.0186
S1:542	−0.216694	0.04443	−4.88	0.0000	0.0369
S1:552	−0.164741	0.04598	−3.58	0.0004	0.0203
S1:592	−0.156593	0.04519	−3.47	0.0006	0.0190
S1:605	0.207590	0.05508	3.77	0.0002	0.0224
S1:622	−0.526948	0.1127	−4.68	0.0000	0.0340
S1:624	0.399025	0.1119	3.57	0.0004	0.0201
S1:642	0.175739	0.03661	4.80	0.0000	0.0358

Sigma	0.108495	RSS	7.30992276
R^2	0.995886	F(82,621) =	1833 [0.000]**
Adj.R^2	0.995342	Log likelihood	608.843
No. of observations	704	No. of parameters	83
Mean(TCBUER)	6.01733	Se(TCBUER)	1.58975
AR 1-2 test		F(2,619) = 4.5777 [0.0106]*	
ARCH 1-1 test		F(1,702) = 4.0782 [0.0438]*	
Normality test		Chi^2(2) = 1.2520 [0.5347]	
Hetero test		F(46,616) = 0.72879 [0.9090]	
RESET23 test		F(2,619) = 1.0400 [0.3541]	
Model saved to C:\OxMetrics\Model UE AR1 LEI12 WkUCL12 IIS SIS.pdf Model saved to C:\OxMetrics\Model UE AR1 LEI12 WkUCL12 IIS SIS.pdf.gwg			

Source: OX Metrics

Table 17 Modeling TCBUER by OLS and estimation of (IIS, SIS, TIIS, and DIIS) saturation variables

The dataset is C:\OxMetrics\USMZTT.csv
The estimation sample is 13–716

	Coefficient	Std. error	t-value	t-prob	Part.R^2
TCBUER_1	0.585264	0.02077	28.2	0.0000	0.5738
WKUNCL_1	0.00202334	0.0002057	9.84	0.0000	0.1409
Constant	6.39888	0.4730	13.5	0.0000	0.2367
TCBLEI_1	0.0255395	0.01156	2.21	0.0276	0.0082
TCBLEI_2	−0.0525240	0.01404	−3.74	0.0002	0.0232
TCBLEI_5	0.0140285	0.008965	1.56	0.1182	0.0041
TCBLEI_9	−0.0392911	0.004735	−8.30	0.0000	0.1045
WKUNCL_6	0.000921127	0.0001905	4.83	0.0000	0.0381
DI:14	−0.436858	0.06555	−6.66	0.0000	0.0700
DI:17	−0.192300	0.06587	−2.92	0.0036	0.0142
DI:22	0.127409	0.07590	1.68	0.0937	0.0048
DI:23	−0.167248	0.07566	−2.21	0.0274	0.0082
DI:26	0.115631	0.06559	1.76	0.0784	0.0052
DI:31	0.256988	0.06559	3.92	0.0001	0.0254
DI:33	0.131747	0.06552	2.01	0.0448	0.0068
DI:47	0.316703	0.07565	4.19	0.0000	0.0288
DI:66	0.160126	0.06548	2.45	0.0148	0.0100
DI:71	−0.175501	0.06556	−2.68	0.0076	0.0120
DI:74	0.254441	0.06551	3.88	0.0001	0.0249
DI:193	0.332019	0.06582	5.04	0.0000	0.0413
DI:255	−0.201872	0.06717	−3.01	0.0028	0.0151
DI:264	−0.234163	0.06571	−3.56	0.0004	0.0211
DI:325	−0.351704	0.06555	−5.37	0.0000	0.0465
DI:352	−0.187410	0.06553	−2.86	0.0044	0.0137

(continued)

Table 17 (continued)

DI:356	0.143838	0.06551	2.20	0.0285	0.0081
DI:588	0.191560	0.07567	2.53	0.0116	0.0107
DI:589	0.150432	0.07569	1.99	0.0473	0.0066
DI:633	0.125241	0.06569	1.91	0.0571	0.0061
I:190	−0.300756	0.1135	−2.65	0.0083	0.0118
I:208	0.283584	0.1037	2.73	0.0064	0.0125
I:237	0.272371	0.1135	2.40	0.0167	0.0097
I:287	0.246427	0.09908	2.49	0.0132	0.0104
I:307	0.221848	0.09525	2.33	0.0202	0.0091
I:592	−0.264380	0.09781	−2.70	0.0071	0.0122
I:597	−0.301450	0.09686	−3.11	0.0019	0.0162
I:649	0.231029	0.09858	2.34	0.0194	0.0092
I:658	0.232573	0.09489	2.45	0.0145	0.0101
I:664	−0.301143	0.09467	−3.18	0.0015	0.0169
I:689	−0.246657	0.09389	−2.63	0.0088	0.0116
I:695	−0.237284	0.09356	−2.54	0.0115	0.0108
I:714	0.240186	0.09544	2.52	0.0121	0.0106
S1:21	−0.402644	0.04772	−8.44	0.0000	0.1077
S1:34	0.313119	0.04260	7.35	0.0000	0.0839
S1:43	−0.311791	0.09779	−3.19	0.0015	0.0169
S1:44	0.315665	0.1099	2.87	0.0042	0.0138
S1:47	−0.256540	0.07580	−3.38	0.0008	0.0190
S1:53	0.150756	0.05764	2.62	0.0091	0.0115
S1:58	−0.282594	0.1014	−2.79	0.0055	0.0130
S1:59	0.273285	0.09528	2.87	0.0043	0.0138
S1:78	0.163259	0.03848	4.24	0.0000	0.0296
S1:132	−0.347695	0.04447	−7.82	0.0000	0.0939
S1:140	−0.134757	0.07461	−1.81	0.0714	0.0055
S1:142	−0.211798	0.09289	−2.28	0.0230	0.0087
S1:144	0.193682	0.07603	2.55	0.0111	0.0109
S1:150	−0.150454	0.07583	−1.98	0.0477	0.0066
S1:152	0.229627	0.09264	2.48	0.0135	0.0103
S1:154	−0.167345	0.07111	−2.35	0.0189	0.0093
S1:166	0.202331	0.06022	3.36	0.0008	0.0188
S1:169	−0.151312	0.05920	−2.56	0.0108	0.0110
S1:188	−0.345706	0.07056	−4.90	0.0000	0.0391
S1:191	−0.279580	0.08215	−3.40	0.0007	0.0193
S1:197	0.401509	0.05881	6.83	0.0000	0.0732
S1:204	0.145572	0.06085	2.39	0.0171	0.0096
S1:209	−0.276356	0.05937	−4.65	0.0000	0.0354
S1:216	0.219691	0.05472	4.01	0.0001	0.0266
S1:221	−0.242372	0.1017	−2.38	0.0175	0.0095
S1:222	0.270737	0.1018	2.66	0.0080	0.0119

(continued)

Table 17 (continued)

S1:227	0.189999	0.05515	3.45	0.0006	0.0197
S1:234	−0.326572	0.09944	−3.28	0.0011	0.0180
S1:235	0.485054	0.1136	4.27	0.0000	0.0300
S1:238	−0.168244	0.07579	−2.22	0.0268	0.0083
S1:244	0.332101	0.1012	3.28	0.0011	0.0179
S1:245	−0.252140	0.09852	−2.56	0.0107	0.0110
S1:257	0.606116	0.1100	5.51	0.0000	0.0489
S1:268	−0.269488	0.07439	−3.62	0.0003	0.0218
S1:270	0.398347	0.1137	3.50	0.0005	0.0204
S1:271	−0.373142	0.1137	−3.28	0.0011	0.0179
S1:273	−0.261831	0.07442	−3.52	0.0005	0.0206
S1:284	−0.225344	0.04910	−4.59	0.0000	0.0345
S1:294	0.445064	0.1004	4.43	0.0000	0.0322
S1:295	−0.597011	0.1314	−4.54	0.0000	0.0338
S1:296	0.520671	0.09794	5.32	0.0000	0.0457
S1:306	−0.0928995	0.03819	−2.43	0.0153	0.0099
S1:339	0.146469	0.03553	4.12	0.0000	0.0280
S1:378	−0.121220	0.03312	−3.66	0.0003	0.0222
S1:395	−0.161087	0.04959	−3.25	0.0012	0.0176
S1:400	−0.304913	0.07827	−3.90	0.0001	0.0251
S1:402	0.216096	0.08054	2.68	0.0075	0.0121
S1:406	−0.170634	0.05544	−3.08	0.0022	0.0158
S1:418	0.122479	0.03767	3.25	0.0012	0.0176
S1:435	−0.382464	0.09579	−3.99	0.0001	0.0263
S1:436	0.342597	0.09644	3.55	0.0004	0.0209
S1:449	0.204884	0.09745	2.10	0.0359	0.0074
S1:450	−0.376628	0.1312	−2.87	0.0042	0.0138
S1:451	0.481347	0.1312	3.67	0.0003	0.0223
S1:452	−0.322578	0.09602	−3.36	0.0008	0.0188
S1:471	0.211413	0.09594	2.20	0.0279	0.0082
S1:472	−0.338881	0.09523	−3.56	0.0004	0.0210
S1:526	−0.238754	0.07017	−3.40	0.0007	0.0192
S1:528	0.242367	0.1137	2.13	0.0335	0.0076
S1:529	−0.261200	0.09594	−2.72	0.0067	0.0124
S1:572	0.235517	0.05179	4.55	0.0000	0.0339
S1:592	−0.101007	0.05369	−1.88	0.0604	0.0060
S1:605	0.187421	0.06313	2.97	0.0031	0.0147
S1:622	−0.509334	0.09727	−5.24	0.0000	0.0444
T1:253	−0.147244	0.02950	−4.99	0.0000	0.0405
T1:257	0.151277	0.02943	5.14	0.0000	0.0429
T1:544	−0.262184	0.04392	−5.97	0.0000	0.0570
T1:545	0.250911	0.04459	5.63	0.0000	0.0509
T1:622	0.611169	0.1130	5.41	0.0000	0.0473

(continued)

Table 17 (continued)

T1:623	−0.660028	0.1218	−5.42	0.0000	0.0474
T1:628	0.0712405	0.01760	4.05	0.0001	0.0270
T1:648	0.0146784	0.006773	2.17	0.0306	0.0079
T1:658	−0.0249817	0.004599	−5.43	0.0000	0.0476

Sigma	0.0925313	RSS	5.05160671
R^2	0.997157	F(113,590) =	1831 [0.000]**
Adj.R^2	0.996612	Log likelihood	738.917
No. of observations	704	No. of parameters	114
Mean(TCBUER)	6.01733	Se(TCBUER)	1.58975

AR 1-2 test	F(2,588) = 2.2860 [0.1026]
ARCH 1-1 test	F(1,702) = 1.9101 [0.1674]
Normality test	Chi^2(2) = 0.24178 [0.8861]
Hetero test	F(116,559) = 0.62076 [0.9990]
RESET23 test	F(2,588) = 2.5819 [0.0765]

Model saved to C:\OxMetrics\Model UE AR1 LEI12 WkUCL12 IIS SIS TIIS DIIS.pdf
Model saved to C:\OxMetrics\Model UE AR1 LEI12 WkUCL12 IIS SIS TIIS DIIS.pdf.gwg

There is a substantial RSS reduction and normal and homoscedastic error terms. We report that the unemployment rate time series of the United States can be modeled with an AR(1) process, one and six-period lags in weekly unemployment claims time series, and 1, 2, 5, 6, and 9 months in The Conference Board LEI and small IIS, SIS, DIIS, and TIS variables, within the *Autometrics* time series modeling and forecasting software. The introduction of Autometrics is a giant step forward in modeling efficiency.

Causality Testing of the US Unemployment, 1959Q2–2018Q2

Causality testing modeling was introduced by Granger (1969) and updated in his modeling (1980, 1986, and 1989). Ashley, Granger, and Schmalensee (1980), Ashley (1998, 2003), Chen and Lee (1990), Chen and Liu (1993a, b), and Thomakos and Guerard (2004) reported statistically significant results of Granger causality tests. Clive granger won the Nobel Prize in Economics in 2003. This section models several variations on the Granger methodology. In our analysis, we use an ARIMA (1,1,1) as a naïve benchmark model. We use several variations of the RWD framework as naïve benchmarks for forecasting comparisons. We will estimate several transfer functions for causality testing.

A rational transfer function model, TF, can be estimated as an extension to a multiple regression model:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 \beta_{2t} + \cdots + \beta_m x_{mt} + \varepsilon_t \tag{20}$$

If $x_t = \phi x_{t-1} + a_t$, an AR(1) model previously discussed, then a simple transfer function can be estimated as

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 \beta_{2t} + \cdots + \beta_m x_{mt} + \frac{1}{1 - \phi_1 B} a_t \tag{21}$$

A simple input transfer function may be written as

$$y_t = \frac{w(B)}{z(B)} X_t + N_t \tag{22}$$

where

$$\begin{aligned} w(B) &= (w_o + w_1 B) + \cdots + w_{h-1} B^{h-1}) B^b \\ \delta(B) &= 1 - \delta_1 B - \cdots - \delta_r B^r \\ \phi(B) &= 1 - \phi B - \cdots - \phi_p B^p \\ \theta(B) &= 1 - \theta_1 B - \cdots - \theta_q B^q \end{aligned}$$

The polynomial $w(B)$ is the delay of the transfer function. We will estimate transfer function models in this analysis using the weekly unemployment claims time series as an input. We also use the leading economic indicators, LEI, as an input to transfer function modeling.

Outliers have long been recognized in economic time series. A very useful outlier analysis and software application can be found in Tsay (1988, 2010) and the SCA software. An additive outlier (AO) is an event that affects a time series for one period only:

$$y_t = z_t + w_A P_t^{(T)} \tag{23}$$

where $P_t^{(T)}$ is a pulse function.

An innovative outlier (IO) is an event that affects the ARIMA model:

$$\begin{aligned} y_t &= z_t + \frac{\phi(B)}{\phi(B)} w_I P_t^{(T)} \\ y_t &= z_t + \frac{\phi(B)}{\phi(B)} \left(a_z + w_I P_t^{(T)} \right) \end{aligned} \tag{24}$$

The IO event affects all Y according to the Ψ weights of the model.

A level shift, LS, is an event that permanently affects the time series:

$$y_t = z_t + w_L S_t^{(T)} \tag{25}$$

where $S_t^{(T)}$ is a step function.

A temporary change event, TC, has an initial impact and an effect that that decays exponentially by δ:

$$y_t = z_t + \frac{1}{1-\delta(B)} w_c P_t^{(T)} \tag{26}$$

The SCA outlier estimation procedures are documented in Liu (1999, 2006).

MZTT estimated threshold autoregressive models, TAR, nonlinear time series models, to forecast the unemployment rate. In Tsay (2010), Tsay made several observations regarding linear and nonlinear time series model forecasting comparisons:

1. The mean square errors, MSE, of linear and TAR models are very close, but TAR models have smaller biases.
2. In economic contractions, TAR models produce lower MSE and bias than the linear models.
3. In economic expansions, the linear models outperform nonlinear models.

In the SCA replication and extension of the MZTT (1998) study of the US unemployment rate, we estimate naïve models and transfer function models with weekly unemployment claims and LEI time series inputs, ARIMA and seasonal ARIMA time series models, and threshold autoregressive models.

The transfer function model forecasts are compared to several naïve models in terms of testing, that is, which model produces the most accurate forecast of real GDP. No-change forecasts of real GDP and random walk with drift models may be useful forecasting benchmarks (Granger & Newbold, 1977; Mincer & Zarnowitz, 1969). The purpose of this analysis is to examine the time series forecasts of composite economic indexes produced by The Conference Board (TCB) and test the hypothesis that the leading economic indicators, LEI, and one of its components, weekly unemployment claims, are useful inputs to transfer function model to model and test causality with the US unemployment rate, 1959–1993 and 1959–2018.

Had one used the SCA system, and tested for causality, as an application of the Chen and Lee (1990) test, one finds statistical significance and causality in the DUE and DLEI relationship. The SCA ARIMA(1,1,0) estimation of the logged change in the US unemployment rate is reported in Table 18. The univariate model residual standard error is 0.193.

Furthermore, the outlier analysis of the DLEI transfer function, modeled with three and two standard deviation criteria for outlier detection, produces residual standard errors of 0.160 and 0.129, respectively (see Table 19). The estimation of a DUE transfer function with 12 months of DLEI lags produces statistically significant DLEI lags of 1,2, 4, and 9 months (see Table 20).

The SCA residual standard error falls from 0.163 in Table 20 to 0.119 in Table 21 with longer lags of the DLEI variable.

The longer lags do not materially change the residual standard error from the one-period DLEI transfer function. The SCA application of the Chen-Lee (1990 causality test finds a feedback relationship between DUE and DLEI during the 1959–1993 time period (see Table 22).

Table 18 Summary for SCA univariate time series model, RWD1, estimation of the logged change of the US unemployment rate, 1959Q2–1993Q4

VARIABLE		TYPE OF VARIABLE		ORIGINAL OR CENTERED			DIFFERENCING	
DUE		RANDOM		ORIGINAL			NONE	
PARAMETER LABEL	VARIABLE NAME	NUM./ DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	C	CNST	1	0	NONE	.0014	.0105	.14
2	DUE	D-AR	1	1	NONE	.1090	.0486	2.24

EFFECTIVE NUMBER OF OBSERVATIONS . .	418
R-SQUARE	0.010
RESIDUAL STANDARD ERROR.	0.193009E + 00
(−2)*LOG LIKELIHOOD FUNCTION	−0.189001E + 03
AIC.	−0.183001E + 03
SIC.	−0.170895E + 03

Table 19 Summary for SCA univariate time series model, RWD1, estimation of the logged change of the US unemployment rate and a transfer function with the logged change of the LEI time series, 1959Q2–1993Q4

VARIABLE	TYPE OF VARIABLE	ORIGINAL OR CENTERED	DIFFERENCING
DUE	RANDOM	ORIGINAL	NONE

PARAMETER LABEL	VARIABLE NAME	NUM./ DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	DUE	MA	1	1	NONE	.7436	.0915	8.12
2	DUE	D-AR	1	1	NONE	.8822	.0663	13.30

TOTAL NUMBER OF OBSERVATIONS.	419
EFFECTIVE NUMBER OF OBSERVATIONS.	418
RESIDUAL STANDARD ERROR (WITHOUT OUTLIER ADJUSTMENT). .	0.185819E + 00
ACF RES	
NAME OF THE SERIES	RES
TIME PERIOD ANALYZED	2 TO 419
MEAN OF THE (DIFFERENCED) SERIES . . .	0.0011
STANDARD DEVIATION OF THE SERIES . . .	0.1858
T-VALUE OF MEAN (AGAINST ZERO)	0.1178

AUTOCORRELATIONS

1–12	−.12	.12	.01	.13	−.02	.01	.01	.04	.09	−.01	−.09	−.19
ST.E.	.05	.05	.05	.05	.05	.05	.05	.05	.05	.05	.05	.05
Q	6.1	12.0	12.0	19.5	19.7	19.7	19.8	19.8	20.5	23.7	27.4	42.6
13–24	.04	−.13	.00	.02	−.02	−.05	.07	−.07	.00	.03	−.00	−.11
ST.E.	.05	.05	.05	.05	.05	.05	.05	.05	.05	.05	.05	.05
Q	43.3	50.7	50.8	50.9	51.0	52.1	54.1	56.4	56.4	56.7	56.8	61.7

```
−1.0  −0.8  −0.6  −0.4  −0.2  0.0  0.2  0.4  0.6  0.8  1.0
 +-----+-----+-----+-----+-----+----+----+-----+-----+----+
```

		I
1	−0.12	X+XI +
2	0.12	+ IX+X
3	0.01	+ I +
4	0.13	+ IX+X
5	−0.02	+ XI +
6	0.01	+ I +
7	−0.01	+ I +
8	0.01	+ I +
9	0.04	+ IX +
10	−0.09	+XXI +
11	0.09	+ IXX+
12	−0.19	XX+XXI +
13	0.04	+ IX +
14	−0.13	XXXI +
15	0.00	+ I +
16	0.02	+ I +
17	−0.02	+ I +
18	−0.05	+ XI +
19	0.07	+ IXX+
20	−0.07	+XXI +
21	0.00	+ I +
22	0.03	+ IX +
23	0.00	+ I +
24	−0.11	XXXI +
--		

(continued)

Table 19 (continued)

The outlier analysis estimation, using a critical three standard deviation criteria for outliers, is:

PARAMETER LABEL	VARIABLE NAME	NUM./DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	DUE	MA	1	1	NONE	.7714	.1054	7.32
2	DUE	D-AR	1	1	NONE	.8752	.0823	10.64

SUMMARY OF OUTLIER DETECTION AND ADJUSTMENT

TIME	ESTIMATE	T-VALUE	TYPE
11	−0.489	−3.72	TC
14	0.752	4.65	IO
21	0.570	3.56	IO
132	0.412	3.17	TC
190	0.624	4.64	TC
192	0.561	3.43	AO
255	0.574	4.41	TC
288	−0.488	−3.05	IO
294	−0.688	−4.30	IO
325	0.547	3.46	AO

TOTAL NUMBER OF OBSERVATIONS.	419
EFFECTIVE NUMBER OF OBSERVATIONS.	418
RESIDUAL STANDARD ERROR (WITHOUT OUTLIER ADJUSTMENT). .	0.186459E + 00
RESIDUAL STANDARD ERROR (WITH OUTLIER ADJUSTMENT) . . .	0.160156E + 00

Table 20 Summary for SCA univariate time series model, RWD1, estimation of the logged change of the US unemployment rate and a transfer function with the longer lags in the logged change of the LEI time series, 1959Q2–1993Q4

VARIABLE	TYPE OF VARIABLE	ORIGINAL OR CENTERED	DIFFERENCING
DUE	RANDOM	ORIGINAL	NONE
DLEI	RANDOM	ORIGINAL	NONE

PARAMETER LABEL	VARIABLE NAME	NUM./DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	C	CNST	1	0	NONE	.0413	.0075	5.47
2	DLEI	NUM.	1	1	NONE	−.1152	.0318	−3.62
3	DLEI	NUM.	1	2	NONE	−.0808	.0373	−2.17
4	DLEI	NUM.	1	3	NONE	.0162	.0369	.44
5	DLEI	NUM.	1	4	NONE	−.0988	.0364	−2.72
6	DLEI	NUM.	1	5	NONE	.0363	.0365	.99
7	DLEI	NUM.	1	6	NONE	−.0184	.0366	−.50
8	DLEI	NUM.	1	7	NONE	−.0533	.0366	−1.45
9	DLEI	NUM.	1	8	NONE	.0340	.0366	.93
10	DLEI	NUM.	1	9	NONE	−.0934	.0366	−2.55
11	DLEI	NUM.	1	10	NONE	.0067	.0372	.18
12	DLEI	NUM.	1	11	NONE	.0097	.0371	.26
13	DLEI	NUM.	1	12	NONE	−.0026	.0316	−.08
14	DUE	D-AR	1	1	NONE	−.1691	.0489	−3.46

(continued)

Table 20 (continued)

EFFECTIVE NUMBER OF OBSERVATIONS . .	406
R-SQUARE	0.293
RESIDUAL STANDARD ERROR.	0.163106E + 00
(−2)*LOG LIKELIHOOD FUNCTION	−0.320265E + 03
AIC.	−0.290265E + 03
SIC.	−0.230170E + 03
--	

Table 21 Summary for SCA univariate time series model, RWD1, estimation of the logged change of the US unemployment rate and a transfer function with the longer lags in the logged change of the LEI time series and outlier analysis, 1959Q2–1993Q4

PARAMETER LABEL	VARIABLE NAME	NUM./DENOM.	FACTOR	ORDER	CONS TRAINT	VALUE	STD ERROR	T-VALUE
1	C	CNST	1	0	NONE	.0388	.0081	4.81
2	DLEI	NUM.	1	1	NONE	−.1157	.0317	−3.65
3	DLEI	NUM.	1	2	NONE	−.0759	.0368	−2.06
4	DLEI	NUM.	1	3	NONE	.0129	.0368	.35
5	DLEI	NUM.	1	4	NONE	−.0894	.0367	−2.44
6	DLEI	NUM.	1	5	NONE	.0311	.0366	.85
7	DLEI	NUM.	1	6	NONE	−.0140	.0365	−.38
8	DLEI	NUM.	1	7	NONE	−.0539	.0363	−1.48
9	DLEI	NUM.	1	8	NONE	.0353	.0362	.98
10	DLEI	NUM.	1	9	NONE	−.0934	.0314	−2.98
11	DUE	MA	1	1	NONE	−.5373	.2375	−2.26
12	DUE	D-AR	1	1	NONE	−.6712	.2144	−3.13

TOTAL NUMBER OF OBSERVATIONS.	419
EFFECTIVE NUMBER OF OBSERVATIONS.	409
RESIDUAL STANDARD ERROR (WITHOUT OUTLIER ADJUSTMENT). .	0.165143E + 00

SUMMARY OF OUTLIER DETECTION AND ADJUSTMENT

TIME	ESTIMATE	T-VALUE	TYPE
17	0.274	2.36	AO
21	0.580	6.44	TC
22	−0.428	−2.95	IO
31	−0.359	−3.01	IO
34	−0.400	−3.45	AO
37	−0.287	−2.41	IO
43	0.318	2.74	AO
46	0.297	2.50	IO
49	0.275	2.31	IO
52	0.274	2.31	IO
58	0.266	2.24	IO
74	−0.295	−2.48	IO
105	0.278	2.33	IO
113	0.266	2.29	AO
130	−0.268	−2.31	AO
132	0.291	3.94	TC
142	0.350	3.02	AO
182	−0.173	−2.34	TC
190	0.445	5.55	TC
192	0.431	3.35	IO
197	−0.348	−4.70	TC
209	0.249	3.37	TC
227	−0.191	−2.58	TC
234	0.358	3.00	IO
255	0.338	2.84	AO
256	0.326	2.68	IO
259	−0.209	−2.78	TC
263	−0.353	−2.91	IO

268	0.285	2.45	AO
270	−0.346	−2.91	IO
274	0.257	3.48	TC
285	0.276	2.32	IO
286	0.455	3.92	AO
288	−0.363	−3.05	IO
294	−0.540	−4.54	IO
296	−0.236	−3.20	TC
306	0.413	3.56	AO
325	0.496	4.28	AO
335	−0.304	−2.55	IO
395	0.196	2.61	TC
400	0.322	3.65	TC
402	−0.218	−2.50	TC

TOTAL NUMBER OF OBSERVATIONS.	419
EFFECTIVE NUMBER OF OBSERVATIONS.	409
RESIDUAL STANDARD ERROR (WITHOUT OUTLIER ADJUSTMENT). .	0.168517E + 00
RESIDUAL STANDARD ERROR (WITH OUTLIER ADJUSTMENT) . . .	0.118993E + 00

Table 22 Causality testing of the logged change of the US unemployment rate and a transfer function with the longer lags in the logged change of the LEI time series, with outliers, 1959Q2–1993Q4

CCM VARIABLES ARE DUE, DLEI			
TIME PERIOD ANALYZED		1 TO 419	
EFFECTIVE NUMBER OF OBSERVATIONS (NOBE). . .		419	
SERIES	NAME	MEAN	STD.
1	DUE	0.0012	ERROR
2	DLEI	0.1081	0.1940 0.3502
NOTE: THE APPROX. STD. ERROR FOR THE ESTIMATED CORRELATIONS BELOW			
IS (1/NOBE**.5) = 0.04885			
SAMPLE CORRELATION MATRIX OF THE SERIES			
1.00			
−0.39 1.00			
SUMMARIES OF CROSS-CORRELATION MATRICES USING +,−,., WHERE			
+ DENOTES A VALUE GREATER THAN 2/SQRT(NOBE)			
− DENOTES A VALUE LESS THAN −2/SQRT(NOBE)			
. DENOTES A NON-SIGNIFICANT VALUE BASED ON THE ABOVE CRITERION			
BEHAVIOR OF VALUES IN (I,J)TH POSITION OF CROSS-CORRELATION MATRIX OVER			
ALL OUTPUTTED LAGS WHEN SERIES J LAGS SERIES I			

	1	2
1	++++++.....−	−−−−−−−−
1	.−.....−	−−..−−.....
2	−−.....+	++++++++++.
2	.+.+.+++++++	.....−−−−

CROSS-CORRELATION MATRICES IN TERMS OF +,−,.

LAGS 1 THROUGH 6

+ −	+ −	+ −	+ −	+ −	+ −
− +	− +	. +	. +	. +	. +

LAGS 7 THROUGH 12

. −	. −	. −	. −	. −	− −
. +	. +	. +	. +	. +	+ .

LAGS 13 THROUGH 18

. −	− −	. .	. .	. −	. −
. .	+ .	. .	+ .	. .	+ .

LAGS 19 THROUGH 24

. .	. .	. .	. .	. .	− .
+ .	+ .	+ −	+ −	+ −	+ −
--					

STEPAR VARIABLES ARE DUE,DLEI. @
ARFITS ARE 1 to 1. rccm 1 to 1.

TIME PERIOD ANALYZED			1 TO 419
EFFECTIVE NUMBER OF OBSERVATIONS (NOBE). . .			419
SERIES	NAME	MEAN	STD. ERROR
1	DUE	0.0012	0.1940
2	DLEI	0.1081	0.3502
NOTE: THE APPROX. STD. ERROR FOR THE ESTIMATED CORRELATIONS BELOW			
IS (1/NOBE**.5) = 0.04885			

SAMPLE CORRELATION MATRIX OF THE SERIES

1.00

−0.39 1.00

SUMMARIES OF CROSS-CORRELATION MATRICES USING +,−,., WHERE

\+ DENOTES A VALUE GREATER THAN 2/SQRT(NOBE)

− DENOTES A VALUE LESS THAN −2/SQRT(NOBE)

. DENOTES A NON-SIGNIFICANT VALUE BASED ON THE ABOVE CRITERION

BEHAVIOR OF VALUES IN (I,J)TH POSITION OF CROSS-CORRELATION MATRIX OVER ALL OUTPUTTED LAGS WHEN SERIES J LAGS SERIES I

	1	2
1	++++++.....−	−−−−−−−−−−−−
1	.−.....−	−−..−−.....
2	−−.....+	++++++++++.
2	.+.+.+++++++	.....−−−−

CROSS-CORRELATION MATRICES IN TERMS OF +,−,.

LAGS 1 THROUGH 6

+ −	+ −	+ −	+ −	+ −	+ −
− +	− +	. +	. +	. +	. +

LAGS 7 THROUGH 12

. −	. −	. −	. −	. −	− −
. +	. +	. +	. +	. +	+ .

LAGS 13 THROUGH 18

. −	− −	. .	. .	. −	. −
. .	+ .	. .	+ .	. .	+ .

LAGS 19 THROUGH 24

. .	. .	. .	. .	. .	− .
+ .	+ .	+ −	+ −	+ −	+ −

DETERMINANT OF S(0) = 0.390986E − 02

NOTE: S(0) IS THE SAMPLE COVARIANCE MATRIX OF W(MAXLAG+1),...,W(NOBE)

AUTOREGRESSIVE FITTING ON LAG(S) 1

SUMMARIES OF CROSS-CORRELATION MATRICES USING +,−,., WHERE

\+ DENOTES A VALUE GREATER THAN 2/SQRT(NOBE)

− DENOTES A VALUE LESS THAN −2/SQRT(NOBE)

. DENOTES A NON-SIGNIFICANT VALUE BASED ON THE ABOVE CRITERION

BEHAVIOR OF VALUES IN (I,J)TH POSITION OF CROSS-CORRELATION MATRIX OVER ALL OUTPUTTED LAGS WHEN SERIES J LAGS SERIES I

	1	2
1	.+.+.....−.−	+−.−..−.−...
1	.−.....	−.....
2	..+.+..+.+.+	−+.+....+...
2	.+.....	+.+.....

CROSS-CORRELATION MATRICES IN TERMS OF +,−,.

LAGS 1 THROUGH 6

. +	+ −	. .	+ −	. .	. .
. −	. +	+ .	. +	+ .	. .

LAGS 7 THROUGH 12

. −	. .	. −	− .	. .	− .
. .	+ .	. +	+ .	. .	+ .

Table 23 Summary for SCA univariate time series model, RWD1, estimation of the logged change of the US unemployment rate and a transfer function with the longer lags in the logged change of the LEI time series and outlier analysis, 1959Q2–2018Q2

PARAMETER LABEL	VARIABLE NAME	NUM./ DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	C	CNST	1	0	NONE	.0258	.0054	4.76
2	DLEI	NUM.	1	1	NONE	−.0483	.0165	−2.92
3	DLEI	NUM.	1	2	NONE	−.0573	.0182	−3.15
4	DLEI	NUM.	1	3	NONE	−.0388	.0185	−2.10
5	DLEI	NUM.	1	4	NONE	−.0122	.0186	−.66
6	DLEI	NUM.	1	5	NONE	.0121	.0186	.65
7	DLEI	NUM.	1	6	NONE	−.0286	.0186	−1.53
8	DLEI	NUM.	1	7	NONE	−.0012	.0186	−.07
9	DLEI	NUM.	1	8	NONE	−.0035	.0186	−.19
10	DLEI	NUM.	1	9	NONE	−.0401	.0186	−2.15
11	DLEI	NUM.	1	10	NONE	−.0053	.0185	−.29
12	DLEI	NUM.	1	11	NONE	−.0319	.0183	−1.74
13	DLEI	NUM.	1	12	NONE	.0181	.0166	1.09
14	DUE	D-AR	1	1	NONE	−.1355	.0374	−3.63

EFFECTIVE NUMBER OF OBSERVATIONS . .	702
R-SQUARE	0.257
RESIDUAL STANDARD ERROR.	0.154294E + 00
(−2)*LOG LIKELIHOOD FUNCTION	−0.631742E + 03
AIC.	−0.601742E + 03
SIC.	−0.533433E + 03
--	

PARAMETER LABEL	VARIABLE NAME	NUM./ DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	C	CNST	1	0	NONE	.0225	.0063	3.59
2	DLEI	NUM.	1	1	NONE	−.0609	.0156	−3.91
3	DLEI	NUM.	1	2	NONE	−.0849	.0158	−5.36
***	DLEI	NUM.	1	3	FIXED	.0000		
***	DLEI	NUM.	1	4	FIXED	.0000		
***	DLEI	NUM.	1	5	FIXED	.0000		
***	DLEI	NUM.	1	6	FIXED	.0000		
***	DLEI	NUM.	1	7	FIXED	.0000		
***	DLEI	NUM.	1	8	FIXED	.0000		
4	DLEI	NUM.	1	9	NONE	−.0728	.0131	−5.54

TOTAL NUMBER OF OBSERVATIONS.	715
EFFECTIVE NUMBER OF OBSERVATIONS.	706
RESIDUAL STANDARD ERROR (WITHOUT OUTLIER ADJUSTMENT). .	0.159075E + 00
--	

ACF RES

NAME OF THE SERIES	RES
TIME PERIOD ANALYZED	10 TO 715
MEAN OF THE (DIFFERENCED) SERIES . . .	0.0000
STANDARD DEVIATION OF THE SERIES . . .	0.1591
T-VALUE OF MEAN (AGAINST ZERO)	0.0000

AUTOCORRELATIONS

1–12	−.13	.08	.05	.09	.04	.05	.01	−.01	−.01	−.06	.06	−.17
ST.E.	.04	.04	.04	.04	.04	.04	.04	.04	.04	.04	.04	.04
Q	12.6	16.8	18.9	24.7	25.7	27.6	27.6	27.7	27.7	30.0	32.4	52.3

(continued)

Table 23 (continued)

13–24	.02	−.09	−.01	.00	−.02	−.03	.05	.00	.00	.01	.03	−.09
ST.E.	.04	.04	.04	.04	.04	.04	.04	.04	.04	.04	.04	.04
Q	52.7	58.7	58.8	58.8	59.1	59.7	61.8	61.8	61.8	61.9	62.7	68.2

−1.0 −0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8 1.0

+- ——+————————+————+————+————+———+———+———+———+———+

		I
1	−0.13	X+XI +
2	0.08	+ IXX
3	0.05	+ IX+
4	0.09	+ IXX
5	0.04	+ IX+
6	0.05	+ IX+
7	0.01	+ I +
8	−0.01	+ I +
9	−0.01	+ I +
10	−0.06	+XI +
11	0.06	+ IX+
12	−0.17	XX+XI +
13	0.02	+ IX+
14	−0.09	XXI +
15	−0.01	+ I +
16	0.00	+ I +
17	−0.02	+XI +
18	−0.03	+XI +
19	0.05	+ IX+
20	0.00	+ I +
21	0.00	+ I +
22	0.01	+ I +
23	0.03	+ IX+
24	−0.09	XXI +
--		

The three standard deviation criteria of outliers produce the following estimated transfer function model

PARAMETER LABEL	VARIABLE NAME	NUM./DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	C	CNST	1	0	NONE	.0088	.0056	1.56
2	DLEI	NUM.	1	1	NONE	−.0422	.0139	−3.03
3	DLEI	NUM.	1	2	NONE	−.0720	.0142	−5.08
4	DLEI	NUM.	1	9	NONE	−.0655	.0118	−5.57

SUMMARY OF OUTLIER DETECTION AND ADJUSTMENT

TIME	ESTIMATE	T-VALUE	TYPE
14	0.598	4.20	AO
21	0.463	4.56	TC
132	0.377	3.71	TC
190	0.554	5.08	TC
192	0.476	3.13	AO
197	−0.314	−3.08	TC
255	0.363	3.56	TC
273	0.334	3.28	TC
286	0.434	3.05	AO
294	−0.622	−4.37	AO
296	−0.337	−3.31	TC
325	0.534	3.75	AO
622	0.431	3.03	AO

TOTAL NUMBER OF OBSERVATIONS.	715
EFFECTIVE NUMBER OF OBSERVATIONS.	706
RESIDUAL STANDARD ERROR (WITHOUT OUTLIER ADJUSTMENT). .	0.160059E + 00
RESIDUAL STANDARD ERROR (WITH OUTLIER ADJUSTMENT) . . .	0.142420E + 00

Table 24 Causality testing of the logged change of the US unemployment rate and a transfer function with the longer lags in the logged change of the LEI time series, with outliers, 1959Q2–2018Q2

CCM VARIABLES ARE DUE, DLEI			
TIME PERIOD ANALYZED			1 TO 715
EFFECTIVE NUMBER OF OBSERVATIONS (NOBE). . .			715
SERIES	NAME	MEAN	STD. ERROR
1	DUE	−0.0029	0.1791
2	DLEI	0.1175	0.4798

NOTE: THE APPROX. STD. ERROR FOR THE ESTIMATED CORRELATIONS BELOW IS (1/NOBE**.5) = 0.03740

SAMPLE CORRELATION MATRIX OF THE SERIES
1.00
−0.35 1.00

SUMMARIES OF CROSS-CORRELATION MATRICES USING +,−,., WHERE
+ DENOTES A VALUE GREATER THAN 2/SQRT(NOBE)
− DENOTES A VALUE LESS THAN −2/SQRT(NOBE)
. DENOTES A NON-SIGNIFICANT VALUE BASED ON THE ABOVE CRITERION

BEHAVIOR OF VALUES IN (I,J)TH POSITION OF CROSS-CORRELATION MATRIX OVER ALL OUTPUTTED LAGS WHEN SER

IES J LAGS SERIES I

	1	2
1	+++++++++.+−	−−−−−−−−−−
1	.....−	−−−−−.−....
2	−−−−.....	++++++++++
2	.....++.+++	+.....−

CROSS-CORRELATION MATRICES IN TERMS OF +,−,.

LAGS 1 THROUGH 6

+ −	+ −	+ −	+ −	+ −	+ −
− +	− +	− +	− +	. +	. +

LAGS 7 THROUGH 12

+ −	+ −	+ −	. −	+ −	− −
. +	. +	. +	. +	. +	. +

LAGS 13 THROUGH 18

. −	. −	. −	. −	. −	. −
. +	. .	. .	. .	. .	. .

LAGS 19 THROUGH 24

. .	. −	. .	. .	. .	− .
+ .	+ .	. .	+ .	+ .	+ −
--					

STEPAR VARIABLES ARE DUE,DLEI. @
ARFITS ARE 1 to 1. rccm 1 to 1.

TIME PERIOD ANALYZED			1 TO 715
EFFECTIVE NUMBER OF OBSERVATIONS (NOBE). . .			715
SERIES	NAME	MEAN	STD. ERROR
1	DUE	−0.0029	0.1791
2	DLEI	0.1175	0.4798

NOTE: THE APPROX. STD. ERROR FOR THE ESTIMATED CORRELATIONS BELOW IS (1/NOBE**.5) = 0.03740

SAMPLE CORRELATION MATRIX OF THE SERIES
1.00
−0.35 1.00

SUMMARIES OF CROSS-CORRELATION MATRICES USING +,−,., WHERE
+ DENOTES A VALUE GREATER THAN 2/SQRT(NOBE)
− DENOTES A VALUE LESS THAN −2/SQRT(NOBE)
. DENOTES A NON-SIGNIFICANT VALUE BASED ON THE ABOVE CRITERION
BEHAVIOR OF VALUES IN (I,J)TH POSITION OF CROSS-CORRELATION MATRIX OVER
ALL OUTPUTTED LAGS WHEN SERIES J LAGS SERIES I

	1	2
1	+++++++++.+−	−−−−−−−−−−−−
1	.....−	−−−−−−.−....
2	−−−−.....	++++++++++
2	.....++.+++	+.....−

CROSS-CORRELATION MATRICES IN TERMS OF +,−,.

LAGS 1 THROUGH 6

+ −	+ −	+ −	+ −	+ −	+ −
− +	− +	− +	− +	. +	. +

LAGS 7 THROUGH 12

+ −	+ −	+ −	. −	+ −	− −
. +	. +	. +	. +	. +	. +

LAGS 13 THROUGH 18

. −	. −	. −	. −	. −	. −
. +	. .	. .	. .	. .	. .

LAGS 19 THROUGH 24

. .	. −	. .	. .	. .	− .
+ .	+ .	. .	+ .	+ .	+ −

DETERMINANT OF S(0) = 0.647282E − 02

NOTE: S(0) IS THE SAMPLE COVARIANCE MATRIX OF W(MAXLAG+1),...,W(NOBE)

AUTOREGRESSIVE FITTING ON LAG(S) 1

SUMMARIES OF CROSS-CORRELATION MATRICES USING +,−,., WHERE
+ DENOTES A VALUE GREATER THAN 2/SQRT(NOBE)
− DENOTES A VALUE LESS THAN −2/SQRT(NOBE)
. DENOTES A NON-SIGNIFICANT VALUE BASED ON THE ABOVE CRITERION
BEHAVIOR OF VALUES IN (I,J)TH POSITION OF CROSS-CORRELATION MATRIX OVER
ALL OUTPUTTED LAGS WHEN SERIES J LAGS SERIES I

	1	2
1	.+++.+....+−	+−−−.−−.−.−.
1	.−.....−	−−...−.−....
2	+.....+.++.	−+++.+..+.+.
2	.....+..	.....

CROSS-CORRELATION MATRICES IN TERMS OF +,−,.

LAGS 1 THROUGH 6

. −	. .	. −	. .	+ −	− .
. .	+ .	. +	+ .	+ +	. .

LAGS 7 THROUGH 12

. −	− −	. .	. .	. .	. −
. .	. .	. .	. .	. .	. .

LAGS 13 THROUGH 18

. −	− −	. .	. .	. .	. −
. .	. .	. .	. .	. .	. .

(continued)

Table 24 (continued)

LAGS 19 THROUGH 24					
. .	. –	. .	. .	. .	– .
. .	. .	. .	+ .	. .	. .

========== STEPWISE AUTOREGRESSION SUMMARY ==========

	I	RESIDUAL	I	EIGENVAL	I	CHI-SQ	I		I	SIGNIFICANCE
LAG	I	VARIANCES	I	OF SIGMA	I	TEST	I	AIC	I	OF PARTIAL AR COEFF.
----	+	----------	+	----------	+	----------	+	---------	+	----------
1	I	.281E – 01	I	.269E – 01	I	341.95	I	–5.510	I	–
	I	.148E + 00	I	.149E + 00	I		I		I	+

NOTE: CHI-SQUARED CRITICAL VALUES WITH 4 DEGREES OF FREEDOM ARE

5%: 9.5 1%: 13.3

NOTE: THE PARTIAL AUTOREGRESSION COEFFICIENT MATRIX FOR LAG L IS THE ESTIMATED PHI(L) FROM THE FIT WHERE THE MAXIMUM LAG USED IS L (I.E., THE LAST COEFFICIENT MATRIX). THE ELEMENTS ARE STANDARDIZED BY DIVIDING EACH BY ITS STANDARD ERROR

--

MTSMODEL ARMATF11. SERIES ARE DUE,DLEI. @
MODEL IS (1-PHI*B)SERIES=C+(1-TH1*B)NOISE.

SUMMARY FOR MULTIVARIATE ARMA MODEL – ARMATF11

VARIABLE DIFFERENCING

DUE

DLEI

PARAMETER		FACTOR	ORDER	CONSTRAINT
1	C	CONSTANT	0	CC
2	PHI	REG AR	1	CPHI
3	TH1	REG MA	1	CTH1
--				

CAUSALTEST MODEL ARMATF11. OUTPUT PRINT(CORR). alpha .01

SUMMARY OF THE TIME SERIES

SERIES	NAME	MEAN	STD DEV	DIFFERENCE ORDER(S)
1	DUE	–0.0029	0.1791	
2	DLEI	0.1175	0.4798	

ERROR COVARIANCE MATRIX

	1	2
1	.032113	
2	–.030341	.230366

ITERATIONS TERMINATED DUE TO:
MAXIMUM NUMBER OF ITERATIONS 10 REACHED
TOTAL NUMBER OF ITERATIONS IS 14

MODEL SUMMARY WITH MAXIMUM LIKELIHOOD PARAMETER ESTIMATES

CONSTANT VECTOR (STD ERROR)

0.052	(0.016)
–0.138	(0.044)

PHI MATRICES
ESTIMATES OF PHI(1) MATRIX AND SIGNIFICANCE

–1.377	–.531	– –
6.484	2.414	+ +

STANDARD ERRORS

.263	.064
.782	.204

THETA MATRICES
ESTIMATES OF THETA(1) MATRIX AND SIGNIFICANCE

−1.323	−.475	− −
6.300	2.112	+ +

STANDARD ERRORS

.264	.066
.780	.209

ERROR COVARIANCE MATRIX

	1	2
1	.025766	
2	−.004718	.125043

==========
SUMMARY OF FINAL PARAMETER ESTIMATES AND THEIR STANDARD ERRORS
==========

PARAMETER NUMBER	PARAMETER DESCRIPTION	FINAL ESTIMATE	ESTIMATED STD. ERROR
1	CONSTANT(1)	0.052124	0.016013
2	CONSTANT(2)	−0.138469	0.044296
3	AUTOREGRESSIVE (1, 1, 1)	−1.377163	0.263322
4	AUTOREGRESSIVE (1, 1, 2)	−0.530786	0.064085
5	AUTOREGRESSIVE (1, 2, 1)	6.484085	0.782178
6	AUTOREGRESSIVE (1, 2, 2)	2.413705	0.204443
7	MOVING AVERAGE (1, 1, 1)	−1.322794	0.263844
8	MOVING AVERAGE (1, 1, 2)	−0.475195	0.065672
9	MOVING AVERAGE (1, 2, 1)	6.300207	0.780190
10	MOVING AVERAGE (1, 2, 2)	2.111769	0.209422

CORRELATION MATRIX OF THE PARAMETERS

	1	2	3	4	5	6	7	8	9	10
1	1.00									
2	−.73	1.00								
3	−.33	−.36	1.00							
4	−.41	−.21	.85	1.00						
5	−.30	−.40	.77	.76	1.00					
6	−.18	−.48	.77	.49	.88	1.00				
7	−.33	−.37	1.00	.85	.79	.78	1.00			
8	−.40	−.24	.88	.98	.78	.55	.88	1.00		
9	−.30	−.40	.78	.76	1.00	.87	.79	.78	1.00	
10	−.18	−.47	.76	.49	.87	.99	.77	.52	.86	1.00

THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX
ALL ELEMENTS IN THE MATRIX PARAMETERS ARE ALLOWED TO BE ESTIMATED
−2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H5) IS −0.26704472E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX
ALL ELEMENTS IN THE MATRIX PARAMETERS ARE ALLOWED TO BE ESTIMATED
−2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H5*) IS −0.26876968E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX
THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO
−2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H4) IS −0.26703238E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX
THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO
−2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H4*) IS −0.26660243E + 04

(continued)

Table 24 (continued)

THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H3) IS −0.26014689E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H3*) IS −0.25922485E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H2) IS −0.25870429E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H1) IS −0.25733307E + 04
CAUSALITY TEST BETWEEN VARIABLES DUE AND DLEI
P, PP, Q, QQ, NP, NAR, NSAR, NMA, NSMA: 1 0 1 0 0 4 0 4 0
−2*(LOG LIKELIHOOD) UNDER H5,H5*,H4,H4*,H3,H3*,H2,H1 ARE: 1 −0.26704472E + 04 2 −0.26876968E + 04 3 −0.26703238E + 04 4 −0.26660243E + 04 5 −0.26014689E + 04 6 −0.25922485E + 04 7 −0.25870429E + 04 8 −0.25733307E + 04
−2*(LOG LIKELIHOOD) UNDER H1,H2,H3,H3*,H4,H4*,H5,H5* ARE: −0.25733306E + 04 −0.25870430E + 04 −0.26014690E + 04 −0.25922485E + 04 −0.26703237E + 04 −0.26660242E + 04 −0.26704473E + 04 −0.26876968E + 04 LR1 TO LR10: 68.97827148437500 0.1235351562500000 14.42602539062500 83.28076171875000 13.71240234375000 83.40429687500000 9.220458984375000 4.299560546875000 −17.24951171875000 97.11669921875000 DDF1 TO DDF10: 2.000000000000000 2.000000000000000 2.000000000000000 2.000000000000000 1.000000000000000 4.000000000000000 1.000000000000000 1.000000000000000 1.000000000000000 5.000000000000000 CHI1 TO CHI10: 9.210340416679674 9.210340416679674 9.210340416679674 9.210340416679674 6.634896640840623 13.27670418742523 6.634896640840623 6.634896640840623 6.634896640840623 15.08627252353956 T1 TO T10: 59.7679329 −9.08680534 5.21568489 74.0704193 7.07750559 70.1275940 2.58556223 −2.33533621 −23.8844090 82.0304260 IT1 TO IT10: 1 −1 1 1 1 1 1 −1 −1 1 I1 TO I6: 1 3 1 3 3 3
RESULT BASED ON THE BACKWARD PROCEDURE (Y:DUE , X: DLEI) DUE <<= DLEI (Y IS STRONGLY CAUSED BY X)
RESULT BASED ON THE FORWARD PROCEDURE (Y:DUE , X: DLEI) DUE <<= DLEI (Y IS STRONGLY CAUSED BY X)

This is also the case for DUE and DLWkUNCL, which we do not report because of space limitations.

Finally, as with *OxMetrics*, we extend our analysis for the 1959–2018 time period. We report an estimated DUE and DLEI transfer function for the longer time period with statistically significant lags at periods 1, 2, 3, and 9 (months). The residual standard error falls from 0.159 to 0.142 (Table 23). Guerard, Thomakos, and Kyriazai (2020) reported statistically significant RSS reductions in the unemployment rate using the LEI input in a transfer function for the 1959–2018 time period with Ox Metrics.

The Chen-Lee test produces a causality finding between DUE and DLEI for the 1959Q2–2018Q2 (see Table 24).

The application of the Chen and Lee (1990) test leads one to find causality in the DUE and DLEI relationship with 4-month DLEI lags for the entire 1959–2018 time period.

Summary and Conclusions

We examined the regression and time series association between real US GDP and the leading economic indicators during the 1959–2018Q2 time period. We report that The Conference Board leading economic indicators are statistically significant, once the regression line addresses outlier estimations in SAS (ETS). We applied the Hendry and Doornik automatic time series PC-Give (*OxMetrics*) methodology to several well-studied macroeconomics series, real GDP, and the unemployment rate. We report that the *OxMetrics* and *Autometrics* system and SCA system substantially reduce the regression sum of squares measures relative to a traditional variation on the random walk with drift model. The modeling process of including the leading economic indicator in forecasting real GDP has been addressed before, but our results are more statistically significant. The use of SAS-estimated transfer functions, Autometrics-estimated time series models with outliers, and SCA-estimated causality models leads researchers to several conclusions: (1) real GDP and the LEI are significantly associated; and (2) the US unemployment are is statistically associated, if not, caused, by The Conference Board leading economic indicators and the weekly unemployment claims time series.

Time Series Analysis Appendix 1: Table 19 (Continued) with Identified Outliers

The outlier analysis estimation, using a critical two standard deviation criteria for outliers, is:

PARAMETER LABEL	VARIABLE NAME	NUM./DENOM.	FACTOR	ORDER	CONSTRAINT	VALUE	STD ERROR	T-VALUE
1	DUE	MA	1	1	NONE	.7714	.1054	7.32
2	DUE	D-AR	1	1	NONE	.8752	.0823	10.64

SUMMARY OF OUTLIER DETECTION AND ADJUSTMENT

TIME	ESTIMATE	T-VALUE	TYPE
8	0.332	3.12	TC
11	−0.532	−4.96	TC
14	0.766	5.86	IO
17	0.325	2.51	IO
21	0.577	4.46	IO
23	0.426	3.34	AO
31	−0.424	−3.28	IO
34	−0.307	−2.92	TC
43	0.339	2.66	AO
46	0.327	2.56	AO
53	−0.314	−2.43	IO
66	−0.307	−2.41	AO
70	−0.324	−2.54	AO
74	−0.429	−3.32	IO
132	0.418	3.99	TC
142	0.352	2.76	AO
166	−0.266	−2.53	TC
188	0.369	2.88	AO
190	0.619	5.64	TC
192	0.595	4.47	AO
194	0.299	2.79	TC

197	−0.361	−3.39	TC
209	0.245	2.34	TC
216	−0.298	−2.31	IO
227	−0.373	−2.89	IO
234	0.363	2.85	AO
247	0.312	2.42	IO
255	0.605	5.75	TC
259	−0.298	−2.84	TC
264	0.346	2.68	IO
268	0.322	2.49	IO
270	−0.354	−2.77	AO
273	0.329	3.12	TC
284	0.316	3.00	TC
288	−0.506	−3.91	IO
294	−0.674	−5.22	IO
296	−0.340	−3.24	TC
306	0.373	2.93	AO
319	−0.308	−2.38	IO
325	0.547	4.29	AO
378	0.316	3.01	TC
405	−0.309	−2.39	IO

TOTAL NUMBER OF OBSERVATIONS.	419
EFFECTIVE NUMBER OF OBSERVATIONS.	418
RESIDUAL STANDARD ERROR (WITHOUT OUTLIER ADJUSTMENT). .	0.186459E + 00
RESIDUAL STANDARD ERROR (WITH OUTLIER ADJUSTMENT) . . .	0.129131E + 00

Time Series Analysis Appendix 2: Table 22 (Continued)

LAGS 13 THROUGH 18					
. −	− .	. .	. .	. .	. .
. +	+ .	. +	. .	. .	. .
LAGS 19 THROUGH 24					
. .	. .	. .	. .	. .	. .
. .	. .	. .	. .	. .	. .

========== STEPWISE AUTOREGRESSION SUMMARY ==========										
	I	RESIDUAL	I	EIGENVAL.	I	CHI-SQ	I		I	SIGNIFICANCE
LAG	I	VARIANCES	I	OF SIGMA	I	TEST	I	AIC	I	OF PARTIAL AR COEFF.
----	+	----------	+	----------	+	---------	+	----------	+	----------
1	I	.320E − 01	I	.297E − 01	I	224.58	I	−6.067	I	−
	I	.743E − 01	I	.766E − 01	I		I		I	+

NOTE: CHI-SQUARED CRITICAL VALUES WITH 4 DEGREES OF FREEDOM ARE
5%: 9.5 1%: 13.3

NOTE: THE PARTIAL AUTOREGRESSION COEFFICIENT MATRIX FOR LAG L IS THE
ESTIMATED PHI(L) FROM THE FIT WHERE THE MAXIMUM LAG USED IS L
(I.E., THE LAST COEFFICIENT MATRIX). THE ELEMENTS ARE
STANDARDIZED BY DIVIDING EACH BY ITS STANDARD ERROR

--

MTSMODEL ARMATF11. SERIES ARE DUE,DLEI. @
MODEL IS (1-PHI*B)SERIES=C+(1-TH1*B)NOISE.

SUMMARY FOR MULTIVARIATE ARMA MODEL – ARMATF11

VARIABLE	DIFFERENCING
DUE	
DLEI	

PARAMETER		FACTOR	ORDER	CONSTRAINT
1	C	CONSTANT	0	CC
2	PHI	REG AR	1	CPHI
3	TH1	REG MA	1	CTH1
--				

CAUSALTEST MODEL ARMATF11. OUTPUT PRINT(CORR). alpha .01

SUMMARY OF THE TIME SERIES

SERIES	NAME	MEAN	STD DEV	DIFFERENCE ORDER(S)
1	DUE	0.0012	0.1940	
2	DLEI	0.1081	0.3502	

ERROR COVARIANCE MATRIX

	1	2
1	.037737	
2	−.026752	.122671

(continued)

ITERATIONS TERMINATED DUE TO:
CHANGE IN (−2*LOG LIKELIHOOD)/NOBE .LE. 0.100E − 03
TOTAL NUMBER OF ITERATIONS IS 12
MODEL SUMMARY WITH MAXIMUM LIKELIHOOD PARAMETER ESTIMATES

CONSTANT VECTOR (STD ERROR)	
0.007	(0.003)
−0.011	(0.007)

PHI MATRICES
ESTIMATES OF PHI(1) MATRIX AND SIGNIFICANCE

.782	−.064	+ −
.511	1.095	+ +

STANDARD ERRORS

.055	.022
.129	.052

THETA MATRICES
ESTIMATES OF THETA(1) MATRIX AND SIGNIFICANCE

.805	.062	+ .
.294	.732	. +

STANDARD ERRORS

.063	.032
.151	.068

ERROR COVARIANCE MATRIX

	1	2
1	.029350	
2	−.006695	.065507

==========
SUMMARY OF FINAL PARAMETER ESTIMATES AND THEIR STANDARD ERRORS
==========

PARAMETER NUMBER	PARAMETER DESCRIPTION	FINAL ESTIMATE	ESTIMATED STD. ERROR
1	CONSTANT(1)	0.007181	0.003101
2	CONSTANT(2)	−0.010655	0.007280
3	AUTOREGRESSIVE (1, 1, 1)	0.781951	0.055383
4	AUTOREGRESSIVE (1, 1, 2)	−0.064217	0.022034
5	AUTOREGRESSIVE (1, 2, 1)	0.510876	0.129085
6	AUTOREGRESSIVE (1, 2, 2)	1.095485	0.051517
7	MOVING AVERAGE (1, 1, 1)	0.804953	0.063443
8	MOVING AVERAGE (1, 1, 2)	0.061788	0.031849
9	MOVING AVERAGE (1, 2, 1)	0.293963	0.150721
10	MOVING AVERAGE (1, 2, 2)	0.732353	0.068170

CORRELATION MATRIX OF THE PARAMETERS

	1	2	3	4	5	6	7	8	9	10

(continued)

1	1.00									
2	−.83	1.00								
3	−.71	.61	1.00							
4	−.79	.62	.89	1.00						
5	.55	−.72	−.81	−.68	1.00					
6	.62	−.79	−.77	−.79	.90	1.00				
7	−.67	.52	.89	.84	−.69	−.65	1.00			
8	−.58	.44	.54	.73	−.49	−.56	.56	1.00		
9	.52	−.68	−.75	−.65	.93	.86	−.73	−.49	1.00	
10	.52	−.65	−.60	−.66	.67	.82	−.53	−.62	.68	1.00

THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX ALL ELEMENTS IN THE MATRIX PARAMETERS ARE ALLOWED TO BE ESTIMATED −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H5) IS −0.17857400E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX ALL ELEMENTS IN THE MATRIX PARAMETERS ARE ALLOWED TO BE ESTIMATED −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H5*) IS −0.17751251E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H4) IS −0.17693824E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H4*) IS −0.17625596E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H3) IS −0.17192010E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H3*) IS −0.17085436E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO FULL MATRIX THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H2) IS −0.17073081E + 04
THE RESIDUAL COVARIANCE MATRIX IS SET TO DIAGONAL MATRIX THE (2,1)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO THE (1,2)TH ELEMENTS IN THE MATRIX PARAMETERS ARE SET TO ZERO −2*(LOG LIKELIHOOD AT FINAL ESTIMATES UNDER H1) IS −0.16916883E + 04
CAUSALITY TEST BETWEEN VARIABLES DUE AND DLEI
P, PP, Q, QQ, NP, NAR, NSAR, NMA, NSMA:
1 0 1 0 0 4 0 4 0
−2*(LOG LIKELIHOOD) UNDER H5,H5*,H4,H4*,H3,H3*,H2,H1 ARE:
1 −0.17857400E + 04
2 −0.17751251E + 04
3 −0.17693824E + 04
4 −0.17625596E + 04
5 −0.17192010E + 04
6 −0.17085436E + 04
7 −0.17073081E + 04
8 −0.16916883E + 04
−2*(LOG LIKELIHOOD) UNDER H1,H2,H3,H3*,H4,H4*,H5,H5* ARE:

(continued)

−0.16916884E + 04
−0.17073081E + 04
−0.17192010E + 04
−0.17085436E + 04
−0.17693824E + 04
−0.17625597E + 04
−0.17857400E + 04
−0.17751251E + 04
LR1 TO LR10: 66.53894042968750 16.35754394531250 11.89294433593750 62.07434082031250 15.61975097656250 78.43188476562500 10.65747070312500 6.822753906250000 10.61486816406250 94.05163574218750
DDF1 TO DDF10: 2.000000000000000 2.000000000000000 2.000000000000000 2.000000000000000 1.000000000000000 4.000000000000000 1.000000000000000 1.000000000000000 1.000000000000000 5.000000000000000
CHI1 TO CHI10: 9.210340416679674 9.210340416679674 9.210340416679674 9.210340416679674 6.634896640840623 13.27670418742523 6.634896640840623 6.634896640840623 6.634896640840623 15.08627252353956
T1 TO T10: 57.3286018 7.14720345 2.68260384 52.8640022 8.98485470 65.1551819 4.02257395 0.187857270 3.97997141 78.9653625
IT1 TO IT10: 1 1 1 1 1 1 1 1 1 1
I1 TO I6: 3 3 3 3 3 3
RESULT BASED ON THE BACKWARD PROCEDURE (Y:DUE , X: DLEI)
DUE <=> DLEI (Y AND X HAVE FEEDBACK RELATION)
RESULT BASED ON THE FORWARD PROCEDURE (Y:DUE , X: DLEI)
DUE <=> DLEI (Y AND X HAVE FEEDBACK RELATION)

References

Ashley, R. (1998). A new technique for postsample model selection and validation. *Journal of Economic Dynamics and Control, 22*, 647–665.

Ashley, R. A. (2003). Statistically significant forecasting improvements: How much out-of-sample data is likely necessary? *International Journal of Forecasting, 19*, 229–240.

Ashley, R., Granger, C. W. J., & Schmalensee, R. (1980). Advertising and aggregate consumption: An analysis of causality. *Econometrica, 48*, 149–1167.

Batchelor, R. (2001). Confidence indexes and the probability of recession: A Markov switching model. *Indian Economic Review, 36*, 107–124.

Box, G. E. P., & Jenkins, G. M. (1970). *Time series analysis: Forecasting and control*. San Francisco: Holden-Day.

Burns, A. F., & Mitchell, W. C. (1946). *Measuring business cycles*. New York: NBER.

Castle, J., Doornik, J. A., & Hendry, D. F. (2013). Model selection in equations with many 'small' effects. *Oxford Bulletin of Economics and Statistics*, 75, 6–22.

Chen, C., & Lee, C. J. (1990). A VARMA test on the Gibson paradox. *Review of Economics and Statistics, 72*, 96–107.

Chen, C., & Liu, L.-M. (1993a). Joint estimation of model parameters and outliers in time series. *Journal of the American Statistical Association, 88*, 284–297.

Chen, C., & Liu, L.-M. (1993b). Forecasting time series with outliers. *Journal of Forecasting, 12*, 13–35.

Clemen, R. T., & Guerard, J. B. (1989). Econometric GNP forecasts: Incremental information relative to naive extrapolation. *International Journal of Forecasting, 5*, 417–426.
Clements, M. P., & Hendry, D. F. (1998). *Forecasting economic time series*. Cambridge: Cambridge University Press.
Dhrymes, P. J. (2017). *Introductory econometrics (Revised Edition)*. New York: Springer.
Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics, 13*, 253–263.
Diebold, F., & Rudebusch, G. D. (1999a). *Business cycles*. Princeton: Princeton University Press.
Diebold, F. X., & Rudebusch, G. D. (1999b). *Business cycles: Durations, dynamics and forecasting*. Princeton: Princeton University Press.
Doornik, J. A. (2009). Autometrics. In J. Castle & N. Shepard (Eds.), *The methodology and practice of econometrics*. Oxford: Oxford University Press.
Doornik, J. A., & Hendry, D. F. (2013a). *PCGIVE*. London: Timberlake Consultants, Ltd.
Doornik, J. A., & Hendry, D. F. (2013b). *Empirical econometric modelling*. London: Timberlake Consultants, Ltd.
Doornik, J. A., & Hendry, D. F. (2015). Statistical model selection with 'big data'. *Cogent Economics & Finance, 3*(2015), 1–15.
Friedman, M., & Schwartz, A. (1963). Money and business cycles. *Review of Economics and Statistics, 45*, 32–64.
Gordon, R. J. (1986). *The American business cycle*. Chicago: University of Chicago Press.
Granger, C. W. J. (1969). Investigating casual relations by economic models and cross-spectral methods. *Econometrica, 37*, 424–438.
Granger, C. W. J. (1980). Testing for causality: A personal viewpoint. *Journal of Economic Dynamics and Control, 2*, 329–352.
Granger, C. W. J. (1986). Developments in the study of cointegrated economic variables. *Oxford Bulletin of Economics and Statistics, 48*, 213–228.
Granger, C. W. J. (1989). *Forecasting in business and economics*. New York: Academic Press.
Granger, C. W. J. (2001). In E. Ghysels, N. R. Swanson, & M. W. Watson (Eds.), *Essays in econometrics*. Two Volumes. Cambridge: Cambridge University Press.
Granger, C. W. J., & Newbold, P. (1977). *Forecasting economic time series*. New York: Academic Press.
Guerard Jr., J. B. (1990). Note: NYSE Volume: An application of outlier analysis. *Journal of Forecasting, 9*, 467–471.
Guerard Jr., J. B. (2001). A note on the forecasting effectiveness of the U.S. leading economic indicators. *Indian Economic Review, 36*, 251–268.
Guerard Jr., J. B. (2004). The forecasting effectiveness of the U.S. leading economic indicators: Further evidence and initial G7 results. In P. Dua (Ed.), *Business cycles and economic growth: An analysis using leading indicators* (pp. 174–187). New York: Oxford University Press.
Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.
Guerard, J., Thomakos, D., & Kyriazai. (2020). Automatic time series modeling and forecasting. A Replication case study of real GDP, the unemployment rate and the impact of leading economic indicators. Cogent economics & finance 8:1759483. https://doi.org/10.1080/2332039.2020.1759483.
Hamilton, J. R. (1994). *Time series analysis*. Princeton: Princeton University Press.
Hamilton, J. D. (2011). Calling recessions in real time. *International Journal of Forecasting, 27*, 1006–1026.
Hendry, D. F. (1986). *Using PC-Give in Econometrics Teaching*. Oxford Bulletin of Economics & Statistics, 48, 87–98
Hendry, D. F. (2000). *Econometrics: Alchemy or science?* Oxford: Oxford University Press.
Hendry, D. F., & Doornik, J. A. (2014). *Empirical model discovery and theory evaluation*. Cambridge: MIT Press.
Hendry, D. F., & Krolzig, H. M. (2005). The properties of automatic gets modelling. *The Economic Journal*, 115, c32–c61.
Hendry, D. F., & Nielsen, B. (2007). *Econometric modeling: A likelihood approach*. Princeton: Princeton University Press.

Koopmans, T. C. (1947). Measure without theory. *The Review of Economic Statistics*, reprinted in R. Gordon & L. Klein. (1965). *Readings in business cycles*. Homewood, IL: Richard A. Irwin, Inc.

Krolzig, H.-M. (2001). Business cycle measurement in the presence of structural change: International evidence. *International Journal of Forecasting, 17*, 349–368.

Krolzig, H.-M., & Hendry, D. F. (2001). Computer automation of general-to-specific model selection procedures. *Journal of Economic Dynamics and Control*, 831–866.

Lahiri, K., & Moore, G. H. (1991). *Leading economic indicators*. Cambridge: Cambridge University Press.

Liu, L.-M. (2006). Time Series Analysis and Forcasting. Scientific Computing Associates. Villa Park, Illinois: Scientific Computing Associates® Corp.

Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika, 65*, 297–303.

Makridakis, S., & Hibon, M. (2000). The M3-competition: Results, conclusions and implications. *International Journal of Forecasting, 16*, 451–476.

McCracken, M. (2000). Robust out-of-sample inference. *Journal of Econometrics, 39*, 195–223.

McLeod, A. I., & Li, W. K. (1983). Diagnostic checking ARMA time series models using squared residual autocorrelations. *Journal of Time Series Analysis, 4*, 269–273.

Mincer, J., & Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (Ed.), *Economic forecasts and expectations*. New York: Columbia University Press.

Mitchell, W. C. (1913). *Business cycles*. New York: Burt Franklin reprint.

Mitchell, W. C. (1951). *What happens during business cycles: A progress report*. New York: NBER.

Montgomery, A. L., Zarnowitz, V., Tsay, R. S., & Tiao, G. C. (1998). Forecasting the U.S. unemployment rate. *Journal of the American Statistical Association, 93*, 478–493.

Moore, G. H. (1961). *Business cycle indicators* (Vol. 2 volumes). Princeton: Princeton University Press.

Nelson, C. R., & Plosser, C. I. (1982). Trends and random walks in macroeconomic time series. *Journal of Monetary Economics, 10*, 139–162.

Reilly, D. R. (1980). Experiences with an Automatic Box-Jenkins Modelling Algorithm. In O. D. Anderson (ed.), *Time series analysis*. Amsterdam: North-Holland.

Tashman, L. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting, 16*, 437–450.

Theil, H. (1966). *Applied economic forecasting*. Amsterdam: North-Holland.

Thomakos, D., & Guerard, J. (2004). Naïve, ARIMA, transfer function, and VAR models: A comparison of forecasting performance. *The International Journal of Forecasting, 20*, 53–67.

Tsay, R. S. (1988). Outliers, level shifts, and variance changes in Time Series. *Journal of Forecasting 7,* 1–20.

Tsay, R. S. (1989). Testing and modeling threshold autoregressive processes. *Journal of the American Statistical Association 84*, 231–249.

Tsay, R. S. (2010). *Analysis of financial time series* (3rd ed.). New York, NY: Wiley.

Vining, R. (1949). Koopmans on the choice of variables to be studied and on methods of measurement. *The Review of Economics and Statistics*, reprinted in R. Gorgon & L. Klein. (1965). *Readings in business cycles*. Homewood, IL: Richard A. Irwin, Inc.

West, K., & McCracken, M. (1998). Regression-based tests of predictive ability. *International Economic Review, 39*, 817–840.

Zarnowitz, V. (1992). *Business cycles: Theory, history, indicators, and forecasting*. Chicago: University of Chicago Press.

Zarnowitz, V. (2001). *The old and the new in the U.S. economic expansion*. The Conference Board. EPWP #01-01.

Zarnowitz, V. (2004). The autonomy of recent US growth and business cycles. In P. Dua (Ed.), *Business cycles and economic growth: An analysis using leading indicators* (pp. 44–82). New York: Oxford University Press.

Zarnowitz, V., & Ozyildirim, A. (2001). On the measurement of business cycles and growth cycles. *Indian Economic Review, 36*, 34–54.

Risk and Return of Equity and the Capital Asset Pricing Model



Individual investors must be compensated for bearing risk. It seems intuitive to the reader that there should be a direct linkage between the risk of a security and its rate of return. We are interested in securing the maximum return for a given level of risk or the minimum risk for a given level of return. The concept of such risk-return analysis is the efficient frontier of Harry Markowitz (1952, 1959). If an investor can invest in a government security, which is backed by the taxing power of the federal government, then that government security is relatively risk-free. The 90-day treasury bill rate is used as the basic risk-free rate. Supposedly the taxing power of the federal government eliminates default risk of government debt issues. A liquidity premium is paid for longer-term maturities, due to the increasing level of interest rate risk. Investors are paid interest payments, as determined by the bond's coupon rate, and may earn market price appreciation on longer bonds if market rates fall or losses vice versa. During the period from 1928 to 2017, treasury bills returned 3.44%, longer-term (10-year treasury) government bonds earned 5.15%, and corporate stocks, as measured by the stock of the S&P 500 Index, earned 11.53% annually, as measured by the mean annual return. The annualized standard deviations are 3.06%, 7.72%, and 19.662%, respectively, for treasury bills, treasury bonds, and S&P stocks.[1] The risk-return trade-off has been relevant for the 1928–2017 period. The correlation coefficient between annual returns for treasury bills and the S&P 500 stock returns was −.030 for the 1928–2017 time period. This was essentially no correlation between treasury bills and large stocks, as measured by the S&P 500 stock. The correlation coefficient between annual returns for treasury bonds and the S&P 500 stock returns was 0.30 for the 1928–2017 time period. Why do corporate stocks offer investors such returns?

First, as a stockholder, one owns a fraction, a very small fraction for many investors, of the firm. When one owns stocks, one is paid a dividend and earns stock price appreciation. That is, an investor buys stock when he or she expects its

[1] Ibbottson and Sinquefield, *Stocks, Bonds, and Bills Yearbook*, 2018.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_14

stock price to increase and compensate the investor for bearing the risk of the stock's price movements. Investors have become aware in recent years that not all price movements are in positive directions. In this chapter, we introduce the reader to (1) the Markowitz and Sharpe risk-return trade-offs; (2) the calculation of historic expected returns and standard deviations of stocks; (3) beta estimations in the Capital Asset Pricing Model; and (4) a series of financial variables and regression-based models of those variables for statistically significant stock selection. One can calculate expected returns on stocks with several different methods. This chapter, and chapter "Multifactor Risk Models and Portfolio Construction and Management," on multifactor risk models, will be the two chapters to allow the reader to understand quantitative modeling and serve as an introduction to financial engineering.

Calculating Holding Period Returns

A very simple concept is the holding period return (HPR) calculation, in which one assumes that the stock was purchased at last (period's) year's price and the investor earns a dividend per share for the current year and a price, appreciation (depreciation) relative to last year's price:

$$HPR_t = \frac{D_t + P_t - P_{t-1}}{P_{t-1}}$$

where

D_t = current year′s dividend,
P_t = current year′s stock price,
P_{t-1} = last year′s stock price
HPR_t = current year′s holding period return.

The assumption of annual returns is arbitrary, but well-known in finance. Markowitz (1959)) used annual returns in his Chapter 6 of *Portfolio Selection* to illustrate holding period returns and investment in the long-run. Guerard and Schwartz (2007) examined three widely held stocks: DuPont, Dominion Resources, and IBM, for 1994–2003 period. The pricing data was taken from the Standard & Poor's *Stock Guide*. The *S&P Stock Guide* presents high and low prices during the calendar year. See Table 1. An average price (AvgP) can be calculated by simply summing the high and low prices and dividing by two.

The IBM holding period returns (HPRs) were consistently positive and rather large during the 1995–1999 period. The reader sees that the HPRs of stocks are often far more dependent upon stock price movements than the dividends received by investors. The HPRs of DuPont range from 40.35% in 1997 to −23.94% in 2001.

Table 1 Annual dividends and prices, 1994–2003

	D				DD				IBM			
Year	Phigh	Plow	Div	AvgP	Phigh	Plow	Div	AvgP	Phigh	Plow	Div	AvgP
2003	65.95	51.74	2.58	58.85	46.00	38.00	1.40	42.00	91.54	73.17	0.63	82.355
2002	67.05	35.40	2.58	51.23	49.80	35.02	1.40	42.41	126.39	54.01	0.59	90.200
2001	69.99	55.13	2.58	62.56	49.88	32.64	1.40	41.26	124.70	83.75	0.55	104.225
2000	67.93	34.81	2.58	51.37	74.00	38.18	1.40	56.09	134.93	80.06	0.51	107.495
1999	49.37	36.56	2.58	42.97	75.18	50.06	1.40	62.62	139.18	80.87	0.47	110.025
1998	48.93	37.81	2.58	43.37	84.43	51.68	1.36	68.06	94.96	47.81	0.44	71.385
1997	42.87	33.25	2.58	38.06	69.75	46.37	1.23	58.06	56.75	31.78	0.39	44.265
1996	44.37	36.87	2.58	40.62	49.68	34.81	1.11	42.25	41.50	20.78	0.33	31.140
1995	41.62	34.87	2.58	38.25	36.50	26.31	1.02	31.41	28.65	17.56	0.25	23.105
1994	45.37	34.87	2.55	40.12	31.18	24.12	0.91	27.65	19.09	12.84	0.25	15.965

Table 2 Holding period returns, 1995–2003

	D	DD	IBM
Year	HPR	HPR	HPR
2003	0.1991	0.0233	−0.0800
2002	−0.1399	0.0618	−0.1289
2001	0.2681	−0.2394	−0.0253
2000	0.2557	−0.0819	−0.0184
1999	0.0501	−0.0593	0.5479
1998	0.2073	0.1956	0.6226
1997	0.0005	0.4035	0.4340
1996	0.1296	0.3805	0.3620
1995	0.0176	0.1727	0.4629

One can estimate an expected return for DuPont by calculating the mean value of the annual HPRs during the 1995–2003 period. The expected return for DuPont during the 1995–2003 period was 9.50%, with a standard deviation of 21.40%. If the annual HPRs of DuPont are normally distributed, that is, the returns fall within the normal "Bell" curve, then 67.6% of the annual observations of DuPont returns should fall within the −11.90 and 30.90 range (one standard deviations). The reader immediately sees how wide the one standard deviation range is for annual returns. See Table 2. We can calculate in a similar manner the expected returns for IBM and Dominion Resources. One sees that the returns are extremely volatile for IBM, having a standard deviation of 30.00%. It may be worthwhile to calculate a coefficient of variation (CV) in which the standard deviation is divided by the expected return. The calculation of the CV is:

Security	Ticker	E(R)	σ	CV
DuPont	DD	.095	.214	2.253
Dominion resources	D	.110	.137	1.245
IBM	IBM	.242	.300	1.240

Source: SAS

which leads the investor to recognize that DuPont, more than IBM or Dominion Resources, produces greater variation for a given level of expected returns.

Markowitz analysis seeks to minimize the risk for a given level of return. In Chapters 7 and 8 of Markowitz (1959), we are taught to estimate portfolios that minimize risk.

The portfolio variance is given by the weighted asset variances and covariances:

$$\begin{aligned} \sigma_p^2 &= x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1(1-x_1)\sigma_{12} \\ \sigma_{12} &= \rho_{12}\sigma_1\sigma_2 \\ \mathrm{p}_{12} &= \text{correlation coefficients of assets } 1,2 \\ \sigma_{12} &= \text{covariance of assets } 1,2 \end{aligned} \tag{1}$$

Minimizing Risk

To minimize risk, one seeks to allocate resources to assets such that the change in risk goes to zero with the change in the percentage invested in the asset. That is, risk minimization implies a first partial derivative of zero, with respect to the change in the asset weight:

$$
\begin{gathered}
\text{Let } x_2 = 1 - x_1 \\
\sigma_p^2 = x_1^2\sigma_1^2 + (1 - x_1)^2\sigma_2^2 + 2x_1(1 - x_1)\sigma_{12} \\
= x_1^2\sigma_1^2 + (1 - x_1)^2\sigma_2^2 + 2x_1(1 - x_1)\rho_{12}\sigma_1\sigma_2 \\
= x_1^2\sigma_1^2 + (1 - x_1)(1 - x_1)\sigma_2^2 + 2x_1\rho_{12}\sigma_1\sigma_2 - 2x_1^2\rho_{12}\sigma_1\sigma_2 \\
\sigma_p^2 = x_1^2\sigma_1^2 + \left(1 - 2x_1 + x_1^2\right)\sigma_2^2 + 2x_1\rho_{12}\sigma_1\sigma_2 - 2x_1^2\rho_{12}\sigma_1\sigma_2 \\
\frac{\partial \sigma p^2}{\partial x_1} = 2x_1\sigma_1^2 - 2\sigma_2^2 + 2x_1\sigma_2^2 + 2p_{12}\sigma_1\sigma_2 - 4x_1p_{12}\sigma_1\sigma_2 = 0 \\
2\sigma_2^2 - 2p_{12}\sigma_1\sigma_2 = 2x_1\sigma_1^2 + 2x_1\sigma_2^2 - 4x_1\rho_{12}\sigma_1\sigma_2 \\
\left(\sigma_1^2 + \sigma_2^2 - 2p_{12}\sigma_1\sigma_2\right)x_1 = \sigma_2^2 - \rho_{12}\sigma_1\sigma_2 \\
x_1 = \frac{\sigma_2(\sigma_2 - p_{12}\sigma_2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}
\end{gathered}
\tag{2}
$$

Equation (2) shows the risk-minimizing weight (percentage invested) of asset one in the portfolio.

Let us assume that the student has access to the CRSP database, the Center for Research in Security Prices database, developed by James Lorie and Lawrence Fisher at the University of Chicago during the 1960–1965 time period. The CRSP database contains, most many data, monthly stock prices, returns, dividends, and shares outstanding for publicly traded US stocks, 1926–present. Instead of tracking down stock prices and dividends, the student can access stock returns for virtually all US stocks traded. Let us continue to use IBM, Boeing, and Dominion Energy, the stocks used in our financial statement and valuation analysis. We can use 60 months of data, from January 2012 to December 2016, to calculate average returns, standard deviations, and correlation coefficients, to build risk-minimizing portfolios (Table 3).

In an equally weighted portfolio, $x_1 = x_2 = .50$. Let x_1 = weight of IBM and x_2 = weight of D. The portfolio expected return is a weighted combination of asset expected returns:

$$
\begin{gathered}
E(Rp) = x_1E(R_1) + x_2E(R_2) \\
= .5(.0017) + .5(.010) \\
= .0009 + .0050 = .0059 \text{ (monthly)}
\end{gathered}
\tag{3}
$$

Table 3 CRSP monthly returns (January 2012–December 2016)

			Rho		
Stock	Mean	STD	IBM	D	BA
IBM	0.0017	0.0503	1.0000	−0.0019	0.1813
D	0.0100	0.0382	−0.0019	1.0000	0.0633
BA	0.0164	0.0574	0.1813	0.0633	1.0000

Source: CRSP database

The annual expected return of IBM and D equally weighted portfolio is 12 * .0059 = .0708, or 7.08%:

$$\begin{aligned}\sigma_p{}^2 &= (.5)^2(.0503) + (.5)^2(.0382) + 2(.5)(.5)(-.0019)(.0503)(.0382)\\ &= .25(.0025) + .25(.0014) + (-.0000)\\ &= .0006 + .0004 + (-.0000) = .0010\\ \sigma p &= \sqrt{.0010} = .032\end{aligned}$$

The expected return on an equally weighted portfolio of IBM and D stock is 7.08%, and its corresponding standard deviation is 11.09%. The portfolio return should lie within the range of −4.25% and 18.17% approximately 67.6% of the time. This range corresponds to the expected return plus and minus one standard deviation of return.

Markowitz analysis sought to minimize risk for a given level of return. Thus, one could construct an infinite number of portfolios, by varying security weights, but the efficient frontier contained securities with weights that maximized return for a given level of risk.

The capital market line (CML) was developed to describe the return-risk trade-off assuming that investors could borrow and lend at the risk-free rate, RF, and that investors must be compensated for bearing risk. Investors seek to hold mean-variance efficient portfolios, invest for a one-period horizon, pay no taxes or transactions costs (we wish), and have homogeneous beliefs. All investors have identical probabilities of the distribution of future returns of securities:

$$E(\widetilde{R}p)\overleftrightarrow{} = R_F + \left[\frac{E(\widetilde{R}_M) - R_F}{\sigma_M}\right]\sigma_P \tag{4}$$

$E(R_P)$ = expected return on the portfolio
$E(R_M)$ = expected return on the market portfolio, where all securities are held relative to their market value
σ_M = standard deviation of the market portfolio
σ_P = standard deviation of the portfolio

The reader notes that as the standard deviation of the portfolio rises, its expected return must rise.

An Introduction to Modern Portfolio Theory

Markowitz created a portfolio construction theory in which investors should be compensated with higher returns for bearing higher risk. The Markowitz framework measured risk as the portfolio standard deviation, its measure of dispersion, or total risk. The Sharpe (1964), Lintner (1965), and Mossin (1966) development of the Capital Asset Pricing Model (CAPM) held that investors are compensated for bearing not total risk, but rather market risk, or systematic risk, as measured by the stock beta. An investor is not compensated for bearing risk that may be diversified away from the portfolio. The beta is the slope of the market model, in which the stock return is regressed as a function of the market return.

The CAPM holds that the return to a security is a function of the security beta:

$$R_{jt} = R_F + \beta_j[E(R_{Mt}) - R_F] + e_j \tag{5}$$

where

R_{jt} = expected security return at time t
$E(R_{Mt})$ = expected return on the market at time t
R_F = risk-free rate
β_j = security beta
e_j = randomly distributed error term

Let us examine the Capital Asset Pricing Model beta, its measure of systematic risk, from the capital market line equilibrium condition:

$$\beta_j = \frac{\mathrm{Cov}(R_j, R_M)}{\mathrm{Var}(R_M)} \tag{6}$$

$$E(R_j) = R_F + \left[\frac{E(R_M) - R_F}{\sigma_M{}^2}\right]\mathrm{Cov}\,(R_j, R_M)$$

$$= R_F + [E(R_M) - R_F]\frac{\mathrm{Cov}(R_j, R_M)}{\mathrm{Var}(R_M)} \tag{7}$$

$$E(R_j) = R_F + [E(R_M) - R_F]\beta_j$$

The security market line, SML, shown in Eq. (7), is the linear relationship between return and systematic risk, as measured by beta.

Let us estimate beta coefficients to be used in the Capital Asset Pricing Model (CAPM), to determine the rate of return on equity. One can regress monthly HPRs against a value-weighted Center for Research in Security Prices (CRSP) index, an index of all publicly traded securities. Most security betas are estimated using 5 years of monthly data, January 2012–December 2016, 60 observations, although one can

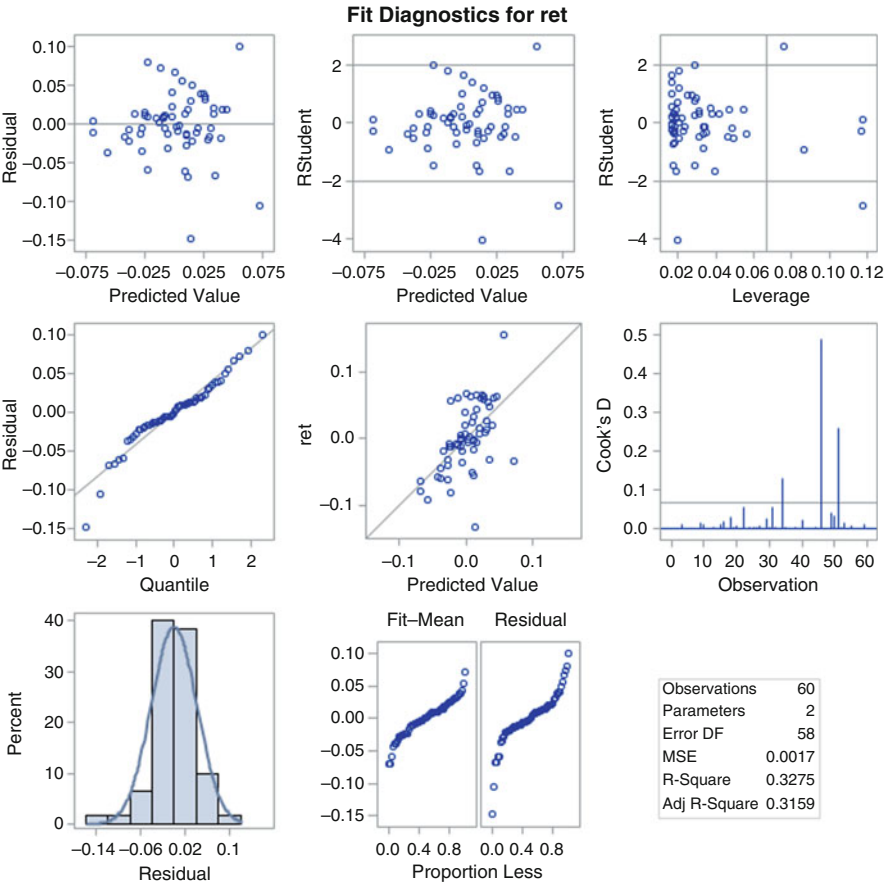
use almost any number of observations.[2] We use Proc Reg in SAS to estimate ordinary least squares (OKS) betas and SAS Proc Robustreg to estimate robust regression stock betas. One can use the Standard & Poor's 500 Index or the CRSP value-weighted stock index. The IBM beta is 0.963 when estimated using the S&P 500 Index and 0.965 when estimated using value-weighted CRSP index. Let us estimate a robust regression beta using the proc robustreg in SAS using the least trimmed squares (LTS) algorithm with the Tukey biweight, weighting function with 99% efficiency criteria and in chapter "Regression Analysis and Estimating Regression Models" and implemented in SAS.

The IBM beta is 0.963 when estimated using the S&P 500 Index and is statistically different from zero, using a t-statistic of 5.31. See Table 4. However, the more important question is whether the IBM beta is statistically different from 1.0, the average of all stocks in the S&P 500. The answer is no. The beta of 0.963 is 0.037 (i.e., 1 − .963) less than one, and the beta's estimated standard error is 0.1812 (as the reader remembers from chapter "Regression Analysis and Estimating Regression Models"), and thus the IBM beta of 0.963 has a t-value of .204 (i.e., (1 − .963)/.1812) from being different from 1.0. The IBM beta is not statistically different from one.

Table 4 OLS SP IBM beta

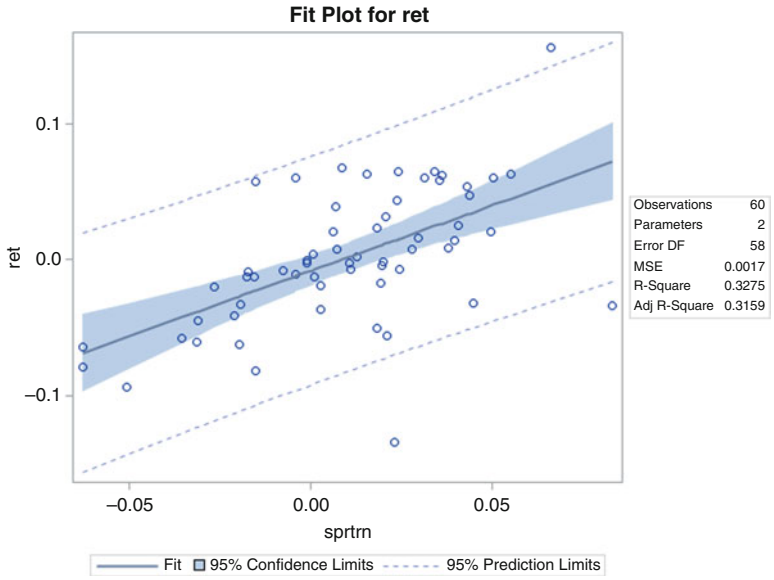
The REG procedure					
Model: MODEL1					
Dependent variable: ret					
Number of observations read					60
Number of observations used					60
Analysis of variance					
Source	**DF**	**Sum of squares**	**Mean square**	**F value**	**Pr > F**
Model	1	0.04880	0.04880	28.24	<.0001
Error	58	0.10022	0.00173		
Corrected total	59	0.14901			
Root MSE		0.04157	**R-Square**		0.3275
Dependent mean		0.00170	**Adj R-Sq**		0.3159
Coeff var		2438.46820			
Parameter estimates					
Variable	**DF**	**Parameter estimate**	**Standard error**	**t value**	**Pr > |t|**
Intercept	**1**	−0.00802	0.00567	−1.41	0.1626
sprtrn	**1**	0.96285	0.18118	5.31	<.0001

[2]One generally needs 30 observations for normality of residuals to occur, from the central limit theorem of statistics.



Source: SAS, Proc Reg

The plot of the student residuals (RStudent) (see below) indicates 4 of 60 observations exceed ±2, indicating possible outliers n the beta regression. There are three observations with large Cook's D leverage statistics, indicating possible regression problems.



Source: SAS, Proc Reg

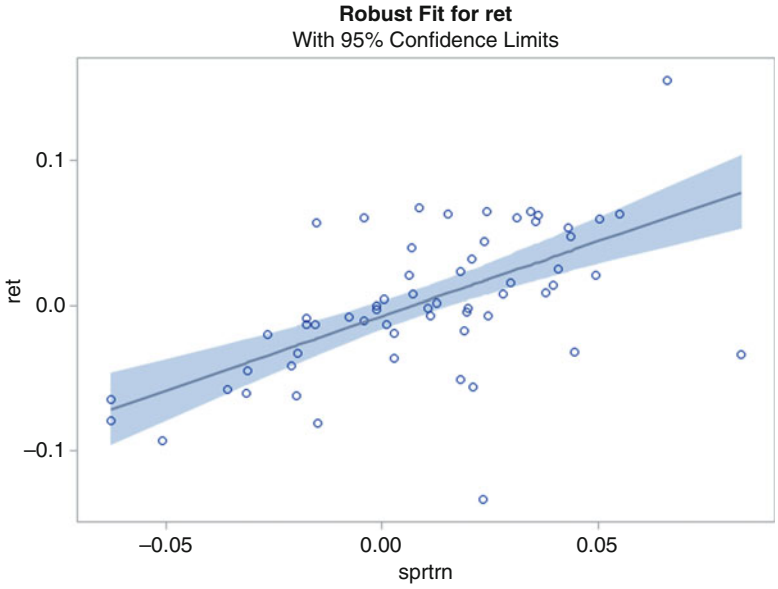
Table 5 Robust CRSP SP IBM beta

The ROBUSTREG procedure							
Model information							
Dataset						WORK.IBM	
Dependent variable						Ret	
Number of independent variables						1	
Number of observations						60	
Method						MM estimation	
Profile for the initial LTS estimate							
Total number of observations						60	
Number of squares minimized						45	
Number of coefficients						2	
Highest possible breakdown value						0.2667	
MM profile							
Chi function						Tukey	
K1						7.0410	
Efficiency						0.9900	
Parameter estimates							
Parameter	**DF**	**Estimate**	**Standard error**	**95% confidence limits**		**Chi-square**	**Pr > ChiSq**
Intercept	1	−0.0071	0.0050	−0.0169	0.0028	1.98	0.1598
Sprtrn	1	1.0285	0.1641	0.7069	1.3501	39.30	<.0001
Scale	0	0.0350					

Source: SAS, Proc Robustreg

The IBM robust regression beta is 1.029 when estimated using the S&P 500 Index and is statistically different from zero, using a chi-squared statistic of 39.30; again, the robust regression IBM beta is not statistically different from 1.0, the average of all stocks in the S&P 500. See Table 5.

Diagnostics summary		
Observation type	**Proportion**	**Cutoff**
Outlier	0.0333	3.0000

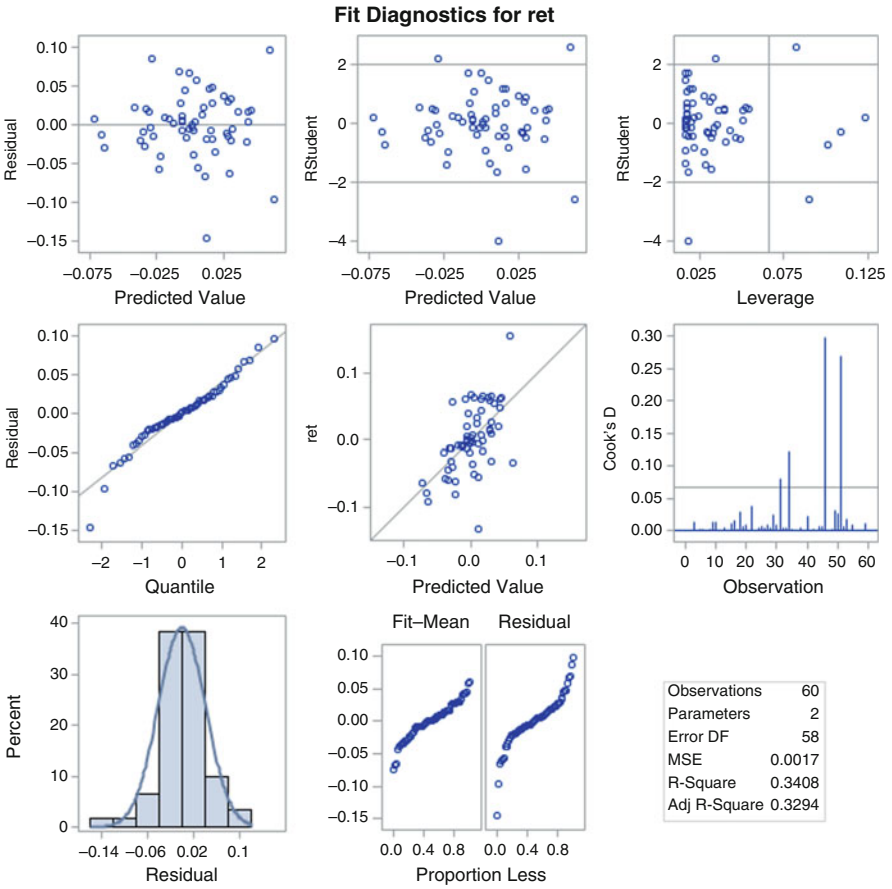


Source: SAS, Proc Robustreg

Goodness of fit	
Statistic	**Value**
R-square	0.3301
AICR	74.1068
BICR	78.5940
Deviance	0.0864

The IBM beta is 0.965 when estimated using value-weighted CRSP index.

The IBM CRSP value-weighted beta, 0.965, is not statistically different from one. See Table 6. One must again use robust regression because student residuals and Cook's D measures again indicate possible outlier issues.



Source: SAS, Proc Reg

Table 6 OLS CRSP VL-weighted IBM beta

The REG procedure
Model: MODEL1
Dependent variable: ret

Number of observations read	60
Number of observations used	60

Analysis of variance

Source	**DF**	**Sum of squares**	**Mean square**	**F value**	**Pr > F**
Model	1	0.05078	0.05078	29.98	<.0001
Error	58	0.09824	0.00169		
Corrected total	59	0.14901			

(continued)

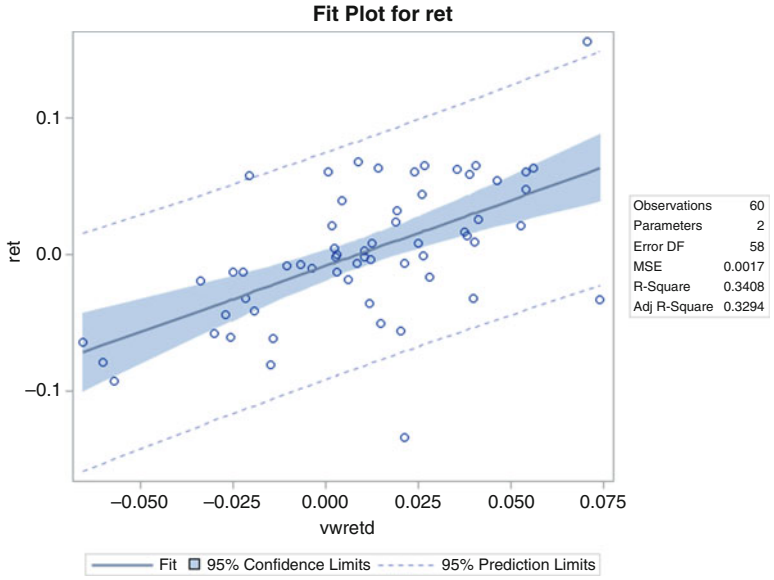
Table 6 (continued)

Root MSE	0.04116	**R-Square**	0.3408
Dependent mean	0.00170	**Adj R-Sq**	0.3294
Coeff var	2414.25685		

Parameter estimates

Variable	DF	Parameter estimate	Standard error	t value	Pr > |t|
Intercept	**1**	−0.00867	0.00564	−1.54	0.1299
Vwretd	**1**	0.96462	0.17617	5.48	<.0001

Source: SAS, Proc Reg



Source: Proc Reg

The IBM robust regression beta is 0.999 when estimated using the CRSP value-weighted index and is statistically different from zero, using a chi-squared statistic of 37.86; again, the robust regression IBM beta is not statistically different from 1.0, the average of all stocks. See Table 7.

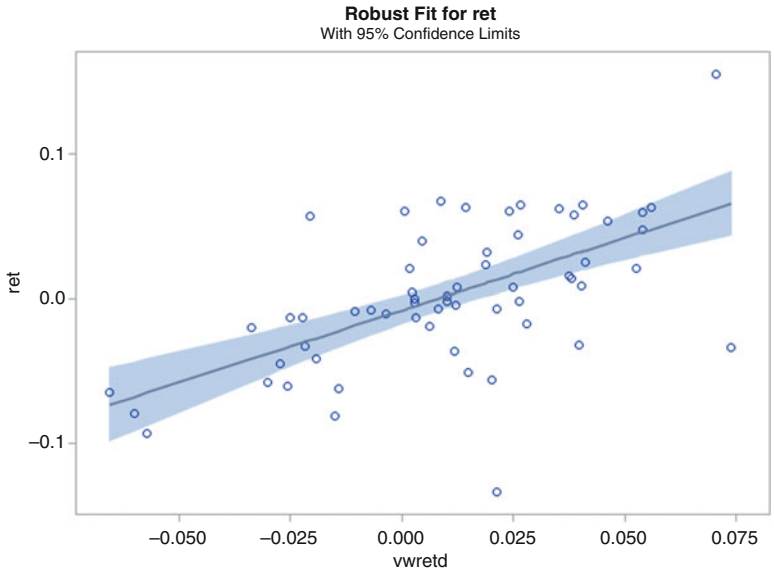


Table 7 Robust CRSP VL-weighted IBM beta

The ROBUSTREG procedure							
Model information							
Dataset	WORK.IBM						
Dependent variable	Ret						
Number of independent variables	1						
Number of observations	60						
Method	MM estimation						
Profile for the initial LTS estimate							
Total number of observations	60						
Number of squares minimized	45						
Number of coefficients	2						
Highest possible breakdown value	0.2667						
MM profile							
Chi function	Tukey						
K1	7.0410						
Efficiency	0.9900						
Parameter estimates							
Parameter	**DF**	**Estimate**	**Standard error**	**95% Confidence limits**		**Chi-square**	**Pr > ChiSq**
Intercept	1	−0.0079	0.0051	−0.0180	0.0022	2.34	0.1261
vwretd	1	0.9999	0.1625	0.6814	1.3183	37.86	<.0001
Scale	0	0.0361					

Source: SAS, Proc Robustreg

Goodness of fit	
Statistic	**Value**
R-square	0.3354
AICR	70.1374
BICR	74.6747
Deviance	0.0864

Source: SAS, Proc Robustreg

Let us estimate the Dominion Energy, D, stock betas are estimated using 5 years of monthly data, January 2012–December 2016, using either the Standard & Poor's 500 Index, or the Dow Jones Industrial Index (DJIA), or many other stock indexes. The D beta is 0.217 when estimated using the S&P 500 Index. The Dominion OLS SP500 beta is not statistically different from zero, with its t-statistic of 1.31, and is statistically different from one. The SAS Proc Reg output is the following (Table 8).

Dominion Energy, like many large (stable) primarily electric utility firms is not very risky and has very low beta, 0.217.

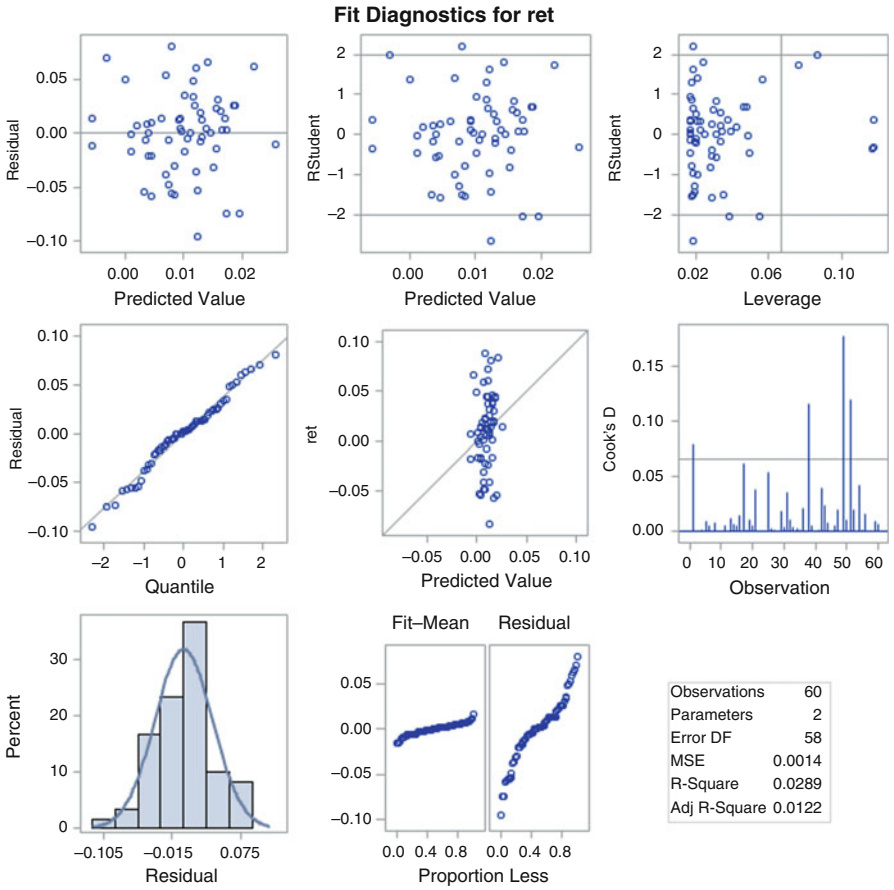


Table 8 OLS SP D beta

The REG procedure
Model: MODEL1
Dependent variable: ret

Number of observations read	60
Number of observations used	60

Analysis of variance

Source	DF	Sum of squares	Mean square	F value	Pr > F
Model	1	0.00249	0.00249	1.73	0.1941
Error	58	0.08354	0.00144		
Corrected total	59	0.08603			

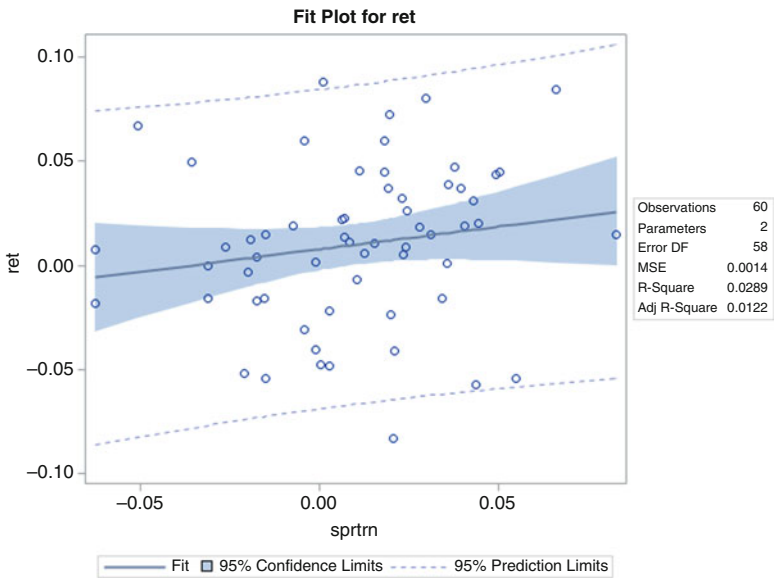
Root MSE	0.03795	**R-Square**	0.0289
Dependent mean	0.00999	**Adj R-Sq**	0.0122
Coeff var	379.96345		

Parameter estimates

Variable	DF	Parameter estimate	Standard error	t value	Pr > |t|
Intercept	**1**	0.00779	0.00518	1.51	0.1376
sprtrn	**1**	0.21732	0.16543	1.31	0.1941

Source: SAS, Proc Reg

The plot of the Dominion student residuals (RStudent) (see below) indicates 5–6 of 60 observations exceed ±2, indicating possible outliers n the beta regression. There are three observations with large Cook's D leverage statistics, indicating possible regression problems.

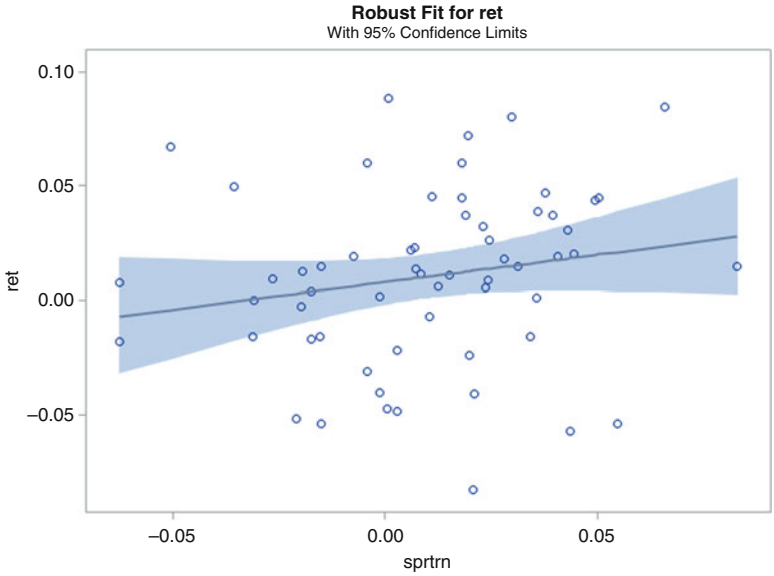


Source: SAS, Proc Reg

Let us estimate a robust regression beta using the OIF99 regression methodology as discussed in Maronna, Martin, Yohai, and Salibian-Barrera (2019) and implemented in SAS (Table 9).

The Dominion robust regression beta is not statistically different from zero, with its chi-squared statistic of 2.06, and is statistically different from one.

Diagnostics summary		
Observation type	**Proportion**	**Cutoff**
Outlier	0.0000	3.0000



Goodness of fit	
Statistic	**Value**
R-square	0.0326
AICR	59.1877
BICR	63.8664
Deviance	0.0788

The Dominion OLS value-weighted CRSP beta, 0.179, is not statistically different from zero, with its t-statistic of 1.10, and is statistically different from one.

The Dominion (ROB) robust regression value-weighted CRSP beta, 0.198, is not statistically different from zero, with its chi-squared statistic of 1.46, and is statistically different from one. See Table 10.

The robust regression beta for D is 0.199 (Tables 11 and 12).

Table 9 Robust SP D beta

The ROBUSTREG procedure							
Model information							
Dataset	WORK.D						
Dependent variable	ret						
Number of independent variables	1						
Number of observations	60						
Method	MM estimation						
Profile for the initial LTS estimate							
Total number of observations	60						
Number of squares minimized	45						
Number of coefficients	2						
Highest possible breakdown value	0.2667						
MM profile							
Chi function	Tukey						
K1	7.0410						
Efficiency	0.9900						
Parameter estimates							
Parameter	**DF**	**Estimate**	**Standard error**	**95% confidence limits**		**Chi-square**	**Pr > ChiSq**
Intercept	1	0.0079	0.0052	−0.0023	0.0181	2.33	0.1271
sprtrn	1	0.2382	0.1660	−0.0872	0.5636	2.06	0.1514
Scale	0	0.0376					

Source: SAS, Proc Robustreg

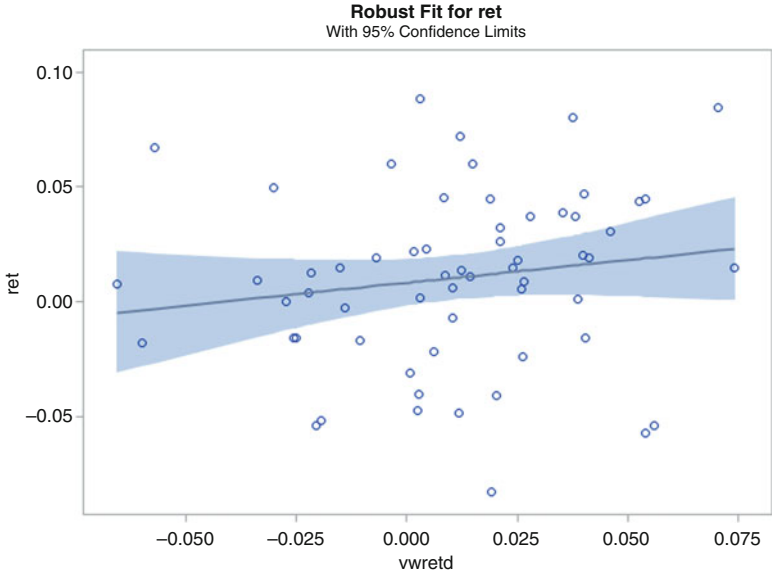


Table 10 OLS CRSP VL-weighted D beta

The REG procedure Model: MODEL1 Dependent variable: ret					
Number of observations read					60
Number of observations used					60
Analysis of variance					
Source	**DF**	**Sum of squares**	**Mean square**	**F value**	**Pr > F**
Model	1	0.00175	0.00175	1.21	0.2765
Error	58	0.08428	0.00145		
Corrected total	59	0.08603			
Root MSE		0.03812	**R-Square**		0.0204
Dependent mean		0.00999	**Adj R-Sq**		0.0035
Coeff var		381.62505			
Parameter estimates					
Variable	**DF**	**Parameter estimate**	**Standard error**	**t value**	**Pr > | t|**
Intercept	**1**	0.00806	0.00522	1.54	0.1283
vwretd	**1**	0.17926	0.16318	1.10	0.2765

Source: SAS, Proc Reg

Goodness of fit	
Statistic	**Value**
R-square	0.0232
AICR	58.5305
BICR	63.2328
Deviance	0.0796

One can run OLS and ROB robust regressions for Boeing, BA. The betas are statistically significant different from zero, but not from 1.0.

If an investor estimates that the security's expected return exceeds the required rate of return from the CAPM and its beta, then the security should be purchased. The incremental purchase of the security drives up its price and lowers its expected return.

The difficulty of measuring beta and its corresponding security market line gave rise to extra-market measures of risk, found in the work of King (1966), Rosenberg (1974), Rosenberg and Marathe (1979), Ross (1976), and Ross and Roll (1980). The Barra risk model was developed in the series of studies by Rosenberg and completely discussed in Grinold and Kahn (1999) that is discussed in the next chapter.

Table 11 Robust CRSP VL-weighted D beta

The ROBUSTREG procedure							
Model information							
Dataset	WORK.D						
Dependent variable	ret						
Number of independent variables	1						
Number of observations	60						
Method	MM estimation						
Profile for the initial LTS estimate							
Total number of observations	60						
Number of squares minimized	45						
Number of coefficients	2						
Highest possible breakdown value	0.2667						
MM profile							
Chi function	Tukey						
K1	7.0410						
Efficiency	0.9900						
Parameter estimates							
Parameter	**DF**	**Estimate**	**Standard error**	**95% confidence limits**		**Chi-square**	**Pr > ChiSq**
Intercept	1	0.0082	0.0053	−0.0021	0.0185	2.43	0.1188
vwretd	1	0.1987	0.1645	−0.1237	0.5211	1.46	0.2271
Scale	0	0.0380					
Diagnostics summary							
Observation type	**Proportion**	**Cutoff**					
Outlier	0.0000	3.0000					

Source: SAS, Proc Robustreg

Table 12 Stock betas

Regression	Stock	S&P 500	V-L CRSP
OLS	IBM	.963	0.965
(t)		(5.31)	(5.48)
ROB	IBM	1.029	0.999
(Chi-squared)		(39.30)	(37.86)
OLS	D	.217	.179
		(1.31)	(1.10)
ROB	D	.238	.199
		(2.06)	(1.46)
OLS	BA	1.053	1.015
		(4.99)	(4.86)
ROB	BA	1.066	1.023
		(26.01)	(23.88)

The total excess return for a multiple factor model, referred to as the MFM, in the Rosenberg methodology for security j, at time t, dropping the subscript t for time, may be written

$$E(R_j) = \sum_{k=1}^{K} \beta_{jk} \widetilde{f}_k + \widetilde{e}_j \tag{8}$$

The nonfactor, or asset-specific, return on security j, is the residual risk of the security, after removing the estimated impacts of the K factors. The term, f, is the rate of return on factor k. A single factor model, in which the market return is the only estimated factor, is obviously the basis of the Capital Asset Pricing Model. We discuss multifactor risk models in the next chapter.

Expected Returns vs. Historic Mean Returns

The expected returns on assets are not completely explained by using only historic means of the securities. One may estimate models of expected return using expectation data and reported financial data. There are several approaches to security valuation and the creation of expected returns. Graham and Dodd (1934) recommended that stocks be purchased on the basis of the price-earnings ratio. Graham and Dodd suggested that no stock should be purchased if its price-earnings ratio exceeded 1.5 times the price-earnings multiple of the market. Thus the "low price-earnings" (PE) criteria was established. It is interesting that Graham and Dodd put forth the low PE model at the height of the Great Depression. Warren Buffet is the greatest student and supporter of Benjamin Graham. In his introduction to the 1973 Graham's The Intelligent Investor edition, Mr. Buffet reported the superior performance records of himself and other Graham students and proteges. Basu (1977) reported evidence supporting the low PE model. Academicians often prefer to test the low P/E approach by testing its reciprocal, the "high EP" approach. The high EP approach specifically addresses the issue of negative earnings per share, which can confuse the low P/E test. Hawawini and Keim (1995) found statistical support for the high EP variable of NYSE and AMEX stocks from April 1962 to December 1989. At a minimum, Graham and Dodd advocated the calculation of a security's net current asset value, NCAV, defined as its current assets less all liabilities. A security should be purchased if its net current value exceeded its current stock price. The price-to-book (PB) ratio should be calculated, but not used as a measure for stock selection, according to Graham, Dodd, and Cottle (1962).

Fundamental Analysis and Stock Selection

Nerlove (1968) estimated a multiple regression to estimate the determinants of stock returns. Wall Street Research departments started to estimate models for identifying stock returns determinants in the 1990s. Fundamental variables such as cash flow and sales have put used in composite valuation models for security selection Ziemba (1990) and Ziemba and Schwartz (1993). Livnat advocated the calculation of free cash flow, which subtracts capital expenditures from the operating cash flow. In addition to the income statement indicators of value, such as earnings, cash flow, and sales, many value-focused analysts also consider balance sheet variables, especially the book-to-market ratio. The income statement measures are dividends, earnings, cash flow, and sales, and the key balance sheet measure is common equity per share outstanding, or book value. Expected returns modeling has been analyzed with a regression model in which security returns are functions of fundamental stock data, such as earnings, book value, cash flow, and sales, relative to stock prices, and forecast earnings per share (Bloch, Guerard Jr., Markowitz, Todd, & Xu, 1993; Fama & French, 1992, 1995; Guerard, 1997; Guerard, Takano, & Yamane, 1993; Haugen and Baker, 1996, 2010; Guerard, Gultekin and Stone, 1997, 2002). Hawawini and Keim (1995) found statistical support for the high cash flow-to-price (CP) and low price-to-book (P/B) variables of NYSE and AMEX stocks from April 1962 to December 1989.

Fama and French (1995) argue that the contrarian investment strategy is inconsistent with market data. Fama and French report results of a three-factor model that incorporates the market factor, size factor, and book-to-market ratio. In the Fama and French analysis of the CRSP database for the 1963–1994 period, the value-weighted market return exceeded the risk-free rate by approximately 5.2%, annually, whereas smaller stocks outperformed larger stocks by 3.2% annually, and higher book-to-market (price) outperformed smaller book-to-market stocks by 5.4%, annually. The Fama and French model may be written as

$$R_j - R_F = \beta_M R_M + \beta_{size} R_{size} + \beta_{BM} R_{BM} \tag{9}$$

where

R_j = stock return
β_M = market beta
R_M = return on market index less the risk-free rate
β_{size} = size beta
R_{size} = return on size variable, defined as the return on small stocks less the return on large stocks
β_{BM} = book-to-market beta
R_{BM} = returns on high book-to-market stocks less the returns on low book-to-market stocks

The authors report the Fama and French three-factor model because of its importance in the literature. In fact, the Fama and French (1992, 1995, 1998, 2008

and 2015) studies have been among the most-cited financial studies of the past zonean. The results of the stock selection model introduced and estimated in the following section does not place a higher weight on the book-to-market (price) ratio than on the cash flow, relative book-to-price (and relative earnings, cash flow, and sales-to-price ratios), and analysts' forecast variables.

The fundamental models were complemented by earnings forecasting models for stock selection. In 1975, a database of earnings per share (EPS) forecasts was created by Lynch, Jones, and Ryan, a New York brokerage firm, by collecting and publishing the consensus statistics of 1-year-ahead and 2-year-ahead EPS forecasts (Brown, 1999). The database has evolved to be known as the Institutional Brokerage Estimation Service (I/B/E/S) database. There is an extensive literature regarding the effectiveness of analysts' earnings forecasts, earnings revisions, earnings forecast variability, and breadth of earnings forecast revisions, summarized in Bruce and Epstein (1994) and Brown (1999). The vast majority of the earnings forecasting literature in the Bruce and Brown references find that the use earnings forecasts do not increase stockholder wealth, as specifically tested in Elton, Gruber, and Gultekin (1981). Reported earnings follow a random walk with drift process, and analysts are rarely more accurate than a no-change model in forecasting earnings per share (Guerard Jr. & Stone, 1992). Analysts become more accurate as time passes during the year, and quarterly data is reported. Analyst revisions are statistically correlated with stockholder returns during the year (Arnott 1985; Hawkins, Chamberlain, & Daniel, 1984). Wheeler (1994) developed and tested a strategy in which analyst forecast revision breadth, defined as the number of upward forecast revisions less the number of downward forecast revisions, divided by the total number of estimates, was the criteria for stock selection. Wheeler found statistically significant excess returns from the breadth strategy. A composite earnings variable, CTEF, is calculated using equally weighted revisions, forecasts, and breadth of FY1 and FY2 forecasts, a variable put forth in Guerard (1997).

Ziemba and Schwartz (1993), and Guerard, Gultekin, and Stone (1997) employed annual fundamental Compustat variables, such as earnings, book value, cash flow, and sales, in addition to the composite earnings forecasting model in a regression model to identify the determinants of quarterly equity returns. The regression models used in the Guerard et al. (1997) studies employed the Beaton-Tukey robust regression procedure and latent root regression techniques to address the issues of outliers and multicollinearity. The reader was introduced to the robust regression and multicollinearity techniques in chapter "Regression Analysis and Estimating Regression Models." Ziemba used capitalization-weighted regressions. Both sets of studies found statistical significance with expectation and reported fundamental data.

Modern Portfolio Theory and GPRD: An Example of Markowitz Analysis

In 1990, Harry Markowitz became the Head of the Global Portfolio Research Department (GPRD) at Daiwa Securities Trust. His department used fundamental data to create models for Japanese and US securities. The basic models tested included the earnings-to-price strategy (EPR), which is the inverse of the low P/E strategy, the book value-to-price (BPR), the cash flow-to-price (CPR), and sales-to-price (SPR) ratios. These variables should be positively associated (correlated) with subsequent returns. Single variable and composite model strategies were tested in Japan and the United States, 1974–1990. The composite models could be created by combining the variables using ordinary least squares (OLS), outlier-adjusted or robust regression (ROB), or weighted latent root regression (WLRR) modeling techniques, in which outliers and the high correlations among the variables are used in the estimation procedures. The Markowitz group tested single variables and composite model strategies, using the outlier-adjustment and multicollinearity modeling techniques shown in chapter "Regression Analysis and Estimating Regression Models." The proprietary models combined the outlier adjustments and multicollinearity techniques to substantially outperform the Japanese equity market, 1990–1994. The Markowitz group found that the use of the more advanced statistical techniques produced higher relative out-of-sample portfolio geometric returns and Sharpe ratios. Statistic modeling is not just fun, but it is consistent with maximizing portfolio returns. The quarterly estimated models outperformed the semiannual estimated models, although the underlying data was semiannual in Japan. The dependent variable in the composite model is total security returns, and the independent variables are the EPR, BPR, CPR, and SPR variables and their respective relative ratios, the current ratio divided by the average of the past 60 months of the ratios. Guerard and Schwartz (2007) reprinted the original Table 1 of Bloch et al. (1993); Table 13 reports backtests underlying the GPRD Daiwa Portfolio Optimization System (DPOS). We continue the table was a benchmark for stock selection modeling.

Further Estimations of a Composite Equity Valuation Model: The Roles of Analysts Forecasts and Momentum in Stock Selection

This section exposes the reader to issues of databases and the inclusion of variables in composite models to identify undervalued securities. The database for this analysis is created by the use of all securities listed on the Compustat database, the I/B/E/S database, and the Center for Research in Security Prices (CRSP) database during the 2003–2018 period. The annual Compustat file contains some 399 data items from the company income statement, balance sheet, and cash flow statement during the 1950–2018 period. The I/B/E/S database contains all earnings forecasts

Table 13 D-POS, Japan (Simulation results: sorted by geometric mean)

SID	OP	TOV	RP	Rtri	ERET	GM	Shrp
900869	3	10.00	1	25.0	PROPRIETARY	25.60	0.89
900867	3	10.00	0	0.0	PROPRIETARY	25.39	0.89
900868	3	10.00	1	20.0	PROPRIETARY	25.32	0.87
900860	3	15.00	1	25.0	PROPRIETARY	24.28	0.84
900853	3	15.00	0	0.0	PROPRIETARY	24.19	0.85
900858	3	10.00	1	25.0	PROPRIETARY	24.04	0.85
900852	3	12.50	0	0.0	PROPRIETARY	23.94	0.85
900855	3	10.00	1	20.0	PROPRIETARY	23.93	0.86
900856	3	12.50	1	20.0	PROPRIETARY	23.90	0.84
900854	3	17.50	0	0.0	PROPRIETARY	23.89	0.84
900859	3	12.50	1	25.0	PROPRIETARY	23.89	0.83
900857	3	15.00	1	20.0	PROPRIETARY	23.81	0.82
900819	3	10.00	0	0.0	REGR(WLRR,4Q,4)	22.74	0.83
900820	3	10.00	1	25.0	REGR(WLRR,4Q,4)	22.68	0.82
900944	3	10.00	0	0.0	BPR	22.43	0.78
900908	3	10.00	1	20.0	REGR(LRR,4Q,9.1)	22.23	0.75
900874	3	10.00	0	0.0	REGR(OLS,4Q,8)	22.16	0.79
900878	3	10.00	0	0.0	REGR(OLS,4Q,9.1)	22.16	0.79
900903	3	10.00	0	0.0	REGR(OLS,4Q,8)	22.16	0.79
900914	3	10.00	0	0.0	REGR(OLS,4Q,9.1)	22.16	0.79
900841	3	10.00	1	25.0	REGR(WLRR,1Q,4)	22.00	0.79
900817	3	10.00	0	0.0	REGR(LRR,4Q14)	21.99	0.76
900983	3	10.00	1	20.0	REGR(WLRR,4Q,9.1)	21.93	0.75
900984	3	10.00	1	20.0	REGR(WLRR,4Q,9.1)	21.86	0.75
900794	3	15.00	1	20.0	REGR(WLRR,1Q,4)	21.84	0.76
900818	3	10.00	1	25.0	REGR(LRR,4Q,4)	21.84	0.75
900877	3	10.00	0	0.0	REGR(WLRR,4Q,8)	21.84	0.78
900906	3	10.00	0	0.0	REGR(WLRR,4Q,8)	21.84	0.78
900985	3	12.50	1	20.0	REGR(WLRR,4Q,9.1)	21.84	0.75
900913	3	10.00	0	0.0	REGR(WLRR,4Q,9.2)	21.83	0.77
900793	3	12.50	1	20.0	REGR(WLRR,1Q,4)	21.78	0.78
900791	3	12.50	0	0.0	REGR(WLRR,1Q,4)	21.75	0.79
900792	3	15.00	0	0.0	REGR(WLRR,1Q,4)	21.68	0.77
900982	3	10.00	1	20.0	REGR(WLRR,4Q,9.1)	21.66	0.75
900842	3	10.00	1	25.0	REGR(WLRR,10,4)	21.55	0.79
900766	3	10.00	1	20.0	REGR(WLRR,1Q,4)	21.49	0.78
900810	3	15.00	0	0.0	REGR(WLRR,1Q,4)	21.47	0.76
900901	3	10.00	0	0.0	REGR(LRR,4Q,9.1)	21.45	0.72
900813	3	10.00	0	0.0	REGR(OLS,4Q,4)	21.42	0.78
900840	3	10.00	1	25.0	REGR(WLRR,1Q,4)	21.41	0.76
900838	3	10.00	1	25.0	REGR(WLRR,1Q,4)	21.40	0.76
900909	3	10.00	1	20.0	REGR(WLRR,4Q,9.1)	21.40	0.75

(continued)

Table 13 (continued)

SID	OP	TOV	RP	Rtri	ERET	GM	Shrp
900910	3	10.00	0	0.0	REGR(LRR,4Q,9.2)	21.34	0.75
900816	3	10.00	1	25.0	REGR(ROB,4Q,4)	21.30	0.76
900839	3	10.00	1	25.0	REGR(WLRR,1Q,4)	21.30	0.75
900912	3	10.00	0	0.0	REGR(LRR,4Q,9.2)	21.29	0.71
900765	3	10.00	0	0.0	REGR(WLRR,1Q,4)	21.24	0.76
900815	3	10.00	0	0.0	REGR(ROB,4Q,4)	21.23	0.76
900902	3	10.00	0	0.0	REGR(WLRR,4Q,9.1)	21.16	0.74
900986	3	15.00	1	20.0	REGRCWLRR,4Q,9.1)	21.09	0.72
900954	3	10.00	0	0.0	REGR(OLS,4Q,4)	20.91	0.72
900876	3	10.00	0	0.0	REGR(LRR,4Q,8)	20.90	0.74
900905	3	10.00	0	0.0	REGR(LRR,4Q,8)	20.90	0.74
900911	3	10.00	0	0.0	REGR(ROB,4Q,9.2)	20.66	0.72
900907	3	10.00	1	20.0	REGR(ROB,4Q,9.1)	20.36	0.74
900763	3	10.00	0	0.0	REGR(LRR,1Q,4)	20.21	0.71
900875	3	10.00	0	0.0	REGR(ROB,4Q,8)	20.15	0.71
900904	3	10.00	0	0.0	REGR(ROB,4Q,8)	20.15	0.71
900787	3	12.50	0	0.0	REGR(LRR,1Q,4)	20.08	0.71
900900	3	10.00	0	0.0	REGR(ROB,4Q,9.1)	20.07	0.72
900781	3	12.50	1	20.0	REGR(OLS,1Q,4)	19.96	0.71
900788	3	15.00	0	0.0	REGR(LRR,1Q,4)	19.92	0.70
900764	3	10.00	1	20.0	REGR(LRR,1Q,4)	19.88	0.70
900790	3	15.00	1	20.0	REGR(LRR,1Q,4)	19.81	0.70
900789	3	12.50	1	20.0	REGR(LRR,1Q,4)	19.78	0.70
900779	3	12.50	0	0.0	REGR(OLS,1Q,4)	19.77	0.67
900786	3	15.00	1	20.0	REGR(ROB,1Q,4)	19.76	0.71
900780	3	15.00	0	0.0	REGR(OLS,1Q,4)	19.72	0.69
900784	3	15.00	0	0.0	REGR(ROB,1Q,4)	19.67	0.71
900782	3	15.00	1	20.0	REG R(OLS,1Q,4)	19.41	0.69
900759	3	10.00	0	0.0	REGR(OLS,1Q,4)	19.40	0.67
900785	3	12.50	1	20.0	REGR(ROB,1Q,4)	19.33	0.69
900760	3	10.00	1	20.0	REGR(OLS,1Q,4)	19.31	0.66
900783	3	12.50	0	0.0	REGR(ROB,1Q,4)	19.10	0.69
900761	3	10.00	0	0.0	REGR(ROB,1Q,4)	19.03	0.68
900931	3	10.00	0	0.0	CPR	19.01	0.68
900762	3	10.00	1	20.0	REGR(ROB,1Q,4)	19.00	0.67
900932	3	10.00	0	0.0	SPR	18.63	0.61
900716	3	10.00	1	20.0	benchmark	17.25	0.60
900927	3	10.00	0	0.0	EPR	16.82	0.57
900826	6	20.00	3	25.0	PROPRIETARY	24.63	0.84
900709	6	20.00	3	20.0	PROPRIETARY	23.61	0.81
900710	6	20.00	3	25.0	PROPRIETARY	23.44	0.82
900733	6	25.00	1	20.0	PROPRIETARY	23.34	0.80

(continued)

Table 13 (continued)

SID	OP	TOV	RP	Rtri	ERET	GM	Shrp
900773	6	17.50	3	20.0	PROPRIETARY	23.26	0.78
900707	6	20.00	3	20.0	PROPRIETARY	23.08	0.79
900847	6	20.00	0	0.0	PROPRIETARY	22.62	0.81
901030	6	20.00	3	20.0	BPR	22.42	0.78
900796	6	20.00	3	20.0	REGR(OLS,2S,4)	22.33	0.79
901047	6	20.00	3	20.0	BPR	22.20	0.77
900770	6	22.50	0	0.0	REGR(OLS,1S,4)	22.17	0.77
900795	6	20.00	0	0.0	REGR(OLS,2S,4)	22.14	0.79
900749	6	20.00	3	25.0	REGR(OLS,1S,4)	22.03	0.78
900800	6	20.00	3	20.0	REGR(LRR,2S,4)	21.98	0.78
900849	6	20.00	0	0.0	REGR(LRR,3S,4)	21.98	0.77
900748	6	20.00	3	20.0	REGR(OLS,1S,4)	21.80	0.77
900754	6	20.00	3	20.0	REGR(LRR,12,4)	21.68	0.74
900747	6	20.00	0	0.0	REGR(OLS,1S,4)	21.65	0.77
900802	6	20.00	3	20.0	REGR(WLRR,2S,4)	21.60	0.79
901029	6	20.00	0	0.0	BPR	21.59	0.76
900755	6	20.00	3	25.0	REGR(LRR,1S,4)	21.52	0.74
900799	6	20.00	6	0.0	REGR(LRR,2S,4)	21.51	0.77
901046	6	20.00	0	0.0	BPR	21.49	0.76
900801	6	20.00	0	0.0	REGR(WLRR,2S,4)	21.40	0.79
900769	6	17.50	0	0.0	REGR(OLS,1S,4)	21.34	0.76
900778	6	22.50	3	20.0	REGR(WLRR,1S,4)	21.30	0.79
900772	6	22.50	0	0.0	REGR(LRR,1S,4)	21.26	0.72
900753	6	20.00	0	0.0	REGR(LRR,1S,4)	21.20	0.72
900756	6	20.00	0	0.0	REGR(WLRR,1S,4)	21.10	0.77
900757	6	20.00	3	20.0	REGR(WLRR,1S,4)	21.06	0.78
900758	6	20.00	3	25.0	REGRCWLRR,1S,4)	21.02	0.78
900777	6	17.50	3	20.0	REGR(WLRR,1S,4)	20.95	0.78
900771	6	17.50	0	0.0	REGR(LRR,1S,4)	20.93	0.71
900848	6	20.00	0	0.0	REGR(ROB,3S,4)	20.74	0.76
900776	6	22.50	3	20.0	REGR(ROB,1S,4)	20.37	0.76
900797	6	20.00	0	0.0	REGR(ROB,22,4)	20.24	0.75
900798	6	20.00	3	20.0	REGR(ROB,2S,4)	20.12	0.76
900752	6	20.00	3	25.0	REGR(RQB,1S,4)	19.56	0.73
900751	6	20.00	3	20.0	REGR(ROB,1S,4)	19.35	0.73
900750	6	20.00	0	0.0	REGR(ROB,1S,4)	19.29	0.72
900775	6	17.50	3	20.0	REGR(ROB,1S,4)	19.26	0.72
901049	6	20.00	3	20.0	CPR	18.99	0.68
901051	6	20.00	3	20.0	SPR	18.69	0.61
901048	6	20.00	0	0.0	CPR	18.65	0.67
901050	6	20.00	0	0.0	SPR	17.87	0.59
901045	6	20.00	3	20.0	EPR	17.55	0.59

(continued)

Table 13 (continued)

SID	OP	TOV	RP	Rtri	ERET	GM	Shrp
901044	6	20.00	0	0.0	EPR	17.34	0.59

Let UL = 2.0 TCR = 2.0 PM = –1 PPar = 0.90 Begin = 7412 End = 9012
SID simulation ID, *OP* period of re-optimization, *TOV* turnover constraint, *Rtri* rebalancing trigger, *ERET* model description, *GM* geometric mean, *Shrp* Sharpe ratio
REGR (technique, period, equation)
Technique = OLS for ordinary least-squares regression analysis
LRR for latent rootregression
ROB forrobust regression
WLRR for weighted latent root regression
Period = 1S for one-period semi-annual analysis
2S for two-period semi-annual analysis
3S for three-period semi-annual analysis
IQ for one-period quarterly analysis
4Q for four-period quarterly analysis
Equation = $4; \mathrm{TRR} = a_0 + a_1\mathrm{EPR} + a_2\mathrm{BPR} + a_3\mathrm{CPR} + a_4\mathrm{SPR} + a_5\mathrm{REPR} + a_6\mathrm{RPBR} + a_7\mathrm{RCPR} + a_8\mathrm{RSPR} + e_t$
$8; \mathrm{TRR} = a_0 + a_1\mathrm{EPR} + a_2\mathrm{BPR} + a_3\mathrm{CPR} + a_4\mathrm{SPR} + e_t$
$9.1; \mathrm{TRR} = a_0 + a_1\mathrm{EPR} + a_2\mathrm{BPR} + a_3\mathrm{CPR} + a_4\mathrm{SPR} + a_5\mathrm{REPR}(2) + a_6\mathrm{RPBR}(2) + a_7\mathrm{RCPR}(2) + a_8\mathrm{RSPR}(2) + e_t,$
where (2) denotes 2-year averages of relative variables
$9.2; \mathrm{TRR} = a_0 + a_1\mathrm{EPR} + a_2\mathrm{BPR} + a_3\mathrm{CPR} + a_4\mathrm{SPR} + a_5\mathrm{REPR}(3) + a_6\mathrm{RPBR}(3) + a_7\mathrm{RCPR}(3) + a_8\mathrm{RSPR}(3) + e_t,$
where (3) denotes 3-year averages of relative variables

made during the 1976–2018 period. The CRSP file contains monthly stock prices, shares outstanding, trading volumes, and returns for all traded securities from 1926 to 2018.

There are a seemingly infinite number of financial variables that may be tested for statistical association with monthly security returns. Bloch et al. (1993) tested a set of fundamental variables in the United States during the 1975–1990 period. Guerard (1997) tested a set of I/B/E/S variables for the 1982–1994 period. In this study, we test the variables of these two studies using both fundamental and expectation data. We initially test the effectiveness of the individual variables using the information coefficients (ICs) rather than the upper quintile excess returns or the excess returns of individual variable portfolio optimizations. The information coefficient is the slope of the regression estimation in which ranked subsequent security returns are a function of the ranked financial strategy. The advantage of the IC approach is that the slope has a corresponding t-statistic that allows one to test the null hypothesis that the strategy is uncorrelated with subsequent returns. In developing a composite model, one seeks to combine variables that are statistically associated with subsequent returns. Let us define the variables tested in this chapter.

EP = earnings per share/price per share

BP = book per share/price per share
CP = cash flow per share/price per share
SP = sales per share/price per share
DY = dividend yield, dividend per share/price per share
NCAV = net current asset value, net current assets per share/price per share
FEP1 = 1-year-ahead forecast earnings per share/price per share
FEP2 = 2-year-ahead forecast earnings per share/price per share
RV1 = 1-year-ahead forecast earnings per share monthly revision/price per share
RV2 = 2-year-ahead forecast earnings per share monthly revision/price per share
BR1 = 1-year-ahead forecast earnings per share monthly breadth/price per share
BR2 = 2-year-ahead forecast earnings per share monthly breadth/price per share

Table 14 Investment factors spread and information coefficients overview

12/31/2002 to 5/29/2015									
Universe return: 1.24									
Benchmark return: 0.74									
					Annualized return				
					Quintiles				
	Annualized								
	Spread		**IC**	**IC Std.**					
Factor	**Return**	**IC**	**T-Stat**	**Dev.**	**1**	**2**	**3**	**4**	**5**
LAR	23.98	0.059	5.08	0.09	32.03	18.00	13.74	10.35	6.33
WLRR	22.31	0.056	4.86	0.09	31.03	16.93	14.26	11.38	6.70
CTEF	20.30	0.055	4.71	0.06	28.26	19.63	15.10	10.58	6.69
CTEFROIC	18.40	0.054	4.64	0.07	27.86	19.35	15.47	9.78	7.70
FY2RV3	12.95	0.041	3.56	0.07	25.40	16.97	15.63	11.02	10.34
PMTREND	14.35	0.039	3.41	0.11	23.98	19.03	15.59	12.77	7.95
BR2	10.63	0.037	3.15	0.05	22.82	16.32	17.73	10.31	11.57
FY1RV3	10.01	0.035	3.03	0.07	23.36	18.67	14.33	11.49	11.46
BR1	10.56	0.035	2.96	0.05	23.00	16.67	17.59	10.51	11.29
FEP1	12.30	0.032	2.73	0.09	26.19	14.79	13.38	12.37	12.49
ALPHA	9.38	0.031	2.69	0.12	21.38	16.97	17.58	13.59	9.26
FEP2	12.21	0.030	2.60	0.10	25.29	16.05	13.60	11.90	12.26
CP	10.50	0.028	2.43	0.08	23.75	17.10	14.62	12.44	11.30
DP	10.55	0.027	2.37	0.09	24.99	15.10	13.57	12.43	12.12
EP	9.70	0.023	2.01	0.08	25.11	13.91	13.00	13.61	13.53
STDEV	−0.81	0.022	1.92	0.15	19.35	13.99	14.47	15.50	14.98
PM71	5.05	0.020	1.71	0.13	19.94	15.88	16.52	14.66	11.60
ROE1YR	1.24	0.017	1.42	0.06	15.90	18.34	16.82	14.48	13.58
ROIC	2.84	0.016	1.39	0.07	17.47	18.96	15.14	14.52	13.08
SP	5.48	0.015	1.32	0.09	17.79	19.99	16.57	13.25	11.53

(continued)

Table 14 (continued)

ROA1YR	−1.07	0.015	1.25	0.07	15.61	15.29	14.71	17.85	15.63
ROA3YR	−0.25	0.014	1.19	0.06	14.86	15.57	15.42	19.26	14.02
ROA5YR	−0.84	0.014	1.16	0.06	15.28	15.39	15.48	17.75	15.21
ROE3YR	0.77	0.013	1.16	0.06	14.91	18.42	17.39	15.09	13.31
NCSR	−0.29	0.013	1.09	0.07	16.45	16.10	17.11	12.26	15.22
ROE5YR	0.85	0.012	1.07	0.06	14.78	18.54	18.01	14.64	13.20
DI	0.30	0.004	0.38	0.09	15.83	15.59	15.52	16.30	15.64
ES	2.82	0.004	0.36	0.07	19.55	15.42	14.32	14.56	15.15
CSR	1.25	0.004	0.34	0.09	17.65	13.85	−0.84	15.88	19.35
BP	7.45	0.004	0.31	0.10	22.56	14.22	14.43	13.75	14.09
PM1	−3.62	0.003	0.32	0.10	13.62	14.84	16.49	15.81	18.04
DR	−3.91	0.002	0.14	0.05	15.16	15.38	13.98	12.88	20.17
NDR	−1.29	−0.004	−0.34	0.08	15.41	15.96	16.36	15.32	15.96
CSI	−0.09	−0.011	−0.92	0.10	14.66	14.55	12.08	16.09	14.79

Source: FactSet Investment Research Systems

where

EP = earnings per share/price per share
BP = book value per share/price per share
CP = cash flow per share/price per share
SP = sales per share/price per share
DP = dividends per share/price per share
PMTREND = price momentum with market effect removed
PM71 = price momentum as Pricet-1/Pricet-7
FEP1 = 1-year-ahead forecast earnings per share/price per share
FEP2 = 2-year-ahead forecast earnings per share/price per share
FY1RV3 = 1-year-ahead forecast earnings per share monthly revision/price per share
BR1 = 1-year-ahead forecast earnings per share monthly breadth
BR2 = 2-year-ahead forecast earnings per share monthly breadth
ROE_1Yr = 1-year return on equity
ROE_5Yr = 5-year return on equity
ROA_1Yr = 1-year return on total assets
ROA_3Yr = 3-year return on total assets
ROA_5Yr = 5-year return on total assets
CTEF = equally weighted FEP1, FEP2, BR1, BR2, RV1, and RV2
LAR = least angle regression of EP, BP, CP, SP, REP, RBP, RCP, RSP, CTEF, and PM
WLRR = weighted latent root regression of EP, BP, CP, SP, REP, RBP, RCP, RSP, CTEF, and PM
ROIC = return on invested capital
CSR = common stock repurchased
CSI = common stock issued
NCSR = net common stock repurchased
DR = debt repurchased
DI = debt issued
NDR = net debt repurchased
ES = corporate exports = Dividends + Interest Paid + Net Equity Repurchased + Net Debt Repurchased
CTEFROIC = proprietary-weighted CTEF

A consensus earnings-per-share I/B/E/S forecast, revisions and breadth variable, CTEF, has been created and tested using I/B/E/S data since January 1976.

The monthly ICs for all traded securities during the January 2003–November 2018 period for these variables are shown in Table 6. The majority of the variables are statistically associated with stockholder returns, a result consistent with the Bloch et al. and Guerard studies. We report the Guerard, Markowitz, and Xu results for the 2003–2018 period in Table 14. The results of Table 14 support the estimation of the composite security valuation model reported in Guerard, Gultekin, and Stone (1997) and Guerard Jr. and Mark (2003). That is, CTEF and WLRR lead a set of 30+ models in having statistically significant IC in the Global universe. The model incorporates reported earnings, book value, cash flow, and sales, the corresponding relative variables, and an equally weighted composite model of earnings forecasts, revisions, and breadth. One notes that the statistical significance of the CTEF variable is quite similar to the 1-year-ahead EPS breadth. The authors stress the statistically significance of financial variables.

We estimate a similar monthly model for the January 2003–November 2018 period:

$$TR_{t+1} = a_0 + a_1EP_t + a_2BP_t + a_3CP_t + a_4SP_t + a_5REP_t + a_6RBP_t + a_7RCP_t + a_8RSP_t + a_9CTEF_t + e_t \quad (10)$$

where

EP = [earnings per share]/[price per share] = earnings-price ratio
BP = [book value per share]/[price per share] = book-price ratio
CP = [cash flow per share]/[price per share] = cash flow-price ratio
SP = [net sales per share]/[price per share] = sales-price ratio
REP = [current EP ratio]/[average EP ratio over the past 5 years]
RBP = [current BP ratio]/[average BP ratio over the past 5 years]
RCP = [current CP ratio]/[average CP ratio over the past 5 years]
RSP = [current SP ratio]/[average SP ratio over the past 5 years]
CTEF = consensus earnings-per-share I/B/E/S forecast, revisions and breadth
e = randomly distributed error term

The monthly ordinary least squares (OLS) regression are plagued with approximately twice the number observations outside the 95% confidence interval as one might expect given a normal distribution of residuals. These aberrant observations, or outliers, lead us to re-estimate the monthly regression lines using a Beaton-Tukey bi-weight (or robust, ROB) regression technique, in which each observation is weighted as the inverse function of its OLS residual. The application of the Beaton-Tukey ROB procedure addresses the issue of outliers. The weighted data is plagued with multicollinearity, the correlation among the independent variables, which may lead to statistically inefficient estimates of the regression coefficients. Bloch et al. (1993) and Guerard, Takano, and Yamane (1993) applied latent root regression (LRR) to the ROB-weighted data, referred to as weighted latent root regression, WLRR, and produced models with higher in-sample F-statistics and

higher out-of-sample geometric means using WLRR than ROB and OLS techniques. We create a composite model weight using the average weight of the positive coefficients of the preceding 12 monthly regressions, a monthly equivalent to the four-quarter averaging techniques used in Guerard, Gultekin, and Stone (1997). One notes the large weighting of the CTEF variable, and the relatively small weighting of the BP variable, a result consistent with Guerard, Gultekin, and Stone (1997). In terms of information coefficients, ICs, the use of the WLRR procedure produces the highest IC for the models during the 1990–2001 period. The WLRR technique produces the largest and most statistically significant IC: a result consistent with the previously noted studies and the GPRD example. The t-statistics on the composite model exceed the t-statistics of its components. The purpose of a composite security valuation model is to identify the determinants of security returns and produce a statistically significant out-of-sample ranking metric of total returns.

We now trace the development of various regression-based models of stock selection modeling. Guerard, working with Markowitz and Xu (1993, 2014, 2015), and Mark (2003, 2020) built and reported statistically significant stock selection models.

REG8 Model

We estimate a monthly model, the original Markowitz model of GPRD DPOS (1993) which we refer to as REG8, for the January 2002–December 2018 period:

$$TR_{t+1} = a_0 + a_1 EP_t + a_2 BP_t + a_3 CP_t + a_4 SP_t + a_5 REP_t + a_6 RBP_t + a_7 RCP_t + a_8 RSP_t + e_t \tag{11}$$

where *EP, BP, CP, SP, REP, RBP, RCP, RSP* are above defined variables at time t and e is randomly distributed error term.

REG9 Model

In Guerard and Schwartz, the authors added CTEF as a ninth factor in the regression, which we do in this chapter. In testing the REG9 model, we use a weighted latent root regression (WLRR) analysis, combining robust regression and multicollinearity corrected techniques on Eq. (11) to identify variables statistically significant at the 10% level. The REG9 model is written as

$$TR_{t+1} = a_0 + a_1 EP_t + a_2 BP_t + a_3 CP_t + a_4 SP_t + a_5 REP_t + a_6 RBP_t + a_7 RCP_t + a_8 RSP_t + a_9 CTEF_t + e_t \tag{12}$$

REG10 Model

The third regression model is a ten-factor model, REG10, publishes as USER with US stocks and GLER with global stocks. The GLER model is estimated monthly using a weighted latent root regression (WLRR) analysis, combining robust regression and multicollinearity corrected techniques on Eq. (13) to identify variables statistically significant at the 10% level. REG10 is written as

$$TR_{t+1} = a_0 + a_1 EP_t + a_2 BP_t + a_3 CP_t + a_4 SP_t + a_5 REP_t + a_6 RBP_t + a_7 RCP_t + a_8 RSP_t + a_9 CTEF_t + a_{10} PM_t + e_t \quad (13)$$

where $\alpha = (a_1, \ldots, a_{10})$ is the rolling average of normalized coefficients of $\alpha_{t-1}, \ldots, \alpha_{t-12}$.

PM = Price Momentum, P(t − 1)/T(t − 7).

The normalization of α_t takes two steps. First, we set non-positive and non-statistical significant components to be zero. Second, the remaining positive significant components are rescaled so that they add up to 1. The 12-month smoothing is consistent with the four-quarter smoothing in Bloch et al. (1993).

While EP and BP variables are significant in explaining returns, the majority of the forecast performance is attributable to other model variables, namely, the relative earnings-to-price, relative cash-to-price, relative sales-to-price, price momentum, and earnings forecast variables. The weighting results are extremely consistent with McKinley Capital Management being a Global Growth specialist. The CTEF and PM variables accounted 40% of the weights in the GLER Model. We refer to using WLRR on the first eight variables, the Markowitz model, as REG8. The model produced out-of-sample statistically significant excess returns in the portfolios. We refer to using WLRR on the ten variables, referred to as GLER in our published articles, as REG10. The REG9 and REG10 model produced out-of-sample statistically significant excess returns in the portfolios, and the excess returns most often exceeded the excess returns of REG8. The GLER model is very similar to the work by Bob Haugen (1996, 2010). In a recent update of Guerard Jr. and Mark (2003) study of CTEF and the REG9 model, Guerard Jr. and Mark (2020) report that the CTEF variable continues to be a positive and statistically significant variable for stock selection in the United States. The CTEF, REG9, and REG10, USER models, have statistically significant ICs, Level I test, and active returns, Level II tests, but active returns and specific returns (stock selection) falls as one expands the models relative to CTEF in the US Factor Contributions increase because of increased earnings yield, value, and industry exposures and returns rise, as we will discuss in chapter "Multifactor Risk Models and Portfolio Construction and Management." The ICs are reported in Table 14. In non-US portfolios, Guerard and Mark report that CTEF variable continues to be a positive and statistically significant variable for stock selection in the non-US portfolios. The CTEF, REG9, and REG10, USER, models have statistically significant ICs, Level I test, and active returns, Level II

Table 15 Portfolio dashboard, mean-variance optimization

Mean-variance optimization **AXIOMA fundament risk model** **1/2002–11/2018**												
Portfolios	**Sharpe ratio**	**Info ratio**	**Risk stock-specific effect**	**Risk stock-specific effect T-Stat**	**Risk factors effect**	**Risk factors effect T-Stat**	**Risk total effect**	**Risk trans-action effect**	**Total effect**	**Earnings yield**	**Growth**	**Medium-term momentum**
R3000_REG9_8TE	0.68	0.38	3.01	**2.04**	0.06	0.55	3.07	−1.20	1.88	**0.29**	−0.15	−0.21
R3000_REG8_6TE	0.60	0.18	1.86	**1.66**	−0.88	−0.24	0.98	−0.94	0.04	**−0.31**	−0.15	−0.28
R3000_CTEF_8TE	0.75	**0.52**	1.61	1.36	3.30	2.24	4.92	−2.26	2.65	**1.81**	0.07	1.41
R3000_REG8_8TE	0.51	0.06	1.77	1.36	−1.53	−0.50	0.24	−1.28	−1.03	**−0.74**	−0.18	−0.41
R3000_REG9_6TE	0.64	0.25	1.35	1.14	0.15	0.55	1.50	−1.12	0.38	**0.43**	−0.13	−0.15
R3000_CTEF_6TE	0.71	0.42	0.35	0.57	2.59	2.26	2.95	−1.55	1.39	**1.66**	0.06	1.07
R3000_CTEF_4TE	0.72	0.42	−0.02	0.20	2.03	2.73	2.02	−0.92	1.10	**1.29**	0.03	0.72
R3000_REG9_4TE	0.62	0.12	0.02	0.05	0.31	0.67	0.33	−0.55	−0.22	**0.53**	−0.11	−0.02
R3000_REG8_4TE	0.58	0.02	−0.06	−0.05	−0.11	0.17	−0.17	−0.45	−0.62	**0.10**	−0.11	−0.19
R3000_REG10_8TE	0.53	0.08	−0.42	−0.14	0.85	0.86	0.43	−1.96	−1.54	**0.61**	−0.17	0.28
R3000_REG10_4TE	0.62	0.12	−0.17	−0.17	0.52	0.89	0.35	−0.81	−0.46	**0.71**	−0.16	0.18
R3000_REG10_6TE	0.53	−0.01	−0.88	−0.62	0.53	0.68	−0.35	−1.33	−1.68	**0.67**	−0.17	0.18
R1000_REG8_6TE	0.69	0.30	2.52	**2.07**	−0.55	0.16	1.97	−0.88	1.10	**0.08**	−0.09	−0.28
R1000_REG8_8TE	0.58	0.13	2.36	1.57	−1.40	−0.18	0.96	−1.10	−0.15	**−0.03**	−0.11	−0.39
R1000_REG8_4TE	0.71	0.27	0.81	1.00	0.44	0.84	1.25	−0.43	0.82	**0.31**	−0.07	−0.21
R1000_REG9_8TE	0.58	0.15	1.25	0.99	0.07	0.59	1.32	−1.15	0.17	**0.55**	−0.15	−0.25
R1000_CTEF_4TE	0.75	0.44	0.63	0.89	1.53	2.34	2.16	−0.69	1.47	**1.18**	0.01	0.62
R1000_REG10_6TE	0.66	0.19	0.75	0.68	0.41	0.70	1.16	−0.95	0.20	**0.75**	−0.09	0.19
R1000_CTEF_6TE	0.73	0.39	0.47	0.66	2.23	2.04	2.70	−1.00	1.70	**1.62**	0.03	0.91

R1000_CTEF_8TE	0.72	0.40	0.29	0.57	2.89	2.09	3.18	−2.08	1.10	**1.72**	0.05	1.21
R1000_REG9_6TE	0.61	0.13	0.54	0.46	0.31	0.71	0.85	−1.04	−0.19	**0.57**	−0.16	−0.15
R1000_REG9F_4TE	0.67	0.19	0.10	0.21	0.73	1.16	0.83	−0.74	0.10	**0.57**	−0.07	−0.03
R1000_REG10_8TE	0.58	0.09	−0.15	0.02	0.62	0.75	0.47	−0.86	−0.39	**0.84**	−0.08	0.46
R1000_REG10_4TE	0.66	0.14	−0.34	−0.29	1.01	1.41	0.66	−0.76	−0.10	**0.71**	−0.08	0.23

Portfolios	**Sharpe ratio**	**Info ratio**	**Risk-specific effect**	**Risk stock effect T-Stat**	**Factors effect**	**Risk effect T-Stat**	**Total effect**	**Riskrisk transaction effect**	**Total effect**	**Earnings yield**	**Growth**	**Medium-term momentum**
ALLXUS_CTEF_6TE	0.85	**1.45**	9.35	**6.89**	5.58	6.80	14.93	−4.16	10.77	**0.69**	−0.18	**3.25**
ALLXUS_CTEF_8TE	0.91	**1.44**	11.51	**6.60**	5.24	5.27	16.74	−4.98	11.76	**0.70**	−0.26	**4.08**
ALLXUS_CTEF_4TE	0.70	**1.26**	6.40	**5.77**	4.66	7.71	11.05	−3.07	7.98	**0.67**	−0.10	**2.05**
ALLXUS_REG10_6TE	0.65	**1.21**	4.16	**3.53**	7.23	6.53	11.39	−4.27	7.12	**0.36**	0.08	**2.22**
ALLXUS_REG10_4TE	0.49	**1.07**	2.66	**3.09**	4.67	5.85	7.33	−3.06	4.26	**0.36**	0.09	**1.27**
ALLXUS_REG10_8TE	0.70	**1.14**	4.41	**2.98**	8.72	6.12	13.13	−4.88	8.26	**0.44**	0.08	**2.87**
ALLXUS_REG8_8TE	0.39	0.45	5.53	**2.93**	5.29	4.65	10.82	−5.44	5.38	**−0.18**	0.42	**0.85**
ALLXUS_REG8_6TE	0.44	**0.61**	2.61	0.59	5.33	4.68	7.94	−4.34	3.60	**0.34**	0.11	**0.15**
ALLXUS_REG9_8TE	0.60	**0.93**	1.03	−0.78	10.85	7.01	11.89	−4.75	7.14	**0.66**	0.08	**0.95**
ALLXUS_REG9_6TE	0.56	**0.95**	1.34	−0.84	8.59	7.25	9.93	−3.92	6.01	**0.63**	0.06	**0.65**
ALLXUS_REG8_4TE	0.44	**0.82**	2.89	−0.89	3.09	3.95	5.98	−3.42	2.56	**0.30**	0.11	**−0.21**
ALLXUS_REG9_4TE	0.48	**1.02**	1.39	−2.08	5.06	6.62	6.45	−2.88	3.56	**0.50**	0.06	**0.43**

(continued)

Table 15 (continued)

Portfolios	Sharpe ratio	Info ratio	Risk stock-specific effect	Stock-specific effect T-Stat	Risk factors effect	Risk factors effect T-Stat	Risk total effect	Risk trans-action effect	Total effect	Earnings yield	Growth	Medium-term momentum
EM_REG8_4TE	0.51	0.31	3.99	**2.38**	1.79	2.11	5.78	−2.05	3.73	**0.25**	0.26	**−0.85**
EM_REG9_4TE	0.55	0.49	3.45	**2.20**	3.28	3.60	6.73	−2.11	4.63	**0.43**	0.16	**−0.19**
EM_CTEF_8TE	0.59	**0.61**	3.54	**2.18**	5.76	3.64	9.30	−3.58	5.72	**0.87**	−0.02	**2.69**
EM_CTEF_4TE	0.58	**0.72**	3.20	**2.16**	3.84	4.42	7.04	−2.10	4.94	**0.63**	−0.09	**1.51**
EM_CTEF_6TE	0.60	**0.65**	3.67	**2.13**	4.67	4.02	8.34	−2.75	5.59	**0.80**	−0.06	**2.13**
EM_REG9_8TE	0.51	0.30	3.40	**1.73**	5.09	3.31	8.50	−3.62	4.88	**0.38**	0.21	**0.25**
EM_REG8_6TE	0.44	0.06	3.30	1.64	2.32	2.17	5.63	−2.79	2.83	**0.13**	0.32	**−0.99**
EM_REG10_4TE	0.54	0.45	2.25	1.58	4.06	4.24	6.31	−1.92	4.39	**0.44**	0.17	**0.53**
EM_REG9_6TE	0.48	0.22	2.62	1.54	4.37	3.49	6.99	−2.80	4.19	**0.41**	0.18	**−0.09**
EM_REG8_8TE	0.43	0.10	3.37	1.36	2.59	2.27	5.96	−3.52	2.44	**0.00**	0.41	**−0.78**
EM_REG10_6TE	0.52	0.33	1.33	1.07	5.63	4.17	6.96	−2.96	4.00	**0.41**	0.23	**0.93**
EM_REG10_8TE	0.49	0.24	0.30	0.59	6.34	3.68	6.64	−3.54	3.10	**0.39**	0.24	**1.32**

Portfolios	Sharpe ratio	Info ratio	Risk stock-specific effect	Stock-specific effect T-Stat	Risk factors effect	Risk factors effect T-Stat	Risk total effect	Risk trans-action effect	Total effect	Earnings yield	Medium-term growth momentum	
JAPAN_REG8_8TE	0.56	**0.54**	5.64	**4.00**	−0.18	0.44	5.46	−2.24	3.26	−0.39	0.28	−0.75
JAPAN_REG8_6TE	0.54	**0.51**	4.73	**3.74**	−0.23	0.20	4.50	−1.49	3.04	−0.34	0.25	−0.72
JAPAN_REG8_4TE	0.50	**0.46**	2.75	**2.88**	0.17	0.63	2.92	−0.79	2.17	−0.16	0.18	−0.58
JAPAN_REG9_8TE	0.42	0.28	2.14	**1.65**	0.74	1.34	2.87	−2.00	0.95	−0.05	0.22	−0.44
JAPAN_REG9_4TE	0.44	0.31	1.54	1.59	0.55	1.45	2.09	−0.82	1.32	0.05	0.13	−0.33
JAPAN_REG9_6TE	0.41	0.25	1.83	1.50	0.44	1.04	2.27	−1.35	0.98	−0.03	0.19	−0.44
JAPAN_REG10_8TE	0.38	0.19	0.23	0.44	1.48	1.98	1.71	−1.80	−0.01	−0.02	0.19	0.11
JAPAN_REG10_6TE	0.37	0.16	−0.13	0.14	1.40	2.01	1.26	−1.48	−0.16	0.03	0.19	0.07
JAPAN_CTEF_8TE	0.35	0.15	−0.93	−0.01	2.20	2.50	1.28	−6.73	−5.40	0.56	−0.25	1.29
JAPAN_CTEF_6TE	0.12	0.09	−0.90	−0.18	2.32	3.13	1.42	−3.87	−2.38	0.56	−0.18	1.19
JAPAN_CTEF_4TE	0.30	0.19	−0.82	−0.38	1.96	3.97	1.13	−1.35	−0.17	0.53	−0.13	0.96
JAPAN_REG10_4TE	0.35	0.07	−0.97	−0.76	1.39	2.70	0.42	−1.06	−0.58	0.08	0.11	0.08

Portfolios	Sharpe ratio	Info ratio	Risk stock-specific effect	Risk stock-specific effect T-Stat	Risk factors effect	Risk factors effect T-Stat	Risk total effect	Risk trans-action effect	Total effect	Earnings yield	Growth	Medium-term momentum
CHINA_A_CTEF_4TE	0.60	**1.23**	7.12	**6.75**	2.22	6.32	9.35	−1.74	7.68	**0.34**	−0.08	0.55
CHINA_A_CTEF_6TE	0.59	**1.05**	7.30	**5.16**	2.54	5.31	9.84	−2.39	7.52	**0.34**	−0.12	0.78
CHINA_A_CTEF_8TE	0.57	**0.95**	7.19	**4.87**	2.75	4.98	9.94	−2.61	7.40	**0.35**	−0.13	0.89
CHINA_A_REG10_4TE	0.51	**0.87**	4.62	**4.43**	2.31	5.60	6.93	−1.78	5.19	**0.22**	0.19	−0.06
CHINA_A_REG9_4TE	0.44	**0.54**	3.45	**3.41**	1.42	3.77	4.88	−1.66	3.24	**0.17**	0.19	−0.51
CHINA_A_REG10_6TE	0.52	**0.72**	4.77	**3.31**	3.06	5.51	7.84	−2.51	5.39	**0.28**	0.24	−0.05
CHINA_A_REG8_4TE	0.42	0.47	3.17	**3.29**	1.29	3.29	4.47	−1.65	2.83	**0.12**	0.20	−0.74
CHINA_A_REG10_8TE	0.50	**0.63**	4.38	**2.89**	3.18	5.05	7.56	−2.79	4.83	**0.30**	0.26	−0.01
CHINA_A_REG8_6TE	0.41	0.39	3.45	**2.67**	1.49	2.84	4.94	−2.41	2.58	**0.11**	0.28	−0.91
CHINA_A_REG8_8TE	0.41	0.37	3.65	**2.56**	1.64	2.75	5.30	−2.67	2.67	**0.10**	0.33	−0.91
CHINA_A_REG9_8TE	0.41	0.38	3.40	**2.40**	1.76	3.06	5.16	−2.63	2.57	**0.18**	0.27	−0.55
CHINA_A_REG9_6TE	0.42	0.42	3.18	**2.37**	1.82	3.52	5.01	−2.37	2.69	**0.18**	0.24	−0.58

REG8 = OIF99 regression(EP,BP,CP,SP,REP,RBP,RCP,RSP)
REG9 = OIF99 regression(EP,BP,CP,SP,REP,RBP,RCP,RSP,CTEF)
REG10 = OIF99 regression(EP,BP,CP,SP,REP,RBP,RCP,RSP,CTEF,PM71)

BOLD IRs exceed 0.50
BOLD Risk Stock Specific T-Stat is statistically significant
Source: FactSet Investment Research Systems

tests; specific returns (stock selection) fall as one expands in the models in the non-US models. Non-US portfolio active returns rise in REG9 and REG10 because of increased size, value, and industry factor contributions. REG9 and REG10, GLER, are more rewarded by active returns in non-US than in US portfolios. Furthermore, the US portfolios require much higher targeted tracking errors, 7–8% to outperform than non-US portfolios, 4–6%. These are extremely important differences. Index-huggers in US portfolios, running targeted tracking errors of less than 5%, will produce far lower specific returns, most likely negative, than Non-US portfolios. Investors are better served with a passive ETF than they are with an "active" portfolio that produces negative specific returns. For regression results of REG8, REG9, and REG10 models, please see Table 15.

Summary and Conclusions

In this chapter, we introduced the reader to estimating stock betas, with OLS and robust regression and estimated variable weights in stock selection models. We report that a stock selection model and an earnings forecasting model reported in Guerard and Schwartz (2007), built on data from 1975 to 2002, continue to be highly statistically significant for December 2002–November 2018 period. We show how forecasted earnings acceleration produces highly statistically significant stock selection in global and US stock universes. CTEF, REG8, REG9, and REG10 models optimized portfolios produce higher active and specific returns in non-US stocks, whereas only CTEF works in US CTEF and PM complement the original eight-factor Markowitz model in non-US stocks. Have markets and stock selection models changed since Guerard Jr. and Mark (2003)? CTEF, REG8, REG9, and REG10 still dominate most other models, including the 36 models tested in Guerard Jr et al. (2018), including the Post-Global Financial Crisis. We concur with our good friend, now deceased, Robert Haugen who stated emphatically that markets are not completely efficiency. The reader is referred to Haugen (1995, 1996, 1999) and Haugen and Baker (1996 and 2010). The most statistically significant variable in identifying security returns is the composite earnings forecast variable. We will use the nine-factor composite model in chapter "International Business Finance" to rank international securities.

Appendix: The Three-Asset Case

Let us now examine a three-asset portfolio construction process using IBM, Dominion Resources, and BA securities:

$$E(Rp) = \sum_{i=1}^{N} x_i E(R_i) \tag{14}$$

$$\sigma_p^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j \sigma_{ij} \tag{15}$$

$$\begin{gathered}
E(Rp) = x_1 E(R_1) + x_2 E(R_2) + x_3 E(R_3) \\
\text{let} \, x_3 = 1 - x_1 - x_2 \\
E(Rp) = x_1 E(R_1) + x_2 E(R_2) + (1 - x_1 - x_2) E(R_3) \\
\sigma_p^2 = {x_1}^2 {\sigma_1}^2 + {x_2}^2 {\sigma_2}^2 + {\sigma_3}^2 {x_3}^2 + 2x_1 x_2 \sigma_{12} + 2x_1 x_3 \sigma_{23} + 2x_2 x_3 \sigma_{23} \\
= {x_1}^2 {\sigma_1}^2 + {x_2}^2 {\sigma_2}^2 + (1 - x_1 - x_2)^2 {\sigma_3}^2 + 2x_1 x_2 \sigma_{12} + 2x_1 (1 - x_1 - x_2) \sigma_{13} \\
+ 2x_2 (1 - x_1 - x_2) \sigma_{23} \\
= {x_1}^2 {\sigma_1}^2 + {x_2}^2 {\sigma_2}^2 + (1 - x_1 - x_2)(1 - x_1 - x_2) {\sigma_3}^2 + 2x_1 x_2 \sigma_{12} + 2x_1 \sigma_{13} \\
- 2{x_1}^2 \sigma_{13} - 2x_1 x_2 \sigma_{13} + 2x_2 \sigma_{23} - 2x_1 x_2 \sigma_{23} - 2{x_2}^2 \sigma_{23} \\
= {x_1}^2 {\sigma_1}^2 + {x_2}^2 {\sigma_2}^2 + \left(1 - 2x_1 - 2x_2 + 2x_1 x_2 + {x_1}^2 + {x_2}^2\right) {\sigma_3}^2 + 2x_1 x_2 \sigma_{12} \\
+ 2x_1 \sigma_{13} - 2{x_1}^2 \sigma_{13} - 2x_1 x_2 \sigma_{13} + 2x_2 \sigma_{23} - 2x_1 x_2 \sigma_{23} - 2{x_2}^2 \sigma_{23}
\end{gathered} \tag{16}$$

$$\frac{\partial \sigma_p^2}{\partial x_1} = 2x_1 \left({\sigma_1}^2 + {\sigma_3}^2 - 2\sigma_{13}\right) + x_2 \left(2{\sigma_3}^2 + 2\sigma_{12} - 2\sigma_{13} - 2\sigma_{23}\right) - 2{\sigma_3}^2 + 2\sigma_{13} = 0$$

$$\frac{\partial \sigma_p^2}{\partial x_2} = 2x_2 \left({\sigma_2}^2 + {\sigma_3}^2 - 2\sigma_{23}\right) + x_1 \left(2{\sigma_3}^2 + 2\sigma_{12} - 2\sigma_{13} - 2\sigma_{23}\right) - 2{\sigma_3}^2 + 2\sigma_{23} = 0$$

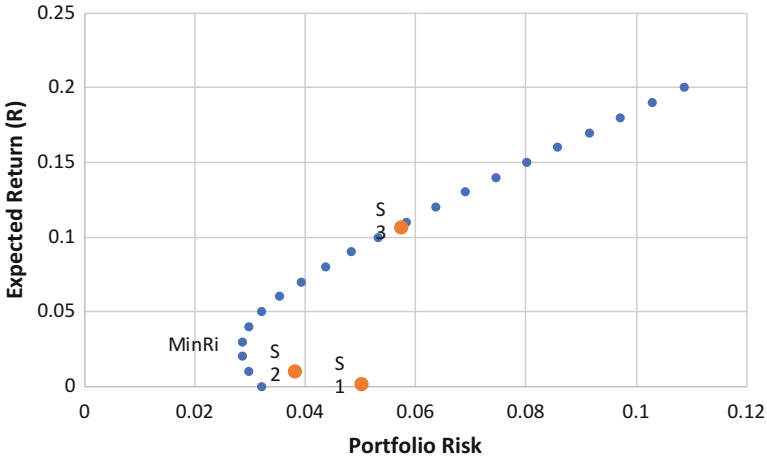
Let's assume

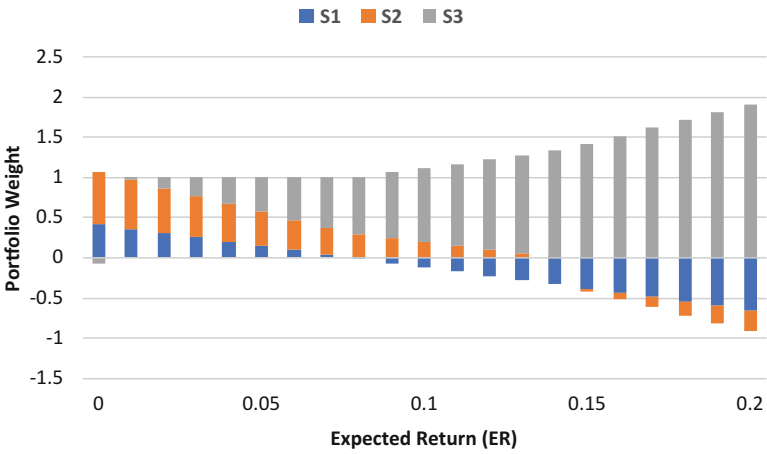
Table 16 Modeling TCBUER by OLS longer lags of LEI and WkUNCL time series and estimation of saturation (IIS and SIS) variables

Security	Security	Mean	STD
IBM	S1	0.17%	5.03%
D	S2	1.00%	3.82%
BA	S3	10.64%	5.74%

Correlation matrix	S1	S2	S3
S1	100%	0%	18%
S2	0%	100%	6%
S3	18%	6%	100%

Source: OX Metrics





The optimal portfolio weights involve selling IBM short and going long on DuPont and Dominion. How does the portfolio using optimal weights compare to an equally weighted portfolio of the three assets?

The equally weighted portfolio has an expected return of 14.9% and a 13.42% standard deviation.

$$E(R_p) = .333(.110 + .095 + .242) = .149$$

$$\sigma_p^2 = (.333)^2(.0167)^2 + (.333)^2(.0407)$$
$$+(.333)^2(.0799) + 2(.333)(.333)(-.0103)$$
$$+2(.333)(.333)(-.0041) + 2(.333)(.333)(.0303)$$

$$\sigma_p^2 = .0019 + .0045 + .0081 + (-.0023)$$
$$+(-.0009) + .0067 = .0180$$

$$\sigma_p = .1342$$

The optimally weighted portfolio has an expected return of

$$E(R_p) = .795(.110) + .2517(.095) + (-.0467)(.242)$$
$$= .0875 + .0244 + (-.0113)$$
$$= .1006$$
$$\sigma_p^2 = (.795)^2(.0167) + (.2517)^2(.0407) + (-.0467)^2$$
$$(.0799) + 2(.795)(.2517)(-.0103)$$
$$+2(.795)(-.0467)(-.0041) + 2(.2517)(-.0467)(.0303)$$
$$= .0106 + .0026 + .0002 + (-.0041) + .0003 + (-.0007)$$
$$\sigma_p^2 = .0089$$
$$\sigma_p = .0943$$

The optimally weighted portfolio has an expected return of 10.06% and a standard deviation of 9.43%. Portfolio risk can be significantly reduced through diversification (Table 16).

References

Arnott, R. (1985). The use and misuse of consensus earnings. *Journal of Portfolio Management*, 18–27.

Basu, S. (1977). Investment performance of common stocks in relation to their price earnings ratios: A test of market efficiency. *Journal of Finance, 32*, 663–682.

Bloch, M., Guerard Jr., J. B., Markowitz, H. M., Todd, P., & Xu, G.-L. (1993). A comparison of some aspects of the U.S. and Japanese equity markets. *Japan and the World Economy, 5*, 3–26.

Brown, L. D. (1999 and 2008). Annotated I/B/E/S Bibliography.

Bruce, B., & Epstein, C. B. (1994). *The handbook of corporate earnings analysis*. Chicago: Probus Publishing Company.

Elton, E. J., Gruber, M. J., & Gultekin, M. (1981). Expectations and share prices. *Management Science, 27*, 975–987.

Fama, E. F., & French, K. R. (1992). Cross-sectional variation in expected stock returns. *Journal of Finance, 47*, 427–465.

Fama, E. F., & French, K. R. (1995). Size and the book-to-market factors in earnings and returns. *Journal of Finance, 50*, 131–155.

Fama, E. F., & French, K. R. (2008). Dissecting anomalies. *Journal of Finance, 63*, 1653–1678.

Graham, B., & Dodd, D. (1934). *Security analysis: Principles and technique* (first ed.). New York: McGraw-Hill Book Company.

Graham, B., Dodd, D., & Cottle, S. (1962). *Security analysis: Principles and technique* (4th ed.). New York: McGraw-Hill Book Company.

Grinold, R., & Kahn, R. (1999). *Active portfolio management*. New York: McGraw-Hill/Irwin.

Guerard Jr., J. B. (1997). Is there a cost to being socially responsible in investing? *Journal of Investing*, 6, 11–18.

Guerard Jr., J. B., Gillam, R. A., Markowitz, H., Xu, G., Deng, S., & Wang, E. (2018). Data mining corrections testing in Chinese stocks. *Interfaces, 48*, 108–120.

Guerard Jr., J. B., Gultekin, M., & Stone, B. K. (1997). The role of fundamental data and analysts' earnings breadth, forecasts, and revisions in the creation of efficient portfolios. In A. Chen, (ed.) *Research in finance 15*. Greenwich, CT: JAI Press

Guerard Jr., J. B., & Mark, A. (2003). The optimization of efficient portfolios: The case for an R&D quadratic term. In A. Chen (Ed.), *Research in finance* (Vol. 20). Bingley, UK: Emerald Group.

Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Guerard Jr., J. B., & Mark, A. (2020). Earnings forecasts and revisions, price momentum, and fundamental data: Further explorations in financial anomalies. In C. F. Lee (Ed.), *Handbook of financial econometrics*. Singapore: World Scientific Publishers.

Guerard Jr., J. B., Markowitz, H. M., & Xu, G. (2015). Earnings forecasting in a global stock selection model and efficient portfolio construction and management. *International Journal of Forecasting, 31*, 550–560.

Guerard Jr., J. B., Rachev, R. T., & Shao, B. (2013). Efficient global portfolios: Big data and investment universes. *IBM Journal of Research and Development, 57*(5), Paper 11.

Guerard Jr., J. B., & Stone, B. K. (1992). Composite forecasting of annual corporate earnings. In A. Chen (Ed.), *Research in finance* (Vol. 10). Greenwich, CT: JAI Press.

Guerard Jr., J. B., Markowitz, H. M. & Xu, G. (2014). The role of effective corporate decisions in the creation of efficient portfolios. *IBM Journal of Research and Development, 58*(4), 6.1– 6.11.

Guerard Jr., J. B., Takano, M., & Yamane, Y. (1993). The development of efficient portfolios in Japan with particular emphasis on sales and earnings forecasting. *Annals of Operations Research, 45*, 91–108.

Haugen, R. A., & Baker, N. (1996). Communality in the determinants of expected results. *Journal of Financial Economics, 41*, 401–440.

Haugen. R. A. (1999). The Inefficient Stock Markets: what pays off and why upper saddle River: Prentice - Hall.

Haugen, R., & Baker, N. (2010). Case closed. In J. B. Guerard (Ed.), *The handbook of portfolio construction: Contemporary applications of Markowitz techniques*. New York: Springer.

Haugen, R. A. (1995). The New Finance: The case Against Efficient Markets. Englewood Cliffs, NJ: Prentice - Hall.

Hawawini, G., & Keim, D. B. (1995). On the predictability of common stock returns: World-wide evidence. In R. A. Jarrow, V. Maksimovic, & W. T. Ziemba (eds.), Handbooks in Operations Research and Management Science. *Finance, 9*, 497–537.

Hawkins, E. H., Chamberlain, S. C., & Daniel, W. E. (1984). Earnings expectations and security prices. *Financial Analysts Journal, 40*(5), 24–38.

King, B. F. (1966). Market and industry factors in stock price behavior. *Journal of Business 39*, 139–191.

Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics, 47*, 51–68.
Markowitz, H. M. (1952). Portfolio selection. *Journal of Finance, 7*, 77–91.
Markowitz, H. M. (1959). *Portfolio selection: Efficient diversification of investment.* Cowles Foundation Monograph No.16. New York: Wiley.
Maronna, R. A., Martin, R. D., Yohai, V. J., & Salibian-Barrera, M. (2019). *Robust Statistics; Theory and Methods (with R)*. New York: Springer.
Mossin, J. (1966). Equilibrium in a capital asset market. *Econometrica, 34*, 768–783.
Nerlove, M. (1968). Factors affecting differences among rates if return on investments in individual common stocks. *Review of Economics and Statistics, 67*, 313–331.
Rosenberg, B. (1974). Extra-market components of covariance in security returns. *Journal of Financial and Quantitative Analysis, 9*, 263–274.
Rosenberg, B., & Marathe, V. (1979). Tests of capital asset pricing hypotheses. In H. Levy (Ed.), *Research in finance* (Vol. 1). Greenwich, CT: JAI Press.
Ross, S. A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory, 13*, 341–360.
Ross, S. A., & Roll, R. (1980). An empirical investigation of the arbitrage pricing theory. *Journal of Finance, 35*(2), 1071–1103.
Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance, 19*, 425–442.
Wheeler, L. B. (1994). Changes in consensus earnings estimates and their impact on stock returns. In B. Bruce & C. B. Epstein (Eds.), *The handbook of corporate earnings analysis*. Chicago: Probus.
Ziemba, W. T. (1990). *Fundamental Factors in US and Japanese Stock Returns*. Berkeley Program in Finance, Santa Barbara.
Ziemba, W.T. & Schwartz, S. (1993). Invest Japan, Chicago, Probus.

Multifactor Risk Models and Portfolio Construction and Management



Moreover, portfolio models by King (1966), Jensen (1968), Elton and Gruber (1970, 1973), Fama and MacBeth (1973), Farrell (1974), Ross and Roll (1980), Fogler, John, and Tipton (1981), and Chen, Roll, and Ross (1986) clearly identified more than one beta. The previous chapter introduced the reader to Markowitz mean-variance analysis and the Capital Asset Pricing Model. The cost of capital calculated in chapter "Debt, Equity, the Optimal Financial Structure, and the Cost of Funds" assumes that the cost of equity is derived from the Capital Asset Pricing Model and its corresponding beta, or measure of systematic risk. The Gordon Model, used for equity valuation in chapter "The Equity of the Corporation: Common and Preferred Stock," assumes that the stock price will fluctuate randomly about its fair market value. The cost of equity is dependent upon the security beta. The stock beta is a regression coefficient usually estimated over 60 months, as we discussed in Chapter 14. Asset testing over the 1926–1966 time period created several issues. The "theoretical" Security Market Line (SML) is too flat and its intercept too high, see Black, Jensen, and Scholes (1972). In this chapter, we address the issues inherent in a multi-beta or multiple factor risk model. The purpose of this chapter is to introduce the reader to multifactor risk models. There are academic –based multifactor risk models, such as those of Cohen and Pogue (1967), Farrell Jr. (1974), Stone (1974), Ross (1976), Ross and Roll (1980), Dhrymes, Friend, and Gultekin (1984), Dhrymes, Friend, Gültekin, and Gültekin (1985), Blin, Bender, and Guerard (1997), and Fama and French (1992, 1995, 2008). There are practitioner multifactor risk models, such as Barra, created during the 1973–1979 time period; Advanced Portfolio Technologies (APT), created in 1987; and Axioma, created in the late 1990s and gaining practitioner acceptance in the 2000–2019 time period. The former academicians who created these practitioner models are Barr Rosenberg, Andrew Rudd, John Blin and Steve Bender, and Sebastian Ceria. We will introduce the reader to the practitioner models and their academician creators in this chapter. Which models are best? We, at McKinley Capital Management, MCM, have tested these models. None of the models are perfect, but the models are generally statistically significant when the statistically significant tilt variables of chapter "Risk and Return

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_15

of Equity and the Capital Asset Pricing Model" are used for portfolio construction. In this chapter, we discuss the MCM Horse Races of the 2010–2019 time period to test stock selection within the commercially available multifactor risk models.[1]

The Barra System

Multifactor risk models were developed in the early 1970s. Rosenberg and McKibben (1973), Rosenberg (1974), Rosenberg and Marathe (1979) and Rudd and Clasing (1982) created the academic support for the creation of the Barra risk model, the primary institutional risk model of the 1975–2005 time period. Accurate characterization of portfolio risk requires an accurate estimate of the covariance matrix of security returns. A relatively simple way to estimate this covariance matrix is to use the history of security returns to compute each variance, covariance, and security beta. The use of beta, the covariance of security and market index returns, is one method of estimating a reasonable cost of equity funds for firms. However, the approximation obtained from simple models may not yield the best possible cost of equity. The simple, single index beta estimation approach suffers from two major drawbacks:

1. Estimating a covariance matrix for the Russell 3000 stocks requires a great deal of data.
2. It is subject to estimation error. For example, in the previous chapter, the reader saw that the estimated correlation between two stocks, such as DuPont and IBM, had a higher correlation than that between DuPont and Dominion Resources. However, one might expect a higher correlation between DuPont and Dow than between DuPont and IBM, given that DuPont and Dow are both chemical firms.

Taking this further, one can argue that firms with similar characteristics, such as firms in their line of business, should have returns that behave similarly. For example, DuPont, IBM, and Dominion Resources will all have a common component in their returns because they would all be affected by news that affects the stock market, measured by their respective betas. The degree to which each of the three stocks responds to this stock market component depends on the sensitivity of each stock to the stock market component.

Additionally, one would expect DuPont and Dow to respond to news affecting the chemical industry, whereas IBM and Dell would respond to news affect the computer industry. The effects of such news may be captured by the average returns of stocks in the chemical industry and the petroleum industry. One can account for industry effects in the following representation for returns:

[1]This chapter reflects our knowledge on January 24, 2020.

$$\begin{aligned}\widetilde{r}_{DD} &= E[\widetilde{r}_{DD}] + \beta \cdot [\widetilde{r}_M - E[\widetilde{r}_M]] \\ &+1 \cdot [\widetilde{r}_{CHEMICAL} - E[\widetilde{r}_{CHEMICAL}]] + 0 \cdot [\widetilde{r}_P - E[\widetilde{r}_{DD}]] + \mu_P\end{aligned} \tag{1}$$

where:

$\widetilde{r}_{DD}$ = DD's realized return
$\widetilde{r}_M$ = the realized average stock market return
$\widetilde{r}_C$ = realized average return to chemical stocks
$\widetilde{r}_P$ = the realized average return to petroleum stocks
E[.] = expectations
β_{DD} = DD's sensitivity to stock market returns
μ_{DD} = the effect of DD specific news on DD returns

This equation simply states that DD's realized return consists of an expected component and an unexpected component. The unexpected component depends on any unexpected events that affect stock returns in general $[\widetilde{r}_M - E[\widetilde{r}_M]]$, any unexpected events that affect the chemical industry $[\widetilde{r}_C - E[\widetilde{r}_C]]$, and any unexpected events that affect DD alone (μ_{DD}). Similar equations may be written for IBM and Dominion Resources.

The sources of variation in DD's stock returns, thus, are variations in stock returns in general, variations in chemical industry returns, and any variations that are specific to DD. Moreover, DD and Dow returns are likely to move together because both are exposed to stock market risk and chemical industry risk. DD, IBM, and D, on the other hand, are likely to move together to a lesser degree because of the only common component in their returns in the market return.

By beginning with our intuition about the sources of co-movement in security returns, Rosenberg (1974) made substantial progress in estimating the covariance matrix of security returns. Rosenberg (1974) is the covariance matrix of common sources in security returns, the variances of security-specific returns, and estimates of the sensitivity of security returns to the common sources of variation in their returns, creating the Barra risk model. Because the common sources of risk are likely to be much fewer than the number of securities, we need to estimate a much smaller covariance matrix, and hence a smaller history of returns is required. Moreover, because similar stocks are going to have larger sensitivities to similar common sources of risk, similar stocks will be more highly correlated than dissimilar stocks.

Barra Model Mathematics

The Barra risk model is a multiple factor model (MFM). MFMs build on single-factor models by including and describing the interrelationships among factors.[2]

[2]The reader is referred to the Barra (US-E3) United States Equity, Version 3, Risk Model Handbook. Guerard Jr. and Mark (2003, 2020) and Miller et al. (2014) used the US-E3 model in their analysis. The reader is also referred to Rosenberg and Marathe (1979), Rudd and Rosenberg (1980), Rudd and Clasing (1982), and Grinhold and Kahn (1999).

For single-factor models, the equation that describes the excess rate of return is

$$\widetilde{r}_j = X_j\widetilde{f} + \widetilde{u}_j \tag{2}$$

where:

$\widetilde{r}_j$ = total excess return over the risk-free rate
X_j = sensitivity of security j to the factor.
$\widetilde{f}_j$ = rate of return on the factor
$\widetilde{u}_j$ = nonfactor (specific) return on security j

We can expand this model to include K factors. The total excess return equation for a multiplier-factor model becomes

$$\widetilde{r}_j = \sum\nolimits_{k=1}^{K} X_{jk}\widetilde{f}_k + \widetilde{u}_j \tag{3}$$

where:

X_{jk} = risk exposure of security j to factor k.
$\widetilde{f}_k$ = rate of return on factor k

Note that when K = 1, the MFM equation reduces to the earlier single-factor version; the CAPM addressed in the previous chapter.

When a portfolio consists of only one security, Eq. 3 describes its excess return. But most portfolios comprise many securities, each representing a proportion, or weight, of the total portfolio. When weights $h_{p1}, h_{p2}, \ldots, h_{pN}$ reflect the proportions of N securities in portfolio P, we express the excess return in the following equation:

$$\widetilde{r}_p = \sum\nolimits_{k=1}^{K} X_{Pk}\widetilde{f}_k + \sum\nolimits_{j=1}^{N} h_{Pj}\widetilde{u}_j \tag{4}$$

where:

$$X_{Pk} = \sum_{j=1}^{N} h_{Pj}X_{jk}$$

This equation includes the risk from all sources and lays the groundwork for further MFM analysis.

Risk Prediction with MFMs

Investors look at the variance of their total portfolios to provide a comprehensive assessment of risk. To calculate the variance of a portfolio, one needs to calculate the covariances of all the constituent components. Without the framework of a multiple factor model, estimating the covariance of each asset with every other asset is

Fig. 1 Stock risk in the Barra system

$$\text{V(I,j)} = \text{Covariance}[r(\tilde{i}), r(\tilde{j})]$$

where V (i,j) = asset covariance matrix, and

i,j = individual stocks.

$$V = \begin{bmatrix} V(1,1) & V(1,2) & \dots & V(1,N) \\ V(2,1) & V(2,2) & \dots & V(2,N) \\ V(3,1) & V(3,2) & \dots & V(3,N) \\ \vdots & \vdots & & \vdots \\ V(N,1) & V(N,2) & \dots & V(N,N) \end{bmatrix}$$

$$\tilde{r} = \text{X}\tilde{\text{f}} + \tilde{u}$$

where $\tilde{r}$ = vector of excess returns,

X = exposure matrix,

$\tilde{f}$ = vector of factor returns, and

$\tilde{u}$ = vector of specific returns.

$$\begin{bmatrix} \tilde{r}\,(1) \\ \tilde{r}\,(2) \\ \vdots \\ \tilde{r}\,(N) \end{bmatrix} = \begin{bmatrix} \text{X}\,(1,1) & \text{X}\,(1,2) & \dots & \text{X}\,(1,\text{K}) \\ \text{X}\,(2,1) & \text{X}\,(2,2) & \dots & \text{X}\ \rightleftarrows\ (2,\text{K}) \\ \vdots & \vdots & & \vdots \\ \text{X}\,(\text{N},1) & \text{X}\ \rightleftarrows\ (\text{N},2) & \dots & \text{X}\ \rightleftarrows\ (\text{N,K}) \end{bmatrix} \begin{bmatrix} \tilde{f}\,(1) \\ \tilde{f}\,(2) \\ \vdots \\ \tilde{f}\,(K) \end{bmatrix} + \begin{bmatrix} \tilde{u}\,(1) \\ \tilde{u}\,(2) \\ \vdots \\ \tilde{u}\,(N) \end{bmatrix}$$

Fig. 2 Barra system factor loadings

computationally burdensome and subject to significant estimation errors. Let us examine the risk structure of the Barra MFM (Fig. 1).

The Barra MFM simplifies these calculations dramatically, replacing individual company profiles with categories defined by common characteristics (factors). The specific risk is assumed to be uncorrelated among the assets, and only the factor variances and covariances are calculated during model estimation. Let us briefly review how Barr Rosenberg initially estimated the Barra factor structure (Fig. 2).

The Rosenberg MFM framework was developed and estimated in Rosenberg's study with McKibben (Rosenberg & McKibben, 1973) and the Rosenberg extra-market component study (1974), in which security-specific risk could be modeled as a function of financial descriptors or financial characteristics of the firm. The financial characteristics that were statistically associated with beta during the 1954–1970 period were:

1. Latest annual proportional change in earnings per share
2. Liquidity, as measured by die quick ratio
3. Leverage, as measured by die senior debt-to-total assets ratio
4. Growth measure of earnings per share
5. Book-to-price ratio
6. Historic beta

7. Logarithm of stock price
8. Standard deviation of earnings per share growth
9. Gross plant per dollar of total assets
10. Share turnover

Rosenberg, Hoaglet, Marathe, and McKibben (1975), Rosenberg and Marathe (1979), and Rudd and Rosenberg (1979, 1980) expanded upon the initial Rosenberg MFM framework.

The statistically significant determinants of the security systematic risk became the basis of the Barra E1 model risk indices. The domestic Barra E3 (US-E3) model uses 13 sources of factor, or systematic, exposures. The sources of extra-market factor exposures are volatility, momentum, size, size nonlinearity, trading activity, growth, earnings yield, value, earnings variation, leverage, currency sensitivity, dividend yield, and non-estimation universe.

The multiple factor model significantly reduces the number of calculations. For example, in the US Equity Model (US-E3), 65 factors capture the risk characteristics of equities. This reduces the number of covariance and variance calculations; moreover, since there are fewer parameters to determine, they can be estimated with greater precision. The Barra US-E3 factor index definitions are shown in Appendix 1. The Barra risk management system begins with the MFM equation:

$$\widetilde{r} = \mathrm{X}\widetilde{\mathrm{f}} + \widetilde{\mathrm{u}} \tag{5}$$

where:

$\widetilde{r}$ = excess return on asset I
X = exposure coefficient on the factor
$\widetilde{f}$ = factor return
$\widetilde{u}$ = specific return

Substituting this relation in the basic equation, we find that

$$\text{Risk} = \text{Var}(\widetilde{r}) \tag{6}$$

$$= \text{Var}\left(\mathrm{X}\widetilde{\mathrm{f}} + \widetilde{u}\right) \tag{7}$$

Using the matrix algebra formula for variance, the risk equation becomes

$$\text{Risk} = \mathrm{XFX}^{\mathrm{T}} + \Delta \tag{8}$$

where:

X = exposure matrix of companies upon factors
F = covariance matrix of factors
X^{T} = transpose of X matrix
Δ = diagonal matrix of specific risk variances

This is the basic equation that defines the matrix calculations used in risk analysis in the Barra equity models.

The Barra Multifactor Model and Analysts' Forecasts, Revisions, and Breadth

Let us address the estimated earnings forecasting components of the CTEF model discussed in the previous chapter for the Russell 3000 universe during the 1990–2001 period. The CTEF model produced not only higher ICs than its components (the reader is referred to chapter "Risk and Return of Equity and the Capital Asset Pricing Model") but also higher and more statistically significant asset selection than its components in the Russell 3000 universe. We retain the Guerard Jr and Mark (2003) analysis of the CTEF variable into the Barra multifactor risk model, as presented in Guerard and Schwartz (2007). See Table 1 for Russell 3000 earnings component results in which portfolios of approximately 100 stocks are produced by tilting on the individual and component CTEF factors. The forecast earnings per share for the 1-year-ahead and 2-year-ahead periods, FEP1 and FEP2, offer negative but statistically insignificant asset selection. The total active returns are positive, and not statistically significant. The asset selection is negative because the FEP variables have positive and statistically significant loadings on the risk indices; particularly the earnings yield index. The factor loading of the FEP variables on the earnings yield risk index is not unexpected, given that the earnings yield factor index in the US-E3 includes the forecast earnings-to-price variable. Thus, there is no multiple factor model benefit to the FEP variables.

The monthly revision variables, the RV variables, offer no statistically significant total active returns or asset selection abilities. The breadth variables, BR, produce statistically significant total active returns and asset selection, despite statistically significant risk index loadings. The breadth variable load on the earnings yield and growth risk indices. Let us take a closer look at the BR1 factor risk index loading.

Table 1 Components of the composite earnings forecasting variable, 1990–2001

Russell 3000 universe								
R3000 earnings analysis	Total active	T-stat	Asset selection	T-stat	Risk index	T-stat	Sectors	T-stat
FEP1	2.14	1.61	−1.18	−1.17	**4.20**	**4.42**	−0.86	−1.34
FEP2	1.21	0.91	−1.43	−1.35	**3.33**	**3.35**	−0.78	−1.15
RV1	0.76	0.69	0.34	0.42	0.92	1.46	−0.34	−0.89
RV2	1.40	1.37	1.09	1.31	0.81	1.42	−0.39	−1.08
BR1	**2.59**	**2.83**	**1.85**	**2.43**	**1.08**	**2.15**	−0.20	−0.51
BR2	**2.43**	**2.36**	**1.51**	**1.75**	**1.09**	**2.04**	−0.01	−0.02
CTEF	**2.87**	**2.81**	**2.07**	**2.66**	**1.19**	**1.70**	−0.26	−0.66

Bold figures denote statistically significant coefficients at the 10% level

Table 2 Risk and return of mean-variance efficient portfolios, 1990–2001

Universe	Total active	t-value	Asset selection	t-value	Risk index	t-value	Sectors	t-value
RMC	1.98	1.37	0.99	0.86	0.97	1.45	−0.88	−0.97
R1000	2.47	2.52	1.85	2.12	0.82	2.13	−.11	−.23
R2500	7.76	4.37	6.48	3.96	1.61	2.85	−0.33	−0.62
R2000	9.68	5.83	8.81	5.57	0.90	2.36	−.02	−.07

where RMC = Frank Russell mid-cap universe
R1000 = Frank Russell largest 1000 stock universe
R2000 = Frank Russell small-cap universe
R2500 = Frank Russell small- and mid-cap universe

The BR1 variable leads a portfolio manager to have a positive average active exposure to the earnings yield index, which incorporates the analyst-predicted earnings-to-price and historic earnings-to-price measures. The BR1 tilt has a negative and statistically significant average exposure to size, nonlinearity, the cube of normalized market capitalization. This result is consistent with analyst revisions being more effective in smaller capitalized securities. The BR1 variable tilt leads the portfolio manager to have a positive and statistically significant exposure to the growth factor index, composed of the growth in the dividend payout ratio, the growth rates in total assets and earnings per share during the past 5 years, recent 1-year earnings growth, and the variability in capital structure. Furthermore, the 1-year-ahead BR is slightly better than the 2-year-ahead BR, a result consistent with Stone, Guerard Jr., Gultekin, and Adams (2002). The CTEF variable produces statistically significant total active returns and asset selection. The CTEF variable loading on the risk index is statistically significant at the 10% level because of its loading on the earnings yield and nonlinear size indices, as was the case with its breadth components. There are no statistically significant sector exposures in the CTEF variable. The CTEF model offers statistically significant asset selection in a multiple factor model framework. See Guerard and Mark for a more complete discussion of the earnings forecasting alpha work within the Barra system.

The Frank Russell large market capitalization (the Russell 1000), middle-market capitalization (mid-cap), small-capitalization (Russell 2000), and small- and middle-market capitalization (Russell 2500) universes are used in the tests in this chapter. See Table 2. One can test the equally weighted composite model, CTEF, of I/B/E/S earnings forecasts, revisions, and breadth, described in the previous section. The portfolio optimization algorithm seeks to maximize the ranking of the CTEF variable while minimizing risk. The underlying CTEF variable is statistically significant, having a monthly information coefficient of 0.049 over the 491,119 observations. The CTEF variable is used as the portfolio tilt variable in the ITG optimization system using the Barra risk model, and statistically significant total excess returns are found in the Frank Russell universes. One can create 100 stock portfolios monthly during the 1990–2001 period. A lambda tilt value of one is initially used in producing efficient portfolios. Active returns rise as the average stock size diminishes a result consistent with the inefficient market literature summarized in Dimson

Table 3 CTEF variable, Russell 1000 universe

Attribution report				
Annualized contributions to total return				
Source of return	**Contribution (% return)**	**Risk (% Std. Dev)**	**Info ratio**	**T-Stat**
Risk-free	4.93	N/A	N/A	N/A
Total benchmark	13.58	14.52		
Market timing	−0.02	0.25	−0.18	−0.61
Risk indices	0.82	1.16	0.62	2.13
Sectors	−0.11	1.01	−0.07	−0.23
Asset selection	1.85	2.76	0.61	2.12
Total exceptional active	2.54	3.07	0.75	2.59
Total active	2.47	3.07	0.73	2.52
Total managed	16.04	14.96		

(1988) and Ziemba (1992). The highest total active returns are found in the Russell 2000 stocks, the smallest stocks in the largest 3000 stocks in the Frank Russell universes, each year; see Table 2. The CTEF tilt variable does produce statistically significant sector exposures in the Russell 2000 stocks, as reported in Table 2, as we previously noted in the Russell 3000 universe.

The CTEF variable produces statistically significant asset selection in the Russell 1000 universe, during the 1990–2001 period. The asset selection produced by CTEF is 1.85 percent annually, with at statistic of 2.12. The reader is referred to Table 3. The risk index exposure is statistically significant in the Russell 1000 universe, producing risk indices returns of 82 basis points, annually, with at statistic of 2.13. The factor exposures of the CTEF variable in the Russell 1000 universe are shown in Table 4. The CTEF variable has statistically significant factor loadings on earnings yield and growth, as was the case in the Russell 3000 universe.

The total active return of CTEF variable for the Russell 2000 universe is shown in Table 5. Active returns to the CTEF variable are much larger in the Russell 2000 stocks than in the Russell 1000 stocks. The CTEF variable produces statistically significant asset selection and significant factor exposures, primarily due to the earnings yield exposure. The earnings forecasting variable produces over 600 basis points annually of greater asset selection in the Russell 2000 universe than it does in the Russell 1000 universe. Earnings forecasts, revisions, and breadth generate greater asset selection in small stock universes than in larger stock universes. Moreover, as the firm size decreases, the CTEF variable is more statistically associated with risk index returns, such as earnings yield. The factor exposures increase as the size of firms decreases. The earning yield variable loading is statistically significant in the Russell 2000 universe. Miller, Xu, and Guerard Jr (2014) updated Guerard Jr and Mark (2003) using US-E3 for the 1979–2009 time period. Miller et al. (2014) reported a "beautiful trade-off curve" and highly statistically significant positive management returns (active returns) and asset selection (stock selection), and a slightly edited form is in Guerard, Xu, and Wang. As one increases the portfolio lambda, decreasing the risk aversion level, the number of

Table 4 CTEF variable factor exposures, Russell 1000 universe

Attribution analysis							
Annualized contributions to risk index return							
Source of return	**Average active exposure**	**Contribution (% return)**			**Total**		
		Average	**Variation**	**Total**	**Risk**		
		[1]	**[2]**	**[1 + 2]**	**(% Std. Dev)**	**Info ratio**	**T-Stat**
Volatility	−0.01	0.01	−0.07	−0.06	0.17	−0.32	−1.12
Momentum	0.12	−0.07	0.08	0.01	0.60	0.03	0.11
Size	−0.20	0.36	−0.09	0.27	0.93	0.24	0.83
Size nonlinearity	−0.02	0.02	0.03	0.05	0.10	0.44	1.52
Trading activity	0.00	0.00	0.01	0.01	0.11	0.11	0.37
Growth	−0.05	0.05	0.03	0.08	0.14	0.48	1.65
Earnings yield	0.13	0.66	−0.12	0.55	0.40	1.20	4.13
Value	0.06	0.03	0.02	0.06	0.17	0.30	1.03
Earnings variation	0.02	−0.02	0.00	−0.03	0.10	−0.21	−0.73
Leverage	0.06	−0.01	−0.04	−0.04	0.17	−0.23	−0.80
Currency sensitivity	−0.02	0.01	−0.05	−0.04	0.11	−0.32	−1.11
Yield	0.04	0.01	−0.04	−0.04	0.14	−0.24	−0.81
Non-est universe	0.00	0.00	0.01	0.01	0.04	0.20	0.68
Total				**0.82**	**1.16**	**0.62**	**2.13**

Table 5 CTEF variable, Russell 2000 universe

Attribution analysis				
Annualized contributions to total return				
Source of return	**Contribution (% return)**	**Risk (% Std. Dev)**	**Info ratio**	**T-Stat**
Risk-free	4.93	N/A	N/A	N/A
Total benchmark	11.73	18.42		
Expected active	−0.11	N/A	N/A	N/A
Market timing	0.09	0.45	0.05	0.17
Risk indices	0.90	1.11	0.68	2.36
Sectors	−0.02	0.95	0.02	0.07
Asset selection	8.81	4.67	1.61	5.57
Total exceptional active	9.79	4.88	1.71	5.90
Total active	9.68	4.88	1.69	5.83
Total managed	21.41	17.67		

assets in the portfolio falls, the managed returns rise, and the information ratio rises versus either the Russell 3000 Growth Index or the S&P 500 Index. Asset selection is statistically significant and substantial, exceeding 350 basis points for the 30-year backtest and produced statistically significant portfolio returns passing the Data Mining Corrections test.

The Barra risk model is an extra-market covariance model to describe the risk behavior of equity securities. Barr Rosenberg and his co-authors developed the Barra system in the mid-to-late 1970s, and the system has been a financial success, went public and was listed on NASDAQ (BARZ), and was acquired by Morgan Stanley in 2004 for over $800 million. The Barra system was no means the only multifactor or extra-market covariance model.[3]

Early Alternative Multi-Beta Risk Models

Bernell Stone (1974) developed a two-factor index model which molded equity returns as a function of an equity index and debt returns.[4] In recent years, Stone et al. (2002) developed a response surface portfolio algorithm to generate portfolios that have similar stock betas (systematic risk), market capitalizations, dividend yield, and sales growth cross sections, such that one can access the excess returns of the analysts' forecasts, forecast revisions, and breadth model, as one moves from low (least preferred) to high (most preferred) securities with regard to your portfolio construction variable (i.e., CTEF). The Stone et al. (2002) response surface risk model identified several risk models of 3-, 6-, and 8-factor that created portfolios producing large excess returns. Excess returns similar to the excess returns to asset selection shown in Table 2 can be produced with Stone algorithm during the 1982–1998 period. Farrell Jr. (1974, 1997) estimated a four-"factor" extra-market covariance model.[5] Farrell took an initial universe of 100 stocks (due to computer limitations) and ran market models to estimate betas and residuals from the market model:

[3]Rudd and Rosenberg (1979) published one of the first academic/commercial portfolio optimization analyses in the Elton and Gruber monograph to honor Harry Markowitz. Miller et al., 2014) used the Barra optimizer.

[4]Bernell Stone is a dear friend, a professor of, and co-author with Guerard. His dissertation *Risk, Return, and Equilibrium: A General Single-Period Theory of Asset Selection and Capital Market Equilibrium* was published by MIT Press. Stone et al. (2002) received an Honorable Mention for the Moskowitz Prize for research in socially responsible investing; please see chapter "Management-Stockholder Relations: Is Optimal Behavior All That Is Necessary?".

[5]Jim Farrell was President of the Institute of Quantitative Research in Finance, "The Q-Group," for over 30 years. The Q-Group sponsors academic research and hosts outstanding seminars.

$$R_{j_t} = a_j + b_j R_{M_t} + e_j \tag{9}$$

$$e_{j_t} = R_{j_t} - \widehat{a}_j - \widehat{b}_j R_{M_T} \tag{10}$$

The residuals of Eq. (10) should be independent variables. That is, after removing the market impact by estimating a beta, the residual of IBM should be independent of Dow, Merck, or Dominion Resources. The residuals should be independent, of course, in theory. Farrell (1974) examined the correlations among the security residuals of Eq. (10) and found that the residuals of IBM and Merck were highly correlated, but the residuals of IBM and D (then Virginia Electric and Power) were not correlated. Farrell used a statistical technique known as cluster analysis to create clusters, or groups, of securities, having highly correlated market model residuals. Farrell found four clusters of securities based on his extra-market covariance. The clusters contained securities with highly correlated residuals that were uncorrelated with residuals of securities in the other clusters. Farrell referred to his clusters as "growth stocks" (electronics, office equipment, drug, hospital supply firms, and firms with above-average earnings growth), "cyclical stocks" (metals, machinery, building supplies, general industrial firms, and other companies with above-average exposure to the business cycle), "stable stocks" (banks, utilities, retailers, and firms with below-average exposure to the business cycle), and "energy stocks" (coal, crude oil, and domestic and international oil firms). In 1976, Ross published his "Arbitrage Theory of Capital Asset Pricing," which held that security returns were a function of several (4–5) economic factors. In 1986, Chen, Ross, and Roll developed an estimated multifactor security return model based on

$$\mathrm{R} = \mathrm{a} + \mathrm{b}_{\mathrm{MP}}\mathrm{MP} + \mathrm{b}_{\mathrm{DEI}}\mathrm{DEI} + \mathrm{b}_{\mathrm{UI}}\mathrm{UI} + \mathrm{b}_{\mathrm{UPR}}\mathrm{UPR} + \mathrm{b}_{\mathrm{UTS}}\mathrm{UTS}\ \mathrm{te}_{\mathrm{t}} \tag{11}$$

where

MP = monthly growth rate of industrial production
DEI = change in expected inflation
UI = unexpected inflation
UPR = risk premium
UTS = term structure of interest rates

Chen, Ross, and Roll (CRR) defined unexpected inflation as the monthly (first) differences of the Consumer Price Index (CPI) less the expected inflation rate. The risk premia variable is the "Baa and under" bond return at time and less the long-term government bond return. The term structure variable is the long-term government bond return less the treasury bill rates, known at time t − 1, and applied to time t. When CRR applied their five-factor model in conjunction with the value-weighted index betas, during the 1958–1984 period, the index betas are not statistically significant, whereas the economic variables are statistically significant. The Dhrymes, Friend, and Gultekin (1984), and Chen, Ross, and Roll multifactor models used 4–5 factors to describe equity security risk. The models used different statistical approaches and economic models to control for risk. The reader may now ask a

simple question: If four or five betas are appropriate, why not estimate 20 betas? The important point is that the betas should be estimated on variables that are independent, or orthogonal, of the other variables. The estimation of 20 betas on orthogonal variables, 10–15 of which are not statistically significant, can produce the quite similar expected returns for securities as 4 or 5 betas estimated on economic variables, or pre-specified variables. Blin and Bender (1987–1997) estimated a 20-(factor) beta model of covariances based on 3.5 years of weekly stock returns data. The Blin and Bender Arbitrage Pricing Theory (APT) model followed the Ross factor modeling theory, but Blin and Bender estimated betas from 20 orthogonal factors. Blin and Bender never sought to identify their factors with economic variables.

This chapter addresses many aspects of estimating and using multifactor models of risk. What are portfolio managers and investors concerned about these risk models? In chapter "The Equity of the Corporation: Common and Preferred Stock," the reader was introduced to equity valuation models, which held that stock prices are functions of discounted expected dividends. The equity discount rate can be determined by the multifactor risk model; see Roll and Ross (1984).

APT Approach

John Blin and Steve Bender founded APT, Advanced Portfolio Technologies, in 1985. Blin, Bender, and Guerard Jr. (1997) and Guerard Jr. (2012) demonstrated the effectiveness of the APT, SunGard APT, and FIS APT systems in portfolio construction and management. Let us review the APT approach to portfolio construction. The estimation of security weights, w, in a portfolio is the primary calculation of Markowitz's portfolio management approach. The issue of security weights will be now considered from a different perspective. The security weight is the proportion of the portfolio's market value invested in the individual security.

$$w_s = \frac{MV_s}{MV_p} \tag{12}$$

where w_s = portfolio weight in security s, MV_s = value of security s within the portfolio, and MV_p = the total market value of portfolio.

The active weight of the security is calculated by subtracting the security weight in the (index) benchmark, b, from the security weight in the portfolio, p

$$w_{sp} - w_{sb} \tag{13}$$

Blin and Bender created the APT, Advanced Portfolio Technologies, Analytics Guide (2005), which built upon the mathematical foundations of their APT system, published in Blin et al. (1997). The following analysis draws upon APT analytics. Volatility can be broken down into systematic and specific risk:

$$\sigma_p^2 = \sigma_{\beta p}^2 + \sigma_{\varepsilon p}^2 \tag{14}$$

where σ_p = total portfolio volatility, $\sigma_{\beta p}$ = systematic portfolio volatility, and $\sigma_{\varepsilon p}$ = specific portfolio volatility. Blin and Bender created a multifactor risk model within their APT risk model based on forecast volatility.

$$\sigma_p = \sqrt{52 * \left(\sum_{c=1}^{C} \sum_{i=1}^{S} w_i \beta_{i,c}\right)^2 + 52 * \sum_{i=1}^{S} w_i^2 \varepsilon_i^2} \tag{15}$$

where

σ_p = forecast volatility of annual portfolio return
C = number of statistical components in the risk model
w_i = Portfolio weight in security i
$\beta_{i,c}$ = loading (beta) of security i on risk component c
$\varepsilon_{i,\ week}$ = weekly specific volatility of security i

The multiplier 52 is the annualization factor since there are 52 weeks per year.

Tracking error is a measure of volatility applied to the active return of funds (or portfolio strategies) against a benchmark, which is often an index. Portfolio tracking error is defined as the standard deviation of the portfolio return less the benchmark return over 1 year.

$$\sigma_{te} = \sqrt{E\big((r_p - r_b) - E(r_p - r_b)\big)^2} \tag{16}$$

where

σ_{te} = annualized tracking error
r_p = actual portfolio annual return
r_b = actual benchmark annual return

Systematic tracking error of a portfolio is a forecast of the portfolio's active annual return as a function of the securities' returns associated with APT risk model components. Portfolio-specific tracking error can be written as a forecast of the annual portfolio active return associated with each security's specific behavior.

$$\sigma_{\varepsilon te} = \sqrt{52 \sum_{i=1}^{N_s} \big(w_{i,p} - w_{i,b}\big)^2 \varepsilon_{i,week}^2} R_P \tag{17}$$

The APT calculated portfolio tracking error versus a benchmark as

$$\sigma_{p-b} = \sqrt{52 \big(w_p - w_b\big)^T \big(b^T b + \varepsilon^T \varepsilon\big)\big(w_p - w_b\big)}, \tag{18}$$

where

σ_{p-b} = forecast tracking error
b = A($N_c \times N_s$) matrix of component loadings; N_c components in the model and N_s securities in the portfolio
ε = a diagonal matrix ($N_c \times N_s$) of the specific loadings
$w_p - w_b$ = the (N_s-dimensional) vector security active weights

$$\sigma_{p-b} = \sqrt{52\left[\left(\sum_{c=1}^{N_c}\sum_{s=1}^{N_s}\left(w_{sp}-w_{sb}\right)b_{sc}\right)^2 + \sum_{s=1}^{N_s}\left(w_{sp}-w_{sb}\right)^2\varepsilon_s^2\right]} \quad (19)$$

where

$w_{sp} - w_{sb}$ = portfolio active weights in security s
b_{sc} = the loading of security s on component c
ε_s = weekly specific volatility of security s

The marginal tracking error is a measure of the sensitivity of the tracking error of an active portfolio to changes in the active weight of a specific security.

$$\partial_s\left[\sigma_{p-b}\right] = \frac{\partial\sigma_{p-b}}{\partial w_{s,p-b}} \quad (20)$$

where $w_{s,\,p-b}$ = active portfolio weights in security s.

The APT calculated contribution-to-risk of a security is

$$\Delta\left[\sigma_p\right] = \frac{52\left(b^T b + s^T s\right)w}{\sqrt{52\left(b^T b + s^T s\right)w}} \quad (21)$$

where $\Delta[\sigma_p] = \mathrm{A}(N_s - \textit{dimensional})$ vector of contribution to volatility for securities in the portfolio with N_s securities.

The portfolio value at risk (VaR) is a measure of the distribution of expected outcomes. If one is concerned with a 95% confidence level, α, and a 30-day time horizon, then the 95%, 30-day VaR of the portfolio is the minimum we would expect to lose is characterized by

$$P[V_T < V_0 - \nu(\alpha, T)] = 1 - \alpha \quad (22)$$

where

$\nu(\alpha, T)$ = VaR at the confidence interval α for time T
V_T = portfolio value at time T
α = required confidence level

If the portfolio returns are normally distributed,

$$\nu_s(\alpha, T) = V_0 - E(V_T) - V_0 \phi^{-1}(\alpha)\sigma_p^T \tag{23}$$

where

$\nu_s(\alpha, T)$ = Gaussian VaR
σ_p^T = forecast portfolio volatility
$\phi^{-1}(\alpha)$ = inverse cumulative normal distributed function

The total VaR is

$$\nu_c(\alpha, T) = V_0 \sqrt{\frac{\alpha}{1-\alpha}\left(\sigma_{\beta p}^T + \left(\phi^{-1}(\alpha)\sigma_{sp}^T\right)^2\right)} \tag{24}$$

The tracking error at risk (TaR) is a measure of portfolio risk estimating the magnitude that the portfolio return may deviate from the benchmark return over time. The maximum deviation of a portfolio return over the time horizon T at given confidence level α is

$$\nu_{p-b}(\alpha, T) = V_0 \sqrt{\left(\frac{1}{\sqrt{1-\alpha}}\sigma_{\beta, p-b}^T\right)^2 + \phi^{-1}\left(\frac{1+\alpha}{2}\right)\sigma_{s,p-b}^2} \tag{25}$$

How relevant is APT and Markowitz mean-variance analysis? We shall see in the next section. Spoiler alert: Harry Marowitz (1952 and 1959) produces statistically significant portfolio active returns and specific returns (stock selection) when used with a statistically significant tilt variable. A great recent book on portfolio risk is Connor, Goldberg, and Korajczyk (2010). Korajcyzk has a direct link to an outstanding bibliography on APT and multifactor risk models, https://www.kellogg.northwestern.edu/faculty/korajczy/htm/aptlist.htm.

Applying the Blin and Blender APT Model

Blin et al. (1997) used an estimated 20-factor beta model of covariances based on 3.5 years of weekly stock returns data. The Blin and Bender Arbitrage Pricing Theory (APT) model followed the Ross factor modeling theory, but Blin and Bender estimated betas from at least 20 orthogonal factors. Empirical support is reported in Guerard Jr (2012), and Guerard Jr., Krauklis, and Kumar (2012) for the application of mean-variance, enhanced index tracking (EIT), and tracking error at risk optimization techniques.

It is well known that as one raises the portfolio lambda, the expected return of a portfolio rises, and the number of securities in the optimal portfolios fall; see Blin et al. (1997). Lambda, a measure of risk aversion, the inverse of the risk aversion acceptance level of the Barra system, is a decision variable to determine the optimal

Table 6 Efficient frontier of the global stock selection model, GLER, with various portfolio optimization technique universe: (1) global I/B/E/S stocks with at least two analysts, 1999–2011, and (2) largest 7500 global stocks (in $USD)

APT risk model and optimization system						
Earnings model or component	Mean-variance methodology	Lambda	Annualized return	Sharpe ratio	Information ratio	Tracking error
GLER	M59	1000	15.84	0.590	0.78	13.11
		500	16.34	0.590	0.82	12.08
		200	16.37	0.610	0.85	12.68
		100	15.90	0.580	0.81	12.66
		5	10.11	0.440	0.51	8.81
Benchmark			5.59	0.240		
GLER	MVTaR	1000	16.10	0.660	0.94	11.18
		500	15.91	0.651	0.90	11.44
		200	16.09	0.691	0.97	10.83
		100	14.18	0.591	0.77	11.23
		5	8.51	0.344	0.33	8.75

Constraints:

1. Turnover is 8% (buys) monthly
2. Long-only holdings (a 4% maximum stock weight)
3. Threshold positions of 35 basis points

number of securities in a portfolio. As lambda increases, the number of stocks in portfolio portfolios falls, and Sharpe ratios rise. A more aggressive, growth-maximizing asset manager uses a larger lambda.

In spite of the Markowitz mean-variance (M59) portfolio construction and management analysis being six decades old, it does very well in maximizing the Sharpe ratio, geometric mean, and information ratio relative to newer approaches. See Table 6 for an adapted version of the Markowitz mean-variance and mean-variance tracking error at risk (MVTaR) simulations of Guerard Jr., Markowitz, and Xu (2015). A lambda of 200, an aggressive risk-tolerance parameter is sufficient in US and global markets to maximize the geometric mean, Sharpe ratio, and information ratio (IR) for the MCM variables. The lambda of 200 MVTaR GLER simulation outperforms the M59 (based on Markowitz's seminal Portfolio Selection, one of the two sleeping partners of the authors) on only the IR basis; 0.97 versus 0.85). MVTaR does not "Beat the Tar" out of the traditional Markowitz mean-variance model, but it does produce lower tracking errors and higher IRs. A lambda of 5 is reported to illustrate an index-enhanced portfolio strategy.[6] The purpose of Guerard Jr. et al.

[6]A very important result of APT portfolio modeling is reported in Guerard Jr. et al. (2012) for the application of mean-variance, enhanced index tracking, and tracking error at risk optimization techniques for the 1997–2009 time period. One can optimize US stocks to create portfolios in the Russell 3000 universe for the 12-year backtest period, 1997–2009, where the absolute value of the stock weight cannot deviate by more than 2%, which Guerard, Krauklis, and Kumar referenced to as an EIT portfolio, or equal active weight of 2% (EAW2). One needs 99 stocks in the mean-variance

(2015) was to update Guerard Jr and Mark (2003) and report additional evidence on the continued statistical significance of earnings forecasting analysis and the 9-factor composite stock selection model introduced in chapter "Risk and Return of Equity and the Capital Asset Pricing Model." In the world of applied investment research, the authors believe that all three levels of portfolio testing must be passed for statistical significance![7]

Applying the Blin and Blender APT Model, Barra, and Axioma: The McKinley Capital Management (MCM) Horse Race Tests

The Axioma Robust Risk Model is a multifactor risk model, in the tradition of the Barra model.[8] Axioma offers both US and world fundamental and statistical risk models. The Axioma Risk Models use several statistical techniques to efficiently estimate factors. The ordinary least squares (OLS) residuals are not homoskedastic; that is, when one minimizes the sum of the squared residuals to estimate factors using OLS, one finds that large assets exhibit lower volatility than smaller assets. A constant variance of returns is not found. Axioma uses a weighted least squares (WLS) regression, which scales the asset residual by the square root of the asset market capitalization (to serve as a proxy for the inverse of the residual variance). Robust regression, using the Huber M-estimator, addresses the issue and problem of outliers. (Asymptotic) principal components analysis (PCA) is used to estimate the statistical risk factors. A subset of assets is used to estimate the factors and the exposures and factor returns are applied to other assets. In 2011, McKinley Capital Management, LLC (MCM), initiated a "Horse Race" testing procedure to test if all optimizers were created equal. The Spring 2012 *Journal of Investing* featured a Quantitative Risk Management Special edition, reporting the MCM US Horse Race

portfolios and 108 stocks in the EAW2 portfolios for a lambda of 200. One needs 159 stocks in the mean-variance portfolios and 158 stocks in the EAW2 portfolios for a lambda of 10. A lambda of 10 is used by an index-hugging investor who is afraid of taking active bets in the portfolio. An even less aggressive manager (a horrid benchmark-hugger) needs 161 stocks in the EAW1 portfolios for a lambda of 10, as opposed to 131 stocks for a Sharpe ratio-maximizing asset manager using a lambda of 100.

[7]Guerard brought the Blin and Bender APT system to MCM in August 2005. The APT system produced real-time portfolios that generated statistically significant Active Returns and Specific Returns; see Guerard, Xu, and Markowitz (2019). The real-time Specific Returns exceeded 300 basis points, very consistent with its backtests. Guerard is highly biased in his assessment of risk models; Blin and Bender offered MCM a great discount and the modeling worked. Long-live APT and the "Blender Boys," as we called them at Drexel, Burnham, Lambert, where Guerard met them in 1988. Blin and Bender sold APT to SunGard, as Rudd sold Barra to Morgan Stanley. Guerard has great respect for entrepreneurs who build great models and sell them to become great commercial successes.

[8]*Axioma Robust Risk Model Handbook*, January 2010

Table 7 APT and axioma statistical risk models, US stocks, 1998–2009

Assumptions and constraints universe: All US stocks in the WRDS database **Time period:** Jan. 1998 to Dec. 2009							
Position constraint							
Minimum position =0.0 Maximum position = 4.0							
Position threshold holding = 0.35% **Turnover** = 8% monthly							
Transaction cost: 1.25% each way							
Risk model							
Axioma—Axioma statistical model APT—APT statistical model							
Risk setting							
Axioma—Tracking error target of 8% APT—Lambda of 200							
Model	Regression weighted EP BP CP SP REP RBP RCP RSP CTEF PM						
Benchmark: R3000G							
	SR	**IR**	**TE**	**SD**	**AR**	**Asset selection**	**T-Stat**
Axioma	0.177	0.69	7.78	21.92	7.06	6.00	5.09
APT	0.117	0.40	10.29	23.02	5.87	6.77	3.92
R3000G	−0.074			19.73	1.73		

and featuring articles by Markowitz; Tsuchida, Zhou, and Rachev (the Cognity system of FinAnalytica); Wormald and van Der Merwe (APT); Saxena and Stubbs (Axioma); and Guerard, Xu, and Gultekin (APT, Barra, and Cognity). They are not. In 2011, APT and Axioma were the winners among several (4–5) optimization systems using the MCM US "Public Models" CTEF and USER. Both APT and Axioma optimization systems produced highly statistically significant asset selection (the Barra attribution system was used as a judge). See Table 7.

What factor contributions of the Axioma portfolio were rewarded by the market, 1998–2009? Small size (the exposure of −1.17 produced 329 basis points of return with a corresponding t-statistic of 3.16) and earnings yield (the exposure of 0.09 produced 41 basis points of return with a corresponding t-statistic of 1.85, statistically significant at the 10% level). Growth, leverage, and yield produced negative statistically significant factor returns in the Barra USER attributions for 1998–2009. APT and Axioma both had statistically based risk models; Axioma and Barra had fundamentally based risk models. Which is better for CTEF and USER analyses?

Why Use the Axioma Statistical Model?

In 2012, MCM ran a second Horse Race competition using a two-I/B/E/S analyst, 7500 largest market-capitalized global universe. The Axioma Statistical Risk Model

Table 8 Axioma analysis of the FactSet Global 7500-stock data universe

Axioma ranked global backtest							
FSGLER model using all country world growth index constituents							
Universe: ACWG							
Simulation period: Jan. 1999–Dec. 2011							
Transactions costs: 150 basis points each way, respectively							
Return model	**Risk model**	**Tracking error**	**Sharpe ratio**	**Information ratio**	**Ann. active return**	**Ann. active risk**	**N**
GLER	STAT	4	0.554	1.475	9.99	6.78	144
		5	0.602	1.385	11.38	8.24	110
		6	0.656	1.409	13.25	9.40	87
		7	0.715	1.454	14.94	10.28	70
		8	0.748	1.451	16.20	11.16	58
	FUND	4	0.382	1.091	6.08	5.57	163
		5	0.460	1.151	7.73	6.72	129
		6	0.521	1.158	9.33	8.06	104
		7	0.582	1.217	11.02	9.06	83
		8	0.647	1.281	12.75	9.95	71

produced higher geometric means, Sharpe ratios, and information ratios using GLER data than its fundamental (WW21AxiomaMH) model (Table 8).

Guerard Jr., Markowitz, and Xu (2014, 2015) reported efficient frontiers using both of the Axioma Risk Models and found that the statistically based Axioma Risk Model, the authors denoted as "STAT," produced higher geometric means, Sharpe ratios, and information ratios than the Axioma Fundamental Risk Model denoted as "FUND." The geometric means and Sharpe ratios increase with the targeted tracking errors; however, the information ratios are higher in the lower tracking error range of 3–6%, with at least 200 stocks, on average, in the optimal portfolios. See Table 8. The Guerard et al. studies assumed 150 basis points, each way, of transaction costs. The use of ITG cost curves produced about 115–125 basis points of transaction costs, well under the assumed costs. The Guerard et al. studies also used the SunGard APT statistical model which produced statistically significant asset selection in US and global portfolios. The dominance of the statistical risk model has been a prevailing result of the MCM Horse Races, 2010–2017.

In 2017, MCM ran the third Horse Race using Axioma with integrated ITG transactions costs, in the MSCI non-US (XUS), Global (GL), and Emerging Markets (EM) universes. The XUS, GL, and EM reported results show much higher active and specific returns in non-US markets than in US stocks. International markets are more inefficient than US markets.

Alpha Alignment Factor

Axioma has pioneered several techniques to address the so-called underestimation of realized tracking errors, particularly during the 2008 financial crisis. This chapter, and Appendix 1, reports the alpha alignment factor (AAF) which recognizes the possibility of missing systematic risk factors and makes amends to the greatest extent that is possible without a complete recalibration of the risk model that accounts for the latent systematic risk in alpha factors explicitly. In the process of doing so, AAF approach not only improves the accuracy of risk prediction but also makes up for the lack of efficiency in the optimal portfolios.

The empirical results in a test-bed of real-life active portfolios based on client data show clearly that the abovementioned unknown systematic risk is a significant portion of the overall systematic risk and should be addressed accordingly. Saxena and Stubbs (2012) reported that the earning-to-price (E/P) and book-to-price (B/P) ratios used in USER model and Axioma Risk Model have average misalignment coefficients of 72% and 68%, respectively. While expected return and risk models are indispensable components of any active strategy, there is also a third component, namely, the set of constraints that is used to build a portfolio. Saxena and Stubbs (2012) proposed that the risk variance-covariance matrix C be augmented with additional auxiliary factors in order to complete the risk model. The augmented risk model has the form of

$$C_{new} = C + \sigma_{\underline{\alpha}}^2 \underline{\alpha} \cdot \underline{\alpha}' + \sigma_{\gamma}^2 \underline{\gamma} \cdot \underline{\gamma}' \tag{26}$$

where $\underline{\alpha}$ is the alpha alignment factor (AAF), σ_α is the estimated systematic risk of $\underline{\alpha}$, $\underline{\gamma}$ is the auxiliary factor for constrains, and σ_γ is the estimated systematic risk of $\underline{\gamma}$. The alpha alignment factor $\underline{\alpha}$ is the unitized portion of the uncorrelated expected return model, i.e., the orthogonal component, with risk model factors. Saxena and Stubbs (2012) reported that the AAF process pushed out the traditional risk model-estimated efficient frontier. Saxena and Stubbs (2015) refer to as alpha in the augmented regression model as the implied alpha. Saxena and Stubbs (2015) report that there is a small increment to specific risk compared to its true systematic risk.

Saxena and Stubbs (2012) applied their AAF methodology to the USER model, running a monthly backtest based on the above strategy over the time period 2001–2009 for various tracking error values of σ chosen from {4%, 5%, … 8%}. For each value of σ, the backtests were run on two setups, which were identical in all respects except one, namely, that only the second setup used the AAF methodology ($\sigma_\alpha = 20\%$). Axioma's fundamental medium-horizon risk model (US2AxiomaMH) is used to model the active risk constraints.

Saxena and Stubbs (2012) analyzed the time series of misalignment coefficients of alpha, implied alpha, and the optimal portfolio and found that almost 40–60% of the alpha is not aligned with the risk factors. The alignment characteristics of the implied alpha are much better than those of the alpha. Among other things, this implies that the constraints of the above strategy, especially the long-only

constraints, play a proactive role in containing the misalignment issue. In addition, not only do the orthogonal components of both the alpha and the implied alpha have systematic risk, but the magnitude of the systematic risk is comparable to that of the systematic risk associated with a median risk factor in US2AxiomMH.

Saxena and Stubbs (2012) showed the predicted and realized active risks for various risk target levels and noted the significant downward bias in risk prediction when the AAF methodology is not employed. The realized risk-return frontier demonstrates that not only does using the AAF methodology improve the accuracy of the risk prediction, it also moves the ex-post frontier upward, thereby giving ex-post performance improvements. In the process of doing so, AAF approach not only improves the accuracy of risk prediction but also makes up for the lack of efficiency in the optimal portfolios. Saxena and Stubbs (2015) extended their 2012 *Journal of Investing* research and reported positive frontier spreads. The reader is referred to Appendix 2 for the technical development of AAF.

Axioma AAF Applications

Saxena and Stubbs (2012) analyzed the time series of misalignment coefficients of alpha, implied alpha, and the optimal portfolio and found that almost 40–60% of the alpha is not aligned with the risk factors. Additionally, not only do the orthogonal components of both the alpha and the implied alpha have systematic risk, but the magnitude of the systematic risk is comparable to that of the systematic risk associated with a median risk factor in US2AxiomMH. Guerard and Chettiappan (2017) reported stock selection in non-US, Global, and EM universes, 2003–2016. We report adapted the Guerard and Chettiappan (2017) results in Table 9 that CTEF and GLER are highly statistically significant in producing statistically significant active and specific returns, which rose with higher targeted tracking errors. EM portfolios dominated XUS and GL portfolios in terms of producing higher geometric means, active returns, and specific returns.

Let us take a deep dive into the CTEF and GLER variables and their effectiveness. Let us examine the use of AAF in two analyses. First, Guerard Jr. (2012) used AAF in creating global portfolios using GLER for the 1999–2009 time period. The Axioma Statistical Risk Model with an AAF of 20% produced realized tracking errors closer to targeted tracking errors and higher information ratios using GLER data than its statistical risk model without AAF model. See Tables 5, 6, and 7 for support for Saxena and Stubbs (2012) position on AAF pushing out the frontiers. Moreover, AAF helps enhance IRs at targeted tracking errors of 5%, 6%, and 7%, respectively, with the Axioma Fundamental Risk Model in the EM universe and targeted tracking errors of 5 and 7% with the Axioma Statistical Risk Model.

The application of the Axioma Fundamental Risk Model to the EM universe produced a real-time portfolio having geometric means, Sharpe ratios, and IRs in the top decile of EM portfolios (see Table 10). See Guerard and Chettiappan (2017).

Table 9 Stock selection modeling in global, non-US, and EM universes

MSCI Index Constituents-only					
Axioma statistical risk model and optimizer					
Mean-variance analysis					
Model: Global CTEF GLER TE					
Period: 2003-01-31 to 2016-12-30 (monthly)		**Benchmark: ACWG**			
	Portfolio	**Active**		**Specific**	
	Return	**Return**	**T-Stat**	**Return**	**T-Stat**
Global_CTEF-TE4	12.44%	3.89%	2.28	1.36%	1.28
Global_CTEF-TE6	15.87%	7.32%	3.32	3.89%	2.89
Global_CTEF-TE8	16.31%	7.77%	3.13	4.13%	2.92
Global_GLER-TE4	13.28%	4.73%	2.95	1.54%	1.42
Global_GLER-TE6	16.03%	7.49%	3.61	3.26%	2.44
Global_GLER-TE8	16.87%	8.32%	3.45	4.27%	2.98
Model: XUS CTEF GLER TE					
Period: 2003-01-31 to 2016-12-30 (monthly)		**Benchmark: ACWXUSG**			
	Portfolio	**Active**		**Specific**	
	Return	**Return**	**T-Stat**	**Return**	**T-Stat**
XUS_CTEF_TE4	14.11%	6.17%	3.59	3.55%	3.10
XUS_CTEF_TE6	16.27%	8.32%	3.89	3.87%	2.85
XUS_CTEF_TE8	18.05%	10.10%	4.28	4.78%	3.15
XUS_GLER_TE4	13.78%	5.83%	3.67	1.93%	1.81
XUS_GLER_TE6	15.11%	7.17%	3.52	3.10%	2.27
XUS_GLER_TE8	16.22%	8.28%	3.82	4.36%	2.94
Model: EM CTEF GLER TE					
Period: 2003-01-31 to 2016-12-30 (monthly)		**Benchmark: EMG**			
	Portfolio	**Active**		**Specific**	
	Return	**Return**	**T-Stat**	**Return**	**T-Stat**
EM_CTEF_TE4	16.15%	5.43%	3.37	4.08%	3.27
EM_CTEF_TE6	17.34%	6.62%	3.00	3.76%	2.27
EM_CTEF_TE8	19.76%	9.04%	3.67	5.17%	2.97
EM_GLER_TE4	17.72%	7.00%	4.90	4.69%	3.97
EM_GLER_TE6	19.81%	9.09%	4.59	6.21%	3.91
EM_GLER_TE8	20.92%	10.20%	4.43	6.55%	3.81

Summary and Conclusions

Multifactor risk models, MFM, may be used to analyze a stock selection or evaluating variable. Blin et al. (1997) reported that properly constructed risk models created portfolios in which tilt variables that have statistically significant ICs produced statistically significant active returns. One must remove the market effect and extra-market covariances to properly estimate the contribution of a variable to the creation of efficient portfolios. It is important to see how quantitative analysis was developed in investment research and analysis. Harry Markowitz, Bill Sharpe, Jan

Table 10 A deep dive into EM portfolio construction

January 2003–December 2016				
Axioma risk model and optimizer				
	FUND risk model			
	Tracking error			
Model: EM MQ	5.00	6.00	7.00	10.00
Ann. port return	16.76	17.34	18.36	20.62
Ann. Std.	22.06	22.19	22.26	22.40
Ann. active return	6.45	7.24	8.05	10.31
Ann. active risk	5.04	5.95	6.65	8.35
N	68.20	63.40	58.50	55.40
ShR	0.682	0.714	0.748	0.844
IR	1.281	1.217	1.208	1.235
	FUND AAF20% risk model			
	Tracking error			
Model: EM MQ	5.00	6.00	7.00	10.00
Ann. port return	16.53	17.19	18.87	10.44
Ann. Std.	22.11	22.09	22.22	22.35
Ann. active return	6.29	6.89	8.57	10.12
Ann. active risk	4.31	5.24	6.94	8.15
N	77.80	69.70	59.80	56.00
ShR	0.671	0.701	0.773	0.838
IR	1.447	1.309	1.234	1.242

Mossin, and Jack Treynor pioneered capital market equilibrium and the creation and estimation of the Capital Asset Pricing Model. In the 1970s, Barr Rosenberg and his colleagues at Barra, John Blin and Steve Bender at APT, and Sebastian Ceria at Axioma developed and estimated multifactor risk models. Recent research and commercialization by APT and Axioma have furthered portfolio optimization with their statistically based risk models. MCM developed its Horse Race risk model philosophy to assess the most effective risk models to maximize the geometric means, Sharpe ratios, and information ratios. APT and Axioma have reigned supreme in our tests. The integration of Axioma and AAF and ITG transactions costs on the FactSet Investment have led to an Axioma portfolio implementation. There is a risk-return trade off. Active returns generally require significant active asset weights is portfolios.

Appendix 1: US-E3 Descriptor Definitions

This Appendix gives the detailed definitions of the descriptors which underlie the risk indices in US-E3. The method of combining these descriptors into risk indices is proprietary to Barra.

Volatility

(a) *BTSG: Beta times sigma*
This is computed as $\sqrt{\beta\sigma_{\varepsilon}}$, where β is the historical beta and σ_ε is the historical residual standard deviation. If β is negative, then the descriptor is set equal to zero.

(b) *DASTD: Daily standard deviation*
This is computed as

$$\sqrt{N_{days}\left[\sum_{t=1}^{T} w_t {r_t}^2\right]}$$

where r_t is the return over day t, w_t is the weight for day t, T is the number of days of historical returns data used to compute this descriptor (we set this to 65 days), and N_{days} is the number of trading days in a month (we set this to 23).

(c) *HILO: Ratio of high price to low price over the last month*
This is calculated as

$$log\left(\frac{P_H}{P_L}\right)$$

where P_H and P_L are the maximum price and minimum price attained over the last 1 month.

(d) *LPRI: Log of stock price*
This is the log of the stock price at the end of the last month.

(e) *CMRA: Cumulative range*
Let Z_t be defined as follows:

$$Z_t = \sum_{s=1}^{t} \log\left(1 + r_{\mathrm{i.s}}\right) - \sum_{s=1}^{t} \log\left(1 + r_{\mathrm{f,s}}\right)$$

where $r_{i,s}$ is the return on stock I in month s and $r_{f,s}$ is the risk-free rate for month s. In other words, Z_t is the cumulative return of the stock over the risk-free rate at the end of month t. Define Z_{max} and Z_{min} as the maximum and minimum values of Z_t over the last 12 months. CMRA is computed as

$$\log\left(\frac{1 + Z_{\max}}{a + Z_{\min}}\right)$$

(f) *VOLBT: Sensitivity of changes in trading volume to changes in aggregate trading volume*
This may be estimated by the following regression:

$$\frac{\Delta V_{\mathrm{i,t}}}{N_{\mathrm{i,t}}} = \mathrm{a} + \mathrm{b}\ \frac{\Delta V_{\mathrm{M,t}}}{N_{\mathrm{m.t}}} + \xi_{\mathrm{i,t}}$$

where $\Delta V_{i,t}$ is the change in share volume of stock I from week t − 1 to week t, $N_{i,t}$ is the average number of shares outstanding for stock I at the beginning of week t − 1 and week t, $\Delta V_{M,t}$ is the change in volume on the aggregate market from week t − 1 to week t, and $N_{M,t}$ is the average number of shares outstanding for the aggregate market at the beginning of week t − 1 and week t.

(g) *SERDP: Serial dependence*

This measure is designed to capture serial dependence in residuals from the market model regressions. It is computed as follows:

$$\mathrm{SERDP} = \frac{\frac{1}{\mathrm{T}-2}\ \sum_{t=3}^{T}(e_t + e_{t-1} + e_{t-2})^2}{\frac{1}{T-2}\ \sum_{t=3}^{T}(e^2{}_t + e^2{}_{t-1} + e^2{}_{t-2})}$$

where e_t is the residual from the market model regression in month t and T is the number of months over which this regression is run (typically, T = 60 months).

(h) *OPSTD: Option-implied standard deviation*

This descriptor is computed as the implied standard deviation from the Black-Scholes option pricing formula using the price on the closest to at-the-money call option that trades on the underlying stock.

Momentum

(a) *RSTR: Relative strength*

This is computed as the cumulative excess return (using continuously compounded monthly returns) over the last 12 months, i.e.,

$$\mathrm{RSTR}_{\overleftarrow{\rightarrow}} = \sum_{t=1}^{T} \log\left(1 + r_{\mathrm{i,t}}\right) - \sum_{t=1}^{T} \log\left(1 + r_{\mathrm{f,t}}\right)$$

where $r_{i,t}$ is the arithmetic return of the stock in month I and $r_{f,t}$ is the arithmetic risk-free rate for month i. This measure is usually computed over the last 1 year—i.e., T is set equal to 12 months.

(b) *HALPHA: Historical alpha*

This descriptor is equal to the alpha term (i.e., the intercept term) from a 60-month regression of the stock's excess returns on the S&P 500 excess returns.

Size

(a) *LNCAP: Log of market capitalization*
This descriptor is computed as the log of the market capitalization of equity (price times number of shares outstanding) for the company.

Size Nonlinearity

(a) *LCAPCB: Cube of the log of market capitalization*
This risk index is computed as the cube of the normalized log of market capitalization.

Trading Activity

(a) *STOA: Share turnover over the last year*
STOA is the annualized share turnover rate using data from the last 12 months—i.e., it is equal to $V_{ann}/\overline{N}_{out}$, where V_{ann} is the total trading volume (in number of shares) over the last 12 months and $\overline{N}_{out}$ is the average number of shares outstanding over the previous 12 months (i.e., it is equal to the average value of the number of shares outstanding at the beginning of each month over the previous 12 months).

(b) *STOQ: Share turnover over the last quarter*
This is computed as the annualized share turnover rate using data from the most recent quarter. Let V_q be the total trading volume (in number of shares) over the most recent quarter and let $\overline{N}_{out}$ be the average number of shares outstanding over the period (i.e., $\overline{N}_{out}$ is equal to the average value of the number of shares outstanding at the beginning of each month over the previous 3 months). Then, STOQ is computed as $4V_q/\overline{N}_{out}$.

(c) *STOM: Share turnover over the last month*
This is computed as the share turnover rate using data from the most recent month—i.e., it is equal to the number of shares traded last month divided by the number of shares outstanding at the beginning of the month.

(d) *STO5: Share turnover over the last 5 years*
This is equal to the annualized share turnover rate using data from the last 60 months. In symbols, STO5 is given by

$$\text{STO5} = \frac{12\left[\frac{1}{T}\sum_{s=1}^{T} V_s\right]}{\sigma_\varepsilon}$$

where V_s is equal to the total trading volume in month s and $\overline{N}_{out}$ is the average number of shares outstanding over the last 60 months.

(e) *FSPLIT: Indicator for forward split*

This descriptor is a 0–1 indicator variable to capture the occurrence of forward splits in the company's stock over the last 2 years.

(f) *VLVR: Volume to variance*

This measure is calculated as follows:

$$\text{VLVR} = \log \frac{\frac{12}{T}\left[\sum_{s=1}^{T} V_s P_s\right]}{\sigma_\varepsilon}$$

where V_s equals the number of shares traded in month s, P_s is the closing price of the stock at the end of month s, and σ_ε is the estimated residual standard deviation. The sum in the numerator is computed over the last 12 months.

Growth

(a) *PAYO: Payout ratio over 5 years*

This measure is computed as follows:

$$\text{PAYO} = \frac{\frac{1}{T}\sum_{t=1}^{T} D_t}{\frac{1}{T}\sum_{t=1}^{T} E_t}$$

where D_t is the aggregate dividend paid out in year t and E_t is the total earnings available for common shareholders in year t. This descriptor is computed using the last 5 years of data on dividends and earnings.

(b) *VCAP: Variability in capital structure*

This descriptor is measured as follows:

$$\text{VCAP}\overleftrightarrow{} = \frac{\frac{1}{T-1}\sum_{t=2}^{T}(|N_{t-1} - N_t|P_{t-1} + |\text{LD}_{t-1} - \text{LD}_t| + |\text{PE}_t - \text{PE}_{t-1}|)}{\text{CE}_T + \text{LD}_T + \text{PE}_T}$$

where N_{t-1} is the number of shares outstanding at the end of time $t-1$; P_{t-1} is the price per share at the end of time $t-1$; LD_{t-1} is the book value of long-term debt at the end of time period $t-1$; PE_{t-1} is the book value of preferred equity at the end of time period $t-1$; and $CE_T + LD_T + PE_T$ are the book values of common equity, long-term debt, and preferred equity as of the most recent fiscal year.

(c) *AGRO: Growth rate in total assets*

To compute this descriptor, the following regression is run:

$$\text{TA}_{it} = a + bt + \xi_{it}$$

where TA_{it} is the total assets of the company as of the end of year t and the regression is run for the period = 1, . . ., 5. AGRO is computed as follows:

$$\text{AGRO}\overleftrightarrow{} = \frac{b}{\frac{1}{T}\sum_{t=1}^{T}\text{TA}_{\text{it}}}$$

where the denominator average is computed over all the data used in the regression.

(d) *EGRO: Earnings growth rate over last 5 years*
First, the following regression is run:

$$\text{EPS}_{\text{t}} = \text{a} + \text{bt} + \xi_{\text{t}}$$

where EPS_t is the earnings per share for year t. This regression is run for the period t = 1, . . ., 5. EGRO is computed as follows:

$$\text{EGRO}\overleftrightarrow{} = \frac{b}{\frac{1}{T}\sum_{t=1}^{T}\text{EPS}_{t}}$$

(e) *EGIBS: Analyst-predicted earnings growth*
This is computed as follows:

$$\text{EGIBS}\overleftrightarrow{} = \frac{(\text{EARN} - \text{EPS})}{(\text{EARN}\overleftrightarrow{} + \overleftrightarrow{}\text{EPS})/2}$$

where EARN is a weighted average of the median earnings predictions by analysts for the current year and next year and EPS is the sum of the four most recent quarterly earnings per share.

(f) *DELE: Recent earnings change*
This is a measure of recent earnings growth and is measured as follows:

$$\text{DELE}\overleftrightarrow{} = \frac{(\text{EPS}_{t} - \text{EPS}_{\text{t-1}})}{(\text{EPS}_{t} + \text{EPS}_{t-1})/2}$$

where EPS_t is the earnings per share for the most recent year and EPS_{t-1} is the earnings per share for the previous year. We set this to missing if the denominator is nonpositive.

Earnings Yield

(a) *EPIBS: Analyst-predicted earnings-to-price*
This is computed as the weighted average of analysts' median predicted earnings for the current fiscal year and next fiscal year divided by the most recent price.

(b) *ETOP: Trailing annual earnings-to-price*
This is computed as the sum of the four most recent quarterly earnings per share divided by the most recent price.

(c) *ETP5: Historical earnings-to-price*
This is computed as follows:

$$\text{ETP5} = \frac{\frac{1}{T}\sum_{t=1}^{T} \text{EPS}_t}{\frac{1}{T}\sum_{t=1}^{T} P_t}$$

where EPS_t is equal to the earnings per share over year t and P_t is equal to the closing price per share at the end of year t.

Value

(a) *BTOP: Book-to-price ratio*
This is the book value of common equity as of the most recent fiscal year end divided by the most recent value of the market capitalization of the equity.

Earnings Variability

(a) *VERN: Variability in earnings*
This measure is computed as follows:

$$\text{VERN} = \frac{\left(\frac{1}{T-1}\sum_{t=1}^{T}\left(E_t - \overline{E}\right)^2\right)^{1/2}}{\frac{1}{T}\sum_{t=1}^{T} E_t}$$

where E_t is the earnings at time t(t = 1, . . ., 5) and $\bar{E}$ is the average earnings over the last 5 years. VERN is the coefficient of variation of earnings.

(b) *VFLO: Variability in cash flows*
This measure is computed as the coefficient of variation of cash flow using data over the last 5 years—i.e., it is computed in an identical manner to VERN, with cash flow being used in pace of earnings. Cash flow is computed as earnings plus depreciation plus deferred taxes.

(c) *EXTE: Extraordinary items in earnings*
This is computed as follows:

$$\text{EXTE} = \frac{\frac{1}{T}\sum_{t=1}^{T} |\text{EX}_t + \text{NRI}_t|}{\frac{1}{T}\sum_{t=1}^{T} E_t}$$

where EX_t is the value of extraordinary items and discontinued operations, NRI_t is the value of nonoperating income, and E_t is the earnings available to common before extraordinary items. The descriptor uses data over the last 5 years.

(d) *SPIBS: Standard deviation of analysts' prediction to price*

This is computed as the weighted average of the standard deviation of I/B/E/S analysts' forecasts of the firm's earnings per share for the current fiscal year and next fiscal year divided by the most recent price.

Leverage

(a) *MLEV: Market leverage*

This measure is computed as follows:

$$\text{MLEV} = \frac{\text{ME}_t + \text{PE}_t + \text{LD}_t}{\text{ME}_t}$$

where ME_t is the market value of common equity, PE_t is the book value of preferred equity, and LD_t is the book value of long-term debt. The value of preferred equity and long-term debt is as of the end of the most recent fiscal year. The market value of equity is computed using the most recent month's closing price of the stock.

(b) *BLEV: Book leverage*

This measure is computed as follows:

$$\text{BLEV} = \frac{\text{CEQ}_t + \text{PE}_t + \text{LD}_t}{\text{CEQ}_t}$$

where CEQ_t is the book value of common equity, PE_t is the book value of preferred equity, and LD_t is the book value of the long-term debt. All values are as of the end of the most recent fiscal year.

(c) *DTOA: Debt-to-asset ratio*

This ratio is computed as follows:

$$\text{DTOA} = \frac{\text{LD}_t + \text{DCL}_t}{\text{TA}_t}$$

where LD_t is the book value of long-term debt, DCL_t is the value of debt in current liabilities, and TA_t is the book value of total assets. All values are as of the end of the most recent fiscal year.

(d) *SNRRT: Senior debt rating*

This descriptor is constructed as a multilevel indicator variable of the debt rating of a company.

Currency Sensitivity

(a) *CURSENS: Exposure to foreign currencies*
To construct this descriptor, the following regression is run:

$$r_{it} = \alpha_I + \beta_i r_{mt} + \varepsilon_{it}$$

where r_{it} is the excess return on the stock and r_{mt} is the excess return on the S&P 500 Index. Let ε_{it} denote the residual returns from this regression. These residual returns are in turn regressed against the contemporaneous and lagged returns on a basket of foreign currencies, as follows:

$$\varepsilon_{it} = c_i + \gamma_{i1}(FX)_t + \gamma_{i2}(FX)_{t-1} + \gamma_{13}(FX)_{t-2} + \mu_{it}$$

where ε_{it} is the residual return on stock I, $(FX)_t$ is the return on an index of foreign currencies over month t, $(FX)_{t-1}$ is the return on the same index of foreign currencies over month $t - 1$, and $(FX)_{t-2}$ is the return on the same index over month $t - 2$. The risk index is computed as the sum of the slope coefficients γ_{i1}, γ_{i2}, and γ_{i3}—i.e., CURSENS = γ_{i1} + γ_{i2} + γ_{i3}.

Dividend Yield

(a) *P_DYLD: Predicted dividend yield*
This descriptor uses the last four quarterly dividends paid out by the company along with the returns on the company's stock and future dividend announcements made by the company to come up with a Barra-predicted dividend yield.

Non-estimation Universe Indicator

(a) *NONESTU: Indicator for firms outside US-E3 estimation universe*
This is a 0–1 indicator variable: It is equal to 0 if the company is in the Barra estimation universe and equal to 1 if the company is outside the Barra estimation universe.

Appendix 2: Factor Alignment Problems and Quantitative Portfolio Management

Until now, we have discussed Markowitz mean-variance optimization (MVO) framework, evolution of factor models in modeling risk and return, and their application in building reliable and robust expected returns models. Next, we move our focus to the application of factor models within the broader aegis of quantitative portfolio managements (QPM).

Factor models play an integral role in quantitative equity portfolio management. Their applications extend to almost every aspect of quantitative investment methodology including construction of alpha models, risk models, portfolio construction, risk decomposition, and performance attribution. Given their pervasive presence in the field and the natural trend toward specialization, it comes as no surprise that different groups of researchers are often involved in developing factor models for each one of the aforementioned applications.

For instance, a team of quantitative portfolio managers (PM) can develop an in-house model for alpha generation and procure a factor model for purposes of risk management from a third-party risk model vendor. Subsequently, they can combine the two models within the framework of Markowitz mean-variance optimization (MVO) framework to construct optimal portfolios. A completely different factor model can then be used for purposes of performance attribution to identify the key drivers and detractors of performance. Notably, the choice of factors in each one of these factor models need not be identical, thereby introducing incongruity in the portfolio management process. To further complicate the matters, the constraints in the quantitative strategy can introduce additional systematic risk exposures that are not captured by the risk model. The problems that arise due to the interaction between the alpha model, the risk model, and constraints in MVO framework are collectively referred to as factor alignment problems (FAP). Detailed theoretical investigation of FAP and subsequent evolution of augmented risk models has become an interesting line of research in recent years. We refer the readers to Saxena and Stubbs (2012, 2013, 2015), Ceria, Saxena, and Stubbs (2012), and Martin, Saxena, and Stubbs (2013) for detailed analysis on these topics. A brief summary of the key themes is presented next.

We initiate the discussion by focusing on the unconstrained MVO problem, namely,

$$\text{Maximize}\, \alpha * h - (1/2) * h^T * Q * h,$$

where α, h, and Q denote the expected returns (referred to as alpha), portfolio holdings, and covariance matrix of returns based on a factor risk model, respectively. Furthermore, let X denote the matrix of risk factors that are used in building the covariance matrix Q. It is instructive to examine the relationship between α and X. Specifically, consider the projection of α on the vector space spanned by the columns of X; the said projection, α_X, is referred to as the spanned component of α,

whereas the residual, $\alpha_O = \alpha - \alpha_X$, is referred to as the orthogonal component. Even though both the spanned and orthogonal components are derived from a common source, namely, α, they have widely different risk characteristics especially when examined through the lens of the covariance matrix Q.

By virtue of being spanned by the risk factors X, the spanned component is deemed to have systematic risk when examined by the risk model. Among other things, this implies that any portfolio exposure to the spanned component is penalized in the MVO optimization framework for both systematic and specific risk. The orthogonal component, on the other hand, is assumed to have no systematic risk since $X^T * \alpha_X = 0$. This implies that any exposure to the orthogonal component gets a free pass since it is penalized only for the specific risk. Furthermore, as the number of holdings in the portfolio increases, specific risk can asymptotically go down to zero due to diversification benefits. This disparity between the risk treatment of the spanned and orthogonal component creates incentives for the optimizer to the load on the orthogonal component at the cost of exposure to the spanned component.

Such an overloading on the orthogonal component is in itself not particularly troublesome. In fact, if the risk model captures "all" possible sources of systematic risk, such an overloading is completely justified by the portfolio construction philosophy of MVO. Unfortunately, risk models are not designed to capture all sources of risk. Instead, they are designed to capture salient sources of systematic risk and often omit systematic risk factors that are deemed to be insignificant. These design choices are motivated by a variety of reasons including but not limited to statistical power concerns, preference for parsimony, aversion to multi-collinearity, etc. Consequently, choices made during the design of a risk model can result in risk "blind spots," namely, systematic risk factors that are missing from the risk model. If the alpha model happens to bet on these risk blind spots, the resulting systematic risk exposures will be left untreated during the portfolio construction phase resulting in latent systematic risk exposures. Next, we discuss some of the practical consequences of such latent risk exposures.

Most QPM employ active or total risk constraints. These constraints are meant to limit the volatility of the portfolio and avoid wild swings in portfolio performance. For instance, a portfolio manager (PM) can include an active risk constraint as part of the portfolio construction process to limit the annualized volatility of active returns to be less than or equal to a predetermined limit, say 3%. The expression of this constraint during portfolio optimization necessarily requires a risk model. If the risk and alpha models suffer from FAP, the optimizer will load up on the orthogonal component of alpha, and the realized risk of the portfolio can be significantly higher than 3%. Saxena and Stubbs (2013) studied this phenomenon in detail and found that the divergences in risk prediction that result from FAP tend to be statistically significant and increase during periods of market stress. In other words, presence of FAP can lead of "volatility" surprises in the realized performance of portfolio, and these surprises tend to aggravate during periods of market turmoil. Additionally, they were able to quantify the amount of latent systematic risk in the orthogonal component of alpha and showed that α_O tends to be more volatile than almost half of the risk factors included in a typical commercial risk model. Ceria et al. examine

potential sources of latent systematic risk factors and argued that proprietary definitions of well-known factors (momentum, value, growth, etc.) can introduce latent systematic risk factors. In other words, a portfolio manager's propensity to curate an alpha factor not only leads to better performance (aka alpha) but also introduces latent systematic risk exposures that often go undiagnosed during the portfolio construction process. Martin et al. (2013) examined the role of constraints in FAP and concluded certain kinds of portfolio constraints can also create unintended systematic risk factor exposures which can expose the portfolio to the vagaries of FAP.

Risk underestimation, as discussed here, is merely a symptom of a bigger issue, namely, compromise of the mean-variance optimality of the resulting portfolio holdings. Recall that the primary objective of portfolio optimization is to create a portfolio having an optimal risk-return trade-off. If a portion of systematic risk exposure of the portfolio is inadequately captured by the risk model, then the resulting portfolio cannot be expected to be optimal ex-post, its ex-ante optimality notwithstanding. Factor alignment problems (FAP) symbolizes the difficulty that a portfolio manager (PM) faces in ensuring consistency between the ex-post characteristics of an optimized portfolio and her ex-ante expectations. Lack of alignment between the alpha and risk factors creates risk "blind spots" that get exaggerated by virtue of employing an optimizer. The resulting optimized portfolios take excessive exposure to systematic factors missing from the risk model, thereby compromising the overarching goal of optimal risk-return trade-off, as originally envisaged by Harry Markowitz. Saxena and Stubbs (2015) demonstrate the usefulness of augmented risk models in addressing this issue. Their results indicate that augmenting the risk model with an appropriate augmenting factor not only remedies the risk underestimation problem but also improves risk adjusted returns, thereby restoring the notion of Markowitz efficiency.

Among other things, their results suggest strong synergistic advantages of integrating alpha and risk research processes. In other words, we need to abandon the "one-size-fits-all" approach to risk management and take a more nuanced approach that is sensitive to the specific requirements of a PM. Ultimately, the primary responsibility of a risk model is to capture all un-diversifiable (i.e., systematic) sources of risk that are relevant to a given investment process. A risk model that is constructed in a manner which is agnostic to the very factors that the PM is betting on cannot be expected to accomplish that goal. Augmented risk models partly accomplish this goal and should act as precursor to fully customized risk models that shed the artificial barrier between alpha and risk research and take a holistic view of the investment process.

References

Black, F., Jensen, M. C., & Scholes, M. (1972). The capital asset pricing model: Some empirical tests. In M. Jensen (Ed.), *Studies in the theory of capital markets*. New York: Praeger.

Blin, J. M., Bender, S., & Guerard Jr., J. B. (1997). Earnings forecasts, revisions and momentum in the estimation of efficient market-neutral japanese and U.S. portfolios. In A. Chen (Ed.), *Research in finance* (Vol. 15). Greenwich, CT: JAI Press, 93–114.

Ceria, S., Saxena, A., & Stubbs, R. A. (2012). Factor alignment problems and quantitative portfolio management. *The Journal of Portfolio Management, 38*(2), 29–43.

Chen, N.-F., Roll, R., & Ross, S. (1986). Economic forces and the stock market. *Journal of Business, 59*, 383–403.

Cohen, K. J., & Pogue, J. A. (1967). An empirical evaluation of alternative portfolio-selection models. *Journal of Business, 40*, 166–193.

Connor, G., Goldberg, L., & Korajczyk, R. A. (2010). *Portfolio risk analysis*. Princeton: Princeton University Press.

Dhrymes, P. J., Friend, I., & Gultekin, N. B. (1984). A critical re-examination of the empirical evidence on the arbitrage pricing theory. *Journal of Finance, 39*, 323–346.

Dhrymes, P. J., Friend, I., Gültekin, M. N., & Gültekin, N. B. (1985). New tests of the APT and their implications. *Journal of Finance, 40*, 659–674.

Dimson, E. (1988). *Stock market anomalies*. Cambridge: Cambridge University Press.

Elton, E. J., & Gruber, M. J. (1970). Homogeneous groups and the testing of economic hypothesis. *Journal of Financial and Quantitative Analysis, 5*, 581–602.

Elton, E. J., & Gruber, M. J. (1973). Estimating the dependence structure of share prices—Implications for portfolio selection. *Journal of Finance, 28*, 1203–1232.

Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy, 81*, 607–636.

Fama, E. F., & French, K. R. (1992). Cross-sectional variation in expected stock returns. *Journal of Finance, 47,* 427–465.

Fama, E. F., & French, K. R. (1995). Size & the book-to-market factors in earnings & returns. *Journal of Finance, 50*, 131–155.

Fama, E. F., & French, K. R. (2008). Dissecting anomalies. *Journal of Finance, 63*, 1653–1678.

Farrell Jr., J. L. (1974). Analyzing covariance of returns to determine homogeneous stock groupings. *Journal of Business, 47*, 186–207.

Farrell Jr., J. L. (1997). *Portfolio management: Theory and applications*. New York: McGraw-Hill/Irwin.

Fogler, R., John, K., & Tipton, J. (1981). Three factors, interest rate differentials, and stock groups. *Journal of Finance, 36*, 323–335.

Grinhold, R., & Kahn, R. (1999). *Active portfolio management*. New York: McGraw-Hill/Irwin.

Guerard Jr., J. B. (2012). Global earnings forecasting efficiency. In J. Kensinger (Ed.), *Research in finance* (Vol. 28).

Guerard Jr., J. B., & Schwartz, E. (2007). *Quantitative corporate finance*. New York: Springer.

Guerard Jr., J. B., & Chettiappan, S. (2017). Active quant: Applied investment analysis in emerging markets. *Journal of Investing, 26*, 138–152.

Guerard Jr., J. B., Krauklis, E., & Kumar, M. (2012). Further analysis of efficient portfolios with the user data. *Journal of Investing, 21*(1), 81–88.

Guerard Jr., J. B., & Mark, A. (2003). The optimization of efficient portfolios: The case for a quadratic R&D term. In A. Chen (Ed.), *Research in finance* (Vol. 20, pp. 213–247). Greenwich, CT: JAI Press.

Guerard Jr., J. B., Markowitz, H. M., & Xu, G. (2014). The role of effective corporate decisions in the creation of efficient portfolios. *IBM Journal of Research and Development, 58*(6), 1–11.

Guerard Jr., J. B., Markowitz, H. M., & Xu, G. (2015). Earnings forecasting in a global stock selection model and efficient portfolio construction and management. *International Journal of Forecasting, 31*, 550–560.

Guerard Jr., J. B., Xu, G., & Gultekin, M. N. (2012). Investing with momentum: The past, present, and future. *Journal of Investing, 21*, 68–80.

Guerard Jr., J. B., Xu, G., & Markowitz, H. M. (2019). A further analysis of robust regression modeling & data mining corrections testing in global stocks. Annals of Operations Research, forthcoming.

Guerard Jr., J. B., & Mark, A. (2020). Earnings forecasts and revisions, price momentum, and fundamental data: Further explorations of financial anomalies. In C. F. Lee (ed.), Handbook of Financial Econometrics, Statistics, and Machine Learning. Singapore, World Scientific Handbook in Financial Economics, 1151–1209.

Jensen, M. C. (1968). The performance of mutual funds in the period 1945-1964. *Journal of Finance, 23*, 389–416.

King, B. F. (1966). Market and industry factors in stock price behavior. *Journal of Business, 39*, 139–191.

Martin, C., Saxena, A., & Stubbs, R. A. (2013). Constraints in quantitative strategies: An alignment perspective. *Journal of Asset Management, 14*(5).

Miller, W., Xu, G., & Guerard Jr., J. B. (2014). Portfolio construction and management in the Barra Aegis system: A case study using the USER data. *Journal of Investing, 23*, 111–120.

Roll, R., & Ross, S. A. (1984). The arbitrage pricing theory approach to strategic portfolio planning. *Financial Analysts Journal, 40*(3), 14–26.

Rosenberg, B. (1974). Extra-market components of covariance in security returns. *Journal of Financial and Quantitative Analysis, 9*, 263–274.

Rosenberg, B., Hoaglet, M., Marathe, V., & McKibben, W. (1975). *Components of covariance in security returns*. Working paper no. 13, Research Program in Finance. Institute of Business and Economic Research. University of California, Berkeley.

Rosenberg, B., & Marathe, V. (1979). Tests of capital asset pricing hypotheses. In H. Levy (Ed.), *Research in finance* (Vol. 1).

Rosenberg, B., & McKibben, W. (1973). The prediction of systematic and unsystematic risk in common stocks. *Journal of Financial and Quantitative Analysis, 8*, 317–333.

Ross, S. A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory, 13*, 341–360.

Ross, S. A., & Roll, R. (1980). An empirical investigation of the arbitrage pricing theory. *Journal of Finance, 35*, 1071–1103.

Rudd, A., & Clasing, H. K. (1982). *Modern portfolio theory: The principles of investment management*. Homewood, IL: Dow Jones-Irwin.

Rudd, A., & Rosenberg, B. (1979). Realistic portfolio optimization. In E. Elton & M. Gruber (Eds.), *Portfolio theory, 25 years after*. Amsterdam: North-Holland.

Rudd, A., & Rosenberg, B. (1980). The 'market model' in investment management. *Journal of Finance, 35*, 597–607.

Saxena, A., & Stubbs, R. A. (2012). An empirical case study of factor alignment using the USER model. *Journal of Investing, 21*(1), 25–43.

Saxena, A., & Stubbs, R. A. (2013). Alpha alignment factor: A solution to the underestimation of risk for optimized active portfolios. *The Journal of Risk, 15*(3), 3–37.

Saxena, A., & Stubbs, R. (2015). Augmented risk models to mitigate factor alignment problems. *Journal of Investment Management, 13*(3), 57–79.

Stone, B. K. (1974). Systematic interest-rate risk in a two-index model of returns. *Journal of Financial and Quantitative Analysis, 9*, 709–721.

Stone, B. K., Guerard Jr., J. B., Gultekin, M., & Adams, G. (2002). *Socially responsible investment screening*. Working Paper. Marriott School of Management. Brigham Young University.

Ziemba, W. T. & Schwartz, S. (1993). *Invest Japan*. Chicago: Probus Publishing Co..

Options



Options can be generalized as contracts that can be bought at a given price enabling one to buy or sell an asset or security at a possible future profit. If the profitable opportunity does not arise, the price paid for the option is foregone. An understanding of the theory and analysis of options is useful to the financial managers in enabling them to estimate trends and may be employed to temporarily secure assets until a decision is made whether to buy or not and to hold on to new projects or innovations until a final decision.

Options are financial contracts that allow one to purchase or sell assets at a predetermined price up to an expiration date. Perhaps the most recognized traded options appear in the financial markets as put and call options on common stocks. Speculators use options because of their low initial cost and possible high returns. Common stock options have been traded on the Chicago Board Options Exchange (CBOE) since 1973. A call option gives its owner the right, not the obligation, to buy a specific number of shares of stocks at a specific price, known as the exercise price, up to the expiration date. A premium paid is the cost of purchasing the option. A buyer of the call option expects the stock price to rise. A seller or writer of the call option expects the stock price to decline or remain the same. The seller retains the premium whether or not the option is exercised. Options help amplify the risk-return of stocks (and other assets) by the use of greater leverage. If a stock rises from \$80 to \$90, a 12.5% return can be enhanced if one purchased a call option (an option to buy stock at \$2 and sell it at \$10) for the stock price rise. Leverage translates to a greater risk-return trade-off. If the stock declines from \$80 to \$60, one loses only the premium (\$2), not the \$20, on the stock. If you are bullish on a stock (expect it to appreciate), then an option can be exciting and profitability.

Let us look at an example of a common stock purchase and an option valuation. The common stock price of IBM was \$134.04 on December 31, 2019. Let us assume that IBM is expected to reach \$155 by mid-June 2020. If a trader believed that IBM would appreciate in value during the coming months, he could purchase 100 shares of stock for \$13,404 plus the brokerage commission. If IBM increases in value to

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_16

$155 per share, then the investor has secured profits of $20.96 per share or $2096 on the 100 shares. The profit on such an investment is 15.645%:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{\$155 - \$134.04}{\$134.04} = .1564$$

On the other hand, the price of IBM may fall or remain constant, and it might rise. If the price of IBM fell to $90.00 per share, then the loss on the stock is −32.86%:

$$R_t = \frac{\$90.00 - 134.04}{134.04} = -.3286$$

A trader could purchase an IBM call option on December 31, 2019, with various exercise or strike prices, shown in Table 1. For call options expiring in June 2020, one notes that the price one must pay for the call option is a function of the strike price. A call option is the right to purchase stock, and that right has value only if one expects the stock price to exceed the exercise price. The intrinsic value of a call option is its common stock price less its exercise price. Let us look at the premia on IBM call options in Table 1.

The premia that one must pay for an IBM call price, an option to purchase stock, declines with the exercise price. As the exercise price rises, there is less of a chance that the stock price will exceed the exercise price and have intrinsic value.

Exercise price	Premium
$125	$12.05
130	8.75
135	6.10
150	1.44
155	0.82
160	0.47

Note that if the common stock price rose to $157 by June 2020 one purchased a call option to purchase IBM at a $150 exercise price, for $1.44, then one could earn a larger return than buying the stock. If the price of IBM rises to $157, then the call option must be worth at least its intrinsic value, or $7.00. The rate of return for the call option is at least 386%:

$$R_t \geq \frac{\$157 - \$150 - 1.44}{1.44} = 3.86$$

or 386%!

The common stockholder return was 17.19% for the similar price movement—options offer investors the advantage of leverage. One may deal in the movement of a $134.04 stock with lower costs by purchasing a call option. However, one may also lose a large portion of the amount at risk by dealing in options. If the price of IBM falls to $90, then the call option has zero value, and the rate of return on the call

Table 1 IBM June 2020 call options (Priced on December 31, 2019)

Strike	Ticker	BSze	Bid	Ask	ASze	Last	IVM	DM	Volm	OInt
19-Jun-20 (171d); CSize 100; IDiv 3.09 USD; R 1.90; IFwd 131.90										
90	IBM 6/19/20 C90	0	41.850	45.300	0	48.5200	0.0000	1.0000	0	0
95	IBM 6/19/20 C95	0	37.000	40.450	0	0.0000	0.0000	1.0000	0	0
100	IBM 6/19/20 C100	0	32.000	35.700	0	33.7000	0.0000	1.0000	0	11
105	IBM 6/19/20 C105	0	27.150	30.600	0	33.2500	0.0000	1.0000	0	0
110	IBM 6/19/20 C110	0	22.950	25.250	0	26.0000	21.5999	0.9645	0	5
115	IBM 6/19/20 C115	0	19.600	20.650	0	19.9200	25.2699	0.8527	1	160
120	IBM 6/19/20 C120	0	15.500	15.950	0	15.1000	22.8308	0.7924	21	510
125	IBM 6/19/20 C125	0	11.800	12.050	0	13.4100	21.6535	0.7001	0	176
130	IBM 6/19/20 C130	0	8.650	8.750	0	8.2500	20.8212	0.5920	10	1074
135	IBM 6/19/20 C135	0	5.950	6.100	0	5.8000	20.0183	0.4785	17	1023
140	**IBM 6/19/20 C140**	**0**	**3.850**	**3.950**	**0**	**3.7100**	**19.1620**	**0.3626**	**34**	**1789**
145	IBM 6/19/20 C145	0	2.370	2.450	0	2.2400	18.6163	0.2566	11	1342
150	**IBM 6/19/20 C150**	**0**	**1.390**	**1.440**	**0**	**1.3200**	**18.2172**	**0.1720**	**352**	**5074**
155	IBM 6/19/20 C155	0	0.790	0.820	0	0.7500	17.9622	0.1084	20	807
160	IBM 6/19/20 C160	0	0.420	0.470	0	0.4400	17.8109	0.0664	13	900
165	IBM 6/19/20 C165	0	0.210	0.290	0	0.3300	17.8518	0.0395	0	390
170	IBM 6/19/20 C170	0	0.120	0.190	0	0.1900	18.2436	0.0256	0	203
175	IBM 6/19/20 C175	0	0.070	0.130	0	0.1200	18.7634	0.0170	0	478
180	IBM 6/19/20 C180	0	0.040	0.110	0	0.0900	19.5545	0.0126	0	46

Source: Bloomberg, L.P.

option is −100%, whereas the common stockholder lost only 32.86%. However there is a limited loss on a call option; one may lose all of your premium, but no more. A call option on a stock whose stock price exceeds its exercise price is referred to as an "in the money" (ITM) option. The intrinsic value of an ITM option is positive, and one profits dollar for dollar on the call option as the stock price rises.

Note that if an investor was "bullish" on IBM, then one might prefer to purchase a call option whose exercise price is $135 approximately the same price as the stock price. One can purchase this option, known as an "at the money" (ATM) call option, for a premium of $6.10. If the price of IBM rises to $150, then the ATM call option return is at least 284.62%:

$$R = \frac{\$150 - \$135 - \$6.10}{6.10} = \frac{8.90}{6.10} = 145.90\%$$

There is a time component to options in which there is time for the stock price to rise and exceed the exercise price, producing a future positive intrinsic value. If IBM rises to $92, then the $90 exercise price call option buyer realizes a rate of return of at least 1900.00% for the month:

$$R_t = \frac{92 - 90 - .10}{.10} = 19.0000$$

Obviously, such a rapid rise of IBM is very unlikely. A call option whose stock price is less than the exercise price is referred to as an "out of the money" (OTM) option. The call option premium for an ITM option exceeds the premium of corresponding ATM or OTM options. For call option buyers, OTM offer the greatest leverage. The reader will note that the rate of return for a $90 exercise price exceeds that of the $85 exercise price, if the stock price rises.

A bullish investor who purchases OTM options also runs the chance that the options will expire useless. Thus, one sees a return and risk trade-off of call options in terms of deciding upon the exercise price. One can draw a graph to illustrate call option buying strategies.

One profits by purchasing call options as stock prices rise to exceed the exercise price and the premium paid for the option (Fig. 1).

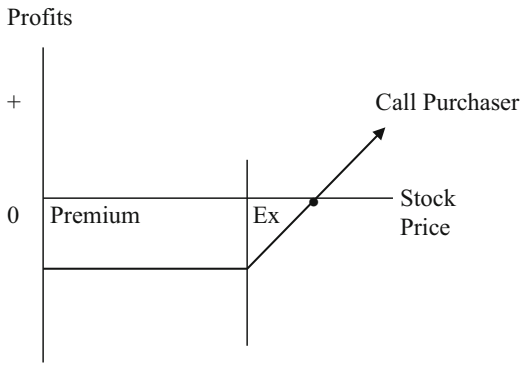


Fig. 1 Profiting on buying a call option

Why would one write or sell a call option? If an investor owns the stock and expects no significant appreciation in the near future, then one can write call options to provide additional income or returns. That is, the investor would receive the premiums. In the case of the previously examined IBM call options, one can write a September 2004 option and receive $1.50. If IBM's stock price is still about $85.00 at the expiration of the November option, then the call writer keeps the premium and the call buyer's option expires without value. The call writer receives a $0.15 premium for writing the $90 exercise or strike price option, and all stock price appreciation above $90 is "traded away" for a $0.15 premium. In essence, the investor who owns the underlying stock and writes calls on the stock is known as a covered call writer, and all gains above the exercise price are called away for the premium. As the call writer writes OTM options with an exercise price that is rising, then the covered call writer returns approximate owning (long) the stock returns. The portfolio manager or writer of this covered call clearly expects the price of IBM to be at or below $90 by the last week of September. The writing of covered calls by portfolio holders can be very profitable.

An investor who buys a put option buys the right to sell the stock. The put option has a specified strike or exercise price for a stated time period, just as does the call option. The put option is of interest to an investor who expects the stock price to decline. For example, if an investor buys an $85 put on IBM, then the investor is paying $1.50 to sell IBM at 85. If the price of IBM falls $70, then the intrinsic value of the put is $15, and there will be an additional premium for time. If the price of IBM falls to $50, then the put option has at least $35 of intrinsic value. An investor who expects the common stock of a firm to fall may either buy a put or sell the stock short. An investor who writes or sells a put expects the stock not to fall and receive the premium.

Options may be used to protect stock gains. Suppose an investor bought Lockheed stock, LMT, on July 21, 2016, in anticipation of the run-up to the Presidential election of November 2016. The investor paid $253.60 a share for the stock in July 2016. Lockheed stock was priced about $390 a share on December 31, 2019. LMT had outperformed the S&P 500, leading to a very happy investor. The investor might be concerned about LMT falling, and the investor could buy a "protective" put, with an exercise price of $375, for $14.60 (see Table 2). Any loss of LMT below $375 would be hedged by the put for a cost of only $14.60. If LMT fell back to $253.60, the investor would maintain a gain of ($375 − 253.60 − $14.60, or $106.80), whereas the stock investor profits would fall to zero! The return on buying LMT and the subsequent protective put would be 38.39% ($106.80/($253.60 + $14.60)).

Options operate like financial leverage on stock prices.

Let us introduce the notation of Malkiel and Quandt that is very useful to address combinations of put and call strategies. The Malkiel-Quandt notation is a bit dated, but in the opinion of the authors, a very useful means to describe the basic premises of buying and writing puts, calls, and straddles, combinations of a call and a put. We will later in the chapter introduce the reader to the binomial options pricing model, the current means of illustrating the pricing structure of calls and puts.

Table 2 LMT put option prices

Strike	Ticker	Bid	Ask	Last	IVM	DM	Volm	OInt
19-Jun-20 (171d); CSize 100; IDiv 4.43 USD; R 1.90; IFwd 387.28								
345	LMT 6/19/20 P345	6.70	7.00	6.90	21.779	−0.199	2	20
350	LMT 6/19/20 P350	7.60	7.90	6.90	21.348	−0.223	0	289
355	LMT 6/19/20 P355	8.70	8.90	9.07	20.902	−0.248	1	64
360	LMT 6/19/20 P360	9.90	10.10	10.00	20.458	−0.277	3	52
365	LMT 6/19/20 P365	11.10	11.50	11.10	20.101	−0.310	6	92
370	LMT 6/19/20 P370	12.60	13.00	12.80	19.682	−0.344	1	226
375	**LMT 6/19/20 P375**	**14.30**	**14.60**	**14.70**	**19.284**	**−0.379**	**13**	**203**
380	LMT 6/19/20 P380	16.20	16.50	16.50	18.945	−0.418	8	50
385	LMT 6/19/20 P385	18.20	18.60	18.40	18.550	−0.458	20	19
390	LMT 6/19/20 P390	20.50	20.80	20.30	18.261	−0.498	15	36
395	LMT 6/19/20 P395	22.80	23.30	24.70	17.778	−0.543	0	2
400	LMT 6/19/20 P400	25.50	26.10	30.50	17.526	−0.585	0	8
405	LMT 6/19/20 P405	28.40	29.00	30.00	17.202	−0.628	0	7
410	LMT 6/19/20 P410	31.60	32.20	30.40	17.033	−0.668	0	7
415	LMT 6/19/20 P415	34.90	35.60	40.80	16.683	−0.711	0	17
420	LMT 6/19/20 P420	38.50	39.10	37.70	16.457	−0.749	0	5
425	LMT 6/19/20 P425	42.30	42.90	39.05	16.241	−0.785	0	13
430	LMT 6/19/20 P430	46.30	47.20	46.20	16.291	−0.814	0	9
435	LMT 6/19/20 P435	49.90	52.50	0.00	16.515	−0.837	0	0

Source: Bloomberg, L.P.

The Malkiel-Quandt Notation

The authors have long admired the options notation put forth by Malkiel and Quandt (1969) in their very enjoyable and stimulating monograph on options markets. One profits from a stock price rise when one purchases a call option and loses only the premium from a stock price decline. As Malkiel and Quandt write, the option profits lose in terms of stock price advances and declines in a 2×1 matrix.

$$\begin{bmatrix} \overline{Pcs} \uparrow \\ \overline{Pcs} \downarrow \end{bmatrix}$$

If one purchases a call option, one profits as the stock price advances, and the Malkiel-Quandt notation would be

$$\begin{bmatrix} +1 \\ 0 \end{bmatrix}$$

Note that the premiums are not included in the analysis and one must be careful in examining option/portfolio strategies. There may be a problem with naked call

writing where the writer does not own the underlying stock and where one loses in the event of a stock price advance in terms of unlimited stock advances and retains only the call premium in the event of a price decline:

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

However, if one is a covered call writer, the Malkiel-Quandt notation is

$$\begin{matrix} \text{Covered} \\ \text{Call} \end{matrix} = \begin{matrix} \text{long in} \\ \text{stock} \end{matrix} + \begin{matrix} \text{call} \\ \text{writer} \end{matrix}$$

$$\text{writer} = \begin{bmatrix} +1 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

It would appear that the covered call writer never wins; however, if one writes a call option with an exercise price in excess of the stock price (OTM), one can substantially profit on the stock price advance and receive a call premium to partially offset the stock price decline.

A put buyer return pattern can be written as

$$\begin{bmatrix} 0 \\ +1 \end{bmatrix}$$

and a put swriter return pattern is

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

The Malkiel-Quandt notation allows the reader to understand the basic concepts of buying and writing calls and puts.[1]

[1]Straddles, straps, and strips, combinations of calls and puts, might pose problems for the Malkiel-Quandt notation. For example, if one purchases one call and one put at the same exercise price on the identical stock, it would appear that the investor will never lose.

$$\begin{matrix} \text{straddle} \\ \text{buyer} \end{matrix} = \begin{matrix} \text{call} \\ \text{buyer} \end{matrix} + \begin{matrix} \text{put} \\ \text{buyer} \end{matrix}$$

$$\begin{bmatrix} +1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ +1 \end{bmatrix} = \begin{bmatrix} +1 \\ +1 \end{bmatrix}$$

One could purchase a 80 September call on IBM for $5.10 and an 80 IBM September put for $0.35. If the investor buys a straddle, one invests $5.45 (or $545 in the 100 share position) in the

It should be obvious to the reader that an investor in IBM stock must have some expectations of future appreciation. Let us assume that IBM stock may rise 20% in the next period from its current price of $55.00 or fall 10%. Let us consider a three-period analysis of possible prices. The possible prices can be determined by the binomial probability distribution (Table 3).

The Binominal Option Pricing Model

The binomial option pricing model (OPM) evolved from Sharpe (1978), Rendleman and Bartter (1979), and Cox, Ross, and Rubinstein (1979). Much easier than the Black and Scholes OPM derivation (the limiting case of the binominal OPM is its

position. The investor purchases volatility in buying a straddle on profits only if the price rises above $59.38 or falls below $50.63. A straddle buyer does not care which event (advance or decline) occurs as long as the volatility is very large. A strike might lead one to buy a straddle on the firm's stock; one could profit on the put if the strike is long and violent and the call if the strike is settled quickly and cheaply. The Malkiel-Quandt positions do not offer the investor an insight into the volatility problems in purchasing a straddle. A straddle writer appears to consistently lose in the Malkiel-Quandt analysis:

$$\begin{bmatrix} -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

However, in reality, the writer wins as long as the price fluctuates between $50.63 and $59.38. The straddle writer wins as long as the stock volatility is low. Anything that reduces the stock volatility enhances the profitability of the straddle writer; for example, a merger, particularly a conglomerate merger, reduces systematic risk and total risk. A merger announcement could lend one to write straddles on the acquiring firm's stock.

A strap is when one has two calls and one put in an option portfolio. A strap buyer is not completely certain about the course of a stock's movement; however, the u = buyer is leaning toward an upward stock movement. One can easily see this in the Malkiel-Quandt notation:

$$\begin{matrix} \text{strip} \\ \text{buyer} \end{matrix} \quad 2\begin{bmatrix} +1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ +1 \end{bmatrix} = \begin{bmatrix} +2 \\ +1 \end{bmatrix}$$

One must be aware of the numerous (3) transaction costs and premiums associated with straps. In our Alcoa example, the 55 October strap would cost $737.50 [2($3) + $1.375 = $7.375]. Therefore, the break-even price range is $47.63 and $58.69 [$55 + $7.28/2]. One profits at a 2:1 ratio as the stock price advances above $55. A strip is two puts and one call. A strip purchaser is somewhat confused but more pessimistic than optimistic about the stock price movement. The Malkiel-Quandt notation for a strip purchaser is

$$\begin{matrix} \text{strip} \\ \text{purchaser} \end{matrix} \quad \begin{bmatrix} +1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 0 \\ +1 \end{bmatrix} = \begin{bmatrix} +1 \\ +2 \end{bmatrix}$$

Straddles, straps, and strips offer investors and portfolio managers opportunities to alter portfolio return distributions.

Table 3 IBM stock price example

t = 0 (current)	t = 1	t = 2	t = 3
$84.71	101.65	121.98	146.38
			109.78
		91.49	109.78
			82.34
	76.24	91.49	109.78
			82.34
		68.62	82.34
			61.76

The expected IBM prices are shown in this table for our three-period analysis

continuous-time partner, the traditional Black and Scholes OPM), the binomial OPM may be developed from a very practical example. Let us assume that the firm's current stock price is P_{cs} and either goes up u in the next period or falls by d. Bookstaber (1981, 1987) is an excellent presenter of the binominal OPM. A call option is equal to the maximum of zero or P_{cs} – Ex where Ex is the exercise price. In the King's English, a call option cannot be negative. In a three-period model, the value of the call option, V_c, may be developed:

t = 0	t = 1	t = 2	t = 3
			$P_{cs}(u)^3 - Ex$
		$P_{cs}(u)^2 - Ex$	$P_{cs}(u^2d) - Ex$
	$P_{cs}(u) - Ex$		$P_{cs}(udu) - Ex$
$P_{cs} - Ex$		$P_{cs}(ud) - Ex$	$P_{cs}(udd) - Ex$
			$P_{cs}(duu) - Ex$
		$P_{cs}(du) - Ex$	$P_{cs}(duu) - Ex$
	$P_{cs}(d) - Ex$		$P_{cs}(dud) - Ex$
		$P_{cs}(d)^2 - Ex$	$P_{cs}(ddu) - Ex$
			$P_{cs}(d)^3 - Ex$

The value of the call option can be found by multiplying the discounted value of the call in each of the eight possible states of nature by the respective probabilities of these states. The probability of an upward price movement, u, is p:

$$p = \frac{r - d}{u - d},$$

where r is one plus the risk-free rate. The probability of a downward movement, d, is q where $q = 1 = p$. Thus the value of the call option:

$$V_c = \frac{p^3[P_{cs}(u^3) - Ex] + 3p^2q[P_{cs}(u^2d) - Ex] + 3pq^2\left[P_{cs}\left(ud^2\right) - Ex\right] + q^3\left[P_{cs}\left(d^3\right) - Ex\right]}{(1+r)^3} \tag{1}$$

This equation may seem rather formidable until one remembers that the reader solved this in sophomore statistics class. If k represents the number of upward movements in n periods:

$$V_c = \frac{1}{r^n}\sum_{k=0}^{n}\frac{k!}{(n-k)!}p^k q^{(n-k)}\left[max\left(0, P_{cs}u^k p^{(n-k)} - Ex\right]\right. \tag{2}$$

Let us work a very small numerical example that will illustrate the analysis. Assume the following:

$$\begin{aligned}
&\mathrm{P_{cs}} = \$84.71\\
&\mathrm{Ex} = \$85.00\\
&r = 1.2\\
&d = .9\\
&r = 1.02\\
&p = \frac{1.02 - .9}{1.2 - .9}\cdot\frac{.12}{.30} = .40
\end{aligned}$$

$$\begin{aligned}
\mathrm{V_c} = \frac{1}{(1.02)^3}\Big\{&(.4)^3\left[\max\left(0, 84.71(1.2)^3 - 85\right)\right]\\
&+3(.4)^2(.6)\left[\max\left(0, 84.71(1.2)^2(.9) - 85\right]\right.\\
&+3(.4)(.6)^2\left[\max\left(0, 84.71(1.2)(.9^2) - 85\right]\right.\\
&+(.4)^3\left[\max\left(0, 84.71(.9^3) - 85\right]\right.\Big\}\mathrm{V_c}\\
&= .942[.064(61.38) + .288(24.79) + .432(0) + .216(0)]\mathrm{V_c}\\
&= .942(3.93 + 7.14)\mathrm{V_c} = \$10.42
\end{aligned}$$

The value of the three-period call is \$10.42. The value of the call option, if only two-period, would be:

$$\begin{aligned}
\mathrm{V_c} = \ &(1.02)^2\Big\{\left[(.4)^2\left(84.71(1.2)^2 - 85\right)\right] + 2(.4)(.6)[\max(0, 84.71(1.2)(.9) - 85)]\\
&+(.6)^2\left[\max\left(0, 84.71(.9)^2 - 85\right]\right\}\\
\mathrm{V_c} &= (.961)[.16(36.98) + .48(6.49) + (.36)(0.00)] = .961(5.92 + 3.12) = \$8.68
\end{aligned}$$

Note, as in the general case of options, as the time to maturity rises, the value of the option increases. The binomial option pricing model convergences to the traditional Black and Scholes Option Pricing Model as the number of periods go to infinity (continuous-time), and the binomial approximation of the stock price distribution becomes normally distributed (for price relatives):

$$\begin{aligned} \log\left(\mathrm{P}^{*}_{\mathrm{cs}}/\mathrm{P}_{\mathrm{cs}}\right) &= k\,\log\left(\mathrm{u}/\mathrm{d}\right) + \mathrm{n}\,\log\,\mathrm{d}, \\ \mathrm{E}\left[\log\left(\mathrm{P}^{*}_{\mathrm{cs}}/\mathrm{P}_{\mathrm{cs}}\right)\right] &= mn, \end{aligned} \tag{3}$$

where * denotes the maturity date and *mn* is the expected mean of the series.

$$\mathrm{Prob}\left[\frac{log\left(P^{*}_{cs}/P_{cs}\right) - mn}{\sigma\sqrt{n}} < Z\right] = \mathrm{N(Z)}$$

The traditional Black and Scholes OPM can be written from the binominal option pricing model as

$$\mathrm{V_c} = \mathrm{P_{cs}N(X)} - \mathrm{Exr^{-t}N}\left(\mathrm{X} - \sigma\sqrt{t}\right) \tag{4}$$

where

$$\mathrm{X} = \frac{log\left(P_{cs}/Exr^{-t}\right)}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}$$

The reader may well remember that the binomial distribution converges to the normal distribution with approximately 30 periods (the central limit theorem).

The More Traditional Black and Scholes Option Pricing Model Derivation

The rudiments of the Black and Scholes OPM were known in the 1960s as shown in the works of Sprenkle (1961), Boness (1964), and Kassouf (1969); however, Black and Scholes (1973) developed their model with a hedge ratio that allowed not only a theoretical value but a risk-free hedging strategy.[2] The Black and Scholes OPM is characterized by rather restrictive assumptions: (1) the stock price series follows a lognormal distribution; that is, the distribution of price relatives is normally distributed; (2) the risk-free is constant; (3) the variance of the price series is constant and known; (4) the stock pays no dividend; and (5) the call option is a European call option that cannot be exercised prior to expiration (maturity).

The traditional Black and Scholes (1973) option pricing model (OPM) finds a theoretical value of the call option, $\mathrm{V_c}$, by creating a risk-free portfolio composed of a call option and shares of the underlying stock. The value of the portfolio, $\mathrm{V_p}$, is found by the sum of the market values of equity and options:

[2]Wall Street quickly adopted the Black and Scholes OPM to the extent that even practitioners into their 50s and 60s who cannot take a derivative, total or partial, have hired "computer jocks" to program the OPM into their computers or use their H-P calculators.

$$V_p = hnV_c + mP_{cs} \tag{5}$$

where

h = hedge ratio
n = number of call options
m = number of shares of stock
P_{cs} = price of common stock

If there are no transactions in the stock and options, n and m are assumed to be constant, and the value of the portfolio may change as a result of fluctuating stock prices and call option prices. The reader is aware that changes in the stock price affect the price of a call; however, the call price is affected by changes in the time to maturity of the option. Although the price of the underlying stock may be unchanged for a trading day, the value of the call will fall as a result of time decay. Without time decay, the change in the portfolio is

$$dV_p = hndP_{cs} + mdV_c \tag{6}$$

If *n* and *m* are selected in this way, the risk free hedge is created:

$$dV_p = hdP_{cs} - dV_c \tag{7}$$

One notices that if the call option is written ($m < 0$), then the stock is purchased ($n > 0$) to create the hedged portfolios. The stock price and option price changes are

$$dP_{cs} = P_{cs_{t+dt}} - P_{cs_t},$$
$$dV_{c'} = V_{c_{t+dt}} - V_{c_t}.$$

The change in the stock price series follows a random walk, that is, the stock price series follows a lognormal distribution and the distribution of price relatives. $(P_{cs_t}/-P_{cs_{t-1}})$ is normally distributed. Black and Scholes (1973) and Smith (1976) write the change in the value of the call option may be written in terms of changes in time and the stock price:

$$dV_c = \frac{\partial V_c}{\partial t}dt + \frac{\partial V_c}{\partial P_{cs}}dP_{cs} + \frac{1}{2}\frac{\partial^2 V_c}{\partial P_{cs^2}}P_{cs}{}^2\sigma^2 \tag{8}$$

where

σ^2 = variance in stock price relatives.

The change in the value of the hedged portfolio is

$$dV_p = hdP_{cs} - \left(\frac{\partial V_c}{\partial t} + \frac{1}{2}\frac{\partial^2 V_{c_2}}{\partial P_{cs}}\sigma^2 P_{cs}^2\right)dt \tag{9}$$

The change in the value of the hedged portfolio should equal the risk-free rate, R_F:

$$R_F V_p = R_F P_{cs}\frac{\partial V_c}{\partial P_{cs}} + \frac{\partial V_c}{\partial^t} + \frac{1}{2}\frac{\partial^2 V_c}{\partial P_{cs^2}}\sigma^2 P_{cs}^2$$

$$0 = R_F P_{cs}\frac{\partial V_c}{\partial P_{cs}} + \frac{\partial V_c}{\partial^t} + \frac{1}{2}\frac{\partial^2 V_{c_2}}{\partial P_{cs}}\sigma^2 P_{cs}^2 - R_F V_P$$

The value of the V_c is found by solving the above partial differential equation subject to

$$\mathrm{V_c} = \max(\mathrm{P_{cs}} - \mathrm{Ex}, 0), 0 < \mathrm{t} < \mathrm{T(maturitydate)}, 0 < \mathrm{P_{cs}}.$$

The Black and Scholes OPM value of the call option is found to be

$$\mathrm{V_c} = \mathrm{P_{cs}N(d_1)} - \frac{Ex}{e^{rt}}N(d_2) \tag{10}$$

where

$$d_1 = \frac{\ln(P_{cs}/Ex) + \left(r + \frac{\sigma}{2}\right)t}{\sigma\sqrt{t}},$$

$$d_2 = \frac{\ln(P_{cs}/Ex) + \left(r - \frac{\sigma}{2}\right)t}{\sigma\sqrt{t}},$$

N(.) = cumulative normal distribution, and $e = 2.71828$.

The optimal hedge ratio, h, to create the risk-free hedge is $\mathrm{N(d_1)}$. If the call option is purchased, $\mathrm{N(d_1)}$ shares of stock are sold short. If the call is over-valued and written, the stock is purchased in the ratio of $\mathrm{N(d_1)}$. The traditional Black and Scholes OPM assumes that no dividends are paid. The dividends may be handled in several manners: (1) the present value of the escrowed dividend may be subtracted from the stock price, and (2) the time to ex-dividend is substituted for the same variable, and the relevant stock price is the price less the present value of the escrowed dividends:

$$0 = \frac{1}{2}\frac{\partial^2 V_{c_2}}{\partial P_{cs}}\sigma^2 P_{cs}^2 + \frac{\partial V_c}{\partial^t} + (R_F - D)P_{cs}\frac{\partial V_c}{\partial P_{cs}} R_F V_c \tag{11}$$

The value of the call option is clearly a function of six variables:

$$V_c = \int \begin{matrix} + & +_{\rightleftarrows} & +_{\rightleftarrows} & + & - & - \\ (P_{cs}, & t, & R_F, & \sigma^2, & D, & Ex) \end{matrix}$$

As the stock price rises, the value of the call increases. This is immediately clear for an "in the money" call in which the stock price exceeds the exercise price. An increase in the exercise price reduces the value of the call. An increase in the time to maturity creates more time for the stock price to exceed the exercise price and thus raise the value of the option. The risk-free rate is used to discount the exercise price in determining the value of the call; an increase in the risk-free rate reduces the present value of the exercise price and increases the value of the call option. The dividend reduces the ex-dividend price and reduces the call. The variance of the price relative series positively affects the value of the call because an increase in the variability serves the probability that stock price will exceed the exercise price and raise the value of the call option.

An investor can access real-time option prices and calculated Black and Scholes implied option volatilities. Let us show an example of a Lockheed Martin call option with an exercise price of $350, priced on March 9, 2020, with data from Bloomberg, L.P. The option expires in over 3 months or 102 days. The stock price of LMT was $354.00, and the 3-month Treasury bill, the risk-free rate, was 0.90%. The historic volatility was 17.90% (the standard deviation). The last traded call was priced at an ask price of $17.60. Is this a fair value of the call option?

Black and Scholes Model Calculation

The calculated Black and Scholes Option Pricing Model value of the LMT June 2020 call with an exercise price of $350 is

$$V_c = N(d_1)P_{cs} - \frac{Ex}{ert}N(d_2)$$

where

$$d_1 = \frac{ln\,\frac{\$354}{\$350} + \left(.009 + \frac{(.179)^2}{2}\right)\frac{102}{252}}{.179\sqrt{102/252}} = \frac{.0215}{.1139} = .189$$

$$d_2 = \frac{ln\left(\frac{\$354}{\$350}\right) + \left(.009 - \frac{(.179)^2}{2}\right)\frac{102}{252}}{.179\sqrt{102/252}} = \frac{.0085}{.1139} = .075$$

The cumulative normal distribution, found in the standard "area under the normal curve" statistical tables, produces cumulative areas of 57.5% and 53.19%, respectively, for d_1 and d_2:

$$V_c = \$354(.575) - \frac{350}{2.71828^{(.009)\left(\frac{102}{252}\right)}}(.5319)$$
$$= \$203.55 - 185.48 = \$18.07$$

The calculated Black and Scholes OPM value of $18.07 is less than the ask price of $34.80. One should (sell) write the LMT call option. One could write the call option at $34.80, buy 57.5 shares of IBM stock to hedge [$N(d_1)$ is the hedge ratio), and earn an excess return of 38.6%. The Black and Scholes LMT call option should sell for $18.07 because its 17.9% is substantially less than the 41.4% implied volatility used by the market to price the option.

The put option price is developed from the put-call parity relationship in which the purchase of a put is equal to the price of the call plus the present value of the strike price less the stock price. That is, the purchase of a put is analogous to the purchase of a call, shorting the stock, and borrowing the discounted exercise price:

$$V_c - P_{cs} - V_P + \frac{Ex}{1 + R_F} = 0$$

$$V_P - V_c + \frac{Ex}{1 + R_F} - P_{cs}$$

The empirical results of Klemkosky and Resnick (1979) and Gould and Galai (1974) substantiate the put-call parity theory and options market efficiency (Galai, 1977; Phillips & Smith, 1980). One may wonder why options trading firms have emerged in options analysis. One can easily find and agree upon the stock price, exercise price, risk-free rate, time to maturity, and dividends. The relevant variance to be used in pricing the option is of great concern and interest as is the role that options and stocks play in a diversified portfolio. Latane and Rendleman (1976) found that if one assumes that the call option is correctly priced and solves the Black and Scholes OPM for the implied variance, the implied variances were superior to historic variances if forecasting future variances. Schmalensee and Trippi (1978) found that variances are not constant but are first-order negatively serially correlated, i.e., mean-reverting. In one of the best studies of options market efficiency, Whaley (1982) found (1) the American call and European call options less the escrowed present values of dividends produced values within 3–5 cents of observed values (far within transactions costs, bid-ask spread) with the goodness-of-fit measures exceeding 0.98. Furthermore, the option pricing errors are due to variance errors and not to dividends, time to maturity (particularly the American call option), or the "in the moneyness" of the call option. Whaley noted the conclusion about the in-the-money nature of option problems. Whaley's option regression results indicated a tendency for "out of the money" ($P_{cs} < Ex$) to be underpriced by the traditional Black and Scholes models at the 10% level (although the regression coefficient = .04–.10). Black (1975)

noted that out of the money options tended to be underpriced. Most options theorists agree that implied variances are relevant variance for options investing and these variances are not constant but rather mean-reverting.

The two major topics of option analysis concern the area of option portfolios and portfolio insurance. Merton, Scholes, and Gladstein (1978, 1982) developed portfolio models to analyze option portfolios. In their earlier price, Merton, Scholes, and Gladstein found using an option simulation period from June 1963 to December 1975 that "in the money" covered calls ($P_{cs} > Ex$) earned less return than merely owning the stock. This is hardly surprising because further stock price appreciation above the exercise price (above the current option premium) is called away by the option buyer. The "out of the money" covered calls earned higher returns than the "in the money" covered calls, but were still less than stock returns. The right tail of the stock price distribution is eliminated by writing calls on the held stock. Although covered call writing reduced returns relative to holding stock, the portfolio standard deviations also were lower involving call writers. The principal contribution of the Merton et al. (1978) call option study was that portfolios composed of 90% commercial paper and 10% purchased out of the money calls outperformed stock returns, also producing much higher portfolio variances. The out of the money calls produced tremendous returns, despite the fact that 70% of the options expired unused, earning approximately 400%. One notes that out of the money option premiums when the exercise price is 1.2 times the stock price (i.e., $Ex = 96$, $P_{cs} = 80$), it should be very small. Therefore, if the stock price raises slightly over the exercise price, the corresponding percentage option returns would be very large.

The Merton et al. (1982) put study found that in the money puts ($Ex > P_{cs}$) provided portfolio insurance that was too costly for investors (reducing portfolio returns), whereas out of the money puts ($Ex < P_{cs}$) offered downside protection (protective puts) at a very reasonable (.4%) cost to the portfolio manager. Writing uncovered puts was far less variable than fully covered call writing, although both strategies had identical Malkiel-Quandt notation; the stock variance led to the higher variability in covered call writing. If one combines protective put purchases with out of the money call writing, one could effectively shape the desired portfolio distribution. Bookstaber and Clarke (1983) developed a portfolio algorithm to set exercise price levels for puts and calls to manage an options portfolio (with as few as five options). The existence of the options markets should not influence the firm's stock price and cost of capital. Stock prices are determined by discounted expected earnings. Option prices are determined by stock prices and stock volatilities.

The OPM and Corporate Liabilities

Black and Scholes (1973) derived an economic value of the European call option, assuming no taxes or transactions costs, log-normally distributed stock prices, and a known and constant risk-free rate:

$$W = XN(d_1) - Ce^{-r}fN(d_2), \tag{12}$$

where

W = price of the call
X = current value of the underlying asset, the stock price
C = exercise price of the call option
t = time to maturity
r_f = instantaneous risk-free rate
N(. . .) = standardized normal cumulative probability density function
$d_1 = \frac{ln\left(\frac{x}{c}\right) + \left(r_f + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$
$d_2 = d_1 - \sigma^2$ = instantaneous variance of the stock returns

The firm's equity can be viewed as a European call option in which stockholders sell the firm to the bondholders with an option to buy back the firm at time period t. The exercise price is the face value of the debt. The equity will have a positive value only if the terminal value of the firm exceeds the face value of the debt.[3] The face value of the debt can be thought of as the "limited liability" of the equity; it is the protection against the depreciation of the firm's value below the face value of the debt.[4]

[3]The authors believe few stockholders view the call option feature as truly meaningful. If the market value of the firm falls below the value of debt, the stockholders' value is zero.

[4]The stockholders own the firm and agree to pay the bondholders the value of the firm until the value of the assets exceeds the face value of the debt. If the value of assets is less than the face value of debt, then the equity holders must declare bankruptcy, and the bondholders can take possession of the firm. When a firm uses leverage, then the risk of the firm increases and the return on equity rises (linearly). The beta of the firm rises with leverage as a linear function:

$$\beta_{jL} = \beta_{ju}\left[1 + \frac{D}{E}(1 - t)\right]$$

where

β_{jL} = firm j beta with leverage
β_{ju} = firm j beta with leverage
D = debt
E = equity
T = corporate tax rate

The cost of equity is a linear function of the firm beta as the reader saw in chapter "Risk and Return of Equity and the Capital Asset Pricing Model" with the Capital Asset Pricing Model (CAPM):

$$Ke = R_F + [E(R_M) - R_F]\beta$$

The beta in the CAPM is a levered beta for the vast majority of firms in our economy. The concept of options and leverage is important because equity is like a call option on the firm. Equity holders can undertake risky projects that may increase the volatility of the firm. The value of the call is a function of volatility, and higher risk increases the value of equity at the expense of debt.

Merton (1974) used the Black and Scholes options model to price corporate liabilities where the value of equity, f, may be expressed:

$$f = h\,(V,\quad B,\quad r_f,\quad \sigma^2,\quad t)\begin{matrix}+ & - & + & + & +\end{matrix}.$$

The value of equity increases with the value of the firm, the risk-free rate, the variance, and the time to maturity of the debt. The equation for the value of equity can be written

$$\text{dist.1} = \frac{\ln\left(\frac{V}{B}\right) + \left(r_f + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$$
$$\text{dist.2} = \text{d}_1 - \sigma\sqrt{t};$$
$$f = \text{V}[\text{N}(\text{d}_1)] - \text{Be}^{-tf\text{t}}[\text{N}(\text{d}_2)].$$

Assume that a firm has a current market value of \$5 million and that the face value of its debt is \$2 million. The firm's variance of return is 10%, and the risk-free rate is 5%. The firm's debt will mature in 20 years. The market values of its debt and equity can be found:

$$\text{dist.1} = \frac{ln\left(\frac{5,000,000}{2,000,000}\right) + \left(.05 + \frac{.10}{2}\right)20}{\sqrt{.10}\left(\sqrt{20}\right)}$$
$$= \frac{.9163 + 2.000}{.3162(4.4721)}$$
$$= 2.0623.$$

From the table of areas under the normal curve,

$$\text{N}(\text{d}_2) = .2422 + .5000 = .7422.$$

The .5000 representing the area under the left-hand side of the normal curve is added as the cumulative normal function is employed. To find the value of equity,

$$S = \$5,000,000(.9803) - \$2,000,000(.7422)e^{-(.05)20}$$
$$= \$4,901,500 - \$546,080.24$$
$$= \$4,355,419.76$$

The market value of the firm's equity, from the options pricing model, is $4,355,419.76. Since its current market value is $5,000,000, the current market value of its debt must be $644,580.24. The market value of debt is the minimum of the value of the firm or the face value of debt.

If the firm engages in a merger that reduces its variance to 8%, the value of its debt will rise and the market value of its equity will decrease:

$$\begin{aligned}\text{dist.1} &= \frac{\ln\left(\frac{5,000,000}{2,000,000}\right) + \left(.05 + \frac{.08}{2}\right)20}{\sqrt{.08}\left(\sqrt{20}\right)} \\ &= \frac{2.7163}{1.2647} \\ &= 2.1478 \\ \text{N(dist.1)} &= .9842 \\ \text{dist.2} &= 2.1478 - \sqrt{.08}\left(\sqrt{20}\right) \\ \text{N(dist.2)} &= .8106\end{aligned}$$

The value of equity is found:

$$\begin{aligned}S &= \$5,000,000(.9842) - \$2,000,000(.8106)e^{-tf\text{t}} \\ &= \$4,324,560.52\end{aligned}$$

The value of equity has fallen from $4,355,419.76 to $4,324,560.52 entirely because of the reduction in the variance resulting from the merger. The value of the debt has risen:

$$\$5,000,000 - \$4,324,560.52 = \$675,439.48$$

Debt's value rises from $644,580.24 to $675,439.48 because of the reduction in the firm's variance.

If investors hold the market portfolio (equal amounts of the firm's debt and equity), a change in the market values of corporate liabilities will not affect the market values of their total holdings. The loss on equity holdings will be offset by the gain on debt holdings. If investors do not hold the market portfolio, wealth transfers from bondholders to stockholders must be made to leave stockholders indifferent to mergers. The merged firm will have a higher debt-to-equity ratio. Because interest on debt is deductible and the value of the firm rises with debt, the higher debt-to-equity ratio increases the value of the firm. The important aspect of this appendix is the stockholder – management relationships are discussed in chapter "Management-Stockholder Relations: Is Optimal Behavior All That Is Necessary?".

References

Black, F. (1975). Fact and fantasy in the use of options. *Financial Analysts Journal, 31*, 36–72.
Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy, 81*, 637–654.
Boness, A. J. (1964). Elements of a theory of stock option value. *Journal of Political Economy, 72*, 163–175.
Bookstaber, R. (1981). *Option pricing and strategies in investing*. Reading, MA: Addison-Wesley Publishing.
Bookstaber, R. (1987). *Options pricing and investing strategies*. Chicago: Probus Publishing.
Bookstaber, R., & Clarke, R. (1983). An algorithm to calculate the return distribution of portfolios with option positions. *Management Science, 29*, 419–429.
Cox, J. C., Ross, S., & Rubinstein, M. (1979). Option pricing: A simplified approach. *Journal of Financial Economics, 6*, 229–263.
Galai, D. (1977). Tests of market efficiency of the Chicago Board of Options Exchange. *Journal of Business, 50*, 167–197.
Gould, J. P., & Galai, D. (1974). Transactions costs and the relationship between put and call prices. *Journal of Financial Economics, 1*, 105–130.
Kassouf, S. T. (1969). An econometric model for option price with implications for investors' expectations and audacity. *Econometrica, 37*, 685–694.
Klemkosky, R. C., & Resnick, B. G. (1979). Put-call parity and market efficiency. *Journal of Finance, 34*, 1141–1157.
Latane, H., & Rendleman, R. (1976). Standard deviations of stock price ratios implied in option prices. *Journal of Finance, 31*, 369–381.
Malkiel, B. G., & Quandt, E. (1969). *Strategies and rational decisions in the securities options market*. Cambridge, MA: MIT Press.
Merton, R. (1974). On the pricing of corporate debt: The risk structure of interest rates. *Journal of Finance, 29*, 449–470.
Merton, R., Scholes, M., & Gladstein, M. (1978). The returns and risks of alternative call option portfolio investment strategies. *Journal of Business, 51*, 183–242.
Merton, R., Scholes, M., & Gladstein, M. (1982). The returns and risks of alternative put-option portfolio investment strategies. *Journal of Business, 55*, 1–55.
Phillips, S., & Smith, C. (1980). Trading costs for listed options: Implications for market efficiency. *Journal of Financial Economics, 8*, 179–201.
Schmalensee, R., & Trippi, R. (1978). Common stock volatility expectations implied by option premia. *Journal of Finance, 33*, 129–148.
Sharpe, W. F. (1978). *Investments*. Englewood Cliffs, NJ: Prentice-Hall.
Smith, C. W. (1976). Option pricing. *Journal of Financial Economics, 3*, 3–51.
Sprenkle, C. M. (1961). Warrant prices and indicators of expectations and preferences. *Yale Economic Essays, 1*, 179–231.
Whaley, R. (1982). Valuation of American call options on dividend-paying stocks. *Journal of Financial Economics, 10*, 29–58.

Real Options



Chapter "Investing in Assets: Theory of Investment Decision-Making" dealt with the capital budgeting process in which a financial manager accepts a project only if the discounted cash flow of that project exceeds the initial costs of the project. The discount rate is the cost of capital. The difference between the discounted cash flow and the initial cash outlay is the net present value, NPV, which should be positive to accept a project. This chapter discusses another application of cash flow and valuation, the application of real option theory.

The application of real option analysis can take several forms. First, one can examine the possible complications of the strict application of the NPV rule to an R&D investment decision, and how the stockholder wealth may be enhanced by the use of real options analysis. An investment project may be viewed as the right to pay an initial cash outlay, or cost, to receive the present value of the project's future cash flow. The second application of real options strategies is the case of abandonment valuation. When one calculates the value of a real option, one equates the investment cost of the project with the exercise price of the real option. The present value of the project is equivalent to the price of the underlying asset. Real options are an application of options theory to the operation and valuation of real investment projects (McDonald, 2003).

Research and development expenditures are capital expenditures involving discounting cash flow such that the net present value is positive. The research and development expenditures leading to the implementation of new technology is the call premium with the present value of the final project being the value of the call option. The R&D costs are the premium paid to acquire the future investment cash flow of the project resulting from the R&D activities. A pharmaceutical firm that engages in R&D expenditures may need to consider abandonment decisions and values. Additionally, current R&D projects uncover options in future and expansive R&D projects. The R&D option is an option to expend the firm when it may not be obvious to management, at the current time, that primary financial decision is profitable. A current or static, negative net present value need not lead management

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_17

to eliminate the R&D project from its consideration. It is possible to reconsider the project at a later date when initial cash outlays of projects change, costs of capital change, and higher estimates of future cash flows are different.

The traditional NPV capital budgeting criteria measures the profitability and acceptability of undertaking a project at the present time and ignores the potential profitability of delaying the project until a later date. McDonald (2003) makes the point that the traditional, or static, NPV criteria assumes that the project will not be undertaken if it is rejected in the current period. The project cost can be thought of as an exercise price, and the present value of the project is the value of the underlying asset. If we delay the investment, we lose the cash flow of the project, very similar to a stock dividend not received. If the project cash flows are less than the interest saved by deferring the payment of the project investment, then it is efficient to wait to invest in the project.

The Option to Delay a Project

Projects are traditionally analyzed using the expected cash flows and discount rates at the time of the analysis; the net present value computed on that basis is a measure of its value at that time. Expected cash flows and discount rates may change over time, however, and then so will the net present value. Information may change over time and the option to wait, or delay a project, may have value. Thus, if expected cash flow rise or the discount rate falls, a project that has a negative net present value now may have a positive net present value in the future. In a competitive market, in which individual firms have no special advantages over their competitors in undertaking projects, this may not seem significant. However, in an environment in which only one firm, such as a firm with a patent, can take a project or barriers to entry, extensive advertising or other frictions prevail to create a change in the project's value over time to give it the characteristics of a call option.

In the abstract, assume that a project requires an initial investment, as the R&D program, C. The present value of expected cash inflows computed right now is PVCF. The net present value of this project is the difference between the two:

$$\text{NPV} = \text{PVCF} - \text{C}$$

Now assume that the firm has exclusive rights to this project for the next n years and that the present value of the cash inflows may change over that time, because of changes in either the cash flows or the discount rate. Thus, the project may have a negative net present value right now, but it may still be a good project if the firm waits. Defining PVCF again as the present value of the cash flows, the firm's decision rule to accept the project should be if $\text{PVCF} > \text{C}$. If the firm does not take the project, it incurs no additional cash flows, though it will lose what it

originally invested in the project. The price of the project, such as an R&D project, is the price of the call option. The exercise price of the call option is the cost of future investments needed when an initial investment is made. The project's expected net present value is analogous to the price of stock in the Black and Scholes formulation. The underlying asset is the project, the strike price of the option is the investment needed to take the project, and the life of the option is the period for which the firm has rights to the project. The variance in this present value represents the variance of the underlying asset. The value of the option is largely derived from the variance in cash flows as the higher the variance, the higher the value of the project delay option. Thus, the value of an option to do a project in a stable business will be less than the value of one in a changing environment. Hamilton (1985), Mitchell and Hamilton (1988), and Hamilton and Mitchell (1990) emphasize that management needs to identify strategic objectives and review the impact of strategic options, such R&D projects directed toward strategic planning, and identify the strategic planning targets of future R&D projects.

Implications of Viewing the Right to Delay a Project as an Option

Several interesting implications emerge from the analysis of the ability to delay a project as an option. First, a project may have an initial negative net present value based upon current expected cash flows, but it may still be a "valuable" project because of the option characteristics. Thus, while a negative net present value should encourage a firm to reject a project, it should not lead it to conclude that the rights to this project are worthless. Second, a project may have a positive net present value but still not be accepted right away. This is because the firm may gain by waiting and accepting the project in a future period, particularly for risky projects. In static analysis, increasing uncertainty increases the riskiness of the project and may make it less attractive. When the project is viewed as an option, an increase in uncertainty may actually make the option more valuable.

Abandonment Value

A project does not always produce its expected cash flow, and the net present value of a project, initially calculated to be positive, does not always produce value for the shareholders. How does management come to grips with cash flow forecasts that turn out to be incorrect or were based on assumptions that are not substantiated? What can the firm's financial managers do to minimize stockholder's losses? A possible solution is abandonment of the project. The option to abandon a project may consist of selling the project's assets and not realizing future cash flow. Van Horne

Table 1 Economic scenarios of the economy and project cash flow

	Year 1	Year 2		
State of the economy	Probabilities	($mm) Cash flow	Probabilities	Cash flow
Recession	.30	$20.0	.30	10
			.40	20
			.30	30
Normal	.30	30.0	.15	20
			.50	30
			.35	50
Boom	.40	50.0	.10	30
			.40	50
			.50	75

(2002) presented an earlier version of this analysis. The abandonment option is a put option, similar in logic to the put options on stocks discussed in chapter "Options."

Let us develop an investment scenario where abandonment value enhances the decision-making process. An R&D project requires the construction of a new building near to, but off, the main corporate grounds. The new building will cost $45,000,000 and can house a small production facility for 3 years even if management decides to forego or postpone the R&D project. Sales of the production facility are dependent upon the state of the economy. The corporate economists have prepared a set of 3-year cash flow forecasts that are first-year probabilities and second-year conditional probabilities. That is, the cash flow forecasts are dependent upon particular states of nature occurring in years one and two. See Table 1. One should calculate the expected net present value of the projected cash flow, assuming a 10% cost of capital. The calculations of the expected net present value and internal rate of return are shown in Table 2. One multiples the cash flow under the various economic scenario, depression, recession, normal, and boom, by the cash flow occurring in that state of the economy. The three scenarios, the two-period analysis produces nine possible states of the economy. The key to the analysis is to calculate the joint probabilities of each possible state. Each state of the economy is conditional upon the previous period's state of the economy. See Table 2 for the calculation of the joint probabilities, the expected present value, and the expected net present value.

The expected present value of cash flow is $65,560,000. Given the cost of the building of $45,000,000, the expected net present value of the new building is $20,560,000. The expected net present value exceeds zero, and the new building can be justified at a cost of capital of 10%. The expected internal rates of return (IRR) of the project for the nine scenarios are shown in Table 3. The expected internal rate of return is 34.84%, far exceeding the cost of capital, and the project is acceptable. The reader is referred to chapter "Investing in Assets: Theory of Investment Decision-Making" in which we stated that in the vast majority of cases, a positive net present value implies that the internal rate of return exceeds the cost of capital.

Table 2 Expected net present value ($ MM)

	Cash flow		PVIF		Present value				
State of nature	Year 1	Year 2	Year 1	Year 2	Year 1	Year 2	Present value	Joint probability	Expected present value
Recession	20	10	0.909	0.826	18.18	8.26	26.44	0.090	2.38
	20	20	0.909	0.826	18.18	16.52	34.70	0.120	4.16
	20	30	0.909	0.826	18.18	24.78	42.96	0.090	3.87
Normal	30	20	0.909	0.826	27.27	16.52	43.79	0.045	1.97
	30	30	0.909	0.826	27.27	24.78	52.05	0.150	7.81
	30	50	0.909	0.826	27.27	41.30	68.57	0.105	7.20
Boom	50	30	0.909	0.826	45.45	24.78	70.23	0.040	2.81
	50	50	0.909	0.826	45.45	41.30	86.75	0.160	13.88
	50	75	0.909	0.826	45.45	61.95	107.40	0.200	21.48
Expected present value									65.56
Expected net present value									20.56

Table 3 Expected internal rate of return

State of nature	Internal rate of return	Joint probability	E(IRR)
Recession	−25.66%	0.090	−2.31%
	−7.50	0.120	−.90
	6.84	0.090	0.62
Normal	7.87	0.045	0.35
	21.53	0.150	3.23
	43.89	0.105	4.61
Boom	33.33	0.040	1.33
	54.31	0.160	8.69
	96.10	0.200	19.22
Expected internal rate of return			34.84%

Table 4 Variance of expected internal rate of return

State of nature	Joint prob	IRR	E(IRR)	Std (IRR)
Recession	0.090	−0.2566	−0.0231	0.0329
	0.120	−0.0750	−0.0090	0.0215
	0.090	0.0684	0.0062	0.0071
Normal	0.045	0.0787	0.0035	0.0033
	0.150	0.2153	0.0323	0.0027
	0.105	0.4389	0.0461	0.0009
Boom	0.040	0.3333	0.0133	0.0000
	0.160	0.5431	0.0869	0.0061
	0.200	0.9610	0.1922	0.0751
			0.3484	0.1494

The expected IRR for the project is 34.84%. The expected IRR exceeds the cost of capital, and hence the expected net present value is positive. The expected variance of the project is 14.94%; the calculations are shown in Table 4.

What is the economic benefit of being able to abandon the new building project after year one at an abandonment value of $24,000,000? If the abandonment value of $24.0 million exceeded the expected present value of cash flow of year two in any scenario or state of the economy, then the expected net present value calculation of the new building should be recalculated. The present value of the abandonment value of $24.0 million exceeds the expected present value of cash flow for year two in the recession ($16.52 million) mode. The abandonment value option increases the project net present value (NPV) to $67.15 MM, see Table 5, and the internal rate of return to 37.77%. The variance of the expected internal rate of return falls to 12.88% with abandonment value.

The recalculated expected net present value of the new building is shown in Table 6. The expected net present value increases by $1.59 million by the presence of the abandonment option. The project should be abandoned after year one.

Table 5 Expected net present value of year two states of nature

	Cash flow					
State of nature	Year 1	Year 2	Year 2	Year 2	Joint probability	Present value of year 2
Recession	20	10	0.826	8.26	0.300	2.48
	20	20	0.826	16.52	0.400	6.61
	20	30	0.826	24.78	0.300	7.43
Recession expected (PV)						16.52
Normal	30	20	0.826	16.52	0.100	1.65
	30	30	0.826	24.78	0.500	12.39
	30	50	0.826	41.30	0.350	14.46
Normal expected (PV)						28.50
Boom	50	30	0.826	24.78	0.100	2.48
	50	50	0.826	41.30	0.400	16.52
	50	75	0.826	61.95	0.500	30.98
Boom expected (PV)						49.97

The presence of an abandonment value of $24.0 million enhances the net present value of the project by $1.59 million because the abandonment value exceeds the expected present value of year two cash flows in the recession scenario. The abandonment analysis may not be complete until one calculates the present value of the cash flow foregone by abandoning the project. The present value of the abandoned cash flow is shown in Table 7 and is $33.74 million.

This chapter follows Copeland and Weston (1992) and calculates the abandonment put value. One uses the present value of the abandoned cash flow as the equivalent of the stock price, the abandonment value as the exercise price, and a 2-year period for the option. If the risk-free rate is 5%, the value of the put option is calculated to be $0.036 million.

$$v_c = P_{cs}N(d_1) - \frac{Ex}{e^{rt}}N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{P_{cs}}{Ex}\right) + rt}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}$$

$$d_2 = d_1 = \sigma\sqrt{t}$$

Table 6 Expected net present value with abandonment value

	Cash flow		PVIF		Present value				
State of nature	Year 1	Year 2	Year 1	Year 2	Year 1	Year 2	Present value	Joint probability	Expected present value
Recession	44	0	0.909	0.826	40.00	0.00	40.00	0.090	3.60
	44	0	0.909	0.826	40.00	0.00	40.00	0.120	4.80
	44	0	0.909	0.826	40.00	0.00	40.00	0.090	3.60
Normal	30	20	0.909	0.826	27.27	16.52	43.79	0.045	1.97
	30	30	0.909	0.826	27.27	24.78	52.05	0.150	7.81
	30	50	0.909	0.826	27.27	41.30	68.57	0.105	7.20
Boom	50	30	0.909	0.826	45.45	24.78	70.23	0.040	2.81
	50	50	0.909	0.826	45.45	41.30	86.75	0.160	13.88
	50	75	0.909	0.826	45.45	61.95	107.40	0.200	21.48
Expected present value with abandonment value									67.15
Expected net present value with abandonment value									22.15

Table 7 Expected net present value of year two cash flow

State of nature	Year 2	Year 2	PV(year 2)	Joint probability	Present value of year 2
Recession	10	0.826	8.26	0.090	0.74
	20	0.826	16.52	0.120	1.98
	30	0.826	24.78	0.090	2.23
Normal	20	0.826	16.52	0.045	0.74
	30	0.826	24.78	0.150	3.72
	50	0.826	41.30	0.105	4.34
Boom	30	0.826	24.78	0.040	0.99
	50	0.826	41.30	0.160	6.61
	75	0.826	61.95	0.200	12.39
Expected (PV) of year 2 cash flow					33.74

$$d_1 = \frac{\ln\left(\frac{33.74}{24}\right) + .05(1)}{.1288\sqrt{1}} + \frac{1}{2}(.1288)$$
$$= \frac{0.3406 + .05}{.1288(1)} + \frac{1}{2}(.1288)\sqrt{1}$$
$$= \frac{.3906}{.1288} + .0644 = 3.097$$

$$d_2 = 3.907 - .1288\sqrt{1} = 3.907 - .1288 = 2.968$$

$$N(d_1) = 1.000$$

$$N(d_2) = 0.9986$$

$$v_c = 33.74(1.000) - \frac{24}{2.71828 - .05(1)}(.9986)$$
$$= 33.74 - \frac{24}{1.051}(.9986)$$
$$= 33.74 - 22.80 = 10.94$$

$$v_{c_0} - P_0 = P_{cs} + \frac{Ex}{e^{rt}}$$

$$P_0 = 10.94 - 33.74 + \frac{24}{1.051}$$
$$= \$10.94 - 33.74 + 22.84 = \$.036$$

The value of the put option to abandon the project is worth \$0.036 MM.

Options in Investment Analysis/Capital Budgeting

In traditional investment analysis, a project or new investment should be accepted only if the returns on the project exceed the hurdle rate, the cost of capital, which leads to a positive net present value. Several additional aspects of real options are embedded in capital budgeting projects. The first is the option to delay a project, especially when the firm has exclusive rights to the project. The second is the option to expand a project to cover new products or markets sometime in the future.

References

Copeland, T., & Weston, J. F. (1992). *Financial policy and corporate policy* (3rd ed.). Reading, MA: Addison-Wesley Publishing Company. Chapter 12.

Hamilton, W. F., & Mitchell, G. R. (1990). R&D in perspective: What is R&D worth? *The McKinsey Quarterly*, 150–160.

McDonald, R. L. (2003). *Derivatives markets*. Boston: Pearson/Addison Wesley.

Mitchell, G. R., & Hamilton, W. F. (1988). Managing R&D as a strategic option. *Research-Technology Management, 31*, 15–22.

Van Horne, J. C. (2002). *Financial management and policy* (12th ed.). Upper Saddle Rive, NJ: Prentice-Hall. Chapter 7.

Mergers and Acquisitions



A company can grow by taking over the assets or facilities of another. The various methods by which one firm obtains or "marries into" the business, assets, or facilities of another company are mergers, combinations, or acquisitions.[1] These terms are not used rigidly. In general, however, a merger signifies that one firm obtains another by issuing its stock in exchange for the shares belonging to owners of the acquired firm, or buys another firm with cash. Company X gives some of its shares to Company Y shareholders for the outstanding Y stock. When the transaction is complete, Company X owns Company Y because it has all (or almost all) of the Y stock. Company Y's former stockholders are now stockholders in Company X. In a combination, a new corporation is formed from two or more companies who wish to combine. The shares of the new company are exchanged for those of the original companies. The difference between a combination and a merger lies more in legal distinctions than in any discernible differences in the economic or financial result. In practice, the terms merger and combination are often used interchangeably. The study of merger profitability is as old as corporate finance itself. Arthur S. Dewing (1921, 1953) reported on the relative unsuccessfulness of mergers for the 1893–1902 time period; and Livermore (1935) reported very mixed merger results for the 1901–1932 time period. The first major merger movement ocurred late in the 19th century and the second major movement ocurred in the 1920s, which we will discuss in this chapter. Manne (1965) framed the mergers and acquisitions decisions as actions in the market for corporate control. Mandelker (1974) put forth the Perfectly Competitive Acquisitions Market (PCAM) hypothesis in which competition equates returns on assets of

[1] Although mergers, combinations, and acquisitions are exciting events of great interest to the financial community, they do not represent the most common method through which even individual companies grow. Most firm growth is direct growth, which takes place quietly and is financed either by internal sources or floating new securities. A study of 74 large companies showed that of their growth from 1900 to 1948, 75% was direct growth and only 25% was accounted for by mergers. See J. F. Weston, *The Role of Mergers in the Growth of Large Firm*, University of California Press, 1953.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_18

similar risk, such that acquiring firms should pay premiums to the extent that no excess returns are realized to their stockholders. The PCAM holds that only the acquired firms' stockholders earn excess returns. Dodd and Ruback (1977), Jensen and Ruback (1983) reported that acquired firms profited handsomely, while acquiring firms lost little money such that wealth was enhanced. Recent evidence by Jarrell, Brickley, and Netter (1988), Ravencraft and Scherer (1989), and Alberts and Varaiya (1989) finds little gain to the acquiring firm and significant merger premiums paid for the acquired firms. The reader is referred to Weston, Mitchell, and Mulhern (2004) for a complete treatment of mergers and corporate governance.

Acquisition usually refers to a transaction in which one firm buys the major assets or the controlling shares of another company. On occasion, one corporation has purchased another corporation's subsidiaries. An acquisition differs from a merger in that generally (but not always) cash is used rather than an exchange of securities.[2]

No new net financial holdings are created in the economy by any of the forms of merger or acquisition (Mossin, 1973). If the transaction involves an exchange of stock, the supply of shares of one company's stock is eliminated and is replaced by the shares of the surviving company. If the transaction is financed by cash, cash holdings by individuals go up, but cash held by corporations goes down; the supply of outstanding securities in the hands of the public goes down, but the amount held by corporations rises. If a corporation floats new securities to obtain funds to finance its acquisition, then process is slightly roundabout, but the net results are the same. One section of the public surrenders its cash for new corporate securities; another group gives up a different issue of securities for cash.

Noneconomic Motives for Combinations

Noneconomic motives may dominate in a number of mergers; sociological or psychological motives, possibly involving the personal well-being of the management, may lie behind some combinations. Management may feel that its prestige, status, or salary will rise if it is in charge of a larger business (even though the *rate* of profit may be no higher than that of the present firm). Management may favor a merger because the acquiring firm has better bonus or retirement provisions than the present company. A merger or combination may take place because the management of one company is getting old and a merger may appear to be the simplest method of obtaining new managerial personnel. A wave of mergers or acquisitions may take place to satisfy the desires for risk of speculative management.

Where an industry is characterized by oligopolistic competition, company growth, whether achieved by combination or direct investment, may be announced

[2]A firm which either by the exchange of securities or purchase of shares controls subsidiary companies is called a *holding* company. The holding company differs from a *parent* company in that the parent company has production functions of its own to perform, whereas a holding company exists *mainly* to control or coordinate its subsidiaries.

as being undertaken to "maintain the company's relative share of the market" or to "prevent deterioration of the company's competitive position." These announced reasons may appear to reflect desires to maintain corporate prestige or status. Actually, they may have considerable economic validity as rule-of-thumb guides to prevent erosion of the earning power of the firm. As noted in chapter "Investing in Assets: Theory of Investment Decision-Making," sometimes the return on a new investment may be computed not on the basis of potential additional earnings but on the basis of losses averted or on a prevented decrease in earnings. Thus, growth can be defensive, a rational move to forestall a downward slide in the company's market position which, once started, may be difficult to halt.

By their nature, the psychological and sociological motives underlying some instances of corporation growth are obscure and complex and do not allow for easy analysis or any reliable quantification. Since economic analysis rests on the supposition of economic rationality on the part of decision-makers, it cannot furnish much light on growth undertaken mainly to satisfy desires for status or power.

Holding Companies

The use of the term "holding company" is not always clear in financial literature. In general, a holding company is a firm that owns controlling shares in other corporations and whose main function is to coordinate and direct the affairs of these subsidiaries. In contrast, many operating companies run subsidiaries and yet are not usually referred to as holding companies, because they have extensive operations of their own.[3] The holding company must also be distinguished from the investment company, which holds shares in other corporations mainly for the investment return and does not guide their internal management.

Among the advantages of the holding company is the flexibility of administration. It enables top management to centralize those functions which can be performed at lower costs in large volumes and to decentralize responsibility for programs and functions better performed or supervised at local levels. Of course, like other forms of organization, the holding company can become too large and thus inefficient in trying to coordinate its various enterprises.[4]

[3]Thus, the Standard Oil of New Jersey, the United States Steel Corporation, and the original Bethlehem Steel Corporation, although they owned many subsidiaries, were not usually thought of as holding companies.

[4]In the 1930s, investors in holding companies suffered heavy losses, and this fact, plus the exposure of legislative and operating scandals involving some promoters of holding companies, led to the passage by Congress of the Public Utility Company Act of 1935. Some of the major provisions of this legislation are:

1. Holding companies were prohibited over the third level-one operating company and two holding companies. That is, a holding company structure could go up to the grandfather level but not beyond. This "great-grandfather death clause" was designed to restrict the excessive leverage possible in the holding company's structure.

A Merger History of the United States

The height of the merger movement was reached in 1901 when 785 plants combined to form America's first billion-dollar firm, the United States Steel Corporation. The series of mergers creating U.S. Steel allowed it to control 65% of the domestic blast furnace and finished steel output. This growth in concentration was typical of the first merger movement. The early mergers saw 78 of 92 large consolidations gain control of 50% of their total industry output, and 26 secure 80% or more.

The beginning of the major merger movement occurred during a period of rapid economic growth. The economic rationale for the large merger movement was the development of the modern corporation, with its limited liability, and the modem capital markets, which facilitated the consolidations through the absorption of the large security issues necessary to purchase firms. Nelson (1959) found the mergers were highly correlated with the period's stock prices (.613) and industrial production (.259) during the 1895–1904 period. The expansion of security issues allowed financiers the financial power necessary to induce independent firms to enter large consolidations. The rationale for the first merger movement was not that of trying to preserve profits despite slackening demand and greater competitive pressures. Nor was the merger movement the result of the development of the national railroad system, which reduced geographic isolation and transportation costs. The first merger movement ended the 1904 depression, with the *Northern Securities* case. Here it was held, for the first time, that antitrust laws could be used to attack mergers leading to market dominance.

A second major merger movement stirred the country from 1916 until the Depression of 1929. This merger movement was only briefly interrupted by the First World War and the recession of 1921 and 1922. Approximately 12,000 mergers of the period coincided with the stock market boom of the 1920s. Although mergers greatly affected the electric and gas utility industry, market structure was not as severely concentrated by the second movement as it had been by the first. Stigler (1950) concluded that mergers during this period created oligopolies, such as

2. Holding company systems were to be consolidated and simplified to develop some aspect of geographical contiguity. If the holding company is to serve an economic function, the location of its operating companies should have some geographical logic.
3. "Upstream" loans were made illegal. An upstream loan is made from a lower-level company to a higher one. A holding company that serves a function should help to finance its operating companies and not the other way around. The request for an upstream loan could not be resisted by a captive operating company even though the purpose of the loan was to bail the holding company out of a financial difficulty, and its net effect might be to deplete the assets of the operating company without necessarily rescuing the holding company.
4. Service and construction companies organized by the holding company system to work for operating companies were required to be nonprofit. In the usual case, there is no felt need to regulate intercompany prices because the resulting profits are only a shift of earnings among the various companies. However, in the public utility fields, excessive service charges or construction costs could affect the rate base and thus may be borne eventually by the consumers.

These rules and other provisions of the Public Utility Company Act of 1935 aroused considerable antagonism in the business community.

Bethlehem Steel and Continental Can. Mergers, primarily vertical and conglomerate in nature as opposed to the essentially horizontal mergers of the first movement, did affect competition adversely. The conglomerate product-line extensions of the 1920s were enhanced by the high cross-elasticities of demand for the merging companies' products (Lintner, 1971). Antitrust laws, though not seriously enforced, prevented mergers from creating a single dominant firm. Merger activity diminished with the Depression of 1929 and continued to decline until the 1940s.

The third merger movement began in 1940; mergers reached a significant proportion of firms in 1946 and 1947. The merger action from 1940 to 1947, although involving 7.5% of all manufacturing and mining corporations and controlling 5% of the total assets of the firms in those industries, was quite small compared to the merger activities of the 1920s. The mergers of the 1940s included only one between companies with assets exceeding $50 million and none between firms with assets surpassing $100 million. The corresponding figures for the mergers of the 1920s were 14 and 8, respectively. Eleven firms acquired larger firms during the mergers of the 1920s than the largest firm acquired during the 1940s.

The mergers of the 1940s affected competition far less than did the two previous merger movements, with the exception of the food and textile industries. The acquisitions by the large firms during the 1940s rarely amounted to more than 7% of the acquiring firms' 1939 assets or to as much as a quarter of the acquiring firm's growth from 1940 to 1947. Approximately $5 billion of assets were held by acquired or merged firms over the period 1940–1947. Smaller firms were generally acquired by larger firms. Companies with assets exceeding $100 million acquired, on average, firms with assets of less than $2 million. The larger firms tended to engage in a greater number of acquisitions than smaller firms. The acquisitions of the larger acquiring firms tended to involve more firms than did those of smaller acquiring firms. Mergers added relatively less to the existing size of the larger acquiring firms. The relatively smaller asset growth of the larger acquiring firms is in accordance with the third merger movement's generally small effects on competition and concentration. Butters, Lintner, and Cary (1951) reported 1990 mergers occurred in the 1940–1947 period represented about 5% of total assets held in 1943 by manufacturing and mining corporations. Butters et al. (1951) reported that larger firms made more acquisitions than smaller firms, and acquired, on average, a larger number of companies than smaller companies engaged in mergers. Firms with over $50 million dollars in assets in 1947, less than 9% of firms listed in Moody's Industrials for 1947, acquired over 33% of the assets of all firms acquired.

The current merger movement is an extension of the 1940s conglomerate movement beginning in 1951 and continuing to the present. Conglomerate mergers involve two firms in different, uncorrelated businesses. Of the nine mergers occurring in 1951 involved acquired firms with assets exceeding $10 million, four were conglomerate mergers, of which three were product-line-extension combinations. The growth of the large conglomerate mergers continued throughout the forecast period. In 1954, 21 of the 37 mergers involving acquired firms with assets exceeding $10 million were conglomerate in nature; 14 of the 21 conglomerate mergers were product-line extensions, while only 2 of the mergers were market-extension

Table 1 Quarterly net announcements (1999–2018)

	First	Second	Third	Fourth	Annual
1999	2413	2833	2826	2628	10,700
2000	3634	3440	3125	2937	13,136
2001	2807	2502	2169	2209	9777
2002	2281	2278	2074	2110	8743
2003	2195	2298	2241	2511	9243
2004	2756	2827	2708	2624	10,915
2005	2774	3076	2988	2846	11,684
2006	2972	3162	3011	2964	12,109
2007	3068	2950	2663	2587	11,268
2008	2339	2145	2206	1705	8395
2009	1597	1746	1851	1978	7172
2010	2294	2306	2310	2191	9431
2011	2517	2585	2371	2392	9865
2012	2345	2377	2423	2890	10,035
2013	2288	2211	2593	2621	9713
2014	2880	3020	3060	3140	12,100
2015	3150	3186	3319	3280	12,935
2016	3073	2929	3101	2943	12,049
2017	3041	2790	2608	2763	11,202
2018	2848	3172	3099	2889	12,008

Source: Mergerstat Review, December 2019

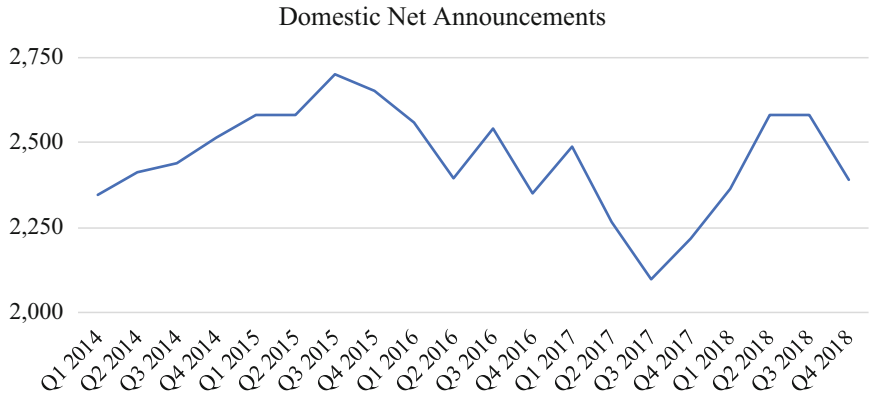


Fig. 1 Domestic transactions

combinations. The mergers of the 1960s, 1970s, and 1980s were almost purely conglomerate in nature (Table 1).

Merger activity continues to be very high, but not increasing as dramatically as in much of the postwar conglomerate merger movement (Fig. 1). The time series merger history has been studied by Nelson (1959, 1966), Melicher, Ledolter, and D'Antonio (1983), Guerard (1985), and Golbe and White (1993).

Domestic mergers continue to be far more numerous than foreign mergers (Fig. 2).

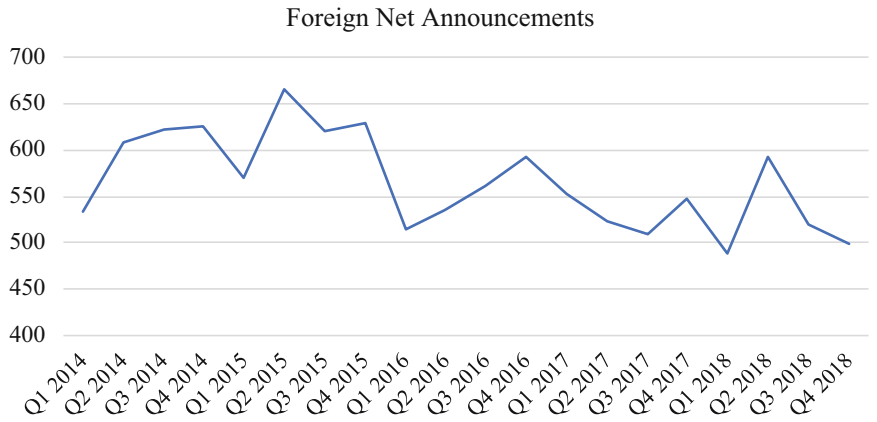


Fig. 2 Foreign transactions

Using an Accounting Basis

If one company purchases another company and only the acquiring company survives, the combination is a merger. Consolidation involves the combination of at least two firms in which a new firm is created.

A company acquires another firm by the *purchase* or accounting method when the acquired firm is valued as the amount actually purchased (Weston, Mitchell, & Mulhern, 2004). The parent firm records the acquired assets at their current market value. If the current value of the acquired firm exceeds its book value, the excess is goodwill that will be amortized over a period of no more than 40 years. The purchase method is the sole acceptable method of accounting for mergers and acquisitions since the Financial Accounting Standards Board (FASB) voted in January 2001 to eliminate the pooling-of-interests method of accounting for mergers and acquisitions.

A second accounting practice, the *pooling-of-interests* method, is no longer allowed because the FASB released its "Business Combinations" statement (No. 141) in June 2001 that had allowed acquisitions in which interests of both firms' stockholders are merged. No assets are revalued and no goodwill was recognized as assets in pooling of interests mergers, and liabilities of the new firms were recorded as such on the unmerged firms' books. The pooling-of-interests method may be used only when one firm issues voting stock for the acquisition; no cash transactions are allowed. The stock issuance for the acquisition must be a single transaction, and the acquiring firm must acquire at least 90% of the acquired firm's stock. The FASB No. 141 Statement superseded the Accounting Principles Board (APB) Opinion 16, "Accounting for Business Combinations" of August 1970, which allowed both purchase and pooling-of-interests methods of accounting for mergers and acquisitions.

In the purchase method of accounting for a merger or acquisition, the purchase price is the basis for recording the assets and liabilities of the acquired firm. The earnings of the acquired firm are reported by the acquiring firm from the date of the acquisition forward. In the pooling-of-interests method of accounting, the original historic cost basis of the assets and liabilities of the acquired firm are carried forward, and the earnings of the combined entity are combined for any reporting periods. Prior year earnings would be the earnings of the pooling-of-interest combination. Assets acquired in a pooling-of-interest M&A could be sold for a higher price than was recorded in the original transaction, and gains in such a transaction would be earned income. The pooling-of-interests method had the potential to manipulate earnings of the firm (Weston et al., 2004).

The Economic Basis for Acquisitions

There are three economic reasons for seeking to acquire other firms: to pursue profits through monopoly power, to generate economies of scale in operations, and to contribute managerial abilities lacking in the acquired firm's management. Expectations of the three economic benefits of mergers can explain horizontal mergers, in which a firm acquires a firm in its own industry. But the economic reasons cannot explain the postwar trend of conglomerate mergers, in which a firm acquires a firm in an unrelated industry. Economists have held that mergers do not generally produce synergism, the "2 + 2 = 5" effect. When synergism is absent, the value of the mergers adheres to the additive property of value in which the value of the merged film is the sum of the market values of the unmerged firms (Mossin, 1973):

$$MV_{ab} = MV_a + MV_b \tag{1}$$

where

MV_{ab} = market value of the merged firm, a and b

MV_a = market value of firm a

MV_b = market value of firm b

If no synergism exists, there would not seem to be any economic rationale for conglomerate mergers.

Theories of Conglomerate Mergers

Conglomerate mergers were explained by Mueller (1969) as resulting from firms' attempts to maximize growth. If a larger, more mature firm wants to consider the more profitable investment opportunities and the higher rates of return on projects of a smaller, younger firm, the larger firm might purchase the smaller firm. By purchasing the smaller firm, the larger firm has internalized the investment

opportunities, and the more profitable marginal investment contributes to the larger firm's growth. Synergism, in which the merged firm's market value exceeds the sum of the unmerged firms' market values, would rarely be found in a conglomerate merger. Mueller hypothesized that the conglomerate merger could be justified only if the managers of the acquiring firm could find investment opportunities currently overlooked by the acquired firm's management. Mueller's theory of mergers has been questioned because its premise is that firms maximize growth. Corporate benefits tend to be distributed on the basis of profits, not growth of sales. Management would maximize profits, in a rational environment, if profits served solely as the basis for benefits.

Gort (1969) put forth a theory of mergers based on valuation discrepancies, or disturbances. Acquisitions of other firms occur when a higher value is placed on the firm's assets by the potential purchaser than by the current owners. Valuation discrepancies tend to occur in times of high stock prices, as purchasers tend to believe the rapid changes in stock prices represent an increased economic disturbance. The higher disturbance increases the distribution of the firm's assets' valuations, raising the probability that an investor will place a higher value on the firm than the current owners. The empirical evidence has revealed that mergers occur during periods of high stock prices; thus, there could be a basis for Gort's disturbance theory.

One of the principal theories of conglomerate mergers is that mergers reduce the probability of bankruptcy for the merged firm. Lewellen (1971) advanced the theory that mergers reduce the variance of the merged firm's cash flow. Mergers may lessen the possibility of default and increase the firm's debt capacity. Debt capacity is increased as lenders are willing to establish a higher corporate lending limit to the merged entity. However, the merger must involve firms with less than perfectly correlated income streams. The merging firms must not have cash flows such that defaults on their borrowings occur simultaneously.

Assume that firms A and B are considering a merger. Given the following probability estimates of the possible states of the world, would the risk of default be lessened by the merger?

		Cash flow	
Probability	State	Firm A	Firm B
0.2	Depression	$100	$920
0.3	Recession	300	700
0.4	Normal	700	240
0.1	Boom	1000	0

It is obvious that the cash flows of firms A and B are negatively correlated. Firm A could be a venture capital firm that profits from a rising economy, while firm B could be a gold mining company that profits when gold prices are high, the economy falling.

Firms A and B have the same expected values E(CF) and essentially the same standard deviations of cash flows, σ_{CF}. The expected cash flow is of the form

$$\sum_{i=1}^{4} p_i CF_i,$$

where p_i = probability of the occurrence of state of nature, I, and the variance is of the form

$$\sum_{i=1}^{4} [CF_i - E(CF)]^2 p_i. \tag{2}$$

We may calculate the expected values of the two cash flows, CF_A and CF_B.

.2($80)	=	$16	.2(920)	=	$184
.3($280)	=	84	.3(700)	=	210
.4($700)	=	280	.4(240)	=	96
.1($1000)	=	100	.1(0)	=	0
$E(CF_A)$	=	490	$E(CF_B)$	=	490

And the variances:

.2($80 − $490)	=	$33,620
.3($280 − $490)	=	13,230
.4($700 − $490)	=	17,640
.1($1100 − $490)	=	37,210
$Var(CF_A)$	=	$101,700
$\sigma(CF_A)$	=	$319
.2($920 − $490)	=	$36,980
.3($700 − $490)	=	13,230
.4($240 − $490)	=	25,000
.1($0 − $490)	=	24,010
$Var(CF_B)$	=	$99,200
$\sigma(CF_B)$	=	$315

The covariance of A's and B's cash flows, the covariance being

$$\sum_{i=1}^{4} p_i [CF_{A_i} - E(CF_A)][CF_{B_i} - E(CF_B)], \tag{3}$$

is calculated:

.2($80 − $490) ($920 − $490)	=	$−35,260
.3($280 − $490) ($700 − $490)	=	−13,230
.4($700 − $490) ($240 − $490)	=	−21,800
.1($1100 − $490) ($0 − $490)	=	−29,890
$Cov(CF_A,CF_B)$	=	$−99,380

The correlation coefficient, r, of the cash flows of firms A and B will establish that these are imperfectly correlated firms:

$$r = \frac{\text{Cov}(CF_A, CF_B)}{\sqrt{\text{Var}(CF_A \times \text{Var}(CF_B)}}$$

$$\frac{\$ - 99,380}{\sqrt{\$101,700 \times \$99,200}} = \frac{\$ - 99,380}{\$100,452} = -.989$$

The correlation coefficient of $-.99$ shows that the cash flows of firms A and B are almost perfectly negatively correlated.

If firms A and B are assumed to have assets of \$10,000 each, a debt of \$6000 each, with each firm's debt having a 4% cost, the unmerged firms should have essentially the same default probability. The interest costs for each firm are \$240, that is, \$6000(.04), and the probabilities of default, given by the area under the normal curve, are essentially equal. The general form is

$$Z = \frac{x - \mu}{\sigma} \tag{4}$$

where

Z = standardized normal variable
X = interest cost, \$240
μ = expected value, \$490 of cash flow
σ = standard deviation of cash flow, \$319 and \$315 for firms A and B

P_r (firm A's cash flow >\$240)

$$Z_A = \frac{\$240 - \$490}{\$319} = -.784$$
$$Z_B = \frac{\$240 - \$490}{\$315} = -.794$$

The Z-values of $-.784$ and $-.794$ correspond to bankruptcy probabilities of .2187 and .2175 for firms A and B, respectively. The Z-value of $-.78$ indicates an area under the normal curve such that there is a .2923 probability that the outcome cash flow will fall between \$240 and \$490. There is a probability of .5000 that the cash flow will exceed \$490. Summing up these probabilities gives .7823, the probability that firm A's cash flow will exceed \$240. Thus, $(1 - .7823) = .2187$, the probability of firm A's cash flow falling below its interest obligations and bankruptcy ensuing.

The merger of firms A and B yields a cash flow of:

Probability	State	(M) Merged firm's cash flow
.2	Depression	$1020
.3	Recession	1000
.4	Normal	940
.1	Boom	1000

The expected value and the variance of the cash flow may be calculated:

.2($1020)	=	$204
.3($1000)	=	300
.4($490)	=	376
.1($1000)	=	100
$E(CF_M)$	=	$980
.2($1020 − $980)	=	$320
.3($1000 − $980)	=	120
.4($940 − $980)	=	640
.1($1000 − $980)	=	40
$Var(CF_M)$	=	$1120

The variance of the merged firm's cash flow is only $1120. Its standard deviation is $33.47, and the probability of bankruptcy goes to zero:

$$Z = \frac{\$480 - \$980}{\$33.47} = -14.94 > -3.0$$

Even though the merged firm's interest cost is $480, the sum of the unmerged firms' interest expenses, the standard deviation is so reduced that the probability of bankruptcy is zero. The Z-value far exceeds −3.0, the minimum value of the normal curve. With less than perfectly correlated cash flows, mergers reduce the merged firm's variance and risk of default.

Another way of examining the reduction in variance is to look at the merger in a portfolio approach. Assume that firms A and B have cash flow variances of 30% and a correlation coefficient of .40. If the merged firm is made up of 50% of firm A and 50% of firm B, the merged firm's variance and standard deviation will be

$$\begin{aligned}
\sigma(CF_M) &= \sqrt{Var(CF_M)} \\
&= \sqrt{W_A^2\sigma_A^2 + W_B^2\sigma_B^2 + 2W_A W_B \sigma_{AB}\sigma_A\sigma_B} \\
&= \sqrt{(.50)^2(.30) + (.50)^2(.30) + 2(.50)(.50)(.40)\left(\sqrt{.30}\right)\left(\sqrt{.30}\right)} \\
&= \sqrt{.075} + .075 + .06 \\
&= \sqrt{.21} \\
&= .4583.
\end{aligned}$$

The unmerged firms A and B had variances of 30%, whereas the merged firm has a variance of 21%. Firms A and B had standard deviations of .5477, that is, $\sqrt{.30}$, and the merged firm's standard deviation is only .4583. The portfolio approach reveals that mergers reduce the risk of the merged firm relative to the risks of the unmerged firms.

Since the variance of the merged firm falls, the merged firm could sustain a higher debt level than the sum of the unmerged firms. The existence of possibly additional debt capacity creates more highly leveraged firms. In the business world of taxation, leverage increases the stock prices and the total market value of the firm.

In contrast to those combinations that exhibit some organic consistency, there are those whose avowed purpose is *diversification.* Diversification has at times been promoted as a cure to almost all managerial difficulties. If one branch of the company runs into difficulties, it is expected that some other operation will show enough profits to assuage the economic hurt. But diversification may also make it possible for a losing division to drain away the returns from the profitable operation. On closer inspection, however, some diversified combinations reveal distinct economic or technical connections. The historic classic case is the ice and fuel company where men and equipment used for delivering ice in the summer were diverted to delivering fuel in the winter. A whiskey distillery may expand into petrochemicals because, among other things, some of the engineering problems are similar. Some combinations have been motivated by historical relationships that may no longer exist. Where there is no logical relationship in the structure of a widely diversified combination, the reasons for the merger may rest on temporary tax advantages or perhaps on speculative manipulations. This type of combination may develop managerial or administrative problems and often does not survive intact for long.

Combinations Correcting Economic or Financial Imbalances

Using the criterion that the economic or financial success of a combination is basically demonstrated by a rise in core earnings, potentially worthwhile merger situations have the following characteristics:

1. One or both of the firms have some imbalance in its financial or economic structure that can be offset by the merger.
2. The combination creates or increases market dominance.
3. The merger enables the firm to take certain tax savings.
4. The merger reduces unused duplicate facilities.

A merger that attempts to correct financial or economic imbalances is perhaps the one type that may lead to socially desirable results. For example, the new firm could lower costs by eliminating duplicate facilities. The merger releases productive factors for more valuable uses elsewhere. Presumably, these factors were previously

underutilized. In other words, the independent firms' output structures were unbalanced—i.e., they did not have an optimum mix of the productive factors.

The following is a list of merger possibilities arising from various imbalances in the companies' production or financial structure. In every case, the firms are not operating with the optimum production or capital mix.

1. A firm with aggressive management and a relatively small asset base might combine with a firm with a large asset base but no new developing managerial talent.
2. A firm with fixed assets and a shortage of net working capital might merge with a firm with abundant working capital.
3. A firm with a heavily leveraged financial structure could combine with one that had an ultraconservative equity position.
4. A firm with a strong marketing position might unite with a firm, with a good production position.
5. A firm with solidly established products could combine with another company that had a topnotch research program.
6. A firm with heavy expansion plans could combine with a company having a strong liquidity position.
7. A firm faced with seasonal demand for its product could combine with another firm whose product showed a reverse seasonal pattern.

A merger is not the only cure possible for a firm with an imbalance. A company can absorb its redundant productive factors by self-expansion (or contraction) in the proper direction. A combination, nevertheless, often is the fastest or cheapest way of making the adjustment.

Combinations Increasing Market Dominance

A merger that increases the company's degree of market dominance can be potentially profitable to the stockholders. Since the merger removes some competition, selling prices may be more easily maintained in the market, or the larger combined company can bargain better with its suppliers.[5] If the combined company can raise revenues or reduce costs, the net earnings available for all the stockholders rise.

Historically, financial promoters seem to have profited by putting together combinations involving increased market power. On the basis of its combined earning power, the new company may be worth more than its component parts and can carry a larger capital structure without loss to the investors. This possibility has, on occasion, been recognized by promoters who have organized the consolidation of

[5] A firm which is able to reduce the prices paid to its suppliers shows no true example of external economies. The result is merely a backward shifting of costs.

two or more companies and as their reward has taken some of the securities in the combination. If the promoters were reasonable, the investors would still obtain increased profits even after allowing for the return on the additional securities floated and given out as a reward or an inducement to the promoters, lawyers, investment bankers, management people, and major stockholders involved. However, sometimes the organizers were too acquisitive or were overly optimistic in forecasting the earnings of the merger and, as a consequence, rewarded themselves excessively with new issues of stocks or bonds. The result was an *overcapitalized* company, a company whose earnings were inadequate to support a market price for its capital structure equal to its *stated* or *par* value.

Although combining to increase market power can profit promoters and stockholders, its overall effect probably reduces the amount of workable competition in the economy. In general, the American tradition has been against private attempts to control market forces. This tradition has been expressed (although not always strongly enforced) in a series of legislative acts including the Sherman Antitrust Act of 1899, the Clayton Antitrust Act of 1914, and the Celler Anti-Merger Amendment of 1950 to the Clayton Antitrust Act. As the law now stands, mergers (or acquisitions of the assets of other firms) that tend substantially to lessen competition are prohibited. What constitutes a lessening of competition is, of course, subject to judicial interpretation. Nevertheless, the rule was construed widely enough in 1958 to prevent the merger of Bethlehem Steel Corporation and Youngstown Sheet and Tube, even though competition between these two geographically separated companies existed in only a small part of their output.[6] The court held the view that both companies were essentially strong and that their merger would eliminate a wide area of *potential* competition. White (2002) found no evidence that the increase in the share of value added in the United States was due to an increase in aggregate concentration. White found that the increase occurred before the large conglomerate merger activity of the late 1960s.

Combinations for Tax Advantages

The present corporation profit tax law allows for a tax loss carryback of 3 years and a tax loss carryforward of 5 years from the year the loss occurred. For example, suppose a corporation shows a deficit on the current year's operations of $1,000,000. It can apply this loss against the profits of the last 3 years, and if it has earned *at least* $1,000,000 taxable profits in these years, the Treasury refunds the company approximately $350,000. If, however, past profits are insufficient to cover the loss, the excess loss may be used to reduce the taxable profits of the next

[6]Bethlehem Steel is now in bankruptcy and no longer an independent company.

5 years. Of course, in any case, the total reduction in tax liability cannot exceed $350,000.

A *merger* for tax savings may take place when a firm with a tax loss carryforward cannot earn enough itself to use its tax credit. Since the law allows the merged companies to consolidate their earnings, the tax loss carryforward enables the new firm to reduce its tax liabilities regardless of which division originated the losses. Thus, a firm with a tax loss credit of $1,000,000 can offer to a potential merger partner the possibility of saving $350,000 in taxes on future income. In a sense, the loss firm brings an extra asset to the union which has to be considered in setting the merger terms.

The potential tax credit never equals the actual loss. In spite of tax credits, it is very difficult to get rich by taking losses. If a company's projected revenues do not cover out-of-pocket costs, its past losses have value in a merger, but merger or not, there is no reason to continue its operations. The company's operations will be closed down, and unless its assets can be turned to something else, it will be liquidated either *before* or after the merger. If it is liquidated before the merger, it becomes what is known as a corporate shell. As a matter of fact, there is occasional trading in these corporation shells-firms which have no assets but do have a charter and tax losses that may be used by other profitable companies.

Under the old tax laws, established stable earning firms could well undertake some risk projects since losses, if any, could be offset for tax purposes against solid current earnings. Some losses would be recovered through reduced taxes. On the other hand, a new firm was at a disadvantage because it would have no established earnings upon which the taxes might be reduced. Under the present tax law, however, a company experiencing losses has two alternatives. It may continue to operate, hoping eventually to make profits against which its loss carryforward will be useful, or, if the situation is hopeless, it will close down and look for a tax merger which will return part of the losses it has suffered. The opportunity of obtaining some return of losses through a merger for tax purposes is thus the sole value of the loss carryforward provision to an unsuccessful new enterprise which discontinues operations. The possibility of a new company obtaining some return of possible losses through a merger helps redress the odds on a risk venture, as opposed to the more favorable conditions for taking chances of older and larger companies. And if the conditions were thus unaltered in favor of established firms, the force of dynamic new competition would be yet weaker, and the forces which encourage the growth of oligopoly and business concentration even stronger.

To sum up, the loss carryforward tax credit was designed (1) to equalize the tax burden for firms having a given rate of return but whose income pattern was erratic with firms having the same rate of return at a more stable level and (2) to improve the expected return for new risk ventures. The latter function is actually attained by allowing a loss company to bring its tax credit into a merger. Harris, Stewart, and Carleton (1982) did not find that the tax loss carryforward variable useful in predicting acquired firms in the 1974–1977 period.

The Larson-Gonedes Exchange Ratio Model

It has been held that mergers often involve firms with differing price-earnings (P/E) multiples. The reader recalls that we discussed the low "P/E", or high EP, investing of Grahman and Dodd in chapter 14. Larson and Gonedes (1969) developed an exchange ratio model that is both theoretical and highly testable. The model can be developed from an assumption that the acquiring firm wants to project a growth image so that the market will capitalize its earnings at a high price-earnings multiple. As the firm acquires smaller firms (in terms of the firms; price-earnings multiples), the earnings of the acquired firms are capitalized at the acquiring firm's higher price-earnings multiple. The merger results in a higher market price for the acquiring firm.

Assume two firms, A and B, can be represented by the following data:

	Firm A	Firm B
Earnings (E)	$20 million	$5 million
Shares outstanding (S)	2 million	1 million
Earnings per share (EPS)	$10	$5
Stock price per share (P)	$160	$50
Price-earnings multiple (P/E)	16	10

Firm A is interested in acquiring firm B and is willing to pay firm B's stockholders $60 per share for their stock. Firm B's stockholders are delighted, because they have immediately profited $10 per share. If firm A is willing to pay $60 per share for firm B, the exchange ratio, ER, is

$$\text{ER} = \frac{\text{P}_\text{B}}{\text{P}_\text{A}} = \frac{\$60}{\$160} = .375.$$

To pay firm B's stockholders, firm A can issue firm B's stockholders (.375) 1,000,000 shares or 375,000 shares of firm A's stock. If a stockholder in firm B before the merger owned 100 shares of stock, worth $5000, he now owns 37.5 shares of firm A's stock. Following the merger, the former stockholder of firm A has holdings worth $6000, that is, 37.5 × ($160). His gain is $1000 because of the merger.

We must now examine whether firm A's stockholders will profit. If firm A pays firm B's stockholders $60 per share and issues 375,000 shares, the merger is very profitable for firm A's stockholders as long as the market allows firm A's price-earnings multiple to remain at 16. Following the merger, firm AB is represented as follows:

Firm AB		
Earnings	=	$20 million + $5 million = $25 million
Shares outstanding	=	2,000,000 + 375,000 = 2,375,000
EPS	=	$10.53
Price	=	P/E (EPS) = 16 ($10.53) = $168.48

Firm A's stockholders have profited $8.48 per share because of the merger. Notice, however, that firm A's stockholders profited only because the market did

not adjust the firm's price-earnings multiple after the merger. A decrease in the price-earnings multiple of the merged firm could well wipe out the acquiring firm's gains.

In the absence of synergy, Larson and Gonedes held that firm A's price-earnings multiple should fall as a result of the merger. The price-earnings multiple of the merged firm, θ, should be the weighted average of the merging firms' price-earnings multiples in the absence of synergy:

$$\begin{aligned}\theta &= \frac{P_A(S_A) + P_B(S_B)}{E_A + E_B} \\ &= \frac{\$160(2,000,000) + \$60(1,000,000)}{\$20,000,000 + \$5,000,000} \\ &= 15.2.\end{aligned}$$

If the market does not adjust for the decline in aggregate real growth after the merger, a fallacious increase in market price can take place. The whole phenomenon may be classed as illusory growth.[7] If the price-earnings multiple of the merged firm is equal to θ, there are no profits for the acquiring firm's stockholders; the price of firm A's stock remains at $160:

$$P_A = \$10.53(15.2) = \$160.06$$

The presence of synergy would be shown in a price-earnings multiple of the merged firm exceeding θ. If the price-earnings multiple of the merged firm is less than θ, the acquiring firm's stockholders suffer a loss and would oppose the merger.

Firm A's shareholders should be willing to pay firm B's stockholders a price dictated by a maximum acceptable exchange ratio, ER_A. Larson and Gonedes derived this to be

$$ER_A = \theta(E_A + E_B) - \left(\frac{P/E_A(E_A)}{(P/E)_A E_A\left(\frac{1}{S_A}\right)(S_B)}\right) \tag{6}$$

For example, if the market were to allow the merged firm's price-earnings multiple to remain at 16, firm A's stockholders could afford to pay firm B's stockholders a price far exceeding $60 per share and still reap merger profits:

$$\begin{aligned}ER_A &= \frac{16(\$25,000,000) - 16(20,000,000)}{16(\$20,000,000)\left(\frac{1}{2,000,000}\right)(1,000,000)} \\ &= .500.\end{aligned}$$

[7]This problem is described in more detail in chapter "Management-Stockholder Relations: Is Optimal Behavior All That Is Necessary?".

An exchange ratio of .500 implies a maximum acceptable price of \$80 per share for firm B's stock:

$$P_B = ER_A(P_A) = .500(\$160) = \$80.$$

If firm A pays firm B stockholders a price of exactly \$80 per share, firm A's stockholders are indifferent to the merger because the price of firm A's stock remains at \$160 per share. Firm A must issue 500,000 shares, that is, .5 (1,000,000), to acquire firm B. Following the merger, firm A is represented as:

	Firm A
Earnings	\$25,000,000
Shares outstanding	2,500,000
EPS	\$10
P/E	16
Price = \$160 =	\$10(16)

Firm A's stockholders profit as long as the exchange ratio is below .500. If the market allows a price-earnings multiple of the merged firm to remain at 16, firm A cannot pay a higher price than \$80 per share for firm B and still profit from the merger. Of course, firm A would prefer to offer firm B's stockholders as small a price as possible for their stock to consummate the merger. If firm A paid firm B's stockholders a price of \$60 per share, the merger premium, MP, would be given in terms of the actual exchange ratio, AER:

$$MP = \left(\frac{AER(P_A) - P_B}{P_B}\right)(100) \tag{7}$$

where

$$AER = \frac{\$60}{\$160} = .375$$

$$MP = \left(\frac{.375(\$160)\text{-}\$50}{\$50}\right)(100) = 20$$

A 20% merger premium would have been paid.[8]

[8]The minimum θ consistent with the profitability constraint, that the merged firm's price must be at least as great as the acquiring firm's price, is given by

$$\theta = \frac{P_A[AER(S_B) + S_A]}{E_A + E_B}$$

$$= \frac{\$160[.375(1,000,000) + 2,000,000]}{\$20,000,000 + 5,000,000}$$

$$= 15.2.$$

Synergism will result only if θ exceeds the weighted average of the merging firm's price-earnings multiples. If

$$\theta > \frac{P_A(S_A) + P_B(S_B)}{E_A + E_B},$$

the market value of the merged firm exceeds the market values of the unmerged firms. If ER $>$ P_B/P_A, the acquired firm profits more proportionally than the acquiring firm. The reverse occurs if ER $<$ P_B/P_A. A firm should engage in a merger only if it benefits its stockholders. The relative bargaining strengths of the acquiring and acquired firms determine the distribution of the merger gains. Note that high price-earnings multiples, generally resulting from stock price, allow more mergers to be profitable. The history of mergers in the United States substantiates the positive association between stock prices and number of mergers.

Valuation of a Merger Candidate

The use of discounted cash flows can value a potential merger candidate. Rappaport (1979, 1981) developed a logical and consistent framework for evaluating the expected cash flow of a merger candidate. The capital budgeting techniques of chapter "Investing in Assets: Theory of Investment Decision-Making," the risk adjustments of a beta estimation illustrated in chapter "Risk and Return of Equity and the Capital Asset Pricing Model," and the cost of capital calculations of chapter "Debt, Equity, the Optimal Financial Structure, and the Cost of Funds" are used in this section. The Rappaport (1979) framework for merger valuation addresses several issues:

1. What is the maximum pay to pay for the target company?
2. What are the primary areas of risk?
3. What are the earnings, cash flow, and balance sheet implications of the acquisition?
4. How should the acquisition be financed?

Notice the minimum θ consistent with the acquiring firm's profitability constraint equals the weighted average price-earnings multiple for the merging firms.

If firm A paid firm B's stockholders $70 per share for their stock, the actual exchange ratio would be .4375 and the merger premium would be 40%. The minimum θ consistent with merger profitability would be

$$\theta = \frac{\$160[(.4375)1,000,000 + 2,000,000]}{\$25,000,000} = 15.6.$$

If firm A is willing to pay firm B's stockholders a 40% merger premium, the market must allow the merged firm to maintain at least a 15.6 price-earnings multiple for the merger to profit the acquiring firm's stockholders.

Let us analyze a potential merger candidate for an acquiring firm with the following capital structure:

Long-term debt	$5647
Equity	9063
Total capital	$14,710

The firm has a bond rating of AA3 (Moody's AA) and a beta of 0.84 versus the S&P 500 Index.

The cost of capital, k_c, can be calculated by using an acceptable market risk premium and the current AA bond yield of 5.6%. The Ibbotson and Sinquefield market risk premium of 8.8% was based on the 1926–1993 period. If we use the Ibbotson and Sinquefield data for the 1951–2002 period, found on WRDS, we find an average annual rate of return on equities of 12.53%, and a corresponding average Treasury bill yield of 5.15%, implying a 7.38% market risk premium. The cost of equity capital for our acquiring firm is

$$k_e = .0515 + (.0738).84 = .1135$$

The cost of equity capital, via the Capital Asset Pricing Model, is 11.35%. The weighted average cost of capital may be calculated as

$$k_c = k_e\left(\frac{E}{D+E}\right) + k_d\left(\frac{D}{D+E}\right)(1-t)$$

$$k_c = .1135\left(\frac{9063}{5647+9063}\right) + .056\left(\frac{5647}{5647+9063}\right)(1-.35)$$

$$= .084$$

The acquiring firm's weighted average cost of capital is the appropriate discount rate for valuating merger candidates.

A potential merger partner has current sales level of $100 million, depreciation of $5 MM, debt value of $50 million, and 10 million shares outstanding. Depreciation is assumed to increase at 20% annually. One can further assume the following scenarios over a 10-year valuation period:

Years	1–5 (%)	6–10 (%)
Growth in sales	10	15
Profit margin	30	32
Tax rate	35	35
f = capital expenditures as a % of sales growth	25	25
w = working capital increase as a % of sales growth	10	10

The forecasted sales and cash flow levels are shown in Table 2 and the discounted cash flows are shown in Table 3. The discounted cash flow of the acquired firm is $281.52 million during the 10-year holding period. One capitalizes the 10-year

Table 2 Forecasted sales, cash flow ($ MM)

	Year of acquisition									
$Sales_0 = 100$	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Sales	110	121	133.1	146.4	161.1	185.3	213.1	245.1	281.8	324.0
Operating expenses	77	84.7	93.2	102.5	112.8	126	144.9	166.7	191.6	220.3
EBIT	33	36.3	39.9	43.9	48.3	59.3	68.2	78.4	90.2	103.7
(t = 0.35) = taxes	11.6	12.7	14.0	15.4	16.9	20.8	23.9	27.4	31.6	36.3
Op. earnings after taxes	21.4	23.6	25.9	28.5	31.4	38.5	44.3	51.0	58.6	67.4
Depreciation	6.0	7.2	8.6	10.4	12.4	14.9	17.9	21.5	25.8	31.0
Δ sales	10	11	12.1	13.0	14.7	24.2	27.8	32.0	36.7	42.2
– CE (f = 0.25)	−2.5	−2.8	−3.0	−3.3	−3.7	−6.1	−7.0	−8.0	−9.2	−10.6
– Δ NWK (w = 0.10)	−1.0	−1.1	−1.2	−1.3	−1.5	−2.4	−2.8	−3.2	−3.7	−4.2
Cash flow	23.9	26.9	30.3	34.3	38.6	44.9	52.4	61.3	71.5	83.6

Table 3 Discounted cash flow

Year	CF	Discount rate	PV(CF)
1	23.9	.9225	22.05
2	26.9	.8571	23.06
3	30.3	.7849	23.78
4	34.3	.7241	24.84
5	38.6	.6680	26.25
6	44.9	.6165	27.68
7	52.4	.5685	29.79
8	61.3	.5247	32.16
9	71.5	.4838	34.59
10	83.6	.4464	37.32
			$281.52

earnings on an infinite period and discounts the result from its 10-year discounted cash flow.

Market valueof the firm	$639.70 MM
−Debt	50.00 MM
Market value of equity	$589.70 MM

Present value of discounted 10-year earnings, earn_{t+10}:

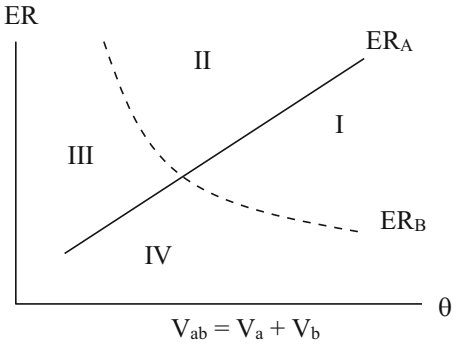
$$\begin{aligned} \text{PV}(\text{Earn}_{t+10}) &= .4464\left(\frac{67.4}{.084}\right) \\ &= \$358.18 \end{aligned}$$

The total present value is the present value cash flow in the 10-year period, $281.52 MM, plus the capitalized 10-year earnings, $358.18, or $639.70 million, see Table 3. One should subtract the assumed debt, $50 MM, from the present value of the firm's cash flow, to derive its estimated value of equity. The market value of equity is $589.70, and given the acquired firm's 10 million shares, the fair value of the acquired firm's equity should be $58.97 per share. The Rappaport merger valuation framework assumes that the acquiring firm can produce a reasonably accurate forecast of projected earnings and cash flow. One pays in an all-cash manner if consistent with the acquiring firm's current liquidity and targeted debt-to-equity ratio.

Testing for Synergism

Research has established that the market anticipates mergers at least 7 months before they occur. Mergers generally occur during periods of rising stock prices, and thus tests for synergy must consider market movements. Adjusting for systematic risk,

Halpern (1973) found that the merger gains before the announcement are divided equally between the acquiring and acquired firms. Conn and Nielsen (1977) tested evidence of merger synergism, as defined by the Larson and Gonedes model, at the month of the merger announcement, the month of the merger consummation, and the month following merger consummation; they found support for the Larson and Gonedes model. Harris et al. (1982) provide evidence that mergers in the 1970s that the price-earnings multiples of the acquired firms were significantly less than that of non-acquired firms.



In terms of the Larson and Gonedes exchange ratio model and the ER-0 diagram, mergers in quadrant 1 profit both firms' stockholders. Above the ER_B rectangular hyperbola, the acquired firm's stockholders win since the merger premium is relatively large. To the right of ER_A, the acquiring firm's stockholders win as the market maintains

$$\theta > \frac{P_A(S_A) + (S_B)}{E_A + E_B}. \tag{8}$$

Conn and Nielsen found that 78 of 131 mergers between 1960 and 1969 occurred in quadrant 1 in the month of announcement, supporting Larson and Gonedes' model.

However, by the time of the merger consummation and the month following the merger consummation, only 72 and 67 of the 131 mergers were profitable to both firms. In the month following consummations, only 51% of the mergers were profitable to both firms' stockholders. The acquired firms' stockholders lost in 24 of the 131 mergers in the month following consummation, whereas the acquiring firms' stockholders lost in 58 cases.[9]

[9]Harris et al. (1982) found that acquired firms had significantly lower price-earnings multiplier than acquiring firms.

Do Mergers Enhance Shareholder Wealth?

The reader probably expects the stockholders of acquired firms to earn positive and highly significant excess returns. After all, merger premiums rose to 25–30% during the 1958–1978 period (Dodd & Ruback, 1977). What about the acquiring firms? If the acquired firms' stockholders profit handsomely from a merger, should not the acquiring firm's stockholders lose? Are mergers zero-sum events? Is wealth created by mergers? Let us examine much of the empirical evidence. Mandelker (1974) put forth the Perfectly Competitive Acquisitions Market (PCAM) hypothesis in which competition equates returns on assets of similar risk, such that acquiring firms should pay premiums to the extent that no excess returns are realized to their stockholders. The PCAM holds that only the acquired firms' stockholders earn excess returns. However, Mandelker studied mergers of 241 acquiring firms during the 1948–1967 period and found that acquiring firms' stockholders earned 5.1% during the 40 months prior to the mergers, but excess returns decreased by 1.7% in the 40 months following the merger. Positive net excess returns (3.7%) were earned by the acquiring firms in the Mandelker study. Thus, Mandelker found no evidence that acquiring firms paid too much for the acquired firms. Moreover, the acquired firms' stockholders realized excess returns of 12% for the 40-month period prior to the merger and 14% for the 7-month period prior to the merger.

The Mandelker results have been substantiated by much of the empirical literature. Dodd and Ruback (1977) found that successful acquiring firms' stockholders gained 2.8% in the month before the merger announcement during the 1958–1978 period, whereas the successful acquired firms' stockholders gained 20.9% excess returns. Dodd and Ruback found that the acquired firms' stockholders gained 19.0% even if the merger was unsuccessful, whereas the acquiring firms' stockholders gained less than 1%. The empirical evidence for the 1973–1998 period is consistent; from 20 months prior to the merger to its close, the combined firms' stockholders gain approximately 1.9% (Andrade, Mitchell, & Stafford, 2001). Moeller, Schlingemann, and Stulz (2003) analyzed 12,023 mergers during the 1980–2001 period and found a 1.1% gain to acquiring firms' shareholders.

Moeller et al. (2003) used a sample of public acquiring firms making acquisitions exceeding $1 million. Private acquired firms accounted for 5583 of the 12,023 acquisitions involving subsidiaries. The private firm acquisitions tended to be more cash-financed (50.56%) than public firm acquisitions (29.57% cash, 55.32% in equity), and the subsidiary acquisitions were predominantly cash (75.92%). The 3-day cumulative average returns (CARS) of the acquiring firms' deals were on average 1.1%; however, the acquisition of private firms produced higher CARS (1.50%) than the public firm acquisitions (−1.02%), but less than the subsidiary acquisitions (2.00%). Smaller-capitalized firms' acquisitions private firm-financed by equity produced the highest CARS, whereas larger-capitalized, equity-financed

acquirers produced the lowest CARS (−2.45%). The equity financing of subsidiaries produced excess returns of 5.40%. In general, Moeller et al. found that cash financing produced higher excess returns than equity deals, led by cash deals for private firms and subsidiaries. The cash acquisitions by small firms produced statistically significant excess returns. Large firms make poorer acquisitions. Rappaport and Sirower (1999) noted that the large 1990s mergers involved more stock and less cash than the large 1980s mergers. Rappaport and Sirower state that the acquiring firm's management asks if its stock is undervalued; issuing new (more undervalued) shares would penalize current stockholders. Acquiring stocks should finance with stock if uncertain of synergy. Rappaport and Sirower (1999) calculate a stockholder value at risk (SVAR) which measures the merger premium divided by the market value of the acquiring firm before the merger announcement is made. The greater the merger premium and the greater the market value of the seller relative to the acquiring firm, then the higher the SVAR (if no post-acquisition) synergies are realized.

Mergers may enhance stockholder wealth; however, whereas Andrade, Mitchell, and Stafford further found that the target or acquired stockholders gained about 23.8% for the 20-month period, consistent across the decades of the 1973–1998 period, the acquiring firms' stockholders lost about 3.8%, during the corresponding 20-month period. For the largest merger in the US history prior to 1983, Ruback (1982) found that DuPont lost 9.89% ($789 million of stockholder wealth) in the month prior to the merger announcement, whereas Conoco stockholders gained 71.2% ($3201.2 million) for the 2-month period prior to the successful DuPont merger announcement.

Do mergers affect the firms' operations? Hall (1993) found that research and development (R&D) activities were not impacted significantly by mergers. Hall found no lessening of R&D spending. Healy, Palepu, and Ruback (1992) reported that mergers seeking strategic takeovers outperformed financially motivated takeover. Strategic takeovers generally involved friendly takeovers financed with stock, whereas financial takeovers were hostile takeovers involving cash payments. During the 1979–1982 period, for the 50 largest mergers, Healy, Palepu, and Ruback found that strategic takeovers made money for the acquiring firms, whereas financial takeovers broke even. Acquiring stockholders of strategic acquisitions made 4.4% for 5 years post-merger, assuming no premiums paid, whereas financial takeovers earned the acquiring stockholders 1.1%. The premiums paid in financial takeovers were higher (45%) than in strategic takeovers (35%), and the synergies were lower in financial takeovers. Trimbath (2002, p. 137) found "no significant merger effect on net profit, operating profit, or market value" when analyzing firms purchased by Fortune 500 firms during the 1981–1995 period. Mergers generate a net gain for stockholders in the US economy, but one prefers to be a stockholder in the acquired, rather than the acquiring, firm.

Divestment and Spinoff

A divestment or spinoff may be considered as a contrast to a merger. Spinoffs reduce the scale of operations. A company may wish to dispose of a subsidiary; such an action is called a divestment.[10] A firm may want to dispose of a subsidiary company for economic or financial reasons, or it may have to divest itself of a given operation as the result of antitrust action. A firm may voluntarily get rid of its interests in a subsidiary if the subsidiary operations do not mesh properly with the parent, or if the parent company needs funds for working capital or to finance other acquisitions or expansion.

A divestment because of the antitrust laws takes place in compliance with court orders or a consent decree where the decision has been made that the ownership of the subsidiary gives the parent company too great an actual or potential dominance in the market. Divestments have also occurred in compliance with actions taken to simplify holding company systems.

Basically, a company can dispose of an unwanted subsidiary in three ways: (1) by the sale of the subsidiaries shares on the security markets, (2) by selling the subsidiary to another company for cash or securities, or (3) by a spinoff. When subsidiaries are sold to another company, payment is often made in the notes, bonds, or the common shares of the purchasing company. A company that has technological know-how but not the necessary development capacity may find it profitable to sell a subsidiary with an experimental product to a market-oriented firm for shares of stock.

A spinoff consists of distributing the ownership shares in the subsidiary on a pro rata basis to the stockholders of the major company. Whereas the sale of the subsidiary (whether on the market so that it becomes an independent company or to another major company) brings the parent company either cash or new security investments, a spinoff decreases the parent company's stated assets with no offsetting increase. On the other hand, a spinoff may be the least upsetting to the management of the subsidiary (a new owning company may begin by shifting personnel around). The spinoff also saves the management of the parent company the effort and difficulty of negotiating a sale or organizing a security flotation. A spinoff spares the parent company worry about whether it best served its own stockholders in the sale price obtained for the subsidiary. Spinoffs have been criticized as an inefficient antimonopoly remedy since they leave the same ownership in charge of both companies. However, history seems to indicate that, in time, the sale and trading of stocks tends to change the ownership complexion in the separate firms. After a while, the management of the formerly related firms develops their own desires to excel, and competition between mother and daughter firm may be as intense as any other in the industry.

[10]Sometimes the subsidiary interest is a branch, a by-product operation, or plant and is not independently incorporated. If so, an initial step of incorporation would have to be taken before the subsidiary can be relinquished.

One factor for divestment is the shedding of an unsatisfactory acquisition or merger. Kaplan and Weisbach (1992) studied such divestitures. Kaplan and Weisbach found that 44% of the large acquisitions completed in the 1970s and early 1980s had been divested in the 1980s. Fifty-six percent of these divestments reported gains or no losses, whereas 44% reported losses on the sales. Kaplan and Weisbach found that the average sale price, deflated by the S&P 500, was 90% of the purchase price. Conglomerate mergers were most often divested.

Summary and Conclusions

Mergers and acquisitions have been a major source of corporate growth and economic concentration during the past 125 years. The empirical evidence is mixed; most acquired firms' stockholders profit handsomely with excess returns exceeding 25%, whereas acquiring firms' shareholders earn excess returns of only about 1–1.50%. We illustrate the Rappaport discounted cash flow model for assessing the market value of the target firm. The financial evidence is consistent with the Perfectly Competitive Acquisitions Market (PCAM) hypothesis in which competition equates returns on assets of similar risk, such that acquiring firms should pay premiums to the extent that no excess returns are realized to their stockholders. The PCAM holds that only the acquired firms' stockholders earn excess returns.

References

Alberts, W. W., & Segall, J. E. (1966). *The corporate merger*. Chicago: The University of Chicago Press.

Alberts, W. W., & Varaiya, N. P. (1989). Assessing the profitability of growth by acquisition. *International Journal of Industrial Organization, 7*, 133–149.

Andrade, G., Mitchell, M., & Stafford, E. (2001). New evidence and perspectives on mergers. *Journal of Economic Perspectives, 15*, 103–120.

Butters, J. K., Lintner, J., & Cary, W. L. (1951). *Corporate mergers*. Cambridge, MA: Graduate School of Business, Harvard University.

Caves, R. E. (1989). Mergers, takeovers, and economic efficiency. *International Journal of Industrial Organization, 7*, 151–174.

Conn, R., & Nielsen, J. (1977). An empirical test of the Larson-Gonedes exchange ratio determination model. *Journal of Finance, 32*, 749–760.

Dewing, A. S. (1921). A statistical test of the success of consolidations. *Quarterly Journal of Economics, 36*, 84–101.

Dewing, A. S. (1953). *The financial policy of corporations* (Vol. 2, 5th ed.). New York: Ronald Press. Chapters 30, 31, 32.

Dodd, P., & Ruback, R. (1977). Tender offers and stockholders returns: An empirical test. *Journal of Financial Economics, 4*, 351–374.

Golbe, D., & White, L. (1993). Catch a wave—The time series behavior of merger. *Review of Economics and Statistics, 75*, 493–499.

Gort, M. (1969). An economic disturbance theory of mergers. *Quarterly Journal of Economics, 83*, 724–742.

Guerard, J. B. (1985). Mergers, stock prices, and industrial production: An empirical test of the Nelson hypothesis. In O. D. Anderson (Ed.), *Time series analysis: Theory and practice* (Vol. 7). Amsterdam: North-Holland Publishing Company.

Hall, B. H. (1993). The effect of takeover activity on corporate research and development. In A. J. Auerback (Ed.), *Corporate takeovers: Causes and consequences*. Chicago: The University of Chicago Press.

Halpern, P. (1973). Empirical estimates of the amount and distribution of gains to companies in mergers. *Journal of Business, 46*, 554–575.

Harris, R. S., Stewart, J. F., & Carleton, W. T. (1982). Financial characteristics of acquired firms. In M. Keenan & L. J. White (Eds.), *Mergers and acquisitions*. Lexington, MA: Lexington Books.

Healy, P. M., Palepu, K. G., & Ruback, R. S. (1992). Does corporate performance improve after mergers. *Journal of Financial Economics, 21*, 135–175.

Healy, P. M., Palepu, K. G., & Ruback, R. S. (1997). Which takeovers are profitable? Strategic or financial. *Sloan Management Review, 37*, 45–57.

Jarrell, G. A., Brickley, J. A., & Netter, J. M. (1988). The market for corporate control: The empirical evidence since 1980. *Journal of Economic Perspectives, 2*, 49–68.

Jensen, M. C., & Ruback, R. S. (1983). The market for corporate control: The scientific evidence. *Journal of Financial Economics, 11*, 5–50.

Kaplan, S. N., & Weisbach, M. S. (1992). The success of acquisitions: Evidence from divestments. *Journal of Finance, 47*, 107–137.

Larson, K., & Gonedes, N. (1969). Business combinations: An exchange ratio determination model. *Accounting Review, 44*, 720–728.

Lewellen, W. G. (1971). A pure financial rationale for the conglomerate mergers. *Journal of Finance, 26*, 521–537.

Lintner, J. (1971). Expectations, mergers, and equilibrium in purely competitive markets. *American Economic Review, 61*, 101–112.

Livermore, S. (1935). The success of industrial mergers. *Quarterly Journal of Economics, 50*, 68–96.

Mandelker, G. (1974). Risk and return: The case of merging firms. *Journal of Financial Economics, 1*, 303–335.

Manne, H. G. (1965). Mergers and the market for corporate control. *Journal of Political Economy, 73*, 110–120.

Melicher, R., Ledolter, J., & D'Antonio, L. J. (1983). A time series analysis of aggregate merger activity. *Review of Economics and Statistics, 65*, 423–430.

Moeller, S. B., Schlingemann, F. P., & Stulz, R. M. (2003). *Do shareholders of acquiring firms gain from acquisitions?* National Bureau of economic research, working paper 9523.

Mossin, J. (1973). *The theory of financial markets*. Englewood Cliffs, NJ: Prentice-Hall.

Mueller, D. C. (1969). A theory of conglomerate mergers. *Quarterly Journal of Economics, 83*, 743–759.

Nelson, R. L. (1959). *Merger movements in American industry 1895–1954*. Princeton: Princeton University Press.

Nelson, R. L. (1966). Business cycle factors in the choice between internal and external growth. In W. W. Alberts & J. E. Segall (Eds.), *The corporate merger*. Chicago: The University of Chicago Press.

Rappaport, A. (1979, July/August). Strategic analysis for more profitable acquisitions. *Harvard Business Review, 57*, 99–110.

Rappaport, A. (1981, May/June). Selecting strategies that create stockholder value. *Harvard Business Review, 59*, 139–149.

Rappaport, A., & Sirower, M. (1999, November–December). Stock or cash? *Harvard Business Review, 77*, 147–158.

Ravencraft, D. J., & Scherer, F. M. (1989). The profitability of mergers. *International Journal of Industrial Organization, 7*, 101–116.

Ruback, R. S. (1982). The Conoco takeover and stockholder returns. *Sloan Management Review, 23*, 13–34.

Ruback, R. S. (1993). Do target stockholders lose in unsuccessful control contests? In A. J. Auerback (Ed.), *Corporate takeovers: Causes and consequences*. Chicago: The University of Chicago Press.

Shugert, W. F., & Tollison, R. D. (1984). The random character of merger activity. *Rand Journal of Economics, 12*, 500–509.

Stigler, G. J. (1950). Monopoly and oligopoly by merger. *American Economic Review, 40*, 23–34.

Trimbath, S. (2002). *Mergers and efficiency: changes across time*. Boston: Kluwer Academic Publishers.

Weston, J. F. (1953). *The role of mergers in the growth of large firms*. Berkeley: University of California Press.

Weston, J. F., Mitchell, M. L., & Mulhern, J. H. (2004). *Takeovers, restructuring, and corporate governance* (4th ed.). New York: Pearson/Prentice-Hall.

White, L. J. (2002). Trends in aggregate concentration in the United States. *Journal of Economic Perspectives, 16*, 137–160.

Corporate Restructing: Leveraged Buyouts, Liquidations, Failures, and Bankruptcies



Not every company justifies the confidence of its original investors placed in it. A free enterprise system is one of profit and loss. There is no guarantee that all capital will earn the "normal" rate of return. In a world of change, where sure knowledge of the future is lacking and decisions are made under conditions of more or less uncertainty, the operation of any business is a calculated risk. The data in Fig. 1 depicts the failure rate per 10,000 firms in the United States during the 1930–1998 period.[1] see Schwartz (1962) and Guerard and Schwartz (2007) for failure rate figures. The reader obviously notes the high level of business failures during the Great Depression. The reader may be surprised by the high level of failures during the 1980s, given the excellent stock market performance. Failures have risen substantially during the past 40 years and were very high during the 1990–1992 period. As noted in chapter "Financing Current Operations and Efficiency Ratio Analysis," there was a deterioration in the Altman Z score, the bankruptcy prediction model, during the 1963–2003 period as a result of falling profit margins and sales efficiency. The large debt issuance of the 1975–2003 period, combined with high-interest rates

[1]The Dun & Bradstreet (D&B) data is used in this chapter. However, the D&B data is not included in the economic report of the President after 1998. A second source of data is the American Bankruptcy Institute, ABI, which carries a 1980–2002 series on its website. The business bankruptcy filings on the ABI database for the 1998–2002 period are:

Year	Business filings
1998	44,367
1999	37,884
2000	35,472
2001	40,099
2002	38,540

The decline in filings may be surprising given the stock market performance of the period. Source: www.abiworld.org.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_19

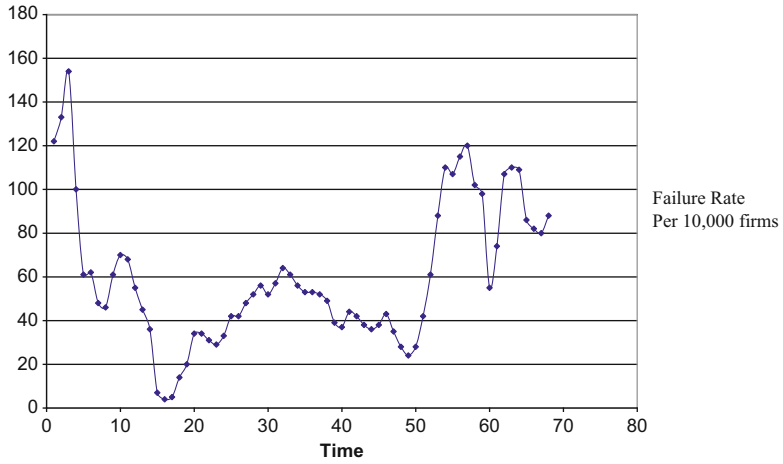


Fig. 1 US failure rate, 1920–1998

during several years of the period, was no doubt an important factor in the rising failure rate. Buell and Schwartz (1981) modeled the failure rate in the United States during the 1950–1978 period as a function of the total debt to total assets ratio and the variation about the time trend of the level of employment. Buell and Schwartz reported the increases in the debt ratio were (highly) statistically associated with the failure rate, whereas the failure rate is negatively associated with employment. An economic downturn is associated with decreasing cash flows and lower operating cash flows of firms. Under an increase in financial leverage, a favorable turn in economic events can bring a firm high earnings; however, on the other hand, a recession in demand or a miscalculation of costs may entail substantial losses.

In this chapter, we address several topics in corporate restructing: leveraged buyouts, liquitutions, failures, and bankruptcies.

In the previous chapter, we discussed traditional mergers. In this chapter, we introduce the reader to leveraged buyouts (LBOs). Leveraged buyouts are primarily financed by debt, much of which is "junk," or debt rated below investment grade. After the LBO, the firm goes private and is no longer publicly traded. Its investors are often known generally as private equity. The target company's assets usually act as collateral for raising the loan. Once the deal is done, the future cash flow from the newly acquired company helps to repay the debt. In some LBOs, the debt is repaid by selling off some assets of the acquired firm. Such deals work out profitable in cases where the return on equity exceeds the cost of debt (Table 1).

Texas power company TXU Corp., TXU, was acquired by a group led by private equity firms Kohlberg Kravis Roberts & Co. (KKR) and Texas Pacific Group (TPG) for $31.8 billion in the largest leveraged buyout in history. TXU is now Energy Future Holdings Corp. The investor group paid $69.25 per share (2/26/2007) for TXU, a 15.4% premium over TXU's closing stock price of $60.02 on Friday. TXU was the second-most active issue on the NYSE that day. The buyout firms paid 8.5 times EBITDA (earnings before interest, tax, depreciation, and amortization) for

Table 1 Largest leveraged buyouts in US history

Firm boughtout	Year of buyout	Price paid	LBO firm
TXU Energy	2007	$32.1 billion	Kohlberg Kravis Roberts & Co. (KKR) and the Texas Pacific Group
First Data	2007	25.7	Kohlberg Kravis Roberts & Co.
Alltel	2007	25.1	Atlantis Holdings
RJR Nabisco	1988	25.1	Kohlberg Kravis Roberts & Co.
Equity Office Properties	2006	24.7	The Blackstone Group
HJ Heinz	2013	23.5	Berkshire Hathaway and 3G Capital Partners
Heathrow Airport Holdings	2006	21.8	Developers
DELL	2013	21.5	Michael Dell and Silver Lake Management
Hospital Corporation of America Healthcare	2006	21.2	Kohlberg Kravis Roberts & Co. and Merrill Lynch Private Equity
Hilton Worldwide	2007	20.2	The Blackstone Group
Alliance Boots	2014	19.6	A. B. Acquisitions

Source: www.cnbc.com

TXU, compared with an average for the utilities industry of 7.9 times EBITDA, TXU said. Additionally, TXU reported that the private equity group included GS Capital Partners, Lehman Brothers Holdings Inc., Citigroup, and Morgan Stanley. The TXU buyout shattered the LBO record of $25.1 KKR paid for RJR Nabisco.

Those bullish fundamentals have also boosted the value of power plants, many of which were nearly worthless in the severe market downturn of 2001–2002 when dozens of new plants came online. Power generators drew renewed takeover interest as electricity demand in many parts of the country is expected to outstrip generation capacity in the coming years, pushing power prices higher. The value of power plants, many of which were nearly worthless in the severe market downturn of 2001–2002.

Several large and more famous leveraged buyouts are:

RJR Nabisco

RJR Nabisco's buyout is the largest, most controversial, hostile and aggressive takeover in the history of the private equity industry. RJR Nabisco Inc, a tobacco and food giant, was bought for $25 billion by Kohlberg Kravis Roberts & Co in 1988 after a stiff takeover fight. The estimated enterprise value of $31 in 1988 is now estimated to be $65 billion.

Alltel Corp

Alltel was handpicked by Goldman Sachs' private equity wing and Texas Pacific Group (TPG Capital) in 2007 for about $27.5 billion. The leveraged buyout of Alltel,

the fifth-largest wireless phone carrier then, was the largest buyout in the US telecommunication space. Alltel was rated among the best-run companies and was thus seen as an attractive target. Goldman Sachs and Texas Pacific Group did not keep Alltel for long, selling it to Verizon Wireless, which was a joint venture between Vodafone and Verizon Communications.

HCA Healthcare Inc.

Hospital Corporation of America, founded in 1968, was acquired by HCA founder Dr. Thomas F. Frist Jr., Kohlberg Kravis Roberts & Co, Bain Capital, and Merrill Lynch Global Private Equity. Hospital Corporation of America, simply known as HCA Healthcare Inc, went later public and is now traded on the NYSE.

First Data Corp

The leveraged buyout of First Data Corp in 2007 by Kohlberg Kravis Roberts & Co is still one of the largest private equity technology deals. The deal was carried out with a provision attached to it, known as "go-shop period" that allowed a company to solicit other proposals for a period of 50 days. The deal valued at $29 billion included costs of restricted shares, stock options, and debt. First Data Corp is a leading electronic transition processing firm. In 2014, Kohlberg Kravis Roberts & Co helped the payment processing company to reduce its debt burden through a $3.5 billion private placement.

Equity Office Properties

The buyout of Equity Office Properties Trust was won by Blackstone Group LP (NYSE: BX) after a bidding contest with Vornado Realty Trust. The contest forced Blackstone to up its offer from $48.50 per share to $55.5 per share. Equity Office Properties Trust had accepted Blackstone's initial offer of $48.5 per share, before Vornado proposed its $52 per share offer, triggering a bidding war between them. The total transaction cost of the deal worked out to be $39 billion, placing it among the largest leveraged buyouts (https://www.investopedia.com/stock-analysis/020915/most-famous-leveraged-buyouts-kkr-cg-bx-vno-apo.aspx).

Bankruptcies fell in the United States during the 1980–2019 time period as the US economy soared. Bankruptcies spiked in 1987, with the "Crash" and again in 2008–2009, the "Global Financial Crisis" (GFC). Overall, bankruptcies fell during the 1980–2019 time period (see Fig. 2).

The Bankruptcy Reform Act of 1978 provides the formal procedures for the resolution of claims against the firm. The bankruptcy act provides for liquidation (chapter "Capital and New Issue Markets") or reorganization (chapter "Investing in Assets: Theory of Investment Decision-Making"). A bankruptcy filing may be voluntary or involuntary.

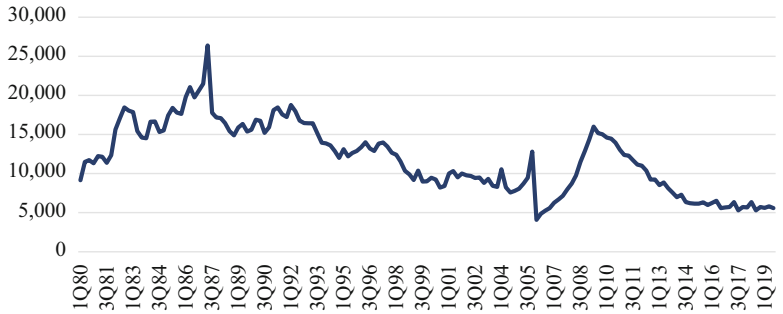


Fig. 2 Quarterly business bankruptcies filings

Voluntary Liquidation

Liquidation is the sale of the operating assets of a firm and their conversion into cash or other liquid assets. In most cases, the cash is returned to the investors according to their legal priorities. The trustee appointed by the court shuts down the firm and sells its assets. Assets are distributed by the absolute priority rule (APR), where the court establishes the seniority, or hierarchy, of claims. No junior claims are paid until senior claims are fully paid.

Liquidations may be voluntary or involuntary. If it cannot properly meet its obligations, an involuntary liquidation may be forced upon the company by its creditors. Involuntary liquidation is one of the remedies in bankruptcy, whereas a voluntary liquidation is undertaken as a decision of the management passed upon by the shareholders. Whether voluntary or involuntary, liquidation is recommended whenever the capitalized value of the reasonably projected cash flow of the firm, i.e., its value as a "going concern," is below the appraised liquidation value of the company's assets.[2] In the case of voluntary liquidation, funds may be obtained from the sale of the assets that could be used to buy market stock or bonds (in which case the liquidated firm will turn into an investment company[3]), or an entirely different type of business may be purchased. A partial liquidation means that some portion of the claims against the company are paid off and the other investors remain with the scaled-down firm.

The level of the cash flow of the firm determines the decision to liquidate. A firm may show accounting losses, and yet the best course of action may be to continue

[2]Sometimes a firm's shares sell on the market consistently below their liquidation value. This indicates that the net liquidating asset value of the firm is greater than its value as a going concern. Of course, the inside holders or the management may strongly resist any proposal to liquidate.

[3]For example, the, Adams Express Company and Berkshire Hathaway are companies which have been turned from operating firms into investment companies.

operations. On the other hand, it may be eminently reasonable to withdraw funds from a firm that shows no explicit accounting loss.

A Liquidation Example

Table 2 shows an example of a firm that might consider dissolution even though its accounting profits are positive.

1. Appraised liquidation value for the sale of the total assets = $13,000,000.
2. 2. Going concern value of the firm on a cash flow basis assuming a necessary overall rate of return of 10% and a time horizon of 25 years, $1,000,000 $a_{25\ |\ 10\%}$ = $9,077,000.
3. Rate of return on the liquid value of the equity on a net basis $\frac{300,000}{6,000,000} = 5.0\%$.

If the stockholders should decide to dissolve the firm, they would receive $6,000,000 (after paying off the creditors) for the $9,000,000 book value of their investment. On the other hand, the gross going concern value of their firm is only $9,077,000, leaving the shareholders with only $2,077,000 as the net ongoing value of their equity.

Viewing the situation from the net basis, let us assume that the going rate of earnings on equity capital for Company A's type of risk and potential return is 10%; then the $6,000,000 obtained from liquidation could be invested to bring in at least $600,000 net instead of the present $300,000 net. Basing the decision on the overall approach (i.e., on the use of EBID) is preferable because it is always possible that an inadequate return on the equity is due to a suboptimal capital structure or an especially high allowance for depreciation. One should emphasize that the book value of the assets (or equity) really is irrelevant to the decision except for the book

Table 2 Company A

Condensed operating statement			
	EBITD	$1,000,000	
	Less depreciation	200,000	
	EBIT	800,000	
	Interest	200,000	
	Earnings before taxes	600,000	
	Less taxes	300,000	
	Income available for common	300,000	
Position statement			
Assets		**Liabilities**	
Current assets	$11,000,000	Current liabilities	3,000,000
Fixed assets	5,000,000	Fixed liabilities	4,000,000
	_____	Common stock equity	9,000,000
	16,000,000		16,000,000

value of certain assets that may give a rough indication of their liquidation value. The book value of fixed assets may diverge widely from liquidation, but the liquidation value and the book value of current assets may run fairly close. The argument for liquidation, in this case, is the assumption that there is no strong reason for forecasting any appreciable improvement in the earnings of the firm for 25 years.

Remaining in Business

Table 3 illustrates the position of a firm that might as well stay in business even though it shows an accounting deficit of $100,000 annually. It can realize only $5,500,000 from the sale of its assets leaving only $500,000 for the equity holders after paying the creditors. On the other hand, it has a gross going concern value of $7,600,000 leaving a net value of $2,600,000 for the equity after allowance is made for liabilities. It may be difficult to understand how this firm can have any value when the operating statement shows a net loss. However, after considering noncash depreciation charges, Company B presently returns $800,000 cash flows annually in available net funds after the payment of interest.

1. Appraised liquidation value for sale of total assets = $5,500,000.
2. Going concern value of the firm on a cash flow basis, assuming a necessary overall rate of return of 10% and a 15-year projection basis; 1,000,000 $a_{15|10\%}$ = $7,600,000.

These funds could be reinvested as risk capital to bring a return of perhaps 10% ($80,000 additional per year) or used to repay the long-term debt, reducing the

Table 3 Company B

Condensed operating statement			
	EBITD	$1,000,000	
	Less depreciation	900,000	
	EBIT	100,000	
	Interest	200,000	
		(100,000)	
	Profit taxes	-0-	
	Deficit on the common equity	(100,000)	
Position statement			
Assets		**Liabilities**	
Current assets	$3,000,000	Current liabilities	1,000,000
Fixed assets	13,000,000	Fixed liabilities	4,000,000
		Common equity	11,000,000
	16,000,000		16,000,000

principal and interest costs. It follows that, in a comparatively brief period, the accounting statements would show a profit.[4]

In addition to the return on the investment of the net new funds of $800,000, the original investment in the firm will, under our assumptions, continue to produce a gross cash flow of $1,000,000 annually for at least 15 more years. The owners of the firm will retrieve more of their investment if they continue to operate than if they liquidate.

To sum up, the decision to liquidate should not be based on the accounting net profits or losses, but on whether the company earns an "adequate" rate of return on the "withdrawable funds" within the firm. A firm can show an accounting profit because all the implicit costs of capital are not recognized. If the firm were explicitly charged for all its funds (including equity) at the going rate, it would show a loss. It might be preferable to liquidate such an enterprise. On the other hand, a company may show an accounting loss because in the past it has misdirected some of its investment. However, the firm may show a good return on the net withdrawable funds. Neither one of these enterprises can be considered "successful," but in the second case, a smaller net loss results if operations continue.

Failure

Failure is the inability of a firm to meet its obligations as they come due.[5] Failure may result from physical losses or disasters, robbery, or embezzlement; usually, however, it occurs because of a more or less prolonged period of operating losses, or because funds have been invested, at too heavy a rate, in assets which are not efficient or easily marketable.

The two main reasons for failure are often given as the lack of adequate working capital, or poor management. But inadequate working capital may be the proximate cause of a firm's failure and not the basic cause. Every business that fails has inadequate working capital at the time of failure. A firm that shows earnings or potential earnings should be able to keep or acquire working capital. Perhaps inadequate working capital is a cause of failure in new small businesses when the organizers fail to provide enough working capital to survive a period of initial losses. Even here, we may doubt whether more working capital would have insured success or merely prolonged the period of duration.

Poor management may be a basic cause for the firm's financial disaster. Mismanagement shows itself in many ways: in a failure to control costs, in the inability to foresee and provide funds for obvious contingencies, or in wasting the firm's resources in unproductive programs. The ability of the management is a

[4]Furthermore, under the present tax loss carryforward provision, Company B would probably not have to pay any profits tax for some time.

[5]Failure, of course, should not be confused with an occasional slow payment on a bill or statement.

qualitative matter; it cannot be adduced from a single mishap, but the management may be suspect if the firm seems subject to a succession of economic errors.

The most important reason for failure is simply bad business conditions. Bad business conditions may be general or centered in a particular industry, or in a particular geographic area. High-interest rates and high unemployment are often associated with bad economic conditions. One could alternatively examine the relationship between failures and the composite index of leading economic indicators discussed and analyzed in chapters "Regression Analysis and Estimating Regression Models" and "Time Series Modeling and the Forecasting Effectiveness of the US Leading Economic Indicators." If demand falls off sharply, even competent management may be unable to keep its company afloat. Of course, some will argue that really good management would have foreseen the economic storm far enough in advance to have a program for saving the firm. But if a firm attempts to cover all risks, it is likely to earn little or no profits. Moreover, the management of a company whose assets are specialized for the requirements of a particular industry may not be able to do much if there is a fall in demand.

Informal Remedies

The failure of a firm puts its future into the hands of its creditors. Since the firm cannot meet its obligations, i.e., cannot pay its debts, the law removes the major control from the owners and gives it to creditors so that they may salvage whatever they can to reduce their losses. The creditors (liability holders) can press their claims against the firm through various devices of differing degrees of legal formality. The debt holders may, without going through the courts, form a creditors' committee to guide and supervise the management of the company through a period of hard times and, by pressing for operational economies, holding sales, collecting accounts, and minimizing purchases, attempt to bring the firm back to a current basis.

Other procedures for dealing with failure, without necessary recourse to the courts, are composition and quasi-reorganization. The firm may be wholly or partially liquidated by a composition of creditors. After the creditors are paid a proportion of their claims on a pro rata basis out of the surrendered assets of the company, a composition relieves the owners of any further liability. In some cases, various sorts of quasi-reorganization may be attempted. For example, the creditors may all take an agreed-upon cut in their claims if the stockholders "put additional funds into the businesses"; if the plan goes through, the stockholders may continue to run the company.

The device likely to be used in more complicated cases (or where perhaps there is distrust among the creditors or between the creditors and the owners) is bankruptcy. Bankruptcy places the affairs of the company under the custody and protection of the courts until the disposition of the various claims can be decided.

No matter what legal device the creditors use to handle the situation of a failed company, there are only two basic functional remedies, reorganization or liquidation. In the next section on bankruptcy, remedies are discussed in more detail.

Bankruptcy

Bankruptcy is a process by which a financially distressed individual or firm can purge itself of debt by surrendering all its assets to the court. These assets are used, as far as they extend, to satisfy the creditors' claims. Once an individual passes through bankruptcy, he can start his economic life anew. He can acquire new holdings, and he need not pay his old debts from these new holdings unless he so volunteers. Bankruptcy is a humane innovation in the law when one considers that it was not so long ago that persons with unpaid debts might be plunged into debtor's prison. The concept of a legal release from hopeless debt was so important to our American forefathers that the right to establish uniform bankruptcy laws was given to Congress in Article I, Section 8, of the Constitution.

The WorldCom Case Looking at current developments, the largest bankruptcies of the 1980–2002 period are shown in Table 4. The largest bankruptcy was that of WorldCom in 2002. WorldCom, Inc. declared bankruptcy in July 2002. Using 2001 data to calculate the Altman Z, as was done in chapter "Financing Current Operations and Efficiency Ratio Analysis" for DuPont and IBM, one finds a value of 1.30 for WorldCom in 2001. The WorldCom value was less than the critical level of 2.000 and indicated potential bankruptcy problems. There can be several criticisms of the Altman Z calculation of this period. First, the end-of-year financials may not have been known much before May, and there may have been little time for persons to calculate the Altman Z score. Second, if one used data of the year 2000 to calculate the Altman Z, one obtains a 1.47 value, less than its critical score, but not a terrible value, such as we saw for Lucent in chapter "Financing Current Operations and Efficiency Ratio Analysis." A third problem is that WorldCom is a telecommunications firm, a utility firm, and as was shown in chapter "Financing Current Operations and Efficiency Ratio Analysis," utility firms had lower Altman Z scores than manufacturing or industrial firms. In fact, the WorldCom Altman Z score exceeded the utility median (1.18) in 2001.

There were several large bankruptcies in 2019, particularly PG&E, the California public utility associated with enormous wildfires in the Los Angeles area (see Table 5).

Table 4 The largest bankruptcies, 1980–2005

Company	Bankruptcy date	Total assets pre-bankruptcy
WorldCom, Inc.	7/21/02	$103,914,000,000
Enron Corporation[a]	12/2/01	63,392,000,000
Conseco, Inc.	12/18/02	61,392,000,000
Texaco, Inc.	4/12/87	35,892,000,000
Financial Corp. of America	9/9/88	33,864,000,000
Global Crossing Ltd.	1/28/02	30,185,000,000
UAL Corp.	12/9/02	25,197,000,000
Adelphia Communications	6/25/02	21,499,000,000
Pacific Gas and Electric Co.	4/6/01	21,470,000,000
MCorp	3/31/89	20,228,000,000
Mirant Corporation	7/14/03	19,415,000,000
First Executive Corp.	5/13/91	15,193,000,000
Gibraltar Financial Corp.	2/8/90	15,011,000,000
Kmart Corporation	1/22/02	14,600,000,000
FINOVA Group, Inc. (The)	3/7/01	14,050,000,000
Homefed Corp.	10/22/92	13,885,000,000
Southeast Banking Corp.	9/20/91	13,390,000,000
NTL, Inc.	5/8/02	13,003,000,000
Reliance Group Holdings, Inc.	6/12/01	12,598,000,000
Imperial Corp. of America	2/28/90	12,263,000,000
Federal-Mogul Corp.	10/1/01	10,150,000,000
First City Bancorp. of Texas	10/31/92	9,943,000,000
First Capital Holdings	5/30/91	9,675,000,000
Baldwin-United	9/26/83	9,383,000,000

Source: http://www.bankruptcydata.com/Research/15_Largest.htm
[a]The Enron assets were taken from the 10-Q filed on 11/19/02

Table 5 Largest bankruptcies in the U.S., 2006–2019

Firm	Date filed	Assets at filing
Lehman Brothers Holdings, Inc.	9/15/08	$691,063,000,000
Washington Mutual	9/26/08	327,913,000,000
General Motors	6/1/09	82,290,000,000
PG&E	1/29/19	71,385,000,000
CIT Group	1/11/09	71,000,000,000
MF Global	8/11/11	41,000,000,000
Chrysler	4/30/09	39,300,000,000
Thornburg Mortgage	1/5/09	36,521,000,000
IndyMac Bancorp	7/31/08	32,734,000,000
General Growth Properties	4/16/09	29,557,000,000
Lyondell Chemical	6/1/09	27,392,000,000

Source: www.bankruptcy.com

Bankruptcy Procedures

Bankruptcy can be either voluntary or involuntary. Voluntary bankruptcy takes place when the owners, seeing the hopelessness of their position, petition for bankruptcy and place the firm's assets into the hands of the court. Involuntary bankruptcy may take place if creditors believe that the situation is deteriorating, that the owners may be attempting to withdraw funds to the eventual detriment of the creditors, or that the owners may be preferring (i.e., favoring) some creditors over the others, they may "throw the firm into bankruptcy." The acts of bankruptcy consist of a list of actions, similar to the ones described above, which legally entitle the creditors to institute bankruptcy proceedings. The purpose of bankruptcy, in this case, is to conserve the firm's resources, such as they may be, for the benefit of all the creditors according to the legal strength of their claims.

Regardless of how the firm came into the hands of the law, the court's first task is to appoint an official entitled a "trustee in bankruptcy." The trustee runs the company until the final decision is made as to how best to dispose of the firm's affairs with the least loss to the creditors. The two major alternatives are liquidation or reorganization.

Just as in the case of voluntary liquidation, the rational economic basis for the choice rests on whether or not the going concern value of the projected funds inflow of the firm exceeds its liquidation value. However, where the firm (e.g., a railroad) is adjudged to perform a necessary public service, the remedy of liquidation may not be allowed; the only reorganization may legally permissible. The creation of Amtrak is an example of a public enterprise created by the failures of many private railroads.

Priorities in Liquidation

If liquidation is the chosen course of action, the claims against the company (the liabilities) must be ranked in order of their legal strength, or priority. First comes the so-called preferred claims, established by law; these consist of such items as all accrued taxes, back salaries, and the actual expenses of the receiver (especially the lawyer) and the bankruptcy proceedings. Second in priority come the holders of liabilities with liens (or mortgages) against specific assets. Whatever funds these assets bring in are applied first to these "sheltered" claims; afterward any unsatisfied portion of the sheltered claim shares equally with the rest of the general creditors. Thus, having a specific lien or pledge or mortgage does not always protect the holder from loss. The protected claims, however, usually fare better than the general creditors, and they can never do worse. After the various protected claims come, the general creditors who share pro rata in the rest of the funds made available by liquidation. Following the debt, holders come the preferred stockholders, if any. If, as rarely happens, anything is left over, it goes to the common stockholders.

Table 6 illustrates the settlement of liabilities (after the preference claims have been paid) under different circumstances. Following the rules of allocation, it indicates how the final settlement for each class of liabilities was calculated. Note that the book value of the assets as such had no bearing on the settlements. The

Table 6 Settling claims in liquidation under varying circumstances

Case 1					
Book value of assets			**Liquidation proceeds**		
Fixed assets	$1,000,000		$424,000		
Current assets	2,000,000		1,500,000		
Total	$3,000,000		$1,924,000		
Liabilities	**Principal**	**Accrued interest or dividends**	**Total claims presented**	**Settlement**	**Percent of claim**
First mortgage bonds (on fixed assets)	$300,000	$6000	$306,000	$306,000	100.0%
Second mortgage bonds (on fixed assets)	300,000	18,000	318,000	268,000	84.3
General creditors	1,800,000	0	1,800,000	1,350,000	75.0
Preferred stock	500,000	100,000	600,000	0	0.0
Common stock (book value)	1,000,000	0	1,000,000	0	0.0
			$4,024,000	$1,924,000	

	Case 2 Liquidation value of assets		
	Fixed assets	$1,500,000	
	Current assets	3,000,000	
	Available funds	$4,500,000	
Liabilities	**Claims presented**	**Settlement**	**Percent of claim**
First mortgage bonds (on fixed assets)	$3,000,000	$2,250,000	75.0%
Second mortgage bonds (on fixed assets)	500,000	250,000	50.0
General creditors	4,000,000	2,000,000	50.0
Common stock	5,000,000	0	0
	$12,500,000	$4,500,000	

	Case 3 Liquidation value of assets		
	Fixed assets	$2,000,000	
	Current assets	5,000,000	
		$7,000,000	
Liabilities	**Claims presented**	**Settlement**	**Percent of claim**
First mortgage bonds (on fixed assets)	$3,000,000	$3,000,000	100.0%
General creditors	3,500,000	3,500,000	100.0
Preferred stock (1,000,000 + 100,000 accumulated dividends)	1,100,000	500,000	45.5
Common stock	8,000,000	0	0.0
		$7,000,000	

accrued interest takes on the same legal priority as the principal of the claim to which it is attached. Liquidations follow the doctrine of absolute priority; i.e., no junior claim is allocated any funds until all the claims senior to it have had their legal due.[6]

Reorganization

If the reasonably capitalized value of the firm's cash flow is greater than the liquidation value, reorganization is the preferable remedy. The purpose of a reorganization is to trim down the existing claims so that the projected cash flow of the bankrupt firm can support its new financial structure. Sometimes, after passing through a reorganization, a firm's operations have taken a favorable turn, and it has moved on to economic success.

The reorganization plan is hammered out by the representatives of all the classes of claimants with the guidance of the SEC or the ICC. It must be passed by a vote of two-thirds of all classes of claims and approved by the court. When all this is accomplished, the reorganized firm is discharged from bankruptcy. Because of the legal intricacies involved, sometimes it may take years for a firm to be discharged from bankruptcy. Improving economic conditions may encourage a junior class of security holders to delay the final reorganization in hopes that the eventual settlement will be more favorable to them. In the meantime, the trustees run the firm, collecting revenues, paying currently incurred obligations, and perhaps servicing old debt that has a strong legal and financial position.

Bankruptcy implies losses for someone. How these losses are to be apportioned is the problem. The doctrine of absolute priority (APR) is followed in liquidation. This doctrine is also the basic rule, reiterated by the Supreme Court,[7] in reorganization. In other words, no value is to be passed on to any junior claim if any class of superior claims has suffered deterioration in the exchange of securities proposed in the reorganization.

In a recent study of 300 cases from the Arizona and New York federal bankruptcy courts from 1995 to 2001, Bris, Welch, and Zhu (2006) found that Chapter 7 liquidations offered few advantages relative to Chapter 11 reorganizations. The Chapter 7 liquidations took almost as long a time to resolve, cost almost as much in fees, and provided creditors with lower recovery rates than Chapter 11 liquidations. Bris, Welch, and Zhu found Chapter 7 firms had lower pre-bankruptcy assets than the corresponding Chapter 11 cases; they had less equity owned by managers and more unsecured debt than Chapter 11 firms. The mean number of days in

[6] A lawyer friend of one of the authors told us that in the world of business, it is not uncommon for a subordinated debt holder to threaten to hold up settlement. In such cases, it may be in the best interests to make a partial payment to the junior debtholders (claims) and allow settlement to proceed.

[7] Case v. Los Angeles Lumber Products Co., 308 U.S. 106 (1939)

bankruptcy exceeded 70 days for both Chapters 7 and 11 firms, with larger bankruptcies taking longer to resolve, and bankruptcy costs (expenses divided by pre-bankruptcy assets) were approximately 8% for Chapter 7 cases and 17% for Chapter 11 cases. The greater the number of creditors, then the lower is the unsecured recovery rate. Bris, Welch, and Zhu also found that APR tends to be violated when there are fewer secured creditors when secured creditors own a larger fraction of total debt and the larger the debt.

A rival to the doctrine of absolute priority is that of relative priority. A reorganization following the lines of relative priority would be considered fair as long as the losses apportioned to each claim were smaller for those of higher senior positions. The first mortgage bonds might accept an adjustment in their claims representing an economic loss of 10%, the second mortgage bonds 30%, the general creditors 50%, and the equity holders 90%. (Under absolute priority, the claims of the first mortgage bonds would have been satisfied in full, and the stockholders would have received nothing as long as the general creditors suffered any loss. Whether or not the second mortgage holders would suffer any loss would depend on the estimate of the value of their specific lien.)

The theory of "joint venturers" has been promulgated in support of the doctrine of relative priority. The notion is that all the investors in a business are together in a risky enterprise. If the firm founders lose more than others, then they should make sacrifices to get the company afloat again. If the initial venture is successful, the debt holders will receive a limited return, and all the extra rewards go to the stockholders. The advocacy of relative priority seems to rest on a certain sympathy to the shareholders—possibly deriving out of some image of the poor farm girl and the rich mortgage holder. In truth, in the modern corporation generally, the shareholders are risk-takers, "men-of-the-world," and the bondholders are mostly pension funds, insurance companies, or trustees, representing basically the holdings of the aged or funds designed for the support of widows and orphans.

An obvious objection to relative priority is that there are an infinite number of solutions within its framework. It becomes quite difficult for any court to decide whether the proposed plan calls for an excessive sacrifice on the part of the senior creditors.

Although absolute priority is, no doubt rightly, the basic rule of law, a small amount of relative priority tends to creep into most approved reorganizations. If a firm has a complicated financial structure, the protective liens may be quite entangled; for example, a railroad bond which has a first mortgage on one section of the track might have a second mortgage on another part of the roadbed and possibly even a third lien over another part of the road. It may be easier in such cases to make some compromises of conflicting claims than to try to make a sure determination of which has the highest priority. Another difficulty arises out of the politics of the reorganization proceeding itself; the junior security holders are often in a strategic position to delay the approval of the plan if they do not obtain some concessions. Furthermore, the trustees in bankruptcy are often recruited from the ranks of the old management, and their latent sympathies may lie with the stockholders. Last, reorganizations often take place during the lows of the business cycle. The courts during such a period may fear that the projected cash flows may be set at

too pessimistic a level. Rather than cut off any claim completely, they prefer to give the junior claimants contingent values, such as warrants which can be exercised profitably if the company makes any recovery of earning power and its new shares rise on the market.[8]

Through the issue of warrants, junior claimants might recoup part of their losses. It should be acknowledged, however, that the existence of the warrants constitutes a drag on the potential appreciation of the shares that may have been given out during the reorganization as part settlement of senior claims. The warrants, then, delay the rate at which the higher priority claimants recover their losses.

The following analysis, found in Table 7, illustrates the problems involved in setting up a reorganization plan. This plan could possibly be rejected by the court; it contains some amount of relative priority.

Financial structure before reorganization[a]

Ranking claims	Principal	Accrued interest
First mortgage bonds (30 years partial sinking fund 1% annually) 4%	$10,000,000	$1,000,000
Debentures 5's	16,000,000	2,000,000
Common stock (200,000's at $100 par value)	20,000,000	0

[a]All the general creditors representing small or preferred claims were settled by the trustee in bankruptcy

The proposed reorganization:

1. The senior bondholders to receive new $100 par value common stock for the accrued interest of $1,000,000, leaving the principal of the bonds intact (total distributed: $10,000,000 bonds and $1,000,000 common).

Table 7 Analysis of a hypothetical reorganization plan of the Hardnox Corporation

1. If existing economic conditions were not to improve significantly, the earnings of the Hardnox Corporation were conservatively expected to average about $1,100,000 per annum before depreciation (on the old base), interest, and taxes. Because of back losses and high depreciation, the company would not have to pay taxes on this section of the cash flow for 15 years. This level of cash flow on the existing plant should prevail for about 15 years with future returns running somewhat lower
2. The liquidation value of the Hardnox Corporation was appraised at about $4,000,000
3. If the cash flow of the firm is conservatively capitalized at 10% for 15 years, the going concern value of Hardnox is a minimum of $8,367,000 ($1,100,000 a $_{15/10\%}$) against a liquidation value of $4,000,000. The reorganization is therefore clearly advantageous

[8]Warrants are options to purchase shares at a fixed price. If a warrant carries an option price of $15, and the shares have a market value of $10, the warrant has little intrinsic value. If the shares should rise above $15, however, the warrants are worth at least the difference between $15 and the share price. Actually, because of their speculative leverage, warrants tend to sell somewhat above their intrinsic value.

2. The debenture holders to receive $500 worth of 4% pfd. and $625 par value of common representing the rest of the principal and accrued interest (total distributed: $8,000,000 pfd. and $10,000,000 common).
3. The common stockholders to receive 1 share of common for every 10 they now held (total distributed: $2,000,000 par value common).

Projected reorganization

Capital structure	
First mortgage bonds (unchanged) 4's	$10,000,000
Pfd. stock 4%	8,000,000
Common stock (130,000's)	13,000,000

Analysis of the Plan

Feasibility

Prior and fixed charges

	Before reorganization	After reorganization
Fixed (interest)	1,200,000	400,000
Contingent	0	320,000
Total	1,200,000	720,000
Sinking fund installment	100,000	100,000
Cash requirements	1,300,000	820,000

If the cash flow runs slightly over $1,000,000 a year, the prior charges on the new financial structure will be met readily. The margin for the repayment of the principal on the bonds in the early years is not great, but if the worst should occur, dividends may be passed on the preferred. Of course, as the bonds are gradually repaid, the bond interest charges drop. If economic conditions improve, some funds may be available to improve the value of the common.

Fairness

Projected operating statement

EBITD	1,100,000
(Estimated true depreciation)[a]	120,000
	980,000
Interest	400,000
	580,000
Preferred dividends	320,000
Earnings on common	260,000
Earnings per common share	$2.00

[a]For tax purposes, depreciation would be kept on the original cost base of the fixed assets. If the ready market value of the fixed plant were $1,800,000, the depreciation on the present investment on a straight-line basis would be $120,000 per annum

Analysis of the Reorganization

1. The maximum likely market value of the new common shares would be $20, representing a price/earnings ratio of 10.
2. The bondholders would get back most of their principal value in the form of their present contract. However, for their accrued interest amounting to $100 a bond, they would receive a share of common having a value at best of $20. Therefore, for each $1100 value of claim, the bondholders would get back securities worth $1020 or 92.7% of their claim.
3. The preferred stock given to the debenture holders has contingent claim against dividends rather than a fixed claim; moreover, it carries only a 4% dividend against the previous 5% on the debentures. It is dubious if the preferred has a value of over $60 per share. Thus, the debenture holders would receive securities valued at $300 worth of preferred and $125 worth of common for each $1125 of principal and accrued interest or a settlement of 37.8%.
4. For each $1000 par value of old common, the shareholders are to get one share of common having a maximum value of $20, or a nominal settlement of 2%. From an aggregate point of view, however, they receive securities having an approximate value of $400,000. This would make up part of the loss suffered by the bondholders or debenture holders.
5. Conclusion: the proposed reorganization plan has a considerable element of relative priority. It is a moot question whether it would pass the scrutiny of the courts.

In practice, reorganization procedures are likely to be even more complex than those shown for the hypothetical Hardnox Corporation. Each class of claimants appoints a protective committee and many proposals are argued back and forth. Nevertheless, the illustration of the Hardnox Corporation case demonstrates most of the essential steps of a corporate reorganization:

1. First, it is determined whether the positive cash of the firm makes reorganization worthwhile.
2. All the claims are assembled and ranked in order of their priority.
3. New securities are proposed to be exchanged for the old claims.
4. Many of the new securities will be given a contingent or variable claim (income bonds, preferred stock, or common stock) in place of the fixed return securities they replace. (A strongly secured senior security may be left entirely untouched.)
5. The object of the reorganization is to reduce the fixed charges and debt of the firm so that it can meet its remaining obligations out of its projected cash flow. (This constitutes the overriding test of "feasibility.")
6. The values remaining in the firm must be divided according to the priority of the claimants. This constitutes the test of legal fairness.

7. If the reorganization is successful, the firm should not fail again, at least in the near future.

Summary

From the viewpoint of the various investors, any reorganization is a necessary unpleasant financial surgery. It distributes losses among the classes of claimants. But the reorganization in itself did not cause loss; it is only a recognition of the economic deterioration that has already occurred. The object of a reorganization is to tailor the financial structure of the firm to fit the going concern value of the remaining earnings.

References

Bris, A., Welch, I., & Zhu, N. (2006). The costs of bankruptcy: Chapter 7 liquidation versus chapter 11 reorganization. *Journal of Finance, 61*, 1253–1303.

Brealey, R. A., Myers, S. C., & Allen, F. (2006). Principles of Corporate Finanace. New York: Mcgraw-Hill/Irwin. 8th edition.

Buell, S. G., & Schwartz, E. (1981). Increasing leverage, potential failure rates, and possible impacts on the macro-economy. *Oxford Economic Papers, 33*, 442–458.

Guerard, J. B. Jr. & Schwartz, E. (2007). *Quantitative Corporate Finance*. New York: Kluwer Academic Press.

Schwartz, E. (1962). *Corporate Finance*, New York: St. Martin's Press.

Van Home, J. C. (2000). Financial Management and Policy upper Saddle River, New Jersey: Prentice Hall.

Corporation Growth and Economic Growth and Stability



Economic growth is usually defined as the rise in total measurable economic output over a given period, i.e., such as in the growth in real GDP modeled in chapters "Regression Analysis and Estimating Regression Models" and "Time Series Modeling and the Forecasting Effectiveness of the US Leading Economic Indicators." This is a definition of gross growth. However, if the population increases significantly in the same period, output per capita may remain constant or even decline. The average citizen of the country is no better off than before, and thus in one very relevant sense, no economic growth has taken place.[1] A net concept of growth (closer to welfare criteria) can be developed by using the increase in average output per capita. Another problem of measuring growth arises if society shows a desire for an increase in leisure. If the average output does not decline while the workweek shortens and vacations lengthen, growth has taken place although its fruits have been absorbed as an increase in leisure. Growth is difficult to discover between the dips and rises of the business cycle. The national per capita output at a peak period may appear to have grown very rapidly compared to that of a preceding recession in business activity. Actually the potential output of the recession period, the output possible were the economy operating at capacity, might be relatively high. Growth should thus be measured from one equilibrium period to another or, if possible, measured in terms of increases in potential capacity product per capita. If a nation's economy falls significantly from its capacity, the problem is essentially one of economic stability and not of economic growth.[2]

[1]Countries such as India or Egypt have increased their total output in the twentieth century and yet have had little or no increase in per capita real income because of the rapid rise in population.

[2]One suggested index of growth is the average product per man hour. This allows for variations in employment and working hours and does a fair job of reflecting growth in potential output.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_20

Factors in Economic Growth

Even if precise measurement is not entirely possible, it can be agreed that economic growth takes place when the potential product of the economy rises sufficiently to allow an increase in the real income per capita. Two factors contribute to a rise in average real income[3]—real capital formation and an increase in the skills of a society, that is, an increase in human capital.

Capital formation is a net increase in the amount of tools, buildings, machinery, stocks of goods, soil drainage projects, port improvements, and other such facilities available to the community. The government may provide capital formation in the form of roads, dams, bridges, public buildings, schools, parks, and other similar items. Educational institutions and foundations may provide new capital in the form of classrooms, laboratories, and hospitals. The major contribution to capital formation by individuals is mostly in the form of house construction. In a modern economy, a large area of net capital formation derives from private businesses whose dominant form is the corporation. Whenever a company invests in new plant or equipment, increases the capacity or efficiency of old plants, builds up inventories, or in any way increases the productive potential of itself and the aggregate economy, it adds to the output capacity, the growth of the economy. Private capital formation or real investment is a dynamic element in our economic system. Most economists consider its behavior the major key to understanding the problems of economic stability and economic growth.

The other element making for economic growth—an increase in the skills of a society (human capital)—is less tangible than physical capital formation. An improvement in productive organization, in communicative ability, and in manual or physical dexterity and the development of new processes can add to the productive potential of a society, even if the total physical capital remains constant. Possibly an increase in the skills and techniques of the population could be subsumed into the general category of capital formation. Resources can be estimated in physical capital and/or in human capital. The director of an economic program in an underdeveloped nation may have to decide whether to use some of his funds to send more of his people away for education and training or whether to build a fish-freezing plant.[4] City fathers may have to decide between a higher school budget and improved storm sewers, and the finance officer of a corporation may have to choose between a new milling machine and a bigger research program.[5]

---

[3] An increase in leisure may be considered a rise in real income or product.

[4] Of course, education is more than a capital investment in human skills. Insofar as it pleasures the spirit, it is also a consumer good.

[5] Although the important role of the private corporation in the formation of capital is clear, less obvious is the part that industry plays in developing the skills of society. Not only are there formal training programs and research, there is a continual "learning on the job," and most importantly inculcating industrial discipline. The crucial significance for economic progress of an understanding and disciplined labor force is evident to economists who have worked in underdeveloped areas.

Savings and Real Investment

Economic growth is made possible only if there is an increase in usable equipment in the society and/or an improvement of the skills and techniques of the populace. If the economy is operating near its capacity, the resources necessary for capital formation or for training programs must come either out of the country's own current production or from the current production of another country in the form of a loan or bridging grant. From a global view, the quantity of goods and services available to mankind is relatively scanty, and the amount of product that can be diverted from current consumption to capital formation of all kinds is not as abundant as one might wish. Yet capital can be made available to society only through voluntary savings or through compelling a decreased production of consumer goods. Thus the classical economists, viewing humanity's material progress historically, did not err in emphasizing the importance of saving in the improvement of mankind's economic environment. The modern emphasis is, however, on the act of real investment itself. Capital formation may slacken, not from a lack of savings but from a lack of profit-making opportunities or dynamic push on the part of the sectors of the economy that make investment decisions.

In summary, nations must rely on increased capital and improved technology for a sustained increase in income. Sacrifice of total possible current consumption can help provide savings to build new plants and machinery. Research may increase its soil fertility, and countries can build dams to irrigate dry areas and provide power. Governments can build schools and train their people, providing the seedbed for a more advanced economy. If the growth of capital exceeds the growth of the population and the decline of natural resources, the economy may continue to return increases in per capita income to its members. If, however, the rate of population growth exceeds the rate of provision of new capital or if the rate of capital growth and inventory is insufficient to make up the loss of natural resources, per capita real income will decline.

A high rate of savings is not a sufficient condition to ensure capital growth. Especially in a rich economy, entrepreneurs and investors may be unwilling to engage in capital formation during periods when the demand for finished products is slack. Thus a growing rate of consumption and individual expenditures may be as necessary for the continuous formation of capital as the passive act of saving, which releases productive resources. The English economist, John Maynard Keynes, held that in normal times a rich economy is more than likely to provide enough savings (or nonconsumption) to enable capital growth to take place (Weintraub, 1978). The major problem may be to provide enough consumption expenditures and other incentives to ensure that sufficient active investment takes place to employ the amount that is saved.

Corporation Investment Spending and Economic Stability

Capital formation is necessary for sustaining economic progress. It is also needed to maintain full employment. According to national income theory, if the amount of investment undertaken is not large enough to offset the amount of money that would be saved at the full employment income level, output and income slide off, and the aggregate economy reaches equilibrium at something less than full employment of labor and other resources.[6] The fiscal and monetary authorities in the government may then initiate various policies to attempt to offset the inadequacy of investment spending, but it is clear that the level of business investment spending in a free enterprise economy is a crucial variable in determining the level of national income and employment.[7]

The capital markets play a most important role in mobilizing consumer savings and allocating them to investment uses. The sale of new securities on the primary capital market taps personal savings and funnels them into capital formation. Even though a good deal of corporation investments are internally financed by business savings (retained earnings and current depreciation charges), the new funds provided by individuals (either directly or indirectly through financial institutions) add a dynamic thrust to the economy. These funds add to the potential growth of existing companies and allow for the formation of new competitive companies. However, dealers and traders must operate fairly, and the management of corporations must be mindful of the interests of their security holders if the capital markets are to function properly. For if the ordinary investor doubts the fairness of the treatment he will receive if he puts his savings into corporation securities, the flow of savings into the capital markets will surely slow down, the cost of risk capital will rise, and there will be a brake on the rate of economic development.

In the United States, the corporation is a major catalyst in private capital formation. As shown in Table 1, in 2000 gross private investment in the United States totaled $2257.2 billion. Of this amount, $1459.8 billion or 64.7% was initiated by business enterprises. The business investment ratio continues to rise. The factors influencing the amount of business investment are complex. The volume and trend of consumer spending, the trend of prices, the costs and risks of obtaining capital, and the optimism or pessimism of entrepreneurs and managers who forecast future demand affect investment decisions.

The forecast of anticipated future returns has more effect in determining new investment than the volume of current profits. Sometimes it is assumed that if firms have high current earnings, they necessarily invest because they have the funds to do so. This assumption confuses the total amount of retained earnings with business

[6]The income theory given here is of course highly simplified.

[7]The responsibility of the government toward maintaining a reasonable level of employment in the economy is not clear-cut. However, passage of the Employment Act in 1946 and subsequent political developments indicate an implicit commitment to the use of stabilizing policies if the economy should veer too sharply from full employment.

Table 1 Business contributions to gross private capital exp. ($1996 billion)

	Real GDP (in billions)	Real gross private domestic investment	Business investment	Percent private business of total investment (%)	Percent private business investment/GDP (%)
1960	3275.8	313.8	68.8	21.9	2.1
1970	4936.6	576.0	131.3	22.8	2.7
1980	6837.6	933.1	429.2	46.0	6.3
1990	9358.3	1252.1	760.5	60.7	8.1
2000	12,924.2	2257.1	1459.8	64.7	11.3
2008	15,672.0	2583.3	1947.4	75.4	12.4
2010	15,415.2	2087.2	1581.4	75.8	10.3
2019Q3	19,121.1	3416.2	2852.4	83.5	14.9

Source: Federal Reserve Economic Data, https://fred.stlouisfed.org, https://fred.stlouisfed.org/help-faq, Economic Research Division, Federal Reserve Bank of St. Louis

investment spending. Retained earnings are more fruitfully regarded as business savings rather than investment. When funds generated internally are used to purchase new productive assets, the firm's savings are offset by capital spending. But if corporate managements generally are pessimistic as to future earnings, they may develop a high "liquidity preference." That is, they may keep a larger proportion of the firm's assets in cash, building up liquid holdings from their current funds flow. If increased spending in other sectors of the economy does not offset this development, the level of economic activity will decline.

On the other hand, if business pessimism can lead to a downturn in economic activity, an aggressive investment policy can contribute to inflation. When the aggregate demand for goods and services is strong, businessmen can create additional effective demand by borrowing funds or drawing down liquid balances and spending on equipment or inventory. This extra demand (financed by an increase in credit or an increase in the velocity of money circulation, or both) pushes the price level upward. However, seldom does this type of demand inflation become severe or dangerous. It can be curbed by effective monetary measures or eventually comes to a halt itself if the central monetary authorities create no net new bank reserves.[8] Runaway or hyperinflations seem to occur only when the fiscal responsibility or financial power of the central government breaks down and it can find no source of funds other than to print money or create monetary credit. At some point in this

[8]Two factors restrain this type of inflation: (1) As more bank credit is utilized, the interest rate rises. This rise in credit costs dampens the net profitability of many projects; (2) as the aggregate price level rises, the increase in the individual firm's dollar volume calls for an increased cash balance to carry the larger amount of sales safely. At this point, the firm's cash balance no longer seems excessive, and the firm is no longer willing to draw down its cash for other assets. When spending units in the aggregate find themselves with minimum cash "transaction" balances, no further increase in the velocity of the circulation of money is likely to take place.

process, the public loses confidence that the official money will ever have any scarcity value. The runaway inflation begins.

Monetary Policy, the Cost of Capital, and the Firm Investment Process

The firm's decision to invest in real assets is determined by the anticipated income possibilities on the one hand and on the other by the cost and availability of funds. What constitutes the cost of funds is a complex matter. Setting the base for the cost structure is the going level of the pure, nonrisk, interest rate. Rising from this are the rates that borrowers of increasing risk position have to pay. Finally, there is the cost of equity funds, influenced by the interest rate, by the risk premium, and, very heavily, by anticipations.[9] The economic feasibility of investment projects is determined by matching the prospective returns against the costs of the capital mix, including the allowances for risks and uncertainties.

Expansion projects are undertaken when their probable profitability exceeds the cost of capital; thus the aggregate amount of investment should increase if the cost of capital decreases. At any given level of anticipations, the volume of investment activity is an inverse function of the basic interest rate, assuming the level of the interest rate helps determine capital costs.[10] (The proportion of increased investment is determined by the elasticity of the investment function to changes in the interest rate.) The possible influence on the volume of investment spending is the rationale for monetary policy. The monetary authorities, by stimulating the money supply and moving the interest rate downward, can hope to stimulate investment and consumption spending. This should lead to a higher level of economic activity. Conversely, holding down monetary growth can lead to a rise in the interest rate, which can be used to dampen an inflationary rise in spending.

According to Keynesian economists if the economy's reaction to monetary policy is not strong enough, the tools of fiscal policy may be used. The government may intervene directly in the income flow of the economy by spending more than it takes in revenue during a recession in business activity or by running a surplus during an inflationary period (Weintraub, 1978). Anti-recessionary policy can be initiated by cutting taxes, by allowing tax revenues to drop, or by increasing government

[9]Estimating the cost of equity funds for new entrants is especially difficult because the supply of ownership funds may be discontinuous. The new entrepreneur may be able to raise a certain amount of capital at risk and then find no further amount available.

[10]Assuming a declining rate of return for successive increments of real capital investment, Keynesian theory constructs the schedule of the marginal efficiency of capital. This hypothetical function gives the amount of aggregate capital investment at various interest rates.

expenditures. Most economists generally agree that fiscal policy should at least be implemented if an economic downturn is sufficiently prolonged and severe.[11]

In contrast to advocates of fiscal policy and/or discretionary Central Bank policies, the monetarists led by Milton Friedman (1968) argue that a reasonably competitive market economy is inherently stable, under the proper financial condition. They argue for a stable forecasted quantity of the money supply as the necessary condition for economic stability. Equilibrating movements of the interest rates, saving and investment will take place in the market.

Economic Growth and Firm Growth

The analysis of growth, both aggregate and that of the firm, has not been completely assimilated into the general corpus of macroeconomic theory. Price theory, although a fascinating subject in its own right, traditionally sets the task of solving the economic relations of output and price with a given set of tastes, usable resources, technology, population, and political, social, and economic institutions. Price theory is most concerned with equilibrium states and the movement from one such state to another. Growth, on the other hand, is a process of continuous disequilibrium.

Growth occurs when demand increases or production functions change. Under these two categories, we can classify shifts in population, tastes, income patterns, and innovations—new products, new techniques, or improved organization. In a free enterprise economy, adjustment to these changes is rewarded by increased returns. Producing to meet an increased demand brings profits; adopting a successful innovation brings profits. Striving to obtain these profits, the entrepreneurs seek out and cajole more capital into potentially profitable activities. The result is capital formation and growth. The leading economic indicators, LEI, disused in chapters "Regression Analysis and Estimating Regression Models" and "Time Series Modeling and the Forecasting Effectiveness of the US Leading Economic Indicators," are measures of economic growth. See Zarnowitz (1992, 2001) and Zarnowitz and Ozyildirim (2001) for complete LEI and economic growth discussions.

Firm Growth and Economic Growth

All instances of aggregate growth do not necessarily beget the growth of individual firms. For example, assume an industry conforming to the purely competitive model, with free entry, with firms already at the optimum size, and with perfect knowledge, including knowledge of the production function. Since all the firms know they are at

[11] See *Managing Our Money, Our Budget, and Our Debt*, by the President's Cabinet Committee on Price Stability for Economic Growth, US Government Printing Office, October 1959, for a "moderate" approach to fiscal policy.

the long-run optimum size, an increase in product demand does bring about growth of the industry by the entry of new firms but not by the growth in the size of individual firms.[12] On the other hand, the economic model will show firm growth if the equilibrium is upset by an innovation which changes the optimum scale of production. The first firm to introduce the change and increase its capacity would show profits. To obtain their share of the returns or to prevent losses, other firms would imitate the innovation.

Since in most industries firms are of disparate size, it must be assumed that complex dynamic factors are at work and that imperfections and frictions hinder a perfectly smooth competitive adjustment. As a matter of fact, Professor Joseph Schumpeter emphasized the need for some competitive imperfections in encouraging economic growth (Schumpeter, 1989). If there were no imperfections or frictions in the competitive structure—no hurdles to entry, for example—a firm adopting an innovation would enjoy its profits for only a brief time. The entrepreneur who initiates an innovation must anticipate a reasonable period of economic advantage if the time, effort, and risk of working up new products, techniques, or expansion projects are to be justified.[13] However, although some competitive imperfection encourages research, development, invention, and innovation, too much tends to hinder widespread adoption of new improvements.

Once the pervasive frictions and imperfections of the real world enter the model, we have forces sheltering the individual firm which grows to meet an increase in demand.[14] Different firms in the same industry may not have the same optimum size. They differ in location, in history, and in the quality of the productive factors available to them; they may use different production techniques, and so they may differ in size. Moreover, in a dynamic society, a constant stream of improvements in communications skills, in organization and administration, and the increase in the engineering efficiency of large productive units tend continually to change the optimal scale of output. Thus the pull of demand and the push of technological change may coincide to produce company growth.

Not all company growth is a response to increases in demand. Firm growth may be motivated by the power or status drives of management rather than by economic considerations. Expansion may be an attempt to acquire market dominance. These types of company growth do not necessarily increase general economic welfare. Not all the elements of a firm's environment favor expansion. Even if the forecasted profits seem substantial, management may reasonably fear the unknown, especially if the firm lacks experience in the new activity. Furthermore, in a dynamic

[12]If we have a perfectly competitive model with imperfect knowledge, i.e., if the firms do not know they are at the lowest cost point for the long run, some may expand in response to increased profits. Those that do so may later suffer losses as new low-cost producers enter the industry.

[13]Thus in a highly competitive industry, research may have to be publicly sponsored. In agriculture, major technological improvements appear to be initiated and disseminated mainly by the various federal agencies, state research and extension services, and the agricultural research activities of state universities and colleges.

[14]It is also possible for an innovation to decrease the scale of operation. In this case firms might shrink in size, and yet the productive potential of the economy would still grow.

environment, the possibility of future competition must be considered. Management may think twice if the present company is profitable and the new project is risky and entails a heavy commitment of fixed costs. However, the rate of future growth may have to be carefully controlled if increases in size are not to raise the cost structure disproportionately. A firm that expands too rapidly may find that its operating and financial structures have become inflexible. A company can be too big, unwieldy, and difficult to manage.

If a company expands by acquiring the already existing facilities or assets of another company through merger, combination, or purchase, the aggregate capacity of the economy has not increased. Corporate expansion can be counted as a net addition to national economic capital formation only when it brings into being new real assets or facilities.

From the viewpoint of macroeconomic analysis, corporation expansion which increases aggregate net investment coincides with either an increase in corporation retained earnings or an increase in the supply of financial assets held by the public. That is, either the book value of stocks increases or new bonds or stocks are floated; if the banking system financed the expansion, the money supply increases. (The organization of a new corporation using new assets also increases the financial holdings of the economy.) Company growth which is also economic growth either taps business savings, the community's pool of personal savings, or it creates an increase in monetary assets. On the other hand, a firm that expands by acquiring the assets of another company adds nothing to the net financial holdings of the economy, since all that is involved is an exchange of various financial assets.

References

Friedman, M. (1968). The role of monetary policy. *American Economic Review, 58*(1968), 1–17.

Schumpeter, J. A. (1989). *Business cycles. Abridged with an introduction by R. Fels*. New York: McGraw-Hill Book Company.

Weintraub, S. (1978). *Keynes, Keynesians and Monetarists*. Philadelphia: University of Pennsylvania Press.

Zarnowitz, V. (1992). *Business cycles: Theory, history, indicators, and forecasting*. Chicago: University of Chicago Press.

Zarnowitz, V. (2001). *The old and the new in the U.S. economic expansion*. The Conference Board. EPWP #01–01.

Zarnowitz, V., & Ozyildirim, A. (2001). On the measurement of business cycles and growth cycles. *Indian Economic Review, 36*, 34–54.

International Business Finance



With the adoption of flexible exchange rates in 1973, international capital markets have become more completely integrated. This chapter discusses portfolio selection of international equities and how international diversification lowers total risk of portfolios. Particular attention is paid to the diversification implications of Asian stocks, other emerging markets, and Latin American securities. The US equity selection model developed and estimated in chapter "Risk and Return of Equity and the Capital Asset Pricing Model" is used to rank global (ex-US) securities and produces statistically significant information coefficients and excess returns. An investor owns foreign stocks because their inclusion into portfolios produces higher Sharpe ratios than using only domestic securities. Global stock Portfolios may produce portfolios of higher returns for a given level of risk than U.S.-only stock portfolios. We report empirical evidence supporting the creation of international, global, and emerging market stock portfolios.

Currency Exchange Rates

One of the additional risks affecting the returns on foreign investment is the possible movement of the exchange rate. The following data would be of interest to a US investor on December 31, 2019:

$$\$1.326\ \text{US} = 1\,£(\text{Pound})$$

The pound sterling rose to $1.326 after the Boris Johnson government won the election and *Brexit* was kept on course.[1] What do these exchange rates mean? Let us

[1]The dollar fell Friday, pressured by a postelection rally in the British pound.

The WSJ Dollar Index, which measures the US currency against a basket of 16 others, was recently down 0.1% at 90.23, pressured by a 1.3% rally in the pound. Investors have piled into the British currency on expectations that a resounding victory by Conservatives in the UK's December

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_21

Table 1 Cross rates, December 31, 2019

Currency cross rates table				
Base currency	Dollar ($)	Euro (€)	Pound (£)	Yen (¥)
US dollar	–	0.8909	0.7543	108.62
EU euro	1.1224	–	0.8465	121.92
UK pound	1.3255	1.1808	–	143.98
Japan yen	0.0092	0.0082	0.0069	–

Source: Bloomberg, L.P

take a trivial, but interesting, example. A buyer can purchase a Honda Odyssey minivan for $25,000 in the United States. If the Honda minivan is priced at 19,000 pounds in London, should our US investor fly to London, buy the minivan, drive around England and Scotland, and ship it back to the United States for $5000? The $25,000 car in the United States should cost 19,000 × $1.326 in London, or $25,194. The US investor should pay the $25,000 domestic cost. The US investor should purchase his Japanese car in the United States, rather than in the United Kingdom.

A more traditional key cross currency rates matrix, see Table 1, is normally presented on a Bloomberg terminal on December 31, 2019:

The US dollar fell from 117.3 yen in 2003 to only 108.86 yen in 2005 and 108.62 yen on December 31, 2019. The United States continued to run persistent balance of trade deficits that led to its continued currency depreciation. The US dollar depreciated versus the euro, falling from 0.917 in 2003 to only 0.8254 in 2005.[2] However, despite the trade deficits, the US dollar ($USD) rose to 0.891 euro on December 31, 2019.

We introduced readers to call and put options on stocks in chapter "Options." We now introduce the reader to calls and puts on the euro. On December 31, 2019, the euro closed at $1.326. If one expects Boris Johnson to be correct and the pound sterling (GBP) appreciate, to, say, $1.35 by June 2020, then one could buy an option on the pound, for the asking price of $1.90; see Table 2. If the price appreciates to $1.35, in the next several weeks, then the price of the call could well exceed to $4.18, the current price of an at-the-money pound call. The gain on the option is 120%

12 election will ensure a smooth, orderly Brexit. https://www.wsj.com/articles/dollar-falls-on-pounds-rally-after-u-k-election-11576257737. WSJ is, of course, *The Wall Street Journal*.

Sterling's rise from multiyear lows in September has made it one of the market's best-performing currencies, having risen by more than 6% against the dollar over the last 12 months. It was recently at $1.3337.

Some market participants believe the pound's rally might not have much further to run.

[2]A key cross currency rates table as of September 6, 2006 would be:

Currency	USD	EUR	JPY	GBP	CAD
USD	. . .	1.2785	.008574	1.8820	.9026
EUR	.7821	. . .	.00671	1.4720	.7060
JPY	116.63	149.11	. . .	219.50	105.27
CAD	1.1079	1.4165	.00950	2.0851	. . .
AUD	1.3046	1.6680	.01186	2.4553	1.1776

Table 2 GBP June 2020 calls (Option quotes. December 31, 2019)

Strike	Ticker	BSze	Bid	Ask	ASze	Last	IVM	DM	Volm	Olnt
Jun-20 (157d 6/5/20); CSize 62,500; BGAMO 1.33										
129	BGAM0C	0	0.000	0.000	0	5.4400	0.000	0.000	0	1
130	BGAM0C	0	0.000	0.000	0	4.7100	0.000	0.000	0	3
131	BGAM0C	65	2.710	5.100	65	4.0300	7.746	0.635	0	0
132	BGAM0C	67	1.490	4.480	65	3.4000	6.757	0.587	0	35
132.5	BGAM0C	65	1.830	4.180	65	3.1100	7.623	0.549	0	7
133	BGAM0C	65	1.570	3.900	65	2.8400	7.588	0.520	0	1
134	BGAM0C	142	2.150	2.350	99	2.3500	7.543	0.460	1	2
135	**BGAM0C**	**142**	**1.730**	**1.900**	**85**	**1.9200**	**7.476**	**0.400**	**0**	**94**
135.5	BGAM0C	65	1.550	1.740	99	1.7300	7.515	0.372	0	1
136	BGAM0C	115	1.380	1.570	99	1.5500	7.515	0.344	1	7
137	BGAM0C	115	1.090	1.270	99	1.2400	7.534	0.292	0	2

Source: Bloomberg, L.P
Contract size (CSize):
100 shares per contract for equities
One futures contract for 62,500 British pound
One futures contract for 125,000 euro
Risk-free rate (R)
Implied forward (IFwd)
On December 31, 2019, the *IVM* Implied volatility based on mid-price, *DM* Delta based on mid-price, *Volm* Amount of options traded this session, *OInt* Open interest − total number of net option contracts outstanding

($4.18–$1.90)/$1.90 = 1.20. The gain on the holding of the (cash) pound is only 1.8% ((1.35–1.326)/1.326). Clearly, leverage enhances returns, as we saw with stocks in chapter "Options."

Economists have long held that in the flexible exchange rate regime in which we live, the purchasing power of currencies should be equated by changing currency rates. That is, if a basket of commodities is cheaper in the United States than in Europe, then US exports should rise, US imports should fall, and the US dollar should rise (appreciate) such that the purchasing power of the US dollar and euro is equal. The purchasing power parity theory (PPPT), put forth by Cassel (1916), holds that

$$dE_x = \frac{P_{US_1}/P_{US_0}}{P_{\in 1}/P_{\in_0}} \quad (1)$$

where

dE_x = change in exchange rate
P_{US_1} = domestic (US) price level at time 1
P_{US_0} = domestic (US) price level at time 0

$P_{\in_1}$ = foreign (euro) price level at time 1
$P_{\in_0}$ = foreign (euro) price level at time 0

The PPPT holds that the rate of change in exchange rates equals the rate of change in price levels, which, of course, equals the relative ratios of inflation rates. See Yeager (1976) for a complete PPPT analysis.

A topic related to the purchasing power parity theory is the interest rate parity theory (IRPT). One may note the relationship between the nominal interest rate and the expected rate of inflation (see chapter "Long-Term Debt"). Interest rates rise as the expected inflation rate rises. Irving Fisher put forth the relationship between interest rates and inflation in his *Theory of Interest* (1911). The Fisher Effect can be stated in terms of the domestic interest rates:

$$\frac{P_{US_0}}{P_{US_1}} = \frac{1 + r}{1 + Rn} \tag{2}$$

where r is the real rate of interest, equaling the marginal productivity of capital (the increase in output, GDP, due to an increase in the capital stock). The Fisher Effect can be stated in terms of the domestic and foreign price levels and interest rates:

$$\frac{E_{x_f}}{E_{x_s}} = \frac{1 + r_{f_0}}{1 + r_{d_0}} \tag{3}$$

E_{x_f} = current forward exchange rate; i.e., € = $1 US
E_{x_s} = current spot exchange rate
r_{d_0} = current domestic interest rate
r_{f_0} = current foreign interest rate

An alternative means of showing interest rate parity is in terms of the premium on the forward. Let E_{x_f} be the 3-month forward exchange rate in dollars per pound, £, and E_{x_s} be the spot (current) exchange rate in dollars per pound. The premium on the forward, p, is

$$p = \frac{E_{x_f} - E_{x_s}}{E_{x_x}}. \tag{4}$$

If r_f is the short-term (3 months) interest rate in London, and r_d is the 3-month interest rate in New York, then

$$1 + r_d = \frac{1}{E_{x_s}}(1 + r_f)E_{x_f} \tag{5}$$

Note that $E_{x_f}/E_{x_s} = H_p$. Interest rate parity holds that

$$\begin{aligned} 1 + r_d &= (1 + p)(1 + r_f) \\ 1 + r_d &= 1 + p + r_f + pr_f \end{aligned} \tag{6}$$

If p and r_f are small, then pr_f becomes almost zero, and

$$p = r_d - r_f \tag{7}$$

The forward premium on the pound can be expressed as a function of the excess of the domestic and foreign interest rates.

On August 26, 2003, the 3-month pound LIBOR (London Interbank Offered Rate) was 3.5663%, whereas the US dollar LIBOR rate was 1.14%. The premium on the pound forward was

$$p = \frac{1.5596 - 1.5733}{1.5733} = -.0087$$

where the spot price of the pound is \$1.5733 and the current December future price of the pound is \$1.5596.[3] The pound, on August 26, had a forward discount (a negative premium) because UK interest rates exceeded US interest rates. Is this forward discount consistent with interest rate differentials? Let us assume that the relevant 3-month rates are consistent for a fourth month, December. An investor could invest \$100 in a 3-month US dollar security and have an ending value of \$100.02. We calculated this terminal value by

$$\$100 \times \left(1 + \frac{0.0114}{4}\right) = \$100.29.$$

An investor could invest \$100 in a pound sterling £ security and have a terminal wealth value of \$100.29, by

$$\$100 \times \left(1 + \frac{0.0357}{4}\right) \times \frac{\$1.5596}{\$1.5733} = \$100.21$$

The US investor will do better to invest in the US security than to invest in the UK security. The interest rate parity theory (IRPT) holds that differences in interest rates determine the forward discount on currency. Most empirical studies, summarized in Cumby and Obstfeld (1984) and Levich (2001), hold that deviations exceeding 3% of IRPT are relatively small in number for the short-term Eurocurrency market.

The spot price of the euro was \$1.0850 on August 26, 2003, and the closing future price for the December euro was \$1.0847. Given the euro LIBOR 3-month rate was

[3] It is very difficult to beat the forward price forecast, Meese and Rogoff (1983). See Levich (2001), Levich and Potti (2015), and Kyriazi and Thomakos (2020) for recent evidence suggesting that statistically significant, 10–15% forecasting error reductions can be achieved.

2.1138%, and the US LIBOR rate was 1.01138%, one could calculate terminal wealth values for a US investor investing in a US dollar security:

$$\$100 \times \left(1 + \frac{.0114}{4}\right) = \$100.29$$

and a euro dollar security:

$$\$100 \times \frac{\$1.0847}{\$1.0850} \times \left(1 + \frac{.02114}{4}\right) = \$100.50$$

The US investor has a slightly higher final wealth investing in the 3-month euro security.

Marston (1995) put forth several reasons why interest rate differentials might become distorted. Banking officials can impose controls of bank interest rates, and these controls can lead to distortions of IRPT. British government controls on resident capital outflows were eliminated in June 1979. Marston (1995) reported that the interest rate differentials between euro sterling and British markets which averaged 1.50% during the flexible exchange rate period with controls fell to −0.03 for the July 1979–March 1991 period without controls. The average differential during a subset of the Bretton Woods period of fixed exchange rates (April 1961–April 1971) was 0.78%. The elimination of controls established a very close IRPT relationship between the euro sterling and British markets. US controls, such as the Voluntary Foreign Credit Restraint Program and the Foreign Direct Investment Program, and the limitation placed on US CD rates, by the Federal Reserve with Regulation Q, led to a 1.33% interest rate differential between the eurodollar and US (CD) markets during the 1966–December 1973 period. The post control interest rate differential during the 1974–March 1991 period was only 0.48%.

International Diversification

Why would US investors want to own foreign or international stocks? First, one can achieve additional diversification by owning foreign stocks, particularly if the US stock returns are not correlated with foreign stock returns. Risk reduction was shown in chapter "Risk and Return of Equity and the Capital Asset Pricing Model" with US stocks, and a US investor can achieve additional diversification by owning foreign stocks. In our first edition, Guerard and Schwartz (2007) we examined the correlation matrix of country stock returns of the G7 (the United States, Canada (CA), France (FR), Germany (GM), Italy (IT), Japan (JP), and the United Kingdom) countries during the 1970–2002 period, shown in Table 3. We use stock indexes of the Morgan Stanley Capital International (MSCI) database. One sees that correlation of US and Canadian MSCI returns are 0.73, which is higher than the correlation between US and Japanese stock returns (0.31). We include the MSCI World Index (WI) in Table 4. A US investor can achieve tremendous diversification

Table 3 MSCI G7 stock market index return correlation matrix, 1970–2002

	Pearson correlation coefficients, N = 396									
	WI	CD	FR	GM	IT	JP	UK	US	N	Mean
WI	1.0000	0.7397	0.6612	0.6320	0.4706	0.6724	0.6948	0.8588	396	0.0052
CD	0.7397	1.0000	0.4810	0.4065	0.3384	0.3250	0.5243	0.7307	396	0.0044
FR	0.6612	0.4810	1.0000	0.6604	0.4844	0.3976	0.5667	0.4908	396	0.0056
GM	0.6320	0.4065	0.6604	1.0000	0.4471	0.3703	0.4671	0.4603	396	0.0050
IT	0.4706	0.3384	0.4844	0.4471	1.0000	0.3548	0.3693	0.2881	396	0.0027
JP	0.6724	0.3250	0.3976	0.3703	0.3548	1.0000	0.3777	0.3010	396	0.0070
UK	0.6948	0.5243	0.5667	0.4671	0.3693	0.3777	1.0000	0.5353	396	0.0052
US	0.8588	0.7307	0.4908	0.4603	0.2881	0.3098	0.5353	1.0000	396	0.0053

Table 4 MSCI G7 stock market index return correlation matrix, 1995–2002

	Pearson correlation coefficients, N = 96									
	WI	CD	FR	GM	IT	JP	UK	US	N	Mean
WI	1.0000	0.8392	0.8250	0.8046	0.6111	0.6368	0.8400	0.9427	96	0.0026
CD	0.8392	1.0000	0.6839	0.6583	0.4674	0.5001	0.6516	0.8062	96	0.0050
FR	0.8250	0.6839	1.0000	0.8469	0.7063	0.4326	0.7649	0.6845	96	0.0043
GM	0.8047	0.6583	0.8470	1.0000	0.6342	0.3097	0.7093	0.7196	96	0.0002
IT	0.6111	0.4674	0.7063	0.6341	1.0000	0.2816	0.5180	0.5103	96	0.0034
JP	0.6368	0.5006	0.4326	0.3097	0.2816	1.0000	0.4676	0.4574	96	−0.0077
UK	0.8400	0.6516	0.7649	0.7093	0.5180	0.4676	1.0000	0.7462	96	0.0027
US	0.9427	0.8062	0.6845	0.7196	0.5103	0.4574	0.7462	1.0000	96	0.0068

Table 5 MSCI regional stock market index correlation matrix, 1995–2002

	Pearson correlation coefficients, N = 96								
	WI	US	EU	NA	EM	Asia	LA	N	Mean
WI	1.0000	0.9427	0.9009	0.9465	0.7536	0.7357	0.7051	96	0.0026
US	0.9427	1.0000	0.7777	0.9995	0.6861	0.5745	0.6473	96	0.0068
EU	0.9001	0.7777	1.0000	0.7828	0.6567	0.5557	0.6417	96	0.0036
NA	0.9465	0.9995	0.7828	1.0000	0.6967	0.5824	0.6553	96	0.0066
EM	0.7536	0.6861	0.6567	0.6967	1.0000	0.6931	0.8941	96	−0.0054
Asia	0.7357	0.5745	0.5557	0.5824	0.6931	1.0000	0.5555	96	−0.0075
LA	0.7051	0.6473	0.6417	0.6553	0.8941	0.5555	1.0000	96	−0.0032

Source: MSCI data

by buying Japanese stocks. The reader is reminded of the excess returns of Japanese stocks reported in chapter "Risk and Return of Equity and the Capital Asset Pricing Model."

Capital markets have become more integrated during the period of flexible exchange rates, and one sees the more complete integration in Table 4 where the MSCI country index correlations of the G7 countries are shown for the 1995–2002 period. The US-Canada country stock return had a 0.81 correlation, and the US-Japan correlation rises to 0.46. One can (still) achieve diversification with the G7 countries.

If a US investor looked to invest within a particular region of the world, as such Europe (EU), North American (the United States and Canada), emerging markets (EM), Asia, or Latin America (LA), where could one minimize the US stock return correlations? The reader is referred to Table 5, the MSCI region return correlation matrix for the 1995–2002 period. It should be obvious that the US and NA correlations are very high (0.99) and offer limited diversification. A US investor could invest in Europe (a correlation coefficient of 0.78), the emerging markets (0.69), Latin America (0.65), or Asia (0.57). It would appear that Asia offered the highest diversification for a US investor, given its relatively low correlation coefficient with US returns. The risk reduction of a US investor owning foreign stocks is shown in Figs. 1 and 2, courtesy of Professor Solnik.

The relatively low correlation between US stocks and UK and Asian stocks creates greater diversification benefits for global investors; see Figs. 1 and 2.

In Table 6, we report USD-based stock market returns for global stock markets for 2019. Note the much lower returns in several countries. Correlations matter, as we will see later in this Chapter.

International Stock Selection and Portfolio Construction and Optimization

In chapter "Risk and Return of Equity and the Capital Asset Pricing Model," we estimated 8-, 9-, and 10-factor model of US equity holding period returns during the 2003–November 2018 period. We can estimate the corresponding nine-factor model

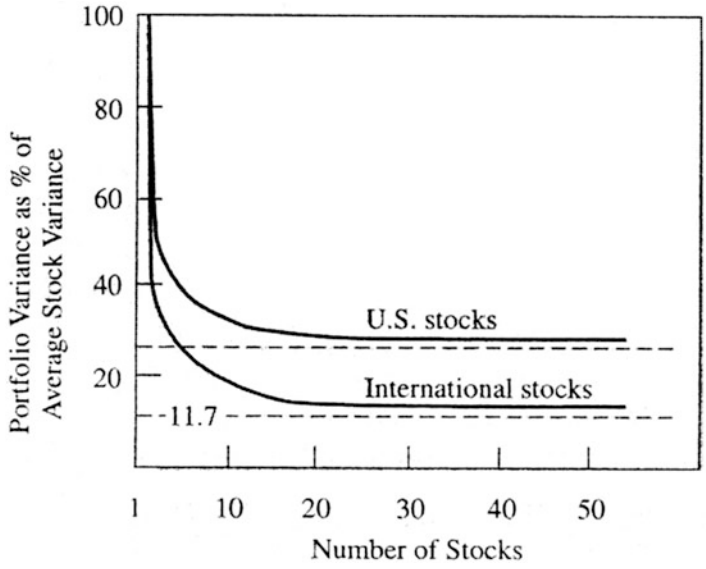


Fig. 1 Risk reduction through domestic and international diversification

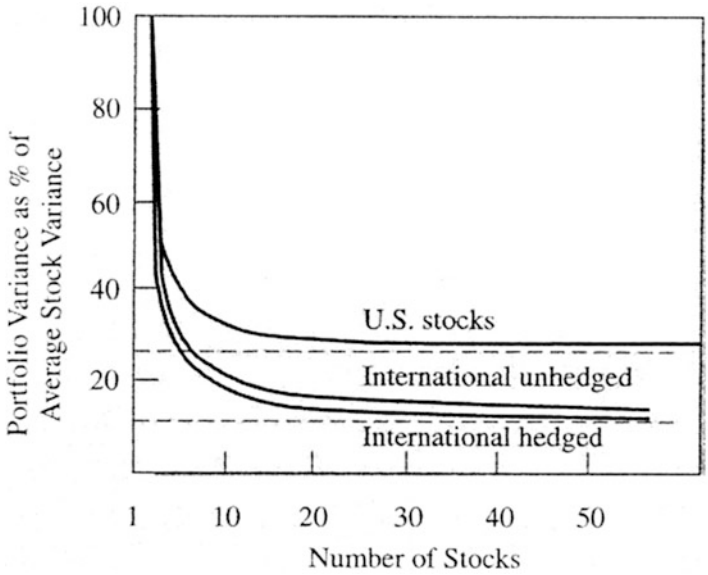


Fig. 2 Risk reduction through domestic and international diversification hedged and unhedged results. Note: The vertical axis of the graph is the ratio $\sigma^2(\overline{R}_N)/\sigma^2(\overline{R}_1)$ where N is the number of stocks in the portfolio. (Source: Bruno Solnik, "Why Not Diversify Internationally Rather Than Domestically" *Financial Analysis Journal* 30, no. 4 (July–August 1974), pp. 48–54)

Table 6 WEI, returns for 2019

1. Americas	Market value	%Change YTD	%YTD Cur (USD)
DOW Jones	28,536.07	+22.33%	+22.33%
S&P 500	3230.49	+28.87	+28.87
NASDAQ	8972.61	+35.23	+35.23
S&P/TSX comp	17,063.43	+19.13	+25.12
S&P/BMV IPC	43,548.32	+4.58	+8.69
2. EMA			
Euro Stoxx 50	3745.15	+24.78	+22.08
FTSE 100	7542.44	+12.10	+16.41
CAC 40	5978.06	+26.37	+23.63
DAX	13,249.01	+25.48	+22.76
IBEX 35	9549.20	+11.82	+9.40
FTSE MIB	23,506.37	+28.28	+25.50
OMX STH30	1771.85	+25.78	+18.89
SWISS MKT	10,616.94	+25.95	+27.84
3. Asia/Pacific			
TOPIX	1721.36	−15.21	+16.42
Nikkei 225	23,656.62	+18.20	+19.44
JPX Nikkei 400	15,376.38	+16.02	+17.24
Hang Seng	28,189.75	+9.07	+9.61
CSI 300	4096.58	+36.07	+34.41
Shanghai comp	3050.12	+22.30	+20.82
Shenzhen comp	1722.94	+35.89	+34.24
HS China Ent	11,168.06	+10.30	+10.86
Taiwan TAIEX	11,997.14	+23.33	+25.63
KOSPI	2197.67	+7.67	+3.86

Source: Bloomberg, L.P

for non-US securities (the World less US) and EM universes, producing statistically significant models for non-US asset managers during the 2003–November 2018 period. The components of the REG10 model are

$$TR_{t+1} = a_0 + a_1 EP_t + a_2 BP_t + a_3 CP_t + a_4 SP_t + a_5 REP_t + a_6 RBP_t + a_7 RCP_t + a_8 RSP_t + a_9 CTEF_t + a_{10} PM_t + e_t \tag{8}$$

where $\alpha = (a_{1,\ \ldots\ ,}a_{10})$ is the rolling average of normalized coefficients of $\alpha_{t\ -\ 1},\ldots, \alpha_{t\ -\ 12}$.

PM = price momentum, P(t − 1)/T(t − 7).

Our portfolio analysis uses the regression-based composite security valuation models shown in Eq. (8). We create monthly optimized portfolios using international stocks which produce higher information ratios (IRs), Sharpe ratios (ShRs), and

geometric means than US. and Japanese stocks for the 2003–November 2018 time period. The portfolios are rebalanced monthly: 8% monthly buys and 8% monthly sells. The maximum weight is 4%; a minimum (threshold) position is 25 basis points, and we use the ITG transaction cost information. REG8 is more effective in Japan stocks, and the CTEF is more effective in US stocks. International stocks, emerging markets stocks, and Chinese A-share stocks produce the highest specific returns with CTEF, whereas REG8, REG9, and REG10 are all statistically significant in XUS, EM, and China A-share stocks (Table 7). Diversification, with the benefits of lower correlations, and the greater opportunity to buy move mis-placed stocks, enhance portfolio returns.

International Corporate Finance Decisions

If access to global capital markets allows multinational enterprises (MNE) to lower its equity costs and costs of capital relative to domestic firms, the MNE could maintain a higher desired debt ratio relative to domestic firms, even with large new capital issues. Eiteman, Stonehill, and Moffett (2004) believe that the marginal cost of capital may be constant for a great range of the MNE capital budgets.[4] Moreover, despite the favorable effect of international diversification of MNE cash flow, bankruptcy risk of MNE is approximately equal to that of domestic firms. If a US firm issues foreign currency-denominated debt, such as a US firm issuing 8% debt in Germany, and the euro appreciates in the coming year, then the US firm's dollar cost of obligations for repaying principal and interest must rise due to the appreciation of the euro.[5] Many firms attempt to manage currency exposures through hedging. Hedging is taking a position such that will change to offset the change in value of an existing position. Currency risk can be defined as the variance in expected cash flow due to unexpected changes in exchange rates. Hedging can reduce the risk of future cash flow and reduce the likelihood that cash flow will fall below a necessary minimum balance. If a firm has an obligation payable in another currency, then it may want to hedge the obligation. Imagine that a US firm purchases an input, such as cocoa from a German firm, that is payable in 90 days. The firm could hedge the obligation, purchasing, for example, 100 million euro of cocoa, and could purchase 100 million euros in the forward market for delivery in 90 days. If the euro increases in value, the potential loss on the transactions (payable) is offset by the gain of the forward exchange purchase.

[4]E. Schwartz put forth a similar position for US firms in chapter "Debt, Equity, the Optimal Financial Structure, and the Cost of Funds."

[5]In the 1990s, German and Japanese firms used greater leverage than US and UK firms, primarily due to lower debt costs and equity costs (in the case of Japan). These relative advantages disappeared in the late 1990s and early 2000s, leading the German and Japanese firms to reduce their leverage. The leverage ratio was relatively constant for US and UK firms (Eun & Resnick, 2004; McCauley & Zimmer, 1994).

Table 7 Portfolio dashboard

Mean-variance optimization of Russell 3000 and international stock selection models									
Axioma fundamental risk model January 2002–November 2018									
Portfolios	**Sharpe ratio**	**Info ratio**	**Risk stock specific effect**	**Risk stock specific effect T-Stat**	**Risk factors effect**	**Risk factors effect T-Stat**	**Risk total effect**	**Risk transaction effect**	**Total effect**
R3000_REG9_8TE	0.68	0.38	3.01	**2.04**	0.06	0.55	3.07	−1.20	1.88
R3000_REG8_6TE	0.60	0.18	1.86	**1.66**	−0.88	−0.24	0.98	−0.94	0.04
R3000_CTEF_8TE	0.75	0.52	1.61	1.36	3.30	2.24	4.92	−2.26	2.65
R3000_REG8_8TE	0.51	0.06	1.77	1.36	−1.53	−0.50	0.24	−1.28	−1.03
R3000_REG9_6TE	0.64	0.25	1.35	1.14	0.15	0.55	1.50	−1.12	0.38
R3000_CTEF_6TE	0.71	0.42	0.35	0.57	2.59	2.26	2.95	−1.55	1.39
R3000_CTEF_4TE	0.72	0.42	−0.02	0.20	2.03	2.73	2.02	−0.92	1.10
R3000_REG9_4TE	0.62	0.12	0.02	0.05	0.31	0.67	0.33	−0.55	−0.22
R3000_REG8_4TE	0.58	0.02	−0.06	−0.05	−0.11	0.17	−0.17	−0.45	−0.62
R3000_REG10_8TE	0.53	0.08	−0.42	−0.14	0.85	0.86	0.43	−1.96	−1.54
R3000_REG10_4TE	0.62	0.12	−0.17	−0.17	0.52	0.89	0.35	−0.81	−0.46
R3000_REG10_6TE	0.53	−0.01	−0.88	−0.62	0.53	0.68	−0.35	−1.33	−1.68
Portfolios	**Sharpe ratio**	**Info ratio**	**Risk stock specific effect**	**Stock specific effect T-Stat**	**Risk factors effect**	**Risk factors effect T-Stat**	**Risk total effect**	**Risk transaction effect**	**Total effect**
JAPAN_REG8_8TE	0.56	0.54	5.64	**4.00**	−0.18	0.44	5.46	−2.24	3.26
JAPAN_REG8_6TE	0.54	0.51	4.73	**3.74**	−0.23	0.20	4.50	−1.49	3.04
JAPAN_REG8_4TE	0.50	0.46	2.75	**2.88**	0.17	0.63	2.92	−0.79	2.17
JAPAN_REG9_8TE	0.42	0.28	2.14	**1.65**	0.74	1.34	2.87	−2.00	0.95
JAPAN_REG9_4TE	0.44	0.31	1.54	1.59	0.55	1.45	2.09	−0.82	1.32
JAPAN_REG9_6TE	0.41	0.25	1.83	1.50	0.44	1.04	2.27	−1.35	0.98
JAPAN_REG10_8TE	0.38	0.19	0.23	0.44	1.48	1.98	1.71	−1.80	−0.01

(continued)

Table 7 (continued)

JAPAN_REG10_6TE	0.37	0.16	−0.13	0.14	1.40	2.01	1.26	−1.48	−0.16
JAPAN_CTEF_8TE	0.35	0.15	−0.93	−0.01	2.20	2.50	1.28	−6.73	−5.40
JAPAN_CTEF_6TE	0.12	0.09	−0.90	−0.18	2.32	3.13	1.42	−3.87	−2.38
JAPAN_CTEF_4TE	0.30	0.19	−0.82	−0.38	1.96	3.97	1.13	−1.35	−0.17
JAPAN_REG10_4TE	0.35	0.07	−0.97	−0.76	1.39	2.70	0.42	−1.06	−0.58
Portfolios	**Sharpe ratio**	**Info ratio**	**Risk specific effect**	**Risk stock effect T-Stat**	**Factors effect**	**Risk effect T-Stat**	**Total effect**	**Risk transaction effect**	**Total effect**
XUS_EP_4TE	0.53	0.62	4.88	**3.67**	1.55	2.79	6.43	−1.38	5.05
XUS_EP_8TE	0.53	0.51	7.28	**3.61**	2.06	2.74	9.34	−2.81	6.53
XUS_EP_6TE	0.54	0.55	6.16	**3.53**	1.61	2.68	7.77	−2.11	5.66
XUS_CTEF_4TE	0.59	0.79	3.98	**3.45**	3.62	4.58	7.60	−1.38	6.22
XUS_CTEF_6TE	0.63	0.79	4.43	**3.33**	5.00	4.18	9.43	−2.10	7.33
XUS_CTEF_8TE	0.65	0.77	4.37	**3.07**	6.33	3.97	10.71	−2.65	8.06
XUS_REG8_4TE	0.39	0.07	2.94	**2.21**	1.12	1.93	4.05	−1.28	2.77
XUS_REG8_6TE	0.35	0.03	3.57	**2.05**	0.99	1.74	4.56	−1.88	2.68
XUS_REG9_6TE	0.40	0.16	2.98	**1.77**	2.78	3.08	5.76	−1.94	3.83
XUS_REG9_4TE	0.42	0.16	2.22	1.63	2.46	3.45	4.67	−1.30	3.38
XUS_REG8_8TE	0.28	−0.04	3.39	1.60	0.68	1.60	4.07	−2.36	1.71
XUS_REG9_8TE	0.38	0.12	2.00	1.29	3.76	2.79	5.77	−2.31	3.46
XUS_REG10_4TE	0.45	0.26	1.45	1.23	3.30	4.17	4.75	−1.40	3.35
XUS_REG10_8TE	0.44	0.28	0.90	0.93	6.02	3.95	6.92	−2.52	4.40
XUS_REG10_6TE	0.44	0.26	1.00	0.89	4.94	4.25	5.94	−2.07	3.88

Portfolios	Sharpe ratio	Info ratio	Specific effect	Effect T-Stat	Factors effect	Effect T-Stat	Total effect	Transaction effect	Total effect
EM_REG8_4TE	0.51	0.31	3.99	**2.38**	1.79	2.11	5.78	−2.05	3.73
EM_REG9_4TE	0.55	0.49	3.45	**2.20**	3.28	3.60	6.73	−2.11	4.63
EM_CTEF_8TE	0.59	0.61	3.54	**2.18**	5.76	3.64	9.30	−3.58	5.72
EM_CTEF_4TE	0.58	0.72	3.20	**2.16**	3.84	4.42	7.04	−2.10	4.94
EM_CTEF_6TE	0.60	0.65	3.67	**2.13**	4.67	4.02	8.34	−2.75	5.59
EM_REG9_8TE	0.51	0.30	3.40	**1.73**	5.09	3.31	8.50	−3.62	4.88
EM_REG8_6TE	0.44	0.06	3.30	1.64	2.32	2.17	5.63	−2.79	2.83
EM_REG10_4TE	0.54	0.45	2.25	1.58	4.06	4.24	6.31	−1.92	4.39
EM_REG9_6TE	0.48	0.22	2.62	1.54	4.37	3.49	6.99	−2.80	4.19
EM_REG8_8TE	0.43	0.10	3.37	1.36	2.59	2.27	5.96	−3.52	2.44
EM_REG10_6TE	0.52	0.33	1.33	1.07	5.63	4.17	6.96	−2.96	4.00
EM_REG10_8TE	0.49	0.24	0.30	0.59	6.34	3.68	6.64	−3.54	3.10
Portfolios	**Sharpe ratio**	**Info ratio**	**Stock specific effect**	**Specific effect T-Stat**	**Risk factors effect**	**Factors effect T-Stat**	**Risk total effect**	**Risk transaction effect**	**Total effect**
CHINAA_CTEF_4TE	0.60	1.23	7.12	**6.75**	2.22	6.32	9.35	−1.74	7.68
CHINAA_CTEF_6TE	0.59	1.05	7.30	**5.16**	2.54	5.31	9.84	−2.39	7.52
CHINAA_CTEF_8TE	0.57	0.95	7.19	**4.87**	2.75	4.98	9.94	−2.61	7.40
CHINAA_REG10_4TE	0.51	0.87	4.62	**4.43**	2.31	5.60	6.93	−1.78	5.19
CHINAA_REG9_4TE	0.44	0.54	3.45	**3.41**	1.42	3.77	4.88	−1.66	3.24
CHINAA_REG10_6TE	0.52	0.72	4.77	**3.31**	3.06	5.51	7.84	−2.51	5.39
CHINAA_REG8_4TE	0.42	0.47	3.17	**3.29**	1.29	3.29	4.47	−1.65	2.83
CHINAA_REG10_8TE	0.50	0.63	4.38	**2.89**	3.18	5.05	7.56	−2.79	4.83
CHINAA_REG8_6TE	0.41	0.39	3.45	**2.67**	1.49	2.84	4.94	−2.41	2.58
CHINAA_REG8_8TE	0.41	0.37	3.65	**2.56**	1.64	2.75	5.30	−2.67	2.67
CHINAA_REG9_8TE	0.41	0.38	3.40	**2.40**	1.76	3.06	5.16	−2.63	2.57
CHINAA_REG9_6TE	0.42	0.42	3.18	**2.37**	1.82	3.52	5.01	−2.37	2.69

Source: FactSet Investment Research Systems, Inc.

Exchange rate variability can affect the firm's financial decisions. In a perfectly integrated international capital market, the equilibrium forward rate is determined such that hedging will not produce a profit. Aliber (1978) found that the average deviation of the forward rate from the future spot rate approached zero as the time horizon expands. There is no need to hedge if long-term nominal foreign currency assets are not exposed in the long run. Commodity price parity (CPP) holds that physical assets are not exposed to exchange rate risk as long as the purchasing power parity theory (PPPT) holds. However, Adler and Bernard (1983) state that the prices of input goods and output may have different exposures such that PPPT and CPP may not be sufficient for sound financial management. In an efficient market, there may be minimal advantages to hedging. A financial manager for Hershey would be extremely naïve (or lucky) if he or she did not hedge their chocolate purchases.

References

Adler, M., & Bernard, D. (1983). International portfolio choice and corporation finance: A synthesis. *Journal of Finance, 38*, 925–984.

Aliber, R. Z. (1978). *Exchange risk and corporate international finance*. New York: Wiley.

Cassel, G. (1916). The present situation of the foreign exchange. *Economic Journal, 26*, 62–65.

Cumby, R. E., & Obstfeld, M. (1984). International interest rate and price level linkages under floating exchange rates: A review of recent evidence. In J. F. O. Bilson & R. C. Marston (Eds.), *Exchange rate theory and practice*. Chicago: University of Chicago Press.

Eiteman, D. K., Stonehill, A. I., & Moffett, M. H. (2004). *Multinational business finance* (10th ed.). Reading, MA: Addison-Wesley.

Eun, C. S., & Resnick, B. G. (2004). *International financial management* (3rd ed.). New York: Irwin.

Guerard, Jr., J. B., Deng, S., Gillam, R. A., Markowitz, H. M. & Xu, G. (2020). *Investing in Global Markets*. Wilmott Magazine, March, 56–73.

Kyriazi, F., & Thomakos, D. D. (2020). Foreign exchange rate predictability: Seek and ye shall and it. In J., Jr. Guerard & W. T. Ziemba (eds.), *Handbook of applied investment research*. Singapore: World Scientific Publishing

Levich, R. M. (2001). *International financial markets: Prices and policies*. New York: McGraw-Hill/Irwin.

Levich, R., & Potti, V. (2015). Predictability and good deals in currency markets. *International Journal of Forecasting, 31*, 454–472.

Markowitz, H. M. & Guerard, J. B. (2019). *The Existence and Persistence of Financial Anomalies*. Presented at the Institute of Quantitate Research in Finance, October 2019. La Jolla, CA.

Marston, R. C. (1995). *International financial integration*. New York: Cambridge University Press.

McCauley, R., & Zimmer, S. (1994). Exchange rates and international differences in the cost of capital. In Y. Amihud & R. Levich (Eds.), *Exchange rates and corporate performance*. New York: Irwin.

Meese, R. A., & Rogoff, K. (1983). Empirical exchange rate models of the seventies: Do they fit out of sample? *Journal of International Economics, 14*, 3–24.

Mincer, J., & Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (Ed.), *Economic forecasts and expectations*. New York: Columbia University Press.

Thomakos, D., & Guerard Jr., J. B. (2004). Naïve, ARIMA, transfer function, and VAR models: A comparison of forecasting performance. *The International Journal of Forecasting, 20*(2004), 53–67.

Yeager, L. B. (1976). *International monetary relations*. New York: Harper & Row.

Management-Stockholder Relations: Is Optimal Behavior All That Is Necessary?



So long as a company is closely held, the control group and the stockholders are identical and seldom is there a conflict of interest between them.[1] However, once a company goes public, it acquires a group of shareholders who depend on the management for the safety and profitability of their investment. In short, an agency-principal relation is established where the management is the agent and the shareholder is the principal. This relation implies a commitment by management that the outside shareholders will be treated fairly in such matters as cash payouts, expansion policies, accounting probity, and the level of executive compensation and that in general the company affairs will be directed vigorously and conscientiously.

Discussions on the problem of the separation of corporate control and ownership date back to the 1930s.[2] Nevertheless, it may be well to note that the shareholder's abdication of a direct concern in managing the firm may be a natural consequence of the organizational development of the large corporation. The average shareholder desires and expects a reasonably competent and just management. Beyond that, he is not much concerned for in most cases his major economic activities lie elsewhere. In this chapter, we continue to stress the importance of the optimality of corporate financial decisions for its stock holders. The goal of management is to maximize the price of the company's common stock and market value Investors want optimality, with the growing exception of socially responsible investing, which we discuss in the chapter appendix.

The history of corporations is full of instances where an inside group abused its position of power to enrich itself at the expense of the shareholders. Boards of

[1]Nevertheless, divergences of interest can exist in family-held companies. For example, a family member holding shares chiefly for their yield and a family shareholder drawing an income mainly from his managerial activity may differ on what is best for the company. Such conflicts are generally settled intramurally, although on occasion they have gone to court.

[2]Adolph Berle and Gardiner C. Means, *The Modern Corporation and Private Property*, MacMillan, 1933.

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9_22

directors, promoters, and management have at times taken a position of caveat emptor ("let the buyer beware") toward the investors in corporations. The exact legal obligation of the board of directors to the shareholders has never been clearly defined. The NYSE requires that the majority of all directors be "independent," as must be all directors on the audit nominating, executive-compensation, and corporate governance committees.[3] Insiders, such as top executives of the firm, cannot be considered independent. Nevertheless, by a process of financial education, and of legal and governmental administrative decisions, and by a voluntary assumption by management themselves, management has come to be viewed as having a quasi-fiduciary responsibility to the shareholders. Successful management is somewhat of a trustee of the stockholders' interests.

The inherent structure of the corporation no doubt permits management possible laxity or indifference to the welfare of the shareholders. But the fact is that shareholders as a group have received substantial economic rewards over time. It is only logical to assume that the management in the majority of corporations possesses a considerable sense of duty and loyalty.[4]

If a firm succeeds in generating exceptional earnings, then how should the profits be divided among the inside managers and the shareholders? The argument is made that the average stockholder does not contribute much to the success of his company; if he buys in after the firm is well established (as most stockholders do), he has not shouldered much of the initial risk. Thus, the argument runs, and the stockholders should be satisfied with a "normal return"; anything in excess of a "normal return" is ascribed to the superior talents or energy of the initial innovating or managerial group and is properly their reward. A normal return can be derived from its beta and the required rate of return from the CAPM. There is some merit to this view insofar as economic incentives should be provided for innovation and good management under the theory of agency. The argument fails to recognize, however, the overall function of risk capital in the economy. The tremendous gains which have on occasion accrued to the shareholders of a spectacularly successful company are in large part offset by the losses taken on many enterprises that fail. If the chance of obtaining potentially large rewards were not available, the supply of venture capital would dry up. The return for risk may seem to come after the risk is over, but to reduce this return ex post is to reduce the supply of new risk-bearers on the next round.

In an appendix to this chapter, we examine a concept of potentially non-optimizing stockholder wealth management when we examine "Socially responsible investing."

[3] J.R. Emshwiller and J.S. Lublin, "Boardrooms, 'Independent' is Debatable," *The Wall Street Journal*, March 3, 2005, C1, C4.

[4] Mr. Benjamin Graham writes "The typical management is honest, competent, and fair-minded." *Intelligent Investor*, Harper, 1959, p. 222.

General Agreement and Potential Conflicts in Management and Control

Before going on to some of the possible conflicts between shareholders and management, one might point out that the interests of these two groups are largely parallel rather than divergent. If the company is profitable, the management obtains increases in income and prestige. Company executives are rewarded with higher salaries, bonuses, stock option plans, and increased respect from their peer groups. Furthermore, although the control group's major income may derive from their salaries and other managerial perquisites and their stock holdings represent only a small fraction of the shares outstanding, these holdings may still be valuable enough to constitute a strong economic incentive to do well for the company. One cannot discount the subtle but powerful psychological motivations toward purposeful activity. If the executive groups have any morale at all, they identify with the company. If their actions have any meaningful social goal, they are directed toward achieving prosperity for the corporation. A management that is indifferent, or at worst is engaged in systematically looting the firm, has reached a state of group disorganization or disintegration. That the majority of managements are fundamentally honest in protecting the interests of the stockholders is an implicit assumption of a workable free enterprise system.[5]

Areas of Potential Conflict

Although there is an identity of major interests between shareholders and management, it is unrealistic not to recognize that these interests are likely to diverge at some important points. Usually, the conflicts are only potential, and the corporate policy (especially if the firm has good earnings) can easily accommodate the differing economic positions of the two groups.

Managerial and Board of Directors' Compensation A strongly entrenched management is able, within wide limits, to set its own compensation. Obviously, the higher the officer and executive salaries, the smaller the earnings available for the common shareholders. On the other hand, a level of managerial compensation that moves up with improved company earnings provides incentives for top-notch

[5]Some people think that all corporation managements are dishonest. Then, of course, others think all government officials are grafters, all policemen bribe takers, all union leaders gangsters, and all professors time servers. People who have these cynical and essentially narrow beliefs rarely suggest meaningful policies for a democratic society.

performance. If a firm wishes to attract exceptional administrative talent, it has to offer a competitive economic reward. One need only read the popular business press to see the wide range of opinions concerning Dick Grasso's compensation as Chairman of the New York Stock Exchange and its implications for management and the board. Nevertheless, at some point, the boundary is crossed between sufficient and too much. Stockholders could contribute to the health of their company by asking intelligent questions about executive compensation at the annual shareholders' meetings. Should the company refuse to moderate grossly unreasonable levels of officers' compensation, the stockholders' suit is a strong weapon.

About 70 years ago, Berle and Means wrote a best seller on the clash of interests involved in the separation of ownership and control in the modern corporation.[6] More recent economic and management *gurus* have attacked the problem under a new name. It is called agency theory (Jensen & Meckling, 1976). The gist of agency theory deals with the construction of a compensation or reward system so that the interests of the agent (the management) largely coincide with those of the principal (the shareholders).

The massive abuse of agency-principal relations is at the heart of the current Enron, WorldCom, and Adelphia debacles.[7] The perceived evidence is that the inside owners and managers raided the till for the enhancement of their personal enjoyment and wealth at the expense of the outside shareholders. Beneath the froth and turmoil, the basic question remains: how is corporate governance to be structured and how is management to be paid so that its interests coincide with the long-run prosperity of the shareholders?

Executive Compensation

Economists since the time of Roberts (1959) and Lewellen (1968) have empirically studied the problem of executive (CEO) compensation. Several issues are involved. Compensation should include a cash salary, a cash bonus, stock option exercises, or better yet restricted grants of stock which must be held for a specified time before they can be sold. The determinants of executive compensation may often be placed within two categories: those reflecting the size of the firm and those reflecting the performance of the firm. Roberts used size variables such as sales, assets, and profits as independent variables in the compensation equation estimation. There is, of course, severe multicollinearity among sales, assets, and profits, for many firms. One will use only the sales variable in this chapter. One can use 5-year least squares (5LS) growth rates in sales (Sales5LS), earnings per share (EPS5LS), and total returns to stockholders (TR5LS), as well as ROE, as performance measures. One can use the Compustat EXECCOMP database from the WRDS system, which

[6]Berle and Means, op. cit.

[7]See chapter "Liquidation, Failure, Bankruptcy, and Reorganization."

Table 1 Mean executive compensation ($ thousands)

Year	N	Salary	Cash bonus	Total direct compensation
1992	433	$622.7	$514.5	$2810.6
1993	1157	542.2	427.9	2159.4
1994	1548	514.2	436.7	1646.4
1995	1600	530.7	490.4	2009.9
1996	1651	548.4	594.7	2587.2
1997	1674	563.9	620.8	3570.4
1998	1731	579.9	604.8	4620.1
1999	1811	582.8	692.8	4187.5
2000	1791	607.1	734.4	6130.7
2001	1637	646.3	664.8	4549.7
2002	538	805.9	1083.7	5801.7

Table 2 Pooled regression determinants of executive compensation 1992–2002

Compensation variable	Constant	Sales	ROE	Sales5LS	EPS5LS	TR5LS	F	$\bar{R}^2$
Salary (t)	.054 (4.84)	.449 (51.23)	.155 (3.35)	−.169 (−6.76)	0.021 (−1.39)	0.019 (−1.24)	554	.249
Bonus	.010 (0.74)	.305 (28.85)	.144 (2.56)	.011 (0.35)	.008 (0.46)	.202 (11.09)	216	.114
Total direct compensation	−.022 (−1.64)	.237 (22.09)	.177 (3.13)	.061 (2.01)	0.008 (0.44)	.229 (12.42)	163	.089

derives its data from proxy statements and 7OK filings with the SEC. The Compustat executive compensation means are shown in Table 1 and indicate that cash bonuses and total direct compensation levels have risen during the 1992–2002 period, particularly relative to a cash salary.

One can pool the data during the 1992–2002 period, and ran regressions for three compensation variables the reader is referred to Table 2 for pooled regression results. CEO salary is positively, and significantly, associated with sales level and the return on equity, ROE. CEO salary is not a function of sales and EPS growth or the 5-year return to stockholders. CEO bonus is positively and significantly associated with sales, ROE, and 5-year return to stockholders. Total direct CEO compensation, including stock options exercised, is a function of sales, ROE, 5-year sales growth, and the 5-year return to stockholders.

The empirical evidence supporting size and performance measures is reassuring, particularly given the growth in bonuses and total direct compensation. A well-known executive compensation study was produced by Lewellen (1968), who analyzed salary and bonus and total compensation of 50 large industrial companies during the 1945–1963 period. Lewellen found that compensation, defined as salary plus bonus, was statistically associated with sales, net income, and the market value of the firm. There was no statistically significant correlation between the performance and size measures and total compensation. Using the EXECCOMP database during the 1992–2002 period, one finds statistically significant correlations between

Table 3 Executive compensation correlations (Lewellen, 1968)

	Salary + bonus (t)	**Total compensation (t)**
Sales	.285 (1.71)	.213 (1.43)
NI	.253 (1.77)	.155 (1.03)
MKTVAL	.275 (1.88)	.203 (1.43)
Executive compensation database		
Sales	.349 (46.4)	.192 (24.4)
NI	.225 (28.7)	.112 (14.0)
MKTVAL	.324 (42.6)	.302 (39.4)

Where *NI* net income, *MKTVAL* market value of firm

salary and bonus and sales, net income, and market value of the firm. Moreover, one finds highly statistically significant correlations between total compensation and size and performance measures. One finds very similar correlation coefficients in the Lewellen (1968) study and the analysis of the EXECCOMP database. Executive compensation is correlated with the size and performance measures of a large sample of firms during the 1992–2002 period (Table 3).

A residual plot reveals no evidence of heteroscedasticity, and the condition number of 2.28 reveals no evidence of multicollinearity in the total direct compensation pooled regression.

Recent CEO compensation research by Dow and Raposo (2005) recommends that shareholders should commit to a policy of high pay, with high-powered initial incentive packages, to provide an atmosphere for CEOs to improve strategy making and curb CEO tendencies for regarding restructuring and other dramatic events. Dow and Raposo further suggest that shareholders never precommit to pay very high compensation packages required to implement dramatic change.

Board of Directors

Supposedly the board of directors are elected by the shareholders; in fact, they are usually nominated by the management and the subsequent election is perfunctory. In general, the members of the board are honorable citizens, competent in their various professions. The managers of TIAA (the largest pension fund in the country) assert that the majority of the board should be outside directors, as distinct from inside directors who hold managerial positions. Most importantly, the outside directors should comprise the majority of the audit and the managerial compensation committees. Given the proper incentives, the directors could perform their job as the unbiased questioners and broad supervisors of managerial policies.

Shareholders should look at the director's cash pay. The members of the board at Enron were getting $350,000 a year, although most directors of publicly traded firms make $35,000–$40,000, annually. Beware when the cash compensation of the directors seems too high! Their job has become too precious, and they are unlikely to question or scrutinize any managerial report.

Of course, the board should be rewarded. They have an important balancing function within the governing framework of the corporation. However, their current annual cash pay and the accompanying perks should be moderate. Their major compensation should be an award of substantial amounts of common stock, restricted, not saleable for 5 years or after retirement whichever comes later.

Compensation in the form of restricted shares would focus the directors on the interests of the long-run stockholders. The board should be concerned about helping shape policies for a viable firm which generates a satisfying return for its shareholders over time.

The compensation scheme for the top executives might differ somewhat from that proposed for the board of directors. Because, in contrast to the directors, the manager's main current income comes from their employment with the company, their cash pay should be higher. It should be set at a level enabling them to maintain a lifestyle proper to their status, perhaps at about the amount of the average pay for a major league ballplayer. Bonuses and other major additional rewards should be in the form of restricted or deferred saleable shares.

Stock Options

The short-term stock option, a most harmful and deleterious form of award, should be abolished. The stock option entitles the holder to buy shares in the near future at a fixed price which is set below the forecasted market. Presumably, the option ties the managers to the shareholder's interest of maximizing the value of the company's stock. But the stock option fixes the manager's attention on the short-term horizon. Unfortunately, the most lucrative way to cash in on an option is to manipulate the books, defer expenses, and anticipate revenues, so that the market is fooled and the price of the shares rises sharply just at the time you prepare to sell your option-acquired stock.

A stock option plan giving the officers an opportunity to purchase company shares at a previously fixed price is a type of compensation whose presumed value is contingent on good economic performance by the company. Unless the management's activities cause an appreciation of the corporation's shares in the market, the executives do not realize any significant gain from their options. However, as noted previously, in many instances, short-term options have proven a disaster. They have encouraged shortsighted managers to indulge in manipulative activities that boomed the speculative market value of the shares at the expense of the longer-run interests of the company.

Moreover, the shareholders may fail to realize how much economic value they may be allowing to the management in the form of stock options. Because no asset leaves the company nor is any explicit liability created, the corporation's accounts do not register the impact of the option. Yet these options represent a potential dilution of the equity of the shareholders, and if the number of shares under option is a large enough fraction of the total shares outstanding, exercising the option could reduce

the earnings per share. The existence of the options can act as a drag on the potential appreciation of the common stock.

Worse yet is the case where large options and saleable shares are granted to a top manager just before his retirement. Here the manager was happy if financial activities could be manipulated so that the stock reached the high point at the time of his leaving. It didn't matter if this price was not likely to be supported by the market over any length of time. As the French expression goes, *aprés moi l' deluge*. The disaster comes after I'm gone.

Bonuses

In addition to their regular salary, many corporation officers receive bonuses based on the company's earnings. Generally, the bonus increases if the firm's profits exceed a base level. A bonus plan has many merits since it lowers managerial costs in poor years, when the company can least afford them, and increases them in years when the company's economic position is better. By making part of the officers' incomes contingent on profitable operations, it furnishes a strong incentive for conscientious management. It is important to re-examine the bonus formula periodically to ascertain whether the base level of earnings has been left at an inappropriately low level without proper consideration to the subsequent growth of the company.

Pension plans are another means of executive remuneration that have come into favor. The promise of a pension may be more valuable to the recipient than an immediate increase in salary of the same amount, since pension payments are likely to be made when his income tax bracket is considerably lower.[8] A generous pension plan gives the firm an advantage in attracting and holding good managerial talent. But pensions may seem unduly high when, as sometimes happens, they are only slightly lower than the active officer's salary and generous pensions that were not voted in advance but are settled upon shortly before the executives' retirement may be somewhat suspect.

A number of sharp changes have taken place in the structure of the economy. Contrary to the Professor Joseph Schumpeter prognosis, management of modern industry may not have devolved into a set of staid, standard procedures. The old standard industries are gone, and the new industries staffed by technicians may be considerably more difficult to manage. (As the expression goes, managing technicians may be compared to herding cats.) We are in an era of innovation and rapid change. Catching the right wave leads to success and big stock returns, while missing the opportunity results in failure. If the risk-return ratio has widened, perhaps we might also expect a rise in the compensation paid for decision-making.

[8]Even a highly paid executive may find it difficult to save as much as he would like. A pension plan represents a method of forced or involuntary saving.

Dividends, Buybacks, and Retained Earnings The theoretical value of common stock is the discounted value of its future cash dividends and net buybacks over purchase price plus a possible final sale or liquidation value. Since the present value of a future dividend or buyback cannot equal that of the immediate payments, the current level of buybacks and dividends is important in setting the intrinsic value of the stock, as we discussed in chapter "The Equity of the Corporation: Common and Preferred Stock." Stockholders are, therefore, extremely interested in the management's dividend or buyback policies. Retained earnings have value to the stockholder only as they can be invested profitably at a rate of return within the company that at a minimum equals the market equity discount rate and thereby generates increased earnings and future dividends.[9]

Management may evaluate the importance of dividends or buybacks versus retained earnings differently than the shareholders. Usually, the management obtains more of its current income from salaries and bonuses than from the cash throw off on the shares it may hold. Salaries are likely to continue as long as the firm is operating. Retained earnings appear to be a source of capital carrying little obligation. Its use increases the safety and survival chances of the firm. To a conservatively oriented management, keeping the firm alive (and assuring continuing salaries) may appear more worthwhile than keeping the return on the shareholder's investment as high as possible.

A conflict between the stockholders and the company officers over the level of retained earnings can also arise if the management's time preference for future income differs from that of the shareholders. Or the management may be niggardly with cash outflow because it overestimates the relative profitability of investment in the company with the alternatives open to the shareholders in other areas of the economy.

Excessively Conservative Financial or Asset Structures An excessively conservative financial structure has a disproportionate amount of equity financing and relatively little debt. Of course, the degree of conservatism, or risk, in the financial structure must be judged in the light of risks the firm faces. An overly conservative asset structure contains a high proportion of liquid, low-earning assets in relation to the needs of the business. Quite possibly, a company could have both an ultraconservative asset structure and an excessively conservative financial structure. Both conditions are the results of a timorous or greedy management.

An extremely conservative financial structure may develop if all financing is accomplished through new common stock issues. More than likely, it is the corollary of a niggardly cash buyback or dividend policy. An ultraconservative financial structure results if internally generated funds are used to finance the firm to the exclusion of other sources.

[9]A complicating factor in valuing future returns may be the differing tax status of the individual shareholders. A stockholder with a high marginal present tax rate may prefer increased future dividends and growth over present returns. If the dividend policies are consistent in pattern, probably each firm eventually acquires stockholders of appropriate tax status.

A financial structure containing a minimal debt component relieves the management of the fear of failure; however, it raises the cost of capital to the firm. Stockholders are deprived of the extra profits that would accrue through the use of judicious leverage. Indeed, some writers have suggested that a firm should carry some debt as a matter of principle since it is a constant reminder to the management that there is a cost of capital. The effort and planning required to service the debt should help to keep the managers from becoming complacent.

On the asset side of the ledger, excessive or redundant net working capital is another symptom of an unenterprising company administration. A large net working capital base helps assure survival of the firm, but survival without profits is hardly desirable. A firm that does not economize on its use of net current funds, other things being equal, has a low rate of return on its total assets.

Excessive net working capital can develop because of a low payout policy, with the retained funds invested in liquid assets. If a high rate of depreciation is covered by operations and the funds obtained do not go back into capital assets, the same condition results. A firm undergoing a decline in its scale of operations is more likely to have redundant net working capital than one that is vigorously expanding. Nevertheless, the managements of static firms are often reluctant to give up any of the funds they have under their control although they may have no plans for their productive use.

Expansion In the shareholders' view, expansion is worthwhile when it increases their reinvested returns. The Gordon model of stock valuation introduced in chapter "The Equity of the Corporation: Common and Preferred Stock" recognizes the importance of reinvested profits and future growth. The use of retained earnings to finance expansion is justified when the company can make the going rate of return on equity on these funds. If the management, however, is not sufficiently concerned about the stockholders, it may pursue expansion for the economic or the psychological rewards that corporation officers receive simply from being associated with a larger firm. The shareholders may well feel that the basic objective of the corporation is sacrificed if the management follows a policy of "dry" or profitless expansion.

In many instances, "dry expansion" is a further reflection of the failure of some corporation administrations to recognize the implicit cost of retained earnings. The company may have a consistently low dividend payout rate and fail to find profitable outlets for its funds. Yet the management may be content because it feels that nothing need be paid on the "surplus" portion of the equity. In other cases, the management may tend to overrate investment opportunities in its own firm, or it may do its shareholders a disservice by acquiring subsidiaries at costs which their earnings potential do not justify.

The management's interests and the welfare of the shareholders are not equally served if the company officers bring about company growth purely in order to enhance management prestige, to raise managerial perquisites, or to assuage their power drives.

Liquidating or Selling the Firm The natural desire of the company officers to continue operations conflicts with the interests of the shareholders if the foreseeable

returns for the firm are not sufficient to cover the value of the capital which can be withdrawn from the firm. The situation is, of course, similar if the company or its assets can be sold as a unit for an amount exceeding the sum of the going market price of its securities. Because a voluntary liquidation or sale eliminates executive incomes, it may well meet with management resistance.

A firm whose common shares sell on the market at a price measurably below their liquidating asset value invites purchase by individuals or syndicates who intend to turn a quick profit by liquidating the company. Such purchasers are often called "raiders"—a term strongly connoting disapproval, especially when used by conservative managements. Nevertheless, the raiders may perform a useful economic function. By their very existence, they serve as a warning to those managements who fail to use their capital productively. However, the shareholders may not do as well under the raiders as they would were the original management to proceed with a careful and orderly liquidation. Nevertheless, a raid and subsequent liquidation may still be an economic improvement over having shareholder funds slowly go to waste under the aegis of a listless management.

A management that does not wish to give up entirely may turn a liquidated operating company into a closed-end investment corporation (i.e., a company which makes its income by holding other companies' stocks and bonds). For a company having substantial carry-forward tax loss credits, becoming a holding company may have some advantages.

Risky Acquisitions

A very speculative management may attempt to make a quick fortune by making many rapid acquisitions, obtaining new properties on low margins, and mortgaging or leveraging the original company to the hilt. Such speculative activities can bring great immediate gains for all, or it can result in the loss of the original equity values in what might have been at the start a small but solid company. The shareholders cannot legitimately complain if they knowingly accept these ventures. However, if the risk-taking proclivities of the management are covered by deceptive practices, the shareholders may have just reason to believe their interests had been breached. A management holding a plethora of exercisable stock options may be especially tempted to play the stock market for short-term speculative gain. Acquisitions must be rationale. Ind that firms with low cash-adjusted leverage, CAL, those attractive as takeover targets, outperform high cash-adjusted leverage firms during the 1980–2003 period by over 11.2% annually. Lower target leverage reduces the liability acquired in the takeover, and large cash holdings increase acquirer liquidity. Cash-adjusted leverage is calculated by subtracting cash holdings from debt liabilities. CAL variable is calculated as (debt − cash)/total assets. The acquired firm's cash holdings can pay off the firm's debt, a concept very similar to the net current asset value (NCAV) concept of Graham and Dodd introduced in chapter "Risk and Return of Equity and the Capital Asset Pricing Model."

Turning Agents into Owners

The prescribed method for dealing with the agency-principal problem is to set up the compensation system so that the interests of the agent are closely tied to that of the principal. Various devices have been developed so that at least part of the managerial compensation is based on the financial performance of the firm. At least a portion of the management's income is based on their being part or quasi-owners in the firm.

Illusory growth takes place when, in evaluating a stock, investors concentrate on an apparent periodic increase in earnings per share without checking on or correcting for the trend of the rate of earnings on total assets or the so-called core business. The true measure of growth is the earnings after the acquisition rise over the projected earnings of the company's independence. The phenomenon of illusory growth is developed by the use of acquisitions or mergers after the acquiring company has somehow been initially recognized as a growth company by the market.

The easiest way to explain illusory growth is by example. Suppose we have Athos Corporation, with 100,000 shares outstanding, earning $1.00 a share, currently selling at $20.00 per share (a P/E ratio of 20 times earnings) because of a market belief in a past record of growth. Athos' total earnings is $100,000. Another firm, Zanox Inc., has 100,000 shares outstanding, has a BPS (earnings per share) of $1.00, and is considered a moderately successful firm selling at $10.00 a share (a P/E ratio of 10 times earnings). Zanox also has total earnings of $100,000.

Athos now acquires Zanox by an exchange of Athos's shares for Zanox's on a value-for-value basis. One share of Athos ($20.00 per share) is exchanged for two $10.00 Zanox shares. (In the process, the Zanox shares are retired.) At the end of the transaction, Athos has 150,000 shares outstanding, but it has added $100,000 to its earnings, bringing the total to $200,000. Dividing through by the new number of shares, $200,000 divided by 150,000 total shares, now gives us an EPS of $1.33. Lo and behold, Athos has raised its earnings per share by 33.0%. It may be now further confirmed in its reputation as a growth company, and its high P/E ratio justified.

Eventually, the illusory growth firm disappoints its investors and the price of its shares drops sharply. As the original growth company merges with more and more average or slow growth firms, the growth rate of earnings on the total asset base must begin to decline. Externally, it becomes more and more difficult to find net new desirable acquisitions, and internally, the managerial problems of the conglomerate become unwieldy. But more basically, a simple mathematical relationship begins to undermine the whole process of illusory growth. When the original company was relatively small, the acquisition of another fair-sized company on the basis of a favorable relationship of P/E ratios raised the EPS of the acquiring company noticeably. When the so-called growth company reaches a large size, it requires larger and larger acquisitions to maintain the picture of rising earnings per share.

Again, a small example may illustrate the problem. Suppose Athos Conglomerate Inc. now has 2,000,000 shares, has $2,000,000 of earnings (EPS of $1.00), and because of a growth reputation sells at $20.00 per share (P/E of 20×). Silon Corporation has 100,000 shares, has earnings of $100,000 (EPS of $1.00), and a

moderate price of $10.00 per share (P/E of 10×). Athos acquires Silon through an exchange of shares, value for value, one share of Athos for two of Silon. After the acquisition, Athos has total earnings of $2,100,000 and total shares outstanding of 2,050,000. The earnings per share goes up from $1.00 to $1.024; but this is hardly a noticeable increase and nowhere near what its followers have expected from Athos in the past. In short, as the illusory growth company becomes larger and larger, it begins to outgrow its environment. It requires larger and larger acquisitions to maintain the appearance of earnings growth, and new gulps of these sources of financial nourishment might not be as available as they were earlier in the conglomerate's career.

Finally, as the investors become disillusioned, a more reasonable P/E ratio is given to the stock and the price of the shares falls sharply. The last round of buyers may experience severe losses on their holdings. Illusionary growth was behind many of the recent financial scandals.

The Diseconomies of Financial Scams

From a broad point of view, management perfections and frailties are but small glitches in the functioning of the financial markets. Yet they do entail a cost. The unnecessary volatility they add to the markets, and the risk of loss because of fraudulent or near-fraudulent behavior, adds to the risk premium carried by the economy's cost of capital. The rise in the cost of capital reduces worthwhile real investment and slows the rate of economic improvement and growth. Moreover, the incidents of wild behavior or outright crookedness bring forth a strong social/psychological reaction of anger and resentment so that some reasonable and normal operations or innovations are mistrusted and hindered. This reaction adds friction and inefficiencies to the function of the financial markets.

The operations of the financial and capital markets are an essential element in the functioning of a free enterprise, price directed economy. Yet in many ways, the financial markets are fragile institutions; in a large part, their operation is made possible only by a considerable degree of trust between the suppliers and users of funds. (In the current scholarly literature, this connection has been extensively examined under the heading of *agency-principal relations*.) Insofar as fake-out devices lead to a waste of resources, a reduction of trust, and the subsequent employment of more elaborate examining and policing devices, they impose an extra cost on the overall cost of capital.

Insider Trading

The chief officers of a corporation, the directors, and major stockholders are in a position to obtain advance information on important developments in the company. Such "inside" information, whether favorable or unfavorable, can easily be turned

into quick trading profits on the stock market.[10] Clearly unethical would be an insider program to create speculative activity by placing tips and financial news in the market or by actually manipulating financial reports and the timing of financial announcements.

The Securities Exchange Act of 1934 substantially restrained the possible speculative activities of management and other insider groups. No longer can such groups operate secretly; any changes in the shareholdings of corporation officers, directors, and major stockholders must be reported, and these changes are published weekly by the Securities and Exchange Commission. Prison terms and/or substantial fines may await insiders guilty of deliberately leaking information or of engaging in other manipulative practices. Bettis, Coles, and Lemmon (2000) surveyed some 1900 member firms of the American Society of Corporate Secretaries in November 1996 with regard to corporate insider trading restrictions. Some 35% of firms (633) responded to the survey. Of the firms responding to the survey, 576, or 92%, had corporate policies regulating or restricting insider trading. The most common policy, involving approximately 75% of the sample, required prior approval of insider trades. Moreover, most companies had blackout periods. Blackout periods are periods during which trading by insiders is not allowed. The most common blackout period was 10 days, generally days +3 through +12 of quarterly earnings announcements. Bettis et al. (2000) found that blackout days reduce the bid-ask spread by 2 basis points, or 8.5%, and insider trading profitability is less in blackout periods than in trading windows during the 1992–1997 period. Insiders earned abnormal profits in allowed trading windows of 0.58%. Bettis, Guerard Jr., and McAuley (2015) reported that (legal) insider trader (LIT) data could be incorporated into traditional stock selection modeling and used with multifactor risk models (see chapter "Multifactor Risk Models and Portfolio Construction and Management"), such that LIT can enhance stockholder returns.

Conflict of Interests No legal rule prohibits company officers or directors from dealing with the corporation on their own account, nor from having the corporation contract with other companies in which they have a personal interest. The interest of the corporation official must be fully disclosed, however, and the transaction must be open. A company official who has a concealed interest in another firm which is a supplier or customer of the corporation is in an especially vulnerable position. A corporation officer who has profited personally at the expense of the company is subject to suit by the corporation or by stockholders on behalf of the company. Major corporation officers should divest themselves of conflicting interests or at least abstain from participating in decisions when these interests are involved.

[10]Trading profits can be made on unfavorable developments, which would tend to depress the price of the stock, by "selling short."

Stockholder Remedies

What can the stockholder do if his management seems incompetent or flouts his basic economic interests? A number of alternatives have been suggested, although none of them appear to be entirely efficacious.

Sell His Shares Some authorities advise the unhappy stockholder to sell his shares. This is the so-called practical approach. It ignores the fact that although this course of action may work for one shareholder, if any large part of the shareholders attempt the same remedy, the price of the stock would drop sharply on the market, entailing a considerable loss for all the stockholders. The management would suffer only insofar as they held any shares themselves.

Institute a Stockholder Suit A stockholder suit is valid only if the management has been guilty of breaching its quasi-fiduciary position. It will not hold for many of the situations described in this chapter. In addition, stockholder suits are difficult and expensive. The mores and culture of the investment community frown upon them; and indeed even shareholders who have benefited by the successful outcome of a "derivative" suit[11] often look askance upon the shareholders who instituted the action. It is true that the remedy of the stockholder suit has often been abused; some shareholders have used it to further their own special interests. Nevertheless, it is a salutary device even if its existence—the possibility of a suit—serves only to remind the management of the limitation of its powers.

Attend and Vote at Annual Stockholders' Meetings The vast majority of shareholders do not attend the annual meetings. But, contrary to common belief, it is possible (assuming a responsive management) to have considerable influence at these meetings. Intelligent, incisive questions force the management to consider their policies carefully. Independent stockholder proposals can be submitted at the annual meetings. These proposals are usually voted down by an overwhelming majority, but if the proposals pick up support from 1 year to the next, they exercise considerable impact on a responsible management.[12] Many such independent proposals are eventually adopted by management itself.

Organize a Proxy Fight to Vote Out the Management If the stockholder is sufficiently outraged by the management, he may attempt to vote in a new board of directors who will appoint a new management at the shareholders' annual meeting. The shareholder can succeed only if he can enlist a majority of the voting stock in his campaign. The management has all the advantages of the "ins" in such a struggle. It has better access to the mailing list of stockholders, to whom it can send

[11]The stockholders' action is called a "derivative" suit because they sue on behalf of the corporation.

[12]Unfortunately, the notion of the "divine right" of management is so imbedded in the investment community's thinking that the questioning, active, independent shareholder must often expect to arouse some hostility—mostly from his fellow shareholders.

proxy requests.[13] It can use company funds and personnel to help run its proxy solicitations and can count on a considerable number of proxies that are almost automatically returned to the management.

Indeed the smaller "independent" stockholder can scarcely hope to win a proxy fight. Even if he finds himself allied with some powerful financial interest seeking to gain control of the management, the outcome is uncertain. And, indeed, in such a case, it may be better to stay with the old management.

"Respectable" entrenched managements look upon the organizers of proxy fights (even when they may represent major, responsible shareholders) with disfavor. The insurgents are called "raiders." Nevertheless, successful management seldom faces a proxy fight. Indeed, an occasional proxy fight probably does the investment community no harm, but rather encourages incumbent managements to seek improvement in their stockholder relations.

To Whom Is Management Responsible?

Social-minded writers have long called for the arrival of a new type of management, one concerned with economic and social justice.[14] Management's primary concern would no longer rest with the maximizing of profits—i.e., with the economic interests of the shareholders. Because the "separation of ownership and control" is largely an accomplished fact, the corporation administrators have acquired tremendous powers independent of any likely stockholder interference. They should, therefore, devote themselves largely to deciding "fair prices," to developing pleasant communities, and to determining "just wages." The exponents of such developments seem not to fear that the corporate managements will abuse their powers. They feel that a properly trained manager would acquire an ethic of self-discipline or, better, of self-restraint, which could be relied upon to moderate any undue selfishness of the managerial groups.

To the more orthodox scholars of economics and finance, this sociological derived thesis seems somewhat romantic. It is doubtful whether the widening area of managerial independence represents a complete divorce from the dominant interests of the risk-taking suppliers of ownership capital, which is to maximize the value to its owners.

It is questionable even if it is granted that benevolent corporation administrations are free to make their own economic policy, whether it is desirable that they do

[13] A proxy is a grant to a designated person to vote one's stock if one cannot attend the meeting. Proxies can be requested for voting on the membership for the board of directors or for voting on specific proposals. The SEC requires full disclosure of the issue on the solicitation of the proxies.

[14] See Adolph Berle and Gardiner C. Means, op. cit. (especially the last chapter). In this book, Berle merely hoped for the day when management would mediate between the stockholders, the public, and labor. In a later book, *Twentieth Century Capitalist Revolution*, Harcourt, Brace, and World, 1954, he is deceptively pleased that that day had arrived.

so. No matter how fine the ethics and social sensibility of the managers, is it wise to encourage them to believe that they should make decisions that transcend the market? This argument should not be misinterpreted. We can agree that managers should be honest, ethical, and law-abiding—that they act within the framework of accepted social standards. As a matter of enlightened self-interest, management should not debase their product, produce unworkable or dangerously defective goods, put men to work under unsafe or unsanitary conditions, cheat on taxes, advertise falsely, or engage in collusive marketing practices. On the positive side, the management of large corporations can practice good community relations, contribute to worthwhile public or educational projects, engage in good labor practices, and encourage cultural and intellectual development and free exchange of ideas in the community. Nevertheless, these are not the main activities of corporation managements. We include Appendix to discuss recent research in socially responsible investing, SRI. They should not consider themselves the dispensers and allocators of benefits to the rest of society, for nothing could be surer to raise the resentment of the rest of the citizenry. Independent men will tend to dislike being beholden to feudalism—even a benevolent feudalism.

Summary and Conclusions

The "new capitalism" implies a denigration of the economic function of management. Maximizing the long-run profits of the firm, in this thesis, becomes a crass money-grubbing activity. There is a considerable lack of appreciation of economic analysis in this view. Management performs important economic functions when it bargains hard but honestly. Management must use the criterion of economic efficiency if factors of production are not to be wasted or misemployed, if productivity is to be improved, if the investment of new capital is to be directed to areas where it is most needed, and if, to sum up, a system of free market prices is to work at all. As we discussed in chapter "Introduction: Capital Formation, Risk, and the Corporation," management must work to maximize stockholder wealth. Clearly, with SRI, "one can do good while doing good." We explore SRI in the chapter "Appendix".

Appendix: Socially Responsible Investing

During the 1990s, many investors sought to invest in securities that were "socially responsible." There are, of course, many definitions of socially responsible investment (SRI). For example, in May 1990, the Domini 400 Social Index (DSI) was created to replicate the returns of the Standard & Poor's 500 (S&P500) Index. The Domini 400 Social Index was created by Kinder, Lydenberg, and Domini (KLD) to select companies with positive social and environmental records: the criteria involved community relations, diversity, employee relations, human rights, safety,

Table 4 Time period returns

Stocks	Year-to-date (%)	One year	Three years	Five years	Ten years	Since inception
DSI	16.46	14.05	−10.96	3.09	10.91	373.0
S&P500	15.93	12.07	−11.41	2.51	10.09	307.9

environment, and (new) corporate governance. The DSI's social criteria led it to exclude stocks that (1) derived more than 2% of gross revenues from military weapons; (2) firms with tobacco, gambling, and alcohol sales; and (3) firms that own or share in nuclear power plants. The returns of the DSI stocks have exceeded the returns of the S&P 500 stocks since 1990, although there is a beta bias (exceeding 1.0) and a growth bias to the DSI portfolio. The relative returns of the DSI and S&P 500 stocks, as of August 30, 2003, have been (Table 4):

The relative returns on SRI portfolios can be volatile.[15] There can be additional social criteria. One may not want to invest in stocks with poor environmental records, or poor employment (unions, safety, pension concerns), or diversity (woman and/or minority CEO, or boards of directors, family or "gay" rights, hiring of disabled persons) records. There can be an almost infinite number of combinations of social criteria exclusions that can be used to create SRI portfolios. Do these social criteria affect portfolio performance and stock prices? The recent empirical evidence, as reported in studies winning the Moskowitz Prize for Research in Socially Responsible Investment, supports the notion that SRI portfolios do not perform statistically different from (traditional) equity portfolios, and stock prices of socially responsible stocks are not determined by different criteria than commons stocks in general. The Moskowitz Prize winning studies are Guerard (1997b), Waddock and Graves (1997), Dowell, Hart, and Yeung (2000). There is no statistically significant cost of capital differences in the SRI stocks and non-SRI stocks.

The academic literature in the last decade has reported both the shadow cost and return enhancement views of ESG-screened investment portfolios. A number of literature reviews and meta-analyses found evidence on both sides of the SRI/ESG performance question. Friede, Busch, and Bassen (2015) review more than 2200 studies and found that "The results show that the business case for ESG investing is empirically very well founded." Sjöström (2011) reviews 21 studies in a meta-

[15]As the Guerard and Schwartz (2007) reported that the relative Domini 400 Social Index returns were trailing the S&P 500 Index as of August 31, 2006.

Time period returns						
Stocks	Year-to-date (%)	One year	Three years	Five years	Ten years	Since inception
DSI	3.43	6.24	9.04	4.07	9.17	11.73
S&P500	5.80	8.88	10.95	4.65	8.92	11.08

Source: www.kld.com/performance

analytic framework and found, contra Friede et al. that "results point in all directions, and . . . there is no clear link between SRI and financial performance."

For stock portfolios, Evans and Peiris (2010) found evidence of a positive relationship between ESG rating criteria and accounting measures of performance including return on assets and market-to-book ratios. Fulton, Kahn, and Sharples (2012) argue that a positive correlation exists between one aspect of ESG, in particular sustainability and superior risk-adjusted returns. Park and Moon use KLD ratings and found that for S&P 500 constituents, top quantile firms outperform by as much as 6.24% over the 1991–2006 period controlling for standard factor model risks. From a global perspective and focusing on corporate social responsibility (CSR), Renneboog, ter Horst, and Zhang (2008) found that some CSR measures are associated with shareholder value, but others are associated with destruction of shareholder value.

Statman analyzes SRI/ESG performance including comparing SRI indexes to the S&P 500 historically, finding little difference in performance. This is echoed in part by Geczy, Stambaugh, and Levin (2005) who in a sophisticated framework create SRI/ESG mutual fund portfolios. They found that index-oriented portfolios under CAPM and zero alpha parameters seem to impose relatively small costs, as little as 1 or 2 basis points per month, compared to non-SRI portfolios. However, when informative alpha beliefs or beliefs in the importance of investment style are present, relative performance deteriorates. Barnett and Salomon argue that a nonlinear relationship exists between social and financial performance, finding that funds with intense screens outperform by weeding out "bad" firms, and funds that have weak screens outperform in risk-adjusted measures accounting for diversification, while funds in the middle underperform.

Corporate governance has been studied. Liang and Renneboog reported that countries with higher social responsibility (sustainability) ratings were more likely to be civil law, rather than common law, countries, with the Scandinavian countries having the highest scores. Corporate social responsibility is positively associated with lower stockholder litigation risk and strong labor regulations. Liang and Renneboog stress that corporate social responsibility reflects the supply of socially responsible behavior by firms and the demand for corporate social responsibility practices by society.

Bialkowski and Starks (2017) reported that SRI funds attracted more flows, on average, than conventional mutual funds between 1999 and 2011. SRI funds had positive net flow in all but two quarters of the 13-year period; and SRI funds had statistically significant inflow following environmental disasters, such as the BP oil spill and the Fukushima Daiichi nuclear meltdown, and accounting scandals, Enron, Tyco, and WorldCom. SRI funds had higher exposure to (MSCI) ESG values that persisted across time. Riedl and Smeets (2017) surveyed some 3382 socially responsible investors and 35,000 randomly selected investors using administrative individual investor data in the Netherlands. Riedl and Smeets reported with an 8% response rate from conventional investors and a 12% response rate from SRI investors that SRI investors contribute more to charity than conventional investors, invest primarily in SRI equity funds than conventional investors, had longer holding periods than

conventional investors, and expected lower (marginally) returns than conventional investors. Investors holding a university degree are more likely to be SRI investors. Our results are consistent with the recent findings on diversity reported in Manconi, Rizzo, and Spalt (2015) and Kim and Starks (2016). Manconi et al. (2015) reported that diversity enhanced returns more than the Fama-French five-factor components and that the market-weighted diversity portfolios are enhanced relative to the equal-weighted diversity portfolios for the 2001–2014 time period. Kim and Starks (2016) tested for unique skills of female members of corporate boards.

How have the costs of SRI changed since the initial studies and enhancements to the database? In this context, we re-examine the KLD database from its 1991 inception through 2015. Over time, the KLD database was subject to enhancements resulting from acquisitions and other methodology changes. For example, in 2000 the human rights category was added (Galema, Plantinga, & Scholtens, 2008), in 2002 governance was added (Statman & Glushkov, 2009), and in 2010 KLD decided to rank companies only on issues relevant for their industry as opposed to all issues. ESG scores are naturally persistent in the short run, as reported and confirmed by Wimmer (2012) in reviewing SRI mutual funds. However, Wimmer (2012) also found that persistence declines after approximately 2 years, making rebalancing crucial for SRI portfolios. Such instability of the dataset results in many versions of the final KLD scores constructed in academic literature, causing potential difficulties in comparing results across papers and time. It also highlights the potential dependency of results on the exact measurement methods used. As a result, recent literature has focused on ways to adjust and normalize the scores to account for the lack of stability in KLD dimensions over time.

From the early KLD studies (Sharfman, 1996) and continuing to the most recent ones (Statman and Glushkov, 2009), there has been an ongoing discussion about the challenges of creating a unique KLD-based score. The simplest way that sums all strengths and subtracts all weaknesses incurs its own set of biases and imbalances driven by data structure rather than companies' ESG attributes. Dorfleitner, Utz, and Wimmer (2014) study the relation between ESG score performance and stock performance in various markets worldwide, reiterating global evidence of positive association between firm ESG ratings and subsequent returns; however, the bias remains. The earlier literature attempted to address the implicit bias arising from weighting each issue equally. For example, in order to avoid treating each ESG strength and weakness as equally important, Waddock and Graves (1997) rely on the issues weighting scheme developed. Because such weightings are highly subjective, they are no longer used in more recent studies. For example, employee relations strengths are evaluated on 10 individual variables, with a maximum score of 8, while human rights strengths are evaluated on 3 variables with a maximum score of 2. Hence, because of the uneven distribution, the raw score will be much more impacted by the employee relations strengths than the human rights strengths. The same issues affect the weights of strengths vs concerns. Depending on which area has a larger number of evaluated metrics, that area will get higher implied weightings in the overall raw calculations.

Another difficulty arises because of the changing nature of the KLD dataset over time. Specifically, as the number of strengths and weaknesses changes in each category, summing the raw strengths and weakness, such as was the earlier practice, creates score dynamics that are influenced by the dataset construction rather than only by the company's changing ESG policies.

Kempf and Osthoff (2007) address this problem by normalizing the net scores within each of the six categories. In addition, they introduce a way to binary transform the weakness into the same direction as the strengths. Using decile spread portfolios, their paper documents statistically positive KLD score alphas for the 1992–2004 period. In addition, they introduce an important methodology to normalize the rankings by the sector to which the company belongs. This reduces the potential sector and resulting factor biases in the rankings. Finally, they create a version of the rank that excludes all companies that have at least one of the six controversial scores. Statman and Glushkov (2009) test a variation of the Kempf and Osthoff (2007) methodology by excluding companies that have a zero value in both the strength and the weakness fields in each category, claiming that those companies were not reviewed by KLD. They exclude the governance component because of short history. Over the 1992–2007 period, they found statistically significant positive alphas for KLD long-short portfolios.

Manescu (2011) adds an additional refinement that normalizes strengths and weaknesses separately. Since in the Kempf and Osthoff (2007) and Statman and Glushkov (2009) methodologies a lack of a weakness is considered a strength, Manescu recognizes that it is important to normalize strengths and weaknesses separately. Unlike prior studies, Manescu does not exclude any categories and uses all seven categories. She found that only community relations scores have statistically significant abnormal positive returns during the 1992–2008 period.

Let us take a new look at SRI/ESG investing with the CTEF variable, featured in chapters 14 and 15, as the stock selection model and the KLD social criteria for the 2000–2014 time period. The appendix draws heavily from Geczy Guerard, and Samonov (2020).

From Raw to Normalized Score Definitions

Raw Score

Inside each of the seven subcategories (governance, community, diversity, employee relations, environment, human rights, and product safety), KLD provides binary ratings on multiple individual measures of strengths and concerns criteria. For each of the seven categories and for each company in each year, the category raw net score is the sum of category strengths minus sum of category weaknesses. The total raw net score is the sum of the strengths across all categories minus the sum of all the weaknesses across all categories. There is total net score only if both strength and weakness exist. If strengths or weaknesses are missing entirely, the net score is NaN for that company in that year.

Normalized Scores

(a) ***Category score***: For each subcategory, for each company in each year, we first normalize strengths (weaknesses) by dividing the sum of strengths (weaknesses) Booleans by the concurrent dimension of strengths (weaknesses), where the dimension is defined as the number of evaluated variables in each category during each year As the number of evaluated variables varies over time and differs across ESG categories, doing this adjustment normalizes the data and allows for cleaner review of company's ESG information rather than KLD's evaluation methodology.

For example, for the strengths of the diversity category, Company A in year 1 is measured by two variables (dimension is 2); its scores are 1 and 1 respectively; hence, its normalized strength score is 1 + 1/2 = 1, while its raw strength score is 2 Then, let's say that next year, Company B has three evaluated diversity strengths criteria with scores of 1, 1, and 0. While the raw score would be the same (2), the normalized score in year 2 is 2/3. This normalization captures the fact that in year 2, the company could have earned a maximum rating of 3, while in year 1, the maximum rating was 2. Manescu (2011) points out that because the dimensions of each sub-strength and sub-weakness are different, this is important.

Next, we deduct normalized weaknesses from the normalized strengths to get the category-normalized net scores. We then subtract the corresponding industry average-normalized net score from each company's normalized score, making scores industry-neutralized. Industries are defined by the ten Fama-French sectors (from Kenneth French's website[16]). By subtracting the average industry score, we neutralize large industry biases present in ESG rankings, which remain even after the normalization. The energy sector has the lowest average scores, while consumer non-durables have the highest.

(b) ***Total score***: Each subcategory-normalized and industry-neutralized score is further ranked from 1 to 100, and then the total score is calculated as the average of all subcategory scores. If one or more subcategory scores is missing, the total score equals the average of all other subcategory scores. Creating a percentile rank in order to combine across subcategories is important, because combining even the normalized scores across seven subcategories is not balanced—as can be seen in panel D of Table 1, the min/max distributions of each score are not even, which would result in varying weightings of each category. Ordinal rankings create an equal contribution to the final score from each subcategory. We suggest that this adjustment process should be the new default, a more robust starting point, for reviewing the ESG information content of the KLD data, making it more comparable over time and across categories, with equal

[16] http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

representation of various ESG issues. If materiality or other reasons are required for varying weightings across ESG issues, these methodologies can be described and tested explicitly and comparted against the robust equally weighted normalized benchmark.

Fama- and French-Weighted Portfolios

For what we term "simply weighted portfolios," we construct both equally weighted and capitalization-weighted portfolios from the normalized KLD rankings by splitting up all the stocks in the KLD universe into high and low groups every year. As the normalized KLD scores range from 1 to 100, the high group holds all the companies with scores equal or greater than 50 and the low group contains the rest. Long-short portfolios are also formed by investing in the high vs low group, and the spread return is analyzed. In addition, we create a version of the overall portfolio that excludes controversial companies, the traditionally labeled as "sin stocks." Portfolios are rebalanced annually. The net score is lagged by 3 months to accommodate KLD's update timing after year-end. Although portfolios are constructed annually, we measure returns on a monthly frequency. The monthly spread portfolio returns and associated factor regressions are presented in Table 1.

The total score as well as several subcategory long-short portfolios have a positive return, although economically small. For example, the capitalization-weighted long-short portfolio that is based on the total score and excludes the controversial companies returns 2.9% per year, statistically significant at the 5% confidence level. The return remains positive in the more recent sample from 2004 to 2015, although lower than in the early history. The human rights capitalization-weighted long-short portfolio has the highest spread of 4.68% per year, which is also statistically significant. The environment long-short portfolio has the lowest return of −0.42% that is not statistically significant. The factor loadings of these normalized and neutralized scores appear understandably different and more neutered from the traditionally reported raw scores because many of the database biases have been restored, taking out implied factor tilts. For example, the overall spread portfolio that excludes controversial companies has a positive loading on value, momentum, and size Fama and French (1992, 1995, 2008, 2016) factors, with a positive intercept, although the R-squared is less than 4%. Importantly, these factor loadings do not appear stable across portfolio weighting schemes and time and hence should not be interpreted as definitive. Instead, these regressions validate the effectiveness of normalization and support a weak positive intercept. In summary, our simple portfolio return analysis strongly supports the no-cost argument associated with the ESG rankings. In addition, it weakly supports some positive association with return in the simple portfolio settings.

The monthly spread portfolio returns and associated factor regressions are presented in Table 5.

The Environment long-short portfolio has the lowest return of −0.42% that is not statistically significant. The factor loadings of these normalized and neutralized scores appear understandably different and more neutered from the traditionally reported raw scores because many of the database biases have been restored, taking out implied factor tilts. For example, the overall spread portfolio that excludes controversial companies has positive loading on Value, Momentum, and Size Fama-French factors, with a positive intercept, although the R-squared is less than 4%. Importantly, these factor loadings do not appear stable across portfolio weighting schemes and time and hence should not be interpreted as definitive. Instead, these regressions validate the effectiveness of normalization and support a weak positive intercept. In summary, our simple portfolio return analysis strongly supports the no-cost argument associated with the ESG rankings. In addition, it weakly supports some positive association with return in the simple portfolio settings.

A Return to Optimized Portfolio Construction and Management

The majority of academic work that measures the costs of ESG investing focuses on the simply weighted portfolios, yet in practice, ESG criteria are often applied alongside expected return and risk models as well as some form of optimized settings. By introducing a realistic multifactor expected return and risk model, along with optimization settings, we can gain deeper insights into the potential costs of ESG investing as the various elements of portfolio construction interact in a more practical setting.

In an optimized setting, we build optimal portfolios using a multifactor expected return model and the APT risk model and optimizer and the Axioma statistical risk model and optimizer as we discussed in chapter "Multifactor Risk Models and Portfolio Construction and Management." We introduce the underlying stock selection model in the coming section, and combined with a KLD-based social score forecasted return, we can build mean-variance tracking error at risk (MVTaR) portfolios such that ESG/SRI values can be effectively brought into the portfolio construction analysis.

In an updated SRI/ESG analysis, Geczy, Guerard, and Samonov (2020) constructed expected returns using the earnings forecasting variable CTEF, the I/B/E/S consensus-based composite variable composed of forecasted earnings yield, earnings revisions, and earnings breadth that we discussed in chapter "Risk and Return of Equity and the Capital Asset Pricing Model." CTEF is the public form of the McKinley Capital Management (MCM) variable representing forecasted

Table 5 KLD ratings ranked long-short portfolios' different periods

March 1992–December 2015								
Total score	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.83%	7.0%	1.6%	−0.03	−0.07	0.08[a]	−0.07[a]	3.6%
Eq. weighted	1.08%	6.2%	0.5%	−0.02	0.13[a]	−0.03	0.07[a]	8.9%
Total ex controversial	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	2.90%[a]	7.1%	2.5%	−0.03	0.11[a]	0.03	0.05	3.4%
Eq. weighted	1.67%	7.1%	0.7%	−0.02	0.25[a]	−0.06	0.09[a]	20.4%
Governance	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	1.63%	6.4%	1.7%	−0.01	−0.01	0.1[a]	−0.02	1.9%
Eq. weighted	0.04%	6.5%	0.3%	−0.04	−0.07	0.07[a]	0.01	3.1%
Community	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	−0.22%	6.5%	0.0%	0.01	0.06	0.06	−0.09[a]	7.4%
Eq. weighted	1.75%	5.0%	1.1%	0.05[a]	0.05	−0.01	0.02	0.9%
Diversity	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.06%	7.5%	1.9%	−0.06	−0.21[a]	−0.31[a]	−0.01	22.2%
Eq. weighted	0.03%	5.6%	0.4%	0.01	−0.04	−0.26[a]	0.03	25.1%
Employee relations	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.54%	6.2%	1.6%	−0.01	−0.12[a]	−0.1[a]	−0.07[a]	7.1%
Eq. weighted	0.98%	4.9%	0.8%	−0.01	0.08[a]	−0.17[a]	0.05	23.5%
Environment	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	−0.42%	6.8%	0.6%	−0.05	−0.03	0.03	−0.1[a]	4.1%
Eq. weighted	0.20%	6.9%	0.2%	−0.09	0.22[a]	−0.03	0.01	19.4%
Human rights	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	4.68%	10.0%	2.7%	0.05	0.18[a]	0.12[a]	0.15[a]	8.3%
Eq. weighted	1.84%	11.0%	1.1%	−0.03	0.05	0.11	0.1[a]	2.9%
Product safety	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.65%	7.7%	1.4%	−0.01	−0.29[a]	0.19[a]	−0.03	32.3%
Eq. weighted	−0.37%	5.8%	−0.5%	−0.02	0.01	0.11[a]	0.00	2.8%
Controversial	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.52%	5.8%	1.0%	−0.04	−0.04	0.05	−0.03	0.8%
Eq. weighted	0.81%	1.9%	0.7%	0.03[a]	0.03[a]	−0.02[a]	0.01	5.0%
March 1992–December 2004								
Total score	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	2.32%	8.4%	3.7%	−0.04	−0.13	0.07	−0.05	4.0%
Eq. weighted	2.09%	7.0%	1.5%	−0.08	0.02	−0.05	0.12[a]	13.9%
Total ex controversial	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	4.17%	8.3%	2.0%	−0.01	0.13	0.04	0.13[a]	10.0%
Eq. weighted	3.32%	8.4%	1.4%	−0.06	0.17[a]	−0.08	0.16[a]	25.0%
Governance	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	1.91%	7.2%	0.8%	0.09	0.12[a]	0.17[a]	−0.07[a]	9.8%
Eq. weighted	−0.02%	6.4%	0.1%	0.01	−0.02	0.1	−0.04	4.3%

(continued)

Table 5 (continued)

March 1992–December 2004								
Community	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.27%	7.2%	1.9%	−0.04	−0.06	0.00	−0.09[a]	2.8%
Eq. weighted	1.02%	4.6%	1.0%	0.01	−0.09[a]	−0.05	0.06[a]	6.3%
Diversity	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.00%	8.1%	2.6%	−0.06	−0.21[a]	−0.26[a]	−0.03	13.1%
Eq. weighted	−0.78%	6.3%	−0.3%	−0.01	−0.06	−0.27[a]	0.06[a]	27.0%
Employee relations	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	1.00%	7.3%	2.9%	−0.02	−0.12[a]	−0.1[a]	−0.08[a]	5.9%
Eq. weighted	0.70%	5.8%	0.8%	−0.06[a]	0.03	−0.18[a]	0.07[a]	31.4%
Environment	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	−0.45%	8.1%	1.0%	−0.05	−0.04	0.05	−0.08[a]	1.3%
Eq. weighted	0.64%	8.2%	1.0%	−0.18[a]	0.12[a]	−0.05	0.05	24.0%
Human rights	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	8.78%	12.1%	5.1%	0.00	0.08	0.03	0.29[a]	21.7%
Eq. weighted	0.95%	12.4%	−0.7%	−0.11	0.02	0.11	0.19[a]	13.2%
Product safety	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	−0.51%	9.4%	0.2%	0.06	−0.29[a]	0.22[a]	−0.02	43.2%
Eq. weighted	1.42%	5.9%	2.0%	−0.05	−0.15[a]	0.06	0.04	15.4%
Controversial	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.65%	6.9%	1.3%	−0.04	−0.01	0.12[a]	−0.05	3.7%
Eq. weighted	1.44%	2.5%	1.0%	0.01	0.04[a]	−0.02	0.02	4.8%
December 2004–December 2015								
Total score	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	−0.89%	5.0%	−0.3%	−0.05	0.06	−0.01	−0.09[a]	7.5%
Eq. weighted	−0.09%	5.1%	0.3%	0.00	0.29[a]	0.21[a]	0.00	25.1%
Total ex controversial	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	1.43%	5.6%	2.0%	−0.04	0.04	−0.08	−0.09[a]	6.3%
Eq. weighted	−0.24%	5.2%	0.1%	0.00	0.31[a]	−0.2[a]	0.00	25.2%
Governance	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	1.30%	5.5%	1.5%	−0.06	−0.04	0.11	0.05	4.8%
Eq. weighted	0.11%	6.7%	0.4%	−0.08	−0.01	0.10	0.09[a]	8.2%
Community	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	−0.79%	5.6%	−0.6%	0.00	0.23[a]	0.08	−0.07[a]	22.9%
Eq. weighted	2.60%	5.5%	2.7%	0.01	0.33[a]	−0.09	0.01	20.2%
Diversity	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.14%	6.8%	1.3%	−0.04	−0.06	−0.53[a]	0.03	45.3%
Eq. weighted	1.04%	4.4%	1.2%	0.04	−0.08	−0.29[a]	0.00	21.6%
Employee relations	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.01%	4.7%	0.3%	−0.01	−0.12[a]	−0.09	−0.07[a]	6.7%
Eq. weighted	1.31%	3.8%	1.2%	0.04	0.04	−0.19[a]	0.01	11.6%

(continued)

Table 5 (continued)

December 2004–December 2015								
Environment	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	−0.38%	4.9%	0.3%	−0.04	0.05	−0.12[a]	−0.11[a]	16.1%
Eq. weighted	−0.30%	5.0%	0.2%	−0.01	0.26[a]	−0.19[a]	−0.03	25.0%
Human rights	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	1.06%	7.5%	0.8%	0.04	0.03	0.19[a]	−0.07	8.8%
Eq. weighted	2.63%	9.6%	2.2%	0.07	−0.10	−0.05	−0.04	−1.1%
Product safety	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	2.00%	5.0%	2.6%	−0.1[a]	−0.07	0.09	−0.01	7.2%
Eq. weighted	−2.43%	5.5%	−1.5%	−0.09	0.4[a]	−0.05	−0.02	33.1%
Controversial	Return	Std.	Intercept	RMRF	HML	SMB	MOM	R^2
Cap. weighted	0.37%	4.1%	0.5%	0.01	0.01	−0.18[a]	−0.01	8.8%
Eq. weighted	0.08%	0.4%	0.1%	0.00	−0.01	−0.02[a]	0.00	32.2%

The tables performance and factor exposures of KLD ratings ranked long-short portfolios during 1992–2015 and the subperiods. To conduct long-short periods, first, high (low) portfolios are formed as top 20%[b] (bottom 20%) of all companies according to the normalized and industry-neutralized net scores. Long-short portfolio is equivalent to long-high portfolio and short-low portfolio. Portfolios are formed each year lagging 3 months of KLD ratings. Monthly portfolio returns are regressed on four Fama-French factors

[a]Significant at 5% level

[b]If at 20%, there are companies with the rating lying outside the 20% range, we include all of them. For equally weighted portfolios, companies are equally weighted. For cap-weighted portfolios, we adjust the weights of the companies that share the same rating of the 20% threshold to only account for the original portion of that rating in the portfolio before including companies that fall outside the 20% range

earnings acceleration, a key variable in our stock selection model. CTEF passes the robustness tests of portfolio construction and transaction cost management tests of statistical significance.

One can create US portfolios with the CTEF model for the Russell 3000 stocks.[17] Guerard Jr. (1997a, 1997b) found that portfolios of US stocks using KLD screens for sin, nuclear, military, and environmental criteria did not significantly underperform unscreened portfolios of US stocks using a statistically based stock selection model. In this section, we replicate the "no cost to being socially responsible in investing" test. Guerard Jr. (1997a, 1997b) treated all KLD criteria as being equally weighted. There is an 8% monthly turnover constraint, a 4% upper bound on stock weight, and a 35 basis point threshold (minimum) stock weight upon initiation, builds upon Markowitz (1952) and Guerard (1997a), and is consistent with Guerard Jr. et al. (2014). The CTEF is simulated for the Russell 3000 stocks using the APT MVTaR portfolio optimization procedure, and its results are reported in Table 6. We replicate Guerard Jr. (1997a) by creating portfolios using a composite score composed of

[17]Guerard Jr., Markowitz, and Xu (2014) used all stocks on the CRSP database in their analysis of the USER model and corporate exports.

Table 6 APT risk model and optimizer and ITG transaction cost

January 2000–December 2014						
Russell 3000 universe						
	Portfolio annualized return	Portfolio annualized STD	Annualized excess returns	Information ratio	Sharpe ratio	Tracking error
CTEF	14.77%	21.01%	9.64%	0.89	0.613	10.81
ECGOV	13.41	17.89	8.28	0.97	0.644	8.53
ECOM	13.03	17.03	7.91	0.97	0.655	8.14
EDIV	12.88	17.16	7.75	0.96	0.641	8.06
EENV	13.11	17.23	7.98	0.98	0.651	8.10
EHUM	15.69	18.41	10.57	1.17	0.750	9.01
EPRO	13.63	17.98	8.50	0.99	0.653	8.57
ETOTAL	15.32	18.24	10.19	1.19	0.736	8.59
Russell 3000	5.13%	15.68%			0.207	

CTEF, 80%, and 20% the KLD raw concerns, for January 2000 to December 2014. The CTEF model produces statistically significant active returns for the 2000–2014 period. The majority of the 924 basis points of total active returns ($t = 3.33$) is composed of asset selection: 501 basis points ($t = 2.84$). The APT MVTaR CTEF XE "CTEF" model total active returns and asset selection portfolios are highly statistically significant. If one creates a composite score composed of CTEF, 80%, and 20% the KLD raw concerns (the KLD concern score = 0 if there is a concern in any of the subcategories), then specific returns (stock selection) asset selection fall for the corporate governance (ECGOV), environmental (EENV), community rights (ECOM), diversity (EDIV), and product (EPRO) KLD criteria from the CTEF specific return of 5.01%. We found that all KLD concerns have costs, except human rights and the total KLD criteria. The introduction of the human rights (EHUM) concern, composed primarily of indigenous people relations and labor strength, and total KLD criteria (ETOTAL) enhances stock selection. Corporate governance, primarily corporate compensation, reduces active returns by 130 basis points; the environment (ENV) concern costs 159 active return basis points; and the product (PRO) concern costs 110 basis points. The CTEF variable produces highly statistically significant active returns and specific returns in the Axioma attribution model with and without the SRI concerns (see Table 2). There continues to be no statistically significant costs to be an SRI investor in the United States associated with total KLD variable concern variables for January 2000 to December 2014, if one tests for equally weighted KLD criteria in an MVTaR model.

CTEF MVTaR with Normalized KLD Criteria

We use the normalized KLD criteria inside the 80/20 CTEF/KLD composite expected return model as an input into the Markowitz MVTaR optimization system, for the 2000–2014 time period. The CTEF model combined with total KLD variables and the KLD human rights criteria in the Russell 3000 analysis produces higher portfolio stock selection, as measured by Axioma specific returns, confirming the simply weighted portfolio results (see Table 6). We produce a Zephyr report that uses the ITG transaction cost model, and we report several major results for Russell 3000 stocks. First, the composite models of CTEF and KLD social criteria substantially reduce tracking errors during the 2000–2014 time period and increase the portfolio Information Ratios. Second, human rights and total KLD criteria enhance annualized excess returns considering ITG transactions costs. Third, the composite models of CTEF and KLD social criteria substantially reduce portfolio standard deviations during the 2000–2014 time period and increase the portfolio Sharpe Ratios (see Table 6). It is absolutely necessary to have a portfolio expected return model that is statistically significant for SRI/ESG integration to enhance the risk - return tradeoff. Thus, a socially responsible investor need not have to bear a lower Information Ratio and Sharpe Ratio simply, to "do good while doing good." An SRI investor can have his/her cake and eat it too!

Analysis of socially responsible investment portfolios, since the introduction of the social investing discipline, has focused on the expected costs of the constraint, but also on the possibility of portfolio design, incorporating social responsibility factors, within which the expected return could be the same or greater than a portfolio without the SRI factors. We found that a number of important investment screens based on KLD social investment variables do not cost investors in our portfolio analyses. Stocks with good KLD human rights, diversity, and total KLD concern do not cost investors holding risk constant and the KLD total social criteria, human rights, and diversity complement CTEF to enhance active returns and specific returns in the Russell 3000 and KLD-only universes for human rights and total KLD criteria. Apparently, it may be possible to have one's cake and eat it too in several KLD-based universes.

KLD STATS[18]

KLD STATS (Statistical Tool for Analyzing Trends in Social and Environmental Performance) is a dataset with annual snapshots of the environmental, social, and governance performance of companies rated by KLD Research and Analytics, Inc. KLD STATS is now sold and serviced by RiskMetrics Group, RMG. KLD covered 650 stocks in annual spreadsheets from 1991 to 2000; 1100 stocks in 2001–2002; and 3100 stocks from 2003.

[18]See "Getting Started: An Introduction to KLD STATS", WRDS.

Strength and Concern (Positive and Negative Indicator) Ratings

RMG covers approximately 80 indicators in seven major qualitative issue areas including community, corporate governance, diversity, employee relations, environment, human rights, and product.

RMG also provides information for involvement in the following controversial business issues: alcohol, gambling, firearms, military, nuclear power, and tobacco KLD STATS presents a binary summary of positive and negative ESG ratings. In each case, if RMG assigned a rating in a particular issue (either positive or negative), this is indicated with a 1 in the corresponding cell. If the company did not have a strength or concern in that issue, this is indicated with a 0.

KLD STATS data is organized by year. Each year, RiskMetrics takes a snapshot of its ratings and index membership to reflect the data at the calendar year-end. Each spreadsheet contains identifying information about the company, index membership, a listing of positive and negative ratings, involvement in controversial business issues, and total counts for each area.

Additionally, at the end of each spreadsheet is a summary count of all strengths and concerns the company received in a general category (either qualitative issue area or controversial business issue) in that year.

ESG Ratings' Definitions

Qualitative Issue Areas

Diversity (DIV-)

Strengths

CEO (DIV-str-A) The company's chief executive officer is a woman or a member of a minority group.

Promotion (DIV-str-B) The company has made notable progress in the promotion of women and minorities, particularly to line positions with profit-and-loss responsibilities in the corporation.

Board of Directors (DIV-str-C) Women, minorities, and/or the disabled hold four seats or more (with no double counting) on the board of directors, or one-third or more of the board seats if the board numbers are less than 12.

Work/Life Benefits (DIV-str-D) The company has outstanding employee benefits or other programs addressing work/life concerns, e.g., childcare, elder care, or flextime. In 2005, KLD renamed this strength from family benefits strength.

Women and Minority Contracting (DIV-str-E) The company does at least 5% of its subcontracting, or otherwise has a demonstrably strong record on purchasing or contracting, with women- and/or minority-owned businesses.

Employment of the Disabled (DIV-str-F) The company has implemented innovative hiring programs; other innovative human resource programs for the disabled, or otherwise, have a superior reputation as an employer of the disabled.

Gay and Lesbian Policies (DIV-str-G) The company has implemented notably progressive policies toward its gay and lesbian employees. In particular, it provides benefits to the domestic partners of its employees.

Concerns

Controversies (DIV-con-A) The company has either paid substantial fines or civil penalties as a result of affirmative action controversies or has otherwise been involved in major controversies related to affirmative action issues.

Non-representation (DIV-con-B) The company has no women on its board of directors or among its senior line managers.

Human Rights (HUM-)

Strengths

Positive Record in South Africa (HUM-str-A) The company's social record in South Africa is noteworthy.

Indigenous Peoples Relations Strength (HUM-str-D) The company has established relations with indigenous peoples near its proposed or current operations (either in or outside the United States) that respect the sovereignty, land, culture, human rights, and intellectual property of indigenous peoples.

Labor Rights Strength (HUM-str-G) The company has outstanding transparency on overseas sourcing disclosure and monitoring, or has particularly good union relations outside the United States, or has undertaken labor rights-related initiatives that KLD considers outstanding or innovative.

Other Strength (HUM-str-X) The company has undertaken exceptional human rights initiatives, including outstanding transparency or disclosure on human rights issues or has otherwise shown industry leadership on human rights issues not covered by other KLD human rights ratings.

Concerns

South Africa (HUM-con-A) The company faced controversies over its operations in South Africa.

Northern Ireland (HUM-con-B) The company has operations in Northern Ireland.

Burma Concern (HUM-con-C) The company has operations or direct investment in, or sourcing from, Burma.

Mexico (HUM-con-D) The company's operations in Mexico have had major recent controversies, especially those related to the treatment of employees or degradation of the environment.

Labor Rights Concern (HUM-con-F) The company's operations have had major recent controversies primarily related to labor standards in its supply chain.

Indigenous Peoples Relations Concern (HUM-con-G) The company has been involved in serious controversies with indigenous peoples (either in or outside the United States) that indicate the company has not respected the sovereignty, land, culture, human rights, and intellectual property of indigenous peoples.

References

Bettis, J. C., Coles, J. L., & Lemmon, M. L. (2000). Corporate policies restricting trading by insiders. *Journal of Financial Economics, 57*, 191–220.

Bettis, C., Guerard Jr., J. G., & McAuley, D. (2015). Is US insider trading still relevant? A quantitative portfolio approach. *Journal of Investment Management, 13*, 33–56.

Bialkowski, J., & Starks, L. T. (2017). *SRI funds: Investor demand, exogenous shocks and ESG profiles*. Working Paper.

Dorfleitner, G., Utz, S., & Wimmer, M. (2014). *Patience pays off—Financial long-term benefits of sustainable management decisions*. Working paper.

Dow, J., & Raposo, C. R. (2005). CEO compensation, change, and corporate strategy. *Journal of Finance, 60*, 2701–2728.

Dowell, G., Hart, S., & Yeung, B. (2000). Do corporate global environmental standards create or destroy value? *Management Science, 46*, 1059–1074.

Evans, J. R., & Peiris, D. (2010). *The relationship between environmental social governance factors and stock returns*. UNSW Australian School of Business Research Paper No. 2010ACTL02. Working paper.

Fama, E. F., & French, K. R. (1992). Cross-sectional variation in expected stock returns. *Journal of Finance, 47*, 427–465.

Fama, E. F., & French, K. R. (2008). Dissecting anomalies. *Journal of Finance, 63*, 1653–1678.

Fama, E. F., & French, K. R. (2016). A five-factor asset pricing model. *Journal of Financial Economics, 118*(1), 1–22.

Friede, G., Busch, T., & Bassen, A. (2015). ESG and financial performance: Aggregated evidence from more than 2000 empirical studies. *Journal of Sustainable Finance and Investment, 5*(4), 210–233.

Fulton, M., Kahn, B. M., & Sharples, C. (2012). *Sustainable investing: Establishing long-term value and performance*. Working paper.

Galema, R., Plantinga, A., & Scholtens, B. (2008). The stocks at stake: Return and risk in socially responsible investment. *Journal of Banking and Finance, 32*, 2646–2654.

Geczy, C., Stambaugh, R. F., & Levin, D. (2005). *Investing in socially responsible mutual funds*. Working paper, 2003.

Geczy, C. C., Guerard, J. B., & Samonov M. (2020). Warning: SRI need not kill your sharpe and Information Ratios—Forecasting earnings and efficient SRI and ESG portfolios. *Journal of Investing, 29*, 110–127.

Guerard Jr., J. B. (1997a). Is there a cost to being socially responsible in investing? *Journal of Forecasting, 16*, 475–490.

Guerard Jr., J. B. (1997b). Is there a cost to being socially responsible in investing? *Journal of Investing, 6*, 11–18.

Guerard Jr., J. B., Markowitz, H. M., & Xu, G. (2014). The role of effective corporate decisions in the creation of efficient portfolios. *IBM Journal of Research and Development, 58*, 6.1–6.11.

Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics, 3*, 305–360.

Kempf, A., & Osthoff, P. (2007). The effect of socially responsible investing on portfolio performance. *European Financial Management, 13*, 908–922.

Kim, D., & Starks, L. T. (2016). Gender diversity on corporate boards: Do women contribute unique skills? *American Economic Review, 106*(5), 267–271.

Lewellen, W. G. (1968). *Executive compensation in large industrial companies*. New York: Columbia University Press.

Manconi, A., Rizzo, A. E., & Spalt, O. G. (2015). *Diversity investing*. Working paper.

Manescu, C. (2011). Stock returns in relation to environmental, social and governance performance: Mispricing or compensation for risk? *Sustainable Development, 19*, 95–118.

Markowitz, H. M. (1952). Portfolio selection. *Journal of Finance, 7*, 77–91.

Renneboog, L., ter Horst, J., & Zhang, C. (2008). Socially responsible investments: Institutional aspects, performance, and investor behavior. *Journal of Banking and Finance, 32*, 1723–1742.

Riedl, A., & Smeets, P. (2017). Why do investors hold socially responsible mutual funds? *Journal of Finance, 62*, 2505–2549.

Roberts, D. (1959). Executive Compensation. Glencol, Illinois : Free Press.

Sharfman, M. (1996). The construct validity of the Kinder, Lydenberg and Domini social performance ratings data. *Journal of Business Ethics, 15*, 287–296.

Sjöström, E. (2011). *The performance of socially responsible investment—A review of scholarly studies published 2008-2010*. Working paper.

Statman, M., & Glushkov, D. (2009). The wages of social responsibility. *Financial Analysts Journal, 65*(4), 33–46.

Waddock, S. A., & Graves, S. B. (1997). The corporate social performance-financial performance link. *Strategic Management Journal, 18*, 303–319.

Wimmer, M. (2012). *ESG-persistence in socially responsible mutual funds*. Working paper.

Index

A
Abandonment option, 488, 490
Abandonment valuation, 485
Accounting period, 32, 257
Accounting reports, 33, 34
Accounting rules, 57
Accounts payables, 35, 36, 43–45, 50, 75, 100, 102, 104
Accounts receivables, 30, 68, 70, 72, 87, 93, 103, 109, 111, 203, 223
Accruals, 32, 36, 101, 104
Accrued interest, 31, 33, 537, 540–542
Accumulated depreciation, 33, 41
Acid test ratios, 99–101
Acquisition costs, 33–35, 128, 159
After-tax profits, 67
Altman Z bankruptcy prediction models, 39, 87
Altman Z scores, 87, 88, 92, 95, 101, 525, 534
America On line (AOL), 2
Amortization, 43–45, 54, 55, 58, 75, 139, 184, 193, 203, 257
Analyst-predicted earnings growth (*EGIBS*), 455
Analyst-predicted earnings-to-price (*EPIBS*), 455
Annual depreciation, 55, 226, 227
tax depreciation schedules (*see* Annual report)
Annual report, 54, 60, 70, 132, 138, 139
Asset selection, 206, 433–435, 437, 445, 446, 598
At-the-money call option, 452

B
Balance sheets, 2, 5, 8, 29–53, 58, 95, 156, 158, 177, 183, 185, 201, 202, 205, 209, 214, 224, 228, 230, 404, 406, 514
Bank loans payable, 36
Bankruptcy prediction, 39, 87, 525
BARRA MFM, 431
BARRA risk model, 401, 428, 429, 434, 437
Beaton-Tukey bi-weight criteria, 413
Best linear unbiased estimators, 269
Beta, 2, 5, 160, 161, 252, 256, 258, 260, 384, 389–402, 404, 427, 428, 431, 437–440, 442, 451, 514, 515, 572, 588
Beta times sigma (*BTSG*), 451
Black-Scholes option pricing formula, 452
Boeing Company, 46, 49, 71, 90
Bond discounts, 55, 193
Bond issues, 30, 36, 105, 107, 125, 177–180, 184, 185, 193, 194, 198, 203, 205–207
Book leverage (*BLEV*), 457
Book-to-price ratio (*BTOP*), 456
Book values, 34, 40–47, 51, 86, 88, 148, 151, 153, 156–158, 195, 209, 255, 404, 405, 412, 413, 454, 456, 457, 501, 530, 531, 537, 553
Boom, 121, 255, 488–493, 498, 503, 506
Breadth of earnings, 405
Brown, R., v
Budgeting
capital budgeting process, 485

C
Call options, 169, 180, 193, 207, 465, 466, 468–471, 473–481, 485–487

© Springer Nature Switzerland AG 2021

J. B. Guerard Jr. et al., *Quantitative Corporate Finance*,
https://doi.org/10.1007/978-3-030-43547-9

Capital Asset Pricing Model (CAPM), 8, 383–420, 430, 481, 572, 589
Capital assets, 33, 160, 203, 244, 251, 252, 257, 427, 438, 450, 515, 580
Capital equipment, 33, 252
Capital expenditures, 55, 67–70, 72, 74, 87, 139, 152, 153, 226, 235, 404, 485, 515
Capital investments, 27, 47, 67, 149, 153, 167, 233, 235, 242, 246, 260, 546, 550
Capital Market Line (CML), 388
Capital stock accounts, 38, 155, 158
Capital structure, 9, 47, 201, 202, 215, 216
Capital surplus accounts, 38
Cash flows, 2, 7–9, 34, 38, 53–77, 79, 80, 85, 90, 91, 107, 112, 116, 139, 147–149, 151, 179, 185, 187, 192, 202, 203, 205, 210, 213, 216, 223, 224, 226, 227, 234, 235, 238, 243–246, 248, 249, 252, 253, 256, 258–260, 404–406, 411–413, 456, 485–488, 490, 491, 503–506, 514, 515, 517, 526, 529–532, 538–542, 566
Center for Research in Security Prices (CRSP), 2, 387–390, 392–396, 399, 401, 402, 404, 406, 410
Closing price, 454, 456, 457
Coefficient of determination, 267
Common equity per share outstanding, 404
Common stocks, 8, 26, 30, 37–50, 53, 56–58, 62, 67–76, 86, 90, 122–124, 127, 128, 138, 145–147, 151, 155, 167–172, 174, 195–197, 204, 207–209, 212, 214, 216, 255, 276, 288, 412, 465, 466, 469, 540, 542, 577–579
Composite earnings forecasting, 405
Composite earnings forecasting variable, 420, 433
Composite Equity Valuation Model, 406–420
Composite Firm Relative Valuation Ratios, 84
Composite security valuation models, 413, 414, 565
Composite valuation models, 404
Compustat, 2, 9, 42, 70, 139, 140, 154, 405, 406, 574, 575
Confidence intervals, 268, 272, 273, 413, 441
Contracts, 6, 7, 14, 15, 17, 20, 37, 39, 83, 184, 204, 207, 209, 210, 216, 465, 557, 584
Copyrights, 35
Correlation coefficients, 267, 269, 302, 383, 387, 505, 506, 563, 576
Correlation matrices, 275, 362, 363, 368, 369, 371, 377, 560, 563
Cost of capital, 2, 5, 117, 165, 167, 168, 215, 216, 218–223, 233–235, 238, 239, 241–244, 246, 247, 251, 253, 259, 427, 480, 485, 488, 490, 494, 514, 515, 550–551, 566, 580, 583, 588
Cost of equity capital, 145, 185, 255, 256, 515
Cost of goods sold, 32, 54, 58
Covariance matrix, 363, 369–372, 376–378, 428, 429, 432, 459, 460
Credit ratings, 31, 104, 155
Cross-sectional return, 282
CTEF, 405, 411–420, 433–437, 445, 448, 449, 566–569, 594, 597–600
Cube of the log of market capitalization (*LCAPCB*), 453
Cumulative range (*CMRA*), 451
Currency sensitivity, 432, 436, 458
Current analysis ratios, 84
Current assets, 30–33, 36, 41, 43–45, 50, 67, 79, 80, 83, 84, 99–101, 157, 158, 201, 203, 235, 245, 249, 403, 411, 531
Current assets-to-total assets (CATA), 42, 46, 47
Current liabilities, 6, 35–37, 41–45, 50, 67, 79, 84, 85, 99–101, 177, 201, 202, 209, 211, 212, 224, 235, 457
Current liabilities-to-total assets (CLTA), 42, 46, 47
Current ratio, 36, 84, 85, 88, 89, 100, 201, 406

D

Daily standard deviation (*DASTD*), 451
Debentures, 36, 105, 130, 139, 142, 177, 181, 195–197, 212, 253, 540, 542
Debt financing, 41, 152, 168, 174, 186, 198, 200, 253
Debt levels, 224, 507
Debt-to-assets ratio, 92
Debt-to-equity ratio, 180, 218, 219, 483, 517
Decision making process, 488
Deferred credits, 37, 193
Dependent variable, 263, 264, 267, 268, 270, 275, 285, 337, 406
Depletion, 35, 54, 203, 223, 224, 257
Depreciation, 33–35, 54, 55, 57, 58, 67–72, 75, 83, 111, 112, 138–140, 156, 179, 184, 199, 200, 203, 223–227, 235, 236, 244–246, 256, 257, 260, 384, 456, 481, 515, 516, 530, 531, 540, 541, 548, 556, 580
Depreciation allowances, 34, 226
Depression, 125, 488, 498, 499, 503, 506
DFBETAs, 273, 274, 290

Discounted cash flows, 233, 235, 252, 259, 485, 514, 515, 517, 522
Discount rates, 151, 166, 183, 217, 233, 234, 238, 241, 244, 245, 247, 249, 250, 258, 439, 485, 486, 515, 579
Dividends, 1, 5, 9, 19, 24, 36, 38–40, 47, 54–59, 67–72, 74–76, 83, 100, 123, 130, 139, 146–164, 168–172, 180, 181, 195, 196, 205, 219, 225, 226, 255, 256, 383–385, 387, 404, 411, 412, 434, 439, 454, 458, 475, 477–479, 486, 537, 541, 542, 579, 580
Dividend yields, 123, 147, 148, 165, 170, 255, 411, 432, 437, 458
Dominion Resources, 2, 42, 47, 146, 159–161, 384, 386, 428, 429, 438
Dow Jones, 397, 565
Debt-to-asset ratio (*DTOA*), 457
DuPont, 24, 42, 84, 87, 88, 146, 384, 386, 428, 520

E
Earned surplus, 39, 40, 53, 57
Earnings, 5–8, 30, 34, 38–40, 54, 56–60, 74, 76, 86, 87, 103, 105, 106, 116, 117, 123, 126, 130, 138, 146–149, 151–153, 155–159, 162–171, 179, 180, 196, 197, 202, 203, 205, 210–218, 223–227, 243, 246, 248, 255–257, 404–406, 413, 415–420, 432–436, 438, 443, 444, 454–457, 480, 497, 502, 507–511, 513–518, 526, 530–532, 540–543, 548, 549, 572, 573, 578–580, 582–584, 594
Earnings before Interest, 55, 58–63, 211, 222, 224, 235
Earnings growth rate over last 5 years (*EGRO*), 455
Earnings per share, 2, 5, 8, 53, 58, 60, 62, 74, 147, 148, 157–160, 162, 164, 168, 212–216, 255, 403–405, 410–413, 431–434, 455–457, 574, 578, 582, 583
Earnings revisions, 405, 594
Earnings variability, 456–457
Earnings yield, 130, 166, 167, 222, 255, 256, 432–435, 445, 455–456, 594
EBIT to interest charges, 212, 224
Economic conditions, 186, 260, 289, 533, 538, 540, 541
Economic value, 34, 156, 184, 480, 577
Economies of scale, 502
Efficient markets, 132, 570
Efficient portfolios, 434, 449
Enron, 21, 23, 93, 535, 574, 576, 589
Equity funds, 9, 58, 122, 157, 167, 168, 170, 171, 196, 197, 201, 215, 255–257, 428, 550, 589
Error terms, 264, 265, 268, 269, 271, 272, 311, 350, 389, 413, 414
Estimation errors, 428, 431
ETP5, 456
Excess returns, 32, 152, 403, 405, 410, 415, 430, 432, 434, 437, 452, 458, 479, 496, 519, 520, 522, 555, 563
Exercise price, 465, 466, 468, 469, 471, 473, 478–481, 485–487, 491
Expected cash flows, 233, 486, 487, 503, 514, 566
Expected net present value, 487, 488, 490
Expected present value, 488, 490, 491
Expected returns, 8, 145, 218, 221, 249, 250, 384, 386–389, 401, 403, 404, 439, 442, 459, 510, 594, 599
Explanatory variables, 302, 336
Exposure to foreign currencies (*CURSENS*), 458
External funds, 86, 140
External risks, 215, 216
Extraordinary items, 61–63, 456, 457

F
Financial analyses, 36, 60, 83, 84, 86, 87, 93, 113, 156, 209, 226, 227, 564
Financial health, 47, 87–89, 93, 95
Financial ratios, 47, 87–89, 185, 187
Financial research, 131
Financial risks, 5, 80, 94, 168, 173, 185, 198, 201, 210–211, 214–218, 224–227, 247
Financial statements, 2, 9, 30, 76, 93, 185, 387
Finished goods, 31
Firm size, 435
First-in, first-out, 32
Fixed assets, 21, 30, 33–35, 41, 47, 54–56, 67, 79, 83, 100, 139, 179, 180, 182, 223, 224, 226, 230, 245, 257, 508, 531
Fixed plant, 4, 33, 223
Flow of funds, 1, 31, 79, 83, 248, 289
Foreign currencies, 259, 458, 570
Franchises, 21, 22, 35, 54, 156
Franco Modigliani, 218
Frank Russell universes, 434, 435
See also Russell 1000; Russell 2000; Russell 3000
Free cash flow, 90, 91, 404
FSPLIT, 454

F-statistic, 318
F-test, 268
Full information, 132
Fundamental analysis, 404–405
Future cash flows, 183, 228, 230, 249, 250, 259, 485–487, 566
F-value, 268, 281

G
General ledger, 87
Global Portfolio Research Department (GPRD), 406, 414
Going concern, 15, 31, 179, 529–531, 536, 540, 543
Goodness of fit, 266, 275, 285
Goodwill, 24, 35, 43–45, 501
Great Depression, 403, 525
Gross national product, 11, 119
Growth rate in total assets (*AGRO*), 454, 455

H
Historical alpha (*HALPHA*), 452
Historical data, 451
Holding period returns (HPR), 156, 384–386, 563
Hurdle rate, 246, 247, 494

I
IBM, 2, 23, 24, 42, 43, 46, 48, 60–63, 67–69, 71, 74, 76, 88, 89, 92, 96, 146, 153–155, 384–388, 390, 392–396, 402, 428, 429, 438, 465–469, 472, 473, 479, 534
Income statements, 2, 29, 32, 33, 53–76, 95, 129, 146, 225, 404, 406
Income taxes, 13–15, 21, 34–36, 50, 55, 56, 59, 61–63, 76, 128, 137, 138, 168, 170, 222, 226, 227, 578
Independent variables, 210, 263, 264, 266, 267, 269, 272–275, 406, 413, 438, 574
Indicator for firms outside US-E3 estimation universe (*NONESTU*), 458
Individual investors, 27, 118, 126, 188, 221, 383, 589
Information coefficients, 410, 414, 434, 555
Insurance companies, 3, 36, 101, 107, 108, 126, 135, 141, 181, 182, 184, 188, 192, 197, 198, 204, 223, 539
Intangible assets, 35, 43–45, 50, 86, 257
Interest expenses, 30, 42, 55, 58, 60–63, 183, 193, 212, 213, 506
Internal rate of return, 234–237, 239, 243, 246, 488, 490
Interest rates, 7, 37, 80–82, 104, 105, 108, 109, 117, 119, 136, 138, 165, 169, 177, 181, 183–196, 198–200, 204, 206, 207, 218, 220, 228, 241, 247, 253, 256–259, 276, 288, 289, 383, 438, 549–551, 558–560
Internal risk, 174, 215, 216
Inventories, 6, 14, 30–32, 39, 41, 43–45, 50, 68, 70, 72, 75, 79, 80, 84, 85, 87, 93, 99–101, 103, 108, 119, 203, 223, 244, 252, 287, 546, 547, 549

J
Joint ventures, 125

L
Lambda, 434, 435, 442, 443, 445
Large market capitalization, 434
Lawsuits, 56
Least squares estimates, 269
Least squares regressions, 264, 266, 269, 270, 274
Lerner-Carleton derivation, 213, 218
Leverage, 9, 61, 86, 88, 95, 148, 152, 170, 210–215, 217, 220, 222, 223, 253, 391, 398, 431, 432, 436, 445, 457, 466, 468, 469, 497, 507, 526, 557, 566, 580, 581
Liabilities, 5, 8, 12–17, 20, 24, 27, 29, 30, 34–41, 43–46, 50, 51, 56, 57, 67, 68, 70, 72, 75, 104, 113, 126, 155, 156, 158, 168, 173, 177, 185, 192, 193, 200, 202, 209, 211–213, 216, 221, 223, 225, 226, 228, 245, 403, 480–483, 498, 501, 502, 510, 530, 531, 533, 536, 537, 577, 581
Linear least squares regression, 264
Liquidations, 6, 15, 40, 130, 157–159, 168, 169, 248, 525–543, 581
Liquidation value, 40, 157, 158, 169, 248, 529–531, 536, 538, 540, 579
Log of market capitalization (*LNCAP*), 453
Log of stock price (*LPRI*), 451
Long-term debt, 6, 36, 37, 42, 46, 51, 55, 69, 70, 72, 76, 80, 83, 142, 177–208, 212, 223, 224, 255, 454, 457, 531
Lucent, 534

M
Market leverage (*MLEV*), 457
Market model regressions, 452

Market values, 1, 29, 35, 38, 40, 116, 128, 156, 159, 160, 162, 170, 184, 210, 214, 216, 218, 222, 224, 388, 427, 439, 457, 475, 481–483, 501–503, 507, 514, 517, 520, 522, 540–542, 575–577
Marketing firms, 223
Markowitz, 8, 384, 386, 388, 389, 406, 414, 415, 420, 427, 439, 442, 443, 445, 449, 459, 461, 599
Mean-variance analysis, 427
Mean-variance efficient portfolios, 153, 388
Mergers, 9, 25, 42, 132, 135, 148, 170, 194, 202, 252, 255, 483, 495–522, 553, 582
Merton Miller, 218
Middle market capitalization, 434
Minority interest, 40, 43–45, 51, 61–63
Minority stock, 17, 40
M&M Proposition, 218
Modern portfolio theory, 389–403, 406
Momentum, 406–420, 432, 436, 452, 461, 565, 593, 594
Monthly returns, 221, 452
Mortgage notes, 36
Multifactor risk models, 384, 403, 427–462
Multiple-factor model, 403, 429, 433, 434

N
Natural disasters, 56
Natural resources, 35, 547
Negative rate of return, 211
Net cash flow, 184, 234, 249, 259
Net current asset value, 158, 403, 411, 581
Net fixed assets, 33, 41
Net future cash flows, 233
Net income, 13, 38, 47, 53, 56, 58–63, 67–72, 74–76, 92, 139, 146, 168, 183, 219, 235, 575, 576
Net present values (NPVs), 194, 228, 233–237, 239–244, 246, 251, 485–488, 490, 491, 494
Net profit rate of return, 86
Net profits, 56, 86, 95, 225, 520, 532
Net receivables, 31, 43–45
Net working capital, 36, 79, 80, 83, 84, 94, 99–101, 113, 180, 181, 186, 201, 212, 235, 508, 580
Net worth, 29, 30, 35, 86, 201, 202
Non-estimation universe indicator, 458
Nonlinear regression, 270
Nonrecurring losses, 56
Normal distribution, 272, 413, 475, 477, 479
Notes payable, 35, 36
Null hypotheses, 219, 267, 268, 270, 276, 315, 316, 337, 410

O
Operating costs, 54, 224, 242, 244, 250, 257
Option-implied standard deviation (*OPSTD*), 452
Ordinary least squares, 264–271, 274, 282, 390, 406, 413, 444
Outlier-adjustment, 354, 356, 359, 361, 365, 367, 375, 406
Owners' equity, 7, 86
Ownership capital, 13, 29, 86, 130, 146, 151, 153, 167, 168, 192, 202, 210, 211, 216, 217, 586
Ownership investment, 211

P
Paid-in surplus, 38, 155
Par values, 17, 38, 40, 155, 156, 158, 169, 187, 198, 199, 203, 509, 540–542
Parent corporations, 124–125
Patents, 35, 40, 54, 125, 486
Payout ratio, 434, 454
Permanent financing, 107, 110, 112, 209
Permanent investments, 35, 87
Petroleum industry, 428
Pierre DuPont, 87
Population regression line, 264
Portfolio optimizations, 406, 410, 434, 437, 450, 460, 461, 597
Portfolio variances, 386, 480
Predicted dividend yield (*P_DYLD*), 458
Preferred stocks, 36, 38, 40, 43–45, 51, 55, 57, 61–63, 87, 130, 145–174, 181, 209, 210, 218, 226, 255, 256, 542
Premiums, 8, 37, 38, 40, 55, 128, 141, 152, 160, 169, 171–173, 188, 193–196, 198, 203, 204, 207, 220, 250, 256, 258, 260, 383, 438, 465, 466, 468–471, 480, 485, 496, 513, 515, 518–520, 522, 550, 558, 559, 583
Prepaid expenses, 31–33
Present values, 34, 55, 162–166, 183, 184, 187, 194, 198, 233–235, 238–240, 242, 244–246, 248, 252, 258, 259, 477–479, 485–487, 490, 491, 517, 579
Price-earnings ratio, 403
Price of common stock, 476
Probability distribution, 472

Profit and loss economy, 211
Profit and loss statement, 32
Profitability ratios, 89, 92
Profits taxes, 36, 55, 57, 59, 532
Put options, 469, 479, 488, 491, 493, 556
PVCF, 486

Q
Quick ratio, 431

R
Random variables, 269, 272, 305
Rate of profits, 80, 214, 247, 257, 496
Rate of return, 5, 8, 27, 55, 86, 87, 117, 145, 148, 149, 151, 156–160, 162–167, 169, 173, 174, 183, 184, 198, 210–212, 216–218, 221, 234, 235, 237, 239, 243–245, 247–251, 253, 255, 383, 389, 401, 403, 430, 466, 468, 510, 515, 525, 530–532, 572, 579, 580
Ratio analysis, 9, 79–96, 104
Ratio of high price to low price over the last month (*HILO*), 451
Raw materials, 31
Real options analysis, 485
Recent earnings change, 455
Recessions, 4, 289, 488–493, 498, 503, 506, 526, 545, 550
Regression analysis, 263–282, 303, 304
Regressions, 2, 219, 263–287, 301, 306, 311, 316, 318, 334, 337, 343, 350, 373, 390, 391, 398, 399, 402, 404, 405, 410, 412–415, 419, 420, 444, 445, 447, 451, 452, 454, 455, 458, 479, 575, 576, 593, 594
Reinvested earnings, 19, 57
Relative book-to-price, 405
Relative cash-to-price, 415
Relative sales-to-price, 415
Relative strength (*RSTR*), 452
Reported earnings, 57, 223, 224, 405, 413
Research and development, 1, 60, 67, 70, 485, 520
Reserve for bad debts, 31
Retained earnings, 15, 37–39, 43–45, 53, 55, 57, 59, 122, 138, 139, 151–153, 156, 173, 177, 202, 223, 256, 548, 549, 553, 579, 580
Return on assets, 59, 86, 90, 91, 213, 214, 432, 589
Return on equities, 41, 59, 90–92, 160, 167, 174, 177, 210, 213, 214, 217, 218, 220, 255, 389, 412, 515, 575, 580
Revision breadth, 405
Risk exposures, 430, 459–461
Risk indexes, 433–435, 453, 458
Risk minimization, 387
Risk models, 384, 403, 416, 427–450, 459–461, 584, 594
Risk prediction, 447, 448, 460
Risk-return analysis, 383
Robust regressions, 273–275, 390, 393, 395, 399–401, 405, 406, 414, 415, 444
Russell 1000, 434–436
Russell 2000, 434–436
Russell 3000, 433, 435, 437, 443, 567, 597–599

S
Sales efficiency ratio, 92, 525
Sales-to-price ratio, 405
Sales to total assets ratio, 86
Securities and Exchange Commission, 135, 138, 139, 584
Security valuation, 160, 403
Sensitivity of changes in trading volume to changes in aggregate trading volume (*VOLBT*), 451
Serial dependence (*SERDP*), 452
Share turnover over the last year (*STOA*), 453
Short-term creditors, 6, 146
Simple, linear, least squares, 264
Single-factor model, 403
Size nonlinearity, 453
Sources of funds, 8, 55, 67, 139, 168, 256, 257
S&P 500, 92, 160, 161, 383, 390, 393, 397, 402, 469, 515, 522, 565, 588, 589
Standard deviation of analysts' prediction to price (*SPIBS*), 457
Standard deviations, 272–274, 352, 354, 356, 365, 367, 383, 384, 386–389, 432, 440, 451, 452, 454, 457, 478, 480, 503, 505–507, 599
Standard & Poor's, 288
Stock Guide, 384
Stockholder wealth, 1, 5, 8, 12, 151, 152, 213, 405, 485, 520, 587
Stock issuance, 76, 501
Stock selection models, 162, 405, 420, 444, 594, 597
STOQ, 453
Strategic planning, 487

Subsidiaries, 20, 31, 40, 41, 54, 56, 59, 74, 104, 108, 124–125, 129, 134, 135, 141, 178, 182, 184, 496, 497, 519–521, 580
Summarization annuity formula, 234
Systematic risk, 5, 160, 250, 252, 389, 427, 432, 437, 447, 448, 459–461, 472, 517

T
Tax rates, 13, 14, 19, 59, 170, 225, 227, 235, 245, 253, 260, 515
Term loans, 7, 36, 177, 181
Time series models, 301–303, 311, 314, 316, 317, 319, 325, 327, 335, 352–354, 357, 359, 364, 373
Time Series of Ratios, 89–92
Times interest earned, 224, 226
Trade acceptances payable, 35
Trade accounts, 211
Trading activity, 432, 453–454
Trailing annual earnings-to-price (*ETOP*), 456
Treasury bills, 105, 108, 160, 383, 438, 478, 515
t-statistics, 219, 326, 336, 414
t-tables, 267
t-test, 267, 304, 312
t-value, 267, 268, 276, 277, 279, 281, 338–340, 344, 345, 347, 354, 356, 360, 365, 367, 374, 390, 434

U
U.S. Equity Model (US-E3), 432

V
Valuations, 2, 3, 8, 31–33, 35, 40, 53, 57, 59, 60, 76, 147, 157, 159, 162–166, 218, 252, 260, 387, 427, 439, 465, 485, 503, 514–517, 580
Value-added ratios, 86
Value investing, 439
Variability in capital structure (*VCAP*), 454
Variability in earnings (*VERN*), 456
Volatilities, 159, 166, 186, 200, 217, 249, 432, 436, 439–442, 444, 451–452, 460, 478–480, 557, 583
Volume to variance (*VLVR*), 454

W
Weighted average cost of capital, 245, 255, 515
Weighted latent root regression (WLRR), 406–409, 411–415
Wharton Research Data Services, 2, 42
Working capital analysis, 83, 93–95

Z
Zero return, 211